FRM PART II • 2026 • EXAM 25 NOVEMBER 2026

# Risk Management and Investment Management

## Notes + Question Workbook

P1 • P2 • P3, in order of importance

60 learning objectives from 17 readings, banded by the Master LO Map v3. P4 is excluded. Built from Vardeez's notes, the GARP 2025 RMIM book (IM-1 to IM-7 and IM-12), the Schweser-Notes 2026 Book 5 (the content authority for IM-8 to IM-11 and IM-13 to IM-17, which are not in the 2025 book), the Tier A and B–D Recall Workbooks and the RMIM Question Bank.

### How to use this book

- The book runs **P1, then P2, then P3**. Inside each part, LOs follow syllabus order. The syllabus index at the front lists all 60 LOs in GARP order with their band and page.
- Each LO is one block: **focus box** (why it sits in its band, Vardeez's pointer and the reported exam shapes), **notes** limited to that focus, a **formula card** or comparison table, **exam traps**, then **6 to 10 practice questions**. **Solutions** sit at the end of each block, so attempt first.
- LO wording is verbatim from the official GARP 2026 learning objectives, as listed in the Master LO Map.
- Type tags: CALC numerical questions expected; CONCEPT statement questions (only the reported workbook calculations appear); BOTH a mix set by the reported shapes and Vardeez's focus.
- Question sources, in priority order: Tier workbook (August 2026 recall and mapped shapes), GARP-tagged bank questions (practice exams, GARP book, mocks), other bank questions, and *New* questions written in GARP style from the readings where an LO was short. Every number has been recomputed.
- Green-edged boxes mark corrections to the source notes or to a bank key. Figures are drawn for black-and-white printing.

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#### IM-3 • Alpha (and the Low-Risk Anomaly)

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| **IM-3 d** | BOTH | Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. | 24 |

#### IM-5 • Portfolio Risk: Analytical Methods

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| **IM-5 a** | CALC | Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. | 36 |
| **IM-5 c** | BOTH | Apply the concept of marginal VaR in making portfolio management decisions. | 52 |
| **IM-5 d** | BOTH | Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. | 63 |

#### IM-7 • Portfolio Performance Evaluation

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| **IM-7 c** | BOTH | Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. | 87 |
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IM-10 • The Rise and Risks of Private Credit

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IM-11 • Private Markets Investing

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| **IM-11 c** | BOTH | Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. | 230 |
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| **IM-11 f** | CONCEPT | Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. | 250 |

IM-14 • Madoff: A Riot of Red Flags

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IM-2 • Factors

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IM-4 • Portfolio Construction

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IM-14 • Madoff: A Riot of Red Flags

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IM-17 • Illiquid Assets

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RISK MANAGEMENT AND INVESTMENT MANAGEMENT • PART

# P1

## Most important

Tier A recall topics and the concepts Vardeez flags in Very Important chapters. 24 LOs from 9 readings.

IM-3 • Alpha (and the Low-Risk Anomaly) • Vardeez: VERY IMPORTANT

# IM-3 b

P1 CALC

Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio.

Workbook: A18 Performance Measures and Attribution • Tier A • August 2026 recall C4 Policy-Mix VaR and Risk Budgeting • Tier C

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic: A18 was on the August 2026 recall ("Jensen's alpha, Sharpe and other metrics theory"), and Map #36 reports "IR, alpha" on 5 of 16 recall sheets. C4 (Map #35, 4 of 16) adds an "IR/2 active risk numerical".

**Vardeez:** no pointer on this LO; the chapter is VERY IMPORTANT. His note: alpha, the information ratio and the Sharpe ratio share one numerator and differ in the denominator.

**Reported exam shapes**

- **S3 (A18)** Jensen's alpha and the information ratio: a plain Jensen's alpha, and the IR of an active manager added on top of an index core.
- **S4 (C4)** IR and active risk numerical: the IR from an expected excess return and a tracking-error volatility (TEV), and the IR of a pool of managers.

The harder Sharpe shapes of A18 (S1, S2) and the "which measure" shape (S6) are drilled in IM-7 b and c and in IM-5 a. Workbook C4.2 (shape S4) is worked as Question IM-6 h.6.

**What the questions look like:** almost all numerical. You get a fund, a benchmark, a risk-free rate and either a beta or a regression output; compute the alpha against the right benchmark, then divide by the right risk. The wrong options use the wrong benchmark (the raw index, which assumes a beta of 1, or the risk-free rate) or the wrong denominator (total volatility instead of tracking error).

## 1. Alpha is a statement about a benchmark

Ang opens with a warning: alpha "is often interpreted as a measure of skill. That is debatable. It is first and foremost a statement about a benchmark." All four quantities in this LO are built on one series, the **excess** (or **active**) return over a benchmark, $r_t^x = r_t - r_t^{\text{bmk}}$ (Eq. 3.1). Calling it "active" assumes the benchmark is *passive*: it can be produced with no investment knowledge, like an S&P 500 or Russell 1000 index fund costing a few basis points.

$$\alpha = \frac{1}{T}\sum_{t=1}^{T} r_t^x \qquad \text{TE} = \bar{\sigma} = \text{stdev}\left(r_t^x\right) \qquad \text{IR} = \frac{\alpha}{\bar{\sigma}} \qquad \text{SR} = \frac{\bar{r}_t - \bar{r}_f}{\sigma}$$

| | What it is | Read it as |
|---|---|---|
| **Alpha (Eq. 3.2)** | Average excess return over the benchmark | Return added beyond the passive alternative |
| **Tracking error (Eq. 3.3)** | Standard deviation of the excess return | How far the manager strays; a TE limit caps the freedom to deviate |
| **Information ratio (Eq. 3.4)** | Alpha divided by tracking error | Alpha per unit of active risk; above 1 is uncommon |
| **Sharpe ratio** | Average return over $r_f$ divided by the volatility of the asset | The IR when the benchmark is the risk-free rate |

- **The Sharpe ratio is a special case.** When the benchmark is $r_f$, alpha is the average return in excess of $r_f$, and the IR "coincides with the Sharpe ratio". The denominator is then the asset's own volatility (the volatilities of raw and excess returns are almost the same unless $r_f$ itself is volatile, as in some emerging markets).
- **Why divide at all.** "Alpha by itself could be produced by a manager taking large amounts of risk." The IR is the average excess return per unit of risk taken. Information ratios above one are not common, although many hedge funds raising money claim them.
- **The discriminator.** Alpha, the IR and the Sharpe ratio share a numerator once the benchmark is fixed. Change the benchmark and all three change; change only the risk measure and you move between IR (tracking error) and Sharpe (total volatility).

## 2. One fund, two benchmarks: Martingale and Jensen's alpha

Martingale's low volatility strategy is built on the Russell 1000 universe. Full sample, against one-month T-bills: mean excess return 8.59%, volatility 12.22%, so a Sharpe ratio of $8.59/12.22 = 0.70$ against 0.45 for the Russell 1000. Its tracking error against the Russell 1000 is 6.16%.

**Naive benchmark (the index itself).** $r_t = 0.0150 + r_t^{R1000} + \varepsilon_t$: the strategy beat the index by $\alpha = 1.50\%$ a year, IR $= 1.50/6.16 = 0.24$. Using the index alone *assumes a beta of 1*.

**Risk-adjusted (CAPM) benchmark.** Regress excess returns on index excess returns:

$$r_t - r_{f,t} = 0.0344 + 0.7272\left(r_t^{R1000} - r_{f,t}\right) + \varepsilon_t$$

$$\iff \qquad r_t = 0.0344 + \underbrace{0.2728\, r_{f,t} + 0.7272\, r_t^{R1000}}_{r_t^{\text{bmk}}} + \varepsilon_t$$

The benchmark is 27% T-bills plus 73% Russell 1000, rebalanced monthly. Against it, $\alpha = 3.44\%$ a year and the IR is a "very high" 0.78. The beta of 0.73 is a *portfolio weight*: the slope tells you how much of the index to hold, the rest goes into T-bills.

**Jensen's alpha** is the alpha against this CAPM benchmark (Jensen, 1968):

$$\boxed{\alpha_P = \bar{r}_P - \left[\bar{r}_f + \beta_P\left(\bar{r}_M - \bar{r}_f\right)\right]} \qquad \beta_P = \frac{\operatorname{cov}(r_P, r_M)}{\sigma_M^2} = \rho_{PM}\frac{\sigma_P}{\sigma_M}$$

- **Direction rule.** With a positive market premium, a naive index benchmark is *riskier* than a fund with $\beta < 1$, so alpha and IR come out **too low** (Martingale: 1.50% against 3.44%). For a fund with $\beta > 1$ the naive benchmark is less risky and flatters the fund: alpha and IR come out **too high**.
- **Beta from a correlation.** When the stem gives $\rho$, $\sigma_P$ and $\sigma_M$, beta is $\rho\sigma_P/\sigma_M$. Using $\rho$ alone, or $\sigma_P/\sigma_M$ alone, are the two classic slips.
- **Ideal benchmarks** (IM-3 c, not in this band) are well defined, tradeable, replicable and adjusted for risk. "Sadly, most benchmarks used in the money management business are not risk adjusted."

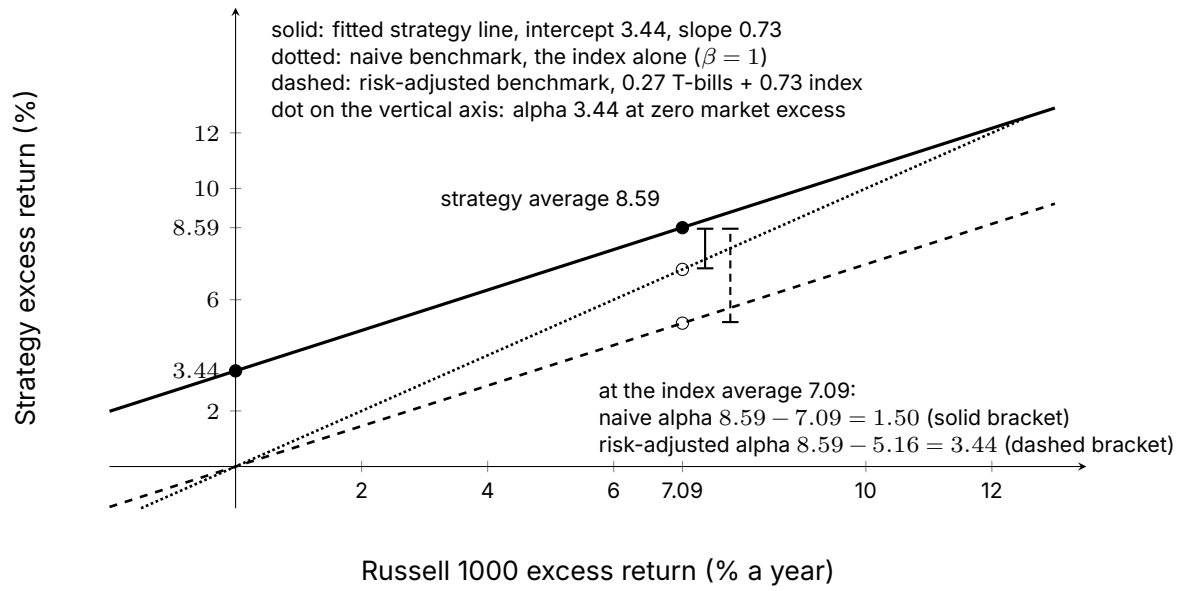



Figure 1. Martingale's low volatility strategy against two benchmarks (GARP Ch.3 numbers; the index's average excess return of 7.09% is implied by the 1.50% naive alpha). At the index's average, the strategy earns 8.59%. The naive benchmark (dotted) predicts 7.09%, so alpha is 1.50; the beta-adjusted benchmark (dashed) predicts only 5.16%, so alpha is 3.44. Same returns, different benchmark, different alpha.

## 3. Which tracking error?

Tracking error is always the standard deviation of the excess return over *the benchmark you used for alpha*. Three practical cases:

1. **From a return history.** Subtract the benchmark year by year, take the mean (alpha) and the *sample* standard deviation (divide by $n - 1$) of the differences. Betas in the table are a red herring when the question asks for the IR "measured against the benchmark".
2. **Against a risk-adjusted (regression) benchmark.** The excess return is $\alpha + \varepsilon_t$, so tracking error is the residual volatility $\sigma(\varepsilon)$, the standard error of the regression. Ang: academics then call tracking error "idiosyncratic volatility". Bodie, Kane and Marcus's $\alpha_P/\sigma(e_P)$ (IM-7 b) is this case.
3. **Against a naive benchmark when $\beta \neq 1$.** The excess return also carries the unhedged market exposure: with a constant $r_f$, $\text{TE}^2 = (1-\beta)^2\sigma_M^2 + \sigma^2(\varepsilon)$.

**Martingale reconciled.** The 0.45 index Sharpe ratio and the 7.09% index excess return imply $\sigma_M = 7.09/0.45 = 15.76\%$. The IR of 0.78 implies a residual volatility of $3.44/0.78 = 4.41\%$. Then $\sqrt{(0.2728 \times 15.76)^2 + 4.41^2} = \sqrt{4.30^2 + 4.41^2} = 6.16\%$: the tracking error against the Russell 1000 is half unhedged market risk (the strategy holds only 73% of the index) and half stock-specific risk. That is GARP's point that "the tracking error is high because Martingale's product has a low beta".

**Correction to the source notes.** SchweserNotes (Reading 85, LO 85.c) computes the risk-adjusted information ratio as $0.0344/0.0616 = 0.5584$. That divides the alpha measured against the beta-adjusted benchmark by the tracking error measured against the *naive* Russell 1000. An IR must use the tracking error against the same benchmark as its alpha. GARP reports 0.78, which implies a tracking error of about 4.4% against the 0.27/0.73 benchmark (Section 3). Schweser's naive-benchmark IR, $0.0150/0.0616 = 0.24$, is consistent with GARP.

## 4. The four measures on one fund (shape S3)

A fund returned 13% with a volatility of 20% and a beta of 0.9; the index returned 11%, T-bills 3%, and the residual volatility of the CAPM regression is 4%.

| Measure | Working | Value |
|---|---|---|
| Sharpe ratio | $(13-3)/20$ | 0.50 |
| Naive alpha (beta assumed 1) | $13-11$ | 2.0% |
| CAPM benchmark return | $3+0.9\times(11-3)$ | 10.2% |
| Jensen's alpha | $13-10.2$ | 2.8% |
| Information ratio (risk-adjusted) | $2.8/4$ | 0.70 |

Here $\beta<1$, so the naive alpha (2.0%) is below Jensen's alpha (2.8%), the Martingale direction. The fund's 20% volatility only enters the Sharpe ratio; the IR never uses it.

## 5. Time units

Means scale with time and volatilities with its square root, so the IR scales with $\sqrt{\text{time}}$:

$$\alpha_{\text{annual}} = 12\,\alpha_{\text{month}}, \qquad \text{TE}_{\text{annual}} = \sqrt{12}\,\text{TE}_{\text{month}}, \qquad \text{IR}_{\text{annual}} = \sqrt{12}\,\text{IR}_{\text{month}}$$

Berkshire's CAPM alpha of 0.72% a month is $0.72\times 12 = 8.6\%$ a year. A monthly alpha of 0.30% with a monthly tracking error of 1.50% (monthly IR 0.20) is 3.60% a year over $1.50\sqrt{12} = 5.20\%$: an annual IR of $0.20\sqrt{12} = 0.69$. Scaling a mean by $\sqrt{12}$, or an IR by 12, are the slips.

## 6. Active risk across managers (shape S4)

For an active manager, the IR is the expected excess return over the benchmark divided by the tracking-error volatility, $\text{IR}_i = \alpha_i/\omega_i$. For a pool of managers whose active returns are **independent**, active returns add with the weights and active variances add with the squared weights:

$$\alpha_P = \sum_i w_i\alpha_i, \qquad \omega_P = \sqrt{\sum_i w_i^2\omega_i^2}, \qquad \text{IR}_P = \frac{\alpha_P}{\omega_P} \le \sqrt{\sum_i \text{IR}_i^2}$$

**Example.** Manager X: $\alpha=2\%$, $\omega=4\%$; Manager Y: $\alpha=3\%$, $\omega=6\%$; both have $\text{IR}=0.50$.

- **50/50:** $\alpha_P = 2.5\%$, $\omega_P = \sqrt{0.25(16)+0.25(36)} = 3.61\%$, $\text{IR}_P = 0.69$.
- **60/40** (weights proportional to $\alpha_i/\omega_i^2$): $\alpha_P = 2.4\%$, $\omega_P = 3.39\%$, $\text{IR}_P = 0.707 = \sqrt{0.5^2+0.5^2}$, the maximum.
- $N$ **equal, independent strategies** with the same IR: $\text{IR}_P = \text{IR}\sqrt{N}$. Four independent IR-0.5 strategies give 1.0. This is Grinold's breadth at work (IM-3 d); the risk-budgeting rules behind the weights are IM-6 h.

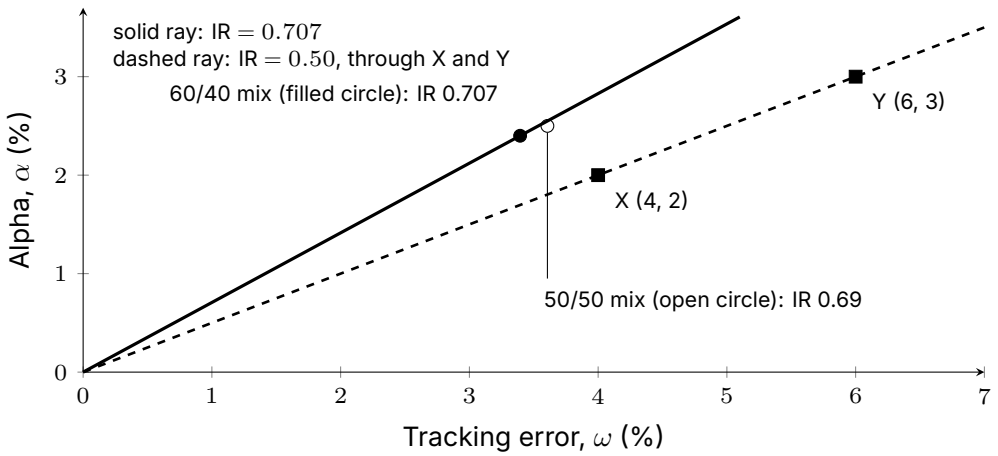



Figure 2. The IR is the slope of a ray from the origin in the (tracking error, alpha) plane. X and Y (squares) sit on the same IR-0.5 ray (dashed). Because their active returns are independent, combining them cuts tracking error by more than alpha, so both mixes sit above the ray; the 60/40 mix (filled circle) reaches the steepest ray (solid).

## Formula card: IM-3 b

| | |
|---|---|
| Excess (active) return | $r_t^x = r_t - r_t^{\text{bmk}}$ |
| Alpha, tracking error, IR | $\alpha = \overline{r^x}$, $\text{TE} = \text{stdev}(r^x)$ (sample, $n-1$), $\text{IR} = \alpha/\text{TE}$ |
| Sharpe ratio | $\text{SR} = (\bar{r}_P - \bar{r}_f)/\sigma_P$: the IR with $r_f$ as the benchmark |
| CAPM benchmark | $r^{\text{bmk}} = (1-\beta)\, r_f + \beta\, r_M$; regression $r - r_f = \alpha + \beta(r_M - r_f) + \varepsilon$ |
| Jensen's alpha | $\alpha_P = \bar{r}_P - [\bar{r}_f + \beta_P(\bar{r}_M - \bar{r}_f)]$, $\beta_P = \rho\, \sigma_P/\sigma_M$ |
| TE against the regression benchmark | $\sigma(\varepsilon)$, the residual (idiosyncratic) volatility; $\text{IR} = \alpha/\sigma(\varepsilon)$ |
| TE against a naive index ($\beta \neq 1$) | $\text{TE}^2 = (1-\beta)^2\sigma_M^2 + \sigma^2(\varepsilon)$ (constant $r_f$) |
| Time scaling | $\alpha \times 12$, $\text{TE} \times \sqrt{12}$, $\text{IR} \times \sqrt{12}$ (monthly to annual) |
| Pool of independent managers | $\alpha_P = \sum w_i\alpha_i$, $\omega_P = \sqrt{\sum w_i^2\omega_i^2}$; $N$ equal: $\text{IR}\sqrt{N}$ |

## Exam traps

- **Naive benchmark (wrong-input twin).** $\bar{r}_P - \bar{r}_M$ assumes $\beta = 1$. With $\beta < 1$ it understates alpha and IR; with $\beta > 1$ it overstates them.
- **Wrong beta.** Correlation used as beta, or $\sigma_P/\sigma_M$ used without $\rho$.
- **Dropped risk-free leg.** $\bar{r}_P - \beta\bar{r}_M$ omits $(1-\beta)r_f$; $\bar{r}_P - \beta(\bar{r}_M - r_f)$ forgets to add $r_f$ back.
- **Sibling denominators.** Sharpe uses total volatility, the IR uses tracking error. Excess return over $r_f$ divided by tracking error is neither.
- **Mixed benchmarks.** Alpha against one benchmark divided by the tracking error against another (Schweser's 0.5584).

- **Population versus sample.** Tracking error from a short history uses $n - 1$.
- **Time units.** Monthly alpha times 12, monthly TE times $\sqrt{12}$; never alpha times $\sqrt{12}$.
- **Adding IRs.** Independent IRs combine as $\sqrt{\sum \text{IR}_i^2}$, not $\sum \text{IR}_i$.

## Practice questions: IM-3 b

### Question IM-3 b.1

Bank Q94 • BT P2.T9.20.5.2 (rebuilt)

Regarding the relationship between alpha, tracking error, the information ratio (IR) and the Sharpe ratio, each of the following is true EXCEPT which is false?

**A.** The Sharpe ratio is a special case of the information ratio in which the benchmark is the risk-free asset.

**B.** The information ratio is the portfolio's average active return (alpha) divided by its tracking error (the standard deviation of that active return).

**C.** If two funds use the same benchmark, the fund with the higher information ratio must have the lower tracking error.

**D.** If the benchmark is risk-adjusted (beta estimated by regressing the fund's excess returns on the benchmark's excess returns), tracking error can be interpreted as idiosyncratic volatility.

### Question IM-3 b.2

Bank Q90 • AnalystPrep Q.2753

A performance analyst has been given the following information for a portfolio:

| | |
|---|---|
| Portfolio return | 10% |
| Beta | 1.2 |
| Standard deviation of the portfolio | 5% |
| Benchmark return | 8% |
| Tracking error | 10% |
| Risk-free rate | 3% |

Which of the following correctly states the Sharpe ratio and the information ratio of the portfolio?

**A.** Sharpe ratio: 0.4; information ratio: 0.2

**B.** Sharpe ratio: 0.4; information ratio: 0.7

**C.** Sharpe ratio: 1.4; information ratio: 0.2

**D.** Sharpe ratio: 1.4; information ratio: 0.7

### Question IM-3 b.3

Bank Q88 • AnalystPrep Q.2411

The performance parameters of two fund managers at an investment firm, measured against the same benchmark, are shown below:

| Parameter | Fund manager 1 | Fund manager 2 |
|---|---|---|
| Return | 12.35% | 13.00% |
| Alpha | 2.35% | 3.00% |
| Tracking error | 25% | 34% |

Which of the following statements is correct?

**A.** Fund manager 1 deviated more from the benchmark index.

**B.** Fund manager 2 deviated more from the benchmark index.

**C.** Fund manager 1 generated a higher excess return.

**D.** Fund manager 2 has a higher information ratio.

### Question IM-3 b.4

Bank Q99 • GARP PE1 Q8 • Mock 3 Q72 (rebuilt)

A portfolio manager at a hedge fund manages an equity portfolio benchmarked to an index. Performance over the last five years is shown below:

| Year | Portfolio return | Benchmark return | Portfolio beta vs. benchmark |
|---|---|---|---|
| 1 | 0.072 | 0.070 | 0.92 |
| 2 | 0.052 | 0.054 | 0.88 |
| 3 | 0.052 | 0.047 | 0.90 |
| 4 | 0.060 | 0.060 | 0.84 |
| 5 | 0.048 | 0.033 | 0.89 |

Using the sample standard deviation, what is the approximate information ratio of the manager measured against the benchmark?

**A.** 0.40

**B.** 0.42

**C.** 0.60

**D.** 1.90

### Question IM-3 b.5

Workbook A18.1 • S3 • Aug-2026 • Map #36

A junior risk analyst at an Indian asset manager is preparing a report on an INR large-cap equity fund. Over the period the fund returned 16% with a standard deviation of 20%. Its correlation with the NIFTY index was 0.9. The index returned 11% with a standard deviation of 15%, and the average T-bill rate was 4%. Which of the following is closest to the fund's Jensen's alpha?

**A.** 2.7%

**B.** 3.6%

**C.** 5.0%

**D.** 5.7%

### Question IM-3 b.6

Bank Q107 • AP Mock 3 Q6 (rebuilt) • re-filed from IM-3 c

A technology fund earned an average return of 14% over the sample period. Over the same period the risk-free rate was 3% and the S&P 500 returned 10%. A regression of the fund's excess returns on the S&P 500's excess returns gives a beta of 1.30, and the fund's tracking error is 5%. The manager reports alpha and the information ratio (IR) by simply comparing the fund with the S&P 500, implicitly assuming a beta of 1.0. Using a benchmark that reflects the fund's true beta, which of the following correctly states the fund's alpha and IR and how the reported figures compare?

**A.** Alpha 1.0% and IR 0.20; the reported 4.0% and 0.80 are overstated because the fund is riskier than the S&P 500.

**B.** Alpha 1.9% and IR 0.38; the reported 4.0% and 0.80 are overstated because the fund is riskier than the S&P 500.

**C.** Alpha 4.0% and IR 0.80; the S&P 500 is the appropriate benchmark, so the reported figures stand.

**D.** Alpha 4.9% and IR 0.98; the reported 4.0% and 0.80 are understated because the fund is riskier than the S&P 500.

### Question IM-3 b.7

Bank Q93 • BT P2.T8.704.1 (rebuilt)

A portfolio's excess returns (return above the risk-free rate) are regressed against the benchmark's excess returns over 60 observations. Key output:

| | |
|---|---|
| Portfolio: average excess return; volatility | 3.29%; 3.67% |
| Benchmark: average excess return; volatility | 0.98%; 1.99% |
| Average difference between portfolio and benchmark returns (active return) | 2.31% |
| Regression intercept; slope | 1.80%; 1.5237 |
| Standard error of the regression | 2.10% |

Which of the following is nearest to the information ratio if the IR is measured as residual return (alpha) per unit of residual risk?

**A.** 0.490

**B.** 0.857

**C.** 0.896

**D.** 1.100

**Question IM-3 b.8** Workbook A18.2 • S3, S6 • Map #36

The CIO of a Singapore university endowment, whose core holding is a passive SGD-hedged global equity index fund, is evaluating an active manager to add on top of that index position. The manager's portfolio data are:

- Average return: 15%; beta relative to the index: 1.2
- Standard deviation of returns: 22%
- Standard deviation of residual returns (tracking error): 6%
- Index average return: 11%; average risk-free rate: 3%

Which of the following is closest to the value of the measure the CIO should use to evaluate this manager?

**A.** 0.40

**B.** 0.55

**C.** 0.67

**D.** 2.00

**Question IM-3 b.9** Bank Q114 • AnalystPrep Q.4706 (rebuilt) • re-filed from IM-3 d • S4

The head of manager research at a multi-manager fund reviews three active equity mandates. Each manager has an expected alpha of 5% a year and a tracking error of 8.4%. The capital is split equally across the three mandates, and the managers' active returns are independent of one another. Which of the following is closest to the information ratio of the combined active portfolio?

**A.** 0.344

**B.** 0.595

**C.** 1.031

**D.** 1.786

## Solutions • IM-3 b

### Solution IM-3 b.1

**Answer: C**

Bank Q94 • BT P2.T9.20.5.2

#### Step 1 – Test A and B against the definitions

With $r_f$ as the benchmark, alpha is the excess return over $r_f$ and the tracking error is the asset's own volatility, so the IR becomes the Sharpe ratio: A is true. B is Eq. 3.4 word for word: true.

#### Step 2 – Test D

Against a regression (beta-adjusted) benchmark, the active return is $\alpha + \varepsilon_t$, so its standard deviation is the residual volatility. Ang: academics then call tracking error "idiosyncratic volatility". True.

#### Step 3 – Test C with a counter-example

IR $= \alpha/\text{TE}$ depends on both inputs. A fund with $\alpha = 2\%$, TE $= 4\%$ has IR 0.50; a fund with $\alpha = 6\%$, TE $= 8\%$ has IR 0.75 and the *larger* tracking error.
$\Rightarrow$ C is the false statement.

#### Interpretation

A higher IR means more alpha per unit of active risk, not less active risk. A manager can raise both alpha and tracking error and still improve the ratio.

#### Why the other options are wrong

**A is incorrect.** True: the Sharpe ratio is the IR with the risk-free rate as benchmark.
**B is incorrect.** True: the definition of the IR.
**D is incorrect.** True: with a risk-adjusted benchmark the tracking error is the regression's residual volatility.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.3 Definition of Alpha, Eq. 3.1 to 3.4; SchweserNotes 2026 Book 5, Reading 85, LO 85.b

### Solution IM-3 b.2

**Answer: C**

Bank Q90 • AnalystPrep Q.2753

#### Step 1 – Sharpe ratio: benchmark is $r_f$, risk is total volatility

$$\text{SR} = \frac{10\% - 3\%}{5\%} = 1.4$$

#### Step 2 – Information ratio: benchmark is the index, risk is tracking error

$$\text{IR} = \frac{10\% - 8\%}{10\%} = 0.2$$

$\Rightarrow$ Each ratio pairs its own benchmark with its own risk measure.

#### Interpretation

The beta of 1.2 is a red herring: neither ratio uses it. The numbers are odd (a 5% total volatility with a 10% tracking error), but the question only tests which inputs go where.

#### Why the other options are wrong

**A is incorrect.** Sharpe computed against the benchmark: $(10 - 8)/5 = 0.4$; IR correct.
**B is incorrect.** Both benchmarks swapped: Sharpe $(10 - 8)/5 = 0.4$ and IR $(10 - 3)/10 = 0.7$.
**D is incorrect.** Sharpe correct, but the IR used the risk-free rate: $(10 - 3)/10 = 0.7$.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.3, Eq. 3.4 and the Sharpe ratio special case

### Solution IM-3 b.3

**Answer: B**

Bank Q88 • AnalystPrep Q.2411

#### Step 1 – Deviation from the benchmark is tracking error

Tracking error is the standard deviation of the excess return: 34% for manager 2 against 25% for manager 1.
$\Rightarrow$ Manager 2 deviated more.

#### Step 2 – Check the other claims

Excess return (alpha): 3.00% for manager 2 against 2.35% for manager 1. IR: $2.35/25 = 0.0940$ for manager 1 against $3.00/34 = 0.0882$ for manager 2.

#### Interpretation

Manager 2 earned more alpha but took proportionally more active risk, so manager 1 has the better IR. Judging by alpha alone ignores the risk taken, which is the reason the IR exists.

#### Why the other options are wrong

**A is incorrect.** Reverses the comparison: manager 1 has the smaller tracking error.
**C is incorrect.** Manager 2's alpha (3.00%) is the higher one.
**D is incorrect.** Manager 2's IR is 0.0882, below manager 1's 0.0940.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.3, Eq. 3.3 and 3.4

### Solution IM-3 b.4

**Answer: C**

Bank Q99 • GARP PE1 Q8 • Mock 3 Q72

#### Step 1 – Excess returns over the benchmark (in %)

$0.2,\ -0.2,\ 0.5,\ 0.0,\ 1.5$; mean (alpha) $= 2.0/5 = 0.40\%$.

#### Step 2 – Sample tracking error

Squared deviations from 0.40: $0.04,\ 0.36,\ 0.01,\ 0.16,\ 1.21$, sum 1.78.

$$\text{TE} = \sqrt{1.78/4} = 0.667\%$$

#### Step 3 – Information ratio

$$\text{IR} = 0.40/0.667 = 0.60$$

$\Rightarrow$ The manager earned 0.60 units of active return per unit of active risk.

#### Interpretation

The beta column is a red herring: the question asks for the IR "measured against the benchmark", so the excess return is the plain difference from the benchmark. With the population standard deviation the IR would be 0.67, which is not offered.

#### Why the other options are wrong

**A is incorrect.** Stopped at the average excess return, 0.40 (in %), without dividing by tracking error.
**B is incorrect.** Divided alpha by the standard deviation of the portfolio's own returns (0.955%): $0.40/0.955 = 0.42$.
**D is incorrect.** Used the betas to form $R_p - \beta R_b$ (mean 1.00%, standard deviation 0.527%): 1.90. This drops the risk-free leg, so it is not an alpha against any benchmark.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.3, Eq. 3.1 to 3.4; GARP Practice Exam 1, Q8

### Solution IM-3 b.5

**Answer: B**

Workbook A18.1 • S3 • Aug-2026

#### Step 1 – Beta from the correlation

$$\beta_P = \rho \frac{\sigma_P}{\sigma_M} = 0.9 \times \frac{20\%}{15\%} = 1.20$$

$\Rightarrow$ The fund moves 1.2% for each 1% move in the index: more systematic risk than the market.

#### Step 2 – Market risk premium

$\bar{r}_M - \bar{r}_f = 11\% - 4\% = 7\%$.

#### Step 3 – CAPM required return

$4\% + 1.20 \times 7\% = 12.4\%$.

#### Step 4 – Jensen's alpha

$$\alpha_P = 16\% - 12.4\% = 3.6\%$$

$\Rightarrow$ After paying for a beta of 1.2, the fund still earned 3.6% a year above the security market line.

#### Interpretation

Positive alpha is necessary for the fund to beat the index on any risk-adjusted basis, but not sufficient: if its nonsystematic risk is large, its Sharpe ratio can still lag the market's.

#### Why the other options are wrong

**A is incorrect.** Used $\sigma_P/\sigma_M = 1.333$ as beta, dropping $\rho$: $16 - (4 + 1.333 \times 7) = 2.7\%$.
**C is incorrect.** Naive benchmark: $16 - 11 = 5.0\%$, implicitly assuming $\beta = 1$. Here $\beta > 1$, so the naive alpha is too high.
**D is incorrect.** Used the correlation as beta: $16 - (4 + 0.9 \times 7) = 5.7\%$.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24, Risk-Adjusted Performance Measures; Ang, Ch.10 (GARP 2025 Ch.3), Benchmarks Matter

### Solution IM-3 b.6

**Answer: B**

Bank Q107 • AP Mock 3 Q6

#### Step 1 – Beta-adjusted benchmark return

$$r^{\text{bmk}} = r_f + \beta(r_M - r_f) = 3\% + 1.30 \times 7\% = 12.1\%$$

$\Rightarrow$ The benchmark is $-30\%$ in T-bills and $+130\%$ in the S&P 500: the fund is levered market exposure.

#### Step 2 – Alpha and IR against it

$$\alpha = 14\% - 12.1\% = 1.9\%, \qquad \text{IR} = 1.9\%/5\% = 0.38$$

#### Step 3 – The reported (naive) figures

$14\% - 10\% = 4.0\%$ and $4.0/5 = 0.80$.

⇒ The naive figures credit the manager with 2.1% that is only payment for extra market risk.

**Interpretation**

This is the Martingale lesson with the sign reversed: a beta above 1 makes the raw index a *less* risky benchmark than the fund, so naive alpha and IR are too high. (The question keeps the 5% tracking error for both versions; strictly, the tracking error against each benchmark differs.)

**Why the other options are wrong**

**A is incorrect.** Applied beta to the total market return and omitted the risk-free leg: $14 - 1.30 \times 10 = 1.0\%$, IR 0.20.
**C is incorrect.** Accepted the naive S&P 500 comparison, which assumes a beta of 1.0.
**D is incorrect.** Subtracted $\beta(r_M - r_f) = 9.1\%$ but forgot to add $r_f$ back: $14 - 9.1 = 4.9\%$, IR 0.98; it also gets the direction wrong.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Benchmarks Matter and Ideal Benchmarks (4. Adjusted for risk)

## Solution IM-3 b.7 — Answer: B

Bank Q93 • BT P2.T8.704.1

**Step 1 – Residual return is the intercept**

Alpha $= 1.80\%$. Check: $3.29 - 1.5237 \times 0.98 = 1.80$.

**Step 2 – Residual risk is the standard error of the regression**

$\sigma(\varepsilon) = 2.10\%$.

**Step 3 – Information ratio**

$$\text{IR} = 1.80/2.10 = 0.857$$

⇒ Against the beta-adjusted benchmark, the portfolio earns 0.86 of residual return per unit of residual risk.

**Interpretation**

With a slope of 1.52 the portfolio is a levered version of the benchmark, so the raw active return (2.31%) overstates the value added: part of it is beta. Labelling the 2.10% "tracking error" only works because the benchmark is the regression benchmark (Section 3).

**Why the other options are wrong**

**A is incorrect.** Divided alpha by the portfolio's total volatility: $1.80/3.67 = 0.490$.
**C is incorrect.** The portfolio's Sharpe ratio: $3.29/3.67 = 0.896$.
**D is incorrect.** Used the unadjusted active return (beta assumed 1): $2.31/2.10 = 1.100$.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.3, Eq. 3.3 and 3.4, Benchmarks Matter

## Solution IM-3 b.8 — Answer: A

Workbook A18.2 • S3, S6

**Step 1 – Choose the measure**

An active portfolio is being mixed with a passive benchmark position, so the relevant measure is the information ratio: alpha per unit of residual risk.

**Step 2 – CAPM required return**

$\bar{r}_f + \beta_P(\bar{r}_M - \bar{r}_f) = 3\% + 1.2 \times (11\% - 3\%) = 12.6\%$.

**Step 3 – Jensen's alpha**

$\alpha_P = 15\% - 12.6\% = 2.4\%$.
$\Rightarrow$ The manager adds 2.4% a year beyond what beta exposure explains.

**Step 4 – Information ratio**

$$\text{IR} = \frac{\alpha_P}{\sigma(e_P)} = \frac{2.4\%}{6\%} = 0.40$$

$\Rightarrow$ Each percentage point of diversifiable risk the endowment takes on by tilting away from the index earns 0.40 points of alpha.

**Interpretation**

By BKM's Eq. 8.2 the gain in the endowment's Sharpe ratio from optimally adding this manager is driven by the IR. An IR of 0.40 is respectable; Ang notes that IRs above one are not common.

**Why the other options are wrong**

**B is incorrect.** Computed the Sharpe ratio, $(15-3)/22 = 0.55$: the total-risk measure, relevant only if the manager were the endowment's entire risky portfolio.
**C is incorrect.** Naive benchmark: $(15-11)/6 = 0.67$, implicitly assuming a beta of 1.
**D is incorrect.** Put the excess return over the risk-free rate in the numerator: $(15-3)/6 = 2.00$, the Sharpe numerator over the IR denominator.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Ch.24, The Information Ratio; Ang, Ch.10 (GARP 2025 Ch.3), Definition of Alpha

**Solution IM-3 b.9** **Answer: C**

Bank Q114 • AnalystPrep Q.4706

**Step 1 – Each manager's IR**

$$\text{IR}_i = \frac{5\%}{8.4\%} = 0.595$$

**Step 2 – The pool's alpha and tracking error (weights $1/3$, independent)**

$$\alpha_P = 3 \times \tfrac{1}{3} \times 5\% = 5\%, \qquad \omega_P = \sqrt{3 \times \left(\tfrac{1}{3}\right)^2 \times 8.4^2} = \frac{8.4\%}{\sqrt{3}} = 4.85\%$$

$\Rightarrow$ Alpha is unchanged by the split, but independent active risks partly cancel.

**Step 3 – Combined IR**

$$\text{IR}_P = \frac{5\%}{4.85\%} = 1.031 = 0.595\sqrt{3}$$

**Interpretation**

Independent IRs add in quadrature: $\sqrt{3 \times 0.595^2} = 1.031$. This is the $\sqrt{N}$ of Grinold's breadth (IM-3 d): three independent sources of alpha raise the IR by $\sqrt{3}$, not by 3. Correlated managers would gain less.

**Why the other options are wrong**

**A is incorrect.** Divided by $\sqrt{3}$ instead of multiplying: $0.595/\sqrt{3} = 0.344$.
**B is incorrect.** The single-manager IR; ignores the diversification of active risk.

**D is incorrect.** Added the three IRs linearly: $3 \times 0.595 = 1.786$, which treats the active risks as perfectly correlated in the denominator but independent in the numerator.

**LO:** IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.3, Eq. 3.4 and Creating Alpha; Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), risk budgeting across active managers

IM-3 • Alpha (and the Low-Risk Anomaly) • Vardeez: VERY IMPORTANT

# IM-3 d

P1 BOTH

Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law.

Workbook: A18 Performance Measures and Attribution • Tier A • August 2026 recall • Vardeez focus

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: Map #36 lists "Grinold's fundamental law and the maximum attainable information ratio (LO 85.d)" next to the A18 recall, and Workbook A18.6 is tagged to the August 2026 sitting. Vardeez also names it first in this VERY IMPORTANT chapter.
**Vardeez:** Grinold's fundamental law (IC, breadth, IR).

**Reported exam shapes**

- **S5 (A18)** Grinold's fundamental law and the maximum attainable information ratio: compute IR from IC and breadth, then judge a statement about its assumptions or limitations.

**What the questions look like:** a mix. About half are one-line calculations: IR from IC and breadth, the IC needed for a target IR, or what happens when breadth changes. Watch the square root and the annual count of bets. The rest are statement questions on what breadth means, the independence assumption, and what happens to IC as assets grow. A18.6 combines both in one item.

## 1. The law

"A portfolio manager creates alpha relative to a benchmark by making bets that deviate from that benchmark. The more successful these bets are, the higher the alpha." Grinold (1989) makes this formal. The **maximum attainable** information ratio (Eq. 3.5) is

$$\boxed{\text{IR} = \text{IC} \times \sqrt{\text{BR}}}$$

- **IC, the information coefficient:** the correlation of the manager's forecasts with the actual returns. It measures how good the forecasts are: *playing well*.
- **BR, breadth:** how many bets are taken, "the number of securities that can be traded and how frequently they can be traded". Count **independent bets per year**: *playing often*.
- **Maximum:** the law "ignores transactions costs, restrictions on trading, and other real-world considerations". It is a ceiling, not a forecast of the realised IR.

Grinold and Kahn's summary: "it is important to play often (high breadth, BR) and to play well (high IC)." The law is a framing tool for active quantitative management: how good forecasters must be, how many bets they must make, or both, to reach a target IR. The IR itself is the one defined in IM-3 b.

## 2. Calculating with it

Three rearrangements cover every numerical question:

$$\text{IR} = \text{IC}\sqrt{\text{BR}} \qquad \text{IC} = \frac{\text{IR}}{\sqrt{\text{BR}}} \qquad \text{BR} = \left(\frac{\text{IR}}{\text{IC}}\right)^2$$

- **GARP's two managers, one IR.** Target IR 0.50. A market timer trades the aggregate market once a quarter: BR $= 4$, so she needs IC $= 0.50/\sqrt{4} = 0.25$, very accurate bets. A stock selector running value, size or momentum across hundreds of stocks makes 400 independent bets a year and needs only $0.50/\sqrt{400} = 0.025$. "You can be very talented at forecasting and make those bets really count, or you can have a very small edge but make a lot of bets."
- **Cross-sectional strategies** (value, size, momentum) "can have very low ICs, often just 2% to 5%, to be highly profitable": with 400 bets those ICs give IRs of 0.40 to 1.00. Schweser's version: 200 bets a year need IC $= 0.50/\sqrt{200} = 0.035$.
- **Required breadth.** A manager with IC $= 0.05$ needs $(0.50/0.05)^2 = 100$ independent bets a year for IR 0.50; with IC $= 0.10$, only 25.
- **Annual breadth.** The IR is annual, so BR counts bets per year. A model making 20 independent forecasts a month has BR $= 240$; with IC $= 0.04$, IR $= 0.04\sqrt{240} = 0.62$, not $0.04\sqrt{20} = 0.18$. The size of the stock universe is not the breadth.
- **Scaling.** Multiplying breadth by $k$ multiplies the IR by $\sqrt{k}$: four times the bets, twice the IR. Multiplying IC by $k$ multiplies the IR by $k$.

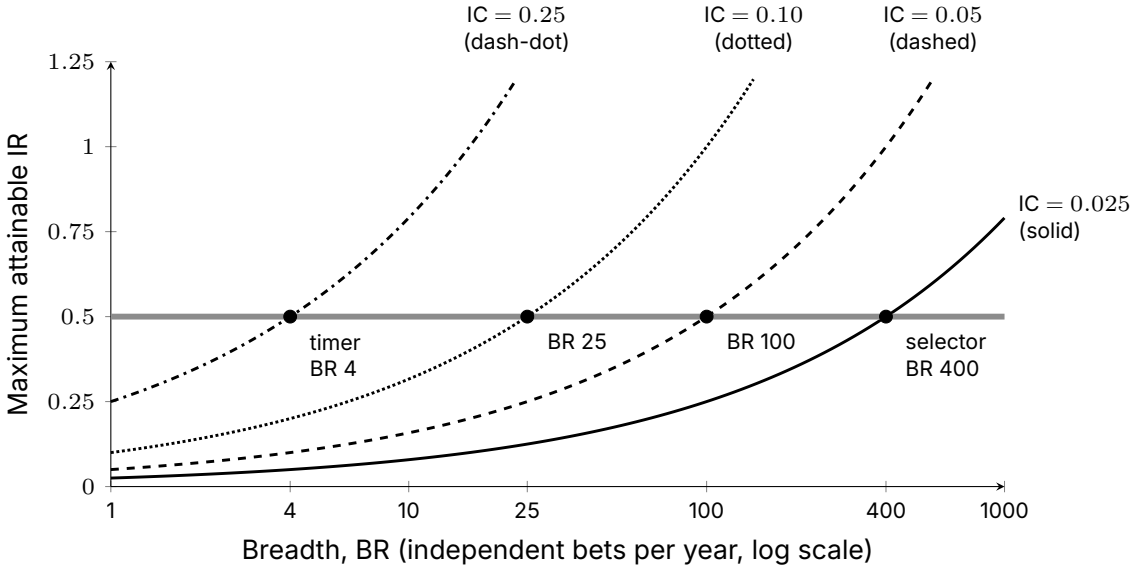



Figure 1. $\text{IR} = \text{IC}\sqrt{\text{BR}}$ for four skill levels. The target IR of 0.50 (thick grey line) is reached at breadths of 4, 25, 100 and 400: each time the required IC halves, the number of independent bets must quadruple. GARP's market timer (IC 0.25, 4 bets) and stock selector (IC 0.025, 400 bets) sit at the two ends of the grey line.

## 3. Assumptions

- **No frictions.** No transaction costs, trading restrictions or other real-world limits: hence "maximum attainable".
- **Mean-variance utility.** The law is derived in that framework, "and so all the shortcomings of mean-variance utility apply": it **ignores downside risk and other higher-moment risk** (skewness, fat tails).
- **Optimal use of information.** It assumes all information is used optimally in building the portfolio.
- **Independent forecasts.** "A crucial assumption is that the forecasts are independent of each other."

The law is itself an approximation within mean-variance (Hallerbach, 2011); extensions allow time-varying signals, correlated forecasts and estimation risk.

## 4. Limitations

Ang names "two very important limitations". Both make the realised IR smaller than $\text{IC}\sqrt{\text{BR}}$.

1. **IC is assumed constant across BR.** "The first manager whom you find may truly have a high IC, but the one hundredth manager whom you hire probably does not." As assets under management rise, the ability to generate high ICs diminishes: decreasing returns to scale in mutual funds, hedge funds and private equity. Some mutual funds close to new investors once they reach an internal size limit for this reason. Buffett on Berkshire: "The huge sums of capital we currently manage eliminate any chance of exceptional performance."
2. **Truly independent forecasts are hard to get.** Correlated bets reduce BR. "An equity manager with overweight positions on 1,000 value stocks offset by underweight positions in 1,000 growth stocks has not placed 1,000 different bets; he's placed just one bet on a value-growth factor." Hiring one hundred fixed income managers who all reach for yield with illiquid bonds is one bet on an illiquidity factor. Correlated factor bets dominate at the portfolio level, which is why top-down factor investing matters.

**No economic content.** The law "says nothing about where positive risk-adjusted opportunities are; it only provides a method of systematically evaluating them". That is why it is influential in industry but makes only a scant appearance in academic work.

**The law as a management style.** Knut Kjær, founding CEO of Norges Bank Investment Management, delegated active decisions widely to specialised managers, "without intervention from superiors", to keep the bets as independent as possible and so raise effective breadth.

> **Correction to the source notes.** (1) Vardeez writes that "a jump in asset prices chosen by asset managers often causes the decline of IC". GARP's point is about size: ICs fall as *assets under management* rise (decreasing returns to scale), which is why some funds close to new money. (2) SchweserNotes (LO 85.d) calls breadth "simply the number of investments deployed". Breadth is the number of *independent* bets per year (securities that can be traded times how often), and correlated positions count as one bet. Neither the number of holdings nor the size of the index is the breadth (Question IM-3 d.2).

**Formula card: IM-3 d**

| | |
|---|---|
| Fundamental law (Eq. 3.5) | $\text{IR} = \text{IC} \times \sqrt{\text{BR}}$, the maximum attainable IR |
| Required skill | $\text{IC} = \text{IR}/\sqrt{\text{BR}}$ |
| Required breadth | $\text{BR} = (\text{IR}/\text{IC})^2$ |
| Annual breadth | independent forecasts per period $\times$ periods per year |
| Scaling | $\text{BR} \times k \Rightarrow \text{IR} \times \sqrt{k}$; $\text{IC} \times k \Rightarrow \text{IR} \times k$ |
| GARP's pair | $0.25\sqrt{4} = 0.50$ (timer) $\quad 0.025\sqrt{400} = 0.50$ (selector) |
| Independent managers | $\text{IR}_P = \sqrt{\sum \text{IR}_i^2}$; $N$ equal: $\text{IR}\sqrt{N}$ (IM-3 b) |

| Assumption | What breaks it in practice | Effect |
|---|---|---|
| **No costs or trading limits** | Transaction costs, restrictions on trading | Realised IR below the maximum |
| **Independent forecasts** | Correlated bets: value against growth, reaching for yield | Effective BR far below the count of positions |
| **IC constant across BR** | Growing AUM; the 100th manager hired | IC falls, so IR falls |
| **Mean-variance utility** | Downside risk, skewness, fat tails | Ignored by the law |
| **Information used optimally** | Imperfect portfolio construction | Realised IR below the maximum |
| **Statistical model only** | No view on where alpha comes from | No guidance on sources of alpha |

**Exam traps**

- **Dropped square root.** IC $\times$ BR inflates the IR by $\sqrt{\text{BR}}$: $0.025 \times 400 = 10$ instead of 0.50.
- **Inverted rearrangement.** IR $\times \sqrt{\text{BR}}$ or IR/BR when solving for IC.
- **Monthly breadth.** Use bets per year; 25 forecasts a month is a breadth of 300.
- **Universe or holdings as breadth.** A 3,000-stock universe is not 3,000 independent bets.
- **Linear scaling.** Tripling breadth multiplies the IR by $\sqrt{3} = 1.73$, not by 3.
- **Maximum read as realised.** The law ignores costs and restrictions; it does not promise the IR after them.
- **Constant IC.** ICs fall as assets under management grow; doubling AUM does not leave the IR unchanged.
- **Higher moments.** The law ignores downside and higher-moment risk; it does not adjust for skewness or kurtosis.

## Practice questions: IM-3 d

**Question IM-3 d.1** Bank Q121 • Schweser Module Quiz 85.1 Q3

Grinold's fundamental law of active management suggests that:

**A.** investors should focus on increasing only their predictive ability relative to stock price movements.

**B.** sector allocation is the most important factor in active management.

**C.** a small number of investment bets decreases the chances of making a mistake and, therefore, increases the expected investment performance.

**D.** to maximize the information ratio, active investors need to either have high-quality predictions or place a large number of investment bets in a given year.

### Question IM-3 d.2

Bank Q117 • BT P2.T9.21.3.1

Grinold's fundamental law of active management states that the expected information ratio of an active strategy is $\text{IR} = \text{IC} \times \sqrt{\text{BR}}$, where IC is the information coefficient and BR is breadth; the extended law adds the transfer coefficient, $\text{IR} = \text{IC} \times \sqrt{\text{BR}} \times \text{TC}$, where TC measures how efficiently constraints and transaction costs allow the actual portfolio to reflect the optimal one. Which of the following is the BEST measure of breadth, BR, in the fundamental law?

**A.** The number of independent bets per year

**B.** The ratio of active return to active risk

**C.** The number of positions (holdings) in the portfolio

**D.** The number of securities in the relevant index, e.g., S&P 1500, Russell 3000, or Wilshire 5000

### Question IM-3 d.3

Bank Q116 • BT P2.T9.21.1.1

A portfolio manager has an information coefficient (IC) of 0.150 and breadth (BR) of 12.0. If she triples her breadth to 36, what happens to the information ratio (IR)?

**A.** The IR drops by one third to 0.346.

**B.** It has no effect on the IR.

**C.** The IR increases by 0.380 to 0.900.

**D.** The IR increases by 1.039 to 1.559.

### Question IM-3 d.4

Bank Q112 • AnalystPrep Q.2761

Which of the following managers has the highest information ratio based on Grinold's fundamental law of active management, $\text{IR} = \text{IC} \times \sqrt{\text{BR}}$?

**A.** A manager with an information coefficient of 0.5 and a breadth of 20.

**B.** A manager with an information coefficient of 0.05 and a breadth of 200.

**C.** A manager with an information coefficient of 0.1 and a breadth of 100.

**D.** A manager with an information coefficient of 0.8 and a breadth of 10.

### Question IM-3 d.5

Bank Q119 • Mock 2 Q54 (rebuilt)

A mutual fund targets an information ratio (IR) of at least 0.60. Analyst 1 argues the fund should place at least 250 independent bets per year; Analyst 2 argues that only 25 independent bets per year suffice. Using the fundamental law (with $\text{TC} = 1$), what information coefficient is required to just reach the target under Analyst 1's breadth and under Analyst 2's breadth, respectively?

**A.** 0.0024 and 0.024

**B.** 0.0379 and 0.120

**C.** 0.120 and 0.0379

**D.** 9.49 and 3.00

### Question IM-3 d.6

Bank Q120 • Mock 2 Q78

Quantitative Equity Design (QED) is a quantitative equity fund manager. QED forecasts monthly alphas for stocks in its 3,000-stock universe. The number of independent forecasts made each month is 25. If the model employed by QED has an information coefficient of 0.15, the annual information ratio for QED is closest to:

**A.** 0.75

**B.** 2.60

**C.** 28.46

**D.** 45.00

### Question IM-3 d.7

Bank Q110 • AnalystPrep Q.2417

An investment management firm allows each fund manager a maximum of 100 transactions per year. A manager assumes IC $= 7.5\%$ and uses IR $=$ IC $\times \sqrt{\text{BR}}$ with BR $= 100$, which implies IR $= 0.75$. When she instead computes the IR from the observed excess return and tracking error, the result is below 0.75. This is most likely due to the:

**A.** correlation between the different bets of the fund manager.

**B.** cap on the number of transactions per year.

**C.** differences in portfolio weights.

**D.** differences in the number of transactions from one manager to another.

### Question IM-3 d.8

Bank Q115 • BT P2.T8.705.3 (rebuilt)

Grinold's fundamental law of active management states that the maximum attainable information ratio is IR $=$ IC $\times \sqrt{\text{BR}}$. Each of the following statements about the law is true EXCEPT:

**A.** As assets under management grow, the information coefficient tends to decline.

**B.** A key advantage of the law is that it incorporates downside risk and higher-moment risk, adjusting for skewness and excess kurtosis.

**C.** An IR of 0.50 can be achieved by a market timer with IC $= 0.25$ making four bets a year, or by a manager with IC $= 0.025$ making 400 bets a year.

**D.** A crucial assumption is that forecasts are independent; overweighting 1,000 value stocks against underweighting 1,000 growth stocks is not 1,000 independent bets.

**Question IM-3 d.9** Workbook A18.6 • S5, S6 • Aug-2026 • Map #36

The CIO of a European pension fund is presenting to the investment committee on two candidate active equity managers. Manager P is a market timer who makes 4 independent bets a year with an information coefficient of 0.20. Manager Q is a stock selector who makes 225 independent bets a year with an information coefficient of 0.03. Which of the following statements should the CIO include in the presentation?

**A.** Q's maximum attainable information ratio is 0.45 versus 0.40 for P, so Q will deliver the higher realized information ratio after transaction costs and trading restrictions.

**B.** Q's maximum attainable information ratio is 0.45 versus 0.40 for P; if Q is added as an active position on top of the fund's passive index holding, BKM would rank it by its Treynor measure.

**C.** Q's maximum attainable information ratio is 0.45 versus 0.40 for P; if Q's 225 positions are overweights in value stocks against underweights in growth stocks, its true breadth and attainable ratio are far lower.

**D.** Q's maximum attainable information ratio is 0.45 versus 0.40 for P; doubling Q's assets under management would leave that ratio unchanged, because the information coefficient is constant across breadth.

## Solutions • IM-3 d

### Solution IM-3 d.1 — Answer: D

Bank Q121 • Schweser Module Quiz 85.1 Q3

**Step 1 – Read the law**

$\text{IR} = \text{IC} \times \sqrt{\text{BR}}$: the IR rises with forecasting skill (IC) and with the number of independent bets (BR).

**Step 2 – The trade-off**

Either input can make up for the other: a high IC with few bets, or a small edge applied many times.

$\Rightarrow$ Only D states the "play well or play often" trade-off.

**Interpretation**

GARP's two managers, a market timer (IC 0.25, 4 bets) and a stock selector (IC 0.025, 400 bets), reach the same IR of 0.50 from opposite ends of this trade-off.

**Why the other options are wrong**

**A is incorrect.** Ignores breadth, which carries as much weight in the law as skill.

**B is incorrect.** Sector allocation is not part of the law; it only counts skill and independent bets.

**C is incorrect.** Reverses the logic: fewer bets lower the attainable IR, because IR rises with $\sqrt{\text{BR}}$.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha, Eq. 3.5; SchweserNotes 2026 Book 5, Reading 85, LO 85.d

### Solution IM-3 d.2 — Answer: A

Bank Q117 • BT P2.T9.21.3.1

**Step 1 – What breadth counts**

Breadth is "the number of securities that can be traded and how frequently they can be traded", and the law assumes the bets are independent.

$\Rightarrow$ The best measure is the number of independent bets (investment decisions) per year.

**Step 2 – Eliminate the look-alikes**

Active return over active risk is the IR, the output of the law. Holdings can be held for years and many can express the same bet. The index size is the opportunity set, not the number of bets made.

**Interpretation**

The transfer coefficient in the stem is an extension from portfolio construction (IM-4); it does not change what BR means. Ang's example: 1,000 value overweights against 1,000 growth underweights are one bet, not 1,000.

**Why the other options are wrong**

**B is incorrect.** That is the information ratio itself.

**C is incorrect.** Holdings are not independent annual decisions: many positions can reflect one factor bet.

**D is incorrect.** The size of the universe is not the number of independent bets actually made.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha, Eq. 3.5

### Solution IM-3 d.3

**Answer: C**

Bank Q116 • BT P2.T9.21.1.1

#### Step 1 – Current IR

$$\text{IR}_0 = 0.150 \times \sqrt{12} = 0.150 \times 3.464 = 0.520$$

#### Step 2 – IR with tripled breadth

$$\text{IR}_1 = 0.150 \times \sqrt{36} = 0.150 \times 6 = 0.900$$

#### Step 3 – Change

$0.900 - 0.520 = 0.380$.
$\Rightarrow$ Tripling breadth multiplies the IR by $\sqrt{3} = 1.73$, not by 3.

#### Interpretation

Breadth enters through its square root, so each extra bet adds less IR than the one before. The same IC is assumed throughout; in practice spreading skill over more bets may lower it.

#### Why the other options are wrong

**A is incorrect.** Wrong direction: cut the IR by a third, $0.520 \times 2/3 = 0.346$.
**B is incorrect.** Ignores that breadth enters the law.
**D is incorrect.** Treated the IR as linear in BR: $0.520 \times 3 = 1.559$, an increase of 1.039.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha, Eq. 3.5

### Solution IM-3 d.4

**Answer: D**

Bank Q112 • AnalystPrep Q.2761

#### Step 1 – Apply the law to each manager

A: $0.5\sqrt{20} = 2.236$. B: $0.05\sqrt{200} = 0.707$. C: $0.1\sqrt{100} = 1.000$. D: $0.8\sqrt{10} = 2.530$.
$\Rightarrow$ D is highest.

#### Interpretation

IR is linear in IC but grows only with the square root of breadth, so a large gap in skill (0.8 against 0.05) beats a twenty-fold gap in breadth. The IRs here are far above anything seen in practice ("information ratios above one are not common"): the question tests the arithmetic only.

#### Why the other options are wrong

**A is incorrect.** 2.24, second highest: breadth of 20 does not compensate for the lower IC.
**B is incorrect.** 0.71: $\sqrt{200}$ is only 14.1, too little to offset an IC of 0.05.
**C is incorrect.** 1.00: a moderate IC with breadth 100 still falls well short of 2.53.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha, Eq. 3.5

### Solution IM-3 d.5

**Answer: B**

Bank Q119 • Mock 2 Q54

#### Step 1 – Rearrange the law

$\text{IC} = \text{IR}/\sqrt{\text{BR}}$.

**Step 2 – Analyst 1 (250 bets)**

$$\text{IC} = 0.60/\sqrt{250} = 0.60/15.811 = 0.0379$$

**Step 3 – Analyst 2 (25 bets)**

$$\text{IC} = 0.60/\sqrt{25} = 0.60/5 = 0.120$$

$\Rightarrow$ Ten times the bets lowers the required skill by $\sqrt{10} = 3.16$ times.

**Interpretation**

More bets allow a lower skill level to reach the same IR; fewer bets demand more skill. The order of the pair matters: the analyst with the larger breadth needs the smaller IC.

**Why the other options are wrong**

**A is incorrect.** Divided the IR by BR instead of $\sqrt{\text{BR}}$: $0.60/250 = 0.0024$ and $0.60/25 = 0.024$.
**C is incorrect.** Right numbers, reversed pairing: implies fewer bets need less skill.
**D is incorrect.** Multiplied the IR by $\sqrt{\text{BR}}$ instead of dividing: $0.60 \times 15.81 = 9.49$ and $0.60 \times 5 = 3.00$.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha, Eq. 3.5

## Solution IM-3 d.6

**Answer: B**

Bank Q120 • Mock 2 Q78

**Step 1 – Annual breadth**

$\text{BR} = 25 \times 12 = 300$ independent forecasts a year.

**Step 2 – Annual IR**

$$\text{IR} = 0.15 \times \sqrt{300} = 0.15 \times 17.321 = 2.60$$

$\Rightarrow$ Only independent forecasts count, and they are counted per year because the IR is annual.

**Interpretation**

The 3,000-stock universe is a red herring: breadth is the number of independent forecasts, not the number of stocks covered. An IR of 2.6 is a theoretical maximum; costs, constraints and correlated forecasts would cut it well down.

**Why the other options are wrong**

**A is incorrect.** Used the monthly breadth of 25: $0.15 \times 5 = 0.75$.
**C is incorrect.** Treated every stock as an independent bet each month: $0.15\sqrt{36{,}000} = 28.46$.
**D is incorrect.** Dropped the square root: $0.15 \times 300 = 45$.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha, Eq. 3.5

## Solution IM-3 d.7

**Answer: A**

Bank Q110 • AnalystPrep Q.2417

**Step 1 – What the 0.75 assumes**

$0.075 \times \sqrt{100} = 0.75$ holds only if the 100 bets are independent: the law's crucial assumption.

**Step 2 – Why the realised IR is lower**

If the bets are correlated, the effective number of independent bets is below 100, so the attainable IR is below 0.75.

$\Rightarrow$ Correlation between bets is the most likely cause.

**Interpretation**

Ang: correlated bets reduce BR; "correlated factor bets tend to dominate at the overall portfolio level". Costs and trading restrictions would also push the realised IR below the maximum, but they are not among the options.

**Why the other options are wrong**

**B is incorrect.** The cap is already in the calculation as BR $= 100$; it cannot explain a shortfall against it.

**C is incorrect.** Portfolio weights are not an input to the law and say nothing about independence.

**D is incorrect.** Other managers' trade counts are irrelevant to this manager's own shortfall.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha (limitations)

## Solution IM-3 d.8 — Answer: B

Bank Q115 • BT P2.T8.705.3

**Step 1 – Test A, C and D**

A: ICs tend to fall as assets under management rise (decreasing returns to scale): true. C: $0.25\sqrt{4} = 0.50$ and $0.025\sqrt{400} = 0.50$: true. D: Ang's own example of correlated bets: true.

**Step 2 – Test B**

The law is derived under mean-variance utility, so it *ignores* downside risk and other higher-moment risk.

$\Rightarrow$ B is the false statement.

**Interpretation**

Each true statement is one of the law's limitations or its signature example; the false one turns a limitation into an advantage (polarity flip).

**Why the other options are wrong**

**A is incorrect.** True: skill tends to diminish as AUM grows.

**C is incorrect.** True: both combinations give 0.50.

**D is incorrect.** True: the independence assumption fails for correlated factor bets.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Creating Alpha (assumptions and limitations)

## Solution IM-3 d.9 — Answer: C

Workbook A18.6 • S5, S6 • Aug-2026

**Step 1 – Fundamental law (Ang Eq. 3.5)**

$$\text{IR}_P = 0.20 \times \sqrt{4} = 0.40, \qquad \text{IR}_Q = 0.03 \times \sqrt{225} = 0.03 \times 15 = 0.45$$

$\Rightarrow$ Q has a tiny edge per bet but plays often; P plays well but rarely. The shared first clause is true in every option.

**Step 2 – Test the second clause**

Breadth counts independent bets. Ang: an equity manager overweight 1,000 value stocks against 1,000 growth stocks "has not placed 1,000 different bets; he's placed just one bet on a value-growth factor". Correlated bets reduce BR, so Q's attainable ratio falls well below 0.45.
⇒ C is correct.

### Interpretation

The fundamental law is a ceiling and a framing tool, not a forecast: it assumes independent forecasts, optimal use of information and mean-variance preferences, and says nothing about where alpha comes from.

### Why the other options are wrong

**A is incorrect.** Scope error: the law gives the maximum IR "since it ignores transactions costs, restrictions on trading, and other real-world considerations"; it does not promise the realised ranking after costs.

**B is incorrect.** Sibling swap of performance measures: an active portfolio mixed with a passive benchmark is evaluated by its information ratio (BKM Eq. 8.2); Treynor applies to one of many subportfolios forming the risky portfolio.

**D is incorrect.** Absolute claim: constant IC across breadth is an assumption Ang lists as a limitation; ICs tend to fall as assets under management rise, so doubling AUM would likely lower the ratio.

**LO:** IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Ang, Ch.10 (GARP 2025 Ch.3), Creating Alpha; Bodie, Kane and Marcus, Ch.24, The Information Ratio

IM-5 • Portfolio Risk: Analytical Methods • Vardeez: VERY IMPORTANT

# IM-5 a

P1 CALC

Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR.

Workbook: A12 Incremental, Marginal and Component VaR • Tier A • August 2026 recall

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic: incremental VaR came up in the August 2026 sitting (1 question), and the A12 shapes were reported on 6 of 16 recall sheets (Map #13).
**Vardeez:** marginal, incremental and component VaR calculations; diversified versus undiversified VaR.

**Reported exam shapes**

- **S1** Incremental VaR of including a platinum position in a silver book: full revaluation against the marginal-VaR shortcut.
- **S3** Marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio.
- **S4** Closed-book formulas: marginal VaR $= (\text{VaR}/W)\beta_i$; component VaR $=$ individual VaR $\times\rho_i$.
- **S6** The intermediate-result trap between marginal and component VaR, and the tangent-versus-curve gap between component and incremental VaR.

Shapes S2 and S5 (where to add using marginal VaR; only beta matters when ranking) are drilled in IM-5 c. The return-to-risk version of S3 is Workbook A12.4 in IM-5 d.

**What the questions look like:** almost all numerical. Either a two- or three-asset portfolio is given as positions, volatilities and a correlation, and you build the VaR measures from scratch; or a risk report is given as weights, betas, marginal or individual VaRs, and you read the other measures off it. The wrong options are the sibling measures.

## 1. Six measures, one portfolio

The reading works in the **delta-normal** setting: positions are linear in normally distributed risk factors, so the portfolio return is normal and every VaR has a closed form. Jorion calls this the covariance matrix approach. Keep three symbols apart, because the sources and the bank mix them:

| Symbol | Meaning | Units |
|---|---|---|
| $x_i$ | position in asset (risk factor) $i$, $x_i = w_i W$ | currency |
| $w_i$ | portfolio weight, $\sum w_i = 1$ | none |
| $W$ | total portfolio value | currency |
| $\alpha$ | standard normal deviate: 1.645 (95%) or 2.326 (99%); GARP's example uses 1.65 | none |

Three of the six measures describe *levels* of risk (individual, undiversified, diversified VaR). The other three are the *VaR tools* that tell you which position is responsible and what a trade would do (marginal, incremental, component VaR).

## 2. Individual, diversified and undiversified VaR

$$\boxed{\text{VaR}_i = \alpha\,\sigma_i\,|x_i|} \qquad \boxed{\text{VaR}_p = \alpha\,\sigma_p W = \alpha\sqrt{x'\Sigma x}} \qquad \boxed{\text{Undiversified VaR} = \textstyle\sum_i \text{VaR}_i}$$

- **Individual VaR** is one position taken in isolation. The absolute value is there because a weight can be negative but risk cannot.
- **Diversified VaR** is the portfolio VaR, "taking into account diversification benefits between components". For two assets,

$$\text{VaR}_p = \alpha\sqrt{x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2\rho_{12}x_1x_2\sigma_1\sigma_2}$$

$$\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho_{12}\,\text{VaR}_1\text{VaR}_2} \qquad \text{(both positions long)}$$

  The second form is the fast one when the stem gives individual VaRs instead of volatilities.
- **Undiversified VaR** is the sum of the individual VaRs: the portfolio VaR when there is no short position and every correlation is 1. With $\rho < 1$, diversified VaR is below it; the gap is the **diversification benefit**. At $\rho = 0$ the VaRs add in quadrature; at $\rho = 1$ they add.
- **Long-short books.** With one leg short, the worst case is $\rho = -1$, not $+1$. Jorion's example is a hedge fund long corporates and short Treasuries: at $\rho = 1$ it has no risk. Undiversified VaR is then the portfolio VaR at the worst correlation.

**GARP's two-currency example (Example 5.1).** USD 2 million in CAD ($\sigma = 5\%$), USD 1 million in EUR ($\sigma = 12\%$), uncorrelated, $\alpha = 1.65$:

$$\Sigma x = \begin{bmatrix} 0.05^2 \times 2 \\ 0.12^2 \times 1 \end{bmatrix} = \begin{bmatrix} 0.0050 \\ 0.0144 \end{bmatrix}, \qquad x'\Sigma x = 2(0.0050) + 1(0.0144) = 0.0244$$

$$\sigma_p W = \sqrt{0.0244} = 0.156205 \text{ (USD million)}$$

$$\text{VaR}_p = 1.65 \times 156{,}205 = \text{USD } 257{,}738$$

Individual VaRs are $1.65 \times 0.05 \times 2\text{m} = 165{,}000$ (CAD) and $1.65 \times 0.12 \times 1\text{m} = 198{,}000$ (EUR), so the undiversified VaR is USD 363,000. The diversification benefit is USD 105,262. These numbers run through the rest of this block.

## 3. Marginal VaR

**Definition.** "The change in portfolio VaR resulting from taking an additional dollar of exposure to a given component." It is the partial derivative of VaR with respect to the position, so it is a **slope** and it is **unitless** (currency of VaR per currency of position).

$$\boxed{\Delta\text{VaR}_i = \frac{\partial\,\text{VaR}}{\partial x_i} = \alpha\,\frac{\text{cov}(R_i, R_p)}{\sigma_p} = \alpha\,\frac{(\Sigma x)_i}{\sqrt{x'\Sigma x}} = \frac{\text{VaR}}{W}\,\beta_i}$$

$$\beta_i = \frac{\text{cov}(R_i, R_p)}{\sigma_p^2} = \frac{\rho_i\,\sigma_i}{\sigma_p} = W\,\frac{(\Sigma x)_i}{x'\Sigma x}$$

- **Beta form (reach for it when a beta is given).** Marginal VaR is VaR per unit of portfolio value times the asset's beta *to the portfolio*. Schweser's example: VaR EUR 400,000 on four equal EUR 1 million positions, $\beta_A = 1.2$: $\Delta\text{VaR}_A = (400{,}000/4{,}000{,}000) \times 1.2 = 0.12$.

- **Covariance form (when a covariance matrix is given).** $\Sigma x$ is already an intermediate step of the VaR run, so $\beta$ and marginal VaR "come almost for free". In Example 5.1:

$$\beta = \frac{3}{0.0244}\begin{bmatrix}0.0050\\0.0144\end{bmatrix} = \begin{bmatrix}0.615\\1.770\end{bmatrix}, \qquad \Delta\text{VaR} = \frac{1.65}{0.156205}\begin{bmatrix}0.0050\\0.0144\end{bmatrix} = \begin{bmatrix}0.0528\\0.1521\end{bmatrix}$$

  Each extra dollar of EUR adds about 15 cents of VaR; each extra dollar of CAD about 5 cents.

- **Ranking.** Under normality marginal VaR is proportional to beta, so ranking positions by marginal VaR needs only the betas. To cut risk, cut where marginal VaR is highest (IM-5 c).

## 4. Incremental VaR

**Definition.** "The change in VaR owing to a new position": a before-and-after comparison. If VaR falls the trade is a hedge; otherwise it is risk-increasing.

$$\boxed{\text{Incremental VaR} = \text{VaR}_{p+a} - \text{VaR}_p} \qquad \boxed{\text{Shortcut: Incremental VaR} \approx (\Delta\text{VaR})'\, a}$$

- **Exact (full revaluation).** Re-run the portfolio VaR with the trade in it. Jorion's bank: 100,000 trades and 10 minutes per run, too slow to price a client's trade on the spot.
- **Shortcut.** Multiply the *existing* portfolio's marginal VaRs by the new exposures, ignoring second-order terms. It is fast because $\Delta$VaR is a by-product of the last VaR run, and it "allows real-time trading limits". It is good when the trade is small relative to the portfolio. It does not need the assets to be uncorrelated: Schweser and Vardeez write "since the two assets are uncorrelated" before applying it, but the zero correlation plays no part.
- **Small trade.** Example 5.1, add USD 10,000 of CAD: shortcut $0.0528 \times 10{,}000 = \text{USD } 528$; full revaluation gives VaR $258{,}267$, an exact increment of USD 529.
- **Large trade.** A USD 100 million portfolio with 10% volatility adds USD 40 million of an asset with 20% volatility and correlation 0.5 to the portfolio ($\alpha = 1.65$). Exact (dollar volatilities 10 and 8):

$$1.65\left(\sqrt{10^2 + 8^2 + 2(0.5)(10)(8)} - 10\right) = 25.77 - 16.50 = \text{USD } 9.27 \text{ million}$$

  Shortcut: $\beta = 0.5 \times 0.20/0.10 = 1.0$, $\Delta\text{VaR} = 0.165$, $\times 40 = \text{USD } 6.60$ million. The shortcut misses 29% of the change.

## 5. Component VaR

**Definition.** "A partition of the portfolio VaR that indicates how much the portfolio VaR would change approximately if the given component was deleted."

$$\boxed{\text{CVaR}_i = \Delta\text{VaR}_i \times x_i = \text{VaR}\,\beta_i\, w_i = \text{VaR}_i\,\rho_i}$$

$$\sum_i \text{CVaR}_i = \text{VaR}\Big(\sum_i w_i\beta_i\Big) = \text{VaR} \qquad \frac{\text{CVaR}_i}{\text{VaR}} = w_i\beta_i$$

- **It adds up exactly**, because the weighted betas sum to 1 (the beta of the portfolio with itself). This is what makes it a decomposition. Adding individual VaRs and taking percentages is not one: it ignores diversification.
- **Individual VaR times correlation (shape S4).** Since $\beta_i = \rho_i\sigma_i/\sigma_p$, $\text{VaR}\,w_i\beta_i = (\alpha\sigma_i w_i W)\rho_i$. Given a correlation with the portfolio, this is the fastest route; the portfolio's total value is then often a red herring.

- **Sign.** A negative component VaR is a hedge for the rest of the portfolio. Barings: Japanese bonds and stocks were negatively correlated $(-0.114)$, so a hedge would have been long both. Leeson was long USD 7.7 billion of Nikkei and short USD 16 billion of JGBs. The bond leg's marginal VaR was negative $(-0.0092$ per USD 1 million$)$ and its position was short, so its component VaR was *positive*, USD 147.15 million: the "hedge" added risk. With USD 688.01 million from the Nikkei, the components sum to the USD 835.16 million monthly 95% VaR.
- **Example 5.1.** $\text{CVaR}_{\text{CAD}} = 0.0528 \times 2\text{m} =$ USD 105,630 (41.0%); $\text{CVaR}_{\text{EUR}} = 0.1521 \times 1\text{m} =$ USD 152,108 (59.0%); total USD 257,738. Read back: $\rho_{\text{CAD}} = 105{,}630/165{,}000 = 0.640$, $\rho_{\text{EUR}} = 152{,}108/198{,}000 = 0.768$.

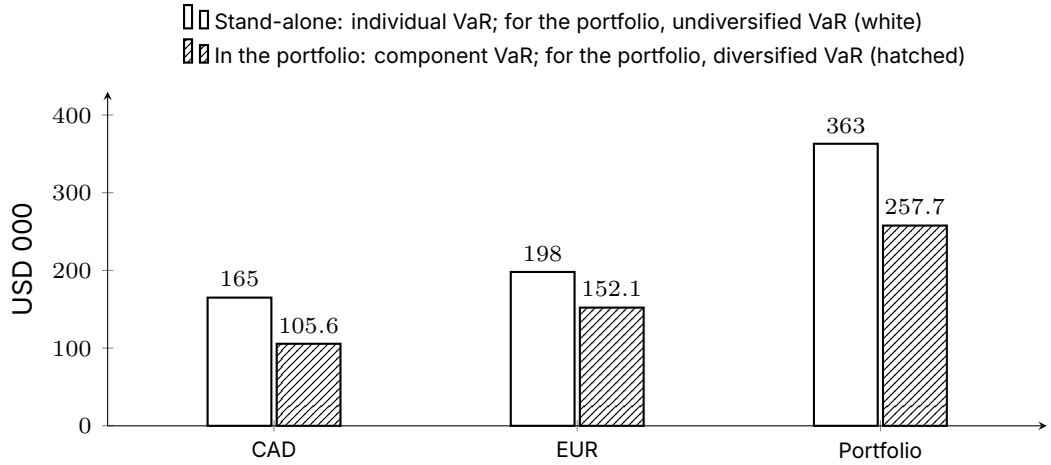



Figure 1. GARP's two-currency portfolio. Each hatched bar is the white bar times the position's correlation with the portfolio (0.640 for CAD, 0.768 for EUR). The hatched bars add up (105.6 + 152.1 = 257.7); the white bars add to 363.0, the undiversified VaR. The difference, 105.3, is the diversification benefit.

## 6. Tangent versus curve (shape S6)

All the tools are readings of one curve: portfolio VaR as a function of the position in one asset (Figure 2, the EUR position of Example 5.1).

- **Marginal VaR** is the slope of the tangent at the current position: 0.1521.
- **Component VaR** is the fall in VaR *along the tangent* if the position goes to zero: $0.1521 \times 1\text{m} =$ 152,108.
- **Incremental VaR of deleting EUR** is the fall *along the curve*: $257{,}738 - 165{,}000 = 92{,}738$. Component VaR is only an approximation of it, and a poor one here because EUR is a third of a two-asset book. The approximation improves when components are small, which is why the decomposition suits large portfolios of many small positions.
- **The rule that settles direction.** Delta-normal VaR is convex in each position, so the tangent always lies on or below the curve. Hence the shortcut $(\Delta\text{VaR})'a$ **understates** the VaR increase from an addition (528 vs 529; 6.60 vs 9.27; adding another USD 1 million of EUR: 152,108 vs 171,262), and component VaR **overstates** the VaR saved by a deletion (152,108 vs 92,738).
- **Best hedge.** Here the curve bottoms out at a zero EUR position (IM-5 d).

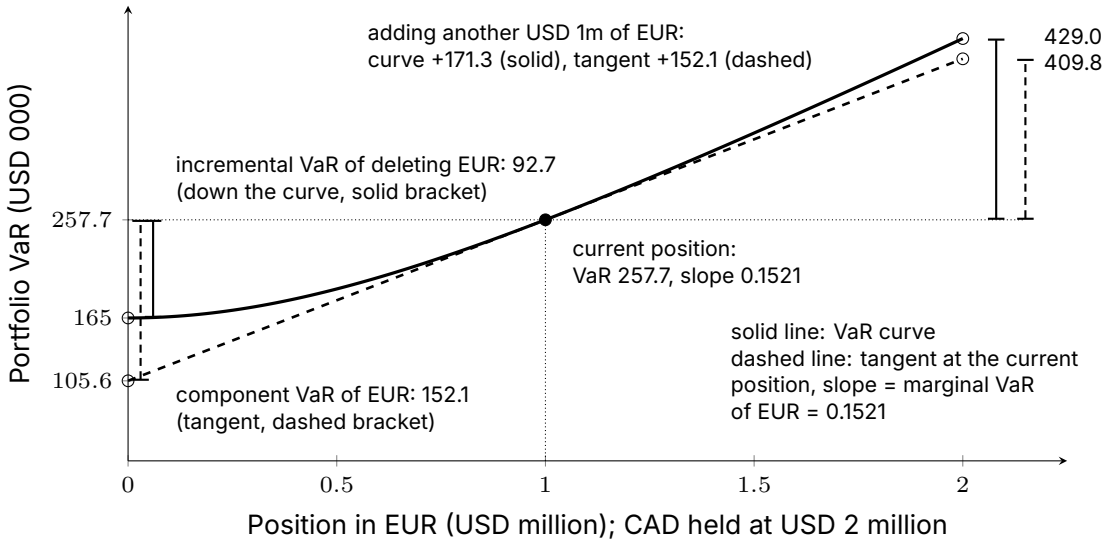



Figure 2. GARP's Figure 5.4, redrawn and extended to the right. The tangent (dashed) lies below the VaR curve (solid) on both sides of the current position, so the marginal-VaR estimate understates an addition and component VaR overstates a deletion. The tangent meets the zero-EUR axis at 105.6, which is the CAD component VaR.

**Correction to the source notes.** GARP Ch.5, Section 5.3, describes Figure 5.4 with "the marginal VaR is the change in VaR owing to an addition of $1 in EUR, or 0.0528; this represents the slope of the straight line that is tangent to the VaR curve". 0.0528 is the CAD marginal VaR. The EUR marginal VaR, and the slope of the tangent in a plot against the EUR position, is 0.1521 (Table 5.1). The component VaR in the same paragraph, USD 152,108 $=$ 0.1521 $\times$ USD 1 million, confirms it. Vardeez's redrawn figure uses 0.1521 correctly.

## 7. Reading a risk report backwards (shapes S3 and S4)

Many questions give a report, not a covariance matrix. Three identities unlock it:

1. **Euler:** $\dfrac{\text{VaR}}{W} = \sum_i w_i \, \Delta\text{VaR}_i$, so $\sigma_p = \dfrac{\text{VaR}/W}{\alpha}$ and $\beta_i = \dfrac{\Delta\text{VaR}_i}{\text{VaR}/W}$.
2. **Correlation from the report:** $\rho_i = \text{CVaR}_i/\text{VaR}_i$.
3. **Confidence rescaling:** under normality every measure, including marginal and component VaR, scales by the ratio of the $\alpha$'s: $\times\, 2.326/1.645 = 1.414$ from 95% to 99%.

**Worked example.** A USD 400 million portfolio, weights 50%, 30%, 20%, 95% marginal VaRs 0.10, 0.16, 0.06. Then $\text{VaR}/W = 0.05 + 0.048 + 0.012 = 0.110$: VaR is USD 44.0 million and $\sigma_p = 0.110/1.645 = 6.69\%$. Betas $0.909,\ 1.455,\ 0.545$ (weighted sum 1.000); components USD 20.0, 19.2 and 4.8 million. At 99% the VaR is $44.0 \times 1.414 =$ USD 62.2 million. With an expected return of 8.2% and a risk-free rate of 2.5%, the Sharpe ratio is $5.7\%/6.69\% = 0.85$. That last step is shape S3: marginal VaR gives $\sigma_p$, and $\sigma_p$ gives the Sharpe ratio.

**Consistency check.** The weighted betas must sum to 1 and the components must sum to the VaR. If they do not, the report cannot describe one portfolio.

**Correction to the source notes.** Vardeez's "component and marginal VaR from a table" example gives weights 0.4 and 0.6, betas 0.5 and 1.2, and volatilities 3.60%, 8.63% and 5.92% (portfolio). The weighted betas sum to $0.4(0.5) + 0.6(1.2) = 0.92$, not 1, so the component VaRs it implies (USD 19,477 and 70,116)

add to USD 89,593 against a VaR of USD 97,384. The stated volatilities imply $\rho = 0.413$ and betas of 0.368 and 1.422. The keyed answer (C: USD 19,477 and 0.1169) is what the formulas give from the inputs as printed, so it stands, but do not use the table to study how the measures fit together.

## Formula card: IM-5 a

| | |
|---|---|
| Individual VaR | $\text{VaR}_i = \alpha\sigma_i\lvert x_i\rvert$ |
| Diversified VaR | $\text{VaR}_p = \alpha\sigma_p W = \alpha\sqrt{x'\Sigma x}$ |
| Two long assets | $\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$ |
| Undiversified VaR | $\sum_i \text{VaR}_i$ (all $\rho = 1$, no shorts); diversification benefit = undiversified − diversified |
| Marginal VaR | $\Delta\text{VaR}_i = \alpha\,\text{cov}(R_i, R_p)/\sigma_p = (\text{VaR}/W)\,\beta_i$ |
| Beta to the portfolio | $\beta_i = \rho_i\sigma_i/\sigma_p = W(\Sigma x)_i/x'\Sigma x$ |
| Incremental VaR | exact $\text{VaR}_{p+a} - \text{VaR}_p$; shortcut $(\Delta\text{VaR})'a$, which understates an addition |
| Component VaR | $\text{CVaR}_i = \Delta\text{VaR}_i\,x_i = \text{VaR}\,w_i\beta_i = \text{VaR}_i\,\rho_i$ |
| Adding up | $\sum_i \text{CVaR}_i = \text{VaR}$; share $\text{CVaR}_i/\text{VaR} = w_i\beta_i$; $\sum_i w_i\beta_i = 1$ |
| Back out from a report | $\text{VaR}/W = \sum_i w_i\Delta\text{VaR}_i$; $\sigma_p = (\text{VaR}/W)/\alpha$ |
| Confidence | 95% to 99%: multiply every measure by $2.326/1.645 = 1.414$ |

| | Question it answers | Units | Adds up to VaR? | Revaluation? |
|---|---|---|---|---|
| **Individual** | Risk of the position on its own | currency | No: sums to undiversified | No |
| **Diversified** | Risk of the whole portfolio | currency | It is the VaR | One VaR run |
| **Undiversified** | Worst case if every correlation goes to 1 (long-only) | currency | No: an upper bound | No |
| **Marginal** | What one more unit of $i$ does | unitless | No | No: a by-product |
| **Incremental** | What a whole new trade (or a deletion) does | currency | No | Yes, if exact |
| **Component** | How much of today's VaR is due to $i$ | currency | Yes, exactly | No |

## Exam traps

- **Marginal VaR reported as money (intermediate result).** Marginal VaR is per unit of position; a "CAD 0.27 million" or "CAD 0.096 million" option is usually a marginal VaR that was never multiplied by the position.
- **Individual VaR as the contribution (dropped term).** $\alpha\sigma_i x_i$ ignores correlation; component VaR multiplies it by $\rho_i$.
- **Shortcut on a large trade (scope error).** $(\Delta\text{VaR})'a$ uses the old book's slope and always understates an addition; a position comparable in size to the book needs full revaluation.
- **Component VaR as the incremental VaR of a deletion.** Component VaR travels down the tangent and overstates the saving; incremental VaR travels down the curve.

- **Diversification benefit offered as incremental VaR (sibling swap).** Undiversified minus diversified VaR is a portfolio-level number, not the effect of a trade.
- **Weight, position or value?** $w_i$ is a weight, $x_i$ a position, $W$ the total. "Five positions of USD 2 million" is a USD 10 million portfolio. The total value is a red herring when the question gives a correlation and asks for component VaR.
- **Confidence level.** Rescale every measure, marginal VaR included, by 2.326/1.645 when the report and the question use different levels.
- **Long-short books.** For a long-short pair, the worst case is $\rho = -1$, not $+1$.

## Practice questions: IM-5 a

### Question IM-5 a.1

Bank Q228 • Schweser Module Quiz 87.1 Q1 (rebuilt)

A junior analyst at a long-only equity fund is preparing the VaR section of the monthly risk report. Every position in the fund is long. Which of the following statements about the fund's diversified and undiversified VaR is correct?

**A.** Undiversified VaR uses the portfolio's covariance structure and is less than diversified VaR.

**B.** Diversified VaR is the sum of the individual VaRs of the positions.

**C.** Diversified VaR can exceed undiversified VaR when correlations are positive.

**D.** Diversified VaR uses the portfolio's covariance structure and is less than or equal to undiversified VaR.

### Question IM-5 a.2

Bank Q231 • AnalystPrep Q.3147

A commodity trader computes the 1-day 99% VaR of a long gold futures position at USD 67 million and the 1-day 99% VaR of a long palladium position at USD 32 million. The correlation between changes in the two commodity prices is 0.4. Which of the following is closest to the combined (diversified) 1-day 99% VaR of the two positions?

**A.** USD 35.00 million

**B.** USD 74.25 million

**C.** USD 85.02 million

**D.** USD 99.00 million

### Question IM-5 a.3

Bank Q225 • GARP PE1 Q18 • Mock 3 Q62

A risk manager is evaluating the risks of a portfolio of stocks. Currently, the portfolio is valued at CAD 248 million and contains CAD 15 million in stock T. The annualized standard deviations of returns of the overall portfolio and of stock T are 16% and 13%, respectively. The correlation of returns between the portfolio and stock T is 0.45. Assuming the risk analyst uses a 1-year 95% VaR and returns are normally distributed, what is the component VaR of stock T?

**A.** CAD 0.096 million

**B.** CAD 1.444 million

**C.** CAD 2.041 million

**D.** CAD 3.948 million

### Question IM-5 a.4

Bank Q226 • Mock 1 Q21 • Mock 2 Q25 (rebuilt)

A risk analyst at a family office reviews a two-asset portfolio worth USD 3.4 million. The portfolio weight of Asset X is 0.40 and of Asset Y is 0.60. The standard deviation of portfolio returns is 9.8%, and the betas of X and Y with respect to the portfolio are 0.70 and 1.20. Assuming normally distributed returns, which of the following is closest to the marginal VaR of Asset X and the component VaR of Asset Y at the 99% confidence level ($z = 2.33$)?

**A.** Marginal VaR (X) 0.1598; component VaR (Y) USD 372,651

**B.** Marginal VaR (X) 0.1598; component VaR (Y) USD 465,814

**C.** Marginal VaR (X) 0.1598; component VaR (Y) USD 558,976

**D.** Marginal VaR (X) 0.2283; component VaR (Y) USD 558,976

### Question IM-5 a.5

New • built from GARP Ch.5 (Jorion), Eq. 5.20 • S3

A risk analyst at a Swiss private bank is preparing the quarterly pack for a CHF 250 million balanced mandate. The delta-normal VaR system reports the following for the mandate's equity sleeve, at the 99% level ($\alpha = 2.326$):

| | |
|---|---|
| Marginal 99% VaR of the equity sleeve (per CHF 1) | 0.1396 |
| Beta of the equity sleeve relative to the mandate | 1.20 |
| Annual volatility of the equity sleeve | 16.0% |
| Expected annual return of the mandate | 7.4% |
| Risk-free rate | 2.0% |

Assuming normally distributed returns, which of the following is closest to the mandate's Sharpe ratio?

**A.** 0.76

**B.** 0.90

**C.** 1.08

**D.** 1.48

### Question IM-5 a.6

Workbook A12.3 • S4, S6 • Map #13

A portfolio manager at a Canadian pension fund reviews the risk report for a CAD 120 million equity portfolio. The VaR system is delta-normal and reports, at the 95% level:

| Position | Value (CAD m) | Marginal 95% VaR (per CAD 1) | Individual 95% VaR (CAD m) |
|---|---|---|---|
| X | 54 | 0.1200 | 7.9 |
| Y | 36 | 0.1875 | 7.2 |
| Z | 30 | 0.1590 | 5.3 |

The 1-day 95% diversified portfolio VaR is CAD 18.0 million. Use $z_{0.95} = 1.645$ and $z_{0.99} = 2.326$. Which of the following is closest to the 1-day 99% component VaR of position Y?

**A.** CAD 0.27 million

**B.** CAD 6.75 million

**C.** CAD 9.54 million

**D.** CAD 10.18 million

### Question IM-5 a.7

Bank Q233 • Mock 1 Q42 (rebuilt)

A manager holds a portfolio consisting of one position: a USD 50 million long investment in Asset P, whose annual return volatility is 7%. She is considering adding a USD 50 million long position in either Asset R or Asset S, and wants to keep the portfolio's annual VaR within USD 13.5 million at the 95% confidence level ($z = 1.645$), assuming normally distributed returns.

| | Annual volatility | Correlation with P |
|---|---|---|
| Asset R | 11% | 0.6 |
| Asset S | 14% | 0.2 |

Which of the two proposed additions will keep the manager within her VaR limit?

**A.** Adding R only

**B.** Adding S only

**C.** Adding either R or S

**D.** Adding neither R nor S

### Question IM-5 a.8

Bank Q224 • BT P2.T9.21.6.3

A risk manager reviews a USD 30.0 million portfolio with two positions, using 95% VaR ($z = 1.645$) and jointly normal returns:

| Position | Size (USD m) | Expected excess return | Volatility | Marginal VaR |
|---|---|---|---|---|
| Consumer ETF | 20.0 | 4.0% | 12.0% | 0.122 |
| Materials ETF | 10.0 | 8.0% | 38.0% | 0.543 |

The correlation between the two positions is $+0.15$. Each of the following statements is true EXCEPT:

**A.** The incremental VaR of the Materials ETF position is larger than the incremental VaR of the Consumer ETF position.

**B.** Each position's component VaR is less than its incremental VaR.

**C.** If the correlation doubles to $+0.30$, each position's component VaR will increase.

**D.** To decrease portfolio VaR or to increase the Sharpe ratio, the manager should reallocate from the Materials ETF to the Consumer ETF.

### Question IM-5 a.9

Workbook A12.1 • S1, S6 • Aug-2026 • Map #13

A senior risk analyst at a UK bank's precious metals desk is presenting VaR tools to newly hired analysts. The desk's book currently holds only silver: a GBP 6 million position with an annual return volatility of 10%. The desk proposes including a GBP 4 million platinum position in the book. Platinum has an annual return volatility of 18% and a correlation of 0.3 with silver. The bank's real-time limit screen estimates the effect of any trade from the marginal VaRs of the current book. The analyst uses delta-normal VaR with $\alpha = 2.33$. Which of the following is closest to the incremental VaR of including the platinum position in the silver book?

**A.** GBP 0.503 million

**B.** GBP 0.590 million

**C.** GBP 0.786 million

**D.** GBP 1.087 million

## Solutions • IM-5 a

### Solution IM-5 a.1

**Answer: D**

Bank Q228 • Schweser Module Quiz 87.1 Q1

#### Step 1 – Write both measures

Diversified VaR $= \alpha\sigma_p W$, where $\sigma_p$ uses every volatility and correlation. Undiversified VaR $= \sum \alpha\sigma_i x_i$, the same formula with every $\rho$ set to 1.

#### Step 2 – Compare them

For long positions, $\sigma_p^2$ rises with each $\rho$, and $\rho \leq 1$, so $\sigma_p$ can be no larger than its value at $\rho = 1$.
$\Rightarrow$ Diversified VaR $\leq$ undiversified VaR, with equality only when every correlation is 1.

#### Interpretation

Undiversified VaR is the long-only worst case: what the portfolio would lose if diversification failed completely. The gap between the two is the diversification benefit that a VaR report shows.

#### Why the other options are wrong

**A is incorrect.** Polarity flip: undiversified VaR ignores the covariance structure and is the upper bound, not the lower.
**B is incorrect.** Sibling swap: the sum of the individual VaRs is the undiversified VaR.
**C is incorrect.** Scope error: with positive correlations below 1, diversified VaR is still below the sum; it reaches it only at $\rho = 1$.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.1, Eq. 5.12 to 5.14; SchweserNotes 2026 Book 5, Reading 87, LO 87.a

### Solution IM-5 a.2

**Answer: C**

Bank Q231 • AnalystPrep Q.3147

#### Step 1 – Use the VaR form of the two-asset formula

$$\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$$

$\Rightarrow$ Valid here because both positions are long and both VaRs are at the same confidence level and horizon.

#### Step 2 – Plug in

$$67^2 + 32^2 = 4{,}489 + 1{,}024 = 5{,}513; \qquad 2(0.4)(67)(32) = 1{,}715.2$$

$$\text{VaR}_p = \sqrt{5{,}513 + 1{,}715.2} = \sqrt{7{,}228.2} = \text{USD } 85.02 \text{ million}$$

#### Interpretation

The answer must sit between the zero-correlation value (74.25) and the perfect-correlation value (99.00). The diversification benefit is $99.00 - 85.02 = \text{USD } 13.98$ million.

#### Why the other options are wrong

**A is incorrect.** Subtracted the VaRs, $67 - 32 = 35.00$: the $\rho = -1$ case for two long positions, which would require the prices to move in opposite directions.
**B is incorrect.** Dropped the cross term ($\rho = 0$): $\sqrt{5{,}513} = 74.25$.
**D is incorrect.** Added the VaRs ($\rho = 1$): $67 + 32 = 99.00$, the undiversified VaR.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.1, Eq. 5.12 to 5.14

### Solution IM-5 a.3

**Answer: B**

Bank Q225 • GARP PE1 Q18

#### Step 1 – Individual VaR of T

$$\text{VaR}_T = 1.645 \times 0.13 \times 15 = \text{CAD } 3.208 \text{ million}$$

#### Step 2 – Component VaR = individual VaR × $\rho$ (shape S4)

$$\text{CVaR}_T = 3.208 \times 0.45 = \text{CAD } 1.444 \text{ million}$$

$\Rightarrow$ T contributes about CAD 1.44 million to the portfolio VaR, well below its stand-alone 3.21 million.

#### Step 3 – Cross-check through beta

$$\beta_T = \frac{0.45 \times 0.13}{0.16} = 0.3656, \qquad \Delta\text{VaR}_T = 1.645 \times 0.16 \times 0.3656 = 0.0962$$

$$\text{CVaR}_T = 0.0962 \times 15 = 1.444$$

#### Interpretation

The portfolio's CAD 248 million value is a red herring: the correlation route never needs it. Only the stock's own volatility, position and correlation with the portfolio matter.

#### Why the other options are wrong

**A is incorrect.** Intermediate result (S6): reported the marginal VaR, 0.0962 per CAD, as if it were CAD millions. It must be multiplied by the CAD 15 million position.
**C is incorrect.** Used the 99% deviate 2.326 for a 95% VaR: $2.326 \times 0.13 \times 15 \times 0.45 = 2.041$.
**D is incorrect.** Used the portfolio's volatility instead of T's and dropped the correlation: $1.645 \times 0.16 \times 15 = 3.948$.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2, Eq. 5.20 and 5.29; GARP Practice Exam 1, Q18

### Solution IM-5 a.4

**Answer: C**

Bank Q226 • Mock 1 Q21 • Mock 2 Q25

#### Step 1 – Portfolio VaR and VaR per dollar

$$\text{VaR}_p = 2.33 \times 0.098 \times 3.4\text{m} = \text{USD } 776{,}356, \qquad \text{VaR}/W = 0.22834$$

#### Step 2 – Check the betas

$0.40(0.70) + 0.60(1.20) = 1.00$.
$\Rightarrow$ The report is internally consistent, so the components will add to the VaR.

#### Step 3 – Marginal VaR of X

$$\Delta\text{VaR}_X = 0.22834 \times 0.70 = 0.1598$$

**Step 4 – Component VaR of Y**

$$\text{CVaR}_Y = \text{VaR} \times w_Y \beta_Y = 776{,}356 \times 0.60 \times 1.20 = \text{USD } 558{,}976$$

$\Rightarrow$ Y carries 72% of the risk on 60% of the capital (and X the other 28%, USD 217,380).

**Interpretation**

Marginal VaR is unitless; component VaR is in dollars. Both come from the same two inputs, VaR per dollar and beta, and the option grid pairs each right half with a wrong one.

**Why the other options are wrong**

**A is incorrect.** Component VaR of Y computed with X's position: $0.27401 \times 1.36\text{m} = 372{,}651$.
**B is incorrect.** Individual VaR of Y, $2.33 \times 0.098 \times 2.04\text{m} = 465{,}814$: this also uses the *portfolio's* volatility and ignores Y's beta.
**D is incorrect.** Reported VaR per dollar, 0.2283, as X's marginal VaR, dropping the beta of 0.70.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2, Eq. 5.20 and 5.27

**Solution IM-5 a.5** **Answer: C**

New • built from GARP Ch.5 (Jorion), Eq. 5.20

**Step 1 – VaR per franc from marginal VaR and beta**

$$\Delta\text{VaR}_i = \frac{\text{VaR}}{W}\beta_i \quad \Rightarrow \quad \frac{\text{VaR}}{W} = \frac{0.1396}{1.20} = 0.11633$$

$\Rightarrow$ The mandate's 99% VaR is 11.6% of its value, about CHF 29.1 million.

**Step 2 – Portfolio volatility**

$$\sigma_p = \frac{\text{VaR}/W}{\alpha} = \frac{0.11633}{2.326} = 5.00\%$$

**Step 3 – Sharpe ratio**

$$\text{SR} = \frac{E(R_p) - R_f}{\sigma_p} = \frac{7.4\% - 2.0\%}{5.00\%} = 1.08$$

**Interpretation**

The sleeve's 16% volatility is a red herring: its beta already contains it ($\beta_i = \rho_i \sigma_i / \sigma_p$). This is the reported chain "marginal VaR, volatility and risk-free rate given: calculate the Sharpe ratio". Every step is one identity: marginal VaR $\rightarrow$ VaR per unit $\rightarrow \sigma_p \rightarrow$ Sharpe.

**Why the other options are wrong**

**A is incorrect.** Used the 95% deviate 1.645 for a 99% marginal VaR: $\sigma_p = 0.11633/1.645 = 7.07\%$, SR $= 0.76$.
**B is incorrect.** Dropped the beta, treating marginal VaR as VaR per franc: $\sigma_p = 0.1396/2.326 = 6.00\%$, SR $= 0.90$.
**D is incorrect.** Forgot to subtract the risk-free rate: $7.4\%/5.00\% = 1.48$.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2, Eq. 5.20; 5.6, Eq. 5.37

### Solution IM-5 a.6

**Answer: C**

Workbook A12.3 • S4, S6

#### Step 1 – Check the report is internally consistent (Euler)

$$\sum_i \Delta\text{VaR}_i\, x_i = 0.1200(54) + 0.1875(36) + 0.1590(30) = 6.48 + 6.75 + 4.77 = 18.0$$

$\Rightarrow$ The components add up to the diversified VaR, so the marginal VaRs belong to this portfolio.

#### Step 2 – 95% component VaR of Y, Eq. 5.27

$$\text{CVaR}_Y = \Delta\text{VaR}_Y \times x_Y = 0.1875 \times 36 = 6.75$$

$\Rightarrow$ Y contributes CAD 6.75 million of the CAD 18.0 million 95% VaR (37.5%) on 30% of the capital.

#### Step 3 – Rescale to 99%

$$\text{CVaR}_Y^{99\%} = 6.75 \times \frac{2.326}{1.645} = 9.54$$

#### Step 4 – Closed-book cross-checks

$$\beta_Y = \frac{\Delta\text{VaR}_Y\, W}{\text{VaR}} = \frac{0.1875 \times 120}{18.0} = 1.25, \qquad \rho_Y = \frac{\text{CVaR}_Y}{\text{VaR}_Y} = \frac{6.75}{7.2} = 0.9375$$

$\Rightarrow$ Component VaR = individual VaR $\times \rho$: $7.2 \times 0.9375 \times 2.326/1.645 = 9.54$. The beta above 1 makes Y a hot spot.

#### Interpretation

Deleting Y would cut the 99% VaR by roughly CAD 9.5 million. That is a linear approximation of the incremental VaR of deletion, and by the convexity of VaR it overstates the true saving (Section 6). The individual VaR column is only needed for the $\rho$ cross-check.

#### Why the other options are wrong

**A is incorrect.** Intermediate result: reported the 99% marginal VaR of Y, $0.1875 \times 2.326/1.645 = 0.265$, as if it were a currency amount. Marginal VaR is unitless and must be multiplied by the position.

**B is incorrect.** Forgot to rescale from 95% to 99%: $0.1875 \times 36 = 6.75$.

**D is incorrect.** Used the individual VaR as the contribution, rescaled: $7.2 \times 2.326/1.645 = 10.18$. This sets $\rho_Y = 1$ and ignores diversification.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2 (Eq. 5.27 to 5.30, Table 5.1) and 5.5 (Eq. 5.33)

### Solution IM-5 a.7

**Answer: A**

Bank Q233 • Mock 1 Q42

#### Step 1 – Weights after the addition

Either way the portfolio becomes USD 100 million, with $w_P = w_{\text{new}} = 0.5$.

#### Step 2 – Diversified VaR with R

$$\sigma^2 = 0.25(0.07)^2 + 0.25(0.11)^2 + 2(0.25)(0.6)(0.07)(0.11) = 0.006560, \quad \sigma = 8.099\%$$

$$\text{VaR} = 1.645 \times 0.08099 \times 100 = \text{USD } 13.32 \text{ million} \leq 13.5$$

### Step 3 – Diversified VaR with S

$$\sigma^2 = 0.25(0.07)^2 + 0.25(0.14)^2 + 2(0.25)(0.2)(0.07)(0.14) = 0.007105, \quad \sigma = 8.429\%$$

$$\text{VaR} = 1.645 \times 0.08429 \times 100 = \text{USD } 13.87 \text{ million} > 13.5$$

$\Rightarrow$ Only R keeps the book inside the limit.

### Interpretation

The lower correlation does not rescue S: its stand-alone VaR (USD 11.52 million) is far larger than R's (USD 9.05 million). The incremental VaRs over the existing USD 5.76 million are 7.57 (R) and 8.11 (S). A full revaluation is needed here: each addition doubles the book.

### Why the other options are wrong

**B is incorrect.** Assumed the lower correlation automatically wins, ignoring S's higher volatility: P + S breaches at 13.87.

**C is incorrect.** Set the correlations to zero: VaRs of 10.72 (R) and 12.87 (S), both within the limit, but the stated correlations raise both.

**D is incorrect.** Used undiversified VaR: $5.76 + 9.05 = 14.81$ and $5.76 + 11.52 = 17.27$, both over the limit, ignoring diversification.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.1, Eq. 5.11 and 5.12; 5.2, Eq. 5.21

## Solution IM-5 a.8 Answer: B

Bank Q224 • BT P2.T9.21.6.3

### Step 1 – The VaR levels

Individual VaRs: Consumer $1.645 \times 0.12 \times 20 = 3.948$; Materials $1.645 \times 0.38 \times 10 = 6.251$. Components: $0.122 \times 20 = 2.44$ and $0.543 \times 10 = 5.43$, which sum to the diversified VaR (USD 7.87 million; 7.88 before the marginal VaRs were rounded).

### Step 2 – Incremental VaR of each position (deletion)

Removing Consumer leaves Materials alone: $7.88 - 6.251 = 1.63$. Removing Materials leaves Consumer alone: $7.88 - 3.948 = 3.93$.

$\Rightarrow$ A is true ($3.93 > 1.63$).

### Step 3 – Component against incremental

Consumer $2.44 > 1.63$; Materials $5.43 > 3.93$.

$\Rightarrow$ Each component VaR is *larger* than its incremental VaR, so B is false. This is the tangent-versus-curve gap (shape S6): component VaR always overstates the saving from a deletion.

### Step 4 – Check C and D

At $\rho = 0.30$ the VaR becomes 8.33 and the components 2.76 and 5.58, both up: C is true. Excess return per unit of marginal VaR is $0.04/0.122 = 0.33$ for Consumer and $0.08/0.543 = 0.15$ for Materials, and Materials also has the higher marginal VaR, so moving money to Consumer lowers VaR and raises the Sharpe ratio: D is true.

### Interpretation

Incremental VaR is measured along the curve; component VaR along the tangent. In a two-asset book the positions are large, so the gap is wide: 0.81 for Consumer and 1.50 for Materials.

### Why the other options are wrong

**A is incorrect.** True: Materials' incremental VaR (3.93) exceeds Consumer's (1.63). Removing

the wrong position when computing the deletion would make it look false.
**C is incorrect.** True: a higher correlation raises each position's covariance with the portfolio, so both components rise.
**D is incorrect.** True: Materials has both the higher marginal VaR and the lower excess-return-to-marginal-VaR ratio (the IM-5 d rule).

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2 and 5.3, Figure 5.4; 5.6, Eq. 5.38

### Solution IM-5 a.9 Answer: D

Workbook A12.1 • S1, S6 • Aug-2026

#### Step 1 – VaR of the current book (silver only)

$$\text{VaR}_p = \alpha\sigma_1 x_1 = 2.33 \times 0.10 \times 6 = 1.398$$

$\Rightarrow$ Before the trade the desk's VaR is GBP 1.398 million.

#### Step 2 – VaR of the book including platinum, Eq. 5.12

$$\sigma_{p+a}W_{p+a} = \sqrt{(6 \times 0.10)^2 + (4 \times 0.18)^2 + 2 \times 0.3 \times (6 \times 0.10)(4 \times 0.18)}$$

$$\sigma_{p+a}W_{p+a} = \sqrt{0.36 + 0.5184 + 0.2592} = \sqrt{1.1376} = 1.06658$$

$$\text{VaR}_{p+a} = 2.33 \times 1.06658 = 2.4851$$

#### Step 3 – Incremental VaR, Eq. 5.21

$$\text{VaR}_{p+a} - \text{VaR}_p = 2.4851 - 1.398 = 1.087$$

$\Rightarrow$ Including platinum raises VaR by GBP 1.09 million: the trade is risk-increasing, not a hedge.

#### Interpretation

The platinum position is two-thirds the size of the existing book, so VaR moves along its curve, not its tangent: the screen's linear estimate (option A) misses more than half of the change, and it errs on the low side, as the convexity rule predicts. Measured along the tangent of the *new* book, platinum's component VaR would be $2.33 \times (0.3 \times 0.10 \times 0.18 \times 6 + 0.18^2 \times 4)/1.06658 \times 4 = 1.416$, again different from the exact 1.087, as in Jorion's Figure 5.4.

#### Why the other options are wrong

**A is incorrect.** Used the screen's shortcut $(\Delta\text{VaR})'a$ on a large trade. In the silver-only book $\Delta\text{VaR}_{\text{plat}} = \alpha\,\text{cov}(R_{\text{plat}}, R_p)/\sigma_p = 2.33 \times 0.3 \times 0.18 = 0.1258$, times 4 gives 0.503. The shortcut ignores second-order terms, valid only for small trades.
**B is incorrect.** Computed the diversification benefit, undiversified minus diversified VaR: $1.398 + 2.33 \times 0.18 \times 4 - 2.4851 = 1.398 + 1.6776 - 2.4851 = 0.590$.
**C is incorrect.** Set the correlation to zero: $\sqrt{1.398^2 + 1.6776^2} - 1.398 = 2.1837 - 1.398 = 0.786$.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2 (Eq. 5.21 to 5.23, Figure 5.4)

IM-5 • Portfolio Risk: Analytical Methods • Vardeez: VERY IMPORTANT

# IM-5 c

P1 BOTH

Apply the concept of marginal VaR in making portfolio management decisions.

Workbook: A12 Incremental, Marginal and Component VaR • A18 Performance Measures and Attribution • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: the A12 shapes were reported on 6 of 16 recall sheets (Map #13), among them "allocation using MVaR" and "MVaR, vol, rf is given: calculate Sharpe ratio". The same Sharpe chain is reported under A18 (Map #36, 5 of 16).
**Vardeez:** no separate pointer for this LO; his reading note (marginal, incremental and component VaR calculations) covers it.

**Reported exam shapes**

- **A12·S2** Allocation using marginal VaR: where to add new exposure (Workbook A12.2), and which position to grow for a better return-to-risk trade-off.
- **A12·S5** Shortcut: when ranking marginal VaRs, only beta matters.
- **A18·S2** Marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio. The full three-asset version is Workbook A12.4 (worked as Question IM-5 d.7); Question IM-5 c.6 is the one-position version.

**What the questions look like:** a decision read off a marginal-VaR report: which position to cut, where to add, which of two equal-return assets to buy, or whether growing a position raises the Sharpe ratio. About half are numerical (estimate the VaR change, back out $\sigma_p$ and the Sharpe ratio); the rest are rule questions in which the wrong options flip the rule (cut versus add), rank by stand-alone volatility instead of beta, or use raw instead of excess returns.

## 1. The slope behind every decision (recap of IM-5 a)

The definitions are in IM-5 a (Section 3 for marginal VaR, Figure 2 for the VaR curve and its tangent). One line carries this LO:

$$\Delta\text{VaR}_i = \frac{\partial\,\text{VaR}}{\partial x_i} = \alpha\,\frac{\text{cov}(R_i, R_p)}{\sigma_p} = \frac{\text{VaR}}{W}\,\beta_i$$

$$\text{change in VaR} \approx \Delta\text{VaR}_i \times a_i$$

where $a_i$ is the change in the position (negative for a cut). Jorion's motivating question is "Which position should I alter to modify my VaR most effectively?". It matters because portfolios are traded incrementally, owing to transaction costs. Marginal VaR answers it for *small* changes: GARP calls marginal and component VaR "useful tools, best suited to small changes in the portfolio". A trade comparable in size to the book moves VaR along the curve, not the tangent, and needs full revaluation (IM-5 a, Section 4).

## 2. Five decisions, one tool (shape A12·S2)

1. **Cut risk.** To lower VaR by reducing one position by a fixed amount, "rank all marginal VaR numbers and pick the asset with the largest $\Delta$VaR because it will have the greatest hedging effect". Example 5.1 (CAD 0.0528, EUR 0.1521): cutting USD 100,000 of EUR lowers VaR by about $0.1521 \times 100{,}000 =$ USD 15,211 (exact 14,880); the same cut in CAD saves only USD 5,282 (exact 5,202). GARP: since EUR's marginal VaR is about three times CAD's, "cutting the position in the EUR will be much more effective".

2. **Add with the least extra risk.** New exposure goes where marginal VaR is *lowest*, and the shortcut gives the VaR increase in real time: $\Delta\text{VaR}_i \times a$ (Workbook A12.2). With equal return forecasts the choice is pure risk: Jorion's analyst choosing between a utility stock (portfolio beta 0.5) and an Internet stock (beta 2.0) for a portfolio already heavy in high-tech, both forecast to return 20%, picks the utility. "Such analysis is only feasible within the context of a portfoliowide VaR system."
3. **Fully invested, lower risk.** With capital fixed, cut where marginal VaR is highest and add where it is lowest. Repeating this leads to the global minimum, where all marginal VaRs are equal (IM-5 d).
4. **Better return for the risk.** Grow the position with the highest *excess* return per unit of marginal VaR, and trim the lowest (Section 4; the end point is the optimal portfolio of IM-5 d).
5. **Price a client trade.** A trader asks whether a client's proposed trade raises or lowers the book's risk. "If the trade is risk-decreasing, then the trader should adjust the bid-offer spread to increase the probability that the client will do the trade"; a risk-increasing trade should be discouraged.

## 3. When ranking, only beta matters (shape A12·S5)

In $\Delta\text{VaR}_i = (\text{VaR}/W)\,\beta_i$ the factor $\text{VaR}/W$ is the same for every position, so

- **the ranking of marginal VaRs is the ranking of the betas** (to the portfolio). With a covariance matrix, rank $(\Sigma x)_i$: $\beta_i = W(\Sigma x)_i / x'\Sigma x$ has the same order (Workbook A12.2, Step 1).
- **it is not the ranking of volatilities.** Since $\beta_i = \rho_i \sigma_i / \sigma_p$, a volatile asset with a low correlation to the book can have a *lower* marginal VaR than a calmer asset that moves with it.
- **Hot spots.** $\Delta\text{VaR}_i > \text{VaR}/W$ exactly when $\beta_i > 1$, which is when the position's share of risk ($w_i\beta_i$) exceeds its share of capital ($w_i$).
- **The confidence level does not change a ranking**: every marginal VaR scales by the same $\alpha$.

**Worked example.** A USD 200 million multi-asset fund, 95% VaR ($\alpha = 1.645$). Correlations: equities/credit 0.6, equities/govies $-0.2$, equities/commodities 0.3, credit/govies 0.3, credit/commodities 0.2, govies/commodities $-0.1$. The fund's volatility is 9.40%, so $\text{VaR}/W = 1.645 \times 9.40\% = 0.1546$ and VaR is USD 30.91 million.

| Sleeve | Weight | $\sigma_i$ | $\rho_i$ (with fund) | $\beta_i$ | $\Delta\text{VaR}_i$ | Share of VaR $w_i\beta_i$ |
|---|---|---|---|---|---|---|
| Equities | 40% | 18% | 0.932 | 1.786 | 0.2760 | 71.4% |
| Credit | 25% | 8% | 0.767 | 0.653 | 0.1010 | 16.3% |
| Government bonds | 25% | 6% | 0.047 | 0.030 | 0.0046 | 0.7% |
| Commodities | 10% | 22% | 0.491 | 1.149 | 0.1776 | 11.5% |

Commodities are the most volatile sleeve, but equities have the highest marginal VaR because their 0.93 correlation with the fund gives a beta of 1.79. Cutting USD 2 million of equities lowers VaR by about $0.2760 \times 2 =$ USD $0.552$ million; the same cut in commodities saves 0.355, in credit 0.202 and in government bonds 0.009 (full revaluation: 0.551, 0.349, 0.201, 0.009). Equities and commodities are the hot spots: equities carry 71% of the risk on 40% of the capital.

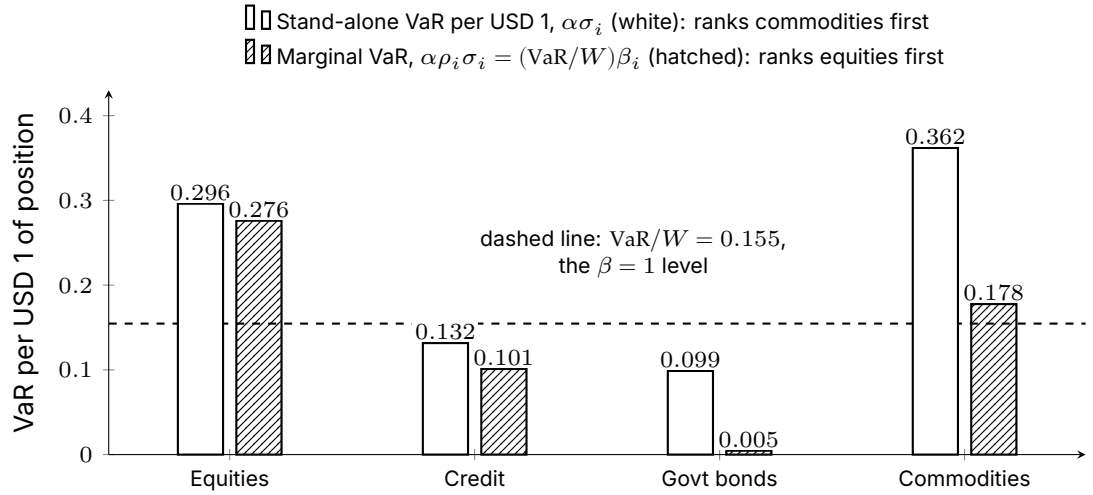



Figure 1. Each hatched bar is the white bar times the sleeve's correlation with the fund. Ranking by stand-alone risk would cut commodities; ranking by marginal VaR (equivalently by beta) cuts equities, which saves more VaR per dollar. Hatched bars above the dashed line ($\beta > 1$) are hot spots.

## 4. Return-to-risk decisions and the Sharpe chain (shapes A12·S2 and A18·S2)

With returns measured in excess of the risk-free rate, the portfolio manager's target is the Sharpe ratio $\mathrm{SR} = (E_p - R_f)/\sigma_p$ (Jorion's Eq. 5.37; it "could be written with VaR in the denominator"). A marginal-VaR report gives $\sigma_p$ without the covariance matrix:

$$\Delta\mathrm{VaR}_i,\ \beta_i \longrightarrow \frac{\mathrm{VaR}}{W} = \frac{\Delta\mathrm{VaR}_i}{\beta_i}\ \left(\text{or } \textstyle\sum_i w_i \Delta\mathrm{VaR}_i\right) \longrightarrow \sigma_p = \frac{\mathrm{VaR}/W}{\alpha} \longrightarrow \mathrm{SR} = \frac{E_p - R_f}{\sigma_p}$$

**Which position to grow.** Add a little of position $i$, funded from cash that earns $R_f$. Differentiating SR with respect to the position gives a change proportional to $(E_i - R_f) - \beta_i(E_p - R_f)$, so

$$\text{growing } i \text{ raises the Sharpe ratio} \iff \frac{E_i - R_f}{\beta_i} > E_p - R_f \iff \frac{E_i - R_f}{\Delta\mathrm{VaR}_i} > \frac{E_p - R_f}{\mathrm{VaR}/W}$$

Between two positions, move money toward the one with the higher excess return per unit of marginal VaR. Two consequences:

- **Use excess returns.** The raw-return test $E_i/\beta_i > E_p$ can give the opposite answer for a high-beta asset (Question IM-5 c.6).
- **Beta, not stand-alone volatility.** A sleeve's own Sharpe ratio $(E_i - R_f)/\sigma_i$ is the wrong lens: what it adds to the fund's risk is $\beta_i\sigma_p$, not $\sigma_i$.

**Worked example (SchweserNotes, LO 87.d).** USD 4 million in A ($\sigma = 6\%$) and USD 2 million in B ($\sigma = 14\%$), uncorrelated, $\alpha = 1.65$; excess returns 6% and 11%. VaR is USD 608,490 and the marginal VaRs are 0.0644 (A) and 0.1754 (B). Excess return per unit of marginal VaR: A $0.06/0.0644 = 0.931$, B $0.11/0.1754 = 0.627$, so money moves from B to A. At USD 4.5 and 1.5 million the ratios are 0.768 and 0.776 (nearly equal) and excess return to VaR rises from $0.460/0.6085 = 0.756$ to $0.435/0.5644 = 0.771$. Moving all the way to USD 5 and 1 million instead (where the marginal VaRs are almost equal, 0.0897 and 0.0977) gives the lowest VaR, USD 546,247, but a lower ratio, 0.751: the minimum-risk position is not the best return-to-risk position (IM-5 d).

**Formula card: IM-5 c**

| | |
|---|---|
| Marginal VaR | $\Delta\text{VaR}_i = \alpha\,\text{cov}(R_i, R_p)/\sigma_p = (\text{VaR}/W)\,\beta_i$ (unitless) |
| VaR change from a small trade | $\approx \Delta\text{VaR}_i \times a_i$ (Eq. 5.23); a cut has $a_i < 0$ |
| Ranking | same order as $\beta_i$, or as $(\Sigma x)_i$; not as $\sigma_i$ |
| Hot spot | $\Delta\text{VaR}_i > \text{VaR}/W \iff \beta_i > 1 \iff w_i\beta_i > w_i$ |
| Marginal VaR to $\sigma_p$ | $\text{VaR}/W = \Delta\text{VaR}_i/\beta_i = \sum_i w_i \Delta\text{VaR}_i$; $\sigma_p = (\text{VaR}/W)/\alpha$ |
| Sharpe ratio | $\text{SR} = (E_p - R_f)/\sigma_p$ |
| Grow position $i$ | raises SR if $(E_i - R_f)/\beta_i > E_p - R_f$; between two, grow the higher $(E_i - R_f)/\Delta\text{VaR}_i$ |

| Goal | Rule with marginal VaR | Stop when |
|---|---|---|
| **Cut risk by a fixed amount** | Cut the position with the largest $\Delta\text{VaR}_i$ | One-off decision |
| **Add with the least extra risk** | Add to the smallest $\Delta\text{VaR}_i$ (equal returns: lowest marginal VaR wins) | One-off decision |
| **Minimise risk, fully invested** | Cut the largest $\Delta\text{VaR}_i$, add to the smallest | All $\Delta\text{VaR}_i$ equal (IM-5 d) |
| **Best return per unit of risk** | Grow the largest $(E_i - R_f)/\Delta\text{VaR}_i$, trim the smallest | All ratios equal (IM-5 d) |
| **Client trade** | Win risk-reducing trades by adjusting the bid-offer spread | Trade done or declined |

**Exam traps**

- **Cut versus add (polarity flip).** Cut the largest marginal VaR; add to the smallest. The option that adds to the largest is the cut rule applied backwards.
- **Stand-alone risk instead of marginal risk (wrong lens).** Largest weight, highest volatility or highest individual VaR does not identify the position to cut; marginal VaR (beta) does.
- **Raw instead of excess returns (dropped term).** The return-to-risk test uses $E_i - R_f$; with raw returns a high-beta position can flip from "grow" to "trim".
- **Highest expected return is not the position to grow.** A high return earned with a high beta can have the lower $(E_i - R_f)/\Delta\text{VaR}_i$.
- **Dropped beta in the Sharpe chain (intermediate result).** Marginal VaR is $(\text{VaR}/W)\beta_i$; dividing it by $\alpha$ without first removing $\beta_i$ gives the wrong $\sigma_p$.
- **Shortcut on a large trade (scope).** $\Delta\text{VaR}_i \times a_i$ is the tangent: it understates the VaR increase from a large addition (IM-5 a).

## Practice questions: IM-5 c

### Question IM-5 c.1

Bank Q257 • BT P2.T8.14.1

A portfolio manager has identified two stocks, a utility stock and a technology stock, each with an expected return of +20% over the next year. She can add only one of them to her portfolio. Which of the following criteria is best for choosing between them?

**A.** The stock with the lower marginal value at risk (VaR)

**B.** The stock with the higher marginal VaR

**C.** The stock with the lower volatility

**D.** The stock with the higher incremental VaR

### Question IM-5 c.2

Bank Q255 • AnalystPrep Q.2468 (rebuilt)

An investor holds three positions and wants to lower portfolio VaR by reducing exactly one position by USD 100,000. The proceeds are held in cash, which has no risk. The risk system reports:

| Position | Weight | Stand-alone volatility | Marginal VaR (per USD 1 of position) |
|---|---|---|---|
| X | 50% | 30% | 0.040 |
| Y | 30% | 20% | 0.065 |
| Z | 20% | 12% | 0.018 |

Which position should be reduced?

**A.** Position X, because it has the largest weight and the highest stand-alone volatility

**B.** Position Y, because it has the largest marginal VaR

**C.** Position Z, because it has the smallest marginal VaR

**D.** Any position, because the reduction is the same USD 100,000 in each case

### Question IM-5 c.3

Workbook A12.2 • S2, S5 • Map #13

A treasury analyst at a Japanese bank manages a two-asset portfolio: JPY 50 billion in asset 1 (annual volatility 8%) and JPY 30 billion in asset 2 (annual volatility 15%). The correlation between the two returns is 0.4. The bank must add JPY 2 billion of new exposure to one of the two assets and wants the choice that increases portfolio VaR the least. To give an answer in real time, the VaR system estimates the change with marginal VaR rather than a full revaluation, using $\alpha = 1.65$. Which of the following correctly identifies the asset to add to and the estimated incremental VaR of the trade?

**A.** Asset 1; JPY 0.148 billion

**B.** Asset 1; JPY 0.215 billion

**C.** Asset 1; JPY 0.264 billion

**D.** Asset 2; JPY 0.424 billion

### Question IM-5 c.4

Bank Q258 • GARP PE1 Q65 • Mock 3 Q16

A money manager who has recently received a small amount of new capital is planning to invest it in an existing fund, which is benchmarked to an index. Rather than investing in a new asset, the manager plans to increase the holding of one of the fund's four assets. Information about these assets and their performance during the most recent evaluation period is given below:

| Asset | Portfolio weight | Return | Volatility of return | Beta to the portfolio |
|---|---|---|---|---|
| BDE | 0.35 | 14% | 19% | 1.20 |
| JKL | 0.30 | 13% | 18% | 0.90 |
| MNO | 0.25 | 13% | 16% | 1.00 |
| STU | 0.10 | 10% | 10% | 0.80 |

The portfolio manager wants to select the asset that has the lowest marginal VaR as long as its Jensen's alpha is greater than or equal to the market risk premium. Assuming the risk-free interest rate is 3% and the market return is 8%, which asset should the portfolio manager select?

**A.** Asset BDE

**B.** Asset JKL

**C.** Asset MNO

**D.** Asset STU

### Question IM-5 c.5

Bank Q266 • Schweser Module Quiz 87.2 Q1 • re-filed from IM-5 e

A portfolio has an equal amount invested in two positions, X and Y. The expected excess return of X is 9% and that of Y is 12%. Their marginal VaRs are 0.06 and 0.075, respectively. To move toward the optimal portfolio, the manager will probably:

**A.** increase the allocation in Y and/or lower that in X.

**B.** increase the allocation in X and/or lower that in Y.

**C.** do nothing, because the information is insufficient.

**D.** not change the portfolio, because it is already optimal.

**Question IM-5 c.6** New • built from GARP Ch.5 (Jorion), Eq. 5.20 and 5.37 to 5.39 • A18·S2

The CIO of a HKD 4,000 million multi-asset fund in Hong Kong is reviewing the fund's equity sleeve. The delta-normal VaR system reports, at the 95% level ($\alpha = 1.645$):

| | |
|---|---|
| Marginal 95% VaR of the equity sleeve (per HKD 1) | 0.1842 |
| Beta of the equity sleeve relative to the fund | 1.40 |
| Annual volatility of the equity sleeve | 14.0% |
| Expected annual return of the equity sleeve | 10.0% |
| Expected annual return of the fund | 7.6% |
| Risk-free rate | 3.0% |

The CIO is considering a small increase in the equity sleeve, funded from the fund's cash, which earns the risk-free rate. Assuming normally distributed returns, which of the following correctly gives the fund's current Sharpe ratio and the effect of the proposed increase?

**A.** Sharpe ratio 0.41; the increase would raise the fund's Sharpe ratio.

**B.** Sharpe ratio 0.58; the increase would lower the fund's Sharpe ratio.

**C.** Sharpe ratio 0.58; the increase would raise the fund's Sharpe ratio.

**D.** Sharpe ratio 0.95; the increase would lower the fund's Sharpe ratio.

## Solutions • IM-5 c

### Solution IM-5 c.1 — Answer: A

Bank Q257 • BT P2.T8.14.1

**Step 1 – Remove what does not discriminate**

Both stocks are forecast to return 20%, so the return side of the decision is a tie.
$\Rightarrow$ Only the effect on portfolio risk is left to decide.

**Step 2 – Measure the effect on portfolio risk**

Adding $a$ of stock $i$ raises VaR by about $\Delta\text{VaR}_i \times a$, and $\Delta\text{VaR}_i = (\text{VaR}/W)\beta_i$ depends on the stock's covariance with the existing book.
$\Rightarrow$ The stock with the lower marginal VaR adds less risk for the same return.

**Interpretation**

This is Jorion's own example: in a portfolio already heavy in high-tech, a utility with portfolio beta 0.5 beats an Internet stock with beta 2.0 when both are forecast to return 20%. Only a portfolio-wide VaR system can supply the betas.

**Why the other options are wrong**

**B is incorrect.** Polarity flip: the higher marginal VaR adds the most risk per dollar for the same return.

**C is incorrect.** Wrong lens: stand-alone volatility ignores correlation with the book; a low-volatility stock can still have the higher beta and marginal VaR.

**D is incorrect.** Higher incremental VaR means a larger increase in portfolio VaR, which is the opposite of the goal.

**LO:** IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.4 Using VaR for the Investment Process; Ch.7 (GARP 2025 Ch.5), 5.2, Eq. 5.20

### Solution IM-5 c.2 — Answer: B

Bank Q255 • AnalystPrep Q.2468

**Step 1 – VaR saved by each cut**

A cut of USD 100,000 lowers VaR by about $\Delta\text{VaR}_i \times 100{,}000$:

$$\text{X: } 0.040 \times 100{,}000 = 4{,}000 \qquad \text{Y: } 0.065 \times 100{,}000 = 6{,}500 \qquad \text{Z: } 0.018 \times 100{,}000 = 1{,}800$$

**Step 2 – Pick the largest saving**

$\Rightarrow$ Y gives the largest reduction, about USD 6,500 of VaR.

**Interpretation**

GARP's rule: to reduce all positions by a fixed amount, "rank all marginal VaR numbers and pick the asset with the largest $\Delta$VaR". X is the largest and most volatile position, but its marginal VaR is lower than Y's, so its co-movement with the rest of the book is weaker. The weight and stand-alone volatility columns are red herrings.

**Why the other options are wrong**

**A is incorrect.** Wrong lens: ranks on size and stand-alone volatility; cutting X saves only about USD 4,000.

**C is incorrect.** Polarity flip: the smallest marginal VaR is where to *add*; cutting Z saves only about USD 1,800.

**D is incorrect.** Equal dollar cuts do not have equal risk effects; each position's VaR per dollar differs.

**LO:** IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2 Marginal VaR (Eq. 5.20 and the ranking rule)

### Solution IM-5 c.3 — Answer: B

Workbook A12.2 • S2, S5

**Step 1 – Covariance vector $\Sigma x$**

$$(\Sigma x)_1 = 0.08^2 \times 50 + 0.4 \times 0.08 \times 0.15 \times 30 = 0.320 + 0.144 = 0.464$$
$$(\Sigma x)_2 = 0.4 \times 0.08 \times 0.15 \times 50 + 0.15^2 \times 30 = 0.240 + 0.675 = 0.915$$

$\Rightarrow$ The shortcut (S5): $\beta_i = W(\Sigma x)_i / x'\Sigma x$, so the marginal VaRs rank exactly as $(\Sigma x)_i$. Asset 1 already has the lower marginal VaR.

**Step 2 – Portfolio volatility and VaR**

$$\sigma_p W = \sqrt{50 \times 0.464 + 30 \times 0.915} = \sqrt{50.65} = 7.1169, \qquad \text{VaR}_p = 1.65 \times 7.1169 = 11.743$$

**Step 3 – Marginal VaRs, Eq. 5.17**

$$\Delta\text{VaR}_1 = 1.65 \times \frac{0.464}{7.1169} = 0.1076, \qquad \Delta\text{VaR}_2 = 1.65 \times \frac{0.915}{7.1169} = 0.2121$$

$\Rightarrow$ Equivalently $\beta_1 = 80 \times 0.464/50.65 = 0.733$ and $\beta_2 = 1.445$. Each extra yen in asset 1 adds 0.11 yen of VaR; in asset 2, 0.21 yen.

**Step 4 – Incremental VaR of adding JPY 2 billion to asset 1, Eq. 5.23**

$$(\Delta\text{VaR})'a = 0.1076 \times 2 = 0.215$$

$\Rightarrow$ VaR rises by about JPY 0.215 billion. Full revaluation gives 0.216, so the shortcut is accurate for a trade this small relative to the JPY 80 billion book.

**Interpretation**

Allocating by marginal VaR: new exposure goes where the marginal VaR (the beta) is lowest; a reduction would come from where it is highest. Here the lower-volatility asset is also the lower-beta asset, so both views agree.

**Why the other options are wrong**

**A is incorrect.** Dropped the covariance term from $(\Sigma x)_1$: $1.65 \times 0.320/7.1169 \times 2 = 0.148$. This ignores the asset's co-movement with the rest of the portfolio.

**C is incorrect.** Used the individual VaR of the new position, $1.65 \times 0.08 \times 2 = 0.264$, which ignores diversification against the existing book.

**D is incorrect.** Polarity flip of the marginal VaR rule: picked the asset with the largest marginal VaR (the one to cut when reducing risk) and added to it: $0.2121 \times 2 = 0.424$.

**LO:** IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2 (Eq. 5.17 to 5.23)

### Solution IM-5 c.4 — Answer: B

Bank Q258 • GARP PE1 Q65

**Step 1 – Apply the alpha screen**

Market risk premium $= 8\% - 3\% = 5\%$. Jensen's alpha $= R_i - [R_f + \beta_i(R_m - R_f)]$:

$$\text{BDE: } 14 - (3 + 1.20 \times 5) = 5.0\% \qquad \text{JKL: } 13 - (3 + 0.90 \times 5) = 5.5\%$$

$$\text{MNO: } 13 - (3 + 1.00 \times 5) = 5.0\% \qquad \text{STU: } 10 - (3 + 0.80 \times 5) = 3.0\%$$

$\Rightarrow$ STU fails the screen ($3.0\% < 5\%$); BDE, JKL and MNO pass.

**Step 2 – Rank the survivors by marginal VaR (shape S5)**

$\Delta\text{VaR}_i = (\text{VaR}/W)\beta_i$, and $\text{VaR}/W$ is common to all four assets, so the lowest beta has the lowest marginal VaR: JKL (0.90), then MNO (1.00), then BDE (1.20).

$\Rightarrow$ Select JKL.

**Interpretation**

No VaR number is given and none is needed: when ranking marginal VaRs, only beta matters. The volatility column is a red herring: STU's low 10% volatility does not rescue it, and a stock's own volatility does not set its marginal VaR. (GARP's solution uses the one beta column both for the alpha screen and for the marginal-VaR ranking.)

**Why the other options are wrong**

**A is incorrect.** BDE passes the screen but has the highest beta (1.20), hence the highest marginal VaR.

**C is incorrect.** MNO passes the screen, but its beta of 1.00 is above JKL's 0.90.

**D is incorrect.** STU has the lowest beta, but its alpha of 3.0% fails the 5% screen; ignoring the screen leads here.

**LO:** IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2, Eq. 5.20; GARP Practice Exam 1, Q65

### Solution IM-5 c.5 — Answer: A

Bank Q266 • Schweser Module Quiz 87.2 Q1

**Step 1 – Excess return per unit of marginal VaR**

$$\text{X: } \frac{0.09}{0.06} = 1.500 \qquad \text{Y: } \frac{0.12}{0.075} = 1.600$$

**Step 2 – Compare**

The ratios differ, so the portfolio is not optimal; Y earns more excess return per unit of risk added.

$\Rightarrow$ Shift money from X to Y until the ratios are equal.

**Interpretation**

The returns given are already in excess of the risk-free rate, so they can be divided directly by the marginal VaRs. Y has both the higher return and the higher marginal VaR; the ratio, not either input alone, decides. Equal marginal VaRs would give the minimum-risk portfolio instead, which is a different destination (IM-5 d).

**Why the other options are wrong**

**B is incorrect.** Reverses the comparison: adds to X, the position with the lower return per unit of marginal VaR.

**C is incorrect.** The information is sufficient: excess returns and marginal VaRs are all the ratio test needs.

**D is incorrect.** The portfolio would be optimal only if the two ratios were equal; 1.5 is not 1.6.

**LO:** IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.6, Eq. 5.38; SchweserNotes 2026 Book 5, Reading 87, LO 87.d

### Solution IM-5 c.6 Answer: C

New • built from GARP Ch.5 (Jorion), Eq. 5.20 and 5.37 to 5.39

#### Step 1 – VaR per HKD of the fund, from the sleeve's marginal VaR

$$\Delta\text{VaR}_i = \frac{\text{VaR}}{W}\beta_i \quad \Rightarrow \quad \frac{\text{VaR}}{W} = \frac{0.1842}{1.40} = 0.13157$$

$\Rightarrow$ The fund's 95% VaR is 13.2% of its value, about HKD 526 million.

#### Step 2 – Fund volatility

$$\sigma_p = \frac{\text{VaR}/W}{\alpha} = \frac{0.13157}{1.645} = 8.00\%$$

#### Step 3 – Sharpe ratio

$$\text{SR} = \frac{7.6\% - 3.0\%}{8.00\%} = 0.575 \approx 0.58$$

#### Step 4 – Does growing the sleeve raise the Sharpe ratio?

$$\frac{E_i - R_f}{\beta_i} = \frac{10.0\% - 3.0\%}{1.40} = 5.00\% \; > \; E_p - R_f = 4.60\%$$

$\Rightarrow$ Growing the equity sleeve, funded from cash, raises the Sharpe ratio. Equivalently, the sleeve earns $0.07/0.1842 = 0.380$ of excess return per unit of marginal VaR against $0.046/0.1316 = 0.350$ for the fund.

#### Interpretation

This is the reported chain "MVaR, vol, rf given: calculate the Sharpe ratio", taken one step further into a decision. The sleeve's 14% volatility is a red herring: its beta already contains it (implied correlation with the fund $= 1.40 \times 8.00/14.0 = 0.80$). Judged on its own Sharpe ratio, $7.0/14.0 = 0.50 < 0.575$, the sleeve looks like a poor addition, but what it adds to the fund's risk is $\beta_i\sigma_p = 11.2\%$, not 14%. The fund size is also not needed.

#### Why the other options are wrong

**A is incorrect.** Dropped the beta: treated the marginal VaR as VaR per HKD, so $\sigma_p = 0.1842/1.645 = 11.20\%$ and SR $= 4.6/11.2 = 0.41$.

**B is incorrect.** Right Sharpe ratio, wrong test: either raw returns, $10.0/1.40 = 7.14\% < 7.6\%$, or the sleeve's stand-alone Sharpe ratio, $0.50 < 0.575$. Both say "lower"; the excess-return test against beta says "raise".

**D is incorrect.** Raw returns throughout: $7.6/8.00 = 0.95$, and the raw test $7.14\% < 7.6\%$ points the wrong way.

**LO:** IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2, Eq. 5.20; 5.6, Eq. 5.37 to 5.39

IM-5 • Portfolio Risk: Analytical Methods • Vardeez: VERY IMPORTANT

# IM-5 d

P1 BOTH

Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk.

Workbook: A12 Incremental, Marginal and Component VaR • Tier A • August 2026 recall

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic: A12 (Map #13, reported on 6 of 16 recall sheets) includes "allocation using MVaR", and Workbook A12.4 drills the return-to-risk version.
**Vardeez:** no separate pointer for this LO. His notes explain both conditions; the worked positions are GARP's Tables 5.4 and 5.5.

**Reported exam shape**

- **A12·S2** Allocation using marginal VaR: where to add, and which position to grow for a better return-to-risk trade-off (Workbook A12.4).

The rules for single moves are in IM-5 c; this LO is about where those moves end: the minimum-risk position and the optimal (maximum return-to-risk) position.

**What the questions look like:** rule questions that set the two end points against each other (equal marginal VaRs versus equal excess-return-to-marginal-VaR ratios), and short calculations: compute the ratios and choose a direction or a mix, find the best hedge, back out a Sharpe ratio from a marginal-VaR report. Question IM-5 d.3 (best hedge) is new: the bank has no calculation of the risk-minimizing position in a single asset.

## 1. Two destinations

IM-5 c gave the single moves. Repeat them until no move helps and you arrive at one of two portfolios, depending on what is being maximised:

| | Minimum-risk portfolio | Optimal portfolio |
|---|---|---|
| **Objective** | Lowest VaR (lowest $\sigma_p$) | Highest SR $= E_p/\sigma_p$ (excess returns) |
| **Condition** | All $\Delta\text{VaR}_i$ equal, so all $\beta_i = 1$ | All $E_i/\Delta\text{VaR}_i$ equal, so $E_i/\beta_i = E_p$ |
| **Who asks** | Risk manager (risk management) | Portfolio manager (portfolio management) |

Vardeez draws the reading as three moves: measure (the VaR measures), decompose (the VaR tools) and act (these two destinations). "You cannot manage what you have not decomposed."

## 2. The risk-minimizing position in one asset: the best hedge

Add $a$ (in currency) of asset $i$ to a portfolio worth $W$. The dollar variance of the new portfolio is (Eq. 5.24)

$$\sigma_{p+a}^2 W_{p+a}^2 = \sigma_p^2 W^2 + 2aW\sigma_{ip} + a^2\sigma_i^2, \qquad \sigma_{ip} = \text{cov}(R_i, R_p)$$

Setting the derivative with respect to $a$ to zero (Eq. 5.25) gives the **best hedge**, "the additional amount to invest in an asset so as to minimize the risk of the total portfolio":

$$a^* = -W\frac{\sigma_{ip}}{\sigma_i^2} = -W\beta_i\frac{\sigma_p^2}{\sigma_i^2} = -W\rho_{ip}\frac{\sigma_p}{\sigma_i}$$

$$\text{remaining dollar volatility} = \sigma_p W\sqrt{1-\rho_{ip}^2}$$

- **Read it as a regression.** $\rho_{ip}\sigma_p/\sigma_i$ is the slope of the portfolio's return on the asset's return: the minimum-variance hedge ratio. The asset's beta to the portfolio, $\beta_i = \rho_{ip}\sigma_i/\sigma_p$, is the *other* regression and is not the hedge ratio.
- **Example 5.1.** For EUR, $a^* = -(\Sigma x)_{\text{EUR}}/\sigma^2_{\text{EUR}} = -0.0144/0.0144 = -$USD 1 million: the best hedge is a net zero euro position, the bottom of the VaR curve in IM-5 a, Figure 2. For CAD it is $-0.0050/0.0025 = -$USD 2 million. Because the two currencies are uncorrelated, the best hedge in each simply closes it.
- **Global equity report (Table 5.2).** The "Best Hedge" column shows that the 4.5% overweight in Japan should be cut by 4.93%, which lowers the tracking-error volatility from 1.82% to 1.48%; the 4.0% overweight in Canada has little impact (best hedge $-0.11\%$, volatility still 1.82%).

## 3. The risk-minimizing position when fully invested

With the total invested fixed, cut the position with the highest marginal VaR and add to the one with the lowest. "This process can be repeated up to the point where the portfolio risk has reached a global minimum. At this point, all the marginal VaRs, or the portfolio betas, must be equal" (Eq. 5.35):

$$\boxed{\Delta\text{VaR}_i = \frac{\text{VaR}}{W}\beta_i = \text{constant}} \qquad \Rightarrow \qquad \beta_i = 1 \text{ for all } i \quad (\text{since } \textstyle\sum_i w_i\beta_i = 1)$$

**Two assets in closed form:** $w_1^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$.

**GARP Table 5.4.** In Example 5.1 EUR has the higher marginal VaR (0.1521 against 0.0528), so EUR is cut and CAD is added. With $\rho = 0$, $w^*_{\text{CAD}} = 0.12^2/(0.05^2 + 0.12^2) = 0.0144/0.0169 = 85.21\%$:

| | Original | | Risk-minimizing | |
|---|---|---|---|---|
| Asset | $w_i$ | $\Delta\text{VaR}_i$ | $w_i$ | $\Delta\text{VaR}_i$ |
| CAD | 66.67% | 0.0528 | 85.21% | 0.0762 |
| EUR | 33.33% | 0.1521 | 14.79% | 0.0762 |
| Diversified VaR | USD 257,738 | | USD 228,462 | |
| Volatility (of the USD 3 million) | 5.21% | | 4.62% | |

At the minimum both betas are 1.000 and both marginal VaRs equal $\text{VaR}/W = 228{,}462/3{,}000{,}000 = 0.0762$.

## 4. The position that maximizes expected return to risk

"The role of the portfolio manager is to choose a portfolio that represents the best combination of expected return and risk." Returns are now in excess of the risk-free rate, which moves the risk-free asset to the origin of Figure 2. Expected returns add up, $E_p = \sum_i w_i E_i$ (Eq. 5.36); risk does not, which is why the VaR tools are needed. The target is the Sharpe ratio, $\text{SR} = E_p/\sigma_p$ (Eq. 5.37), or with VaR in the denominator. At its maximum (Eq. 5.38 and 5.39):

$$\boxed{\frac{E_i}{\Delta\text{VaR}_i} = \text{constant} \iff \frac{E_i}{\beta_i} = \text{constant} = E_p}$$

- **The constant is** $E_p$, because $E_p = \sum w_i E_i = c\sum w_i\beta_i = c$. So at the optimum $E_i = \beta_i E_p$: "simply a restatement of the capital asset pricing model". Roll (1977): the efficiency of any portfolio implies that each asset's expected return is proportional to its beta relative to that portfolio.

- **How to move:** increase a position whose $E_i/\beta_i$ exceeds $E_p$; trim one whose ratio falls short (IM-5 c, Section 4). The optimum is reached when "there is no reason to deviate".
- **GARP Table 5.5** ($E_{\text{CAD}} = 8\%$, $E_{\text{EUR}} = 5\%$, excess). Original: $E_p = 7.00\%$, $E/\beta$ is $0.08/0.615 = 0.1301$ for CAD and $0.05/1.770 = 0.0282$ for EUR, so CAD is increased. Optimum: CAD 90.21%, EUR 9.79%, betas 1.038 and 0.649, both ratios $0.0771 = E_p = 7.71\%$, VaR USD 230,720. With $\rho = 0$ the weights are proportional to $E_i/\sigma_i^2$: $0.08/0.0025 = 32.0$ and $0.05/0.0144 = 3.47$, so $w_{\text{CAD}} = 32.0/35.47 = 90.21\%$.
- **Optimal is riskier than minimum risk**: VaR 230,720 against 228,462; volatility 4.66% against 4.62%. The extra risk buys a higher expected return (7.71% against 7.56%) and the higher Sharpe ratio.

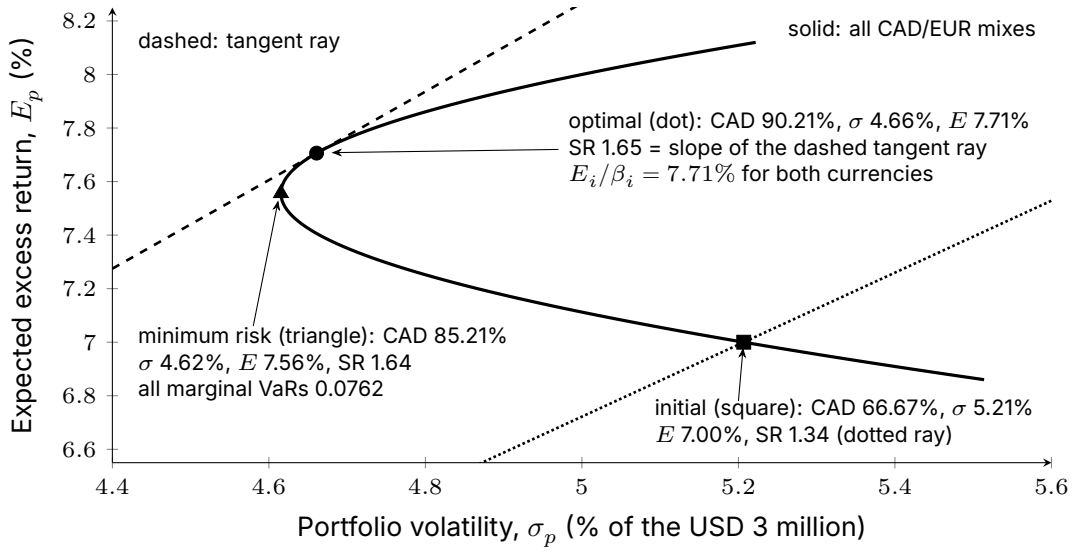



Figure 2. GARP's two-currency portfolio (Tables 5.4 and 5.5; Vardeez's figure, redrawn in consistent units). Cutting EUR on marginal VaR alone lands on the minimum-risk portfolio (triangle). Taking expected return into account lands on the optimal portfolio (dot), slightly riskier but on the steepest ray from the origin. Rays from the origin (the risk-free asset) have slope equal to the Sharpe ratio; only part of each ray is inside the plotted window.

**Correction to the source notes.** GARP's Tables 5.4 and 5.5 report the "standard deviation" as 15.62%, 13.85% and 13.98%, and the Sharpe ratios as 0.448 and 0.551. Those three figures are the dollar volatilities in USD million (USD 156,205, 138,462 and 139,829), not percentages of the USD 3 million portfolio: $1.65 \times 156{,}205 = 257{,}738$ is the VaR. As volatilities of the portfolio they are 5.21%, 4.62% and 4.66%, and the Sharpe ratios are $7.00/5.21 = 1.344$ (original), $7.56/4.62 = 1.637$ (minimum risk) and $7.71/4.66 = 1.653$ (optimal). Every ratio is three times GARP's, so the ranking, the weights and the conclusions are unchanged. Vardeez's figure and the A12 revision notes carry GARP's figures; use the corrected ones if a question gives the volatility in percent.

## 5. Risk management versus portfolio management

Risk management asks only how to make the portfolio less risky; its answer is the global minimum, where all marginal VaRs are equal. Portfolio management balances increasing risk against increasing expected return; its answer is the optimal portfolio, where all excess-return-to-marginal-VaR ratios are equal. Marginal VaR is the instrument in both; what changes is what it is divided into (Vardeez: "nothing in the first case, and the position's excess expected return in the second"). Whichever destination you work toward, check that $\sum_i w_i\beta_i = 1$ before trusting a table of betas.

## Formula card: IM-5 d

| | |
|---|---|
| Best hedge (Eq. 5.26) | $a^* = -W\beta_i\sigma_p^2/\sigma_i^2 = -W\rho_{ip}\sigma_p/\sigma_i$; remaining dollar volatility $\sigma_p W\sqrt{1-\rho_{ip}^2}$ |
| Minimum risk (Eq. 5.35) | $\Delta\text{VaR}_i = (\text{VaR}/W)\beta_i$ equal for all $i$, so every $\beta_i = 1$ |
| Two assets, minimum risk | $w_1^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$ |
| Expected return (Eq. 5.36) | $E_p = \sum_i w_i E_i$ (additive; risk is not) |
| Sharpe ratio (Eq. 5.37) | $\text{SR} = E_p/\sigma_p$ with excess returns, or $E_p/\text{VaR}$ |
| Optimum (Eq. 5.38, 5.39) | $E_i/\Delta\text{VaR}_i$ constant $\iff E_i/\beta_i = E_p$ for all $i$ |
| Direction | increase $i$ if $E_i/\beta_i > E_p$ (excess); trim it if below |
| Two assets, $\rho = 0$, optimum | $w_i \propto E_i/\sigma_i^2$ |
| Check | $\sum_i w_i\beta_i = 1$ at every portfolio |

## Exam traps

- **Minimum risk is not optimal (sibling swap).** Equal marginal VaRs, or equal betas, define the minimum-risk portfolio. The optimal portfolio needs equal excess-return-to-beta (or to marginal VaR) ratios.
- **The optimum must be less risky (absolute claim).** It is usually slightly riskier than the minimum-risk portfolio (VaR 230,720 against 228,462).
- **Raw instead of excess returns (dropped term).** The ratios $E_i/\beta_i$ use returns in excess of $R_f$; subtract it before comparing.
- **Highest expected return wins (wrong lens).** The position to grow has the highest return *per unit of beta*, not the highest return.
- **Best hedge with the wrong slope (wrong input).** The hedge ratio is $\rho\sigma_p/\sigma_i$ (portfolio on asset), not the asset's beta to the portfolio and not $\rho$ alone.
- **Remaining risk after the best hedge.** Volatility falls to $\sigma_p\sqrt{1-\rho^2}$, not $\sigma_p(1-\rho^2)$.
- **Percent or dollar volatility (units).** GARP's tables print dollar volatilities as percentages; a Sharpe ratio needs $\sigma_p$ as a percentage of portfolio value.

## Practice questions: IM-5 d

### Question IM-5 d.1

Bank Q262 • BT P2.T8.3.4

A risk manager wants to shift a portfolio to its global risk minimum. Which of the following is most effective?

**A.** Add (trim) positions with low (high) Sharpe ratios until the Sharpe ratios are equal

**B.** Add (trim) positions with low (high) marginal VaR until the marginal VaRs are equal

**C.** Add (trim) positions with low (high) ratios of excess return to beta until the ratios are equal

**D.** Add (trim) positions with low (high) ratios of individual VaR to portfolio VaR until the ratios are equal

### Question IM-5 d.2

Bank Q261 • BT P2.T9.21.6.1 (rebuilt)

Emily manages a USD 20.0 million portfolio allocated between a Technology ETF and a REIT ETF. Annual volatilities are 20% (Tech) and 30% (REIT); the correlation between them is $-0.20$. She uses a 95% confidence level ($z = 1.645$) and compares four allocations:

| Mix | #1 | #2 | #3 | #4 |
|---|---|---|---|---|
| Weight in Tech ETF | 35.0% | 50.0% | 65.0% | 80.0% |
| Weight in REIT ETF | 65.0% | 50.0% | 35.0% | 20.0% |

If her goal is to minimize the portfolio's diversified VaR, which allocation is best?

**A.** Mix #1, with 35.0% allocated to the Tech ETF

**B.** Mix #2, with 50.0% allocated to the Tech ETF

**C.** Mix #3, with 65.0% allocated to the Tech ETF

**D.** Mix #4, with 80.0% allocated to the Tech ETF

### Question IM-5 d.3

New • built from GARP Ch.5 (Jorion), Eq. 5.24 to 5.26

A risk analyst at a US university endowment manages a USD 50 million equity portfolio with an annual return volatility of 12%. Before a board meeting, the endowment wants to cut risk as far as possible with a single position in an equity index futures contract. The futures' annual return volatility is 20% and its beta relative to the current portfolio is 1.25. The analyst uses delta-normal VaR at the 95% confidence level ($\alpha = 1.645$) over one year. Which of the following is closest to the futures position that minimizes the portfolio's VaR (the best hedge) and the resulting 1-year 95% VaR?

**A.** Sell USD 22.5 million of futures; VaR of USD 4.32 million

**B.** Sell USD 22.5 million of futures; VaR of USD 6.53 million

**C.** Sell USD 37.5 million of futures; VaR of USD 8.18 million

**D.** Sell USD 62.5 million of futures; VaR of USD 14.69 million

### Question IM-5 d.4

Bank Q265 • GARP PE2 Q16 • Mock 4 Q17

A portfolio manager wants to reach the optimal portfolio on the portfolio's efficient frontier by replacing one stock with a new stock not currently in the portfolio, keeping the portfolio value constant. The manager considers the following actions:

- Action 1: Sell the stock with the highest marginal VaR and buy an equivalent value of a new stock that would have the lowest marginal VaR in the portfolio.
- Action 2: Sell a stock and buy an equivalent value of a new stock, which would cause the ratio of expected excess return to portfolio beta to be equal for all stocks in the portfolio.
- Action 3: Sell a stock and buy an equivalent value of a new stock, which would cause the portfolio betas of all stocks to be equal.
- Action 4: Sell a stock and buy an equivalent value of a new stock, which would significantly decrease the portfolio standard deviation without changing the average excess portfolio return.

Which action would create the optimal portfolio?

**A.** Action 1

**B.** Action 2

**C.** Action 3

**D.** Action 4

### Question IM-5 d.5

Bank Q259 • AnalystPrep Q.2473 (rebuilt)

A manager holds a two-currency portfolio; the risk-free rate is 2.0%. Each beta is the position's beta relative to the portfolio.

| Position | Weight | Expected return | Beta |
|---|---|---|---|
| CAD | 66.67% | 8.00% | 0.615 |
| EUR | 33.33% | 5.00% | 1.770 |

Which of the following correctly gives the portfolio's expected excess return and the adjustment that moves the portfolio toward the position that maximizes the ratio of expected excess return to risk?

**A.** Expected excess return 5.0%; increase CAD and decrease EUR

**B.** Expected excess return 5.0%; increase EUR and decrease CAD

**C.** Expected excess return 7.0%; increase EUR and decrease CAD

**D.** Expected excess return 7.0%; increase CAD and decrease EUR

### Question IM-5 d.6

Bank Q264 • BT P2.T9.21.6.2 (rebuilt)

Peter manages a USD 10.0 million portfolio allocated between an Energy ETF and a Healthcare ETF, and compares four allocations. The Energy ETF offers an expected excess return (over the risk-free rate) of 6.0% and the Healthcare ETF 12.0%. Betas are each ETF's beta relative to the portfolio.

| Mix | #1 | #2 | #3 | #4 |
|---|---|---|---|---|
| Weight in Energy ETF | 45.0% | 60.0% | 75.0% | 90.0% |
| Marginal VaR (Energy) | 0.051 | 0.085 | 0.126 | 0.158 |
| Beta (Energy, portfolio) | 0.215 | 0.441 | 0.787 | 1.032 |
| Weight in Healthcare ETF | 55.0% | 40.0% | 25.0% | 10.0% |
| Marginal VaR (Healthcare) | 0.391 | 0.353 | 0.263 | 0.110 |
| Beta (Healthcare, portfolio) | 1.642 | 1.838 | 1.639 | 0.716 |

Peter's goal is not risk minimization but to maximize the ratio of expected excess return to risk. Which allocation is best?

**A.** Mix #1, with 45.0% allocated to the Energy ETF

**B.** Mix #2, with 60.0% allocated to the Energy ETF

**C.** Mix #3, with 75.0% allocated to the Energy ETF

**D.** Mix #4, with 90.0% allocated to the Energy ETF

### Question IM-5 d.7

Workbook A12.4 • S2, S3, S4, S5 • Map #13

A portfolio manager at an Australian superannuation fund runs an AUD 500 million portfolio of three assets. Returns are normal and the risk system reports 1-year 95% marginal VaRs ($\alpha = 1.645$):

| Asset | Weight | Expected return | Marginal 95% VaR (per AUD 1) |
|---|---|---|---|
| A | 50% | 9.0% | 0.12 |
| B | 30% | 12.0% | 0.20 |
| C | 20% | 4.5% | 0.05 |

The risk-free rate is 3%. The manager wants to move the portfolio toward the highest ratio of expected excess return to risk, starting by increasing one position funded from another. Which of the following correctly identifies the position to increase and the portfolio's current Sharpe ratio?

**A.** Increase C; Sharpe ratio 0.76

**B.** Increase B; Sharpe ratio 0.76

**C.** Increase A; Sharpe ratio 1.14

**D.** Increase A; Sharpe ratio 0.76

## Solutions • IM-5 d

### Solution IM-5 d.1

**Answer: B**

Bank Q262 • BT P2.T8.3.4

**Step 1 – What the risk manager minimizes**

Only VaR (portfolio risk); expected returns play no part.

**Step 2 – The condition at the minimum**

If two marginal VaRs differ, moving money from the higher one to the lower one lowers VaR, so the portfolio is not yet at the minimum. At the global minimum "all the marginal VaRs, or the portfolio betas, must be equal" (Eq. 5.35).

$\Rightarrow$ Add to low-marginal-VaR positions and trim high ones until the marginal VaRs are equal.

**Interpretation**

This is the risk manager's destination. The portfolio manager's destination (option C, with the direction corrected) uses expected returns as well and ends at a different, slightly riskier portfolio.

**Why the other options are wrong**

**A is incorrect.** Equalising Sharpe ratios brings in expected returns, so it is not a pure risk minimization, and position-level Sharpe ratios ignore correlation.

**C is incorrect.** Sibling swap: equal excess-return-to-beta ratios define the optimal portfolio, not the minimum-risk one; the stated direction is also reversed (high ratios should be added to).

**D is incorrect.** Individual VaR ignores correlation with the rest of the book, so equalising it does not minimize portfolio VaR.

**LO:** IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.6, Eq. 5.35, Table 5.4

### Solution IM-5 d.2

**Answer: C**

Bank Q261 • BT P2.T9.21.6.1

**Step 1 – Locate the minimum-variance weight**

$$w^*_{\text{Tech}} = \frac{\sigma_R^2 - \rho\sigma_T\sigma_R}{\sigma_T^2 + \sigma_R^2 - 2\rho\sigma_T\sigma_R} = \frac{0.09 + 0.012}{0.04 + 0.09 + 0.024} = \frac{0.102}{0.154} = 66.2\%$$

$\Rightarrow$ The minimum lies just above Mix #3; VaR rises on either side of it.

**Step 2 – Diversified VaR of each mix (USD million)**

$$\text{VaR} = 1.645\sqrt{(4w)^2 + \big(6(1-w)\big)^2 + 2(-0.20)(4w)\big(6(1-w)\big)}$$

where $4w = 20w \times 0.20$ and $6(1-w) = 20(1-w) \times 0.30$ are the dollar volatilities. Mix #1: 6.368; Mix #2: 5.356; Mix #3: 4.931; Mix #4: 5.239.

**Step 3 – Cross-check with marginal VaR (the IM-5 d condition)**

Marginal VaRs (Tech, REIT): #1 0.053 and 0.461; #2 0.141 and 0.394; #3 0.239 and 0.260; #4 0.306 and 0.087.

$\Rightarrow$ Only at Mix #3 are the two marginal VaRs nearly equal: it is the closest to the global minimum.

**Interpretation**

The negative correlation makes the more volatile REIT ETF a hedge for the Tech ETF, so the minimum keeps 34% in REIT despite its 30% volatility. Between #1 and #4 the marginal VaRs cross over, which is the signal that the minimum lies in between.

**Why the other options are wrong**

**A is incorrect.** Puts 65% in the more volatile REIT ETF: VaR USD 6.37 million, with marginal VaRs far apart (0.053 against 0.461).

**B is incorrect.** Equal weighting is a naive guess: VaR USD 5.36 million, marginal VaRs still unequal (0.141 against 0.394).

**D is incorrect.** Overshoots: Tech's marginal VaR (0.306) now exceeds REIT's (0.087), and VaR is back up to USD 5.24 million.

**LO:** IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.6, Eq. 5.35

## Solution IM-5 d.3

**Answer: B**

New • built from GARP Ch.5 (Jorion), Eq. 5.24 to 5.26

### Step 1 – Covariance and correlation of the futures with the portfolio

$$\sigma_{ip} = \beta_i \sigma_p^2 = 1.25 \times 0.12^2 = 0.0180, \qquad \rho_{ip} = \frac{0.0180}{0.12 \times 0.20} = 0.75$$

### Step 2 – Best hedge, Eq. 5.26

$$a^* = -W\beta_i \frac{\sigma_p^2}{\sigma_i^2} = -50 \times 1.25 \times \frac{0.0144}{0.04} = -22.5$$

$\Rightarrow$ Sell USD 22.5 million of futures: a hedge ratio of $\rho_{ip}\sigma_p/\sigma_i = 0.75 \times 0.12/0.20 = 0.45$ per dollar of portfolio.

### Step 3 – Remaining risk and VaR

$$\sigma W = \sqrt{(50 \times 0.12)^2 + 2(-22.5)(50)(0.0180) + 22.5^2 \times 0.04}$$

$$\sigma W = \sqrt{36 - 40.5 + 20.25} = 3.969$$

$$\text{VaR} = 1.645 \times 3.969 = \text{USD } 6.53 \text{ million}$$

$\Rightarrow$ The same answer comes from $\sigma_p W\sqrt{1-\rho^2} = 6 \times \sqrt{0.4375} = 3.969$. VaR falls from USD 9.87 million to USD 6.53 million.

### Interpretation

The best hedge cannot remove more than the part of the portfolio's risk that the futures explain ($\rho^2 = 56\%$ of the variance); the rest stays. Any position beyond USD 22.5 million starts adding risk back (USD 62.5 million leaves VaR higher than doing nothing).

### Why the other options are wrong

**A is incorrect.** Right hedge, but cut the volatility by $(1-\rho^2)$ instead of $\sqrt{1-\rho^2}$: $9.87 \times 0.4375 = 4.32$.

**C is incorrect.** Used the correlation as the hedge ratio: $0.75 \times 50 = 37.5$, which overhedges; VaR is then $1.645\sqrt{36 - 67.5 + 56.25} = 8.18$.

**D is incorrect.** Used the futures' beta to the portfolio, 1.25, as the hedge ratio: $1.25 \times 50 = 62.5$. That is the slope of the wrong regression; VaR rises to $1.645\sqrt{36 - 112.5 + 156.25} = 14.69$.

**LO:** IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.2, Eq. 5.24 to 5.26 (best hedge)

### Solution IM-5 d.4 — Answer: B

Bank Q265 • GARP PE2 Q16

#### Step 1 – The condition for the optimal portfolio

The optimal portfolio maximizes the Sharpe ratio. At that point the ratio of expected excess return to marginal VaR is the same for every holding (Eq. 5.38).

#### Step 2 – Translate marginal VaR into beta

$\Delta\text{VaR}_i = (\text{VaR}/W)\beta_i$, so the condition is equivalently equal $(E_i - R_f)/\beta_i$ for all stocks.
$\Rightarrow$ Action 2 imposes exactly this.

#### Interpretation

If the ratios differ, moving money from the low-ratio stock to the high-ratio stock raises the return per unit of risk, so the portfolio is not yet optimal. Equal ratios mean no reallocation can help; Jorion notes this is a restatement of the CAPM relative to the portfolio.

#### Why the other options are wrong

**A is incorrect.** Action 1 is the risk-minimizing rule; it ignores expected returns, and the minimum-risk portfolio is not the optimal one.

**C is incorrect.** Equal betas is the minimum-risk condition (all marginal VaRs equal), not the optimal one.

**D is incorrect.** Lowering volatility at the same excess return is an improvement, but it does not make the ratios equal, so the result need not be optimal.

**LO:** IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.6, Eq. 5.37 to 5.39; GARP Practice Exam 2, Q16

### Solution IM-5 d.5 — Answer: A

Bank Q259 • AnalystPrep Q.2473

#### Step 1 – Expected and excess return

$$E_p = 0.6667(8\%) + 0.3333(5\%) = 7.0\%, \qquad E_p - R_f = 7.0\% - 2.0\% = 5.0\%$$

#### Step 2 – Excess return per unit of beta

$$\text{CAD: } \frac{8\% - 2\%}{0.615} = 9.76\% \qquad \text{EUR: } \frac{5\% - 2\%}{1.770} = 1.69\%$$

$\Rightarrow$ CAD's ratio exceeds the portfolio's 5.0%, EUR's falls short: increase CAD, decrease EUR.

#### Interpretation

These are GARP's Table 5.5 weights and betas, here with total returns and a 2% risk-free rate. The direction is the same as in GARP (CAD 0.1301 against EUR 0.0282 on excess returns of 8% and 5%). Note that the weighted betas sum to 1, a quick check that the table describes one portfolio.

#### Why the other options are wrong

**B is incorrect.** Right excess return, wrong direction: EUR has the lower ratio and should be trimmed.

**C is incorrect.** Reports the total return, 7.0%, and moves in the wrong direction.

**D is incorrect.** Right direction, but 7.0% is the expected return before the 2.0% risk-free rate is subtracted.

**LO:** IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.6, Eq. 5.38 and 5.39, Table 5.5

### Solution IM-5 d.6 — Answer: C

Bank Q264 • BT P2.T9.21.6.2

**Step 1 – The optimality condition**

At the optimum, expected excess return divided by beta (or by marginal VaR) is equal across the two ETFs.

**Step 2 – Excess return per unit of beta (Energy, Healthcare)**

#1: 0.279, 0.073 #2: 0.136, 0.065 #3: 0.076, 0.073 #4: 0.058, 0.168

**Step 3 – Same check with marginal VaR**

#1: 1.176, 0.307 #2: 0.706, 0.340 #3: 0.476, 0.456 #4: 0.380, 1.091

$\Rightarrow$ Only Mix #3 has the two ratios nearly equal on both measures.

**Interpretation**

The table is internally consistent: in every mix the weighted betas sum to 1, and marginal VaR divided by beta (VaR/W) is the same for both ETFs. It matches volatilities of 10% (Energy) and 25% (Healthcare), uncorrelated, at 95%; with those inputs the exact optimum is 75.8% Energy and Mix #3 has the highest Sharpe ratio of the four (0.768). Between #3 and #4 the ratios cross, so the optimum lies just beyond #3.

**Why the other options are wrong**

**A is incorrect.** Energy's ratio is almost four times Healthcare's: far too much in Healthcare.

**B is incorrect.** Still far apart (0.136 against 0.065): an in-between weight is not the equal-ratio point.

**D is incorrect.** Healthcare's ratio (0.168) now far exceeds Energy's (0.058): Energy is overweight. Choosing it follows minimum-risk logic (the lowest Healthcare marginal VaR), not equal ratios.

**LO:** IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.6, Eq. 5.38 and 5.39

### Solution IM-5 d.7 — Answer: D

Workbook A12.4 • S2, S3, S4, S5

**Step 1 – VaR per AUD of portfolio from the marginal VaRs (Euler)**

$$\frac{\text{VaR}}{W} = \sum_i w_i \, \Delta\text{VaR}_i = 0.5(0.12) + 0.3(0.20) + 0.2(0.05) = 0.06 + 0.06 + 0.01 = 0.13$$

$\Rightarrow$ The 95% VaR is 13% of value, AUD 65 million.

**Step 2 – Portfolio volatility**

$$\sigma_p = \frac{\text{VaR}/W}{\alpha} = \frac{0.13}{1.645} = 7.90\%$$

**Step 3 – Sharpe ratio (excess returns, Eq. 5.37)**

$$E_p - R_f = 0.5(9\%) + 0.3(12\%) + 0.2(4.5\%) - 3\% = 9.0\% - 3\% = 6.0\%$$

$$\text{SR} = \frac{6.0\%}{7.90\%} = 0.76$$

**Step 4 – Which position to grow (Eq. 5.38)**

Betas: $\beta_i = \Delta\text{VaR}_i/(\text{VaR}/W)$: 0.923, 1.538, 0.385 (they sum to 1 when weighted). Excess return per unit of marginal VaR:

$$\frac{E_A}{\Delta\text{VaR}_A} = \frac{6\%}{0.12} = 0.50, \qquad \frac{E_B}{\Delta\text{VaR}_B} = \frac{9\%}{0.20} = 0.45, \qquad \frac{E_C}{\Delta\text{VaR}_C} = \frac{1.5\%}{0.05} = 0.30$$

$\Rightarrow$ Only beta and excess return matter. A has the highest ratio and C the lowest, so A is increased, funded from C. Check: $E_A/\beta_A = 6.5\%$ exceeds $E_p = 6.0\%$, while B (5.85%) and C (3.9%) fall short.

### Interpretation

At the optimum every $E_i/\Delta\text{VaR}_i$ is equal (Jorion's restatement of the CAPM). Shifting 1% from C to A raises the Sharpe ratio from 0.759 to about 0.761. Growing C would lower risk but also the Sharpe ratio: that is risk management (minimum risk), not portfolio management (best return per unit of risk).

### Why the other options are wrong

**A is incorrect.** Applied the risk-minimizing rule (add where marginal VaR is lowest) instead of the return-to-risk rule. (Using raw expected returns, $4.5\%/0.05 = 0.90$, would also point to C.)
**B is incorrect.** Chose the highest expected return without dividing by marginal VaR: B's high return comes with the highest beta, and $E_B/\Delta\text{VaR}_B = 0.45 < 0.50$.
**C is incorrect.** Right position, but the Sharpe ratio used the raw expected return: $9.0\%/7.90\% = 1.14$. The numerator must be in excess of the risk-free rate.

**LO:** IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. IM-5 e: Explain the difference between risk management and portfolio management and describe how to use marginal VaR in portfolio management. • **Reference:** Jorion, Value at Risk 3e, Ch.7 (GARP 2025 Ch.5), 5.6 (Eq. 5.35 to 5.39, Tables 5.4 and 5.5)

IM-7 • Portfolio Performance Evaluation • Vardeez: VERY IMPORTANT

# IM-7 b

P1 CONCEPT

Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant.

Workbook: A18 Performance Measures and Attribution • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: "Jensen's alpha, Sharpe and other metrics theory" came up in the August 2026 sitting, and the A18 shapes were reported on 5 of 16 recall sheets (Map #36). **Vardeez:** no pointer on this LO (his focus in the reading is IM-7 a, c and e).

**Reported exam shapes**

- **S1** Sharpe as the wrong lens for one of many subportfolios (Workbook A18.3, Question IM-7 b.7). The harder form, Sharpe paired with $M^2$, is Workbook A18.4 (worked as Question IM-7 c.3).
- **S2** Marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio. This is a calculation drilled in IM-5 c, in Workbook A12.4 (worked as Question IM-5 d.7) and in Question IM-5 a.5; from this LO it needs only the Sharpe definition.
- **S3** Jensen's alpha and the information ratio. The numerical forms are Workbook A18.1 and A18.2 (worked as Questions IM-3 b.5 and IM-3 b.8); here the concepts (Questions b.1 and b.6).
- **S6** Which measure suits which use: Sharpe, Treynor, information ratio (Questions b.4, b.5 and b.7; Workbook A18.6, worked as Question IM-3 d.9, adds the fundamental-law version).

**What the questions look like:** CONCEPT. A role is described (the investor's entire risky portfolio, one of many managers, an active add-on to an index core, a benchmark-relative mandate) and you name the measure; or two measures rank funds differently and you explain why. The only numbers to work come from Workbook A18.3, which goes in as it is.

## 1. Why average return is not enough

The simplest evaluation ranks a fund within a **comparison universe** of funds with a similar style (growth equity, high-yield bonds and so on) on its (usually time-weighted) average return, and gives it a percentile: the ninth-best of 100 funds is the 90th percentile manager. It is a useful first step, with two weaknesses:

- **Risk still differs inside a universe.** One equity manager concentrates on high-beta growth stocks; one bond manager runs a longer duration.
- **The median fund is not an investable benchmark.** A passive investor cannot lock in the median return in advance, whereas an index fund is available to anyone.

Hence the risk-adjusted measures. GARP adds a caveat for all of them: their reliability needs a long history of consistent management, covering bull as well as bear markets.

## 2. Four measures, three denominators

| | Definition | Risk measure | What it measures |
|---|---|---|---|
| **Sharpe** | $S_P = \dfrac{\bar{r}_P - \bar{r}_f}{\sigma_P}$ | total, $\sigma_P$ | reward to total volatility; slope of P's capital allocation line |
| **Treynor** | $T_P = \dfrac{\bar{r}_P - \bar{r}_f}{\beta_P}$ | systematic, $\beta_P$ | excess return per unit of beta; slope of P's T-line |
| **Jensen's alpha** | $\alpha_P = \bar{r}_P - \left[\bar{r}_f + \beta_P(\bar{r}_M - \bar{r}_f)\right]$ | none (a difference) | average return above the CAPM prediction for P's beta: height above the SML |
| **Information ratio** | $\mathrm{IR}_P = \dfrac{\alpha_P}{\sigma(e_P)}$ | residual, $\sigma(e_P)$ | alpha per unit of the risk that holding the index would diversify away |

- **Sharpe and Treynor** share the numerator and differ in the denominator. **Sharpe and the information ratio** share the shape (a reward over a standard deviation) and differ in what the reward is measured against.
- **Jensen's alpha** is the only one that is not a ratio. With a multifactor model, the alpha also nets out the factor loadings, for example $\alpha_P = r_P - r_f - \beta_P(r_M - r_f) - s_P\, r_{\text{SMB}} - h_P\, r_{\text{HML}}$ under Fama-French. Schweser adds two practical points: alpha is estimated by regression on each period's returns, while Sharpe and Treynor use averages; and the beta should be the one measured over the evaluation period.
- **Two definitions of the information ratio.** GARP (Bodie, Kane and Marcus) defines it as alpha over the residual standard deviation, called "tracking error" in the industry, and uses that definition throughout. Schweser, Ang (IM-3) and Vardeez use the return in excess of a benchmark over the standard deviation of that excess return; some reserve "appraisal ratio" for $\alpha/\sigma(e)$. The two coincide when the portfolio's beta to the benchmark is 1. In a question, the inputs tell you which one is meant: an alpha and a residual standard deviation, or a benchmark return and a tracking error.

## 3. The circumstances: which measure for which role

The right measure depends on how the portfolio fits into the investor's overall plan, because that decides which risk the investor actually bears.

1. **The portfolio is the investor's entire risky portfolio: Sharpe.** The investor bears all of its volatility, so total risk is the right denominator. The benchmark is the Sharpe ratio of the market index, because the investor can always choose an indexed fund instead. Its return form is $M^2$ (IM-7 c).
2. **The portfolio is one of many sub-portfolios: Treynor.** GARP's example is CalPERS, about $350 billion in early 2019, run as a fund of funds across many managers. Each manager's nonsystematic risk is largely diversified away in the total, so only systematic risk counts. Its return form is $T^2$ (IM-7 c).
3. **The portfolio is an active position added to a passive index core: information ratio.** A university adds a hedge fund to its indexed holdings. When the active portfolio $H$ is combined optimally with the index, the Sharpe ratio of the combination is (Eq. 8.2)

$$S_P^2 = S_M^2 + \left[\frac{\alpha_H}{\sigma(e_H)}\right]^2$$

so the improvement is set by the information ratio. With an index Sharpe ratio of 0.40 and a hedge fund IR of 0.30, the combination reaches $\sqrt{0.16 + 0.09} = 0.50$. The IR quantifies the trade-off between the alpha and the diversifiable risk the tilt takes on.

4. **Jensen's alpha: necessary, not sufficient.** GARP finds no situation in which alpha alone is the criterion for choosing one fund over another. But every one of the three measures above needs a positive alpha for the portfolio to beat the passive index, which is why alpha is the most widely used measure (Section 5).

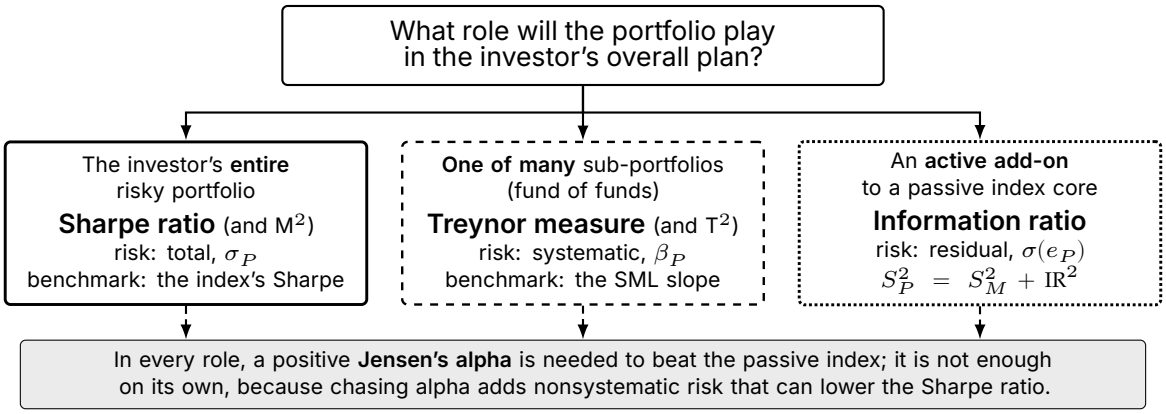



Figure 1. GARP's summary table as a decision. The box outline marks the measure: solid for Sharpe, dashed for Treynor, dotted for the information ratio. The grey box applies to all three.

## 4. One pair of portfolios, three answers

GARP's 12-month example (Tables 8.2 and 8.3, monthly excess returns, in %) shows that "the appropriate way to evaluate a portfolio depends in large part on how the portfolio fits into the investor's overall investment plan":

| | Sharpe | $M^2$ | Alpha | Beta | Treynor | $T^2$ | $\sigma(e)$ | IR |
|---|---|---|---|---|---|---|---|---|
| Portfolio P | 0.43 | 2.16 | 1.63 | 0.70 | 3.97 | 2.34 | 2.02 | **0.81** |
| Portfolio Q | **0.49** | **2.66** | 5.26 | 1.40 | **5.38** | **3.74** | 9.81 | 0.54 |
| Market M | 0.19 | 0.00 | 0.00 | 1.00 | 1.64 | 0.00 | 0.00 | 0.00 |

- **Entire risky fund:** Q, on its higher Sharpe ratio (0.49 against 0.43) and $M^2$.
- **One of many sub-portfolios:** Q again, on its higher Treynor measure (5.38 against 3.97).
- **Active portfolio mixed with the index:** P, on its higher information ratio (0.81 against 0.54). P is far better diversified (residual standard deviation 2.02% against 9.81%, $R^2$ 0.91 against 0.64).

Twelve months is far too short for statistical significance (IM-7 d), but the logic carries over.

## 5. Why the measures disagree

- **Sharpe against Treynor is about diversification.** For a well-diversified portfolio total risk is close to systematic risk, so the two rank alike. A poorly diversified portfolio carries extra nonsystematic risk: Sharpe penalises it, Treynor does not, so it ranks higher on Treynor than on Sharpe. Used together, the two reveal the degree of diversification; the bigger the gap in rankings, the less diversified the portfolio.

- **Every measure runs through alpha (GARP's table).**

$$T_P = \frac{\alpha_P}{\beta_P} + T_M, \qquad S_P = \frac{\alpha_P}{\sigma_P} + \rho\, S_M, \qquad T_P - T_M = \frac{\alpha_P}{\beta_P}, \qquad S_P - S_M = \frac{\alpha_P}{\sigma_P} - (1-\rho)\, S_M$$

  where $\rho < 1$ is the correlation between P and the market. A positive alpha is therefore necessary to beat the index on Treynor, and on Sharpe whenever $\rho \leq 1$; on Sharpe it is not sufficient, because the second term is negative.

- **GARP's Concept Check 8.2 shows it.** P: return 35%, beta 1.2, standard deviation 42%, residual 18%; market 28% and 30%; T-bills 6%. P beats the market on alpha $(35 - [6 + 1.2(22)] = 2.6\%)$, Treynor (24.2 against 22.0) and the information ratio (0.144), but its Sharpe ratio is 0.69 against 0.73. Here $\rho = 1.2 \times 30/42 = 0.857$: the alpha term adds $2.6/42 = 0.062$ and the diversification term subtracts $(1 - 0.857) \times 0.733 = 0.105$.

- **Alpha says nothing about diversification.** Like Treynor, it sees only beta. And the highest alpha is not a selection rule: in GARP's Table 8.1 the portfolio with the lower alpha has the steeper T-line (IM-7 c).

| | Risk adjusted for | Most relevant when (GARP) | Graph |
|---|---|---|---|
| **Sharpe** | Total, $\sigma_P$ | The portfolio is the investor's entire risky investment, competing for the overall risky portfolio | Slope of the CAL (return against $\sigma$) |
| **Treynor** | Systematic, $\beta_P$ | Ranking many portfolios that will be mixed to form the overall risky portfolio | Slope of the T-line (return against $\beta$) |
| **Information ratio** | Residual, $\sigma(e_P)$ | Evaluating an active portfolio to be mixed with the passive benchmark | Not drawn in GARP |
| **Jensen's alpha** | Systematic, through the CAPM benchmark | Never the criterion on its own; positive alpha is required under all three | Height above the SML at P's beta |

**Exam traps**

- **Wrong lens (sibling swap).** Sharpe for one of many managers (should be Treynor); Treynor for an active add-on to an index (should be the information ratio); Treynor or alpha for a stand-alone fund (should be Sharpe). Name the role before computing anything.
- **Denominators swapped.** Sharpe divides by $\sigma$, Treynor by $\beta$, the information ratio by the residual standard deviation. "Treynor uses standard deviation" is the classic flipped option.
- **Diversification read the wrong way (polarity).** A fund that ranks high on Treynor and low on Sharpe is the *poorly* diversified one. Adding stocks with the same beta raises Sharpe and leaves Treynor unchanged.
- **"Always agree" (absolute qualifier).** Sharpe and alpha can disagree; alpha and Treynor agree on whether a portfolio beats the market, not necessarily on how two portfolios rank.
- **Highest alpha wins.** Alpha is necessary, not sufficient, and not a ranking rule.
- **Which information ratio?** GARP's is $\alpha/\sigma(e)$; the industry's is excess return over the benchmark divided by tracking error. Read the inputs.
- **Lower is better.** For every ratio here, higher is better.

## Practice questions: IM-7 b

### Question IM-7 b.1

Bank Q381 • Mock 2 Q80

Money managers are routinely evaluated using risk-adjusted performance measures including Sharpe's measure, Treynor's measure, Jensen's alpha, and the information ratio. Which of the following statements is most likely correct?

**A.** Jensen's alpha gives no indication of the diversification in the portfolio.

**B.** A lower information ratio indicates better performance.

**C.** When ranking portfolios, Sharpe will always give the same conclusion as Jensen's alpha.

**D.** The Treynor measure is very similar to the Sharpe ratio except that it uses standard deviation as the measure of risk.

### Question IM-7 b.2

Bank Q357 • AnalystPrep Q.2757 (rebuilt)

A risk analyst at a family office is reviewing a portfolio manager who increases the number of stocks in his portfolio from ten to twenty, spread across a range of industries. The added stocks have the same average beta as the existing holdings, so the portfolio's beta and excess return are unchanged. What is the most likely effect on the portfolio's Sharpe and Treynor ratios?

**A.** Sharpe decreases; Treynor is unaffected.

**B.** Sharpe is unaffected; Treynor increases.

**C.** Sharpe increases; Treynor is unaffected.

**D.** Sharpe is unaffected; Treynor decreases.

### Question IM-7 b.3

Bank Q356 • AnalystPrep Q.3030 (rebuilt)

A performance analyst at a Dubai family office compares two equity funds over the same period. The risk-free rate was 3% and the market return was 8%.

| Fund | Return | Beta | Standard deviation |
|---|---|---|---|
| X | 11% | 0.80 | 20% |
| Y | 10% | 1.00 | 12% |

Fund X ranks first on the Treynor measure but second on the Sharpe ratio. Which of the following statements best explains the reversal?

**A.** Fund X is better diversified than Fund Y, which lowers its Sharpe ratio.

**B.** Fund Y has the higher Jensen's alpha, so the Treynor ranking is misleading.

**C.** Fund X carries a larger share of unsystematic risk relative to its systematic risk, which Treynor ignores but Sharpe penalizes.

**D.** The Sharpe ratio measures risk with beta and the Treynor measure uses standard deviation, so the two must disagree.

**Question IM-7 b.4** Bank Q360 • AP Mock 2 Q26 (rebuilt)

A newly established hedge fund is a stand-alone investment that will represent the investor's entire risky portfolio. It has no market or index benchmark and pursues absolute returns. Which measure is most appropriate to evaluate its performance?

**A.** Sharpe ratio

**B.** Treynor measure

**C.** Information ratio

**D.** Jensen's alpha

**Question IM-7 b.5** Bank Q359 • AP Mock 1 Q58

An investment firm has hired an active manager for its pension fund. Her benchmark is the Russell 2000 Growth index. Which statistics are most suitable for evaluating the manager's performance and risk?

**A.** VaR and information ratio

**B.** Tracking error and Sharpe ratio

**C.** Tracking error and information ratio

**D.** VaR and Sharpe ratio

**Question IM-7 b.6** New • built from GARP Ch.8 • S3

The investment committee of a US public pension plan is reviewing an active equity manager. Over the evaluation period the manager's Jensen's alpha was $+2.1\%$ a year, its beta 0.95 and its residual standard deviation 7.0%. The plan holds a passive index fund as its core and is considering adding the manager on top of it. Which of the following statements is correct?

**A.** The positive alpha guarantees that the manager's Sharpe ratio exceeded the index's Sharpe ratio over the period.

**B.** The information ratio divides the alpha by the manager's total standard deviation, so it rewards well-diversified portfolios.

**C.** The improvement in the plan's Sharpe ratio from adding the manager optimally depends on the manager's information ratio, not on the alpha alone.

**D.** Because its alpha is positive and its beta is below 1, the manager should be preferred to any candidate with a lower alpha, whatever role it plays.

**Question IM-7 b.7** Workbook A18.3 • S1, S6 • Map #36

A portfolio manager at a large EUR-denominated pension fund allocates assets across more than 20 external equity managers using a funds-of-funds approach, so each manager's nonsystematic risk is largely diversified away. She must choose between two candidate managers. Annual data for the evaluation period are:

| | Fund X | Fund Y | Market index |
|---|---|---|---|
| Average return | 17% | 13% | 10% |
| Standard deviation | 18% | 24% | 15% |
| Beta | 1.4 | 0.8 | 1.0 |

The average risk-free rate was 4%. Which of the following correctly identifies the preferred fund and the supporting measure?

**A.** Fund X, because its Sharpe ratio is 0.72 versus 0.38 for Fund Y

**B.** Fund Y, because its Treynor measure is 11.25 versus 9.29 for Fund X

**C.** Fund Y, because its Treynor measure is 16.25 versus 12.14 for Fund X

**D.** Fund X, because its Jensen's alpha is 4.6% versus 4.2% for Fund Y

## Solutions • IM-7 b

### Solution IM-7 b.1 — Answer: A

Bank Q381 • Mock 2 Q80

**Step 1 – Which risk enters each measure**

Sharpe: total risk $\sigma_P$. Treynor: beta. Jensen's alpha: beta, through the CAPM return it subtracts. Information ratio: residual risk.

**Step 2 – Where diversification shows up**

Diversification removes nonsystematic risk. A measure that sees only beta cannot see it.
$\Rightarrow$ Jensen's alpha, like Treynor, gives no indication of how diversified the portfolio is.

**Interpretation**

Schweser states it directly: like the Treynor measure, Jensen's alpha "only takes into account the systematic risk of the portfolio and, hence, gives no indication of the diversification". Sharpe is the measure that sees it.

**Why the other options are wrong**

**B is incorrect.** Polarity flip: a higher information ratio (more reward per unit of active risk) is better.

**C is incorrect.** Absolute qualifier: Sharpe (total risk) and alpha (systematic risk) can disagree. In GARP's Concept Check 8.2, P has a positive alpha of 2.6% but a lower Sharpe ratio than the market.

**D is incorrect.** Sibling swap: Treynor uses beta; it is Sharpe that uses standard deviation.

**LO:** IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1 Risk-Adjusted Performance Measures; SchweserNotes 2026 Book 5, Reading 89, LO 89.b

### Solution IM-7 b.2 — Answer: C

Bank Q357 • AnalystPrep Q.2757

**Step 1 – Treynor**

$T_P = (\bar{r}_P - \bar{r}_f)/\beta_P$. The excess return and the beta are unchanged.
$\Rightarrow$ Treynor is unaffected.

**Step 2 – Sharpe**

Adding stocks across industries diversifies away nonsystematic risk, so $\sigma_P$ falls while the numerator stays the same.
$\Rightarrow$ Sharpe rises.

**Interpretation**

This is the mechanism behind every Sharpe-Treynor disagreement: the two differ only in whether nonsystematic risk is charged. Diversifying moves the Sharpe ratio towards the value implied by the portfolio's Treynor measure.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: lower total risk raises Sharpe.

**B is incorrect.** Treynor depends on beta, not on diversifiable risk, so it does not rise; and Sharpe does change.

**D is incorrect.** Wrong on both: Sharpe rises with the lower standard deviation, and Treynor does not move.

**LO:** IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1; SchweserNotes 2026 Book 5, Reading 89, LO 89.b (Treynor measure)

## Solution IM-7 b.3 — Answer: C

Bank Q356 • AnalystPrep Q.3030

### Step 1 – Same numerator, different denominators

Both ratios divide the excess return over the risk-free rate (X 8%, Y 7%); Treynor divides by beta, Sharpe by total standard deviation. A reversal can only come from the denominators.

### Step 2 – Read the risk profile

X has the lower beta (0.80) but the higher standard deviation (20% against 12%): far more total risk per unit of systematic risk. Treynor: X $8/0.80 = 10.0$, Y $7/1.00 = 7.0$. Sharpe: X $8/20 = 0.40$, Y $7/12 = 0.58$.

$\Rightarrow$ Most of X's risk is unsystematic. Treynor ignores it; Sharpe charges for it.

### Step 3 – Check the alpha claim

X: $11 - (3 + 0.8 \times 5) = 4.0\%$; Y: $10 - (3 + 1.0 \times 5) = 2.0\%$. X has the higher alpha, consistent with its Treynor lead.

### Interpretation

The reversal is a diagnosis of poor diversification in X. Which ranking to use depends on X's role: as one of many sub-portfolios, Treynor; as the whole risky portfolio, Sharpe.

### Why the other options are wrong

**A is incorrect.** Polarity flip: a better-diversified fund would rank relatively better on Sharpe, not worse. X ranks worse on Sharpe because it is less diversified.

**B is incorrect.** False fact: X's alpha (4.0%) is the higher, and alpha ranks these two in line with Treynor.

**D is incorrect.** Definitions swapped, and "must disagree" is false: for well-diversified funds the two rank alike.

**LO:** IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1; SchweserNotes 2026 Book 5, Reading 89, LO 89.b and 89.c

## Solution IM-7 b.4 — Answer: A

Bank Q360 • AP Mock 2 Q26

### Step 1 – Name the role

The fund is the investor's entire risky portfolio, held on its own.

$\Rightarrow$ The investor bears all of its volatility: total risk is the relevant risk.

### Step 2 – Check what each measure needs

Sharpe needs only the fund's returns, its standard deviation and the risk-free rate. Treynor and alpha need a beta against a market portfolio; the information ratio needs a benchmark.

$\Rightarrow$ The Sharpe ratio fits both the role and the data.

### Interpretation

GARP: Sharpe is the measure "when choosing among portfolios competing for the overall risky portfolio". The market's Sharpe ratio is still the hurdle, because the investor could hold an index

fund instead.

### Why the other options are wrong

**B is incorrect.** Treynor charges only systematic risk, which suits one of many sub-portfolios, not a fund held alone.

**C is incorrect.** The information ratio needs a benchmark to measure active return and tracking error; this fund has none, and it is not an add-on to an index core.

**D is incorrect.** Alpha is measured against the CAPM line using beta, so it ignores the diversifiable risk the investor bears here.

**LO:** IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1 The Sharpe Ratio for Overall Portfolios and summary table

## Solution IM-7 b.5 Answer: C

Bank Q359 • AP Mock 1 Q58

### Step 1 – Name the role

The mandate is benchmark-relative: the manager is paid to beat the Russell 2000 Growth index.
$\Rightarrow$ Both risk and performance should be measured against that benchmark.

### Step 2 – Pick the matched pair

Tracking error is the standard deviation of returns relative to the benchmark: the active risk. The information ratio is the active return per unit of that risk.
$\Rightarrow$ Tracking error and information ratio.

### Interpretation

Under GARP's definition the IR is $\alpha/\sigma(e)$ with $\sigma(e)$ called "tracking error"; under the industry definition it is excess return over the benchmark divided by tracking error. Either way the pair is the same, and it answers the question the pension fund is asking: how much does the manager add over the benchmark, per unit of risk taken relative to it.

### Why the other options are wrong

**A is incorrect.** VaR measures absolute loss risk, not deviation from the benchmark, so it does not pair with the information ratio.

**B is incorrect.** Sharpe is measured against the risk-free rate with total risk; it ignores the benchmark.

**D is incorrect.** Both are absolute measures that ignore the benchmark entirely.

**LO:** IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1 Risk-Adjusted Performance Measures (item 4) and The Information Ratio; SchweserNotes 2026 Book 5, Reading 89, LO 89.b

## Solution IM-7 b.6 Answer: C

New • built from GARP Ch.8 • S3

### Step 1 – Name the role

The manager would be an active position on top of a passive index core.
$\Rightarrow$ GARP's measure for that role is the information ratio, $\alpha/\sigma(e)$.

### Step 2 – Why the IR, not the alpha

When the active portfolio is combined optimally with the index, $S_P^2 = S_M^2 + [\alpha/\sigma(e)]^2$ (Eq. 8.2). The gain depends on the alpha *per unit of residual risk*: the same 2.1% alpha is worth more with 3% residual risk than with 7%.
$\Rightarrow$ C is correct.

### Interpretation

Here the IR is $2.1/7.0 = 0.30$; with an index Sharpe ratio of 0.40, the optimal combination would reach $\sqrt{0.40^2 + 0.30^2} = 0.50$. The beta and the sign of the alpha alone do not tell you that.

### Why the other options are wrong

**A is incorrect.** Absolute qualifier: positive alpha is necessary but not sufficient for a higher Sharpe ratio, since $S_P - S_M = \alpha/\sigma_P - (1-\rho)S_M$ can still be negative (Concept Check 8.2).
**B is incorrect.** Wrong denominator: GARP's information ratio divides by the residual (nonsystematic) standard deviation, not by total volatility.
**D is incorrect.** Absolute claim: alpha is never the selection rule on its own, and the right measure depends on the role; a lower-alpha fund can have the higher IR or Treynor measure.

**LO:** IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1 The Information Ratio (Eq. 8.2) and The Role of Alpha in Performance Measures

## Solution IM-7 b.7

**Answer: B**

Workbook A18.3 • S1, S6

### Step 1 – Choose the measure

Each manager is one of many components of the overall risky portfolio and nonsystematic risk diversifies away, so only beta matters: the Treynor measure.

### Step 2 – Excess returns

$$\bar{r}_X - \bar{r}_f = 17\% - 4\% = 13\%, \qquad \bar{r}_Y - \bar{r}_f = 13\% - 4\% = 9\%$$

### Step 3 – Treynor measures

$$T_X = \frac{13\%}{1.4} = 9.29\%, \qquad T_Y = \frac{9\%}{0.8} = 11.25\%$$

$\Rightarrow$ Per unit of beta, Y delivers 11.25% of excess return and X 9.29%; the market's Treynor is 6%, so both beat the SML, Y by more.

### Step 4 – Sharpe ranking (to see the flip)

$$S_X = \frac{13\%}{18\%} = 0.72, \qquad S_Y = \frac{9\%}{24\%} = 0.375, \qquad S_M = \frac{6\%}{15\%} = 0.40$$

$\Rightarrow$ On total risk X is far better: Y carries a lot of nonsystematic risk, which washes out in a multi-manager structure.

### Interpretation

The rankings flip because Y's volatility is mostly nonsystematic (high $\sigma$, low $\beta$). Sharpe ranks a stand-alone overall portfolio; Treynor ranks one of many.

### Why the other options are wrong

**A is incorrect.** Right numbers, wrong lens (sibling swap): the Sharpe ratio uses total risk, relevant only if the fund were the investor's entire risky portfolio.

**C is incorrect.** Used raw returns instead of excess returns: $13\%/0.8 = 16.25$ and $17\%/1.4 = 12.14$. Right fund, wrong values.
**D is incorrect.** The alphas are right ($17-(4+1.4\times 6)=4.6\%$; $13-(4+0.8\times 6)=4.2\%$), but highest alpha is not a selection rule: GARP's Table 8.1 shows the lower-alpha portfolio winning on Treynor.

**LO:** IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), The Treynor Ratio (Table 8.1, Figure 8.3)

IM-7 • Portfolio Performance Evaluation • Vardeez: VERY IMPORTANT

# IM-7 c

P1 BOTH

Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures.

Workbook: A18 Performance Measures and Attribution • Tier A • Vardeez focus: M-squared

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic (A18, Map #36, reported on 5 of 16 recall sheets: "Sharpe ratio (this was bit hard)"), and Vardeez flags the $M^2$ calculation in a Very Important chapter.
**Vardeez:** M-squared calculation.

**Reported exam shapes**

- **S1** Sharpe ratio, the harder form: Sharpe paired with $M^2$ (Workbook A18.4, Question IM-7 c.3). Sharpe as the wrong lens for one of many subportfolios is Workbook A18.3 (worked as Question IM-7 b.7).
- **S6** Which measure suits which use: comparing two managers with $M^2$ against Treynor (Questions c.6 and c.7).

**What the questions look like:** BOTH. Compute $M^2$ (mix the portfolio with T-bills, or borrow, until it has the index's standard deviation) or $T^2$ (until it has a beta of 1) and compare two portfolios. The wrong options are the adjusted portfolio's return before the market return is subtracted, the unscaled Sharpe difference, the weight inverted (levering the wrong way), Jensen's alpha, or the right number attached to the wrong measure. Graph questions ask where $M^2$, $T^2$ and alpha sit on the two charts.

## 1. The problem $M^2$ and $T^2$ solve

From IM-7 b: the Sharpe ratio is the slope of a portfolio's capital allocation line (CAL) in the return against standard deviation plane, and the Treynor measure is the slope of its T-line in the return against beta plane. Slopes rank portfolios, but their size is hard to read. In GARP's Concept Check 8.2, $S_P = 0.69$ against $S_M = 0.73$: is a difference of 0.04 economically meaningful? $M^2$ and $T^2$ answer in percentage points of return, by moving the portfolio along its own line until it has the market's risk.

## 2. $M^2$: matching the market's standard deviation

Mix the active portfolio P with T-bills so that the adjusted portfolio $P^*$ has the index's standard deviation:

$$w = \frac{\sigma_M}{\sigma_P} \text{ in P}, \quad 1 - w \text{ in T-bills}, \qquad r_{P^*} = w\,\bar{r}_P + (1-w)\,\bar{r}_f = \bar{r}_f + S_P\,\sigma_M$$

$$\boxed{M^2 = r_{P^*} - \bar{r}_M = \sigma_M\,(S_P - S_M)} \qquad \text{(Eq. 8.1)}$$

- $\sigma_P > \sigma_M$**: de-lever.** $w < 1$, the rest in T-bills. GARP's Example 8.1 (P: 35%, $\sigma$ 42%; M: 28%, 30%; T-bills 6%): $w = 30/42 = 0.714$ in P and 0.286 in bills, $r_{P^*} = 0.286 \times 6\% + 0.714 \times 35\% = 26.7\%$, so $M^2 = 26.7\% - 28\% = -1.3\%$. P earned more than the market but did worse for its risk, consistent with its lower Sharpe ratio. Check: $30 \times (0.690 - 0.733) = -1.3\%$.
- $\sigma_P < \sigma_M$**: lever.** $w > 1$: borrow at the risk-free rate and invest the proceeds in P. A portfolio returning 9% with $\sigma = 10\%$ against a market at 11% and 15%, T-bills 3%: $w = 1.5$, $r_{P^*} = 1.5 \times 9\% - 0.5 \times 3\% = 12\%$, so $M^2 = +1.0\%$.

- **Same ranking as Sharpe, always.** $M^2$ is the Sharpe difference times the fixed number $\sigma_M$. It adds interpretation, not information: like Sharpe, it measures total risk.
- **Definitions.** Some sources call $r_{P^*}$ itself the $M^2$; rankings are the same either way (Schweser). There is no squared term: the name refers to Leah and Franco Modigliani.

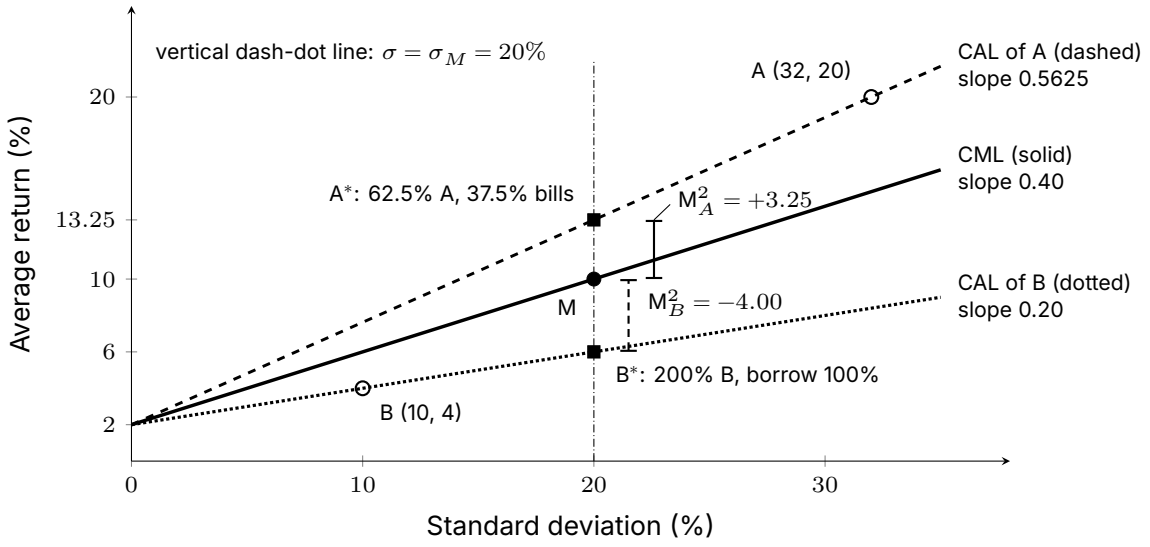



Figure 1. $M^2$ on the return against standard deviation chart (market M at $\sigma_M = 20\%$ and 10%, T-bills 2%). A is de-levered along its CAL (dashed) and B is levered along its CAL (dotted) until each has the market's 20% volatility. $M^2$ is the vertical distance at $\sigma_M$ between the adjusted portfolio (filled square) and the market (filled circle): $+3.25\%$ for A (solid bracket), $-4.00\%$ for B (dashed bracket). A steeper CAL than the CML (solid) means a positive $M^2$.

## 3. Treynor and $T^2$: matching the market's beta

When beta is the relevant risk, plot return against beta. The SML runs from the risk-free rate through the market at $\beta = 1$; its slope is the market's Treynor ratio. The line from $\bar{r}_f$ through any portfolio is that portfolio's **T-line**: every beta-return combination available by mixing it with T-bills. Its slope is the portfolio's Treynor measure, and the steepest T-line wins. GARP's Table 8.1 (T-bills 6%, market 16%):

| | Beta | Return | Excess | Alpha | Treynor | Weight to reach $\beta = 1$ | $T^2$ |
|---|---|---|---|---|---|---|---|
| Q | 1.25 | 22.0% | 16.0% | **3.5%** | 12.80% | $w_Q = 0.80$ (20% bills) | 2.80% |
| U | 0.80 | 17.0% | 11.0% | 3.0% | **13.75%** | $w_U = 1.25$ (borrow 25%) | **3.75%** |
| M | 1.00 | 16.0% | 10.0% | 0.0% | 10.00% | | 0 |

$$r_{Q^*} = 0.8 \times 22\% + 0.2 \times 6\% = 18.8\%, \qquad r_{U^*} = 1.25 \times 17\% - 0.25 \times 6\% = 19.75\%$$

$$\boxed{T^2 = r_{P^*} - \bar{r}_M \text{ with } w = 1/\beta_P} \qquad \Longrightarrow \qquad T^2 = T_P - T_M = \frac{\alpha_P}{\beta_P}$$

Q has the higher alpha, but **U is preferred**: for any level of beta it offers the higher return (Figure 2). $T^2$ does for Treynor what $M^2$ does for Sharpe: same ranking, read in return units. Jensen's alpha is a third distance on the same chart: the height of the portfolio above the SML at its *own* beta, before any rescaling.

**Correction to the source notes.** GARP's Table 8.1, as printed in our copy, gives portfolio Q's beta as 1.3. The table's own alpha for Q ($3.5\% = 16\% - 1.25 \times 10\%$) and the text that follows ("$\beta_{Q^*} = w_Q \times 1.25$", so $w_Q = 0.8$, $r_{Q^*} = 18.8\%$, $T^2 = 2.8\%$) all require 1.25. With 1.3 the alpha would be 3.0%, tying U, and $w_Q$ would be 0.769. Use 1.25.

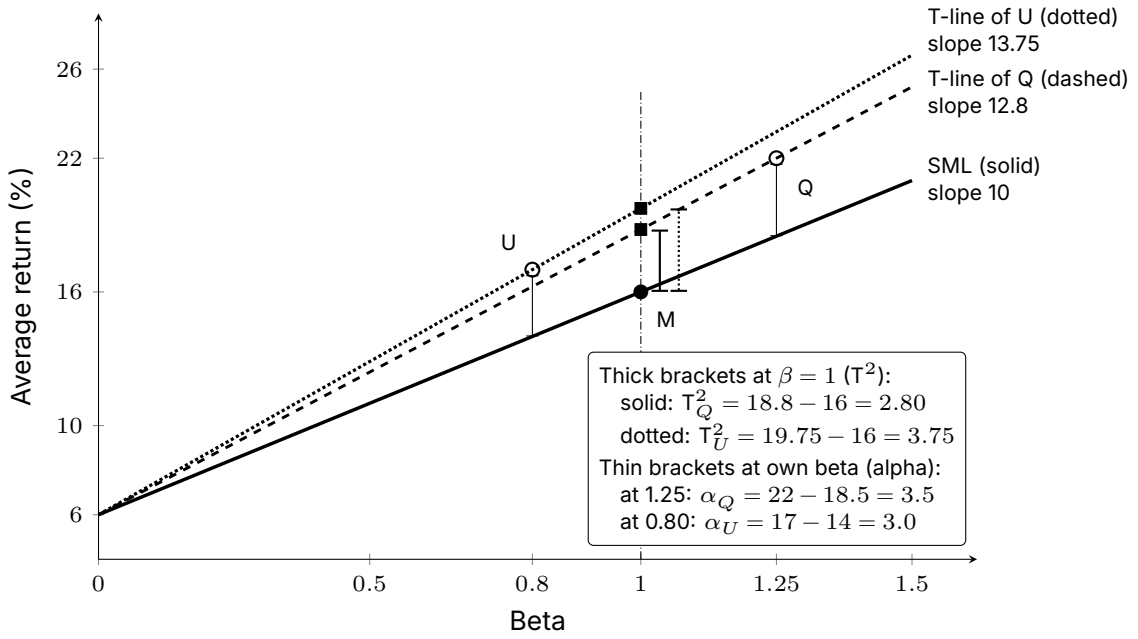



Figure 2. GARP's Figure 8.3 redrawn with Table 8.1 (open circles: Q and U; filled circle: M; all lines start at the 6% T-bill rate). U's T-line (dotted) is steeper than Q's (dashed), so at a beta of 1 (dash-dot line) U* (upper filled square, 19.75%) beats Q* (lower filled square, 18.8%), even though Q stands higher above the SML at its own beta. $T^2$ is measured at $\beta = 1$; alpha at the portfolio's own beta.

## 4. Uses in comparing two portfolios

- **$M^2$ when each portfolio would be the investor's entire risky holding.** Total risk is what the investor bears, so compare the two after scaling each to the market's volatility. GARP Practice Exam 2 frames it as two managers whose portfolios "represent the investor's entire holdings".
- **$T^2$ (or Treynor) when each portfolio is one of many sub-portfolios.** Their nonsystematic risk will be diversified away, so compare them at the market's beta.
- **The difference is directly comparable.** Both portfolios are moved to the same risk, so the market return cancels: $M_1^2 - M_2^2 = \sigma_M(S_1 - S_2)$ and $T_1^2 - T_2^2 = T_1 - T_2$. A gap of 1.75 points in $M^2$ means one manager would have earned 1.75% more at the market's volatility.
- **The two can disagree.** A portfolio with a lot of nonsystematic risk can win on $T^2$ and lose on $M^2$; which one counts depends on the role, exactly as in IM-7 b.

## 5. Which measures always rank alike

- **Sharpe and $M^2$:** identical rankings, because $M^2 = \sigma_M(S_P - S_M)$ rises one-for-one with $S_P$.
- **Treynor and $T^2$:** identical rankings, because $T^2 = T_P - T_M$.
- **Jensen's alpha and Treynor:** the same *verdict against the market*, since $\alpha_P = \beta_P(T_P - T_M)$ and $\beta_P > 0$: alpha is positive exactly when the T-line is steeper than the SML. Not necessarily the same *ranking* of two portfolios: dividing by different betas can reorder them (Table 8.1).

- **Sharpe/M$^2$ against Treynor/T$^2$:** can disagree whenever the portfolios carry different amounts of nonsystematic risk.

**Correction to the source notes.** Vardeez writes "Jensen's alpha and the Treynor measure give the same rankings" and makes it a trap ("Jensen and Treynor rank alike"). They give the same verdict on whether a portfolio beats the market, not the same ranking across portfolios. GARP's own Table 8.1 is the counterexample: alpha ranks Q first (3.5% against 3.0%), Treynor ranks U first (13.75% against 12.80%). The Sharpe and M$^2$ half of the statement is correct.

## Formula card: IM-7 c

| | |
|---|---|
| Weight for M$^2$ | $w = \sigma_M/\sigma_P$ in P, $1-w$ in T-bills ($w > 1$: borrow) |
| Adjusted return | $r_{P^*} = w\bar{r}_P + (1-w)\bar{r}_f = \bar{r}_f + S_P\,\sigma_M$ |
| M$^2$ | $M^2 = r_{P^*} - \bar{r}_M = \sigma_M(S_P - S_M)$ |
| Weight for T$^2$ | $w = 1/\beta_P$ in P, $1-w$ in T-bills ($\beta_P < 1$: borrow) |
| T$^2$ | $T^2 = r_{P^*} - \bar{r}_M = T_P - T_M = \alpha_P/\beta_P$, with $T_M = \bar{r}_M - \bar{r}_f$ |
| Two portfolios | $M_1^2 - M_2^2 = \sigma_M(S_1 - S_2)$; $T_1^2 - T_2^2 = T_1 - T_2$ |
| Beta from correlation | $\beta_P = \rho_{PM}\,\sigma_P/\sigma_M$ |

| | M$^2$ | T$^2$ |
|---|---|---|
| **Built on** | Sharpe ratio (total risk) | Treynor measure (systematic risk) |
| **Rescale P until** | $\sigma_{P^*} = \sigma_M$ | $\beta_{P^*} = 1$ |
| **Chart** | Return against standard deviation: CAL and CML | Return against beta: T-line and SML |
| **Read off** | Vertical gap at $\sigma_M$ between P$^*$ and M | Vertical gap at $\beta = 1$ between P$^*$ and M |
| **Use when P is** | The investor's entire risky portfolio | One of many sub-portfolios |
| **Ranks like** | Sharpe | Treynor (and the verdict of alpha) |

## Exam traps

- **Forgot to subtract the market (intermediate result).** $r_{P^*}$ is a real and offerable number; M$^2$ is $r_{P^*} - \bar{r}_M$.
- **Unscaled Sharpe difference.** $S_P - S_M$ is in Sharpe units; multiply by $\sigma_M$, not by $\sigma_P$.
- **Weight inverted.** $w = \sigma_M/\sigma_P$, not $\sigma_P/\sigma_M$; the inverted weight levers a high-volatility portfolio further instead of de-levering it.
- **Raw returns.** Dropping the T-bill leg, or dividing raw returns, changes the answer and can flip a ranking.
- **Jensen's alpha offered as M$^2$ or T$^2$ (sibling swap).** Alpha is the gap at the portfolio's own beta; T$^2$ is the gap at $\beta = 1$, which is $\alpha/\beta$.
- **Right number, wrong measure.** For an investor's entire holdings use M$^2$, even when the Treynor gap looks bigger.
- **"Alpha and Treynor rank alike".** Only their verdict against the market always agrees.

## Practice questions: IM-7 c

### Question IM-7 c.1

New • built from GARP Ch.8 (Figures 8.2 and 8.3)

A performance analyst at a Toronto pension consultant is preparing training slides on the graphical representation of risk-adjusted performance measures. One chart plots average return against standard deviation and shows the capital market line (CML) and the capital allocation line (CAL) of an active portfolio P. A second chart plots average return against beta and shows the security market line (SML) and the T-line of P. Which of the following statements about the two charts is correct?

**A.** $M^2$ is the difference in slope between P's capital allocation line and the CML, so it is measured in Sharpe-ratio units.

**B.** $M^2$ is the vertical distance, at the market's standard deviation, between the market portfolio and the point on P's capital allocation line.

**C.** $T^2$ is the vertical distance, at P's own beta, between portfolio P and the security market line.

**D.** On the beta chart, the portfolio with the steeper T-line is always the one with the higher Jensen's alpha.

### Question IM-7 c.2

Bank Q390 • Mock 1 Q61 (rebuilt)

A portfolio has a mean return of 11% and a volatility of 25%. The market portfolio has a mean return of 9.95% and a volatility of 21%. The risk-free rate is 5%. Defining $M^2$ as the difference between the return of the risk-adjusted portfolio and the market return, which of the following is most accurate?

**A.** $M^2 = -0.09\%$; the portfolio underperformed the market

**B.** $M^2 = 0.09\%$; the portfolio outperformed the market

**C.** $M^2 = 1.05\%$; the portfolio outperformed the market

**D.** $M^2 = 2.19\%$; the portfolio outperformed the market

### Question IM-7 c.3

Workbook A18.4 • S1 • Map #36

An investment consultant at a USD family office is presenting to newly hired analysts how to express the risk-adjusted performance of a concentrated portfolio P that the family holds as its entire risky portfolio. Over the period, P returned an average of 22% with a standard deviation of 28% and a beta of 1.3. The market index returned 12% with a standard deviation of 16%. The average T-bill rate was 4%. Which of the following correctly states the Sharpe ratio of P and its $M^2$?

**A.** Sharpe ratio 0.64; $M^2$ 0.14%

**B.** Sharpe ratio 0.64; $M^2$ 2.29%

**C.** Sharpe ratio 0.64; $M^2$ 4.00%

**D.** Sharpe ratio 0.79; $M^2$ 0.57%

### Question IM-7 c.4

Bank Q387 • BT P2.T8.17.3 (rebuilt)

An analyst compares two portfolios, A and B, with the market M:

| | Portfolio A | Portfolio B | Market M |
|---|---|---|---|
| Average return | 11% | 22% | 9.0% |
| Beta | 0.70 | 1.20 | 1.00 |
| Standard deviation | 10% | 20% | 16% |

The risk-free rate is 4.0%. The Treynor-squared ($T^2$) measure of a portfolio is the return of the portfolio after it is re-mixed with the risk-free asset to have a beta of 1.0, minus the market return. What are, respectively, the $T^2$ measures of Portfolio A and Portfolio B?

**A.** $A = 3.5\%$ and $B = 12.0\%$

**B.** $A = 5.0\%$ and $B = 10.0\%$

**C.** $A = 6.2\%$ and $B = 9.4\%$

**D.** $A = 10.0\%$ and $B = 15.0\%$

### Question IM-7 c.5

Bank Q388 • BT P2.T9.21.10.3

Below is the performance of two portfolios, P and Q, in comparison with the market. The risk-free rate is 3.0%.

| | Portfolio P | Portfolio Q | Market M |
|---|---|---|---|
| Beta | 0.70 | 1.20 | 1.00 |
| Excess return | 10.0% | 14.0% | 5.0% |
| Volatility | 12.0% | 27.0% | 16.0% |

In regard to Portfolio P as it compares with Portfolio Q, which of the following statements is TRUE?

**A.** Portfolio P has a higher alpha than Q but is equal to Q in regard to both $M^2$ and $T^2$.

**B.** Portfolio P has a higher alpha than Q but is inferior to Q in regard to both $M^2$ and $T^2$.

**C.** Portfolio P has a lower alpha than Q but is superior to Q in regard to both $M^2$ and $T^2$.

**D.** Portfolio P has a lower alpha than Q and is also inferior to Q in regard to both $M^2$ and $T^2$.

### Question IM-7 c.6

Bank Q386 • BT P2.T8.17.2

The following data compare two portfolios, A and B, with the market M:

| | Portfolio A | Portfolio B | Market M |
|---|---|---|---|
| Average return | 15% | 24% | 12% |
| Beta | 0.80 | 1.50 | 1.00 |
| Standard deviation | 30% | 24% | 16% |
| Tracking error | 18% | 15% | 0 |

The risk-free rate is 3.0%. If the portfolios are ranked by the Modigliani-squared ($M^2$) measure and by the slope of the T-line (Treynor measure), how do they rank?

**A.** Portfolio A offers both a higher $M^2$ and a steeper T-line than Portfolio B.

**B.** Portfolio A offers a higher $M^2$, but Portfolio B has a steeper T-line.

**C.** Portfolio B offers a higher $M^2$, but Portfolio A has a steeper T-line.

**D.** Portfolio B offers both a higher $M^2$ and a steeper T-line than Portfolio A.

### Question IM-7 c.7

Bank Q389 • GARP PE2 Q8 • Mock 4 Q9

An investor is comparing the performances of two portfolio managers who have been allocated an equal amount of investment funds. The managers apply the same strategy with the same constraints, and their portfolios represent the investor's entire holdings (they are not diversified). The investor gathers the following data about the two managers and the market index:

| Measure | Manager 1 | Manager 2 | Market index |
|---|---|---|---|
| Average return | 32% | 28% | 22% |
| Beta with respect to market index | 1.2 | 1.4 | 1.0 |
| Standard deviation of returns | 18% | 14% | 10% |
| Tracking error | 8% | 6% | 0% |

The risk-free rate of interest is 3%. Which of the following is an appropriate measure to use and the correct conclusion to reach when comparing the performances of the two managers?

**A.** The Modigliani-squared measure, which shows that Manager 1 outperforms Manager 2 by 2%

**B.** The Modigliani-squared measure, which shows that Manager 2 outperforms Manager 1 by 2%

**C.** Treynor's measure, which shows that Manager 1 outperforms Manager 2 by 6%

**D.** Treynor's measure, which shows that Manager 2 outperforms Manager 1 by 6%

## Solutions • IM-7 c

### Solution IM-7 c.1

**Answer: B**

New • built from GARP Ch.8 (Figures 8.2 and 8.3)

#### Step 1 – Where $M^2$ sits

P* is formed by moving along P's CAL (mixing P with T-bills, or borrowing) until its standard deviation equals the market's. GARP: "$M^2$ is the vertical distance between points M and P*".
$\Rightarrow$ B is correct: a vertical distance at $\sigma_M$, in return units.

#### Step 2 – Test the beta-chart claims

At P's own beta, the gap between P and the SML is Jensen's alpha, not $T^2$. $T^2$ is the gap at $\beta = 1$ between P* and M. And in Table 8.1 the steeper T-line (U) belongs to the portfolio with the lower alpha.

#### Interpretation

Three distances, three measures: $M^2$ at $\sigma_M$ on the standard deviation chart; $T^2$ at $\beta = 1$ on the beta chart; alpha at the portfolio's own beta on the beta chart. The slopes themselves are the Sharpe and Treynor ratios.

#### Why the other options are wrong

**A is incorrect.** The slope difference is $S_P - S_M$; $M^2$ multiplies it by $\sigma_M$ to turn it into a return.
**C is incorrect.** Sibling swap: the vertical distance to the SML at P's own beta is Jensen's alpha; $T^2 = \alpha/\beta$.
**D is incorrect.** Absolute claim: GARP's U has the steeper T-line (13.75 against 12.80) but the lower alpha (3.0% against 3.5%).

**LO:** IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), Example 8.1 and Figure 8.2; The Treynor Ratio, Table 8.1 and Figure 8.3

### Solution IM-7 c.2

**Answer: B**

Bank Q390 • Mock 1 Q61

#### Step 1 – Weight that matches the market's volatility

$$w = \frac{\sigma_M}{\sigma_P} = \frac{21}{25} = 0.84 \text{ in the portfolio}, \quad 0.16 \text{ in the risk-free asset}$$

#### Step 2 – Return on the risk-adjusted portfolio

$$r_{P^*} = 5\% + 0.84 \times (11\% - 5\%) = 10.04\%$$

#### Step 3 – $M^2$

$$M^2 = 10.04\% - 9.95\% = +0.09\%$$

$\Rightarrow$ At the market's volatility the portfolio would have beaten the market by 9 basis points.

#### Interpretation

Check with Sharpe ratios: $S_P = 6/25 = 0.240$, $S_M = 4.95/21 = 0.2357$, and $21 \times (0.240 - 0.2357) = 0.09\%$. A positive $M^2$ always goes with a higher Sharpe ratio than the market's.

#### Why the other options are wrong

**A is incorrect.** Subtracted in the wrong order (market minus portfolio): the sign and the conclusion flip.

**C is incorrect.** Raw return difference, $11\% - 9.95\% = 1.05\%$, with no risk adjustment.
**D is incorrect.** Inverted the weight, $25/21 = 1.19$: $5\% + 1.19 \times 6\% - 9.95\% = 2.19\%$, levering a riskier portfolio up instead of de-levering it.

**LO:** IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), Eq. 8.1 and Example 8.1; SchweserNotes 2026 Book 5, Reading 89, LO 89.c

### Solution IM-7 c.3 — Answer: B

Workbook A18.4 • S1

#### Step 1 – Sharpe ratios

$$S_P = \frac{22\% - 4\%}{28\%} = 0.643, \qquad S_M = \frac{12\% - 4\%}{16\%} = 0.500$$

$\Rightarrow$ P earns 0.64% of excess return per 1% of total volatility, against 0.50% for the index.

#### Step 2 – Weight in P that matches the index's volatility

$$w_P = \frac{\sigma_M}{\sigma_P} = \frac{16\%}{28\%} = 0.5714$$

$\Rightarrow$ Hold 57.14% in P and 42.86% in T-bills so P* has a 16% standard deviation.

#### Step 3 – $M^2$

$$r_{P^*} = 0.5714 \times 22\% + 0.4286 \times 4\% = 14.286\%, \qquad M^2 = 14.286\% - 12\% = 2.29\%$$

#### Step 4 – Check

$$(S_P - S_M)\,\sigma_M = 0.143 \times 16\% = 2.29\%$$

#### Interpretation

At the index's volatility P would have beaten the index by 2.29 points a year. Because P is the family's whole risky portfolio, total risk is the right measure; the beta of 1.3 is a red herring (it would feed $T^2$).

#### Why the other options are wrong

**A is incorrect.** Reported $S_P - S_M = 0.143$ as a return of 0.14%; the Sharpe difference must be scaled by $\sigma_M$.
**C is incorrect.** Scaled the Sharpe difference by $\sigma_P$ instead of $\sigma_M$: $0.143 \times 28\% = 4.00\%$, which matches P's volatility rather than the index's.
**D is incorrect.** Used raw returns throughout: $S_P = 22/28 = 0.79$, $S_M = 12/16 = 0.75$, $M^2 = (0.786 - 0.75) \times 16\% = 0.57\%$, equivalent to dropping the T-bill leg $(1 - w)\bar{r}_f$.

**LO:** IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), The $M^2$ Measure and the Sharpe Ratio (Eq. 8.1, Example 8.1)

### Solution IM-7 c.4 — Answer: B

Bank Q387 • BT P2.T8.17.3

**Step 1 – Treynor measures**

$$T_A = \frac{11-4}{0.70} = 10.0\%, \qquad T_B = \frac{22-4}{1.20} = 15.0\%, \qquad T_M = 9-4 = 5.0\%$$

**Step 2 – T² = $T_P - T_M$**

$$T_A^2 = 10.0 - 5.0 = 5.0\%, \qquad T_B^2 = 15.0 - 5.0 = 10.0\%$$

**Step 3 – Check by re-mixing A to a beta of 1**

$w_A = 1/0.70 = 1.4286$ (borrow 42.86%): $r_{A^*} = 4\% + 1.4286 \times 7\% = 14.0\%$, and $14.0\% - 9\% = 5.0\%$.

$\Rightarrow$ B beats the market by 10 points at a beta of 1, A by 5.

**Interpretation**

Re-mixing to $\beta = 1$ makes the excess return equal to the Treynor measure, which is why T² reduces to $T_P - T_M$. The standard deviations are not needed; they would give M².

**Why the other options are wrong**

**A is incorrect.** Jensen's alphas, $7 - 0.7 \times 5 = 3.5\%$ and $18 - 1.2 \times 5 = 12.0\%$: gaps at the portfolios' own betas, not rescaled to $\beta = 1$.

**C is incorrect.** M² values (re-mixing to 16% volatility): $7 \times 1.6 - 5 = 6.2\%$ and $18 \times 0.8 - 5 = 9.4\%$. Total risk instead of beta.

**D is incorrect.** The Treynor measures themselves; forgot to subtract the market's 5.0%.

**LO:** IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), The Treynor Ratio, Example 8.2

### Solution IM-7 c.5 — Answer: C

Bank Q388 • BT P2.T9.21.10.3

**Step 1 – Jensen's alpha (excess returns given, market excess 5%)**

$$\alpha_P = 10 - 0.7 \times 5 = 6.5\%, \qquad \alpha_Q = 14 - 1.2 \times 5 = 8.0\%$$

$\Rightarrow$ P has the lower alpha.

**Step 2 – M² (scale to the market's 16% volatility)**

$$M_P^2 = 10 \times \frac{16}{12} - 5 = 8.33\%, \qquad M_Q^2 = 14 \times \frac{16}{27} - 5 = 3.30\%$$

$\Rightarrow$ P is levered up ($w = 1.33$), Q de-levered ($w = 0.59$); P wins.

**Step 3 – T² (scale to a beta of 1)**

$$T_P^2 = \frac{10}{0.70} - 5 = 9.29\%, \qquad T_Q^2 = \frac{14}{1.20} - 5 = 6.67\%$$

$\Rightarrow$ P wins again.

**Interpretation**

Alpha is not scaled for the amount of risk taken; M² and T² are. P is the more efficient portfolio on both total and systematic risk, despite its smaller alpha. Note also $T^2 = \alpha/\beta$: $6.5/0.7 = 9.29$ and $8.0/1.2 = 6.67$.

### Why the other options are wrong

**A is incorrect.** P's alpha is lower, not higher, and its $M^2$ and $T^2$ differ from Q's.
**B is incorrect.** Wrong on both: P's alpha is lower and P is superior on $M^2$ and $T^2$.
**D is incorrect.** Right on alpha, but judging by raw excess return (10% against 14%) instead of risk-adjusted return gets the $M^2$ and $T^2$ comparison backwards.

**LO:** IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), Eq. 8.1, Example 8.2

## Solution IM-7 c.6

**Answer: C**

Bank Q386 • BT P2.T8.17.2

### Step 1 – $M^2$ of each portfolio

A: $w = 16/30 = 0.5333$, $r_{A^*} = 0.5333 \times 15\% + 0.4667 \times 3\% = 9.4\%$, $M^2_A = 9.4\% - 12\% = -2.6\%$.
B: $w = 16/24 = 0.6667$, $r_{B^*} = 0.6667 \times 24\% + 0.3333 \times 3\% = 17.0\%$, $M^2_B = 17.0\% - 12\% = +5.0\%$.
$\Rightarrow$ B has the higher $M^2$.

### Step 2 – Slope of the T-line

$$T_A = \frac{15 - 3}{0.80} = 15.0\%, \qquad T_B = \frac{24 - 3}{1.50} = 14.0\%$$

$\Rightarrow$ A has the steeper T-line.

### Interpretation

A's volatility is mostly nonsystematic (30% standard deviation on a beta of 0.8), which $M^2$ charges and Treynor ignores. If A and B compete to be the investor's whole risky portfolio, B wins; if each is one of many sub-portfolios, A wins. The tracking errors are not needed.

### Why the other options are wrong

**A is incorrect.** A's $M^2$ $(-2.6\%)$ is below B's $(+5.0\%)$.
**B is incorrect.** Reverses both rankings.
**D is incorrect.** B's Treynor (14.0%) is below A's (15.0%); comparing excess returns (21% against 12%) without dividing by beta gives this.

**LO:** IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), Example 8.1 and The Treynor Ratio; SchweserNotes 2026 Book 5, Reading 89, LO 89.c

## Solution IM-7 c.7

**Answer: B**

Bank Q389 • GARP PE2 Q8 • Mock 4 Q9

### Step 1 – Choose the measure

Each portfolio is the investor's entire holding and is not diversified, so total risk is the relevant risk: $M^2$ (equivalently, Sharpe), not Treynor.

### Step 2 – Sharpe ratios

$$S_1 = \frac{32 - 3}{18} = 1.611, \qquad S_2 = \frac{28 - 3}{14} = 1.786, \qquad S_M = \frac{22 - 3}{10} = 1.900$$

### Step 3 – Risk-adjust each to the market's 10% volatility

$$r_{1^*} = 3\% + 1.611 \times 10\% = 19.11\%, \qquad r_{2^*} = 3\% + 1.786 \times 10\% = 20.86\%$$

$\Rightarrow M_1^2 = -2.89\%$ and $M_2^2 = -1.14\%$: Manager 2 beats Manager 1 by $10 \times (1.786 - 1.611) = 1.75\%$, about 2%.

**Step 4 – Treynor, for comparison only**

$T_1 = 29/1.2 = 24.17$, $T_2 = 25/1.4 = 17.86$: a gap of about 6 in favour of Manager 1, but on the wrong measure for undiversified holdings.

**Interpretation**

Both managers trail the market on a risk-adjusted basis (negative $M^2$), and Manager 2 trails it by less. The market return cancels when two $M^2$ values are compared, so only the Sharpe gap and $\sigma_M$ matter. The tracking errors and betas are distractors for this role.

**Why the other options are wrong**

**A is incorrect.** Right measure, reversed ranking: Manager 1's higher raw return (32% against 28%) ignores its much higher volatility.

**C is incorrect.** Treynor suits a portfolio that is one component of a diversified whole; it does favour Manager 1 by about 6 (24.17 against 17.86), but it is the wrong measure here.

**D is incorrect.** Wrong measure and wrong direction: Treynor favours Manager 1.

**LO:** IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), The $M^2$ Measure and the Sharpe Ratio; GARP Practice Exam 2, Q8

IM-7 • Portfolio Performance Evaluation • Vardeez: VERY IMPORTANT

# IM-7 i

P1 BOTH

Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution.

Workbook: A18 Performance Measures and Attribution • Tier A • Map #36

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic: "portfolio manager asset allocation and selection attributes" was reported on the A18 recall sheets (Map #36, 5 of 16).
**Vardeez:** no pointer on this LO; his notes give the two contribution formulas and the sum check.

**Reported exam shapes**

- **S4** Attribution split into asset allocation and security selection effects (Workbook A18.5, Question IM-7 i.8; also Questions i.2 to i.7).

**What the questions look like:** almost all numerical. A table gives bogey and manager weights, index and manager returns by market (sometimes by sector or country). You compute the allocation effect, the selection effect, the total, or the split of one market's selection into sector allocation and security selection. The wrong options use the wrong weights (bogey instead of manager in selection), the wrong returns (the manager's instead of the index's in allocation), the interaction term, a market-level figure never multiplied by the market's weight, or the total offered as one component.

## 1. The bogey and the decision levels

Attribution asks which decisions produced the result, rather than adjusting it for risk. The managed portfolio P is compared with a **bogey** B, the return the manager would have earned with a completely passive strategy. "Passive" has two parts:

- **Neutral weights across asset classes,** set with the client according to its risk tolerance: perhaps 75% equity, 15% bonds and 10% cash for a risk-tolerant client, 45/35/20 for a risk-averse one. Departures must be justified by a view that a market will over- or underperform.
- **An index within each class,** such as the S&P 500 for equity.

Any difference $r_P - r_B$ must then come from asset allocation bets (departures from the neutral weights) or selection bets (departures from the index within a class). A common system splits it into three levels: (1) allocation across equity, fixed income and money markets; (2) sector (industry) choice within each market; (3) security choice within each sector.

**Three portfolios, two steps.** The cleanest way to see the split:

$$\underset{\text{Bogey}}{\sum_i w_{B,i} r_{B,i}} \xrightarrow{\text{asset allocation}} \underset{\text{Manager's weights, index returns}}{\sum_i w_{P,i} r_{B,i}} \xrightarrow{\text{selection}} \underset{\text{Actual portfolio}}{\sum_i w_{P,i} r_{P,i}}$$

$$\boxed{\text{Allocation} = \sum_i (w_{P,i} - w_{B,i})\, r_{B,i}} \qquad \boxed{\text{Selection} = \sum_i w_{P,i}\, (r_{P,i} - r_{B,i})}$$

$$\text{Allocation} + \text{Selection} = \sum_i w_{P,i} r_{P,i} - \sum_i w_{B,i} r_{B,i} = r_P - r_B$$

The $w_P r_B$ terms cancel, so the two contributions always add exactly to the excess return (Vardeez's sum check). If the manager holds the bogey's weights, allocation is zero; if the manager earns the index return in every class, selection is zero.

## 2. Asset allocation: GARP's example

Neutral weights 60/30/10. The bogey returned $0.60(5.81) + 0.30(1.45) + 0.10(0.48) = 3.97\%$ and the managed portfolio 5.34%, an excess return of 1.37%. The manager held 70/7/23:

| Market | Actual weight | Benchmark weight | Active weight | Index return | Contribution |
|---|---|---|---|---|---|
| Equity | 0.70 | 0.60 | $+0.10$ | 5.81% | $+0.5810\%$ |
| Fixed income | 0.07 | 0.30 | $-0.23$ | 1.45% | $-0.3335\%$ |
| Cash | 0.23 | 0.10 | $+0.13$ | 0.48% | $+0.0624\%$ |
| Contribution of asset allocation | | | | | **0.3099**% |

Overweighting equity in a month when the S&P 500 returned 5.81% was the main source: 31 basis points.

**A per-class trap.** The active weights sum to zero, so $\sum(w_P - w_B)(r_{B,i} - r_B)$ gives the same total; this "relative" form is used by some systems. The class-by-class numbers differ: bonds contribute $-0.3335\%$ in GARP's form but $-0.23 \times (1.45 - 3.97) = +0.58\%$ in the relative form, because underweighting a class that lagged the bogey is rewarded there. Use GARP's form unless the stem defines the effect against the total benchmark return.

## 3. Selection within markets

| Market | Portfolio return | Index return | Excess | Portfolio weight | Contribution |
|---|---|---|---|---|---|
| Equity | 7.28% | 5.81% | 1.47% | 0.70 | 1.03% |
| Fixed income | 1.89% | 1.45% | 0.44% | 0.07 | 0.03% |
| Contribution of selection within markets | | | | | **1.06%** |

$0.31\% + 1.06\% = 1.37\%$, the whole excess return. The selection effect is weighted by the *manager's* weights. Some systems (Brinson, Hood and Beebower) weight selection by the bogey's weights and report the remainder as an **interaction** term, $\sum(w_P - w_B)(r_P - r_B)$. GARP's selection effect contains that interaction: here $0.60(1.47) + 0.30(0.44) = 1.014\%$ at bogey weights plus an interaction of $0.10(1.47) - 0.23(0.44) = 0.046\%$ gives the same 1.06%.

## 4. Sector and security selection within a market

The equity component beat the S&P 500 by 1.47%. GARP splits it again, with the same two formulas one level down:

- **Sector allocation within equity** $= \sum_s (w_{P,s} - w_{B,s})\, r_s$, the sector's active weight times the sector's return (GARP Table 8.8, eight sectors): $1.29\%$. The big source is a 19.97-point overweight in consumer noncyclicals, which returned 10.0% (contribution 2.00%).
- **Security selection within sectors** = equity excess − sector allocation $= 1.47\% - 1.29\% = 0.18\%$.
- **To the portfolio:** multiply both by the equity weight, 0.70: sector allocation $1.29 \times 0.70 = 0.90\%$, security selection $0.18 \times 0.70 = 0.13\%$, together $1.47 \times 0.70 = 1.03\%$ (102.9 basis points).

**Correction to the source notes.** GARP's Table 8.8 prints the consumer cyclical sector's allocation contribution as $+0.3546$. Its active weight is $8.47 - 12.5 = -4.03$ points and its return 8.8%, so the contribution is $-0.3546$. The table's total of 1.2898 only adds up with the negative sign (with $+0.3546$ the eight entries sum to 1.9990).

## 5. The aggregate contribution

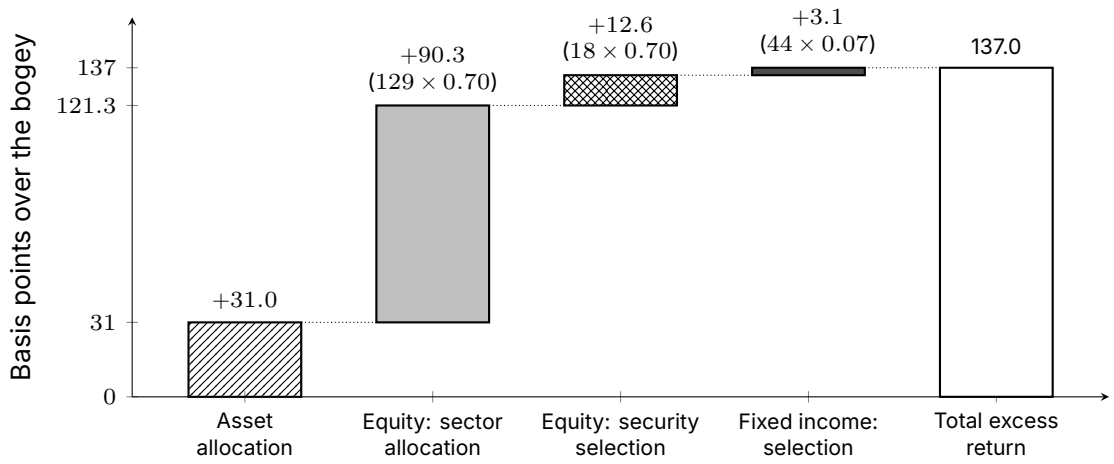



Figure 1. GARP Table 8.9 as a waterfall: bogey 3.97% to portfolio 5.34%. Hatched: asset allocation; grey: sector allocation inside equity; cross-hatched: security selection inside equity sectors; thin dark bar: fixed-income selection; white: the total. The two equity bars are the equity component's 147 basis points times the 0.70 equity weight.

In GARP's month every facet of the process added value. The sector analysis can be repeated inside fixed income in the same way.

**What moves the answer (GARP Concept Check 8.5).** The allocation effect depends on the neutral weights: with a 70/25/5 bogey (return 4.45%), the same 70/7/23 portfolio shows an allocation effect of $0(5.81) - 0.18(1.45) + 0.18(0.48) = -0.17\%$. The selection effect depends on the index returns: had the S&P 500 returned 5.00%, equity selection would be $0.70 \times 2.28 = 1.60\%$ and total selection 1.63%.

**Worked example: a global equity manager (GARP Problem 12).** Benchmark: MSCI country weights; the manager may hold countries in any proportion.

| Country | MSCI weight | Manager weight | Manager return | Index return | Allocation | Selection |
|---|---|---|---|---|---|---|
| U.K. | 0.15 | 0.30 | 20% | 12% | +1.80 | +2.40 |
| Japan | 0.30 | 0.10 | 15% | 15% | −3.00 | 0.00 |
| U.S. | 0.45 | 0.40 | 10% | 14% | −0.70 | −1.60 |
| Germany | 0.10 | 0.20 | 5% | 12% | +1.20 | −1.40 |
| Total | | | 12.5% | 13.8% | **−0.70** | **−0.60** |

The manager lost 1.30% against the index: $-0.70\%$ from country allocation (mainly the 20-point underweight in Japan, which returned 15%) and $-0.60\%$ from stock selection (good in the U.K., poor in the U.S. and Germany). The two add to $-1.30\%$.

## Formula card: IM-7 i

| | |
|---|---|
| Bogey return | $r_B = \sum_i w_{B,i}\, r_{B,i}$ (neutral weights, index returns) |
| Excess return | $r_P - r_B = \sum_i w_{P,i} r_{P,i} - \sum_i w_{B,i} r_{B,i}$ |
| Asset allocation | $\sum_i (w_{P,i} - w_{B,i})\, r_{B,i}$: active weight times index return |
| Selection | $\sum_i w_{P,i}\, (r_{P,i} - r_{B,i})$: manager's weight times return over the index |
| Sum check | Allocation + Selection $= r_P - r_B$ exactly |
| Inside one market | market excess = sector allocation $\sum_s (w_{P,s} - w_{B,s}) r_s$ + security selection |
| To the portfolio | multiply each market-level piece by the market's *portfolio* weight $w_{P,i}$ |
| Interaction (other systems) | $\sum (w_P - w_B)(r_P - r_B)$; GARP's selection = selection at bogey weights + interaction |

## Exam traps

- **Manager's returns in the allocation effect.** Allocation uses *index* returns; using the manager's returns mixes selection into allocation.
- **Bogey weights in the selection effect.** Selection uses the *manager's* weights; bogey weights drop the interaction term.
- **The interaction term offered as a component.** $\sum (w_P - w_B)(r_P - r_B)$ is neither GARP's allocation nor its selection effect.
- **Market-level figure not weighted (intermediate result).** A selection effect inside equity must be multiplied by the equity weight before it enters the portfolio total (GARP: $147 \times 0.70$).
- **Whole market selection as security selection.** The equity component's excess still contains sector allocation; security selection is what is left after it.
- **One component offered as the total, or the total as one component.** Check that allocation plus selection equals $r_P - r_B$.
- **Sign.** An underweight in a market with a positive index return gives a negative allocation contribution in GARP's form, even if that market lagged the bogey.

## Practice questions: IM-7 i

### Question IM-7 i.1

New • built from GARP Ch.8, Section 8.5

A consultant at a Johannesburg asset manager is explaining the firm's performance attribution system to new trustees of a client pension fund. The system uses a bogey that holds neutral weights across equity, bonds and cash, and an index within each market. Which of the following statements about the system is correct?

**A.** The asset allocation contribution multiplies each market's active weight by the manager's own return in that market.

**B.** The security selection contribution multiplies each market's return in excess of its index by the bogey's weight in that market.

**C.** The neutral weights are set with the client according to its risk tolerance, so the same manager decisions can show different allocation contributions for two clients.

**D.** A manager who holds exactly the bogey's market weights can still show a nonzero asset allocation contribution when the market indexes move.

### Question IM-7 i.2

Bank Q416 • AnalystPrep Q.3186 (rebuilt)

A fund began the year with USD 100,000 and had no contributions or withdrawals. Its actual return was 16% (ending value USD 116,000). A performance attribution gives the following ending values:

| Portfolio | Ending value (USD) |
|---|---|
| Passive policy benchmark (benchmark weights, benchmark index returns) | 107,000 |
| Actual asset allocation held at benchmark index returns | 110,000 |
| Actual portfolio (actual allocation and actual security selection) | 116,000 |

What was the contribution of the manager's asset allocation decision to the fund's return?

**A.** 3%

**B.** 6%

**C.** 9%

**D.** 10%

### Question IM-7 i.3

Bank Q421 • GARP PE1 Q61 • Mock 3 Q20

A fund of funds (FOF) manager is conducting a performance attribution analysis for a portfolio consisting of equity and fixed-income securities, to evaluate the effects of the portfolio manager's asset allocation and security selection decisions. The FOF manager uses the following information:

| Asset class | Portfolio weight | Benchmark weight | Portfolio return | Benchmark return |
|---|---|---|---|---|
| Equity | 58% | 50% | 8% | 11% |
| Fixed income | 42% | 50% | 6% | 7% |

What is the contribution of the portfolio manager's asset allocation decision to the portfolio's overall excess return?

**A.** $-2.16\%$

**B.** $-1.84\%$

**C.** $-0.16\%$

**D.** $0.32\%$

### Question IM-7 i.4

Bank Q419 • BT P2.T9.21.11.2 (rebuilt)

Peter manages a portfolio measured against a bogey with 50% equity, 30% bonds and 20% cash.

| Component | Bogey weight | Index return | Actual weight | Actual return |
|---|---|---|---|---|
| Equity (S&P 500) | 50% | 8.0% | 80% | 11.0% |
| Bonds (bond index) | 30% | 4.0% | 10% | 5.0% |
| Cash (money market) | 20% | 1.0% | 10% | 1.0% |
| Return | 5.4% | | | 9.4% |

Peter's portfolio outperformed the bogey by 4.0%. Of this excess return, how much is attributable to security selection?

**A.** 1.50%

**B.** 1.80%

**C.** 2.50%

**D.** 3.40%

### Question IM-7 i.5

Bank Q420 • BT P2.T9.21.8.2

A fund's benchmark (bogey) is a paper portfolio equally weighted between equities (S&P 500) and a bond index. After one year the benchmark's excess return was $+3.50\%$.

| Component | Benchmark weight | Benchmark excess return | Fund weight | Fund excess return |
|---|---|---|---|---|
| Equity (S&P 500) | 50% | 5.00% | 80% | 7.00% |
| Bonds | 50% | 2.00% | 20% | 1.00% |
| Total | | 3.50% | | 5.80% |

The fund's excess return was $+5.80\%$, so it outperformed its benchmark by 2.30%. Asset allocation refers to the fund's decision to vary from the 50/50 mix; security selection refers to performance within an asset class. Which of the following statements about the fund's performance is TRUE?

**A.** Both asset allocation and security selection (active management) contributed positively to the fund's outperformance.

**B.** Equities were a drag on the fund's performance in regard to both asset allocation and security selection, but bonds over-compensated.

**C.** The fund's outperformance is due to a reduction in both policy-mix risk and active-management risk.

**D.** Neither asset allocation nor security selection (active management) contributed positively to the fund's outperformance.

### Question IM-7 i.6

Bank Q414 • AP Mock 3 Q7 (rebuilt)

A portfolio manager runs a portfolio benchmarked to the XYZ index. The composition of the index and the portfolio's performance are given below:

| Sector | Index weight | Index return | Portfolio weight | Portfolio return |
|---|---|---|---|---|
| Infrastructure | 0.20 | 10.00% | 0.15 | 12.00% |
| Financial services | 0.15 | 15.00% | 0.10 | 15.00% |
| Utilities | 0.20 | 20.00% | 0.25 | 22.00% |
| Energy | 0.10 | 8.00% | 0.05 | 5.00% |
| IT | 0.20 | 18.00% | 0.25 | 10.00% |
| Fixed income | 0.15 | 15.00% | 0.20 | 12.00% |

Which of the following is closest to the excess return (portfolio minus index) generated by the portfolio manager?

**A.** $-1.95\%$

**B.** $-0.95\%$

**C.** $+0.95\%$

**D.** $+1.00\%$

### Question IM-7 i.7

New • built from GARP Ch.8, Table 8.8

A performance analyst at a Swiss pension fund is attributing the monthly return of a balanced mandate. The bogey holds 50% in equity; the manager holds 60%. Inside the equity component, the manager's sector weights and returns, and the equity index's sector weights and returns, were:

| Sector | Manager weight | Index weight | Manager return | Index sector return |
|---|---|---|---|---|
| Technology | 40% | 30% | 8.0% | 6.0% |
| Financials | 25% | 30% | 4.0% | 3.0% |
| Energy | 10% | 20% | −1.0% | −2.0% |
| Consumer | 25% | 20% | 5.0% | 4.0% |

Which of the following correctly states the sector allocation effect measured within the equity component, and the contribution of security selection within equity sectors to the total portfolio's excess return?

**A.** Sector allocation 0.85%; security selection contribution 0.70%

**B.** Sector allocation 0.95%; security selection contribution 0.78%

**C.** Sector allocation 0.85%; security selection contribution 0.84%

**D.** Sector allocation 0.85%; security selection contribution 1.40%

### Question IM-7 i.8

Workbook A18.5 • S4 • Map #36

A fund-of-funds analyst at a Canadian pension plan is preparing a monthly attribution report for a balanced manager. The neutral (bogey) weights, the manager's weights, and the returns are:

| Market | Bogey weight | Manager weight | Index return | Manager's return in market |
|---|---|---|---|---|
| Equity | 55% | 65% | 4.0% | 5.2% |
| Fixed income | 35% | 20% | 1.2% | 1.6% |
| Cash | 10% | 15% | 0.4% | 0.4% |

A separate sector analysis of the equity component shows that the manager's sector weights within equity contributed 0.80% of the equity component's excess return over the equity index. Which of the following correctly states the contribution of the asset allocation decision to the portfolio's excess return, and the contribution of security selection within equity sectors to the portfolio's excess return?

**A.** Asset allocation 0.24%; security selection within equity 0.40%

**B.** Asset allocation 0.24%; security selection within equity 0.78%

**C.** Asset allocation 0.30%; security selection within equity 0.26%

**D.** Asset allocation 0.24%; security selection within equity 0.26%

## Solutions • IM-7 i

### Solution IM-7 i.1 — Answer: C

New • built from GARP Ch.8, Section 8.5

**Step 1 – What the bogey is**

Neutral weights across markets, "determined in consultation with the client" according to its risk tolerance, and an index within each market.

**Step 2 – Why it changes the allocation effect**

The allocation effect is $\sum(w_P - w_B)r_B$. A different neutral $w_B$ changes every active weight, so identical holdings can show a different allocation contribution for each client.

$\Rightarrow$ C is correct. GARP's Concept Check 8.5 shows it: moving the bogey from 60/30/10 to 70/25/5 turns the same portfolio's allocation effect from $+0.31\%$ into $-0.17\%$.

**Interpretation**

The attribution result is always relative to a benchmark chosen with the client. Before reading an allocation effect, ask what the neutral weights were.

**Why the other options are wrong**

**A is incorrect.** Wrong returns: allocation uses the *index* return of each market. Using the manager's returns mixes selection into allocation.

**B is incorrect.** Wrong weights: GARP's selection effect uses the *manager's* weights (the bogey-weight version leaves out the interaction).

**D is incorrect.** If $w_P = w_B$ in every market, every active weight is zero and the allocation effect is zero, whatever the indexes do.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5 Performance Attribution Procedures, Concept Check 8.5

### Solution IM-7 i.2 — Answer: A

Bank Q416 • AnalystPrep Q.3186

**Step 1 – Returns of the three portfolios**

Policy benchmark $107{,}000/100{,}000 - 1 = 7\%$; actual allocation at index returns $10\%$; actual portfolio $16\%$.

**Step 2 – Allocation: second step minus first**

$10\% - 7\% = 3\%$: the effect of moving from benchmark weights to the manager's weights, holding index returns.

**Step 3 – Selection and the total**

Selection $16\% - 10\% = 6\%$; total excess $16\% - 7\% = 9\% = 3\% + 6\%$.

**Interpretation**

This is the three-portfolio ladder of Section 1 in currency form. The middle portfolio, the manager's weights at index returns, is what separates the two decisions.

**Why the other options are wrong**

**B is incorrect.** The selection contribution, $16\% - 10\%$.

**C is incorrect.** The total excess return over the benchmark, allocation plus selection.

**D is incorrect.** The return of the actual allocation at index returns, not its excess over the benchmark.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector

and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5 Asset Allocation Decisions

### Solution IM-7 i.3 — Answer: D

Bank Q421 • GARP PE1 Q61 • Mock 3 Q20

**Step 1 – Active weights**

Equity $58\% - 50\% = +8$ points; fixed income $42\% - 50\% = -8$ points.

**Step 2 – Allocation effect: active weight times benchmark return**

$$0.08 \times 11\% + (-0.08) \times 7\% = 0.88\% - 0.56\% = +0.32\%$$

$\Rightarrow$ Tilting towards the asset class whose benchmark returned more was worth 32 basis points.

**Step 3 – Cross-check the whole decomposition**

$r_P = 0.58(8\%) + 0.42(6\%) = 7.16\%$, $r_B = 9.00\%$, excess $-1.84\%$. Selection $0.58(-3\%) + 0.42(-1\%) = -2.16\%$. $0.32\% - 2.16\% = -1.84\%$.

**Interpretation**

Good allocation, poor selection: the manager picked the right asset class and then underperformed inside both. Because the active weights sum to zero, the relative form gives the same total: $0.08 \times (11 - 9) + (-0.08) \times (7 - 9) = 0.32\%$.

**Why the other options are wrong**

**A is incorrect.** The selection effect, $\sum w_P(r_P - r_B) = -2.16\%$.
**B is incorrect.** The total excess return, $7.16\% - 9.00\%$.
**C is incorrect.** The interaction term, $0.08(-3\%) + (-0.08)(-1\%) = -0.16\%$: weight differences times return differences.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5, Table 8.7; GARP Practice Exam 1, Q61

### Solution IM-7 i.4 — Answer: C

Bank Q419 • BT P2.T9.21.11.2

**Step 1 – Selection uses the manager's weights**

$$0.80(11 - 8) + 0.10(5 - 4) + 0.10(1 - 1) = 2.40\% + 0.10\% + 0 = 2.50\%$$

**Step 2 – Allocation, for the sum check**

$$(0.30)(8\%) + (-0.20)(4\%) + (-0.10)(1\%) = 2.40\% - 0.80\% - 0.10\% = 1.50\%$$

$\Rightarrow 1.50\% + 2.50\% = 4.00\%$, the whole outperformance.

**Interpretation**

Peter added value both ways: the 30-point equity overweight in a strong equity year (1.50%) and beating the index inside equity and bonds (2.50%).

**Why the other options are wrong**

**A is incorrect.** The asset allocation contribution, not selection.
**B is incorrect.** Weighted the return differences by the bogey's weights: $0.50(3) + 0.30(1) = 1.80\%$. That is selection at bogey weights, which leaves out the 0.70% interaction.

**D is incorrect.** Added the unweighted bond return difference to the weighted equity effect: $2.40 + 1.0 = 3.40\%$.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5, Tables 8.6 and 8.7

### Solution IM-7 i.5

**Answer: A**

Bank Q420 • BT P2.T9.21.8.2

#### Step 1 – Asset allocation

$$(0.80 - 0.50)(5\%) + (0.20 - 0.50)(2\%) = 1.50\% - 0.60\% = +0.90\%$$

#### Step 2 – Security selection

$$0.80(7\% - 5\%) + 0.20(1\% - 2\%) = 1.60\% - 0.20\% = +1.40\%$$

#### Step 3 – Sum check

$0.90\% + 1.40\% = 2.30\%$, the outperformance.
$\Rightarrow$ Both effects are positive.

#### Interpretation

Equity drove both: the overweight added 1.50% and equity selection 1.60%. Bonds were the drag on both counts ($-0.60\%$ and $-0.20\%$). The decomposition works the same on excess returns as on total returns, because the risk-free rate cancels.

#### Why the other options are wrong

**B is incorrect.** Reversed: equities contributed positively to both effects; bonds were the drag.
**C is incorrect.** The fund took more policy-mix risk (80/20 against 50/50) and active risk, not less.
**D is incorrect.** Contradicted by the numbers: both effects are positive.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5 Performance Attribution Procedures

### Solution IM-7 i.6

**Answer: B**

Bank Q414 • AP Mock 3 Q7

#### Step 1 – Index return

$$0.20(10) + 0.15(15) + 0.20(20) + 0.10(8) + 0.20(18) + 0.15(15) = 14.90\%$$

#### Step 2 – Portfolio return

$$0.15(12) + 0.10(15) + 0.25(22) + 0.05(5) + 0.25(10) + 0.20(12) = 13.95\%$$

#### Step 3 – Aggregate contribution

$13.95\% - 14.90\% = -0.95\%$.

#### Step 4 – Split it

Allocation $\sum(w_P - w_B)r_B = -0.50 - 0.75 + 1.00 - 0.40 + 0.90 + 0.75 = +1.00\%$; selection $\sum w_P(r_P - r_B) = 0.30 + 0 + 0.50 - 0.15 - 2.00 - 0.60 = -1.95\%$; together $-0.95\%$.

**Interpretation**

The weights were well chosen, but the IT overweight earned 10% against 18% for the index's IT sector, costing 2.00% on its own.

**Why the other options are wrong**

**A is incorrect.** The selection effect alone.
**C is incorrect.** Right size, wrong sign: the portfolio underperformed.
**D is incorrect.** The allocation effect alone.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5 Summing Up Component Contributions

**Solution IM-7 i.7** **Answer: C**

New • built from GARP Ch.8, Table 8.8

**Step 1 – Equity returns**

Manager: $0.40(8) + 0.25(4) + 0.10(-1) + 0.25(5) = 5.35\%$. Equity index: $0.30(6) + 0.30(3) + 0.20(-2) + 0.20(4) = 3.10\%$. Equity excess $2.25\%$.

**Step 2 – Sector allocation within equity**

$$\sum_s (w_{P,s} - w_{B,s})\, r_s = 0.10(6) - 0.05(3) - 0.10(-2) + 0.05(4)$$

$$= 0.60 - 0.15 + 0.20 + 0.20 = 0.85\%$$

$\Rightarrow$ The tilt towards technology and away from energy added 85 basis points inside equity.

**Step 3 – Security selection within sectors**

$2.25\% - 0.85\% = 1.40\%$. Check: $\sum_s w_{P,s}(r_{P,s} - r_s) = 0.40(2) + 0.25(1) + 0.10(1) + 0.25(1) = 1.40\%$.

**Step 4 – Contribution to the portfolio**

$1.40\% \times 0.60 = 0.84\%$, using the manager's equity weight.

**Interpretation**

At portfolio level, sector allocation contributes $0.85 \times 0.60 = 0.51\%$ and security selection $0.84\%$, together the equity component's $2.25 \times 0.60 = 1.35\%$. The bogey's 50% equity weight is a red herring here: it matters only for the asset allocation effect.

**Why the other options are wrong**

**A is incorrect.** Multiplied by the bogey's equity weight: $1.40 \times 0.50 = 0.70\%$.
**B is incorrect.** Used the manager's sector returns in the sector allocation effect: $0.10(8) - 0.05(4) - 0.10(-1) + 0.05(5) = 0.95\%$, leaving $2.25 - 0.95 = 1.30\%$, times $0.60 = 0.78\%$.
**D is incorrect.** Stopped at the equity-level figure, 1.40%, without multiplying by the equity weight.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5 Sector and Security Selection Decisions, Tables 8.8 and 8.9

**Solution IM-7 i.8** **Answer: D**

Workbook A18.5 • S4

**Step 1 – Bogey, portfolio and excess return**

$$r_B = 0.55(4.0\%) + 0.35(1.2\%) + 0.10(0.4\%) = 2.66\%$$

$$r_P = 0.65(5.2\%) + 0.20(1.6\%) + 0.15(0.4\%) = 3.76\%, \qquad r_P - r_B = 1.10\%$$

**Step 2 – Asset allocation contribution (active weight times index return)**

$$(0.10)(4.0\%) + (-0.15)(1.2\%) + (0.05)(0.4\%) = 0.40\% - 0.18\% + 0.02\% = 0.24\%$$

$\Rightarrow$ Overweighting equity in a month when the equity index returned 4% added 24 bp.

**Step 3 – Equity component: total selection, then split**

Equity excess $= 5.2\% - 4.0\% = 1.20\%$; security selection within sectors $= 1.20\% - 0.80\% = 0.40\%$.

**Step 4 – Contribution to the portfolio (multiply by the equity weight)**

$$0.40\% \times 0.65 = 0.26\%$$

$\Rightarrow$ Stock picking inside equity sectors added 26 bp to the whole portfolio.

**Step 5 – Check: the full decomposition**

Sector allocation in equity $0.80\% \times 0.65 = 0.52\%$; bond selection $0.20 \times (1.6\% - 1.2\%) = 0.08\%$; total $0.24 + 0.52 + 0.26 + 0.08 = 1.10\%$.

**Interpretation**

All decisions added value this month, with most of it coming from sector bets inside equity rather than from the asset mix or stock picking.

**Why the other options are wrong**

**A is incorrect.** Stopped at the equity-level figure $1.20\% - 0.80\% = 0.40\%$ without multiplying by the 65% equity weight (GARP multiplies 147 bp by 0.70).

**B is incorrect.** Intermediate result: took the whole equity selection effect, $1.20\% \times 0.65 = 0.78\%$, which still contains the sector allocation decision.

**C is incorrect.** Asset allocation computed with the manager's own returns instead of index returns: $0.10(5.2\%) - 0.15(1.6\%) + 0.05(0.4\%) = 0.30\%$, which mixes selection into allocation.

**LO:** IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.5 Performance Attribution Procedures (Tables 8.6 to 8.9)

IM-8 • Hedge Fund Investment Strategies • Vardeez: VERY IMPORTANT

# IM-8 a

P1 CONCEPT

Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: hedge funds came up twice in the August 2026 sitting, and Map #34 records the A24 section on 3 of 16 recall sheets. Vardeez rates the reading Very Important. **Vardeez:** equity-based and macro strategies (the third of his three IM-8 priorities, after convertible arbitrage and merger arbitrage).

**Reported exam shapes**

- **S1** Hedge fund strategies across the families (equity, macro, arbitrage, event-driven, distressed): four managers or four brochures, one correct description. Workbook A24.1 is the August item and is worked here, last. Its other options belong to IM-8 b, c and d, so Section 1 maps the whole reading.

**What the questions look like:** all conceptual. A mandate or a marketing claim is matched to a strategy, or four descriptions are tested and one is right. The wrong options swap a feature between siblings (the liquidity of global macro against emerging markets, the net exposure of long/short against nonhedged equity, who earns an interest rebate) or flip a polarity (markets tend to rise, so short sellers face a headwind).

**Source.** No GARP book covers IM-8. SchweserNotes 2026 Reading 90 (Stowell and Stowell, Ch.12) is the authority; Vardeez is checked against it.

## 1. The hedge fund edge and the strategy map

Hedge funds can do four things that long-only mutual funds generally cannot: **short sell**, **use leverage**, **use derivatives** and **take concentrated positions** when opportunities arise. Every strategy in the reading is one way of using that freedom. Stowell groups them into four families (Figure 1): equity-based and macro strategies (this LO), arbitrage strategies (IM-8 b), and event-driven strategies, which include merger arbitrage (IM-8 c) and distressed securities (IM-8 d).

Two questions sort any strategy, and the exam's wrong options are built by getting one of them wrong:

1. **What are the legs?** Long, short or both, and in what: related stocks, a convertible and its stock, a target and its acquirer, senior and junior claims.
2. **Where does the return come from?** Security selection (alpha), a macro view, the convergence of a mispricing, or a corporate event. The risk profile follows: whatever stops the return from arriving is the main risk.

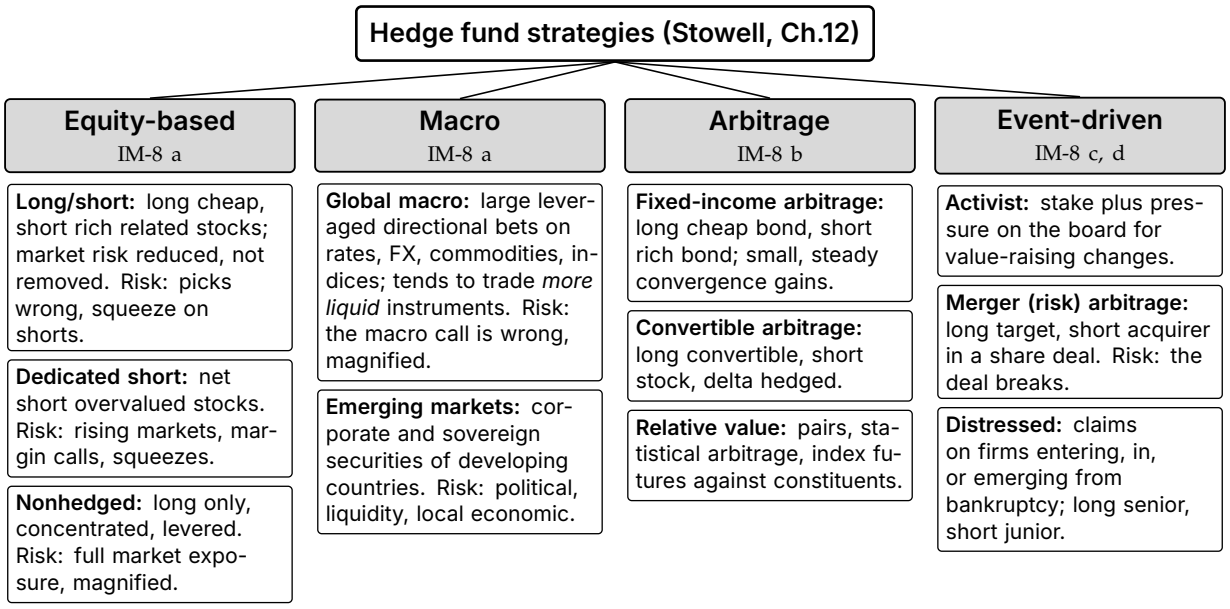



Figure 1. The strategy map of the reading. Grey boxes are the four families; each white box gives the legs and the main risk. The left two columns are this LO; the right two are IM-8 b to d.

## 2. Equity long/short

**Mechanics.** Fundamental analysis of a firm, its industry, its geography and the wider macroeconomic environment. The manager goes **long the undervalued and short the overvalued** among related securities, to exploit mispricings while hedging overall market risk. Done well, security selection earns alpha on *both* legs.

- **Investment horizon.** Each position has one: the period over which its valuation should converge to the forecast. Positions that revalue faster are worth more. Example: building long positions in likely takeover targets.
- **Shorts are on the shortest leash.** Rising prices can trigger costly margin calls or a short squeeze that forces the position closed at a large loss. Shorts are therefore diversified so that no single short exceeds **2% to 3% of fund capital**.
- **Market risk is reduced, not removed.** The primary goal is lower market risk; managers then use factor models to mitigate the factor exposures that remain beyond beta. A long/short fund still carries some directional and factor exposure.

| Sources of return | Costs |
|---|---|
| Alpha on the long and the short positions | Financing charges on margin loans |
| Interest rebate: interest on the cash proceeds of short sales, net of broker and securities-lender fees | Fees paid to borrow the shares sold short, and dividends owed on the borrowed shares |
| Interest on the liquidity buffer posted at the start to cover daily mark-to-market | Transaction costs from trading and rebalancing |

## 3. Dedicated short (short-biased)

- **The opportunity.** Mutual funds look for positive-alpha stocks. They rarely short; at most they *underweight* a negative-alpha stock. Overvalued securities therefore get little attention, which leaves room for short specialists. Their skill is spotting overvaluation: aggressive accounting, potential legal problems, management obscuring bad news.
- **The risks.** Margin calls, short squeezes and forced covering driven by risk limits.
- **The headwind.** Market prices tend to *rise* over time, so short positions carry a built-in bias toward negative returns. Without specialised expertise, a short-selling fund is likely to lose money while taking significant risk.

## 4. Nonhedged equity

- **Long only**, fundamental stock picking in individual firms or industries, much like a traditional stock picker. It keeps **full market exposure**: there is no short leg.
- **How it differs from a mutual fund.** A mutual fund diversifies and may ignore an idea whose expected return is too small to justify the analysis. A hedge fund can concentrate, analyse the upside *and* the downside of each name in depth, and use **leverage** to make a small mispricing worth having.
- **Bull call spread.** Buy a call with a lower strike and sell a call with a higher strike. The sold call cheapens the trade and caps the upside, so the position earns an enhanced return even if the stock rises by only 10% to 15%.

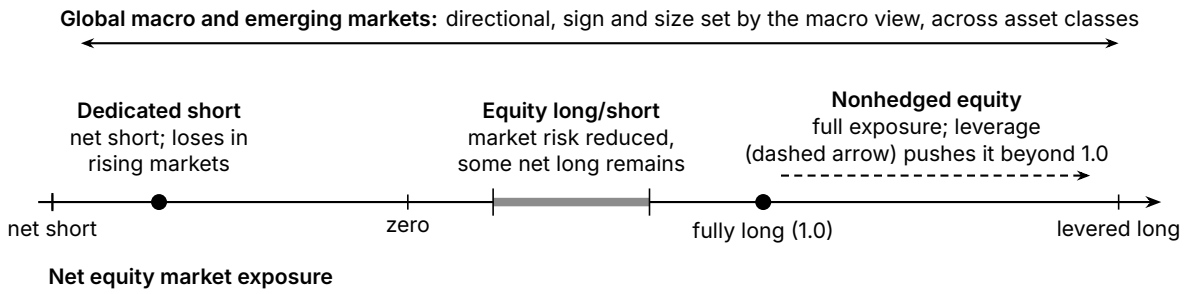



Figure 2. Where the equity strategies sit on net market exposure (schematic, not to scale). The grey band is long/short: the hedge lowers exposure but does not take it to zero. The double-headed arrow on top marks the macro strategies, whose exposure can sit anywhere, in any asset class.

## 5. Macro strategies

### Global macro

- **Large, leveraged, directional bets** on anticipated moves in broad equity or fixed-income markets, interest rates, foreign exchange and commodities, often through options, swaps and futures on a wide range of asset classes and indices.
- **Broad discretion.** A mutual fund is held to its mandated area; a global macro fund can move capital across sectors, regions and asset classes wherever it sees an opportunity.
- **Inputs to the view:** economic forecasts, international capital flows, interest rate trends, political developments, relations between countries and governments, economic policies.

- **Liquidity:** global macro funds tend to trade in **more liquid** instruments.
- **Return and risk.** The return is being right on the macro call, magnified by leverage; the risk is being wrong on it, magnified the same way.

### Emerging markets

- Invests in the securities of developing countries: **both corporate securities and sovereign debt**. These economies typically show rapid growth and ongoing industrialisation.
- Needs fundamental valuation skill *and* deep insight into each specific market.
- Seeks higher returns, against heightened risks unique to emerging markets: **political conditions, liquidity constraints and local economic factors**.

**The one structural difference between the two macro strategies is liquidity**: global macro leans to liquid instruments, emerging markets accepts liquidity constraints. Both are directional.

> **Correction to the source notes.** Vardeez writes that global macro "concentrates on the most liquid instruments in the market" while emerging markets "accepts some of the least liquid". Schweser's wording is comparative: global macro funds "tend to trade in more liquid instruments", and emerging markets strategies carry "liquidity constraints". Learn the comparative. An option claiming that global macro trades *only* the most liquid instruments is an absolute qualifier, not the reading's position.

**Comparison card: IM-8 a**

| | Positions | Return comes from | Main risks |
|---|---|---|---|
| **Long/short** | Long underpriced, short overpriced related stocks | Alpha on both legs; interest rebate; interest on the liquidity buffer | Stock picks wrong; margin calls and squeezes on shorts; residual factor exposure |
| **Dedicated short** | Net short overvalued stocks | Spotting overvaluation (accounting, legal red flags) | Rising markets (a built-in headwind); squeezes; forced covering |
| **Nonhedged** | Long only, concentrated, levered | Stock selection; leverage; bull call spreads | Full market exposure, magnified by leverage |
| **Global macro** | Directional, any asset class, derivatives | Correct macro calls, magnified by leverage | Wrong macro call; leverage. Trades more liquid instruments |
| **Emerging markets** | Corporate and sovereign securities of developing countries | Growth and industrialisation; local insight | Political, liquidity and local economic risks |

**Exam traps**

- **Liquidity swapped between the macro siblings.** Global macro trades *more* liquid instruments; emerging markets carries liquidity constraints.
- **"Market neutral" long/short.** The goal is to *reduce* market risk; some directional and factor exposure remains, which factor models then try to mitigate.

- **The headwind reversed.** Markets tend to rise over time. That is the problem for a dedicated short fund, not an advantage.
- **Interest rebate on a long-only fund.** The rebate is interest on short-sale proceeds, so only strategies with shorts earn it. Nonhedged equity has none.
- **Cost or return?** The rebate and the interest on the liquidity buffer are returns; borrow fees, dividends on borrowed shares and margin financing are costs.
- **Nonhedged equity as low risk.** It is long only with leverage: full market exposure, magnified.
- **Emerging markets as sovereign only.** The universe is corporate *and* sovereign securities.

## Practice questions: IM-8 a

### Question IM-8 a.1

Bank Q426 • BT P2.T9.410.1 (adapted)

A private wealth adviser is explaining to a family office client why it might allocate part of its equity budget to a hedge fund rather than to a mutual fund. Based on how hedge funds differ from long-only mutual funds, which of the following is the best argument in favour of the hedge fund?

**A.** The client wants to minimise the total fees charged, which reduce its net return.

**B.** The client values the investor protections that come from the heavier regulation of hedge funds.

**C.** The client wants a manager able to short sell, use leverage and derivatives, and concentrate positions when opportunities arise.

**D.** The client prefers a simple strategy with a high degree of transparency about positions.

### Question IM-8 a.2

Bank Q427 • Schweser Module Quiz 90.1 Q1

An operational due diligence analyst at a fund of hedge funds is reconciling the income of four managers. Which of the following hedge fund strategies is most likely to earn an interest rebate as a source of income?

**A.** Global macro

**B.** Equity long/short

**C.** Emerging markets

**D.** Nonhedged equity

**Question IM-8 a.3** Bank Q428 • Schweser Module Quiz 90.1 Q2

A pension fund's investment committee is reviewing a manager that runs a dedicated short (short-biased) strategy. Which of the following statements best describes the primary risk associated with this strategy?

**A.** A lack of leverage prevents meaningful alpha generation.

**B.** Short sale proceeds generate insufficient interest rebates to offset costs.

**C.** Market prices tend to decline over long periods, limiting upside potential.

**D.** Rising share prices can trigger margin calls and forced covering of positions.

**Question IM-8 a.4** Bank Q422 • BT P2.T8.412.1 (adapted)

An allocator receives a marketing presentation from a manager that describes itself as an equity long/short fund. The allocator is sceptical. Each of the following claims in the presentation is inconsistent with an equity long/short strategy EXCEPT:

**A.** "We let our highest-conviction short grow to 10% of fund capital, because concentration on the short side is where our alpha comes from."

**B.** "We are non-directional: after our hedges, the fund has virtually no exposure to the stock market."

**C.** "You might find us net long small-cap and emerging market stocks while short large-cap stocks."

**D.** "Because markets are efficient, we avoid the idiosyncratic risk taken by so-called stock pickers."

**Question IM-8 a.5** Bank Q429 • AnalystPrep Q.3033 (rebuilt)

A consultant is writing a one-paragraph profile of an emerging markets hedge fund for a client's investment committee. Which of the following statements best describes the strategy and its return and risk profile?

**A.** It invests in corporate and sovereign securities of developing countries, seeking higher returns but bearing added political, liquidity and local economic risks.

**B.** It invests only in the sovereign debt of developing countries, and its risk is limited because rapid economic growth removes liquidity and political risk.

**C.** It takes long and short positions in related developed-market equities to hedge market risk, so its returns come almost entirely from security selection.

**D.** It makes large leveraged directional bets across currencies, commodities and fixed income based on macroeconomic forecasts, trading mainly very liquid instruments.

**Question IM-8 a.6** Bank Q423 • AP Mock 3 Q8 (rebuilt)

A fund manager has recently started his own hedge fund. The fund's investors want it to take large, leveraged, discretionary positions across equity, fixed-income, currency and commodity markets, with a highly active asset allocation that bets on a diverse range of macroeconomic risk factors over time. Which of the following is the most suitable hedge fund style for this manager?

**A.** Dedicated short

**B.** Merger arbitrage

**C.** Nonhedged equity

**D.** Global macro

**Question IM-8 a.7** Workbook A24.1 • S1 • Aug-2026 • Map #34

A senior analyst in the alternatives team of a Dutch pension fund is presenting four hedge fund managers to the investment committee. Which of the following descriptions is correct?

**A.** Manager W runs fixed-income arbitrage: it buys a convertible bond and sells short the issuer's stock, rebalancing the hedge as the bond's delta changes in order to monetize volatility.

**B.** Manager X runs merger arbitrage on a share-for-share deal: it buys the target and sells short the acquirer, accepting that losses if the deal fails can far exceed gains if it closes.

**C.** Manager Y runs global macro: it makes large, leveraged directional bets across rates, currencies and commodities, and it tends to trade less liquid instruments than other strategies.

**D.** Manager Z runs passive distressed investing: it sits on creditor committees and litigates to steer the workout, accepting that the restructuring may take longer than expected.

## Solutions • IM-8 a

### Solution IM-8 a.1 — Answer: C

Bank Q426 • BT P2.T9.410.1 (adapted)

**Step 1 – Name what separates a hedge fund from a mutual fund**

The reading's starting point: hedge funds can short sell, use leverage, use derivatives and take concentrated positions when opportunities arise. Long-only mutual funds generally cannot.
⇒ The case for a hedge fund is access to a manager with the skill to use that flexibility.

**Step 2 – Test the other arguments against the hedge fund model**

Fees, regulation and transparency all point the other way: hedge funds charge management and performance fees, operate under lighter oversight than mutual funds (IM-9 a), and keep their strategies secret.

**Interpretation**

Every strategy in Figure 1 is a particular use of the four freedoms. When a stem asks why an investor would choose a hedge fund, the answer is the freedom and the skill to use it, never lower cost or more protection.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: hedge funds typically charge more (a management fee plus a performance fee), so fees argue against them.
**B is incorrect.** Polarity flip: hedge funds face significantly less regulatory oversight than open-end mutual funds, which is why investors must do their own due diligence.
**D is incorrect.** Polarity flip: hedge fund strategies are often complex and deliberately secretive; limited transparency is one of the incremental risks of hedge fund investing.

**LO:** IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, Exam Focus and LO 90.a)

### Solution IM-8 a.2 — Answer: B

Bank Q427 • Schweser Module Quiz 90.1 Q1

**Step 1 – Define the interest rebate**

The rebate is interest earned on the cash proceeds of short sales, net of broker and securities-lender fees.
⇒ Only a strategy that sells short generates the cash on which a rebate is paid.

**Step 2 – Find the strategy with a short book**

Equity long/short holds short positions as a defining leg, and Stowell lists the interest rebate among its sources of return, next to alpha on both legs and interest on the liquidity buffer.

**Interpretation**

The rebate is a return; the fees to borrow shares and the dividends owed on them are costs. A dedicated short fund would also earn a rebate, but it is not among the options.

**Why the other options are wrong**

**A is incorrect.** Global macro makes leveraged directional bets, mostly through derivatives; short-sale proceeds are not a defining income source.
**C is incorrect.** Emerging markets funds buy developing-country corporate and sovereign securities; a rebate is not part of their return profile.
**D is incorrect.** Nonhedged equity is long only, so it has no short-sale proceeds to earn a rebate

on.

**LO:** IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.a, Equity Long/Short Strategy)

### Solution IM-8 a.3 Answer: D

Bank Q428 • Schweser Module Quiz 90.1 Q2

#### Step 1 – What hurts a short position

A short loses when the price rises, and the loss has no upper bound. A rising price brings margin calls, can set off a short squeeze, and can force the manager to cover under risk limits, locking in the loss.

#### Step 2 – Add the structural headwind

Market prices tend to rise over time, so a dedicated short fund is fighting the drift of the market.
⇒ The primary risk is rising prices and the forced covering they cause.

#### Interpretation

This is why Stowell says that without specialised expertise in spotting overvaluation (aggressive accounting, legal issues, management hiding bad news), a short seller is likely to lose money while taking significant risk.

#### Why the other options are wrong

**A is incorrect.** Absolute claim: hedge funds can use leverage; a lack of it is not the strategy's defining risk.
**B is incorrect.** Sibling swap: the rebate is a source of return; its size is a cost question, not the primary risk.
**C is incorrect.** Polarity flip: prices tend to *rise* over long periods, which is exactly the headwind for short sellers.

**LO:** IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.a, Dedicated Short Strategy)

### Solution IM-8 a.4 Answer: C

Bank Q422 • BT P2.T8.412.1 (adapted)

#### Step 1 – Recall what long/short is designed to do

Long underpriced and short overpriced related stocks, earning alpha on both legs. The primary goal is to *reduce* market risk; factor models then mitigate the factor exposures that remain beyond beta.
⇒ Residual tilts are expected; a claim of zero exposure is not.

#### Step 2 – Test each claim

A breaks the risk rule for shorts (no single short above 2% to 3% of capital). B claims the market risk is gone. D denies the stock-picking that is the strategy's source of return. C describes a residual size and region tilt: long small-cap and emerging market names against large-cap shorts.
⇒ C is the only claim consistent with a stock picker who reduces but does not eliminate market and factor exposure.

#### Interpretation

The tilt in C is exactly the kind of exposure "beyond beta" that Stowell says managers try to control with factor models. Its presence is plausible; its size is what the allocator should ask

about.

### Why the other options are wrong

**A is incorrect.** Scope error: shorts are diversified so that no single position exceeds 2% to 3% of capital, because rising prices can force a short to be closed at a large loss.

**B is incorrect.** Absolute claim: the strategy reduces market risk; it does not remove it, so "virtually no exposure" overstates the hedge.

**D is incorrect.** Polarity flip: alpha comes from security selection, which means taking stock-specific risk on purpose; a manager who believed markets were efficient would have no reason to run the strategy.

**LO:** IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.a, Equity Long/Short Strategy)

## Solution IM-8 a.5 Answer: A

Bank Q429 • AnalystPrep Q.3033

### Step 1 – The universe

An emerging markets strategy invests in the securities of developing countries: both corporate securities and sovereign debt.

### Step 2 – The return and risk profile

The aim is higher returns from rapid growth and industrialisation. The price is a set of risks unique to emerging markets: political conditions, liquidity constraints and local economic factors. Managers need valuation skill and deep insight into each specific market.

$\Rightarrow$ A states the full universe and the matching risk profile.

### Interpretation

Keep the macro siblings apart on liquidity: global macro tends to trade more liquid instruments; emerging markets accepts liquidity constraints. Option D is a faithful description of the sibling.

### Why the other options are wrong

**B is incorrect.** Scope error and polarity flip: the universe includes corporate securities, and growth does not remove political or liquidity risk; those risks are heightened.

**C is incorrect.** True fact about the wrong strategy: this is equity long/short.

**D is incorrect.** True fact about the wrong strategy: this is global macro, which leans toward liquid instruments.

**LO:** IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.a, Emerging Markets Strategy)

## Solution IM-8 a.6 Answer: D

Bank Q423 • AP Mock 3 Q8

### Step 1 – Read the mandate

Multiple asset classes (equities, fixed income, currencies, commodities); large, leveraged, discretionary positions; active allocation across macroeconomic risk factors.

### Step 2 – Match it

Global macro makes large, leveraged bets on broad markets, rates, FX and commodities, uses derivatives on many asset classes, and has broad discretion to move capital wherever it sees opportunity.

⇒ Only global macro spans every market in the mandate.

### Interpretation

The key word is *across*: every other option is confined to one opportunity set. Global macro behaves like a highly active, leveraged asset allocator.

### Why the other options are wrong

**A is incorrect.** A dedicated short fund sells overvalued securities, mainly equities; it is not a multi-asset allocator.

**B is incorrect.** Merger arbitrage is event-driven and bets on one narrow risk, deal completion.

**C is incorrect.** Nonhedged equity is long-only stock picking; it cannot express views on rates, currencies or commodities.

**LO:** IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.a, Global Macro Strategy)

## Solution IM-8 a.7 Answer: B

Workbook A24.1 • S1 • Aug-2026

### Step 1 – Test each mechanic against the reading

Merger (risk) arbitrage: in a share-for-share merger the arbitrageur "may short the acquirer's stock while simultaneously purchasing the target's stock". If the deal fails, the target falls back to (or below) its pre-announcement price and the acquirer may rise, and "the losses from a failed deal can far exceed the gains earned when a deal is successfully completed".

⇒ B is correct.

### Interpretation

Each strategy is identified by its legs and its payoff shape. Merger arbitrage resembles a short put on deal completion: a small, likely gain against a large, less likely loss. That is why the hedge fund's edge is legal and antitrust expertise in judging the probability that the deal closes (IM-8 c).

### Why the other options are wrong

**A is incorrect.** Sibling swap: the mechanic described (long convertible, short stock, delta hedging, monetizing volatility) is convertible arbitrage. Fixed-income arbitrage trades yield divergences between similar bonds, such as on-the-run against off-the-run Treasuries (IM-8 b).

**C is incorrect.** Polarity flip: global macro funds "tend to trade in more liquid instruments".

**D is incorrect.** Role misassignment: sitting on creditor committees, supervising the workout and litigating describe *active* distressed investors; passive investors stay uninvolved and prefer less complex situations (IM-8 d).

**LO:** IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90)

IM-8 • Hedge Fund Investment Strategies • Vardeez: VERY IMPORTANT

# IM-8 b

P1 CONCEPT

Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic (A24, hedge funds twice in August 2026, Map #34 on 3 of 16 sheets), and convertible arbitrage is the first of Vardeez's three IM-8 priorities.
**Vardeez:** convertible arbitrage.

**Reported exam shapes**

- **S1** Hedge fund strategies across the families. The arbitrage version: name the strategy from its legs, or set the legs from the mispricing. Workbook A24.1 (worked as Question IM-8 a.7) uses this LO in its option A: long convertible, short stock, delta hedged, is convertible arbitrage, not fixed-income arbitrage.

**What the questions look like:** conceptual. Which legs, in which direction; which cash flows are returns and which are costs; what the delta hedge does when the stock moves; why a hedged convertible gains on a fall. The verb "describe the mechanics" means the trade, the rebalancing and the payoff asymmetry, not the definition of a convertible bond.

## 1. What the arbitrage strategies look for

An arbitrage strategy looks for one of three situations:

1. the **same asset** trades at different prices in different markets;
2. **two assets with similar cash flows** are priced differently;
3. an asset with a **known future price** trades today at a value that is not its risk-free discounted price.

The core trade is always the same: **buy the cheap asset, sell the rich one, profit when the prices converge**. The strategies differ in what is paired with what (Figure 1 of IM-8 a): bonds with bonds (fixed-income arbitrage), a convertible with its own stock (convertible arbitrage), or related equities and index products (relative value arbitrage).

## 2. Fixed-income arbitrage

A long/short approach to pricing inefficiencies among fixed-income securities: the yields of similar bonds should converge.

- **Cross-market.** Hedged yields on Italian government bonds temporarily diverge from US Treasury yields. If the Italian yield is too high (price too low), buy the hedged Italian bonds and short the Treasuries; profit as the yields converge.
- **On-the-run against off-the-run.** The newly issued (on-the-run) 30-year Treasury trades at a premium to the off-the-run 29½-year bond issued six months earlier. Short the expensive on-the-run bond, buy the cheaper off-the-run bond; as the on-the-run premium fades, the trade earns a small positive profit.

The profile is **small, steady convergence gains per trade**. Leverage is what makes those small gains worth having, which is why relative value funds carry the highest average leverage of any strategy (IM-9 a, Figure 1).

## 3. The convertible bond

A convertible bond contains an **embedded equity call option**: the holder may convert the bond into a fixed number of shares (the *conversion ratio*), and will do so when the value of the shares received exceeds the bond's cash redemption value. Until then it pays coupons like any bond.

| The price is sensitive to | Matching component of value |
|---|---|
| Changes in interest rates; the issuer's credit risk | **Investment value**: the value as a straight bond, a floor set by rates and credit |
| Movements in the issuer's share price | **Conversion value**: conversion ratio × share price |
| The share price's volatility | **Option value**: the value of the right to convert, above both floors |

In Schweser's example the conversion ratio is 21.2037 shares and the stock trades at \$41.54, so the conversion value is $21.2037 \times 41.54 = \$880.80$. The bond trades at 101.375% of par, \$1,013.75: the excess over both floors is the option value (Figure 3).

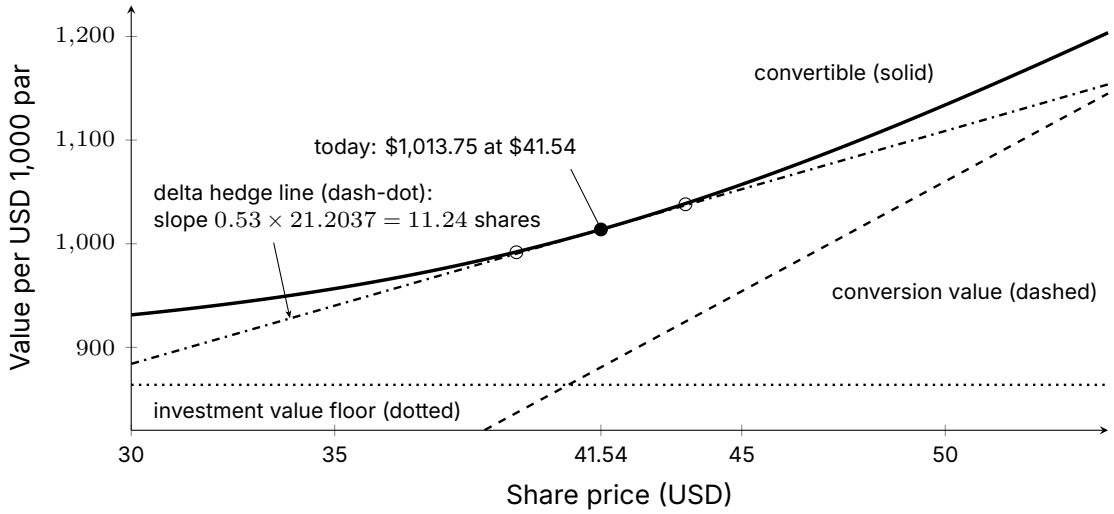



Figure 3. A stylised convertible price curve fitted to Schweser's example (price \$1,013.75 and delta 53% at \$41.54). The open circles are the pricing model's values after a 5% fall (\$991.78) and a 5% rise (\$1,038.07). The curve is convex and lies above the dash-dot hedge line on both sides, so the long convertible, short stock position gains whichever way the stock moves.

## 4. The convertible arbitrage trade

**Legs. Buy the convertible; sell short the issuer's stock.** The short is sized by the convertible's delta: $0.53 \times 21.2037 = 11.24$ shares per bond in the example. The arbitrageur aims to profit from the bond's income and from the stock's volatility.

**Rebalancing (delta hedging).** The hedge is designed to neutralise equity risk, but the convertible's delta changes with the share price, so the short must be resized continuously:

- **Stock rises** ⇒ delta rises ⇒ **short more shares** (sell after a rise).
- **Stock falls** ⇒ delta falls ⇒ **buy back shares** (buy after a fall).

Selling high and buying low, over and over, is how the hedge **monetizes volatility**: because the convertible's price is a convex (nonlinear) function of the share price, the long convertible outperforms its delta hedge in both directions (Figure 3).

**Where the money comes from, and what it costs**

| Sources of return | Costs |
|---|---|
| The hedge itself, monetizing volatility through convexity | Borrowing fees paid to the stock lender |
| The bond's coupon payments | Dividends that must be paid to the lender on the borrowed shares |
| Interest earned on the short sale proceeds | |
| "Cheapness": the convertible itself is undervalued, for example when the embedded call is priced below fair value | |

**Three sensitivities, one hedge.** The short stock offsets the share-price sensitivity only. The bond's exposure to interest rates and to the issuer's credit remains in the position, and the strategy usually runs with leverage, which made funding the main danger in 2007 to 2009 (Question IM-9 a.4).

## 5. The worked example: hedged against unhedged

Inputs: bond at \$1,013.75, stock at \$41.54, delta 53%, conversion ratio 21.2037, annual coupon \$2.50. A pricing model revalues the bond at \$1,038.07 if the stock rises 5% and \$991.78 if it falls 5% (the model values would be given on the exam).

| | Stock $-5\%$ | Stock $+5\%$ |
|---|---|---|
| Convertible price change | $-21.97$ | $+24.32$ |
| Coupon | $+2.50$ | $+2.50$ |
| **Unhedged (long convertible only)** | $\mathbf{-19.47}$ | $\mathbf{+26.82}$ |
| Short 11.24 shares: share move \$2.077 | $+23.34$ | $-23.34$ |
| **Hedged: price change + short, before carry** | $\mathbf{+1.37}$ | $\mathbf{+0.98}$ |

- **The hedge changes the sign on the downside**: a 5% fall goes from a \$19.47 loss to a \$1.37 gain. It does not merely shrink the loss.
- The price is a much smaller gain on the upside (\$0.98 against \$26.82). The unhedged position is a leveraged bet on the stock; the hedged one is a bet on movement.
- Both hedged outcomes are positive because the curve lies above the hedge line (Figure 3): that is "monetizing volatility" in numbers.

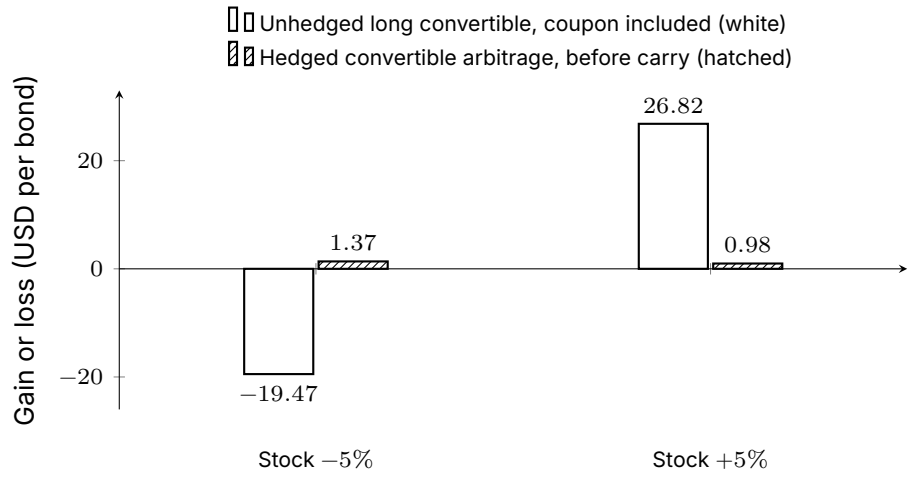



Figure 4. Schweser's convertible arbitrage example. The hedge turns the downside into a gain and gives up most of the upside. The \$1.37 is Schweser's figure; the \$0.98 is rebuilt from the stated inputs with the same short of 11.24 shares.

**Correction to the source notes.** (1) Schweser, and Vardeez after it, say that whether the stock rises or falls, "the gain or loss on the short stock position tends to exceed the corresponding loss or gain on the convertible". That holds only on the way down. On the way up the convertible's gain exceeds the short's loss: \$24.32 against \$23.34 in the example. Convexity means the long convertible *outperforms* its hedge in both directions; that is what makes the hedged position gain either way. (2) Schweser's unhedged figures include the \$2.50 coupon but its hedged \$1.37 does not ($-21.97 + 23.34 = 1.37$). Like for like, with the coupon in both, the hedged outcomes are $+\$3.87$ (fall) and $+\$3.48$ (rise). The conclusion, a downside loss turned into a gain, is unchanged.

## 6. Relative value arbitrage

Relative value arbitrage looks for pricing inefficiencies across related assets and asset classes.

- **Pairs trading.** Two peers in the same industry whose prices have historically moved closely together diverge with no fundamental news. Buy the undervalued one, short the overvalued one, and profit when the spread reverts to its prior relationship.
- **Statistical arbitrage.** The pairs idea across a large number of long and short positions, using factor models to find temporary mispricings. One source of return is **liquidity provision**: when an institution sells a large block, the price may fall temporarily; the fund buys the depressed stock, shorts a related stock to hedge market risk, and holds until the price normalises.
- **Index arbitrage.** Index futures against their constituents when the two are mispriced, for example long S&P 500 futures and short the constituents. Shorting all 500 stocks is impractical, so algorithms weigh transaction and other costs against the mispricing and pick an **optimal subset**. Firms that already hold the constituents elsewhere can source the shares internally at lower cost, a competitive advantage.

**Comparison card: IM-8 b**

| | Long | Short | Earns when / main risk |
|---|---|---|---|
| **Fixed-income arbitrage** | The cheap bond (higher yield; off-the-run) | The rich bond (on-the-run; Treasuries) | Yields converge; small, steady gains that need leverage |
| **Convertible arbitrage** | The convertible bond | The issuer's stock, delta × conversion ratio | Volatility (rebalancing), coupon, short interest, cheapness; rates, credit and funding risk remain |
| **Pairs trading** | The undervalued peer | The overvalued peer | The spread reverts to its historical relationship |
| **Statistical arbitrage** | Many temporarily cheap stocks | Many related stocks | Temporary mispricings close, including after block sales |
| **Index arbitrage** | The cheap side (e.g. futures) | The rich side (a subset of constituents) | Futures and constituents realign; internal share sourcing lowers cost |

**Exam traps**

- **Legs reversed.** Convertible arbitrage is long the convertible, short the stock. Fixed-income arbitrage shorts the rich bond (on-the-run), not the cheap one.
- **Strategy label swapped.** Long convertible, short stock is convertible arbitrage, not fixed-income arbitrage; pairs trading is relative value arbitrage.
- **Rebalancing direction.** After a rise, delta is higher, so short *more* shares; after a fall, buy shares back. Reversing it buys high and sells low.
- **The hedge only shrinks the loss.** It turns the 5% downside into a gain ($19.47 loss to $1.37 gain).
- **Coupon dropped.** The unhedged downside loss is $19.47 because the $2.50 coupon is added back to the $21.97 price fall.
- **Cost or return?** Borrow fees and dividends on the borrowed shares are costs; coupons and the short-proceeds interest are returns.
- **Fully hedged.** The short stock hedges equity risk only; rate, credit and funding risk stay in the trade.
- **Value components as a sum.** Investment value and conversion value are two alternative floors; the option value is the premium above the higher one. They are not three amounts to add.
- **Index arbitrage trades every constituent.** Algorithms pick an optimal subset.

## Practice questions: IM-8 b

### Question IM-8 b.1

Bank Q434 • Schweser Module Quiz 90.2 Q2

A hedge fund analyst buys one regional bank's stock and sells short a peer bank whose share price has historically moved closely with it, after the two prices diverged with no fundamental news. This pairs trade is best classified as which type of arbitrage?

**A.** Merger arbitrage

**B.** Convertible arbitrage

**C.** Fixed-income arbitrage

**D.** Relative value arbitrage

### Question IM-8 b.2

Bank Q431 • AnalystPrep Q.2576 (rebuilt)

The on-the-run 30-year Treasury bond trades at a premium to the off-the-run 29½-year Treasury issued six months earlier. Which trade would a fixed-income arbitrageur most likely establish, and from what source does it profit?

**A.** Buy the on-the-run bond and short the off-the-run bond; profit as the premium widens.

**B.** Short the on-the-run bond and buy the off-the-run bond; profit as the on-the-run premium fades.

**C.** Buy both bonds; profit from the higher coupon income.

**D.** Short both bonds; profit from a rise in Treasury yields.

### Question IM-8 b.3

Bank Q433 • Schweser Module Quiz 90.2 Q1

A consultant is writing strategy definitions for a fund of hedge funds' manager database. Which of the following characteristics best defines a convertible arbitrage strategy?

**A.** Buying undervalued sovereign bonds and shorting corporate bonds.

**B.** Taking offsetting long and short positions across different yield curves.

**C.** Buying a convertible bond and shorting the underlying stock to hedge equity exposure.

**D.** Buying stocks with high dividend yields while shorting stocks with relatively low dividend yields.

### Question IM-8 b.4

Bank Q432 • AnalystPrep Q.2577

A company has issued bonds with an option to convert them into equity after three years. Because of an error in the conversion factor, the convertible bonds appear undervalued when compared with the projected market price of the company's stock at the time of conversion. A hedge fund manager wants to exploit this mispricing. The most appropriate strategy is:

**A.** Going long the convertible security and short the common stock of the company.

**B.** Going short the convertible security and long the common stock of the company.

**C.** Going short the convertible security and short the common stock of the company.

**D.** Going long the common stock and long the convertible security.

### Question IM-8 b.5

New • built from SchweserNotes 2026 Reading 90

A convertible arbitrage fund buys USD 50 million of a convertible bond and sells short the issuer's stock against it. The fund's controller is classifying the expected cash flows of the position for the next quarter. Which of the following is correctly classified as a cost of the strategy rather than a source of return?

**A.** The coupons received on the convertible bond.

**B.** The interest earned on the cash proceeds of the short sale.

**C.** The gains from rebalancing the delta hedge as the share price moves.

**D.** The dividends paid to the stock lender on the borrowed shares.

### Question IM-8 b.6

New • built from SchweserNotes 2026 Reading 90

A convertible arbitrageur is long a convertible bond with a delta of 53% and short the corresponding number of the issuer's shares. Over the next week the share price rises 8% and the bond's delta rises to 61%. Which action keeps the position delta neutral, and what does the pattern of such adjustments achieve over time?

**A.** Buy back some shares, because the higher delta means the short is now too large; the adjustments lock in the bond's coupon.

**B.** Sell more shares short; adjustments of this kind sell stock after rises and buy it back after falls, which monetizes volatility.

**C.** Leave the short unchanged, because the hedge ratio is fixed by the conversion ratio at inception; the profit comes from the coupon alone.

**D.** Sell more shares short; adjustments of this kind buy stock after rises and sell it after falls, which earns the interest rebate.

**Question IM-8 b.7** New • built from SchweserNotes 2026 Reading 90

A portfolio manager compares two ways of holding a convertible bond bought at USD 1,013.75 per USD 1,000 par when the issuer's stock trades at USD 41.54. The bond's delta is 53%, its conversion ratio is 21.2037 shares and its annual coupon is USD 2.50. The desk's pricing model gives:

| Share price move | $-5\%$ | $+5\%$ |
|---|---|---|
| Convertible price (USD) | 991.78 | 1,038.07 |

Strategy 1 holds the bond unhedged. Strategy 2 also sells short the delta-equivalent number of shares (about 11.24 per bond). Which of the following statements is correct?

**A.** Strategy 2 shrinks both the gain on a rise and the loss on a fall, but it still loses money when the stock falls 5%.

**B.** Strategy 2 removes all risk from the position, including the bond's interest rate and credit risk.

**C.** Strategy 2 gains whether the stock rises or falls 5%, because the convertible's price is convex in the share price and lies above the hedge line.

**D.** Strategy 2 gains more than Strategy 1 when the stock rises 5%, because the short position adds to the bond's gain.

**Question IM-8 b.8** New • built from SchweserNotes 2026 Reading 90

A risk analyst at a multi-strategy fund is documenting the firm's relative value arbitrage book. Which of the following descriptions is correct?

**A.** Statistical arbitrage can earn a return by providing liquidity: when an institution sells a large block, the fund buys the temporarily depressed stock and shorts a related stock until prices normalise.

**B.** Index arbitrage requires trading every constituent of the index against the futures contract, so firms that already hold the constituents have no cost advantage.

**C.** Pairs trading buys the stock that has risen and shorts the stock that has fallen, betting that the divergence between the two peers will continue.

**D.** Fixed-income arbitrage earns large returns per trade by taking outright directional positions on the level of interest rates.

## Solutions • IM-8 b

### Solution IM-8 b.1 — Answer: D

Bank Q434 • Schweser Module Quiz 90.2 Q2

**Step 1 – Read the legs**

Long one bank, short a historically correlated peer, after a divergence with no news. The bet is that the spread reverts.

**Step 2 – Classify**

Pairs trading is the long-established relative value arbitrage strategy; statistical arbitrage extends it across many positions.

$\Rightarrow$ Relative value arbitrage.

**Interpretation**

Arbitrage strategies are told apart by what is paired with what: peer equities (relative value), a convertible and its own stock (convertible arbitrage), similar bonds (fixed-income arbitrage). Merger arbitrage pairs a target with its acquirer and is event-driven.

**Why the other options are wrong**

**A is incorrect.** Sibling swap: merger arbitrage needs an announced deal and pairs the target with the acquirer.

**B is incorrect.** Sibling swap: convertible arbitrage pairs a convertible bond with the issuer's stock.

**C is incorrect.** Sibling swap: fixed-income arbitrage pairs bonds, not peer equities.

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Relative Value Arbitrage)

### Solution IM-8 b.2 — Answer: B

Bank Q431 • AnalystPrep Q.2576

**Step 1 – Identify the rich and the cheap bond**

The on-the-run bond trades at a premium, so it is the rich one; the off-the-run bond, six months older and almost identical in cash flows, is the cheap one.

**Step 2 – Set the legs and the source of profit**

Short the rich on-the-run bond, buy the cheap off-the-run bond. As the on-the-run issue ages, its premium fades and the two prices converge.

$\Rightarrow$ A small, hedged, positive convergence profit.

**Interpretation**

This is arbitrage situation 2 of Section 1: two assets with similar cash flows priced differently. The gain per trade is small, which is why fixed-income arbitrage depends on leverage.

**Why the other options are wrong**

**A is incorrect.** Legs reversed: buying the rich bond and shorting the cheap one bets on the premium widening, against the expected fade.

**C is incorrect.** A long-only position in both bonds is a duration bet, not an arbitrage, and the coupons are nearly identical.

**D is incorrect.** A short in both bonds is a directional bet on rising yields, not a relative value trade.

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strate-

gies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Fixed-Income Arbitrage)

### Solution IM-8 b.3 — Answer: C

Bank Q433 • Schweser Module Quiz 90.2 Q1

#### Step 1 – The defining legs

Convertible arbitrage buys a convertible bond and simultaneously sells short the issuer's stock.

#### Step 2 – The purpose of the short

The short stock offsets the equity (delta) exposure of the embedded call, and it is resized as the delta changes. What remains is the bond's income and the ability to monetize volatility.
$\Rightarrow$ C.

#### Interpretation

The other three options are real long/short trades, each in a different market. None involves a convertible, which is the one ingredient the strategy cannot do without.

#### Why the other options are wrong

**A is incorrect.** True fact about the wrong strategy: a sovereign against corporate bond trade is a credit spread (fixed-income) position.
**B is incorrect.** True fact about the wrong strategy: offsetting positions across yield curves is fixed-income arbitrage.
**D is incorrect.** True fact about the wrong strategy: a dividend-yield long/short is an equity long/short tilt.

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Convertible Arbitrage)

### Solution IM-8 b.4 — Answer: A

Bank Q432 • AnalystPrep Q.2577

#### Step 1 – Buy what is cheap

The convertible is undervalued, so the arbitrageur buys it. This is the "cheapness" source of return: the embedded call is priced below its fair value.

#### Step 2 – Hedge what you do not want

The convertible carries equity exposure through its embedded call. Selling the issuer's stock short neutralises that exposure, leaving the mispricing (and the income) as the position's return.
$\Rightarrow$ Long the convertible, short the stock: convertible arbitrage.

#### Interpretation

The question is the mechanics in miniature: the long leg captures the mispricing, the short leg removes the share-price risk. Without the short, the trade would be a directional bet on the stock.

#### Why the other options are wrong

**B is incorrect.** Legs reversed: shorting the undervalued convertible loses as its price corrects, and the long stock doubles down on equity risk instead of hedging it.
**C is incorrect.** Shorting the cheap convertible bets against the mispricing; a short stock does not hedge a short convertible, it adds to the exposure.
**D is incorrect.** Long both legs doubles the equity exposure; nothing is hedged.

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Convertible

Arbitrage)

### Solution IM-8 b.5 Answer: D

New • built from SchweserNotes 2026 Reading 90

#### Step 1 – List the sources of return

The hedge itself (monetizing volatility through rebalancing), the bond's coupons, the interest earned on the short sale proceeds, and any cheapness in the convertible.

#### Step 2 – List the costs

Borrowing fees paid to the stock lender, and the dividends that must be paid to the lender on the borrowed shares.

⇒ The dividends owed on the borrowed shares are a cost.

#### Interpretation

A short seller owes the lender every dividend the shares pay while they are on loan. A high-dividend issuer makes the short leg expensive to carry, which cuts into the coupon and rebate income.

#### Why the other options are wrong

**A is incorrect.** Sibling swap: coupons are income earned by the long bond leg.

**B is incorrect.** Sibling swap: the interest on short proceeds (the rebate) is a source of return.

**C is incorrect.** Sibling swap: rebalancing gains are how the strategy monetizes volatility; they are the hedge's return, not a cost.

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Convertible Arbitrage)

### Solution IM-8 b.6 Answer: B

New • built from SchweserNotes 2026 Reading 90

#### Step 1 – Which way must the hedge move?

The number of shares to short is delta times the conversion ratio. Delta has risen from 53% to 61%, so the hedge needs more shares: sell more short.

⇒ After a rise, the arbitrageur sells stock.

#### Step 2 – What the pattern does

After a fall, delta drops and the arbitrageur buys shares back. Repeatedly selling after rises and buying after falls is buying low and selling high: the rebalancing harvests the stock's movement. Because the convertible's price is convex in the share price, the hedged position gains on moves in either direction.

⇒ This is how the hedge monetizes volatility.

#### Interpretation

The hedge ratio is a moving target precisely because the embedded call is nonlinear. A fixed hedge would leave the fund long equity after rises and short equity after falls.

#### Why the other options are wrong

**A is incorrect.** Direction reversed: a higher delta needs a larger short, not a smaller one, and coupons are earned whether or not the hedge is adjusted.

**C is incorrect.** The hedge ratio depends on delta, which moves with the share price; continuous rebalancing is the defining mechanic.

**D is incorrect.** Right action, wrong mechanism: the adjustments sell after rises and buy after

falls (not the reverse), and their payoff is the volatility gain, not the rebate.

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Convertible Arbitrage)

### Solution IM-8 b.7 — Answer: C

New • built from SchweserNotes 2026 Reading 90

**Step 1 – Strategy 1, unhedged**

The bond's price moves by $-21.97$ or $+24.32$; with the \$2.50 coupon the outcomes are $-\$19.47$ and $+\$26.82$. It is a leveraged bet on the stock.

**Step 2 – Strategy 2, hedged**

The short of 11.24 shares gains or loses $11.24 \times 2.077 = \$23.34$. On a fall the bond loses \$21.97 and the short gains \$23.34: $+\$1.37$. On a rise the bond gains \$24.32 and the short loses \$23.34: $+\$0.98$.

$\Rightarrow$ The hedged position gains in both cases (before carry).

**Step 3 – Why**

The convertible's price curve is convex: it gains more than the hedge loses on a rise and loses less than the hedge gains on a fall (Figure 3).

**Interpretation**

The hedge does not merely shrink the downside; it changes its sign, and it gives up most of the upside to do so. This is Schweser's example (Figure 4).

**Why the other options are wrong**

**A is incorrect.** Sign error: the hedged position *gains* \$1.37 on a 5% fall; it does not merely lose less.

**B is incorrect.** Absolute claim: the short stock hedges share-price risk only; the bond's interest rate and credit sensitivities remain.

**D is incorrect.** Polarity flip: on a rise the short position loses \$23.34, so Strategy 2 gains far less than Strategy 1 (\$0.98 against \$26.82).

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Example: Convertible arbitrage)

### Solution IM-8 b.8 — Answer: A

New • built from SchweserNotes 2026 Reading 90

**Step 1 – Statistical arbitrage**

Stat arb spreads the pairs idea over many long and short positions using factor models. Schweser names liquidity provision as one source of return: after a large block sale depresses a price, the fund buys it, shorts a related stock to hedge market risk, and holds until the price normalises.

$\Rightarrow$ A is correct.

**Step 2 – Check the other descriptions**

Index arbitrage trades an optimal subset of constituents chosen by algorithm, and firms that already hold the shares source them internally at lower cost. Pairs trading buys the undervalued peer and shorts the overvalued one, betting on reversion. Fixed-income arbitrage is a relative value trade with small, steady gains.

**Interpretation**

Every arbitrage strategy bets on convergence. An option that bets on a divergence continuing, or that takes an outright directional view, has left the arbitrage family.

### Why the other options are wrong

**B is incorrect.** Absolute claim and polarity flip: trading all constituents is impractical, so a subset is used, and internal sourcing is a competitive advantage.

**C is incorrect.** Polarity flip: pairs trading buys the laggard (undervalued) stock and shorts the leader (overvalued), expecting convergence, not continued divergence.

**D is incorrect.** Polarity flip: fixed-income arbitrage earns small, consistent profits from convergence between similar bonds; it is not an outright bet on rates.

**LO:** IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.b, Relative Value Arbitrage)

IM-8 • Hedge Fund Investment Strategies • Vardeez: VERY IMPORTANT

# IM-8 c

P1 CONCEPT

Describe the mechanics and risks of event-driven strategies such as merger arbitrage.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic (A24, hedge funds twice in August 2026, Map #34 on 3 of 16 sheets), and merger arbitrage is the second of Vardeez's three IM-8 priorities.
**Vardeez:** merger arbitrage.

**Reported exam shapes**

- **S1** Hedge fund strategies across the families. The event-driven version: the legs of a merger arbitrage and what happens if the deal closes or breaks. The correct answer of Workbook A24.1 (worked as Question IM-8 a.7) is this LO: long the target, short the acquirer, with losses on a failed deal that can far exceed the gains on a completed one.

**What the questions look like:** conceptual. Name the strategy from the event; set the legs; say who wins and who loses if the deal fails; place an idea in the right event category; explain why hedge funds, not mutual funds, run the trade.
**TYPE note.** The annualised expected-return formula is in Section 5 because it shows how the arbitrageur judges the bet. Under the CONCEPT tag no question asks you to compute it, so the bank's numerical item (Schweser Module Quiz 90.3 Q1, Bank Q440) is left out. Know what each input means.

## 1. What counts as an event

Event-driven strategies focus on specific corporate events that create pricing discrepancies and market inefficiencies. Stowell sorts the opportunities into five categories:

| Category | Opportunities |
|---|---|
| **Strategic (hard catalysts)** | Risk arbitrage; evaluating a firm's strategic alternatives and taking a position; potential spin-offs of divisions or business units; likely takeover targets; proxy contests and activist campaigns; **stub trades**, which isolate the discount between a holding company's share price and the value of its underlying assets |
| **Financial** | Price moves around credit re-ratings or liquidity events; changes in accounting methods; financial recapitalisations; primary offerings of debt or equity; bankruptcy reorganisations |
| **Operational** | Synergies from mergers; corporate turnarounds or restructurings; senior management changes |
| **Legal / regulatory** | Litigation outcomes; regulatory changes; anticipated legislation |
| **Technical** | **Broken risk arbitrage** situations; secondary offerings of equity or equity-linked notes |

Note the pair that separates two rows: risk arbitrage on a live deal is *strategic*; a deal that has collapsed (broken risk arbitrage) is a *technical* opportunity.

## 2. Activist strategy

- **Mechanics.** Take an equity or derivative position in a company, then seek to influence management or the board to pursue initiatives the activist believes will raise the share price. Activists may solicit proxy votes to pressure the board.
- **What they push for:** changes in corporate governance, spin-offs of business units, cost reductions, share repurchases, higher dividends, increased leverage, or other value-raising actions.
- **Success and failure.** Success needs two things: management adopts the proposals *and* the share price responds. Failure is a campaign that produces no meaningful change.

## 3. Merger (risk) arbitrage: the trade and its risks

**Goal.** Profit from the spread between the acquirer's offer and the target's share price after an acquisition or merger is announced.

- **Legs.** Buy (long) the target's stock. In a **share-for-share** deal, also short the acquirer's stock, at the exchange ratio; the position is unwound when the merger completes. In a **cash** deal (a tender offer at a fixed price) there is no acquirer stock to deliver, so the trade is the long target position.
- **If the deal closes**, the arbitrageur can earn: the arbitrage spread (the target's price at the announcement against the final bid price at closing); dividends on the target's stock; interest on the short-sale proceeds, net of borrowing costs and any dividends owed on the shorted acquirer stock; and a higher bid, especially if competing bidders emerge.
- **If the deal fails**, the target's price typically falls back to its pre-announcement level, or lower, and the acquirer's price may rise, creating a loss on the short. **Both legs lose at once.**
- **The asymmetry.** The losses from a failed deal can far exceed the gains from a completed one. A small, likely gain against a large, less likely loss is the payoff of a seller of insurance, a short put on deal completion (Figure 1).

**Why hedge funds run it.** Long-only investors such as mutual funds are not attracted: the potential gain (the acquisition price less the post-announcement price) is small, and the deal may not close. Hedge funds can apply **legal and antitrust expertise** to judge the probability of completion, and they can use the **short leg** to generate incremental return.

## 4. Worked example: Arielle Way and Cockrell Industries

Arielle Way (AWI) offers **two AWI shares for each Cockrell (CI) share**. Before the announcement AWI traded at \$52 and CI at \$74, so the offer was worth $2 \times 52 = \$104$, a premium of $104/74 - 1 = 40.5\%$. Neither pays dividends; the deal should close within two months. After the announcement AWI trades at \$50 and CI at \$95.

- **The trade.** Buy 100 CI at \$95 (cost \$9,500); sell short 200 AWI at \$50 (proceeds \$10,000). The short is set by the exchange ratio: 2 AWI for each CI share bought.
- **Deal closes.** The 100 CI shares become 200 AWI shares, which are delivered to close the short. Profit $= 10{,}000 - 9{,}500 = \$500$, or **\$5 per CI share** $(2 \times 50 - 95)$. Because the shares received exactly cover the short, the \$5 is locked in whatever AWI trades at on closing. A competing bid could raise CI and lower AWI, adding to the profit.
- **Deal breaks (illustration).** Suppose both prices return to their pre-announcement levels: CI falls \$21 to \$74 and AWI rises \$2 to \$52. The long loses $100 \times 21 = \$2{,}100$ and the short loses $200 \times 2 = \$400$: $-\$25$ **per CI share**, five times the gain if the deal closes.

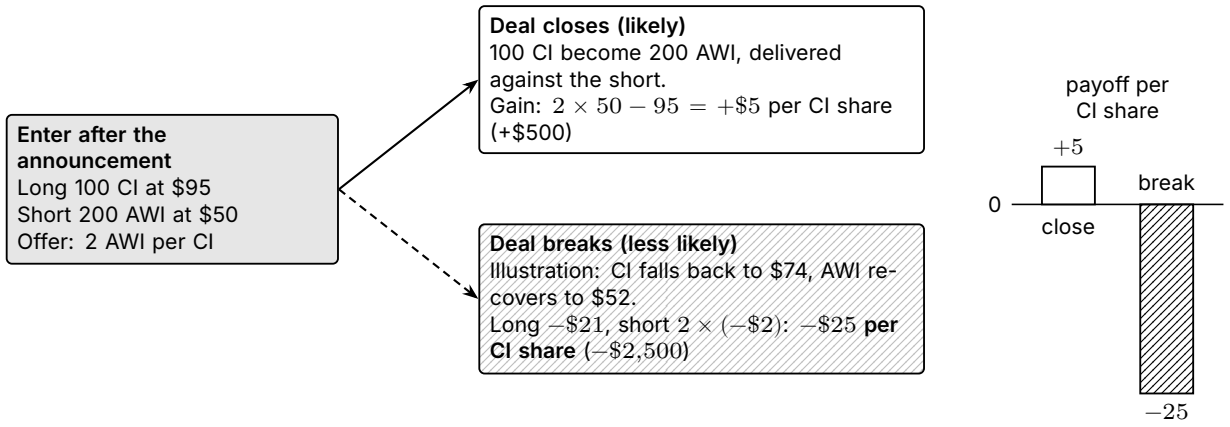



Figure 1. The AWI and CI merger arbitrage. Solid arrow and white bar: the deal closes. Dashed arrow and hatched box and bar: the deal breaks, with both prices assumed to return to their pre-announcement levels. A small, likely gain against a large, unlikely loss is the payoff of selling insurance.

## 5. How the arbitrageur judges the bet

Schweser gives the annualised expected return on a merger arbitrage position:

$$\text{annualised expected return} = \frac{C \times G - (1 - C) \times L}{Y \times P}$$

where $C$ is the probability that the deal succeeds, $G$ the gain if it does (offer price less current price), $L$ the loss if it fails (current price less the **pre-announcement** price, not a fall to zero), $Y$ the expected time to closing in years, and $P$ the **current** price (not the offer or the pre-announcement price).

**Example (cash tender offer).** Global Semiconductors offers $12 a share for Local Semiconductors, which traded at $9 and rose to $10.50 on the news. Closing is expected in six months with a 90% probability: $C = 0.90$, $G = 1.50$, $L = 1.50$, $Y = 0.5$, $P = 10.50$.

$$\frac{0.90(1.50) - 0.10(1.50)}{0.5 \times 10.50} = \frac{1.20}{5.25} = 22.86\%$$

The formula makes the mechanics explicit: the arbitrageur is paid for being right about $C$, which is where the legal and antitrust expertise comes in, and a modest fall in $C$ can turn the expected return negative because $L$ is usually several times $G$.

**Comparison card: IM-8 c**

| | Position | Earns when | Main risk and the hedge fund's edge |
|---|---|---|---|
| **Activist** | Equity or derivative stake; proxy votes to pressure the board | Management adopts the initiatives and the price responds | No meaningful change. Edge: willingness to engage and campaign |
| **Merger arbitrage** | Long target; short acquirer at the exchange ratio in share deals | The deal closes: spread, target dividends, short interest, a higher bid | The deal breaks and both legs lose; losses far exceed gains. Edge: legal and antitrust judgement of completion |
| **Distressed (IM-8 d)** | Claims on firms entering, in or emerging from bankruptcy; long senior, short junior | Recovery exceeds the price paid; capital-structure mispricing corrects | Valuation, illiquidity, a longer process, legal outcomes. Edge: restructuring and legal expertise |

**Exam traps**

- **Legs reversed.** Merger arbitrage is long the *target*, short the *acquirer*. Shorting the target is a bet that the deal fails.
- **Short leg in a cash deal.** The acquirer short exists only in share-for-share deals, sized by the exchange ratio; a cash tender offer is just the long target position.
- **Asymmetry reversed.** Small likely gain, large unlikely loss: selling insurance, not buying it.
- **"The short hedges a break."** On a break the target falls *and* the acquirer may rise: both legs lose.
- **Category swap.** Stub trades are strategic; broken risk arbitrage is technical; credit re-ratings are financial; merger synergies are operational.
- **Formula inputs.** $L$ runs back to the pre-announcement price; $P$ is the current price.
- **Activism succeeds when proposed.** It succeeds only if management adopts the changes and the share price responds.

## Practice questions: IM-8 c

### Question IM-8 c.1

Bank Q437 • AnalystPrep Q.2760

A consultant is sorting the managers on a fund of hedge funds' approved list into strategy buckets. Which of the following hedge fund strategies can be described as event-driven?

**A.** Trend followers

**B.** Equity long/short

**C.** Global macro

**D.** Distressed securities

### Question IM-8 c.2

New • built from SchweserNotes 2026 Reading 90

An event-driven portfolio manager files each new idea under one of the five categories of event-driven opportunity in Stowell's framework. Which of the following pairings is correct?

**A.** A position taken after a merger agreement collapses, exploiting the dislocation in the target's price (a broken risk arbitrage situation): technical opportunity.

**B.** A stub trade that isolates the discount between a holding company's share price and the value of its underlying assets: technical opportunity.

**C.** A position taken ahead of an anticipated credit rating change: operational opportunity.

**D.** A position based on the synergies expected from a merger: legal/regulatory opportunity.

### Question IM-8 c.3

Bank Q436 • AnalystPrep Q.2574 (rebuilt)

Quasar Hedge Fund (QHF) wants an event-driven strategy that profits from the price discrepancy between the current price of a target firm's shares and the final purchase price offered by the acquirer after a deal is announced. Management accepts the risk that a deal may be cancelled or modified, but does not want a strategy that is mainly a bet on interest rate movements or one that requires deep expertise in distressed securities. Which hedge fund strategy should QHF adopt?

**A.** Convertible arbitrage

**B.** Merger arbitrage

**C.** Fixed-income arbitrage

**D.** Distressed securities

**Question IM-8 c.4** New • built from SchweserNotes 2026 Reading 90

An activist hedge fund has built a 6% stake in a conglomerate and is campaigning for the spin-off of a division and a share buyback. Which of the following best describes the strategy and how it succeeds or fails?

**A.** The fund profits from the spread between an announced offer price and the target's current price, and loses if the deal breaks.

**B.** The fund must stay passive, because only creditors sitting on a creditor committee may seek to influence management.

**C.** The fund's return comes mainly from the interest rebate on its short sale proceeds, so the campaign itself is a hedge.

**D.** The fund seeks to influence the board, for example by soliciting proxy votes; it succeeds if management adopts the initiatives and the share price responds, and fails if no meaningful change is made.

**Question IM-8 c.5** Bank Q435 • Mock 4 Q73 (rebuilt)

A research analyst is preparing a report on hedge fund strategies. Which of the following statements would be correct for the analyst to include?

**A.** Global macro funds are backward looking and make small, unleveraged bets on a single market.

**B.** In a merger arbitrage strategy, the target's stock is sold short in the expectation that the deal fails.

**C.** In a merger arbitrage strategy, the arbitrageur buys the target's stock (and may short the acquirer's stock), profiting from the spread if the deal closes.

**D.** An equity long/short strategy aims to maximise market (beta) exposure rather than to exploit security mispricings.

**Question IM-8 c.6** Bank Q439 • Mock 1 Q33

A general partner of Quicksilver Capital Partners, an up-and-coming hedge fund based in Geneva, identifies Diamond Technologies, Inc., as a strong candidate for a takeover by a larger player, Aztec Mining Corporation. Which of the following is incorrect in relation to implementing a merger arbitrage strategy?

I. Buy Aztec and sell Diamond Technologies.

II. The payoff is large profits if the merger goes ahead, and small losses if it does not.

III. The strategy is likened to selling insurance.

**A.** I and II

**B.** I and III

**C.** II and III

**D.** I, II and III

**Question IM-8 c.7** New • built from SchweserNotes 2026 Reading 90

After a share-for-share acquisition is announced, a long-only mutual fund and a hedge fund both review the target's stock. Which of the following statements is most accurate?

**A.** Long-only funds are usually drawn to the trade, because the gap between the offer value and the post-announcement price is large and the outcome is certain.

**B.** If the deal fails, the hedge fund's short position in the acquirer offsets the loss on the target, so a failed deal is roughly neutral for the arbitrageur.

**C.** The hedge fund can apply legal and antitrust expertise to judge the probability that the deal closes, and its short position in the acquirer adds incremental return.

**D.** The hedge fund would short the acquirer at the exchange ratio even if the offer were in cash, because the short leg is what defines merger arbitrage.

## Solutions • IM-8 c

### Solution IM-8 c.1 — Answer: D

Bank Q437 • AnalystPrep Q.2760

**Step 1 – What makes a strategy event-driven**

The return depends on a specific corporate event that creates a pricing discrepancy: a merger, a restructuring, a recapitalisation, a bankruptcy reorganisation.

**Step 2 – Find the event in the options**

Distressed securities strategies trade the claims of firms entering, in, or emerging from bankruptcy; the bankruptcy and restructuring are the event. Stowell lists bankruptcy reorganisations among the financial event-driven opportunities and treats distressed investing in the event-driven module.

$\Rightarrow$ Distressed securities.

**Interpretation**

Ask what has to *happen* for the position to pay off. For the other three strategies the answer is a price trend, good stock selection or a correct macro view, never a corporate event.

**Why the other options are wrong**

**A is incorrect.** Trend followers ride price momentum across markets; no corporate event is involved.

**B is incorrect.** Equity long/short earns alpha from security selection on both legs (IM-8 a).

**C is incorrect.** Global macro bets on economies, rates, currencies and policy, top down (IM-8 a).

**LO:** IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.c and 90.d)

### Solution IM-8 c.2 — Answer: A

New • built from SchweserNotes 2026 Reading 90

**Step 1 – Recall the five categories**

Strategic (hard catalysts), financial, operational, legal/regulatory and technical (Section 1).

**Step 2 – Place each idea**

Broken risk arbitrage situations are listed under technical opportunities, next to secondary offerings. Stub trades are strategic; credit re-ratings are financial; merger synergies are operational.

$\Rightarrow$ A is the only correct pairing.

**Interpretation**

The live deal and the collapsed deal sit in different rows: risk arbitrage on an announced deal is a strategic opportunity, while the dislocation left behind when a deal breaks is a technical one.

**Why the other options are wrong**

**B is incorrect.** Sibling swap: stub trades are a strategic (hard catalyst) opportunity.

**C is incorrect.** Sibling swap: price moves around credit re-ratings are a financial opportunity.

**D is incorrect.** Sibling swap: synergies from mergers are an operational opportunity; legal/regulatory covers litigation, regulatory change and legislation.

**LO:** IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.c, event-driven opportunity categories)

### Solution IM-8 c.3 — Answer: B

Bank Q436 • AnalystPrep Q.2574

#### Step 1 – Read the target return

The gap between the target's current price and the acquirer's final offer, after a deal is announced, is the merger arbitrage spread.

#### Step 2 – Read the accepted risk and the exclusions

The accepted risk, a deal cancelled or modified, is merger arbitrage's principal risk. The exclusions rule out a rates bet and a strategy needing distressed expertise.
$\Rightarrow$ Merger (risk) arbitrage.

#### Interpretation

If the deal fails, the target typically falls back to its pre-announcement price and the acquirer may rise, so the cancellation risk QHF accepts is where the large losses come from.

#### Why the other options are wrong

**A is incorrect.** Convertible arbitrage pairs a convertible bond with the issuer's stock; it does not depend on an announced deal.
**C is incorrect.** Fixed-income arbitrage trades yield differences between similar bonds and is rate and spread sensitive, which management excluded.
**D is incorrect.** Distressed securities is event-driven, but it targets firms in or near bankruptcy and needs deep restructuring expertise, which management excluded.

**LO:** IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.c, Merger Arbitrage)

### Solution IM-8 c.4 — Answer: D

New • built from SchweserNotes 2026 Reading 90

#### Step 1 – The mechanics of activism

Take an equity or derivative position, then seek to influence management or the board, for example by soliciting proxy votes, toward governance changes, spin-offs, cost cuts, buybacks, higher dividends or more leverage.

#### Step 2 – Success and failure

Success needs management to adopt the initiatives and the share price to respond; failure is a campaign that produces no meaningful change.
$\Rightarrow$ D.

#### Interpretation

A spin-off and a buyback are both on Stowell's list of activist demands. The event is created by the fund itself, which is what separates activism from merger arbitrage, where the event (the offer) is given.

#### Why the other options are wrong

**A is incorrect.** True fact about the wrong strategy: an offer spread and deal-break risk describe merger arbitrage.
**B is incorrect.** Polarity flip and sibling swap: activists engage on purpose; creditor committees belong to active distressed investing.
**C is incorrect.** True fact about the wrong strategy: the interest rebate is a return source of strategies with short books, such as equity long/short.

**LO:** IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.c, Activist Strategy)

### Solution IM-8 c.5 — Answer: C

Bank Q435 • Mock 4 Q73 (key corrected in the bank; confirmed)

**Step 1 – Merger arbitrage legs**

The arbitrageur buys the target and, in a share-for-share deal, may short the acquirer; the position is unwound at completion, ideally at a profit. The spread is earned if the deal closes.
⇒ C is correct.

**Step 2 – Check the other statements against Reading 90**

Global macro is forward looking and makes large, leveraged bets across markets. Equity long/short seeks alpha from mispricings while reducing market risk. Shorting the target is the opposite of merger arbitrage.

**Interpretation**

The main risk of the correct trade is a failed deal: the target falls back toward its pre-announcement price, and the acquirer short may also lose.

**Why the other options are wrong**

**A is incorrect.** Polarity flip on every feature: global macro funds anticipate macro trends and take large, leveraged positions across many markets.
**B is incorrect.** Legs reversed: shorting the target profits only if the deal fails; the strategy is long the target to capture the spread.
**D is incorrect.** Polarity flip: long/short aims to *reduce* market exposure and earn alpha from mispricings.

**LO:** IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.a and 90.c)

### Solution IM-8 c.6 — Answer: A

Bank Q439 • Mock 1 Q33

**Step 1 – Statement I**

The target is Diamond Technologies and the acquirer is Aztec Mining. The arbitrageur buys the target and, in a share-for-share deal, shorts the acquirer. Statement I reverses this: incorrect.

**Step 2 – Statement II**

If the merger goes ahead, the gain is the (small) spread; if it fails, the target falls back and the losses can far exceed the gains. Statement II reverses the asymmetry: incorrect.

**Step 3 – Statement III**

Small, steady gains with occasional large losses is the payoff of a seller of insurance (a short put on deal completion): correct.
⇒ The incorrect statements are I and II.

**Interpretation**

Figure 1 shows the shape: \$5 per CI share if the AWI deal closes against about \$25 if both prices return to their pre-announcement levels.

**Why the other options are wrong**

**B is incorrect.** Marks III as incorrect, but the selling-insurance analogy is the standard description; it also misses that II is wrong.
**C is incorrect.** Marks III as incorrect and misses that I reverses the legs.
**D is incorrect.** Includes III, which is correct.

**LO:** IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.c, Merger Arbitrage)

**Solution IM-8 c.7** **Answer: C**

New • built from SchweserNotes 2026 Reading 90

### Step 1 – Why long-only funds stay away

The potential gain, the acquisition price less the post-announcement price, is relatively small, and there is uncertainty about whether the deal closes. Mutual funds are therefore not attracted.

### Step 2 – What the hedge fund brings

Specialised legal and antitrust expertise to assess the likelihood of completion, and the short position in the acquirer, which generates incremental return (interest on the short proceeds, and extra profit if a competing bid lifts the target and depresses the acquirer).
$\Rightarrow$ C.

### Interpretation

The edge is judgement about $C$, the probability of completion. In the return formula of Section 5, $L$ is usually several times $G$, so a small error in $C$ swings the expected return sharply.

### Why the other options are wrong

**A is incorrect.** Polarity flip: the spread is small and the outcome uncertain, which is exactly why long-only funds are not attracted.
**B is incorrect.** Polarity flip: on a break the acquirer may *rise*, so the short loses too; both legs lose at once.
**D is incorrect.** Scope error: the acquirer short belongs to share-for-share deals, where the target shares convert into acquirer shares; a cash offer is captured with the long target position alone.

**LO:** IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.c, Merger Arbitrage)

IM-8 • Hedge Fund Investment Strategies • Vardeez: VERY IMPORTANT

# IM-8 d

P1 CONCEPT

Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: A24 covers all four IM-8 LOs, and hedge funds came up twice in the August 2026 sitting (Map #34 on 3 of 16 sheets). Vardeez does not flag this LO among his three IM-8 priorities.
**Vardeez:** no pointer for this LO (his notes cover it, with the Naples Ocean Views example).

**Reported exam shapes**

- **S1** Hedge fund strategies across the families. The distressed version: active against passive investors, where the opportunity comes from, the capital-structure trade, and the core risk. Option D of Workbook A24.1 (worked as Question IM-8 a.7) swaps the active and passive roles.

**What the questions look like:** conceptual. Why distressed claims are cheap (and a reversed reason as the distractor), which leg to buy and which to short, who sits on the creditor committee, and which development hurts the trade. Schweser's quiz names the core risk: the restructuring takes longer than expected.

## 1. What the strategy buys, and why it is cheap

A distressed securities strategy takes speculative positions in the securities of firms that are **entering, in, or emerging from bankruptcy**. The opportunity exists because the natural holders are forced or reluctant sellers, so the claims can be bought at substantial discounts to their risk-adjusted values:

- **Equity held through emergence**, and **non-investment-grade bonds** that are cheap because many institutional investors and pension funds are *prohibited* from owning such debt.
- **Bank loans**: banks sell parts of their distressed loans to clean up their balance sheets.
- **Trade claims**: the original holders (suppliers) often lack the expertise or the desire to stay involved through the bankruptcy process.

## 2. Two ways to profit

1. **Buy and hold through the restructuring**, expecting the securities to gain once the firm emerges.
2. **Trade the capital structure.** Analyse the pricing of senior and junior claims and look for mispricings among the issuer's securities. During a restructuring, senior securities may appreciate relative to junior ones, so **buy the senior claim and short the junior claim** before prices adjust. Shorting equity that is expected to lose most of its value is the extreme case.

## 3. Worked example: Naples Ocean Views

Naples Ocean Views (NOV), a real estate holding company, may be forced into bankruptcy within two years. It has \$1 billion of 5% debt due in 10 years, trading at \$30 per \$100 of face value; its stock trades at \$3. In liquidation the bonds are estimated to be worth \$35 per \$100 of face. An investor buys one \$1,000 bond for \$300 and shorts 100 shares at \$3. NOV is liquidated in two years.

| | |
|---|---|
| Bond: coupons, $5\% \times 1{,}000 \times 2$ years | $+100$ |
| Bond: liquidation value, $35/100 \times 1{,}000$ | $+350$ |
| Bond: price paid | $-300$ |
| **Gain on the bond** | $\mathbf{+150}$ |
| Short: proceeds $100 \times \$3$; the stock becomes worthless and is never bought back | $\mathbf{+300}$ |
| **Total gain** | $\mathbf{+450}$ |

**Priority is the whole trade.** The bond's recovery of \$35 against a price of \$30 adds only \$50 of principal gain; the coupons do as much again. The leg that pays is the equity short: in liquidation the bondholders' claims rank ahead of the shareholders', so the stock goes to zero and all \$300 of proceeds are kept.

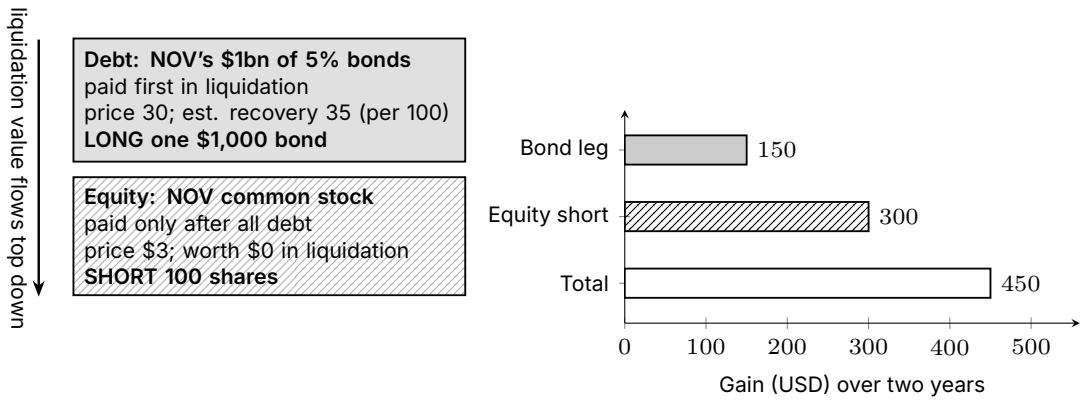



Figure 1. The NOV capital-structure trade. Left: the bond (grey) ranks ahead of the stock (hatched), so liquidation value reaches the bondholders and leaves nothing for the shareholders. Right: the bond leg gains \$150 (coupons \$100 plus recovery \$350 less \$300 paid), the equity short \$300; total \$450.

## 4. Active and passive investors

| | |
|---|---|
| **Active** | Try to influence the restructuring: (1) participate on a **creditor committee**; (2) supervise the **workout process**, out of court or in court, to protect their interests; (3) engage in **legal proceedings** when that could improve their position. |
| **Passive** | Choose not to get involved; prefer distressed investments that are **less complex and less time intensive** to analyse and monitor. |

## 5. Challenges, risks and considerations

- **Valuation is hard.** Distressed firms have unstable cash flows, limited liquidity and complex capital structures; recovery estimates may be unreliable.
- **Time.** The restructuring or bankruptcy can take much longer than expected. Schweser's quiz calls this the core risk: a longer process erodes the return on capital that is tied up.

- **Illiquidity.** Distressed securities are frequently illiquid, so positions are hard to adjust or exit without moving prices.
- **Legal and regulatory outcomes:** court rulings, creditor negotiations, debtor-in-possession (DIP) financing, and possible changes in bankruptcy law.
- **Erosion of recovery:** competition from other distressed investors, unexpected deterioration in the business, or new senior debt ranking ahead of existing claims.
- **Expertise.** The strategy needs specialised skill in financial analysis, legal processes and corporate restructuring.

**Comparison card: IM-8 d**

| Challenge or risk | What it does to a trade like NOV |
|---|---|
| **Process takes longer** | The same $450 arrives later, and capital stays tied up in an illiquid position: the return per year falls |
| **Valuation uncertainty** | The $35 recovery is an estimate; unstable cash flows and a complex structure can make it wrong |
| **Illiquidity** | The bond cannot be sold or resized without moving its price |
| **Legal and regulatory** | Court rulings, creditor negotiations, DIP financing and bankruptcy-law changes can change who recovers what |
| **New senior debt or DIP financing** | A claim ranking ahead of the bonds reduces their recovery |
| **Competition from distressed investors** | Bids up the claims, so the entry discount shrinks |
| **Business deterioration** | Lowers the liquidation value available to every claim |

**Exam traps**

- **Why junk bonds are cheap, reversed.** Many institutions are *prohibited* from holding non-investment-grade debt, which forces sales; they are not required to hold it.
- **Active and passive swapped.** Creditor committees, workout supervision and litigation are active; passive investors stay out and choose simpler situations.
- **Legs reversed.** The capital-structure trade is long senior, short junior: seniority recovers first.
- **Seniority reversed.** Senior secured creditors recover the most, not the least.
- **The core risk.** A restructuring that takes longer than expected; bankruptcy outcomes are not predictable, and distressed claims trade thinly, not heavily.
- **Example slips.** Coupons run for two years ($100); the $35 recovery is per $100 of face, so $350 on a $1,000 bond.

## Practice questions: IM-8 d

### Question IM-8 d.1

Bank Q441 • AnalystPrep Q.2573 (rebuilt)

A newly formed hedge fund is choosing one strategy. It is willing to accept illiquid positions, default risk and a restructuring process of uncertain length, and it expects to profit from securities of firms entering, in, or emerging from bankruptcy that are bought at deep discounts to risk-adjusted value. It does not want a strategy whose returns depend mainly on leverage applied to small mispricings, nor one that requires continuously hedging net market exposure. Which strategy best fits?

**A.** Equity long/short

**B.** Global macro

**C.** Distressed securities

**D.** Convertible arbitrage

### Question IM-8 d.2

Bank Q442 • Mock 2 Q20 (rebuilt)

A consultant is reviewing the pitch book of a distressed securities manager. Which of the following is NOT a recognised source of return opportunity in a distressed securities strategy?

**A.** Buying shares in a company and holding them until it emerges from bankruptcy.

**B.** Buying loans that banks sell to clean up their balance sheets.

**C.** Buying trade claims from original holders who lack the expertise or desire to stay involved through the bankruptcy process.

**D.** Buying non-investment-grade bonds that are cheap because pension funds and institutions are required to hold them.

### Question IM-8 d.3

Bank Q443 • Schweser Module Quiz 90.3 Q2

A risk manager is briefing the board of an endowment on a proposed allocation to a distressed securities fund. Which of the following scenarios best reflects the core risk faced by investors in distressed securities?

**A.** A restructuring process takes longer than expected.

**B.** Senior secured creditors receive the lowest recoveries.

**C.** Bankruptcy guarantees predictable outcomes for all creditors.

**D.** High trading volume in distressed debt limits position building.

### Question IM-8 d.4

New • built from SchweserNotes 2026 Reading 90

Two hedge funds buy the bonds of a retailer that has just filed for bankruptcy. Fund 1 joins the creditor committee, supervises the workout and is prepared to go to court. Fund 2 avoids involvement and prefers distressed situations that are less complex and less time intensive to analyse and monitor. Which of the following statements is correct?

**A.** Fund 1 is an active investor that seeks to influence the restructuring and protect its position; Fund 2 is a passive investor.

**B.** Fund 1 is an active investor, but its involvement must end if the restructuring moves out of court, because workouts are reserved for the debtor's banks.

**C.** Fund 2 is an active investor, because choosing simpler situations is itself a way of influencing the restructuring.

**D.** Both funds are active investors, because every holder of a distressed bond automatically sits on the creditor committee.

### Question IM-8 d.5

New • built from SchweserNotes 2026 Reading 90

A distressed debt analyst expects that, as a struggling issuer restructures, its senior secured loans will be repaid in full while its subordinated bonds will recover little. The market currently prices both at similar discounts to face value. Which trade best exploits this view?

**A.** Buy the subordinated bonds and short the senior loans, expecting junior claims to gain relative to senior claims.

**B.** Buy the senior loans and short the subordinated bonds, expecting senior claims to appreciate relative to junior claims.

**C.** Buy both the senior loans and the subordinated bonds, since the whole capital structure trades at a discount.

**D.** Short only the equity, because a restructuring always leaves the equity worthless.

### Question IM-8 d.6

New • built from SchweserNotes 2026 Reading 90

A fund holds the 5% bonds of a real estate company expected to file for bankruptcy. It bought them at 30 per 100 of face value against an estimated liquidation recovery of 35, and it is short the company's stock. Which of the following developments would most directly threaten the bond leg of the trade?

**A.** The company's stock falls to zero in liquidation.

**B.** Restricted institutions were forced to sell the bonds before the fund bought them, depressing its entry price.

**C.** The equity holders receive nothing until the bondholders' claims are met in full.

**D.** The court approves new debtor-in-possession financing that ranks ahead of the existing bonds.

# Solutions • IM-8 d

## Solution IM-8 d.1

**Answer: C**

Bank Q441 • AnalystPrep Q.2573

### Step 1 – Match the target

Securities of firms entering, in, or emerging from bankruptcy, bought at deep discounts to risk-adjusted value: the definition of a distressed securities strategy.

### Step 2 – Match the accepted risks and the exclusions

Illiquidity, default and a restructuring of uncertain length are the typical distressed risks. Returns come from recovery and capital-structure mispricing, not from leverage on small mispricings or from continuous market hedging.

$\Rightarrow$ Distressed securities.

### Interpretation

The two exclusions are the fingerprints of the arbitrage family: leverage on small mispricings (fixed-income and convertible arbitrage) and continuous hedging (delta hedging, long/short).

### Why the other options are wrong

**A is incorrect.** Equity long/short targets mispricings among ordinary stocks and hedges market risk; it is not about bankrupt firms.

**B is incorrect.** Global macro takes leveraged directional views on rates, currencies and economies.

**D is incorrect.** Convertible arbitrage relies on leverage and continuous delta hedging, both of which the fund wants to avoid.

**LO:** IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.d)

## Solution IM-8 d.2

**Answer: D**

Bank Q442 • Mock 2 Q20 (key corrected in the bank; confirmed)

### Step 1 – List the recognised sources

Schweser's examples: holding equity until the firm emerges from bankruptcy; buying non-investment-grade bonds that are cheap because many institutions and pension funds are *prohibited* from owning them; buying loans that banks sell to clean up their balance sheets; buying trade claims from holders who lack the expertise or desire to stay involved.

### Step 2 – Find the reversed mechanism

D says the institutions are *required* to hold the bonds. A requirement to hold would support the price; the discount comes from prohibitions that force them to sell.

$\Rightarrow$ D is not a recognised source.

### Interpretation

Every source of opportunity in this strategy is a forced or reluctant seller. A distractor that turns the seller into a forced *holder* removes the reason for the discount.

### Why the other options are wrong

**A is incorrect.** A recognised source: holding shares through emergence from bankruptcy.

**B is incorrect.** A recognised source: banks sell distressed loans to clean up their balance sheets.

**C is incorrect.** A recognised source: trade claims trade at discounts because their holders do not want to stay through the bankruptcy.

**LO:** IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.d, Overview and Return Opportunities)

### Solution IM-8 d.3 Answer: A

Bank Q443 • Schweser Module Quiz 90.3 Q2

#### Step 1 – List the risks Schweser names

Unstable cash flows and complex capital structures make valuation difficult and recovery estimates unreliable; the restructuring or bankruptcy can take much longer than expected; the securities are illiquid; outcomes depend on courts, creditor negotiations and DIP financing.

#### Step 2 – Pick the one the quiz calls core

Schweser's answer key: distressed investing carries significant timing risk, because prolonged restructuring or bankruptcy proceedings erode expected returns.
$\Rightarrow$ A.

#### Interpretation

Time is the risk that touches every distressed position, active or passive: the recovery may be right and still arrive too late to earn an adequate return.

#### Why the other options are wrong

**B is incorrect.** Polarity flip: senior secured creditors usually recover the *most*.
**C is incorrect.** Polarity flip: bankruptcy outcomes are uncertain and depend on courts and negotiations.
**D is incorrect.** Polarity flip: distressed securities are typically illiquid, with low volume, which makes exiting hard.

**LO:** IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.d, Challenges, Risks, and Key Considerations)

### Solution IM-8 d.4 Answer: A

New • built from SchweserNotes 2026 Reading 90

#### Step 1 – What active investors do

They try to influence the restructuring: they participate on a creditor committee, supervise the workout process (out of court or in court) to protect their interests, and engage in legal proceedings when that could improve their position. Fund 1 does all three.

#### Step 2 – What passive investors do

They choose not to get involved and prefer distressed investments that are less complex and less time intensive to analyse and monitor. That is Fund 2.
$\Rightarrow$ A.

#### Interpretation

This is the swap tested in option D of Workbook A24.1: a "passive" investor who sits on the committee and litigates is a contradiction.

#### Why the other options are wrong

**B is incorrect.** Scope error: active investors supervise the workout whether it is an out-of-court restructuring or an in-court bankruptcy.
**C is incorrect.** Role misassignment: choosing simpler situations and staying uninvolved is the definition of passive.

**D is incorrect.** Absolute claim: committee membership is a choice of active investors, not automatic for every bondholder.

**LO:** IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.d, Investor Approaches)

## Solution IM-8 d.5 Answer: B

New • built from SchweserNotes 2026 Reading 90

### Step 1 – Read the view

Senior claims will be repaid in full; junior claims will recover little. The market prices them alike, so the senior claim is cheap relative to the junior claim.

### Step 2 – Set the legs

Buy the senior loans and short the subordinated bonds. As the restructuring proceeds, the senior claim should appreciate relative to the junior claim, and the spread between them should widen toward the recovery difference.
⇒ B: the capital-structure trade.

### Interpretation

This is the structure of the NOV example one level up the ladder: long the claim that ranks first, short the claim that ranks behind it. Priority decides recovery, so priority decides the trade.

### Why the other options are wrong

**A is incorrect.** Legs reversed: it bets that junior claims gain on senior claims, the opposite of the analyst's view.
**C is incorrect.** A long position in both is a bet on the whole firm's recovery, not on the mispricing between the two claims; the subordinated leg loses under the analyst's view.
**D is incorrect.** Absolute claim: equity is not always worthless (firms can emerge from restructuring), and a short in the equity alone does not express the senior against junior view.

**LO:** IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.d, ways to profit)

## Solution IM-8 d.6 Answer: D

New • built from SchweserNotes 2026 Reading 90

### Step 1 – What the bond leg depends on

Its profit comes from the liquidation recovery (estimated at 35 per 100) exceeding the price paid (30), plus any coupons received. Anything that cuts the recovery threatens the leg directly.

### Step 2 – Test each development

New debtor-in-possession financing that ranks ahead of the existing bonds takes its share of the liquidation value first, so less is left for the bonds. Schweser lists DIP financing and new senior debt among the developments that can erode recovery values.
⇒ D.

### Interpretation

A recovery estimate is only as good as the claims ranked above it. The distressed investor must watch the whole capital structure, not just its own claim, which is why the strategy needs legal and restructuring expertise.

### Why the other options are wrong

**A is incorrect.** This is the gain on the short leg: the stock going to zero is what the equity short is for.
**B is incorrect.** Forced selling by restricted institutions is the source of the discount the fund bought at, not a threat to the recovery.
**C is incorrect.** This is the priority that the trade relies on: bondholders are paid before shareholders.

**LO:** IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. • **Reference:** Stowell and Stowell 4e, Ch.12 (SchweserNotes 2026 Book 5, Reading 90, LO 90.d, Challenges, Risks, and Key Considerations)

IM-9 • Risk, Regulation and Organizational Structure • Vardeez: IMPORTANT

# IM-9 a

P1 CONCEPT

Describe different risks faced by investors in hedge funds.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: the A24 section carries this reading's shape S2, hedge funds came up twice in the August 2026 sitting, and Map #34 records the section on 3 of 16 recall sheets. **Vardeez:** hedge fund risks.

**Reported exam shapes**

- **S2** Hedge fund risks, systemic risk, regulation and structure: one "which statement is correct" item across all four IM-9 LOs. Workbook A24.2 is the August item and is worked here, last. Its options draw on IM-9 b (the leverage spiral), c (regulation) and d (feeders and partners), so Section 1 previews the whole reading.

Survivorship bias, a portfolio-level risk here, is tested against its sibling biases in Workbook A24.3 (worked as Question IM-17 c.10).

**What the questions look like:** conceptual. Classify a risk (portfolio-level or investment-level; which of the five incremental risks), or name the risk behind a story: fee incentives point to risk tolerance, strategies falling together to correlation spikes, financing withdrawn to leverage.
**Source.** No GARP book covers IM-9. SchweserNotes 2026 Reading 91 (Stowell and Stowell, Ch.14) is the authority; Vardeez is checked against it.

## 1. The reading on one page

Hedge funds are lightly regulated directly, so their investors carry more of the burden of judging risk, and the system relies partly on the banks that lend to them. The four LOs follow one thread:

| LO | Question it answers | Core content |
|---|---|---|
| **IM-9 a** | What can go wrong for the investor? | Portfolio-level and investment-level risks; five incremental risks: leverage, regulation, short selling, transparency, risk tolerance |
| **IM-9 b** | How can hedge funds hurt the system? | Fire sales under leverage (the leverage spiral); losses to banks through revenue, credit and counterparty risk; LTCM, Amaranth, Archegos |
| **IM-9 c** | What do regulators do about it? | US exemptions and Dodd-Frank Title IV; EU, Singapore, Hong Kong, China; investor protection, market integrity, financial stability |
| **IM-9 d** | How is a fund built, and why? | Domicile, the limited partnership, the master-feeder structure, service providers, the tax treatment of carried interest |

## 2. Two ways to classify the risks

Stowell offers two lenses, and questions use both:

1. **Portfolio-level against investment-level.** Portfolio-level risks arise from holding hedge funds in a portfolio at all; investment-level risks arise from how a particular fund is run.
2. **Five incremental risks** that are particularly pronounced in hedge funds: **leverage, regulation, short selling, transparency and risk tolerance**.

## 3. Portfolio-level risks

- **Benchmarking issues.** Strategies that go long and short across markets have no natural benchmark, so judging performance and fitting the fund into an allocation is harder.
- **Survivorship bias.** Funds that fail or stop reporting drop out of the databases, so the surviving funds' history overstates the returns an investor could have expected. (The sibling biases are drilled in IM-17 c.)
- **Unrelated business taxable income (UBTI).** Debt-financed income can become taxable for investors who are otherwise tax exempt, such as endowments and pension funds. **Investing through an offshore fund** can mitigate it; this is one reason for the offshore feeder in IM-9 d.

Schweser's summary adds the effect of leverage on the portfolio's overall performance and volatility.

## 4. Investment-level risks

| Risk | What it looks like |
|---|---|
| **Counterparty risk** | A prime broker or derivatives counterparty fails, putting cash, collateral and contracts at risk |
| **Conflicts of interest** | The manager's interests diverge from the investors', for example through fees or preferential terms for some investors |
| **Key person risk** | The fund depends on one or a few individuals; their departure undermines the strategy |
| **Strategy failure** | The edge disappears or the model stops working |
| **Style drift** | The manager moves away from the stated strategy |
| **Regulatory changes** | New rules, such as limits on short selling or leverage, change what the strategy can do |
| **Correlation spikes** | Strategies that normally behave independently move together in stress, so diversification fails when it is needed |

Schweser lists the seven risks (its Figure 91.1); the glosses in the right-hand column are this workbook's.

## 5. The five incremental risks

### Leverage

- Two forms: **on-balance-sheet** (direct borrowing) and **off-balance-sheet** (for example, derivatives).
- "When returns are positive, leverage is a powerful ally; when returns are negative, it becomes a difficult adversary": it amplifies losses just as it amplifies gains.
- Leverage raises a fund's **sensitivity to liquidity shocks**: when prices fall, a levered fund may have to sell (the leverage spiral of IM-9 b: $100 of stock on a $75 loan must sell $30 after a 10% fall to get back to 3.0x).
- **Most hedge fund leverage is collateralised** with cash or assets held by the fund; true uncollateralised borrowing is rare.
- By strategy (Figure 1), relative value carries the most leverage and managed futures the least. Relative value trades earn small returns per trade (IM-8 b): the leverage is what makes them worth doing, and why they are not low risk at the fund level.

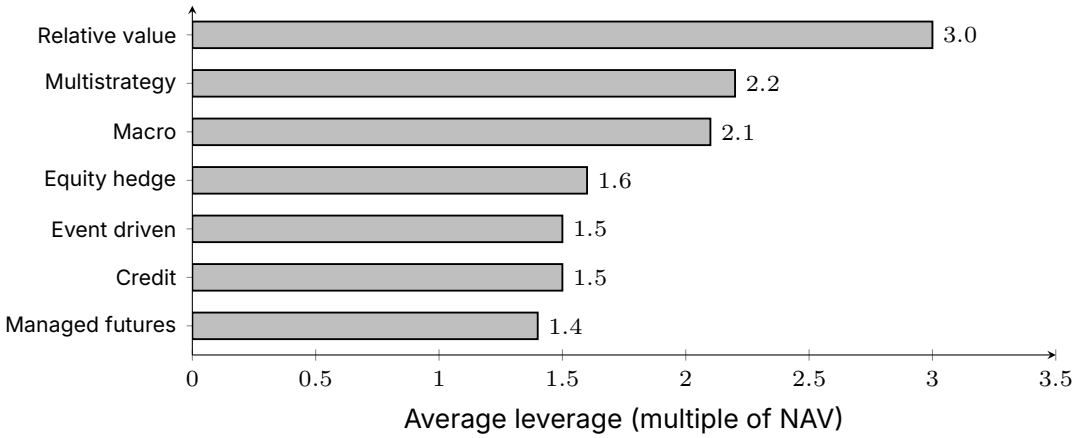



Figure 1. Hedge fund leverage by strategy, from the 2020 findings of the US Treasury's Office of Financial Research (Schweser's Figure 91.2). Schweser's text layer does not carry the values; they are taken from Vardeez's redraw of the printed figure. Relative value carries about twice the leverage of managed futures.

## Regulation

Hedge funds operate with **significantly less regulatory oversight** than open-end mutual funds, so prospective investors must **conduct their own due diligence**. Before 2010 a manager advising fewer than 15 clients was exempt from SEC registration. The global financial crisis of 2007 to 2009 showed that hedge funds are regulated *indirectly*, through their highly regulated counterparties, above all banks; the Dodd-Frank Act then required registration. Rules differ widely across jurisdictions (IM-9 c).

## Short selling

A short sale means borrowing shares, posting collateral, adding collateral as required, and returning the shares when the position is closed. A long position's loss is limited (an asset cannot go below zero); a short position's loss is **theoretically unlimited**, because a price has no upper bound. In a **short squeeze**, the price of a heavily shorted stock rises quickly, short sellers are forced to buy to close their positions, and the forced buying pushes the price higher still (GameStop, 2021).

## Transparency

Many hedge funds arbitrage perceived inefficiencies; when a trade becomes **crowded**, the profit disappears. Funds are therefore secretive about their strategies, which makes it harder for investors to monitor risks and check that the strategy is the one they bought. **Exit gates** and other liquidity limits can stop investors leaving when they suspect a strategy shift.

## Risk tolerance

Hedge fund managers generally have a **higher risk tolerance** than most open-end mutual fund managers, and their **compensation** often rewards taking more risk: a performance fee shares in the gains but not the losses. They also have a wider range of opportunities, including newer strategies, and use derivatives, which can reduce risk but make it harder to assess.

**Comparison card: IM-9 a**

| Risk | Lens and level | The tell in a question stem |
|---|---|---|
| **Benchmarking, survivorship bias, UBTI** | Portfolio-level | "No suitable benchmark"; "dead funds drop out of the database"; "tax-exempt investor, debt-financed income" |
| **Counterparty, conflicts, key person, strategy failure, style drift, regulatory change, correlation spikes** | Investment-level | "Prime broker fails"; "manager departs"; "fund drifts from its mandate"; "unrelated strategies lose together in a crisis" |
| **Leverage** | Incremental | Borrowing or derivatives; financing withdrawn; forced selling |
| **Regulation** | Incremental | Lighter oversight than mutual funds; own due diligence needed |
| **Short selling** | Incremental | Unlimited loss; squeeze; collateral calls |
| **Transparency** | Incremental | Secret strategy; gates; cannot monitor |
| **Risk tolerance** | Incremental | Performance fee shares gains, not losses |

**Exam traps**

- **Level swapped.** Benchmarking, survivorship bias and UBTI are portfolio-level; counterparty, key person, style drift and correlation spikes are investment-level.
- **UBTI fix.** An offshore fund mitigates UBTI for tax-exempt investors, not an onshore one.
- **Leverage as uncollateralised borrowing.** Most hedge fund leverage is collateralised; it can be off-balance-sheet through derivatives.
- **Highest leverage.** Relative value (about 3x), not managed futures (about 1.4x) or macro.
- **Short and long reversed.** The short position has the unlimited loss.
- **Fee incentive mislabelled.** A performance fee that rewards risk-taking is the risk tolerance risk, not leverage or transparency.
- **Correlation spike against style drift.** Funds that stay on strategy but lose together in a crisis show a correlation spike; style drift is a manager leaving its mandate.

## Practice questions: IM-9 a

### Question IM-9 a.1

New • built from SchweserNotes 2026 Reading 91

A consultant to a US university endowment classifies the risks of a proposed hedge fund allocation as portfolio-level or investment-level, following Stowell. Which of the following is a portfolio-level risk?

**A.** Style drift

**B.** Key person risk

**C.** Unrelated business taxable income (UBTI)

**D.** Correlation spikes

### Question IM-9 a.2

Bank Q445 • AnalystPrep Q.2585 (rebuilt)

Justin Boucher is considering an investment in a hedge fund whose managers receive 20% of the profits generated by the fund. He worries that this compensation structure will entice the managers to take large, risky bets, because they share in the gains but not in the losses. Which of the five incremental risks of hedge fund investing does Boucher's concern most directly describe?

**A.** Leverage

**B.** Regulation

**C.** Transparency

**D.** Risk tolerance

### Question IM-9 a.3

Bank Q446 • AnalystPrep Q.2588 (rebuilt)

A hedge fund manager runs several funds and uses different strategies across them to achieve diversification. During a severe market crisis the funds suffer losses at the same time, even though their strategies normally behave independently and none of them departed from its mandate. Which investment-level risk best explains why the diversification failed?

**A.** Key person risk

**B.** Correlation spikes

**C.** Style drift

**D.** Unrelated business taxable income (UBTI)

**Question IM-9 a.4** Bank Q444 • AnalystPrep Q.3876 (rebuilt)

Convertible arbitrage hedge funds suffered large losses during the 2007 to 2009 crisis. Which of the following was the most likely cause?

**A.** The loss of financing from prime brokers forced leveraged funds to deleverage and sell convertibles at depressed prices.

**B.** A rise in issuers' stock prices caused losses on the short stock hedge that were not offset by the convertibles.

**C.** A surge in demand for convertibles from long-only investors pushed prices well above model values.

**D.** A collapse in equity volatility reduced the value of the embedded call options.

**Question IM-9 a.5** New • built from SchweserNotes 2026 Reading 91

A risk analyst summarises hedge fund leverage for an investment committee, using Stowell's discussion and the 2020 findings of the US Treasury's Office of Financial Research. Which of the following statements is correct?

**A.** Most hedge fund leverage is uncollateralised borrowing, which is why leverage makes funds sensitive to liquidity shocks.

**B.** Leverage can be on-balance-sheet (direct borrowing) or off-balance-sheet (for example, derivatives), and relative value funds carried the highest average leverage in the 2020 data.

**C.** Managed futures funds carried the highest average leverage in the 2020 data, because their individual trades earn the smallest returns.

**D.** Leverage amplifies gains but not losses, because lenders are repaid only out of the fund's profits.

**Question IM-9 a.6** New • built from SchweserNotes 2026 Reading 91

A pension fund trustee asks why short selling and limited transparency appear among the incremental risks of hedge fund investing. Which response is correct?

**A.** A short position's loss is capped at the sale proceeds, while a long position's loss is unlimited.

**B.** Once collateral is posted at the start of a short sale the lender cannot ask for more, so a squeeze cannot force a fund to close a short.

**C.** Hedge funds publish their positions to investors daily, so transparency risk arises only from exit gates.

**D.** Short sellers can be caught in squeezes in which forced covering pushes prices higher, and funds keep strategies secret to protect trades from crowding, which makes their risks harder to monitor.

**Question IM-9 a.7** Workbook A24.2 • S2 • Aug-2026 • Map #34

The CRO of a US university endowment is preparing a report for the board on the risks of investing in hedge funds and on how hedge funds are regulated and structured. Which of the following statements is correct to include in the report?

**A.** A fund holding USD 100 of stock on a USD 75 loan that sees a 10% price fall must sell USD 30 of stock to return to its target leverage, the forced-selling channel through which hedge funds can spread losses.

**B.** In a master-feeder structure, US tax-exempt investors such as pension funds invest through the onshore feeder, which keeps debt-financed income from creating unrelated business taxable income for them.

**C.** Because hedge funds can transmit losses to banks and markets, the reading recommends the strictest feasible regulation of hedge funds as the most effective way to reduce their systemic risk.

**D.** In the limited partnership, investors act as the general partners and the manager as a limited partner, and the performance fee can be taxed at capital gains rates if assets are held at least a year.

# Solutions • IM-9 a

## Solution IM-9 a.1

**Answer: C**

New • built from SchweserNotes 2026 Reading 91

### Step 1 – Recall the two lists

Portfolio-level: benchmarking issues, survivorship bias, UBTI. Investment-level: counterparty risk, conflicts of interest, key person risk, strategy failure, style drift, regulatory changes, correlation spikes.

⇒ UBTI is the portfolio-level risk.

### Step 2 – Why UBTI matters to this investor

An endowment is tax exempt, but debt-financed income earned through a levered fund can become taxable to it. Investing through an offshore fund (the offshore feeder of IM-9 d) can mitigate the problem.

### Interpretation

The test for the level: does the risk come from holding hedge funds in a portfolio (a tax, a benchmark, a biased database), or from how one fund is run (its people, its counterparties, its strategy)?

### Why the other options are wrong

**A is incorrect.** Sibling swap: style drift, a manager leaving its stated strategy, is an investment-level risk.

**B is incorrect.** Sibling swap: dependence on one or a few individuals is an investment-level risk.

**D is incorrect.** Sibling swap: correlation spikes, strategies moving together in stress, are an investment-level risk.

**LO:** IM-9 a: Describe different risks faced by investors in hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a)

## Solution IM-9 a.2

**Answer: D**

Bank Q445 • AnalystPrep Q.2585

### Step 1 – List the five incremental risks

Leverage, regulation, short selling, transparency and risk tolerance.

### Step 2 – Read the concern

A 20% share of profits rewards the upside but does not charge the manager for the downside, so the manager is paid to take more risk. Stowell: hedge fund managers generally have a higher risk tolerance than mutual fund managers, and their compensation structures often incentivise additional risk.

⇒ Risk tolerance.

### Interpretation

The manager's payoff resembles a call option on the fund's performance: its value rises with volatility. Leverage may be the tool used to take the extra risk, but the incentive behind it is the risk tolerance risk.

### Why the other options are wrong

**A is incorrect.** Confuses the tool with the cause: leverage amplifies outcomes, but the stem is about why the manager wants bigger bets.

**B is incorrect.** Regulation concerns lighter oversight and rule changes, not fee incentives.

**C is incorrect.** Transparency concerns secrecy that hinders monitoring; nothing in the stem concerns disclosure.

**LO:** IM-9 a: Describe different risks faced by investors in hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a, Risk Tolerance)

### Solution IM-9 a.3 Answer: B

Bank Q446 • AnalystPrep Q.2588

#### Step 1 – Read what failed

Strategies that normally behave independently lost money together in a crisis, while each stayed within its mandate.

#### Step 2 – Name the risk

That is a correlation spike: in extreme stress, assets and strategies that usually move independently become highly correlated (the LTCM lesson in IM-9 b).
$\Rightarrow$ Diversification that relies on low correlations disappears exactly when it is needed.

#### Interpretation

Leverage makes this worse: funds that face margin calls sell at the same time, and the common selling pushes their returns together (Question IM-9 b.3).

#### Why the other options are wrong

**A is incorrect.** Key person risk is dependence on particular individuals; it does not explain simultaneous losses across strategies.
**C is incorrect.** Style drift is a manager departing from its stated strategy; the stem says none did.
**D is incorrect.** UBTI is a portfolio-level tax issue for tax-exempt investors, unrelated to diversification.

**LO:** IM-9 a: Describe different risks faced by investors in hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a and 91.b)

### Solution IM-9 a.4 Answer: A

Bank Q444 • AnalystPrep Q.3876

#### Step 1 – Recall how the strategy earns

Convertible arbitrage earns small returns per trade and relies on leverage to make them worth having (IM-8 b). Its delta hedge protects against share-price moves, not against losing its financing.

#### Step 2 – Apply the leverage risk

Leverage raises a fund's sensitivity to liquidity shocks. In the crisis, prime brokers withdrew financing and demanded collateral; levered funds had to deleverage, selling convertibles into a market with few buyers, at prices below model values.
$\Rightarrow$ A.

#### Interpretation

This is the leverage incremental risk at the fund level, and the first step of the systemic channel in IM-9 b: lenders force selling, and the selling spreads losses.

#### Why the other options are wrong

**B is incorrect.** A delta-hedged position is largely protected from a stock rise, because the convertible gains too; this was not the main driver.
**C is incorrect.** Polarity flip: demand from long-only investors would raise convertible prices and help the long leg; in the crisis that demand disappeared.

**D is incorrect.** Polarity flip: volatility rose sharply in the crisis, which raises the value of the embedded options.

**LO:** IM-9 a: Describe different risks faced by investors in hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a, Leverage; LO 91.b)

### Solution IM-9 a.5 Answer: B

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – The forms of leverage

Stowell: on-balance-sheet leverage (direct borrowing) or off-balance-sheet leverage (for example, derivatives).

#### Step 2 – The 2020 data

Relative value funds carried the highest average leverage, about 3.0 times NAV; managed futures the lowest, about 1.4 times (Figure 1).
⇒ B.

#### Interpretation

Relative value trades earn small, steady returns per trade, so they need leverage; that same leverage makes them vulnerable to liquidity shocks. Most hedge fund leverage is collateralised with cash or assets held by the fund.

#### Why the other options are wrong

**A is incorrect.** Polarity flip: most hedge fund leverage is collateralised; true uncollateralised borrowing is rare. The sensitivity to liquidity shocks comes from leverage itself.
**C is incorrect.** Sibling swap: managed futures carried the *lowest* average leverage in the data.
**D is incorrect.** Polarity flip: leverage amplifies losses just as it amplifies gains; the loan must be repaid whatever the fund earns.

**LO:** IM-9 a: Describe different risks faced by investors in hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a, Leverage and Figure 91.2)

### Solution IM-9 a.6 Answer: D

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – Short selling

A long position cannot lose more than its value; a short position's loss is theoretically unlimited, because a price has no upper bound. In a squeeze, a heavily shorted stock rises quickly, short sellers are forced to buy to close, and the forced buying pushes the price up further.

#### Step 2 – Transparency

When a trade becomes crowded the profit disappears, so funds keep their strategies secret. Investors then find it harder to monitor risks and to check that the strategy has not shifted; gates and other liquidity limits can stop them leaving.
⇒ D.

#### Interpretation

Both risks come from the hedge fund edge itself: the freedom to short creates the squeeze risk, and the arbitrage nature of many strategies creates the need for secrecy.

#### Why the other options are wrong

**A is incorrect.** Short and long reversed: the long loss is limited; the short loss is unlimited.
**B is incorrect.** Polarity flip: short sellers must add collateral as required; that is part of how a squeeze forces covering.

**C is incorrect.** Polarity flip: hedge funds are often highly secretive; gates add to the transparency problem, they are not its only source.

**LO:** IM-9 a: Describe different risks faced by investors in hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a, Short Selling and Transparency)

### Solution IM-9 a.7 Answer: A

Workbook A24.2 • S2 • Aug-2026

#### Step 1 – Verify the leverage spiral

Before: stock 100, loan 75, equity 25, debt to equity 3.0x. After a 10% fall: stock 90, loan 75, equity 15, so 5.0x. To get back to 3.0x with equity of 15 the loan must fall to 45: sell $75 - 45 = 30$ of stock, leaving stock 60, loan 45, equity 15.

$\Rightarrow$ A 10% price fall forces the sale of 30% of the original position, with no fundamental reason to sell: this is how forced selling becomes contagion.

#### Interpretation

Stowell's two systemic channels are fire-sale contagion from a failing strategy and losses imposed on lenders and counterparties (revenue, credit and counterparty risk for banks, as in Archegos). The spiral in A is the first channel (IM-9 b, Figure 1).

#### Why the other options are wrong

**B is incorrect.** Sibling swap: the offshore feeder is used for non-US investors and US tax-exempt investors such as pension funds; investing through an offshore fund is the stated mitigant for UBTI.

**C is incorrect.** Absolute claim: the mitigants are conservative bank lending, lower leverage with more diversification, and reasonable regulation; excessively strict regulation can reduce the liquidity hedge funds supply as investors of last resort and may raise systemic risk.

**D is incorrect.** Role misassignment: the manager is the general partner and the investors are the limited partners. The tax clause is true (carried interest).

**LO:** IM-9 a: Describe different risks faced by investors in hedge funds. IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91)

IM-9 • Risk, Regulation and Organizational Structure • Vardeez: IMPORTANT

# IM-9 b

P1 CONCEPT

Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic (A24 shape S2; hedge funds twice in August 2026; Map #34 on 3 of 16 sheets).
**Vardeez:** systemic risk.

**Reported exam shapes**

- **S2** Hedge fund risks, systemic risk, regulation and structure. The correct answer of Workbook A24.2 (worked as Question IM-9 a.7) is this LO's leverage spiral, and its option C is the "strictest regulation" trap of Section 5. Workbook A24.4 (worked as Question IM-14 b.10) uses this LO's LTCM lesson as a distractor.

**What the questions look like:** conceptual. Name the channel; read the leverage spiral (its direction, not its arithmetic); separate revenue, credit and counterparty risk; match LTCM and Amaranth to their lessons; spot the absolute claim about strict regulation.
**TYPE note.** The spiral's numbers are in the notes. The bank's numerical version (Bank Q451, the forced sale in USD) is left out under the CONCEPT tag; Workbook A24.2 keeps its numbers because workbook questions always go in.

## 1. Systemic risk and the two channels

**Systemic risk** is the threat of disruption to the broader financial system or economy. Hedge funds can add to it in two ways:

1. **Contagion through forced selling.** A failing strategy may have to unwind its positions at **fire-sale prices**, pushing prices down for every other holder of similar assets.
2. **Losses to financial institutions** that lend to hedge funds or act as their counterparties.

The first is about what hedge funds *do* (their activities); the second about who is *exposed* to them. The LO wording names both.

## 2. The forced-selling channel: the leverage spiral

A fund holds \$100 of stock financed with 25% margin: a \$75 loan and \$25 of equity, a debt-to-equity ratio of 3.0x (Schweser calls this 3x leverage).

- **The price falls 10%.** The stock is worth \$90; the loan is still \$75; equity falls to \$15. Leverage jumps to $75/15 = 5.0$x.
- **Restoring 3.0x.** With \$15 of equity, the loan can be only $3 \times 15 = \$45$: the fund must **sell \$30 of stock** and repay \$30 of the loan. It is left with \$60 of stock, a \$45 loan and \$15 of equity.
- **The rule.** With target leverage $k$ (debt to equity), the forced sale is $L - kE'$. Since the starting loan was $k$ times the starting equity, this equals $k$ times the loss: a \$10 loss at 3x forces \$30 of selling.

The selling has no fundamental reason; the leverage constraint alone makes it unavoidable. As prices fall, larger and larger amounts must be sold: a **downward spiral** that spreads to everyone holding the same assets.

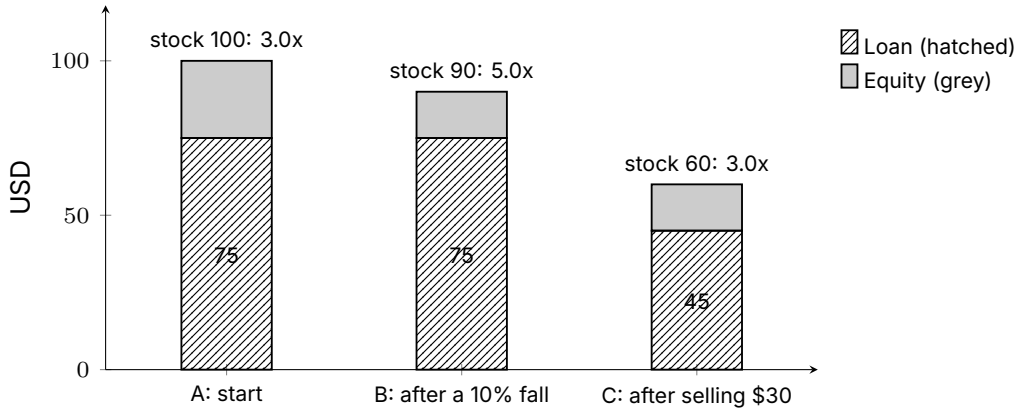



Figure 1. Schweser's Figure 91.3, redrawn. Each bar is the stock position, split into the loan (hatched) and the equity (grey); leverage is debt to equity. A $10 fall cuts equity from $25 to $15 and lifts leverage to 5.0x; returning to 3.0x forces the sale of $30 of stock, three times the loss.

**What amplifies it.** Computer-driven selling; margin calls from counterparties; investor redemptions; falling risk appetite in stress. **Funds of hedge funds** make it worse: their redemptions cascade to the underlying hedge funds and create a waterfall of additional selling.

## 3. Three episodes

| Episode | Lesson |
|---|---|
| **LTCM (1998)** | Monitor **liquidity and leverage**. Assets that typically behave independently can become **highly correlated** and move together in extreme stress. |
| **Amaranth Advisors (2006)** | Raised the question of how systemic hedge funds are. The **Bank of England** concluded that key intermediaries contribute far more to systemic risk than hedge funds, and that hedge funds may even reduce it by facilitating risk transfer. The view is not universal: critics argue that leverage and complex, nonlinear strategies increase systemic risk. |
| **Global financial crisis (2007 to 2009)** | Lenders forced hedge funds to post more collateral by unwinding positions; the fire sales spread contagion and large losses into asset classes that were otherwise uncorrelated. |

## 4. The exposure channel: banks

Banks face three distinct risks when they deal with hedge funds:

1. **Revenue risk.** Banks earn fee and service income from trading, clearing, custody, securities lending, financing and reporting. If several hedge funds fail, that income shrinks. *No default on the bank is needed*, only the loss of the business.
2. **Credit risk.** Margin loans are secured by collateral, but the collateral can fall in value faster than expected, leaving the bank **undersecured**.
3. **Counterparty risk.** Derivatives with hedge funds, such as credit default swaps and other OTC contracts, lose if the fund defaults.

**Archegos (2021).** Archegos Capital was a **family office, not a hedge fund**, but its **total return swaps** created exposures much like those of a highly leveraged hedge fund. When it collapsed, Credit Suisse

lost about $5 billion and other banks about $5 billion more. Banks then required hedge funds to reduce leverage. The lesson is that the exposure comes from the structure of the trades, not the legal form of the client.

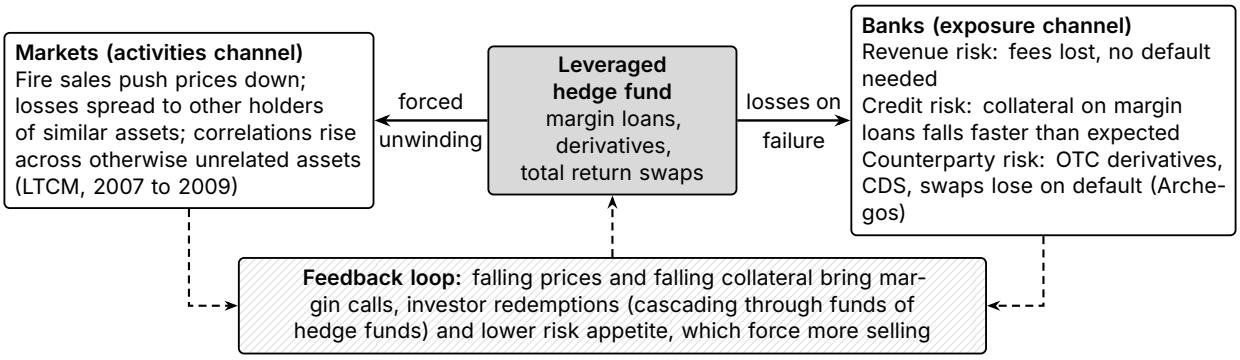



Figure 2. The two systemic channels. Solid arrows: the fund's forced selling hits markets (left) and its failure hits banks (right). Dashed arrows: the feedback that turns one fund's problem into a spiral.

## 5. Mitigation, and the cost of overdoing it

Three approaches mitigate the systemic risk from hedge funds:

1. **more conservative lending practices by banks**;
2. **lower leverage and greater diversification** within hedge fund portfolios;
3. **reasonable regulation**.

The qualifier matters. Excessively strict regulation of hedge funds and private equity firms could **reduce the liquidity** they provide: hedge funds often act as **investors of last resort**, taking risks (such as distressed securities) that traditional investors will not. Overly strict rules can therefore *raise* systemic risk. The reading's position is "reasonable", not "as strict as possible".

**Comparison card: IM-9 b**

| Bank risk | How it arises | Needs a fund default? | Tell in the stem |
|---|---|---|---|
| **Revenue risk** | Fees from trading, clearing, custody, securities lending, financing and reporting disappear if several funds fail | No | "Prime brokerage income falls" |
| **Credit risk** | Collateral behind margin loans falls faster than expected; the loan becomes undersecured | Not necessarily: undersecured first | "Margin loan", "collateral" |
| **Counterparty risk** | OTC derivatives (CDS, swaps) with the fund lose value if it defaults | Yes | "Derivative contract", "total return swap" |

**Exam traps**

- **Spiral direction.** A price fall *raises* debt-to-equity leverage (the loan is unchanged); restoring the target means selling $k$ times the loss, more than the price fall itself.

- **Leverage measure.** Schweser's 3x and 5x are loan over equity, not assets over equity (which would be 4x and 6x).
- **Bank risks swapped.** Revenue risk needs no default; counterparty risk does; credit risk turns on collateral value.
- **Archegos as a hedge fund.** It was a family office; the exposure came from total return swaps.
- **Case swap.** Correlated assets in stress and the leverage-and-liquidity lesson is LTCM; the Bank of England's "intermediaries matter more" verdict followed Amaranth.
- **Absolute regulation claim.** "Strictest feasible" regulation is not the reading's view: overly strict rules drain liquidity from investors of last resort.
- **Amaranth verdict as settled.** The Bank of England's view is not universal; critics point to leverage and nonlinear strategies.

## Practice questions: IM-9 b

### Question IM-9 b.1

New • built from SchweserNotes 2026 Reading 91

A financial stability analyst at a central bank is writing a briefing on how hedge funds can add to systemic risk. Which of the following best summarises the two channels identified by Stowell?

**A.** Hedge funds add to systemic risk mainly through their high fees and through the survivorship bias in their reported returns.

**B.** A failing strategy can be forced to unwind at fire-sale prices, spreading losses to other holders of similar assets, and a fund's failure can impose large losses on the institutions that lend to it or trade with it.

**C.** Hedge funds add to systemic risk only when they are larger than the largest banks, because size is the only way losses can be transmitted.

**D.** Hedge funds add to systemic risk mainly by lending to banks, so that a fund's failure cuts off the banks' funding.

### Question IM-9 b.2

New • built from SchweserNotes 2026 Reading 91

A prime broker's risk manager is explaining to a new analyst why leveraged hedge funds can amplify price declines. A client fund finances its stock positions with margin loans and keeps a fixed target ratio of debt to equity. Which of the following statements is correct?

**A.** When prices fall the loan is unchanged while equity shrinks, so leverage rises; restoring the target requires selling stock worth a multiple of the loss, and the sales can push prices down further.

**B.** When prices fall leverage falls, because the collateral falls faster than the loan; the fund can then add to its positions.

**C.** When prices fall the fund needs to sell only an amount of stock equal to the loss on its position, so forced selling cannot exceed the market move.

**D.** Forced selling of this kind happens only when the assets' fundamental value has fallen, so it does not spread to other markets.

**Question IM-9 b.3** Bank Q448 • BT P2.T8.412.2 (rebuilt; re-filed from LO 91.a)

An investor holds a hedge fund portfolio spread across several strategies that show low historical correlations. Many of the funds use collateralised leverage. During a market-wide liquidity shock, the investor is most concerned that the diversification benefit will disappear. Which of the following best explains why?

**A.** Key person risk, because diversification depends on each fund's lead manager remaining in place.

**B.** Unrelated business taxable income (UBTI), because debt-financed income becomes taxable for tax-exempt investors during stress.

**C.** Survivorship bias, because reported historical correlations include only the funds that survived.

**D.** Correlation spikes, because leveraged funds facing margin calls and forced liquidation sell similar assets at the same time, pushing returns together.

**Question IM-9 b.4** Bank Q452 • new bank question (GARP practice-exam style)

A bank's chief risk officer is assessing the bank's prime brokerage business after the failure of a highly leveraged investment vehicle that used total return swaps. Which of the following statements best describes the credit risk the bank faces on its margin loans to hedge funds?

**A.** Credit risk is negligible because margin loans are collateralised, so losses can arise only from derivatives transactions.

**B.** Credit risk arises mainly from the loss of fee income if several hedge fund clients fail at the same time.

**C.** Credit risk arises because the value of the collateral can decline faster than anticipated, leaving the bank undersecured.

**D.** Credit risk is eliminated once the hedge fund reduces its leverage, because only highly leveraged funds can default.

**Question IM-9 b.5** New • built from SchweserNotes 2026 Reading 91

After the 2021 collapse of Archegos Capital, the CRO of a global bank reviews the bank's exposure to hedge fund clients. Which of the following statements is correct?

**A.** Archegos was a hedge fund that defaulted on exchange-traded futures, so its lesson concerns clearinghouses rather than bank exposure.

**B.** Archegos was a family office, but its total return swaps created exposures resembling those of a highly leveraged hedge fund; Credit Suisse lost about USD 5 billion and other banks about USD 5 billion more.

**C.** Revenue risk arises only when a hedge fund client defaults, whereas counterparty risk can arise even if no client defaults.

**D.** Because the bank's margin loans are collateralised, its only exposure to hedge fund clients is the loss of fee income.

**Question IM-9 b.6** New • built from SchweserNotes 2026 Reading 91

A risk consultant is preparing case studies on hedge fund failures and systemic risk for a bank's board. Which of the following pairings of case and lesson is correct?

**A.** LTCM (1998): the Bank of England concluded that key intermediaries contribute far more to systemic risk than hedge funds do.

**B.** Amaranth Advisors (2006): assets that normally behave independently became highly correlated in extreme stress, showing the need to monitor liquidity and leverage.

**C.** Amaranth Advisors (2006): its failure settled the debate by showing that hedge funds are the main source of systemic risk.

**D.** LTCM (1998): the need to monitor liquidity and leverage, because assets that typically behave independently can become highly correlated in extreme stress.

**Question IM-9 b.7** Bank Q450 • Schweser Module Quiz 91.1 Q2

A policy adviser is ranking proposals to reduce the systemic risk associated with hedge funds. Which of the following actions is least likely to reduce systemic risk associated with hedge funds?

**A.** Strict new regulations

**B.** Lower leverage in hedge fund portfolios

**C.** More conservative lending practices by banks

**D.** Greater diversification in hedge fund portfolios

## Solutions • IM-9 b

### Solution IM-9 b.1 — Answer: B

New • built from SchweserNotes 2026 Reading 91

**Step 1 – The activities channel**

The failure of a fund's strategy can trigger contagion if it is forced to unwind positions at fire-sale prices: prices fall for every other holder of similar assets.

**Step 2 – The exposure channel**

Hedge funds can impose large losses on the financial institutions that lend to them or act as their counterparties (revenue, credit and counterparty risk for banks).

$\Rightarrow$ B states both channels.

**Interpretation**

The LO wording itself splits the two: "the activities of hedge funds" (fire sales) and "exposure to hedge funds" (banks).

**Why the other options are wrong**

**A is incorrect.** True facts about the wrong topic: fees and survivorship bias concern investors (IM-9 a), not the financial system.

**C is incorrect.** Absolute claim: a fund far smaller than a bank can transmit losses through fire sales and counterparty exposures, as LTCM and Archegos showed.

**D is incorrect.** Roles reversed: banks lend to hedge funds, through margin loans and derivatives, not the other way round.

**LO:** IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. •
**Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.b)

### Solution IM-9 b.2 — Answer: A

New • built from SchweserNotes 2026 Reading 91

**Step 1 – What a price fall does to leverage**

The loan does not change when prices fall; the whole loss comes out of equity. Debt to equity therefore rises (3.0x to 5.0x in Figure 1).

**Step 2 – What restoring the target requires**

The permitted loan is $k$ times the reduced equity, so the fund must sell $L - kE'$, which equals $k$ times the loss: at 3x, a \$10 loss forces \$30 of selling.

$\Rightarrow$ The forced sale is a multiple of the loss, and the selling itself can push prices lower and start the cycle again.

**Interpretation**

The selling is driven by the leverage constraint, not by any change in the assets' fundamental value, which is exactly why it spreads to other holders and to otherwise unrelated assets.

**Why the other options are wrong**

**B is incorrect.** Polarity flip: leverage rises after a fall, because equity absorbs the whole loss while the loan stays put.

**C is incorrect.** Intermediate result: selling only the amount of the loss (\$10 in the example) leaves leverage at $65/15 = 4.3$x, above the target.

**D is incorrect.** Polarity flip: forced selling happens for no fundamental reason; that is what makes it contagious.

**LO:** IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. •

**Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.b, Figure 91.3)

### Solution IM-9 b.3 — Answer: D

Bank Q448 • BT P2.T8.412.2 (key corrected in the bank; confirmed)

#### Step 1 – Identify the mechanism

Leverage makes funds sensitive to liquidity shocks. In a market-wide shock, margin calls and collateral demands force many levered funds to deleverage at the same time, and they sell similar, liquid assets.

#### Step 2 – What that does to correlations

Common forced selling pushes the returns of otherwise unrelated strategies together: a correlation spike. Schweser describes the same thing in the crisis: fire sales spread contagion and losses into asset classes that were otherwise uncorrelated.

$\Rightarrow$ D.

#### Interpretation

Seen from the investor, this is the investment-level risk of correlation spikes (IM-9 a.3). Seen from the system, it is the activities channel: one fund's forced selling becomes everyone's loss.

#### Why the other options are wrong

**A is incorrect.** Key person risk concerns one fund's manager; it cannot explain a simultaneous, market-wide loss of diversification.

**B is incorrect.** UBTI is a tax issue for tax-exempt investors, unrelated to how returns move together in a shock.

**C is incorrect.** Survivorship bias distorts measured history; it is not a mechanism that makes correlations rise during a shock.

**LO:** IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. •
**Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a and 91.b)

### Solution IM-9 b.4 — Answer: C

Bank Q452 • new bank question

#### Step 1 – Separate the three bank risks

Revenue risk is the loss of fee income; credit risk is on margin loans; counterparty risk is on derivatives.

#### Step 2 – Define the credit risk

Margin loans are secured by collateral, but the collateral can lose value faster than anticipated, leaving the bank undersecured.

$\Rightarrow$ C.

#### Interpretation

Collateral reduces credit risk; it does not remove it. In a spiral the collateral falls just when the borrower is weakest, which is why the bank's credit loss and the market's fire sale happen together.

#### Why the other options are wrong

**A is incorrect.** Absolute claim: collateral reduces but does not eliminate credit risk, and derivatives losses are a separate, counterparty risk.

**B is incorrect.** Sibling swap: the loss of fee income is revenue risk.

**D is incorrect.** Absolute claim: lower leverage reduces default risk but does not eliminate it, and collateral can still fall in value.

**LO:** IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.b, Systemic Risk From Bank Exposure to Hedge Funds)

### Solution IM-9 b.5 Answer: B

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – What Archegos was

A family office, not a hedge fund. Its total return swaps gave it exposures that closely resembled those of a highly leveraged hedge fund.

#### Step 2 – What it cost

Credit Suisse lost about USD 5 billion and other banks about USD 5 billion more; banks then required hedge funds to reduce leverage.
$\Rightarrow$ B.

#### Interpretation

The case is in the reading because the systemic exposure came from the structure of the trades, not from the legal form of the client. A bank that monitored only its "hedge fund" clients would have missed it.

#### Why the other options are wrong

**A is incorrect.** Definition error: Archegos was a family office using total return swaps with banks; the losses fell on the banks.
**C is incorrect.** Roles reversed: revenue risk needs no default (only the loss of business); counterparty risk arises when a counterparty defaults.
**D is incorrect.** Absolute claim: collateral does not remove credit risk, and the bank also has counterparty exposure on derivatives such as swaps.

**LO:** IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.b, Archegos)

### Solution IM-9 b.6 Answer: D

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – LTCM

The 1998 failure underscored the importance of monitoring liquidity and leverage, and showed that assets that typically behave independently can become highly correlated in extreme stress.

#### Step 2 – Amaranth

The 2006 failure raised the question of how systemic hedge funds are. The Bank of England concluded that key intermediaries contribute far more to systemic risk than hedge funds; critics disagree, pointing to leverage and nonlinear strategies.
$\Rightarrow$ D is the correct pairing.

#### Interpretation

Two cases, two verdicts: LTCM is the warning about leverage and correlation; Amaranth produced the Bank of England's view that the intermediaries, not the hedge funds, are the bigger systemic risk.

#### Why the other options are wrong

**A is incorrect.** Case swap: the Bank of England's conclusion followed Amaranth.
**B is incorrect.** Case swap: the correlation-in-stress lesson is LTCM's.
**C is incorrect.** Polarity flip and absolute claim: the Bank of England reached the opposite view,

and the debate is not settled.

**LO:** IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.b)

### Solution IM-9 b.7 — Answer: A

Bank Q450 • Schweser Module Quiz 91.1 Q2

#### Step 1 – List the three approaches

More conservative lending by banks; lower leverage and greater diversification within hedge fund portfolios; reasonable regulation.
⇒ B, C and D are on the list.

#### Step 2 – Read the qualifier

Excessively strict regulation can reduce the liquidity hedge funds provide as investors of last resort (for example in distressed securities), with unintended negative effects; it may even raise systemic risk.
⇒ Strict new regulation is the least likely to help.

#### Interpretation

The judgement in the paragraph is the word "reasonable". Any option that equates more regulation with less systemic risk has removed it, as option C of Workbook A24.2 does.

#### Why the other options are wrong

**B is incorrect.** Lower leverage reduces forced deleveraging and fire-sale contagion.
**C is incorrect.** Conservative lending (tighter margins and collateral) limits banks' credit and counterparty losses.
**D is incorrect.** Greater diversification lowers the chance that one strategy's failure forces an unwind.

**LO:** IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.b, mitigation)

IM-9 • Risk, Regulation and Organizational Structure • Vardeez: IMPORTANT

# IM-9 c

P1 CONCEPT

Summarize different approaches taken by regulators to mitigate risks related to hedge funds.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic (A24 shape S2; hedge funds twice in August 2026; Map #34 on 3 of 16 sheets).
**Vardeez:** regulatory approaches.

**Reported exam shapes**

- **S2** Hedge fund risks, systemic risk, regulation and structure. Option C of Workbook A24.2 (worked as Question IM-9 a.7), "the strictest feasible regulation", is the absolute claim this LO and IM-9 b teach you to reject.

**What the questions look like:** conceptual, with facts. Which law did what (Title IV of Dodd-Frank removed the private adviser exemption); which jurisdiction does what (China restricts short selling most; ESMA interprets the AIFMD, member states enforce it); the thresholds (100 investors, USD 5 million, 500 investors, 25%, 35 nonaccredited investors, USD 150 million and USD 1.5 billion for Form PF, USD 250 million in Singapore, EUR 100 million in the EU); and the three regulatory concerns.
**Filing note.** Vardeez puts the US exemptions under IM-9 d; Schweser files them under LO 91.c. They are taught here and recapped in IM-9 d as the structure's regulatory motive.

## 1. What regulators worry about

At a high level, regulators have three concerns, and each has its own tool:

| Concern | The worry and the tool |
|---|---|
| **Investor protection** | That hedge funds defraud investors. Mitigated by **restricting participation to investors presumed sophisticated** enough to evaluate hedge fund risks (accredited investors, qualified purchasers). |
| **Market integrity** | Insider trading and other abuses. Hedge funds are held to the **same rules and enforcement mechanisms** as the rest of the financial markets. |
| **Financial stability** | That hedge funds destabilise markets (IM-9 b). Addressed by **limiting leverage** and requiring **timely and accurate posting of collateral**. |

Regulation also works *indirectly*: the 2007 to 2009 crisis showed that hedge funds are regulated through their highly regulated counterparties, above all banks.

## 2. The United States: living inside the exemptions

The **Investment Company Act of 1940** requires mutual funds to register with the SEC, which brings regular reporting and limits on leverage, short selling and performance fees. Those limits would be impractical for hedge funds, which avoid registration through exemptions:

| Rule | Exemption or threshold |
|---|---|
| **'40 Act, Section 3(c)(1)** | Fewer than **100 investors** |
| **'40 Act, Section 3(c)(7)** | Only **qualified purchasers** (individuals with more than **USD 5 million** in investment assets); but a fund with **500 or more investors** must register under the Securities Exchange Act of 1934 |
| **Advisers Act 1940, private adviser exemption (until Dodd-Frank)** | A manager advising **fewer than 15** clients (funds) need not register; many firms structured themselves to stay below the threshold |
| **ERISA (1974)** | Benefit-plan participation kept **below 25%** of total fund assets |
| **Regulation D** | Private placements, no advertising to the public; only **accredited investors** (net worth of at least USD 2.5 million, or annual income of USD 250,000 in each of the prior two years with a reasonable expectation of maintaining it); **Rule 506** permits up to **35 nonaccredited** investors |

The United States and the United Kingdom have generally kept a favourable environment for hedge funds, but both take a strict view of **side letters** that could advantage some investors over others.

## 3. Waves of US regulation and Dodd-Frank Title IV

Regulatory interest came in waves (Figure 1). The SEC tried to require hedge fund registration in 2004; the US Court of Appeals for the District of Columbia overturned the rule in 2006. In 2007 the President's Working Group on Financial Markets rejected broad new regulation and encouraged voluntary guidelines. The Madoff fraud and the 2007 to 2009 crisis renewed scrutiny and led to the **Dodd-Frank Act of 2010**.

- **Title IV** (the Private Fund Investment Advisers Registration Act) **eliminated the private adviser exemption**: the 15-client threshold no longer matters, and hedge fund managers must **register with the SEC**.
- Advisers managing at least **USD 150 million** must register, maintain extensive records and make them available to the SEC, appoint a **chief compliance officer**, and undergo periodic SEC examinations.
- Funds report to the SEC *and* the **Financial Stability Oversight Council (FSOC)**, so regulators can monitor AUM, leverage, counterparty exposures and strategies.
- **Form PF** (operational and portfolio detail): filed **annually** by advisers with at least USD 150 million of AUM and **quarterly** by those with more than USD 1.5 billion.

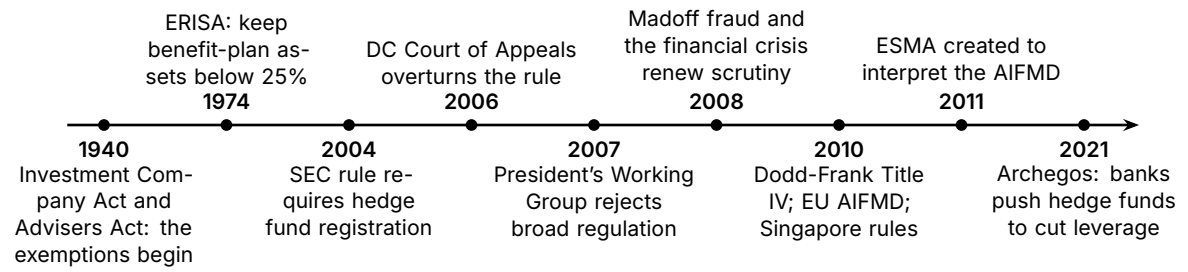



Figure 1. Waves of hedge fund regulation (schematic, equal spacing, not to scale in time). Labels alternate above and below the line. The US moved from exemptions (1940s) to a failed registration rule (2004 to 2006) to mandatory registration under Dodd-Frank (2010).

## 4. Other jurisdictions

- **European Union.** Much more restrictive than the US and UK. The **Alternative Investment Fund Managers Directive (AIFMD, 2010)** brought comprehensive rules for hedge funds and private equity for the first time: **mandatory registration, limits on leverage**, and detailed reporting, disclosure and marketing rules for EU and non-EU funds. The **European Securities and Markets Authority (ESMA, 2011) interprets** the AIFMD; **enforcement stays with the member states**. Funds managing more than **EUR 100 million** face extensive reporting on leverage, strategies, risk exposures and marketing. EU-domiciled funds use the **EU Passport** to manage and market across member states. Also: disclosure of net short positions, restrictions on naked short selling, mandatory central clearing of derivatives, reporting of all derivatives trades, measures to slow high-frequency trading. Emphasis varies: **Portugal** regulates derivatives markets closely, **France** stresses leverage limits, **Germany** makes it harder for banks to lend to highly leveraged vehicles.
- **Singapore (2010).** Hedge funds with more than **USD 250 million** must register with the Monetary Authority of Singapore (MAS), give investors and the MAS quarterly unaudited and annual audited reports, and hold a capital markets services licence.
- **Hong Kong.** The Securities and Futures Ordinance (SFO) defines hedge fund activities (dealing in securities, leveraged foreign exchange, futures); funds need a **licence for each activity**; the SFO sets out recommended reporting and disclosure and limits marketing to potential clients.
- **China.** Funds are government supported or private (the private segment is young). The China Securities Regulatory Commission (CSRC) **prohibits leverage** and restricts short selling to **50 securities of the CSI 300 Index**; brokers may not use client shares for short sales, so shorting costs roughly **10% a year**. The China Insurance Regulatory Commission (CIRC) has encouraged insurers to set up asset managers that run hedge-fund-like strategies.

**Correction to the source notes.** (1) Vardeez writes that China's CSRC "limits leverage and restricts short selling to certain CSI 300 securities; naked short selling is not allowed". Schweser: the CSRC *prohibits* the use of leverage, short selling is restricted to *50* CSI 300 securities, and shorting is expensive (about 10% a year) because brokers may not use client shares. The naked-short-selling restriction is in Schweser's EU paragraph, not its China paragraph. (2) Schweser states the private adviser exemption both as "fewer than 15 clients" (LO 91.a) and "fewer than 15 funds" (LO 91.c and its quiz key), and Vardeez repeats both. They describe one rule: the exemption counted clients, and each fund could be counted as a single client. Either wording can appear in a stem.

## 5. Mitigating systemic risk, and self-regulation

- **The three approaches** of IM-9 b: more conservative bank lending; lower leverage and more diversification in hedge funds; **reasonable** regulation. Overly strict rules can drain the liquidity hedge funds supply as investors of last resort.
- **Self-regulation.** The Alternative Investment Management Association (AIMA) published its *Guide to Sound Practices for Hedge Fund Valuation*.
- **Public listing.** When managers such as Man Group (London) or Blackstone (US) are publicly traded, the extra scrutiny gives investors more information and indirect access to the industry.

**Comparison card: IM-9 c**

| Jurisdiction | Instrument and threshold | What it does |
|---|---|---|
| **United States** | '40 Act exemptions; Dodd-Frank Title IV (2010); Form PF annual from USD 150m, quarterly above USD 1.5bn | Regulates the *adviser*: SEC registration, records, chief compliance officer, examinations, reports to the SEC and FSOC |
| **European Union** | AIFMD (2010), ESMA (2011); extensive reporting above EUR 100m; EU Passport | Registration, leverage limits, reporting, marketing rules; ESMA interprets, member states enforce |
| **Singapore** | MAS registration above USD 250m (2010) | Quarterly unaudited and annual audited reports; capital markets services licence |
| **Hong Kong** | Securities and Futures Ordinance | Licence for each activity; reporting and disclosure; limits on marketing |
| **China** | CSRC | No leverage; shorting limited to 50 CSI 300 names; brokers cannot lend client shares (about 10% a year) |

**Exam traps**

- **Which law removed the exemption.** Title IV of Dodd-Frank removed it; the Advisers Act of 1940 is where it lived. Reg D governs private placements; the AIFMD is EU law.
- **3(c)(1) against 3(c)(7).** 3(c)(1): fewer than 100 investors. 3(c)(7): qualified purchasers, with 500 or more investors triggering '34 Act registration.
- **ERISA direction.** Benefit-plan assets kept *below* 25%.
- **Rule 506.** Up to 35 nonaccredited investors, not any number.
- **Form PF frequency.** Annual from USD 150 million; quarterly above USD 1.5 billion.
- **ESMA's role.** It interprets the AIFMD; the member states enforce.
- **China.** Leverage is prohibited, not capped; short selling is limited to 50 names, not banned outright.
- **More is better.** The reading wants reasonable regulation; overly strict rules can raise systemic risk.

## Practice questions: IM-9 c

### Question IM-9 c.1

New • built from SchweserNotes 2026 Reading 91

A securities regulator's policy paper groups its hedge fund objectives under the three high-level concerns identified in the reading. Which of the following correctly matches each concern with the way it is addressed?

**A.** Investor protection: limiting leverage; market integrity: restricting participation to sophisticated investors; financial stability: monitoring insider trading.

**B.** Investor protection: restricting participation to investors presumed sophisticated enough to evaluate the risks; market integrity: the same insider-trading rules and enforcement as other markets; financial stability: limiting leverage and requiring timely, accurate posting of collateral.

**C.** Investor protection: mandatory central clearing; market integrity: Form PF filings; financial stability: a ban on short selling.

**D.** Investor protection: registration under the Investment Company Act of 1940; market integrity: caps on performance fees; financial stability: a cap on the number of investors.

### Question IM-9 c.2

Bank Q454 • Schweser Module Quiz 91.2 Q1 (rebuilt)

Before 2010, many US hedge fund managers avoided SEC registration under the private adviser exemption (advising fewer than 15 funds). Which of the following eliminated that exemption and now requires US-based hedge fund managers to register with the SEC?

**A.** Regulation D

**B.** The Investment Advisers Act of 1940

**C.** The Alternative Investment Fund Managers Directive

**D.** Title IV of the Dodd-Frank Act

### Question IM-9 c.3

Bank Q453 • Schweser Module Quiz 91.1 Q1

A global macro manager is deciding where to run a strategy that depends on short selling. In which of the following jurisdictions do hedge funds encounter the most significant restrictions on short selling?

**A.** China

**B.** Singapore

**C.** Hong Kong

**D.** The European Union

**Question IM-9 c.4** New • built from SchweserNotes 2026 Reading 91

The general counsel of a US hedge fund is reviewing how the fund stays outside the Investment Company Act of 1940 and the related US rules. Which of the following statements is correct?

**A.** A fund relying on Section 3(c)(1) may accept any number of investors, provided each is a qualified purchaser.

**B.** Under Regulation D, a fund sold by private placement may accept any number of nonaccredited investors, provided it does not advertise.

**C.** A fund relying on Section 3(c)(7) is limited to qualified purchasers (individuals with more than USD 5 million in investment assets) and must register under the Securities Exchange Act of 1934 if it has 500 or more investors.

**D.** To stay outside ERISA, a fund must keep benefit-plan investors above 25% of total fund assets.

**Question IM-9 c.5** New • built from SchweserNotes 2026 Reading 91

A US adviser manages USD 2 billion across several hedge funds. Before 2010 it advised fewer than 15 funds and relied on the private adviser exemption. Under the Dodd-Frank Act, which of the following best describes its obligations?

**A.** It remains exempt from SEC registration as long as it advises fewer than 15 funds.

**B.** It must register with the SEC and file Form PF annually; quarterly filing applies only to advisers below USD 150 million.

**C.** It reports to ESMA, which both interprets and enforces the rules for hedge funds marketed in the United States.

**D.** It must register with the SEC, appoint a chief compliance officer, keep records available to the SEC, face periodic examinations and file Form PF quarterly; its reports also reach the FSOC.

**Question IM-9 c.6** New • built from SchweserNotes 2026 Reading 91

A global allocator compares hedge fund regulation in four jurisdictions. Which of the following statements is correct?

**A.** Under the AIFMD, ESMA interprets the directive's provisions, while enforcement remains the responsibility of the individual EU member states.

**B.** Singapore requires every hedge fund, whatever its size, to register with the Monetary Authority of Singapore and to publish audited quarterly reports.

**C.** Hong Kong regulates hedge funds under a single licence covering all activities, with no limits on marketing to potential clients.

**D.** China's CSRC permits leverage of up to three times capital but bans short selling entirely.

## Solutions • IM-9 c

### Solution IM-9 c.1 — Answer: B

New • built from SchweserNotes 2026 Reading 91

**Step 1 – Investor protection**

The worry is fraud; the tool is restricting participation to investors presumed sophisticated enough to evaluate hedge fund risks.

**Step 2 – Market integrity and financial stability**

Market integrity: insider trading, policed with the same rules and enforcement as the rest of the markets. Financial stability: limiting leverage and requiring timely and accurate posting of collateral.

$\Rightarrow$ B.

**Interpretation**

Each concern has its natural tool: who may invest (protection), how trading is policed (integrity), how much leverage and collateral (stability). The distractors shuffle the tools.

**Why the other options are wrong**

**A is incorrect.** Tools swapped: leverage limits serve financial stability; restricting participation serves investor protection; insider-trading rules serve market integrity.

**C is incorrect.** True facts about the wrong concerns: central clearing and Form PF are stability and monitoring tools, and short selling is restricted (in China, to 50 names), not banned, in the reading.

**D is incorrect.** Polarity flip: hedge funds avoid '40 Act registration through exemptions; that registration is what they are structured to stay outside.

**LO:** IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.c, Regulatory Objectives)

### Solution IM-9 c.2 — Answer: D

Bank Q454 • Schweser Module Quiz 91.2 Q1

**Step 1 – Where the exemption lived**

The private adviser exemption was part of the Investment Advisers Act of 1940: managers advising fewer than 15 funds (clients) need not register.

**Step 2 – What removed it**

Title IV of the Dodd-Frank Act (the Private Fund Investment Advisers Registration Act) eliminated the exemption, so hedge fund managers must register with the SEC, report to the SEC and FSOC, and file Form PF.

$\Rightarrow$ D.

**Interpretation**

Keep the verbs apart: the Advisers Act *contained* the exemption; Dodd-Frank *removed* it.

**Why the other options are wrong**

**A is incorrect.** Regulation D governs private placements and who may invest (accredited investors); it says nothing about adviser registration.

**B is incorrect.** The Advisers Act is the law in which the exemption sat, not the provision that eliminated it.

**C is incorrect.** The AIFMD is the EU framework; it does not govern SEC registration of US managers.

**LO:** IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.c, Title IV)

### Solution IM-9 c.3 — Answer: A

Bank Q453 • Schweser Module Quiz 91.1 Q1

#### Step 1 – China's rules

The CSRC restricts short selling to only 50 securities of the CSI 300 Index, and brokers may not use client shares to facilitate short sales, which makes shorting cost roughly 10% a year. It also prohibits leverage.

#### Step 2 – Compare the others

Singapore's rules centre on registration (above USD 250 million), reporting and licensing; Hong Kong's on activity licences, reporting and marketing limits; the EU's on registration, leverage limits, reporting and disclosure of net short positions.
$\Rightarrow$ China.

#### Interpretation

China regulates the trade itself, by limiting which securities may be shorted at all. The US regulates the adviser; the EU regulates the manager and its leverage.

#### Why the other options are wrong

**B is incorrect.** Singapore's regime is about registration, reporting and licensing, not short-selling limits.
**C is incorrect.** Hong Kong's SFO focuses on licensing, reporting and marketing limits.
**D is incorrect.** The EU requires disclosure of net short positions and restricts naked short selling, which is far less restrictive than a 50-name list.

**LO:** IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a, Regulation)

### Solution IM-9 c.4 — Answer: C

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – The two '40 Act exemptions

Section 3(c)(1): fewer than 100 investors. Section 3(c)(7): only qualified purchasers (individuals with more than USD 5 million in investment assets), but 500 or more investors trigger registration under the Securities Exchange Act of 1934.
$\Rightarrow$ C is correct.

#### Step 2 – Regulation D and ERISA

Reg D allows only accredited investors, with Rule 506 permitting up to 35 nonaccredited investors. Funds stay outside ERISA by keeping benefit-plan participation below 25% of fund assets.

#### Interpretation

Every number here is a ceiling the fund is organised to stay under, which is why IM-9 d reads the structure as a set of constraints.

#### Why the other options are wrong

**A is incorrect.** Sibling swap: 3(c)(1) is the fewer-than-100-investors exemption; the qualified purchaser test belongs to 3(c)(7).
**B is incorrect.** Absolute claim: Rule 506 permits up to 35 nonaccredited investors, not any number.
**D is incorrect.** Polarity flip: benefit-plan assets must stay *below* 25% of the fund.

**LO:** IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.c, U.S. Regulatory Framework)

## Solution IM-9 c.5

**Answer: D**

New • built from SchweserNotes 2026 Reading 91

### Step 1 – Registration

Title IV eliminated the private adviser exemption, so the 15-fund threshold no longer helps. An adviser managing at least USD 150 million must register with the SEC, maintain extensive records and make them available, appoint a chief compliance officer and undergo periodic examinations.

### Step 2 – Reporting

Reports go to the SEC and the FSOC. Form PF is filed annually from USD 150 million of AUM and quarterly above USD 1.5 billion; at USD 2 billion the adviser files quarterly.
⇒ D.

### Interpretation

Dodd-Frank turned hedge fund oversight into a systemic-risk tool: the FSOC receives data on AUM, leverage, counterparty exposures and strategies.

### Why the other options are wrong

**A is incorrect.** Outdated: the exemption was eliminated by Title IV.
**B is incorrect.** Thresholds reversed: quarterly filing applies *above* USD 1.5 billion; USD 150 million is the threshold for annual filing.
**C is incorrect.** Jurisdiction swap: ESMA interprets the EU's AIFMD, and enforcement there sits with member states; it has no role for US advisers.

**LO:** IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a, Regulation; LO 91.c, Title IV and Form PF)

## Solution IM-9 c.6

**Answer: A**

New • built from SchweserNotes 2026 Reading 91

### Step 1 – The EU

The AIFMD (2010) brought registration, leverage limits and reporting, disclosure and marketing rules. ESMA, created in 2011, interprets its provisions; enforcement remains the responsibility of the individual member states.
⇒ A is correct.

### Step 2 – Check the others

Singapore: registration above USD 250 million, quarterly *unaudited* and annual *audited* reports. Hong Kong: a licence for each activity, and limits on marketing. China: leverage prohibited; short selling restricted to 50 CSI 300 securities.

### Interpretation

The EU row is a who-does-what distinction: one body interprets, many enforce. That split is why enforcement emphasis differs by country (Portugal, France, Germany).

### Why the other options are wrong

**B is incorrect.** Scope error: the requirement applies above USD 250 million, and the quarterly reports are unaudited (the audited report is annual).
**C is incorrect.** Polarity flip: licences are activity-specific, and the SFO places limits on marketing.
**D is incorrect.** Both halves wrong: the CSRC prohibits leverage, and short selling is restricted

to 50 securities, not banned.

**LO:** IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.a, Regulation; LO 91.c, International Regulatory Approaches)

IM-9 • Risk, Regulation and Organizational Structure • Vardeez: IMPORTANT

# IM-9 d

P1 CONCEPT

Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure.

Workbook: A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: A24 shape S2 covers structure, hedge funds came up twice in August 2026, and Map #34 records the section on 3 of 16 sheets. Vardeez does not flag this LO. **Vardeez:** no pointer for this LO.

**Reported exam shapes**

- **S2** Hedge fund risks, systemic risk, regulation and structure. Two options of Workbook A24.2 (worked as Question IM-9 a.7) are this LO: B puts pension funds in the wrong feeder, D swaps the general and limited partners.

**What the questions look like:** conceptual. Who is the general partner; which feeder a pension fund uses and why; why the limited partnership (carried interest); what domicile does and does not achieve; which service provider does what.

## 1. Domicile

Hedge funds choose their domicile to **minimise taxes and reduce regulatory oversight**. In the United States, **Delaware** is common for its favourable legal environment. Offshore funds are most often in the **Cayman Islands (55%)**, the British Virgin Islands (15%) and Bermuda (10%). Domicile can reduce taxes *at the fund level*, but the manager still pays tax on the fees it earns, and investors are still taxed on their gains in their home jurisdictions.

## 2. The limited partnership

- The usual legal form is a **limited partnership**: the **manager is the general partner**; the **investors are the limited partners**.
- It is normally **open-ended**: the fund can issue new interests to incoming investors and redeem those who want to leave, **subject to the lockups and gates** in the partnership agreement.
- The structure gives the investors liability protection, and it is designed to minimise both regulatory oversight and taxation.

## 3. The master-feeder structure

The **master fund** is where all capital is pooled and all trading takes place. It is funded by **feeder funds**, each built for a group of investors, typically one **onshore** and one **offshore**. For a US-domiciled fund the **offshore feeder** serves **non-US investors and US tax-exempt investors such as pension funds**; US taxable investors use the onshore feeder. The design avoids multiple layers of taxation and lets tax-exempt investors participate tax-efficiently: the offshore feeder is the UBTI mitigant of IM-9 a.

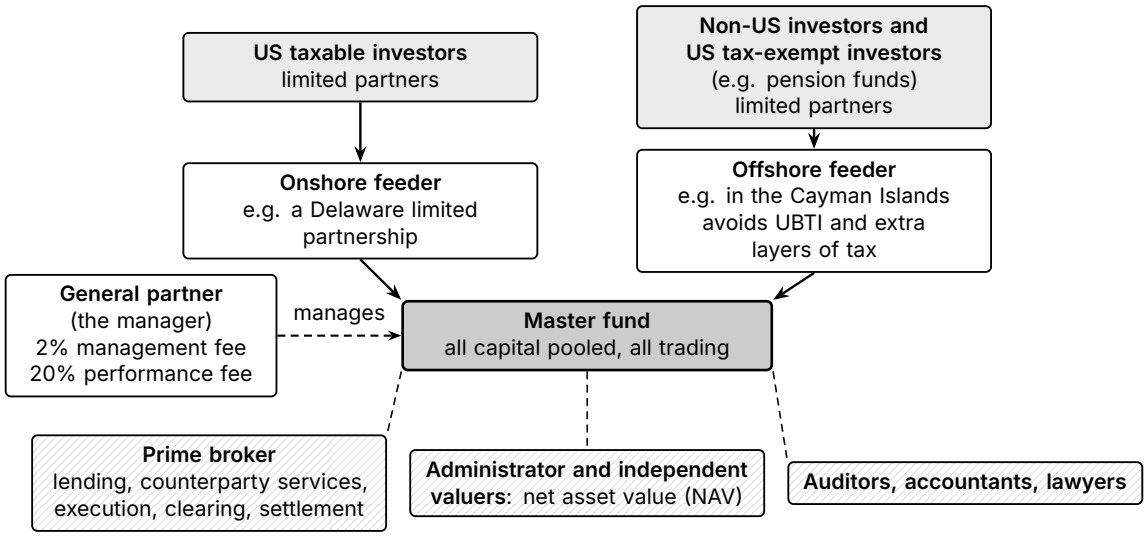



Figure 1. A typical master-feeder structure for a US-domiciled fund. Solid arrows carry capital from the investors (limited partners) through the feeders to the master fund, where all trading happens. The dashed arrow is the general partner's management role; dashed lines link the service providers (hatched).

## 4. Service providers

Running the structure needs specialised support: **accountants, lawyers, auditors, administrators** and **independent valuation professionals** to determine the fund's **net asset value** (**NAV**). A **prime broker** provides lending, counterparty services, trade execution, clearing and settlement.

## 5. Motivations: tax and regulation

**Tax.** The typical **2 and 20** fee model is a 2% management fee and a 20% performance fee.

- The **management fee** is taxed as **ordinary income**.
- The **performance fee** (**carried interest**) may be taxed at the lower **capital gains** rate if the underlying assets were held for at least one year. This "carried interest loophole" cuts the manager's tax on it by roughly half, and is a key incentive for the partnership form.
- Before 2008, managers could also **defer** tax on management fees under Internal Revenue Code **Section 457**, choosing when to receive them; that loophole was **eliminated in 2008**.
- Attempts to remove the carried interest treatment **failed in 2008 and again in 2021**; it remains in place but could change, and managers watch it closely.

**Regulation.** The structure keeps the fund inside the US exemptions of IM-9 c: fewer than 100 investors (3(c)(1)) or only qualified purchasers (3(c)(7)), private placement to accredited investors (Reg D), benefit-plan assets below 25% (ERISA), and, before Dodd-Frank, fewer than 15 funds per adviser. Read this way, the thresholds explain features that otherwise look arbitrary.

> **Correction to the source notes.** Vardeez's IM-9 d section carries the US exemptions ('40 Act, private adviser exemption, ERISA, Regulation D) and the Dodd-Frank history. Schweser files those under LO 91.c (they are taught in IM-9 c here). Vardeez's section omits the structure this LO asks for: domicile, the limited partnership, the master-feeder design, the service providers and the tax motivation. Sections 1 to 5 above are built from Schweser LO 91.d.

**Comparison card: IM-9 d**

| Feature | What it is | Motivation |
|---|---|---|
| **Domicile** | Delaware onshore; Cayman (55%), BVI (15%), Bermuda (10%) offshore | Lower fund-level tax, lighter oversight (managers and investors are still taxed) |
| **Limited partnership** | Manager is the general partner; investors are limited partners; open-ended, with lockups and gates | Liability protection; minimal oversight and tax; carried interest at capital gains rates |
| **Master-feeder** | Feeders (onshore and offshore) invest in a master that pools capital and trades | Avoid multiple layers of tax; tax-efficient access for non-US and US tax-exempt investors (no UBTI) |
| **Service providers** | Administrator and independent valuers (NAV); auditors, accountants, lawyers; prime broker | Independent NAV and audit; financing, execution, clearing and settlement |
| **2 and 20** | 2% management fee (ordinary income); 20% performance fee (carried interest) | Capital gains treatment if assets held at least a year; Section 457 deferral ended in 2008 |

**Exam traps**

- **Partners swapped.** The manager is the general partner; investors are the limited partners.
- **Feeder swapped.** US tax-exempt investors such as pension funds use the *offshore* feeder; US taxable investors use the onshore feeder.
- **Which fee gets capital gains treatment.** Only the performance fee (carried interest), and only for assets held at least a year; the management fee is ordinary income.
- **Deferral.** Section 457 deferral of management fees ended in 2008; it is not available now.
- **Domicile removes all tax.** It can reduce fund-level tax only; managers and investors are still taxed.
- **Where trading happens.** In the master fund, not the feeders.
- **Who computes NAV.** Administrators and independent valuation professionals, not the prime broker.

## Practice questions: IM-9 d

### Question IM-9 d.1

New • built from SchweserNotes 2026 Reading 91

A prospective investor asks how a typical hedge fund is organised legally. Which of the following descriptions is correct?

**A.** It is a closed-end corporation: the manager owns the voting shares and investors hold notes that cannot be redeemed.

**B.** It is an open-ended limited partnership in which the investors are the general partners and the manager is a limited partner.

**C.** It is an open-ended limited partnership: the manager is the general partner and the investors are limited partners, who can enter and redeem subject to the lockups and gates in the partnership agreement.

**D.** It is a limited partnership that cannot admit new investors after launch, because the partnership agreement fixes the number of partners.

### Question IM-9 d.2

Bank Q456 • new bank question (GARP practice-exam style)

The general partner of a US-domiciled hedge fund is explaining the master-feeder structure to a prospective investor, a US public pension plan. Which of the following statements best describes how the pension plan would most likely participate, and why?

**A.** It invests through the onshore feeder, because tax-exempt investors benefit from the onshore partnership's pass-through treatment.

**B.** It invests directly in the master fund, because only the master fund holds the pooled capital and executes trades for all investors.

**C.** It invests through the offshore feeder, because US investors are required to use offshore vehicles to avoid registration with the SEC.

**D.** It invests through the offshore feeder, which helps tax-exempt investors participate in a tax-efficient manner.

### Question IM-9 d.3

Bank Q455 • Schweser Module Quiz 91.2 Q2

A tax adviser is explaining to a new manager why most hedge funds are structured as limited partnerships. Which of the following is most likely an incentive for hedge funds to be structured as limited partnerships?

**A.** Management fees are taxed at capital gains rates.

**B.** Performance fees can be taxed at capital gains rates.

**C.** All fees paid to the general partner are treated as capital gains.

**D.** Management fees can be deferred indefinitely for tax purposes.

**Question IM-9 d.4** New • built from SchweserNotes 2026 Reading 91

A manager is choosing the domicile for a new family of funds. Which of the following statements about domicile is most accurate?

**A.** An offshore domicile can reduce taxes at the fund level, but the manager is still taxed on its fees and investors on their gains in their home jurisdictions.

**B.** An offshore domicile removes all tax: neither the manager's fees nor the investors' gains are taxed anywhere.

**C.** Bermuda hosts the majority of offshore hedge funds, followed by the British Virgin Islands.

**D.** The main purpose of an offshore domicile is to qualify the fund for the EU Passport.

**Question IM-9 d.5** New • built from SchweserNotes 2026 Reading 91

The chief operating officer of a new hedge fund is appointing service providers. Which of the following pairings of provider and role is correct?

**A.** Independent administrator: acts as the general partner and makes the investment decisions.

**B.** Auditor: provides margin financing for the fund's leveraged positions.

**C.** Prime broker: provides lending, counterparty services, trade execution, clearing and settlement.

**D.** Prime broker: determines the fund's net asset value independently of the manager.

**Question IM-9 d.6** New • built from SchweserNotes 2026 Reading 91

A consultant explains to a pension fund trustee why hedge funds are organised the way they are. Which of the following best summarises the motivations?

**A.** The structure is chosen to maximise transparency to regulators and investors, in exchange for lighter tax treatment.

**B.** The structure is chosen so that the fund can register under the Investment Company Act of 1940 and market to the general public.

**C.** The structure is chosen mainly so that the manager's management fee can be taxed at capital gains rates.

**D.** The partnership gives investors liability protection and, with the domicile and feeders, minimises oversight and tax; funds keep benefit-plan assets below 25% to stay outside ERISA.

## Solutions • IM-9 d

### Solution IM-9 d.1 — Answer: C

New • built from SchweserNotes 2026 Reading 91

**Step 1 – The legal form and the roles**

A limited partnership: the hedge fund manager is the general partner; the investors are the limited partners.

**Step 2 – Open-ended**

The partnership can issue new interests to incoming investors and redeem investors who want to leave, subject to the lockups and gates in the partnership agreement.

$\Rightarrow$ C.

**Interpretation**

The general partner runs the fund and earns the fees; the limited partners provide the capital and have limited liability. Lockups and gates are how an open-ended fund protects itself from redemptions it cannot meet.

**Why the other options are wrong**

**A is incorrect.** Wrong form: a hedge fund is typically an open-ended limited partnership, not a closed-end corporation.

**B is incorrect.** Roles swapped: the manager is the general partner and the investors are limited partners (option D of Workbook A24.2).

**D is incorrect.** Polarity flip: the fund is open-ended; it admits new investors and redeems existing ones, within the agreement's terms.

**LO:** IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.d, Organizational and Structural Design)

### Solution IM-9 d.2 — Answer: D

Bank Q456 • new bank question

**Step 1 – How investors reach the master**

The master fund pools the capital and trades; investors enter through feeders, each built for one group of investors.

**Step 2 – Which feeder**

For a US-domiciled fund, the onshore feeder serves US taxable investors; the offshore feeder serves non-US investors and US tax-exempt investors such as pension funds. The structure avoids multiple layers of taxation and lets tax-exempt investors participate tax-efficiently.

$\Rightarrow$ The pension plan uses the offshore feeder, for tax efficiency.

**Interpretation**

This is the UBTI mitigant of IM-9 a seen from the structure side: investing through an offshore fund keeps debt-financed income from becoming taxable to a tax-exempt investor.

**Why the other options are wrong**

**A is incorrect.** Feeder swapped: the onshore feeder is for US taxable investors (option B of Workbook A24.2).

**B is incorrect.** Investors do not normally invest in the master directly; the feeders exist to tailor access to each group.

**C is incorrect.** Right feeder, wrong reason: the motive is tax efficiency; SEC registration is a

manager-level requirement under the Advisers Act.

**LO:** IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.d, master-feeder structure)

### Solution IM-9 d.3 Answer: B

Bank Q455 • Schweser Module Quiz 91.2 Q2

#### Step 1 – The two fees

Under 2 and 20, the 2% management fee is taxed as ordinary income; the 20% performance fee (carried interest) may be taxed at the lower capital gains rate if the underlying assets were held for at least one year.

#### Step 2 – The deferral history

Deferral of management fees under Section 457 was eliminated in 2008.
$\Rightarrow$ The capital gains treatment of the performance fee is the incentive.

#### Interpretation

The "carried interest loophole" roughly halves the tax on the performance fee. Attempts to close it failed in 2008 and 2021, which is why managers still watch it closely.

#### Why the other options are wrong

**A is incorrect.** Sibling swap: the management fee is ordinary income; it is the performance fee that can get capital gains treatment.
**C is incorrect.** Absolute claim: only the performance fee can qualify, and only if the assets were held at least a year.
**D is incorrect.** Outdated: Section 457 deferral ended in 2008, and it was never indefinite.

**LO:** IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.d, Tax Motivations and Regulatory Considerations)

### Solution IM-9 d.4 Answer: A

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – Why funds choose a domicile

To minimise taxes and reduce regulatory oversight: Delaware onshore; the Cayman Islands (55%), the British Virgin Islands (15%) and Bermuda (10%) offshore.

#### Step 2 – What it cannot do

Domicile can reduce taxes at the fund level, but managers still pay tax on the fees they earn and investors are still taxed on their gains at home.
$\Rightarrow$ A.

#### Interpretation

Domicile shifts where the fund itself is taxed; it does not make the people behind it tax free. Tax-exempt access is the feeders' job (Section 3).

#### Why the other options are wrong

**B is incorrect.** Absolute claim: managers and investors remain taxable.
**C is incorrect.** Ranking error: the Cayman Islands host the majority (55%), ahead of the British Virgin Islands (15%) and Bermuda (10%).
**D is incorrect.** True fact about the wrong topic: the EU Passport is a benefit of EU-domiciled funds under EU regulation, not the purpose of an offshore domicile.

**LO:** IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.d, Organizational and Structural Design)

### Solution IM-9 d.5 — Answer: C

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – The prime broker

Stowell: the prime broker provides lending, counterparty services, trade execution, clearing and settlement.
⇒ C is correct.

#### Step 2 – The others

Administrators and independent valuation professionals determine NAV; auditors audit; the general partner (the manager) makes the investment decisions.

#### Interpretation

Independence is the point of the service providers: the NAV comes from administrators and independent valuers, not from the manager or the firm financing it. Madoff kept these functions in-house (IM-14).

#### Why the other options are wrong

**A is incorrect.** Role misassignment: the general partner is the manager; an administrator is an independent service provider.
**B is incorrect.** Role misassignment: financing comes from the prime broker; an auditor's job is the audit.
**D is incorrect.** Role misassignment: NAV is determined by administrators and independent valuation professionals.

**LO:** IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.d, service providers)

### Solution IM-9 d.6 — Answer: D

New • built from SchweserNotes 2026 Reading 91

#### Step 1 – The partnership's purpose

Schweser: the limited partnership provides liability protection and is designed to minimise both regulatory oversight and taxation. Domicile and the master-feeder design serve the same ends.

#### Step 2 – The regulatory thresholds

The fund is organised to stay inside the US exemptions: benefit-plan assets below 25% (ERISA), fewer than 100 investors or qualified purchasers only ('40 Act), accredited investors (Reg D).
⇒ D.

#### Interpretation

Read every structural feature as a motive: liability (limited partners), tax (carried interest, feeders, domicile) and regulation (the exemption thresholds).

#### Why the other options are wrong

**A is incorrect.** Polarity flip: the structure is designed to minimise oversight, and hedge funds are typically secretive.
**B is incorrect.** Polarity flip: hedge funds use exemptions precisely to avoid '40 Act registration, and they market through private placements, not to the public.

**C is incorrect.** Sibling swap: the management fee is ordinary income; only the performance fee can be taxed at capital gains rates.

**LO:** IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. • **Reference:** Stowell and Stowell 4e, Ch.14 (SchweserNotes 2026 Book 5, Reading 91, LO 91.c and 91.d)

IM-10 • The Rise and Risks of Private Credit • Vardeez: VERY IMPORTANT

# IM-10 c

P1 CONCEPT

Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic: "private credit" was reported from the August 2026 sitting, listed twice on the same consolidated recall sheet (A28, Map #49, 2 of 16 sheets). The material named there is the growth story, the valuation opacity and the financial-stability concerns: "cheap marks if read; zero if skipped".
**Vardeez:** private credit risks, in a VERY IMPORTANT chapter.

**Reported exam shapes**

- **S3** Valuation opacity: stale marks, deferred losses, NAV-based metrics.
- **S4** Financial-stability concerns and the IMF's policy fixes. The policy half is drilled in IM-10 d; Workbook A28.2, worked here, joins the two halves.

Shapes S1 and S2 of A28 (growth drivers and the BIS regression; bank versus BDC cost of capital) are CI-3 material in the Current Issues book. S5 and S6 belong to IM-11.

**What the questions look like:** all conceptual. A short scenario (a fund growing fast, retail money arriving through a semiliquid vehicle, a borrower paying interest in kind, NAVs that do not move while listed loan prices fall) and a "which risk" or "which statement is correct" stem. The wrong options name the neighbouring risk: credit for liquidity, leverage for interconnectedness, valuation for conduct.

## 1. Private credit in brief (background for IM-10 c and d)

Private credit is **nonbank corporate credit extended through direct, bilateral agreements**, excluding public securities and commercial bank lending. IM-10 a and b are P4 in this book; these are the facts the risk discussion leans on.

- **Lenders and vehicles.** Long-horizon institutions (pension funds, insurers, sovereign wealth funds, family offices) through closed-end funds (over 80% of the market), business development companies (BDCs, about 14%) and CLOs (about 5%). Retail money increasingly arrives through BDCs and semiliquid funds.
- **Borrowers.** Middle-market firms (annual revenues USD 100 million to 1 billion), often with weak earnings or high leverage and unable to get bank finance: riskier than high-yield or leveraged-loan issuers, but usually backed by a private equity (PE) sponsor.
- **Terms.** Floating rate; more collateral and covenants than a bank loan (limits on dividends, buybacks, asset sales); tailored terms such as deferred interest; a higher all-in cost.
- **Size and growth.** About USD 2.1 trillion of loans and undeployed capital in 2023, not far below the global high-yield bond market. It grew after 2007 to 2009 on low rates and stricter bank regulation (insurers entered because their capital charges are lighter than banks'), and now also lends to larger firms.

## 2. The assessment: moderate today, untested

**The shift.** Credit has moved from regulated, transparent channels (heavily regulated banks; public bonds with extensive disclosure) to lightly regulated funds whose loans are **nonrated, nontraded and nonstandard**. Risks are hard to spot in time, and there is not yet enough data to assess the full impact on financial stability.

**Today: moderate,** because of **reasonable matching of asset and liability durations** and **relatively low leverage**; liquidity and interconnectedness risks appear low. **Why it still matters:** the asset class has grown fast and has not been through a serious market correction; competition for deals invites weaker underwriting, thinner risk premiums and looser covenants; and rapid growth with little transparency and light oversight is what could make the risks systemic. An "assess" stem wants both halves: options calling the risks negligible, or an imminent systemic crisis, take one side.

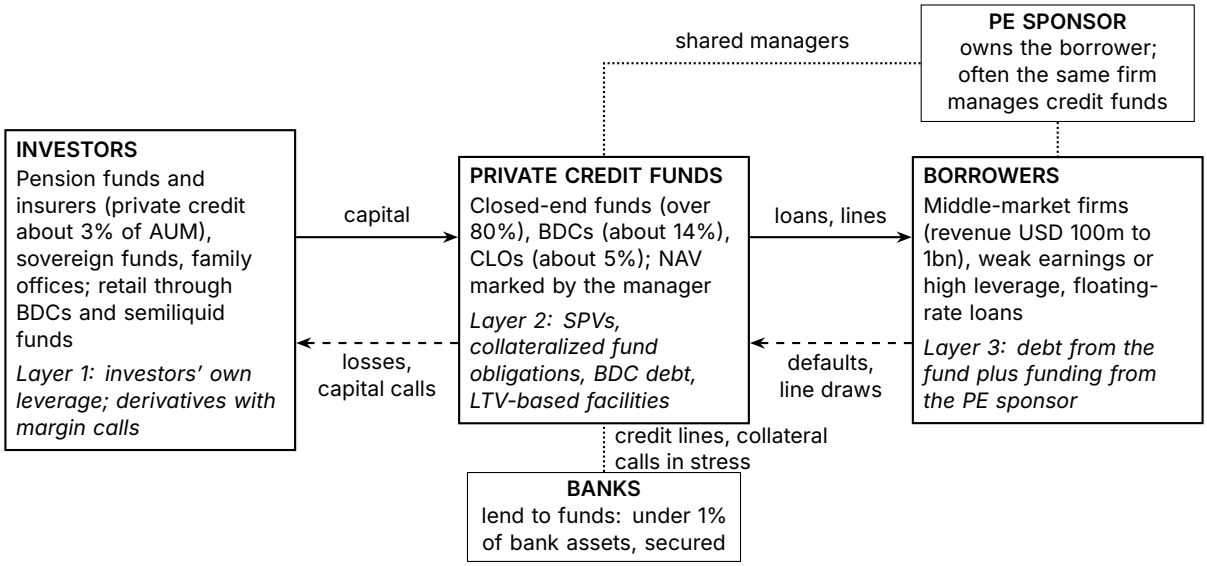



Figure 1. Where the risk sits. Solid arrows: money going in. Dashed arrows: how stress comes back (borrower defaults and simultaneous drawings on credit lines hit the fund; the fund passes losses and capital calls to its investors). Dotted lines: the links that make it an interconnectedness problem. Each box carries its own layer of leverage, and nobody sees all three.

## 3. The vulnerabilities, one at a time

### Credit

- **Mechanism.** Small, highly leveraged borrowers on floating-rate loans: rate rises go straight into interest bills, and a wave of defaults would cut lender profitability. Struggling borrowers are often given relief by adding unpaid interest to the debt (payment in kind, PIK): the claim grows, and if cash flow stays short the lender faces default on a larger balance.
- **Floating rate is a transfer, not a hedge:** it protects the lender's coupon by moving rate risk onto the borrower, where it returns as credit risk.
- **Track record.** Losses **lower than high-yield bonds and similar to leveraged loans** (PE sponsor support, collateral). Cyclicality is unsettled: funding stayed available in March 2020, but it rises and falls with equity returns and shrinks when banks tighten standards.

### Liquidity

- **Mitigants.** Long lockups; closed-end funds and CLOs do not allow withdrawals, which "greatly mitigates" the risk.
- **Semiliquid evergreen funds** take money continuously and redeem at set dates, with gates (a cap on outflows) and suspensions (deferred payment). Attracting retail money tends to need *more frequent* windows, which raises the risk again.
- **Credit lines to borrowers.** If many borrowers draw at once, the fund calls capital: the liquidity risk is **shifted to the investors**, whose uncalled commitments may exceed what they have invested, and who (pension funds) also face margin calls on derivatives.

### Leverage

Fund-level leverage looks modest, but it sits at **three levels at once** (Figure 1) and the data to add it up do not exist. *Investors* (pension funds, insurers) risk downgrades, and their margin and collateral calls in stress spill over. *Funds* lever through SPVs and collateralized fund obligations (limited partner interests moved into an SPV that issues debt); most closed-end funds are unlevered, BDCs are capped but have levered up, and loan-to-value limits bring collateral calls when asset values fall. *Borrowers* borrow from the fund and, in effect, from their PE sponsor. In a downturn defaults become fund losses and then investor losses, and large capital calls can make every lender cut back at once.

### Valuation (shape S3)

**Why the marks are weak:** few comparables (each loan is bespoke), infrequent valuations, no secondary market, subjective external valuers, and no accounting standard or regulation prescribing a method. **How they behave:** prices are less sensitive to shocks than high-yield bonds and leveraged loans but carry extra valuation discounts that grow in downturns; values move less and later than public debt, so price and NAV can differ for several quarters (Figure 2).

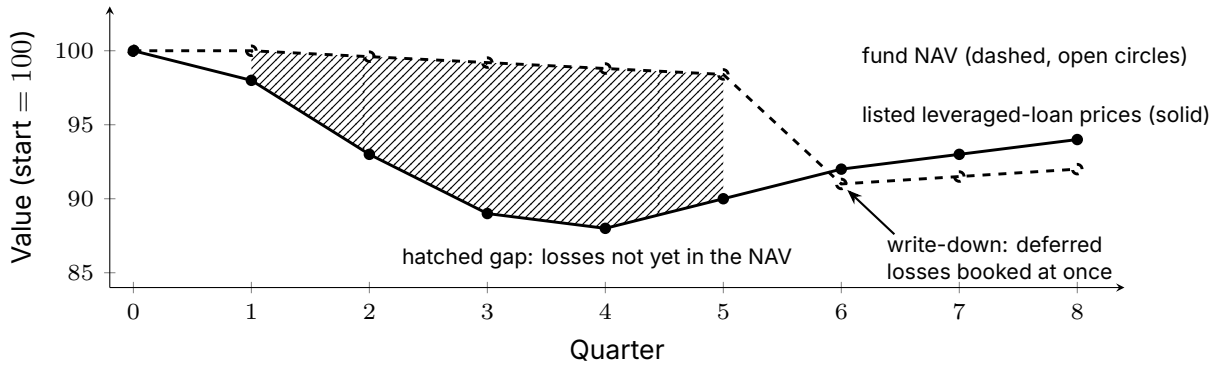



Figure 2. Stale marks, a stylised illustration (not data). While public loan prices fall, the manager's NAV barely moves; investors who leave in quarters 2 to 5 exit at an overstated value, returns and NAV-based fees are overstated, and the loss arrives in one quarter when it is finally recognised.

**What stale marks do.** (1) Overvaluation: better-informed investors can leave before write-downs (redemption limits keep this run risk small). (2) Overstated returns, useful for raising money and misleading to investors. (3) Overpaid fees, charged on overstated values. (4) Understated, deferred losses that can arrive all at once and trigger a wave of defaults. (5) Every NAV-based number inherits the problem, including RVPI and TVPI in IM-11 c. **More frequent marks are not automatically better**: the estimation error can add volatility and short-termism, which is why the IMF's fix (IM-10 d) is targeted external valuation and audit.

### Interconnectedness, underwriting and conduct

- **PE and private credit are intertwined:** many credit managers are PE managers and most borrowers are PE-backed, which lowers credit risk but lets weakness spill between the two markets and multiplies conflicts. Banks' exposure is under 1% of total assets and secured; pension funds and insurers hold about 3% of AUM. **PE-influenced life insurers** hold *more* illiquid assets and have *weaker* solvency ratios than peers, so a jump in defaults could wipe out their regulatory capital quickly.
- **The domino:** because borrowers, funds and investors all use leverage, one shortfall on a large capital call in stress can cascade into a cut in funding across the economy.
- **Underwriting:** the race for larger borrowers against banks and bond markets has produced loans priced too low for their risk and looser covenants, with large losses in store if defaults rise.
- **Conduct:** newer, less sophisticated investors may not be told the risks and redemption limits; conflicts arise when the fund is on both sides of a transaction.

## 4. From fund losses to financial stability

The IMF's channels read as a chain: fragile, floating-rate borrowers default together; insurers and pension funds with large, opaque holdings take sudden losses; less liquid funds face mass withdrawals from a wider investor base; layered leverage turns one capital-call shortfall into a domino; losses deferred by stale marks are recognised at once; and because many parts of the financial sector hold the asset, regulators cannot trace the spillovers.

**Correction to the source notes.** (1) Vardeez's trap summary says interconnectedness "is about overlapping parties, not overlapping borrowings". Schweser, reporting the IMF, ties them together: interconnectedness risk exists "due to the labyrinth of leverage with numerous participants (e.g., borrowers, funds, investors), each engaging in leverage transactions". Separate them this way: *leverage risk* is how much borrowing there is, hidden in layers; *interconnectedness risk* is who is linked to whom and how a loss travels along that leverage. Question IM-10 c.7 follows Schweser.

(2) Schweser's Module Quiz 92.2 Q2 asks which risk is "currently of greatest concern" and keys interconnectedness, although the reading says liquidity and interconnectedness risks "appear to be low". The bank rebuilt it to ask for the source of interconnectedness risk (Question IM-10 c.7).

(3) Schweser writes that funds "could be subject to minimum loan-to-value ratios". A fall in collateral value *raises* the loan-to-value ratio, so the limit that triggers a collateral call is a *maximum* loan-to-value ratio.

| Clue in the stem | Risk it points to | Not to be confused with |
|---|---|---|
| Rates rise; small, highly leveraged borrowers; floating-rate loans | Credit | Liquidity (funds and investors) |
| Unpaid interest added to the loan | Credit: relief that compounds the debt | Lower expected loss |
| Many borrowers draw their credit lines at once; capital calls | Liquidity, shifted to investors | Credit |
| Redemption windows, gates, suspensions, retail inflows | Liquidity | Conduct (how the product is sold) |
| SPVs, collateralized fund obligations, BDC debt, LTV triggers | Leverage | Fund-level leverage alone |
| PE sponsor also runs the credit fund; insurers and pensions; domino effect | Interconnectedness | Credit (sponsors *lower* it) |
| Flat NAV while listed loan prices fall; fees on NAV | Valuation | Liquidity |
| More capital chasing deals; thin spreads; loose covenants | Underwriting | Credit (the later consequence) |
| Retail investors who do not understand redemption limits | Conduct | Liquidity |

**Exam traps**

- **Absolute assessment.** The IMF view is "moderate now, untested, potentially systemic". Options saying the risks are negligible, or already systemic, take one side.
- **Two comparisons in one sentence.** Losses are *lower* than high-yield bonds and *similar* to leveraged loans. Cyclicality is left unsettled: evidence points both ways.
- **Closed-end is the mitigant, not the source,** of liquidity risk, but it does not remove the capital-call channel from borrowers' credit-line drawings.
- **Floating rates and PIK.** Floating rates turn the lender's rate risk into the borrower's credit risk; PIK relief raises the lender's exposure and expected loss.
- **Low fund-level leverage** is not low system leverage: investors and borrowers lever too.
- **Numbers.** Banks under 1% of assets, secured; pension funds and insurers about 3% of AUM; PE-influenced life insurers hold more illiquid assets *and* have weaker solvency ratios.
- **Stale marks** overstate value, returns and fees and understate (defer) losses. More frequent marks are not automatically the cure.

## Practice questions: IM-10 c

### Question IM-10 c.1

Bank Q479 • AP Mock 1 Q74

A private credit fund is evaluating a potential loan to a company in a cyclical industry. Which loan feature is particularly important to consider in order to mitigate potential downside risk?

**A.** Floating interest rates.

**B.** Closed-end fund structure.

**C.** Enhanced covenants.

**D.** Short loan maturity.

### Question IM-10 c.2

Bank Q472 • AnalystPrep Q.6384 (rebuilt)

Private credit borrowers tend to be small and highly leveraged, and most private credit loans are floating rate. Which risk is most directly raised by this borrower profile when interest rates rise?

**A.** Credit risk: higher debt-service burdens could trigger a large number of borrower defaults.

**B.** Liquidity risk: investors may rush to withdraw from semiliquid funds.

**C.** Valuation risk: the lack of price discovery for unquoted loans.

**D.** Conduct risk: unsophisticated investors may not understand fund restrictions.

### Question IM-10 c.3

Bank Q474 • AnalystPrep Q.6389 (rebuilt)

Frontier Investments, a private credit fund, is considering lending to a highly leveraged private-equity-backed company on a payment-in-kind (PIK) basis, so unpaid interest is added to the loan principal. Which statement best describes the main credit risk from this PIK structure?

**A.** The outstanding debt grows, and the lender faces default if cash flow stays insufficient to pay interest and reduce the debt.

**B.** Cash interest is received earlier, which accelerates the lender's recovery of capital.

**C.** Deferral of interest lowers expected loss because the borrower gets temporary relief.

**D.** The borrower's leverage declines over the life of the loan.

### Question IM-10 c.4

Bank Q477 • Schweser Module Quiz 92.2 Q1

Historically speaking, which of the following asset classes has experienced the highest level of losses?

**A.** Leveraged loans.

**B.** High-yield bonds.

**C.** Private credit loans.

**D.** Broadly syndicated loans.

**Question IM-10 c.5** Bank Q470 • AP Mock 2 Q47

Vanguard Credit, a private credit fund investing primarily in loans to technology startups, is experiencing rapid growth in assets under management. Which potential risk is most likely to increase for the fund in this scenario?

**A.** Stale valuations due to a lack of market prices.

**B.** Deteriorating underwriting standards due to increased competition.

**C.** Increased exposure to interest rate risk due to floating rate loans.

**D.** Liquidity risk due to the use of closed-end fund structures.

**Question IM-10 c.6** New • built from SchweserNotes 2026 Reading 92

A risk analyst at a Canadian public pension fund reviews the fund's commitment to a closed-end private credit fund. About 40% of the commitment is still uncalled. The credit fund has granted revolving credit lines to many of its middle-market borrowers. The pension fund's investment committee asks how the credit fund's liquidity risk could reach the pension fund. Which answer is most consistent with the IMF's analysis of private credit?

**A.** It cannot: because the closed-end fund permits no withdrawals, liquidity risk is eliminated for the fund and for its investors.

**B.** If many borrowers draw on their credit lines at the same time, the fund may need to call capital, shifting the liquidity strain to investors such as the pension fund.

**C.** Mainly through the floating-rate coupons on the loans, which reset lower when policy rates rise and cut the cash the fund distributes.

**D.** Mainly through other limited partners redeeming early, because closed-end private credit funds offer quarterly redemption windows.

**Question IM-10 c.7** Bank Q478 • Schweser Module Quiz 92.2 Q2 (rebuilt, key corrected)

Which of the following best describes the main source of interconnectedness risk in corporate private credit?

**A.** Multiple layers of leverage among borrowers, funds and investors, combined with close ties between private equity and private credit.

**B.** The predominance of floating-rate loans in fund portfolios.

**C.** The use of gates and suspensions on semiliquid fund redemptions.

**D.** Infrequent valuation of unquoted loans.

**Question IM-10 c.8** Bank Q471 • AnalystPrep Q.6394

Regulators are concerned about the growing participation of retail investors in private credit, particularly through business development companies (BDCs). Which of the following is most likely a key risk associated with this trend?

**A.** Retail investors may not fully understand private credit risks.

**B.** Lower returns for institutional investors.

**C.** Their participation will increase the demand for more standardized loan terms.

**D.** A lower amount of dry powder in the market.

**Question IM-10 c.9** Bank Q475 • AnalystPrep Q.6390 (rebuilt)

A regulator worries that a sudden downturn will hit institutional investors with large private credit allocations. Which of the following is a key vulnerability that could amplify the downturn's effects by delaying the recognition of losses?

**A.** Infrequent, model-based valuations that lag the true value of the loans.

**B.** Diversification of private credit portfolios across sectors.

**C.** Closed-end fund structures that do not permit withdrawals.

**D.** Strong covenants that give lenders early warning.

**Question IM-10 c.10** Workbook A28.2 • S3, S4 • Aug-2026 • Map #49

A senior risk consultant is advising a Brazilian securities regulator on a draft framework for private credit funds, which are increasingly sold to retail investors through semiliquid evergreen structures with quarterly redemption windows. The funds' loans have no secondary market and are marked by the managers each quarter, and their reported NAVs have barely moved through a period in which listed leveraged-loan prices fell. Which of the following recommendations is consistent with the IMF's analysis of private credit risks?

**A.** Permit more frequent redemption windows for retail investors, since more frequent redemptions reduce both the liquidity risk and the conduct risk of these funds.

**B.** Where valuations are significant or prone to error, require more frequent external valuations and audits of valuations, since stale marks can defer losses and overstate fees.

**C.** Rely on leverage reported at the fund level, since most closed-end private credit funds use little leverage and leverage across the private credit system is therefore low.

**D.** Leave monitoring of the loans' credit quality to external rating agencies, so that funds holding large private credit exposures need not run bank-like monitoring of their own.

## Solutions • IM-10 c

### Solution IM-10 c.1 — Answer: C

Bank Q479 • AP Mock 1 Q74

**Step 1 – Name the risk**

A cyclical borrower's earnings fall in a downturn, so the lender's exposure is credit risk: the borrower may be unable to service the loan.

**Step 2 – Find the loan-level protection**

Private credit loans carry more, and stricter, covenants than bank loans: limits on dividends, share buybacks and asset sales. They keep cash in the business and give the lender an early warning and control rights before losses build.

$\Rightarrow$ Enhanced covenants are the lender's contractual defence against a deteriorating borrower.

**Interpretation**

Covenants and collateral, with PE sponsor support, are why private credit losses have run below high-yield bonds. The other side of the same coin is IM-10 c's underwriting risk: competition for deals has been loosening covenants.

**Why the other options are wrong**

**A is incorrect.** Floating rates protect the lender's coupon against rising rates, but they raise the borrower's interest bill in exactly the stress the question describes: they add credit risk, they do not mitigate it.

**B is incorrect.** A closed-end structure is a fund feature that mitigates liquidity risk. It is not a loan feature and does nothing for this borrower's default risk.

**D is incorrect.** A short maturity forces a cyclical borrower to refinance, possibly in a downturn: it adds refinancing risk.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.a and 92.c

### Solution IM-10 c.2 — Answer: A

Bank Q472 • AnalystPrep Q.6384

**Step 1 – Read the profile**

Small, highly leveraged borrowers have thin cushions. Floating-rate loans pass every rate rise straight into their interest bill.

**Step 2 – Follow it to a loss**

Higher debt service on weak earnings means more defaults, which cut lender profitability.

$\Rightarrow$ The profile raises credit risk, the first item on the IMF's list of possible systemic risks.

**Interpretation**

Floating rates are a transfer of rate risk from lender to borrower, and it returns to the lender as default risk. Borrower PIK relief (Question IM-10 c.3) is how the stress often shows up first.

**Why the other options are wrong**

**B is incorrect.** Liquidity risk comes from the fund's redemption terms and credit lines, not from the borrower's leverage.

**C is incorrect.** Valuation risk comes from the lack of price discovery and transparency; it would hide these losses, not cause them.

**D is incorrect.** Conduct risk is about selling to unsophisticated investors, unrelated to the bor-

rower's balance sheet.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c (Credit Risks)

### Solution IM-10 c.3 — Answer: A

Bank Q474 • AnalystPrep Q.6389

**Step 1 – What PIK does**

The unpaid interest is added to the outstanding debt. The borrower gets temporary relief; the lender gets a larger claim instead of cash.

**Step 2 – Where the credit risk sits**

The balance, and the borrower's leverage, grow over time. If cash flow stays too low to pay interest and reduce the debt, the lender faces default on the larger balance.
$\Rightarrow$ PIK postpones the problem and enlarges it.

**Interpretation**

The IMF notes that borrowers hit by higher floating rates are often given exactly this relief. It also feeds valuation risk: a loan that accrues PIK interest can be carried at par while its cash coverage deteriorates.

**Why the other options are wrong**

**B is incorrect.** Reverses the mechanism: PIK defers cash interest, so cash arrives later, not earlier.
**C is incorrect.** Relief for the borrower is not relief for the lender: a bigger claim on the same weak cash flow raises expected loss.
**D is incorrect.** Accrued interest increases the debt, so leverage rises.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c (Credit Risks)

### Solution IM-10 c.4 — Answer: B

Bank Q477 • Schweser Module Quiz 92.2 Q1

**Step 1 – Recall the ranking**

Private credit losses have been lower than those on high-yield bonds and similar to those on leveraged loans.

**Step 2 – Explain it**

PE sponsors support their borrowers and most private credit loans require collateral, both of which have reduced defaults and losses.
$\Rightarrow$ High-yield bonds have the highest losses of the four.

**Interpretation**

Keep the two comparisons apart: *lower* than high yield, *similar* to leveraged loans. Schweser adds that broadly syndicated loans, with a relatively deep secondary market and investment-grade borrowers among them, likely have the lowest losses of the four. (Leveraged loans are themselves syndicated loans to highly leveraged borrowers.)

**Why the other options are wrong**

**A is incorrect.** Leveraged loans have losses similar to private credit, below high yield.
**C is incorrect.** Private credit losses are below high yield, thanks to collateral and sponsor support.

**D is incorrect.** Broadly syndicated loans are likely the lowest of the four, not the highest.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c (Credit Risks)

### Solution IM-10 c.5 Answer: B

Bank Q470 • AP Mock 2 Q47

#### Step 1 – What rapid growth changes

More money under management means more capital to deploy into a limited pool of borrowers.

#### Step 2 – The consequence the IMF names

Competition for deals leads lenders to cut spreads, loosen covenants and relax credit standards: loans priced too low for their risk.

⇒ Underwriting risk rises with growth, and with it expected losses in the next downturn.

#### Interpretation

The other three features are structural: they are there whether the fund grows or not. Only underwriting quality is a function of how much capital is chasing deals, which is why the IMF ties the rapid rise of private credit to it.

#### Why the other options are wrong

**A is incorrect.** Stale valuations are a permanent feature of unquoted loans; growth does not make them worse.

**C is incorrect.** Floating-rate loans reduce the lender's interest rate risk; the risk moves to the borrower. Growth does not change that.

**D is incorrect.** Closed-end funds do not permit withdrawals, so they mitigate liquidity risk.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c (Underwriting Standards)

### Solution IM-10 c.6 Answer: B

New • built from SchweserNotes 2026 Reading 92

#### Step 1 – What the closed-end structure does

Closed-end funds and CLOs do not permit withdrawals, which greatly mitigates the redemption side of liquidity risk.

#### Step 2 – The channel it does not close

Revolving credit lines to borrowers are a call on the fund's cash. If many borrowers draw at the same time, the fund may need extra funding and can call it from investors' uncalled commitments.

⇒ The liquidity risk is shifted to the investors, here a pension fund with 40% of its commitment still uncalled.

#### Interpretation

The IMF adds that investors' uncalled commitments may exceed their current private credit holdings, and that pension funds already face margin and collateral calls on their derivatives. A capital call arriving in the same stress is how a fund's liquidity problem becomes an investor's.

#### Why the other options are wrong

**A is incorrect.** Absolute qualifier: closed-end structures mitigate redemption risk, but they do not eliminate the capital-call channel.

**C is incorrect.** Floating-rate coupons reset *higher* when rates rise, and their effect runs through the borrowers' credit risk, not through the fund's liquidity.
**D is incorrect.** Closed-end funds do not offer redemption windows; scheduled redemptions are a feature of semiliquid evergreen funds.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c (Liquidity Risks; Interconnectedness Risks)

### Solution IM-10 c.7 — Answer: A

Bank Q478 • Schweser Module Quiz 92.2 Q2

#### Step 1 – Where the links are

Many private credit managers are PE managers and many borrowers are PE-backed, so weakness in one market can spill into the other.

#### Step 2 – What carries the shock

Borrowers, funds and investors each use leverage. A large capital call in stress, and a shortfall at one participant, can have a domino effect that reduces funding across the economy.
⇒ Interconnectedness risk is the layered leverage plus the PE and private credit ties.

#### Interpretation

Sponsor backing lowers credit risk while raising interconnectedness risk. The original Schweser quiz asked which risk is "currently of greatest concern" and keyed interconnectedness, which the reading does not support (it calls that risk low today); see the correction box in Section 4.

#### Why the other options are wrong

**B is incorrect.** Floating-rate loans drive borrower credit risk, not the links between participants.
**C is incorrect.** Gates and suspensions are liquidity-management tools.
**D is incorrect.** Infrequent valuation is valuation risk.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c (Interconnectedness Risks)

### Solution IM-10 c.8 — Answer: A

Bank Q471 • AnalystPrep Q.6394

#### Step 1 – Who is arriving

BDCs and semiliquid funds open private credit to retail investors, who are less sophisticated than the institutions the asset class was built for.

#### Step 2 – Name the risk

The credit and liquidity risks, and the limits on withdrawals, may not be properly explained to them.
⇒ This is conduct risk, the last item on the IMF's list.

#### Interpretation

Conduct risk sits next to liquidity risk: the same retail inflows that raise conduct risk also push funds toward more frequent redemption windows. IM-10 d's fix is disclosure and suitability controls.

#### Why the other options are wrong

**B is incorrect.** No link between retail inflows and institutional returns is drawn.
**C is incorrect.** Standardised loan terms would be a benefit, not a risk, and retail demand does

not set loan terms.

**D is incorrect.** Dry powder is not one of the IMF's risks; if anything, retail inflows add to it.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c

### Solution IM-10 c.9 — Answer: A

Bank Q475 • AnalystPrep Q.6390

#### Step 1 – Why the marks lag

No comparables, no secondary market, infrequent valuations and minimal regulatory monitoring: loan values move less and later than public debt.

#### Step 2 – Why the lag amplifies a downturn

Managers can defer unrealised losses as long as possible. When they must be realised they can arrive all at once, which could trigger a wave of defaults.

$\Rightarrow$ Infrequent, model-based valuations delay loss recognition and concentrate it (Figure 2).

#### Interpretation

This is shape S3: stale marks, deferred losses. Meanwhile investors misjudge their risk, returns look better than they are, and fees are charged on overstated values.

#### Why the other options are wrong

**B is incorrect.** Diversification spreads losses; it is a mitigant.

**C is incorrect.** Closed-end funds restrict redemptions and mitigate liquidity risk; they do not delay loss recognition.

**D is incorrect.** Strong covenants give earlier warning, which works against delay.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c (Valuation Risks)

### Solution IM-10 c.10 — Answer: B

Workbook A28.2 • S3, S4 • Aug-2026

#### Step 1 – Diagnose the symptom

Flat NAVs while public loan prices fall are stale valuations: few comparables, infrequent marks, no secondary market. Private credit values change less and respond more slowly than public debt, so price and NAV can diverge for several quarters.

$\Rightarrow$ Stale marks may overstate asset values and returns, inflate manager fees, and defer losses that are later recognised all at once.

#### Step 2 – Match the IMF policy fix

Valuation: greater regulatory oversight of valuation methods; more frequent external valuations where valuations are significant and/or prone to error; audits of valuations to fix deficiencies in methods and controls.

#### Step 3 – Check the retail angle

The liquidity fixes are closed-end structures, less frequent redemptions, longer notice and settlement, liquidity tools and stress tests, and clear disclosure. The conduct fix is to disclose conflicts and make sure retail investors understand the credit, liquidity and redemption restrictions.

#### Interpretation

B is correct. The regulator's framework should pair the valuation fix with the liquidity and con-

duct fixes, because evergreen retail funds combine all three risks. Note that the fix is targeted (significant or error-prone valuations), not "mark everything more often": more frequent marks full of estimation error can add volatility and short-termism.

### Why the other options are wrong

**A is incorrect.** Polarity flip: more frequent redemptions increase liquidity risk, and can raise conduct risk because managers may manipulate valuations to discourage redemptions.

**C is incorrect.** Scope error: low leverage in closed-end funds does not extend to the system; leverage is layered at investor, fund (SPVs, collateralized fund obligations) and borrower level, and fund-level reports can obscure it.

**D is incorrect.** Role misassignment: the IMF wants funds with substantial private credit to run more rigorous, bank-like monitoring themselves; reliance on internal valuations and rating agencies is the weakness, not the fix.

**LO:** IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; Schweser-Notes 2026 Book 5, Reading 92, LO 92.c and 92.d

IM-10 • The Rise and Risks of Private Credit • Vardeez: VERY IMPORTANT

# IM-10 d

P1 CONCEPT

Assess potential policy recommendations that could help mitigate the risks associated with private credit.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: the IMF's policy fixes are half of shape S4 in A28, from the August 2026 recall of "private credit" (Map #49, 2 of 16 sheets).
**Vardeez:** no separate pointer; the objective sits in a VERY IMPORTANT chapter, and Vardeez gathers it from the "how to mitigate it" blocks.

**Reported exam shapes**

- **S4** Financial-stability concerns and the IMF's policy fixes. The risks themselves are in IM-10 c; Workbook A28.2 (worked as Question IM-10 c.10) asks which recommendation is consistent with the IMF's analysis.

**What the questions look like:** a regulator, supervisor or bank credit committee facing one of the six risks, and four candidate responses. One matches the IMF; the others reverse it (more frequent redemptions), stretch it (caps now, everywhere), or hand the job to the wrong party (rating agencies, the fund's own unaudited marks).

## 1. The overall approach

Recap from IM-10 c: the risks look moderate today but untested, opaque and spread across sectors and borders. The policy response follows from that:

- **Stronger oversight, globally.** The US and the EU (and jurisdictions in Asia) have expanded their rules, focusing on **conflicts, conduct, valuation and disclosures**.
- **The most effective method** is stronger regulation and supervision on a **cross-sectoral and cross-border** basis: authorities in different jurisdictions working together to understand the risks of private credit and its effects on other asset classes and on systemic institutions.
- **Six risks to address:** credit, liquidity, leverage, valuation, interconnectedness and conduct. (Underwriting, a vulnerability in IM-10 c, is covered under credit: scrutiny of underwriting policies.)

## 2. Six risks, six fixes

Each row pairs the gap the IMF sees with its recommendation; Figure 1 then shows who carries out each fix.

| Risk | The gap the IMF sees | Recommendation |
|---|---|---|
| **Credit** | Insurers and pension funds have no requirement to monitor the loans' credit quality; they rely on internal valuations and rating agencies, which give little confidence. | Funds with substantial private credit adopt more rigorous, **bank-like monitoring** of loan performance in structured products and direct lending; careful scrutiny of underwriting policies and credit risk. |

| Risk | The gap the IMF sees | Recommendation (continued) |
|---|---|---|
| **Liquidity** | Open-ended structures and generous redemption terms on illiquid assets; unsophisticated investors may all try to leave in stress. | Regulators allow **only closed-end funds** or **less frequent redemptions**; **longer notice** and **more time to settle**; mandatory **liquidity management tools and stress tests** where mismatches are large; thorough, understandable **disclosure** of risks and redemption limits. |
| **Leverage** | Reported leverage omits or obscures SPVs and feeder or master funds; data gaps and inconsistent reporting across countries and sectors. | **Banks lending to private credit firms** require proof of strong liquidity management: **stress tests** of constrained funding, of write-downs on leveraged assets, and of large, abrupt drawings by the firms' borrowers. More **reporting** within and across borders, regulator cooperation, and **leverage caps if** leverage becomes systemically dangerous. |
| **Valuation** | Regulation does not address valuation; each investor values its own assets, so one asset can carry very different values. | **Oversight of valuation methods**; **more frequent external valuations** where valuations are significant and/or error-prone; **audits** of valuations and controls; write-downs of illiquid and semiliquid assets recognised accurately and in the right period. |
| **Interconnect-edness** | Regulatory arbitrage: risk migrates to the countries and sectors with the laxest rules, which may explain large insurer and pension holdings; little data on the links. | **Harmonise risk assessment** across sectors and borders; cross-border **cooperation** where risks are concentrated; regulators within each country cooperate across sectors on data, to reduce contagion. |
| **Conduct** | More unsophisticated investors; conflicts where the fund is on both sides of a transaction, without market pricing as a check. | Conduct supervisors identify the risks and require **disclosure of conflicts**; put investors' interests first (fair sales practices, suitability); controls so that retail investors understand credit and liquidity risks and redemption limits. |

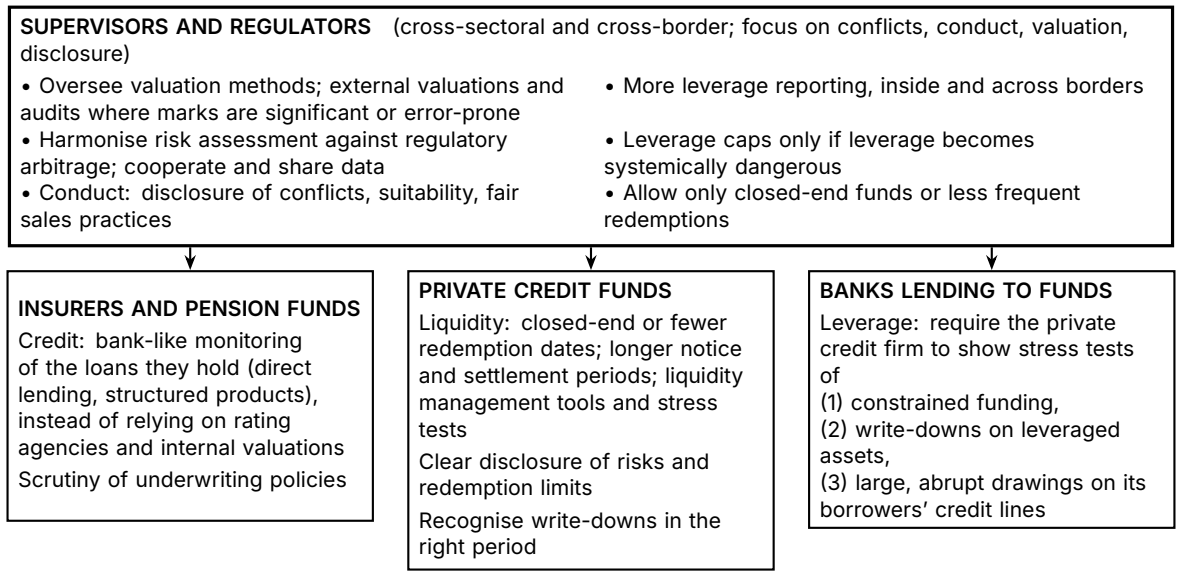



Figure 1. Who does what. The supervisors (top box, heavy outline) set the frame; the three boxes below show the party that carries out each fix. A wrong option usually gives the job to the wrong box, for example credit monitoring to rating agencies or valuation to the fund's own unaudited marks.

## 3. Assessing the fixes

The LO verb is "assess": know what each fix does, what it costs, and what is *not* recommended.

- **Most fixes are about seeing the risk** (reporting, cooperation, harmonisation, audits, disclosure), the answer to IM-10 c's root problem: loans that are not traded, rated or observed.
- **The liquidity fixes are structural.** Closed-end only, fewer redemption dates, longer notice and settlement change the product; the cost is less liquidity for the retail investors semiliquid funds were built to attract.
- **More frequent redemptions push two risks the wrong way:** liquidity mismatch, and conduct risk (managers may manipulate valuations to discourage redemptions, or make redemptions conditional on reinvesting in their other products).
- **Caps and marks are targeted.** Leverage caps only if leverage becomes systemically dangerous; external valuations where marks are significant or error-prone, plus audits. Marking everything more often is not the recommendation: noisy marks add volatility and short-termism.
- **Not on the list:** banning private credit, or pushing it back onto bank balance sheets or into public markets.

**Correction to the source notes.** Vardeez's IM-10 d table has five rows and *no valuation row*, although valuation is one of the IMF's six key risks and its fix is examined (Question IM-10 d.5; Workbook A28.2). Its credit and liquidity rows describe the problem without the recommendation: the fixes are bank-like monitoring by funds with substantial private credit (credit), and closed-end or less frequent redemptions, longer notice and settlement, liquidity tools, stress tests and clear disclosure (liquidity). Its summary that "every recommendation is about seeing the risk" also understates the liquidity fixes, which restrict the product's structure, not only its visibility.

**Exam traps**

- **Redemption frequency, polarity.** More frequent redemptions *increase* liquidity and conduct risk; fewer redemption dates and longer notice are the fix.
- **Wrong party.** Credit monitoring by the investing institutions, not by rating agencies; valuation checked externally and audited, not left to the fund's own marks; leverage stress tests demanded by the *banks* that lend to private credit firms.
- **Caps now, everywhere.** Leverage caps are reserved for systemically dangerous leverage; the first step is data.
- **Fund-level data are enough.** Reports omit SPVs and feeder or master funds; the fix is more reporting, within and across borders.
- **Arbitrage as diversification; national action as enough.** Arbitrage concentrates risk; the fix is harmonisation and cross-border cooperation.

## Practice questions: IM-10 d

### Question IM-10 d.1

Bank Q480 • Schweser Module Quiz 92.2 Q3 (rebuilt)

Which of the following statements regarding policy recommendations on private credit risks is most accurate?

**A.** Allowing more frequent redemptions in private credit funds can create conduct risk, because managers may be tempted to manipulate valuations to discourage redemptions.

**B.** Given that private credit risks are largely mitigated, authorities should avoid an overly regulatory approach that would limit the growth of this emerging asset class.

**C.** Regulatory arbitrage across borders and sectors has helped mitigate excessive concentration of private credit risk in certain jurisdictions and sectors.

**D.** Because current regulations already account for asset valuation, no further regulatory oversight of the valuation methods used by private credit investors is needed.

### Question IM-10 d.2

New • built from SchweserNotes 2026 Reading 92

The chief risk officer of a European life insurer tells the board that private credit is now a material share of the general account, held through direct lending funds and private credit CLO tranches. The insurer judges the loans' credit quality from the managers' internal valuations and from external credit ratings. Which action is most consistent with the IMF's policy recommendations on private credit credit risk?

**A.** Cap the allocation at 3% of assets under management, the level the IMF reports for insurers and pension funds, since exposures at that level carry no credit risk.

**B.** Put in place more rigorous, bank-like processes to monitor the performance of the underlying loans, with close scrutiny of the managers' underwriting policies.

**C.** Rely on the private equity sponsors that back most borrowers, since sponsor support has kept private credit losses below those of high-yield bonds.

**D.** Require every borrower to obtain a public credit rating before the insurer invests, so that the rating agencies take over the monitoring of the loans.

### Question IM-10 d.3

New • built from SchweserNotes 2026 Reading 92

A securities regulator in Singapore is consulting on rules for private credit funds that are now sold to retail investors through semiliquid structures. The funds hold illiquid corporate loans with no secondary market. Which package of measures is most consistent with the IMF's recommendations on the liquidity risk of these funds?

**A.** Require daily dealing at a daily NAV, so that the funds' liquidity terms match those of retail bond funds.

**B.** Permit only closed-end structures or less frequent redemptions, require longer notice and settlement periods, and mandate liquidity management tools and stress testing.

**C.** Allow frequent redemption windows as long as the loans are floating rate, since floating-rate loans hold their value when rates rise.

**D.** Address the risk mainly through leverage caps on the funds, since liquidity mismatches in private credit arise from fund-level borrowing.

**Question IM-10 d.4** New • built from SchweserNotes 2026 Reading 92

A bank's credit committee is reviewing a request for a revolving facility from a private credit manager whose funds lend to PE-backed middle-market firms. The manager points out that its funds report low leverage. Which condition is most consistent with the IMF's recommendations on private credit leverage risk?

**A.** Require the manager to show strong liquidity management, including stress tests of constrained funding, of write-downs on leveraged assets, and of large, abrupt drawings on its borrowers' credit lines.

**B.** Rely on the regulatory leverage caps that apply to business development companies, since those caps limit leverage across the private credit system.

**C.** Decline the facility, since the IMF recommends that banks stop lending to private credit firms until binding leverage caps are in force everywhere.

**D.** Require the manager's borrowers to refinance at fixed rates, since fixed-rate loans remove the leverage of the borrowers.

**Question IM-10 d.5** Bank Q481 • New question written for this bank

A financial stability regulator is reviewing how pension funds and insurers value their private credit holdings. Each institution currently selects its own valuation method, and the regulator has found that the same loan is carried at materially different values across institutions. Several loans are large and have valuations that are prone to estimation error. Which of the following policy recommendations is most consistent with mitigating the valuation risk in this situation?

**A.** Leave valuation to each institution because it has the best information about its own loans, and rely on external credit rating agencies to confirm the carrying values.

**B.** Require more frequent external valuations for the significant, error-prone positions and audits of valuation methods and internal controls, with write-downs recognized in the appropriate period.

**C.** Prohibit write-downs on semiliquid loans until the loans are realized, so that temporary valuation noise does not trigger investor redemptions.

**D.** Require the fund managers to produce internal valuations at a higher frequency, because the same managers can then be relied on without any external review or audit.

**Question IM-10 d.6** Bank Q482 • New question written for this bank

A macroprudential supervisor observes that insurers and pension funds in jurisdictions with lenient rules hold substantially larger private credit positions than their peers in stricter jurisdictions. The supervisor also lacks consistent data on leverage and on the interconnections between funds, banks and institutional investors. Which of the following is the most appropriate policy response?

**A.** Have each national regulator separately tighten its own rules and reporting, since cross-border coordination would add little given that private credit is bilateral and nonbank.

**B.** Allow regulatory competition to continue, because moving risk to lenient jurisdictions disperses it and reduces the concentration of private credit risk in any one jurisdiction.

**C.** Harmonize risk assessment processes across sectors and borders to reduce regulatory arbitrage, and strengthen cooperation and reporting requirements among regulators to close the data gaps.

**D.** Rely on the leverage reported by each fund to set leverage caps, since fund-level reporting already captures leverage at SPVs, feeder and master funds, investors and borrowers.

## Solutions • IM-10 d

### Solution IM-10 d.1 — Answer: A

Bank Q480 • Schweser Module Quiz 92.2 Q3

**Step 1 – Where conduct risk comes from**

Conduct risk arises from conflicts of interest and from selling to unsophisticated investors.

**Step 2 – Why more frequent redemptions feed it**

With more frequent redemptions, managers have a reason to manipulate valuations to discourage redemptions, or to make redemptions conditional on reinvesting in other products they control.
⇒ A states the IMF's own example of a conduct risk created by a liquidity choice.

**Interpretation**

The point links valuation and conduct into one mechanism: stale or managed marks are not only a measurement problem, they are something a manager may prefer. The fix is disclosure of conflicts and investor-first supervision.

**Why the other options are wrong**

**B is incorrect.** The reading calls for stronger, cross-sectoral and cross-border oversight; it does not say the risks are largely mitigated.
**C is incorrect.** Regulatory arbitrage concentrates risk where the rules are lax; harmonisation is recommended to reduce it.
**D is incorrect.** Regulation currently does *not* address valuation; the reading recommends oversight of valuation methods, external valuations and audits.

**LO:** IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; SchweserNotes 2026 Book 5, Reading 92, LO 92.d (Conduct Risks)

### Solution IM-10 d.2 — Answer: B

New • built from SchweserNotes 2026 Reading 92

**Step 1 – The gap**

Insurers and pension funds have no requirement to monitor the creditworthiness of the underlying loans, so they lean on internal valuations and rating agencies, which "may not provide much confidence".

**Step 2 – The IMF's fix**

Funds that hold a substantial amount of private credit should adopt more rigorous processes, similar to banks', to monitor loan performance in structured products and direct lending, and underwriting policies and credit risks need careful scrutiny.
⇒ B is the credit-risk recommendation.

**Interpretation**

Monitoring sits with the institution that holds the risk. The insurer's current approach (internal valuations plus ratings) is the weakness the IMF describes, not a substitute for the fix.

**Why the other options are wrong**

**A is incorrect.** Misuses a number: about 3% of AUM describes current exposure, not a policy cap, and no level of exposure is free of credit risk.
**C is incorrect.** A true fact used as a wrong policy: sponsor support has reduced defaults, but it is not monitoring, and the PE ties are themselves an interconnectedness risk.
**D is incorrect.** Wrong party: reliance on rating agencies is the problem; the IMF does not propose

mandatory public ratings for private borrowers.

**LO:** IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; SchweserNotes 2026 Book 5, Reading 92, LO 92.d (Credit Risks)

### Solution IM-10 d.3 — Answer: B

New • built from SchweserNotes 2026 Reading 92

#### Step 1 – The risk

Illiquid loans inside a fund that lets retail investors out at regular dates: a liquidity mismatch, made worse if many unsophisticated investors try to leave together in stress.

#### Step 2 – The IMF's remedies

Limit the outflows (only closed-end funds, or less frequent redemptions), slow them (more notice, more time to settle), prepare for them (mandatory liquidity management techniques and stress testing where mismatches are large), and explain them (thorough, understandable disclosure of risks and redemption limits).
⇒ B is the package.

#### Interpretation

These are the structural fixes in the list: they change the product, at the cost of the liquidity that made semiliquid funds attractive to retail investors. Disclosure completes the package and doubles as a conduct fix.

#### Why the other options are wrong

**A is incorrect.** Polarity flip: daily dealing on illiquid loans maximises the mismatch.
**C is incorrect.** Wrong mechanism: floating rates protect coupons, not the ability to sell a loan; they do nothing for a fund facing redemptions.
**D is incorrect.** Sibling swap: leverage caps are the contingency fix for leverage risk; the liquidity mismatch here comes from redemption terms.

**LO:** IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; SchweserNotes 2026 Book 5, Reading 92, LO 92.d (Liquidity Risks)

### Solution IM-10 d.4 — Answer: A

New • built from SchweserNotes 2026 Reading 92

#### Step 1 – Why the fund-level number is not enough

Reported leverage may omit or obscure SPVs and feeder or master funds, and leverage also sits with investors and borrowers.

#### Step 2 – What the IMF asks of banks that lend to private credit firms

Require the firm to show strong liquidity management processes, including stress tests for (1) funding sources becoming constrained, (2) write-downs on leveraged assets, and (3) large and abrupt use of credit by the firm's borrowers.
⇒ A is the recommendation, aimed at exactly this committee.

#### Interpretation

The three stress tests mirror the channels of IM-10 c: funding drying up, stale marks being written down, and simultaneous credit-line drawings forcing capital calls. Leverage caps come later, if leverage is judged systemically dangerous.

#### Why the other options are wrong

**B is incorrect.** Scope error: BDC caps exist, but BDC leverage has risen over time, and the caps say nothing about leverage at investors, other fund structures or borrowers.
**C is incorrect.** Absolute: the IMF does not ask banks to stop lending; bank exposure is small (under 1% of assets) and secured, and caps are a contingency.
**D is incorrect.** Irrelevant mechanism: fixed rates change the borrowers' rate exposure, not their leverage, and are not an IMF recommendation.

**LO:** IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; SchweserNotes 2026 Book 5, Reading 92, LO 92.d (Leverage Risks)

### Solution IM-10 d.5 — Answer: B

Bank Q481 • New question written for this bank

#### Step 1 – The gap

Regulation does not address asset valuation: each institution values its own holdings, so the same asset carries different values depending on the method.

#### Step 2 – The IMF's fix

Greater regulatory oversight of valuation methods; more frequent external valuations where valuations are significant and/or prone to error; audits to find deficiencies in methods and internal controls; write-downs of illiquid and semiliquid assets recognised accurately and in the appropriate period.
⇒ B combines every element.

#### Interpretation

Two words carry the answer: *external* (the party producing the mark is not the party paid on it) and *targeted* (significant or error-prone positions, not every asset every month). Workbook A28.2 (IM-10 c.10) picks the same fix out from among four risk areas.

#### Why the other options are wrong

**A is incorrect.** The status quo that creates the problem, plus reliance on rating agencies, which the IMF says give little confidence.
**C is incorrect.** Polarity: the key valuation risk is write-downs that are *not* recognised; delaying them worsens stale marks.
**D is incorrect.** Frequency without independence: internal marks produced more often still lack external verification and audit.

**LO:** IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; SchweserNotes 2026 Book 5, Reading 92, LO 92.d (Valuation Risks)

### Solution IM-10 d.6 — Answer: C

Bank Q482 • New question written for this bank

#### Step 1 – Name the two problems

Larger holdings where rules are lenient is regulatory arbitrage; missing data on leverage and links is the data deficiency behind leverage and interconnectedness risk.

#### Step 2 – Match the fixes

Concentrated risk may require harmonising risk assessment across sectors and borders to reduce or eliminate arbitrage. Data gaps call for more reporting within and across borders and closer cooperation among regulators, internationally and between sectors at home.

⇒ Only C addresses both.

### Interpretation

The IMF warns that the many interconnections between insurers and pension funds, and their potential systemic effects, "should be a warning sign to regulators". Cooperation is the recurring word in its fixes for leverage and interconnectedness alike.

### Why the other options are wrong

**A is incorrect.** Purely national action leaves the arbitrage and the cross-border data gaps in place; the most effective approach is cross-sectoral and cross-border.

**B is incorrect.** Arbitrage concentrates risk in lax jurisdictions and sectors; it does not disperse it.

**D is incorrect.** Fund-level reports may omit or obscure leverage at SPVs, feeder and master funds, investors and borrowers, so caps set on them would miss total leverage.

**LO:** IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. • **Reference:** IMF, Global Financial Stability Report, April 2024, Ch.2; SchweserNotes 2026 Book 5, Reading 92, LO 92.d (Interconnectedness and Leverage Risks)

IM-11 • Private Markets Investing • Vardeez: IMPORTANT

# IM-11 b

P1 CONCEPT

Describe different investment structures that can be used to invest in private assets and compare their characteristics.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: section A28 was reported from the August 2026 sitting (Map #49, 2 of 16 sheets). The recall named private credit; the workbook carries A28 into IM-11 with shape S6 and its own question A28.4. Vardeez flags the investment structures as a focus of this IMPORTANT reading.

**Vardeez:** investment structures ("Private Market Investment Structures").

**Reported exam shapes**

- **S6** Private markets structures, co-investing and intermediated vehicles. The structures are drilled here, co-investing in IM-11 e and intermediated vehicles in IM-11 f. Workbook A28.4, worked here, joins all three.

**What the questions look like:** all conceptual. Either an investor profile (experience, size, wish for control or for diversification, need for a tailored mandate or for one asset) and "which structure fits", or one structure and "which statement is correct". The wrong options lend one structure's attribute to its neighbour: the control of direct investment to a fund of funds, the carried interest of a discretionary mandate to an advisory one, the diversification of a fund to an SPV.

## 1. Private markets in brief (background for IM-11 b to f)

The reading is Demaria et al., *Asset Allocation and Private Markets* (2021), Ch.3, examined only up to Section 3.3.10. No GARP text covers it; SchweserNotes 2026 Reading 93 is the content authority in this book.

- **Three asset types.** Private equity (PE): equity in private firms, which may later list. Private debt (PD): loans and other debt to private firms. Private real assets (PRA): privately held real assets. Listed firms with thin trading that raise capital privately do so through private investments in public equities (PIPEs).
- **Why investors come (IM-11 a, P4 in this book).** Performance and diversification; control and value creation; a broader universe (more than 300 million private companies against about 52,000 listed ones, at every stage of the lifecycle); potential for absolute returns; wider geographic reach (less home bias); and specialised structures such as funds of funds and co-investments.
- **The fund vocabulary used in every IM-11 objective.** The **general partner** (GP) manages the fund and decides; **limited partners** (LPs) *commit* capital, which the GP *calls* (paid-in capital) as deals are found. Exits produce **distributions**; what the fund still holds is its **residual value**, the net asset value (NAV) set by the manager. The LPs pay a **management fee** plus **carried interest** (a performance fee). A closed-end fund has no redemption rights: an LP who needs cash must sell in a limited secondary market, often at a discount to NAV.

## 2. Six structures

An investment structure "determines how capital is pooled, managed, and governed" and defines the rights and obligations of managers and investors. Schweser lists six.

- **Direct investment.** The investor owns an interest in a private company or asset. *High control*, but significant expertise, active involvement and strong networks are needed (private investing often depends on local presence and relationships), and information asymmetry with the company is a challenge. *High concentration risk*: diversification stays limited unless many positions are held. Users: entrepreneurs, family offices, buyout investors; crowdfunding is also direct investment. Some strategies, such as leveraged buyouts and distressed debt, are generally not accessible in this format.
- **Private funds.** Capital is pooled and specialised managers (GPs) make the investment and oversight decisions: *diversification*, but *limited control* for each investor. The most common institutional structure, usually a **closed-end limited partnership**. Users: institutions and large private investors, pension funds; retail investors in some jurisdictions.
- **Private funds of funds (FoFs).** A fund that invests in a portfolio of private funds: *more diversification* than one fund, at the cost of *an extra layer of management fees*. The entry point for investors with limited experience and for those who put diversification first; the **closest approximation to a passive approach** in private markets.
- **Managed accounts.** A separate account tailored to one investor's mandate. *Discretionary*: the manager makes all investment decisions, and carried interest is typical. *Advisory*: the investor keeps veto authority, and carried interest is often absent. Customisation and transparency for large investors, but terms rest on negotiation and usually need a substantial commitment. Users: institutions seeking tailored solutions.
- **Special purpose vehicles (SPVs).** An entity created to hold a specific asset or group of assets, with equity or debt; often used for private real assets. Users: more sophisticated investors who want *targeted exposure to one deal or asset*.
- **Securitization.** Assets, typically debt, are pooled and cut into **tranches** with different risk and return. The securities can trade publicly, so liquidity is *higher than in private funds*, but the structure adds complexity and reduces transparency. Users: investors who want a specific risk profile, or who want to transfer certain risks.

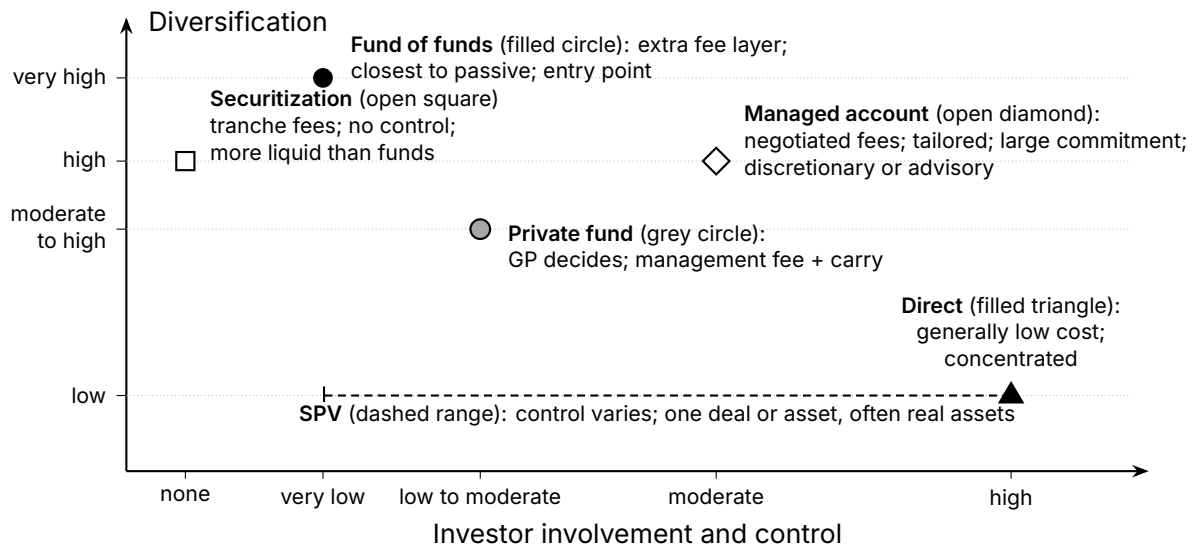



Figure 1. Schweser's comparison (Figure 93.1, rebuilt from its LO 93.b summary). Moving right buys control and costs diversification; the fund of funds sits top left and pays twice in fees for it. Every structure has very low liquidity except securitization, whose securities can trade. The SPV is drawn as a dashed range because its involvement and control vary.

| Structure | Involvement | Diversification | Liquidity | Costs | Control | Typical user |
|---|---|---|---|---|---|---|
| **Direct investment** | High | Low (unless many positions) | Very low | Generally low | High | Entrepreneurs, family offices, buyout investors |
| **Private fund** | Low to moderate | Moderate to high | Very low | Management fee plus carried interest | Limited | Institutions, pension funds, large private investors |
| **Fund of funds** | Very low | Very high | Very low | Higher than a fund (extra layer) | Very low* | Inexperienced investors; diversification first |
| **Managed account** | Moderate | High | Very low | Negotiated | Moderate | Institutions wanting a tailored mandate |
| **SPV** | Varies | Low | Very low | Set-up and maintenance | Varies | Sophisticated investors targeting one deal or asset |
| **Securitization** | None after purchase | High | Higher than private funds | Tranche-based | None | Investors wanting a risk profile, or to transfer risk |

*Schweser's summary gives no control rating for funds of funds; it follows from two layers of delegation and from "closest approximation to a passive investment approach".

## 3. How the instruments themselves are structured

Inside any structure, the claim can be shaped.

- **Equity:** common or preferred shares. Preferred shares can be *negotiated* to include specific rights and protections, so not all preferred shares offer the same features or the same level of influence.
- **Debt:** convertible or nonconvertible, with senior, subordinate or junior rights that set the priority of claims. **Convertible debt** gives downside protection through its debt component and upside through the option to convert into equity.

## 4. Reading an investor profile

Most questions describe an investor and ask which structure fits. The profile words point straight at a row of the table:

- **"Limited experience", "diversification first", "passive":** fund of funds, and the cost to name is the extra layer of fees.
- **"Pooled", "professional managers decide", "closed-end":** private fund.
- **"Tailored", "separate account", "veto":** managed account; a veto means an advisory mandate (carried interest often absent), full delegation a discretionary one (carried interest typical).
- **"One asset", "one deal", "infrastructure or real estate project":** SPV.
- **"Tranches", "a specific risk profile", "transfer risk", "some liquidity":** securitization.
- **"Control", "networks", "local presence":** direct investment, with concentration as the price.

Two structures from the rest of the reading sit between these rows. A **co-investment** is a direct stake in one deal taken alongside a fund (IM-11 e); the **intermediated fund** is the private fund seen from the LP's side, with its benefits and challenges (IM-11 f).

**Correction to the source notes.** (1) Vardeez's structure table has four rows (direct investment, private funds, funds of funds, securitization). Schweser's Reading 93 has six: managed accounts and SPVs are missing from Vardeez, and the bank tests both (Questions IM-11 b.2 to b.4).
(2) Vardeez says securitization "does not trade control for diversification but slices risk". In Schweser's comparison it sits at the end of that same trade-off: no control after purchase and high diversification (Figure 1). What sets it apart is liquidity, higher than in private funds because the securities can trade, bought at the price of complexity and lower transparency.

**Exam traps**

- **Fund of funds as the low-cost or high-control route.** It is the most passive route and pays an extra layer of fees. High control and generally low cost describe direct investment.
- **Discretionary and advisory swapped.** Discretionary: the manager decides and carried interest is typical. Advisory: the investor keeps a veto and carried interest is often absent.
- **Managed accounts for small investors.** They need negotiation and a substantial commitment; they suit institutions.
- **SPV as a diversifier.** An SPV holds one asset or group of assets for targeted exposure: low diversification.
- **Securitization as simple or controllable.** It adds complexity, reduces transparency and gives no control; its edge is liquidity.
- **Direct investment as the route to every strategy.** Leveraged buyouts and distressed debt are generally not accessible in this format.
- **"Preferred" read as standard.** Preferred share rights are negotiated; convertible debt is the instrument that pairs downside protection with upside.

## Practice questions: IM-11 b

### Question IM-11 b.1

Bank Q486 • Schweser Module Quiz 93.1 Q2

Which of the following private market investment structures requires the highest level of investor expertise and involvement and offers high control, but results in concentrated risk and low diversification unless many positions are held?

**A.** Private funds.

**B.** Direct investment.

**C.** Managed accounts.

**D.** Private funds of funds.

### Question IM-11 b.2

Bank Q487 • New question written for this bank

A family office with limited experience in private markets wants an entry point that places a high priority on diversification and comes closest to a passive approach. The office accepts that it will not control individual investment decisions. Which of the following structures is most appropriate, and what is its main added cost?

**A.** Special purpose vehicle; it concentrates exposure in one asset or deal.

**B.** Private fund of funds; it adds an extra layer of management fees.

**C.** Managed account; it usually requires a large capital commitment and negotiation.

**D.** Direct investment; it requires substantial local networks to source deals.

### Question IM-11 b.3

Bank Q488 • New question written for this bank

A large pension plan negotiates a separate managed account with a private equity manager under an advisory mandate. Which of the following is most likely true of this arrangement?

**A.** The manager makes all investment decisions, and the arrangement typically involves carried interest.

**B.** The plan's capital is pooled with other investors in a closed-end limited partnership managed by the general partner.

**C.** The plan retains veto authority over investments, and the arrangement often does not involve carried interest.

**D.** The arrangement is mainly suited to retail investors because it requires only a small capital commitment.

**Question IM-11 b.4** New • built from SchweserNotes 2026 Reading 93

A risk analyst at a Dutch insurer compares two routes into private debt: (1) a mezzanine tranche of a securitization of middle-market loans, and (2) an SPV set up to hold a single infrastructure loan. The investment committee asks how the two structures differ. Which of the following statements is correct?

**A.** The SPV offers high diversification, because an SPV can be structured with either equity or debt.

**B.** The securitization lets the insurer choose which loans enter the pool, while the SPV leaves every decision to a general partner.

**C.** The tranche is likely to be more liquid than a private fund interest, but securitization adds complexity, reduces transparency and gives no control.

**D.** The securitization removes the complexity of private debt, which makes it the usual entry point for investors new to private markets.

**Question IM-11 b.5** New • built from SchweserNotes 2026 Reading 93

The investment director of a Singapore family office is negotiating an investment in a fast-growing private logistics company. She wants protection if the company underperforms and a share of the gains if it succeeds. Which of the following instrument structures best meets her objective?

**A.** Senior nonconvertible debt, because seniority gives both priority over other creditors and a share in equity gains.

**B.** Convertible debt, because the debt component protects the downside and the option to convert gives upside potential.

**C.** Junior nonconvertible debt, because its lower priority earns a higher coupon that replicates the equity upside.

**D.** Preferred shares, because all preferred shares carry the same standard rights and protections for the holder.

**Question IM-11 b.6** Workbook A28.4 • S6 • self-generated

The CIO of a Japanese life insurer is preparing a JPY 30 billion first allocation to private markets. Three routes are on the table: a fund of funds, a primary commitment as a limited partner in a blind-pool buyout fund, and co-investments offered by that fund's GP alongside its deals. An investment committee member summarises the trade-offs. Which of the following statements is correct?

**A.** The fund of funds gives the insurer the highest control and the lowest total fees of the three routes, making it the structure closest to a direct investment.

**B.** Co-investment deal flow tends to be countercyclical, arriving when valuations are low, which largely offsets the adverse selection risk of co-investing.

**C.** In the blind-pool fund, the limited partners choose which assets are acquired within the strategy guidelines, so blind pool risk falls mainly on the GP.

**D.** Co-investing reduces or removes management fees and carried interest and allows deal-level targeting, but brings adverse selection, compressed due diligence and concentrated positions.

## Solutions • IM-11 b

### Solution IM-11 b.1 Answer: B

Bank Q486 • Schweser Module Quiz 93.1 Q2

**Step 1 – Match the profile**

Highest expertise and involvement, high control, concentrated risk, low diversification unless many positions are held.

**Step 2 – Find the structure**

In a direct investment the investor owns the asset and does the sourcing, due diligence and oversight itself. Each position is one company or asset, so diversification needs many separate positions.

⇒ Direct investment fits every attribute (the bottom-right corner of Figure 1).

**Interpretation**

Direct investing also needs strong networks and local presence, and it faces information asymmetry with the company. Its costs are generally low; its price is concentration and the work it takes.

**Why the other options are wrong**

**A is incorrect.** Private funds pool capital and delegate decisions to the GP: involvement is low to moderate and diversification moderate to high.

**C is incorrect.** Managed accounts give moderate involvement and control with high diversification; negotiated control is not full ownership control.

**D is incorrect.** Funds of funds are the opposite profile: very low involvement and very high diversification.

**LO:** IM-11 b: Describe different investment structures that can be used to invest in private assets and compare their characteristics. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.b

### Solution IM-11 b.2 Answer: B

Bank Q487 • New question written for this bank

**Step 1 – Read the profile**

Limited experience, diversification as the priority, close to passive, no control over individual decisions.

**Step 2 – Match it**

A fund of funds invests in a portfolio of private funds: more diversification than a single fund, the closest approximation to a passive approach, and the usual entry point for investors with limited experience.

**Step 3 – Name the cost**

The investor pays the underlying funds' fees and the fund of funds' own fees: an extra layer of management fees.

⇒ B gives the right structure and the right cost.

**Interpretation**

The fund of funds buys diversification and convenience with fees, the mirror image of direct investment, which buys control with concentration and effort.

**Why the other options are wrong**

**A is incorrect.** An SPV is for targeted exposure to one asset or deal, used by sophisticated in-

vestors; it is the opposite of a diversified entry point.
**C is incorrect.** A managed account suits a large institution with a tailored mandate; it is neither passive nor an entry point.
**D is incorrect.** Direct investment requires expertise, networks and active involvement and gives low diversification.

**LO:** IM-11 b: Describe different investment structures that can be used to invest in private assets and compare their characteristics. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.b

### Solution IM-11 b.3 — Answer: C

Bank Q488 • New question written for this bank

#### Step 1 – What a managed account is

A separate account tailored to one investor's mandate, set up by negotiation and usually needing a substantial commitment.

#### Step 2 – Advisory against discretionary

Advisory: the investor keeps veto authority over investments, and carried interest is often not involved. Discretionary: the manager makes all decisions, and carried interest is typical.
⇒ An advisory mandate means the plan keeps a veto and often pays no carry.

#### Interpretation

The veto is the reason the performance fee drops away: the manager is advising, not deciding. The plan also gets the customisation and transparency that make managed accounts attractive to large investors.

#### Why the other options are wrong

**A is incorrect.** Describes a discretionary mandate (sibling swap).
**B is incorrect.** Pooling in a closed-end limited partnership is a private fund, not a separate account.
**D is incorrect.** Managed accounts need negotiation and a substantial commitment; they suit institutions, not retail investors.

**LO:** IM-11 b: Describe different investment structures that can be used to invest in private assets and compare their characteristics. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.b

### Solution IM-11 b.4 — Answer: C

New • built from SchweserNotes 2026 Reading 93

#### Step 1 – The securitization

Debt is pooled and divided into tranches with different risk and return. The tranches can trade publicly, so they are more liquid than an interest in a private fund. The investor has no involvement or control after purchase, and the structure adds complexity and reduces transparency.

#### Step 2 – The SPV

An entity created to hold a specific asset or group of assets: targeted exposure, low diversification, very low liquidity, set-up and maintenance costs. Involvement and control vary with how it is set up.
⇒ Only C describes either structure correctly.

#### Interpretation

The two routes answer different needs. The tranche gives a chosen slice of a diversified pool's

risk with some liquidity; the SPV gives a concentrated, deal-specific exposure that a sophisticated investor can shape.

### Why the other options are wrong

**A is incorrect.** A true fact used for the wrong conclusion: an SPV can hold equity or debt, but it holds one asset or group, so diversification is low.
**B is incorrect.** Role reversal: securitization gives no control after purchase; in an SPV, involvement and control vary with how the vehicle is set up.
**D is incorrect.** Polarity: securitization adds complexity; the entry point for inexperienced investors is the fund of funds.

**LO:** IM-11 b: Describe different investment structures that can be used to invest in private assets and compare their characteristics. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.b

## Solution IM-11 b.5 — Answer: B

New • built from SchweserNotes 2026 Reading 93

### Step 1 – Translate the objective

"Protection if it underperforms" is a debt claim with priority. "A share of the gains" is equity upside.

### Step 2 – Find the instrument that has both

Convertible debt: the debt component gives downside protection, and the option to convert into equity gives upside potential.
$\Rightarrow$ B meets both halves of the objective.

### Interpretation

Private deals are negotiated, so the claim is designed rather than bought off the shelf. That is also why preferred shares vary: their rights and protections are negotiated case by case.

### Why the other options are wrong

**A is incorrect.** Seniority sets priority among creditors; nonconvertible debt has no equity upside.
**C is incorrect.** Junior debt has *less* protection, and a fixed coupon is not equity upside.
**D is incorrect.** Absolute qualifier: preferred shares can be negotiated to include specific rights, so not all preferred shares offer the same features or influence.

**LO:** IM-11 b: Describe different investment structures that can be used to invest in private assets and compare their characteristics. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.b

## Solution IM-11 b.6 — Answer: D

Workbook A28.4 • S6 • self-generated

### Step 1 – Place each route in the structure spectrum

Fund of funds: very low involvement, very high diversification, an additional layer of fees; the closest approximation to a passive approach and a common entry point for inexperienced investors. Primary fund: pooled capital, the GP decides, diversification with limited LP control.

### Step 2 – Intermediated vehicle trade-offs

Benefits: expertise, diversification, deal flow, cost sharing. Challenges: management fee plus carried interest create a hurdle, principal-agent risk, opacity, blind pool risk (capital committed before the assets are known; the manager decides how it is deployed), illiquidity, due diligence demands, conflicts.

**Step 3 – Co-investing trade-offs**

Advantages: reduced or no management fees or carry (transaction and due diligence costs still shared), greater control at deal level, higher net returns, learning and access. Disadvantages: adverse selection (procyclical flow, deals outside the GP's core expertise), speed and complexity, resource needs, concentration, fewer legal protections and a passive role.

### Interpretation

D is correct. For a first allocation the committee would typically weigh a fund of funds or a primary commitment for diversification, adding co-investments only once the insurer has the resources to evaluate deals quickly (IM-11 e).

### Why the other options are wrong

**A is incorrect.** Sibling swap: high control and low cost describe direct investment; a fund of funds is the most passive route with an extra fee layer.

**B is incorrect.** Polarity flip: co-investment deal flow can be procyclical, appearing when valuations are already high, which worsens rather than offsets adverse selection.

**C is incorrect.** Role misassignment: in a blind pool the fund manager (GP) ultimately decides how capital is deployed, and the LPs bear the risk of exposures they would not have chosen (IM-11 f).

**LO:** IM-11 b: Describe different investment structures that can be used to invest in private assets and compare their characteristics. IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.b, 93.e and 93.f

IM-11 • Private Markets Investing • Vardeez: IMPORTANT

# IM-11 c

P1 BOTH

Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic: shape S5 of section A28, the section reported from the August 2026 sitting (Map #49, 2 of 16 sheets). The workbook's question on it, A28.1, is self-generated. Vardeez names IRR, MOIC and PME as a focus of this IMPORTANT reading.
**Vardeez:** IRR, MOIC, PME ("Private Equity Performance Measures").

**Reported exam shapes**

- **S5** Private equity performance measures: IRR against TVPI, and DPI against PME (Workbook A28.1, Question IM-11 c.6). A28.1 also carries shape S3 of IM-10 c, NAV-based metrics, seen here from the investor's side.

**What the questions look like:** two kinds. *Concept*: which metric uses timing, which ignores it, which a credit line or a generous NAV can inflate, which needs a benchmark. *Calculation*: a table of calls, distributions, NAV and index levels, and you compute TVPI, DPI, RVPI, an IRR or a PME. The wrong numbers come from dividing by committed capital, discounting only one leg of the PME, adding DPI to TVPI, or treating a credit line as free.

## 1. The cash flows behind every metric

Recap from IM-11 b: LPs *commit* capital; the GP *calls* it as deals are found (paid-in capital, PI); exits produce *distributions* ($D_t$); what is unsold is the *residual value*, the NAV set by the manager. Every metric in this objective is built from three things: calls $C_t$, distributions $D_t$ and, if the fund is not finished, the latest NAV.

**Running example.** A buyout fund, commitment 100 (in millions). The benchmark index level is shown for the PME.

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| Capital called $C_t$ | 30 | 30 | 25 | 0 | 0 | 0 | 0 |
| Distributions $D_t$ | 0 | 0 | 5 | 20 | 40 | 45 | 30 |
| Net cash flow to the LP | −30 | −30 | −20 | 20 | 40 | 45 | 30 |
| Cumulative net cash flow | −30 | −60 | −80 | −60 | −20 | 25 | 55 |
| Index level $I_t$ | 100 | 108 | 103 | 115 | 124 | 130 | 141 |

Paid-in capital is 85: 15 of the commitment is never called. Distributions total 140, and the fund is fully realised at year 6 (NAV zero). Plotted, the cumulative cash flow falls while capital is called and rises as exits arrive: the J-shape that makes timing so important for the IRR (Figure 1).

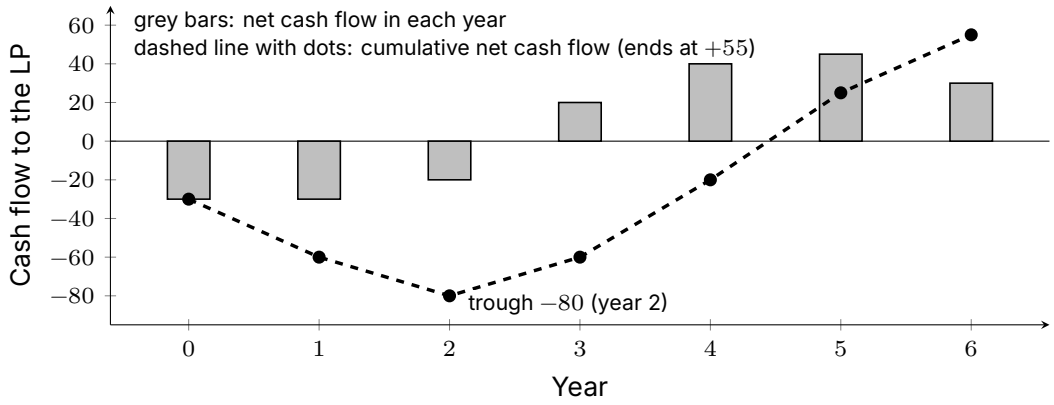



Figure 1. The running example's J-curve. Calls come first (grey bars below zero), distributions later; cumulative cash (dashed) bottoms out at $-80$ in year 2 and turns positive between years 4 and 5. Any metric that weights time, the IRR above all, depends on where these bars fall.

## 2. Internal rate of return (IRR)

$$\sum_t \frac{D_t - C_t}{(1+r)^t} + \frac{\text{NAV}_T}{(1+r)^T} = 0$$

The IRR is the rate $r$ that sets the present value of all calls (negative), distributions (positive) and, for a live fund, the NAV treated as a final inflow, to zero. It is "the annualized return on invested capital", and the most commonly used PE metric. For the running example (NAV zero): $r = 15.07\%$.

- **Advantages.** Intuitive and easy to interpret; incorporates the time value of money; widely accepted, suited to multi-year projects and to comparing competing investments.
- **Disadvantages.** Measures only deployed capital, not committed; not adjusted for risk (it can encourage managers to pursue riskier deals); highly sensitive to the timing of cash flows, so open to manipulation; assumes reinvestment at the IRR; can be overstated when credit lines are used; no explicit benchmark or market context.

### Early distributions and the reinvestment assumption

Two funds each invest 100 at year 0 and return 200 in total (a 2.0x multiple). Fund P returns 50 at year 1 and 150 at year 6; fund Q returns 200 at year 6.

$$\text{IRR}_Q = 2^{1/6} - 1 = 12.25\% \qquad \text{IRR}_P = 17.36\% \ \text{(solves } 100 = 50/(1+r) + 150/(1+r)^6\text{)}$$

P's IRR is five points higher on the same multiple. It quietly assumes the 50 received at year 1 is reinvested at 17.36% for five years: $50(1.1736)^5 + 150 = 261.3 = 100(1.1736)^6$. Early cash "can artificially inflate IRR" and may push managers toward short-term gains over higher long-term returns. Schweser's fix is a time-zero IRR, which it also calls a modified IRR (MIRR): reinvest the interim distributions at a stated rate instead. At 6%:

$$\text{MIRR}_P = \left(\frac{50(1.06)^5 + 150}{100}\right)^{1/6} - 1 = \left(\frac{216.91}{100}\right)^{1/6} - 1 = 13.78\%$$

### Subscription credit lines

A fund can buy assets with a bank credit line and call the LPs' capital later. The LPs' money is then at work for a shorter time, so the measured IRR rises; nothing about the assets has changed. In the

running example, bridge the year-0 call of 30 for one year at 6%: the LPs pay $31.8 + 30 = 61.8$ at year 1 and nothing at year 0.

| | LPs' paid-in | IRR | MOIC (TVPI) |
|---|---|---|---|
| No credit line | 85.0 | 15.07% | 1.65x |
| Year-0 call bridged one year at 6% | 86.8 | 16.26% | 1.61x |

The IRR goes up while the multiple goes *down*: the line's interest is paid by the LPs. Schweser lists both versions of the trick: delaying capital calls by drawing on credit lines, and using credit lines to accelerate distributions.

## 3. Multiple of invested capital (MOIC) and its submetrics

$$\text{MOIC} = \frac{\text{total distributions} + \text{value of remaining assets}}{\text{capital invested}} \qquad (1.0\text{x} = \text{breakeven})$$

$$\boxed{\text{DPI} = \frac{\sum D_t}{\text{PI}} \qquad \text{RVPI} = \frac{\text{NAV}}{\text{PI}} \qquad \text{TVPI} = \frac{\sum D_t + \text{NAV}}{\text{PI}} = \text{DPI} + \text{RVPI}}$$

- **TVPI (total value to paid-in)** is the primary MOIC measure. It includes residual value, so it can be manipulated if managers inflate valuations.
- **DPI (distributed to paid-in)** rests entirely on realised cash flows, so it is not subject to valuation manipulation; it is often preferred once a fund has had enough exits. Its limitation: realised cash flows may reflect economic conditions different from those affecting the assets still held.
- **RVPI (residual value to paid-in)** rises naturally as the fund's assets appreciate, and like TVPI it can be manipulated by overstated residual values.

**Year-3 snapshot of the running example.** The fund has paid in 85, distributed $5 + 20 = 25$, and reports a NAV of 95:

$$\text{DPI} = \frac{25}{85} = 0.29\text{x} \qquad \text{RVPI} = \frac{95}{85} = 1.12\text{x} \qquad \text{TVPI} = 0.29 + 1.12 = 1.41\text{x}$$

79% of the reported multiple $(1.12/1.41)$ is the manager's mark. That gap between DPI and TVPI is the reporting-risk signal of IM-11 d. (The interim IRR on the same numbers, with the NAV as a year-3 inflow, is 18.2%, above the 15.1% the fund finally delivered: the mark of 95 was generous.) At year 6 the NAV is zero and DPI = TVPI $= 140/85 = 1.65$x.

- **Advantages.** Focuses on absolute performance; independent of dollar size, so comparisons across funds are straightforward; less susceptible to manipulation than the IRR; cash-based submetrics such as DPI are robust.
- **Disadvantages.** Measures only deployed capital; not adjusted for risk; ignores the time value of money, including holding periods; cannot be annualised; no explicit benchmark or market context.

## 4. Public market equivalent (PME)

The PME compares the fund with a public index by discounting every fund cash flow at the index's return: the sum of discounted distributions over the sum of discounted calls. For a fund still holding assets, the NAV enters as a final distribution.

$$\boxed{\text{PME} = \frac{\sum_t D_t \dfrac{I_0}{I_t} + \text{NAV}_T \dfrac{I_0}{I_T}}{\sum_t C_t \dfrac{I_0}{I_t}}} \qquad \text{PME} > 1 : \text{ the fund beat the index}$$

**Running example.** Discount factors $I_0/I_t$: 1.0000, 0.9259, 0.9709, 0.8696, 0.8065, 0.7692, 0.7092.

$$\text{calls: } 30(1) + 30(0.9259) + 25(0.9709) = 30.00 + 27.78 + 24.27 = 82.05$$

$$\text{distributions: } 5(0.9709) + 20(0.8696) + 40(0.8065) + 45(0.7692) + 30(0.7092) = 110.40$$

$$\text{PME} = \frac{110.40}{82.05} = 1.35$$

In year-0 index terms the fund paid back 1.35 for every 1.00 it called: it beat its benchmark (the index compounded at 5.9% a year against the fund's 15.1% IRR). At year 3, with the NAV of 95 as a final distribution, the interim PME is $(4.85 + 17.39 + 95 \times 0.8696)/82.05 = 1.28$: like the TVPI, it leans on the manager's mark.

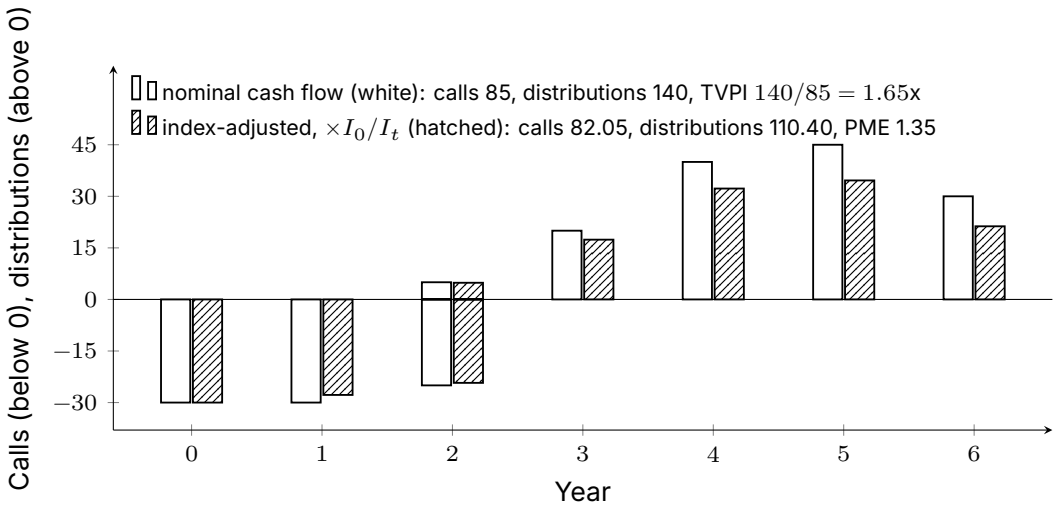



Figure 2. The running example's cash flows, nominal (white) and restated in year-0 index terms (hatched). The later the cash flow, the more the index's growth shrinks it, so distributions lose more than calls. TVPI divides the white bars; PME divides the hatched ones.

- **Advantages.** Explicit benchmarking gives market context; it helps neutralise the effect of market cycles on the evaluation; it measures the manager's value creation in a way that is less prone to manipulation.
- **Disadvantages.** Measures only deployed capital; sensitive to the choice of benchmark; a suitable benchmark may not exist; most reliable for fully realised funds. Accurate implementation requires matching each distribution with its associated capital call, to reflect the correct holding period; Cambridge Associates' **modified PME (mPME)** approximates this and is a more robust, less manipulable measure of manager value creation.

- **Timing still matters** in a PME: each cash flow is restated with the index return from its own date. Move the year-2 distribution of 50 in Workbook A28.1 to year 3 and its PME falls from 0.96 to 0.92 while its TVPI stays at 1.42x. What PME does not do is annualise: it is a ratio, not a rate.

## 5. The limitation all three share: deployed, not committed

IRR, MOIC and PME all measure **deployed** capital. They ignore the cash drag of the uncalled commitment: in the running example, 15 of the 100 committed was never called, and on a commitment basis the fund returned only $140/100 = 1.40$x. Because funds often do not call their full commitment, investors overcommit (Schweser's example: commit 110% of the desired allocation, expecting only about 60% to be called). That helps them reach their target exposure, but brings the risk of overallocation, or even default, if the full commitment is called after all.

**Correction to the source notes.** (1) Vardeez writes that the IRR "is the only measure that uses the timing of cash flows". The PME also depends on timing: it discounts each cash flow at the index return from its own date (Section 4: one distribution in A28.1 moved a year later takes its PME from 0.96 to 0.92). What only the IRR does is annualise. The bank's own explanations agree (Questions IM-11 c.1 and c.2).
(2) Vardeez's trap summary says a credit line that delays capital calls "does nothing to MOIC". It does not raise MOIC, but it is not neutral when the line charges interest that the LPs pay: in Section 2 the IRR rises from 15.07% to 16.26% while the multiple falls from 1.65x to 1.61x (also Question IM-11 c.5).
(3) Vardeez's PME disadvantages give only "works best with fully realized funds" and the need to match each distribution with its capital call. Schweser's list is: deployed capital only, sensitivity to the choice of benchmark, suitable benchmarks may not exist, and most reliable for fully realised funds. Distribution matching is how the PME is implemented accurately, and the mPME approximates it.

**Formula card: IM-11 c**

| | |
|---|---|
| Paid-in capital | $\text{PI} = \sum_t C_t$ (capital called, not committed) |
| DPI, RVPI, TVPI | $\text{DPI} = \sum D_t/\text{PI}$; $\text{RVPI} = \text{NAV}/\text{PI}$; $\text{TVPI} = \text{DPI} + \text{RVPI}$; 1.0x is breakeven |
| IRR | $r$ solving $\sum_t (D_t - C_t)/(1+r)^t + \text{NAV}_T/(1+r)^T = 0$ |
| One call, one distribution | $\text{IRR} = (D_T/C_s)^{1/(T-s)} - 1$ for a call at $s$ and a distribution at $T$ |
| MIRR (one call at 0) | $\left(\sum_t D_t(1+k)^{T-t}/C_0\right)^{1/T} - 1$, distributions reinvested at $k$ |
| PME | $\left[\sum_t D_t\, I_0/I_t + \text{NAV}_T\, I_0/I_T\right] \Big/ \sum_t C_t\, I_0/I_t$; above 1: the fund beat the index |
| Reporting-risk read | large gap between DPI and TVPI: most of the value is the manager's NAV |

| | IRR | MOIC (TVPI; DPI; RVPI) | PME |
|---|---|---|---|
| **Question it answers** | What annual rate did the deployed capital earn? | How many times has the money come back (in cash, DPI; including NAV, TVPI)? | Did the fund beat the public index? |
| **Uses timing** | Yes, heavily | No: ignores holding periods | Yes, through index discounting |
| **Annualised** | Yes | No: cannot be | No: a ratio |
| **Benchmark** | None | None | Explicit |
| **Risk-adjusted** | No | No | No |
| **Inflated by** | Early distributions, credit lines, generous NAV | Generous NAV (TVPI, RVPI); DPI cannot be | Less prone; choice of benchmark |
| **Capital base** | Deployed | Deployed | Deployed |

**Exam traps**

- **Committed for paid-in (wrong input).** Every multiple divides by capital *called*. $140/100 = 1.40$x instead of $140/85 = 1.65$x.
- **One leg of the PME (dropped term).** Both calls and distributions are index-adjusted. Discounting only the distributions understates the PME; only the calls, overstates it.
- **A good TVPI or IRR is not outperformance.** Only the PME brings in the market: A28.1 has TVPI 1.42x and PME 0.96.
- **Credit lines.** They raise the IRR by shortening the LPs' holding period; they do not raise the multiple, and their interest lowers it.
- **DPI and TVPI added (sibling formula).** TVPI already contains DPI: RVPI = TVPI − DPI.
- **Which one is manipulable.** DPI rests on cash and cannot be marked up; TVPI and RVPI rest on the manager's NAV.
- **Timing words.** Only the IRR annualises; MOIC ignores timing; PME uses timing without annualising. None of the three is risk-adjusted, and all three measure deployed capital only.

## Practice questions: IM-11 c

### Question IM-11 c.1

Bank Q490 • Schweser Module Quiz 93.2 Q2 (rebuilt)

Which of the following private market performance metrics is most useful for assessing absolute performance (total value relative to capital invested) without regard to holding periods?

**A.** Time-zero IRR (modified IRR).

**B.** Internal rate of return (IRR).

**C.** Public market equivalent (PME).

**D.** Multiple of invested capital (MOIC).

**Question IM-11 c.2** Bank Q489 • Schweser Module Quiz 93.2 Q1

Which of the following performance metrics for private equity best incorporates the timing of cash flows but can be artificially inflated by early distributions or the use of credit lines?

**A.** Internal rate of return (IRR).

**B.** Residual value to paid-in (RVPI).

**C.** Public market equivalent (PME).

**D.** Multiple of invested capital (MOIC).

**Question IM-11 c.3** New • built from SchweserNotes 2026 Reading 93

A member of a sovereign wealth fund's investment committee makes four statements about the public market equivalent (PME) of the fund's buyout commitments. Which of the following statements is correct?

**A.** A PME above 1.0 shows that the fund earned a positive IRR, so it carries the same information as a TVPI above 1.0x.

**B.** Because PME discounts every cash flow at index returns, it captures the cash drag on the part of the commitment never called.

**C.** PME adds market context, but its result depends on the benchmark chosen, a suitable benchmark may not exist, and it is most reliable for fully realized funds.

**D.** PME annualizes the fund's excess return over the index, so it can be compared directly with the fund's IRR.

**Question IM-11 c.4** Bank Q492 • New question written for this bank • re-filed from IM-11 d

A risk manager reviews a private equity fund with a DPI of 0.55x and a TVPI of 1.65x. The fund's net asset value is determined by the manager. Which of the following conclusions is most appropriate?

**A.** RVPI is 0.55x, so the gap between DPI and TVPI indicates selection risk relative to the category pooled average.

**B.** RVPI is 1.10x, so about two-thirds of reported value rests on the manager-determined NAV, which signals potential reporting risk.

**C.** RVPI is 1.10x, so the fund is largely realized and DPI is more exposed to valuation manipulation than TVPI.

**D.** RVPI is 2.20x, so the fund has distributed most of its value and reporting risk is low.

**Question IM-11 c.5** New • built from SchweserNotes 2026 Reading 93

A private equity analyst at a Dutch pension fund is checking the returns a buyout fund reports on one of its deals. The fund bought the company for EUR 200 million at date 0, financing the purchase with a subscription credit line. One year later it called EUR 214 million from its limited partners (LPs) to repay the line with 7% interest. Four years after the purchase the fund sold the company and distributed EUR 330 million to the LPs. The LPs had committed EUR 250 million to the fund. Ignoring management fees and carried interest, which of the following is closest to the IRR and the TVPI the LPs earned on this deal?

**A.** IRR 13.3%; TVPI 1.65x

**B.** IRR 15.5%; TVPI 1.54x

**C.** IRR 15.5%; TVPI 1.65x

**D.** IRR 18.2%; TVPI 1.65x

**Question IM-11 c.6** Workbook A28.1 • S5, S3 • self-generated

A private markets analyst at a Canadian pension plan is evaluating a fully liquidated buyout fund before re-upping with the same manager. The plan committed CAD 150 million; the fund's cash flows to and from the plan (CAD million) and the level of the benchmark public equity index were:

| Year | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Capital called | 80 | 40 | 0 | 0 | 0 |
| Distributions | 0 | 0 | 50 | 60 | 60 |
| Index level | 100 | 125 | 140 | 160 | 175 |

The manager's marketing deck reports a fund-level IRR of 17.5%, boosted by a subscription credit line that delayed capital calls to investors. No assets remain in the fund. Which of the following correctly states the fund's PME and TVPI, and the conclusion the analyst should draw?

**A.** PME 0.90 and TVPI 1.42x: the fund underperformed the index

**B.** PME 0.96 and TVPI 1.42x: the fund underperformed the index

**C.** PME 0.96 and TVPI 1.13x: the fund underperformed the index

**D.** PME 1.52 and TVPI 1.42x: the fund outperformed the index

## Solutions • IM-11 c

### Solution IM-11 c.1 — Answer: D

Bank Q490 • Schweser Module Quiz 93.2 Q2

**Step 1 – What the stem asks for**

Total value relative to capital invested, with no time dimension: a multiple, not a rate.

**Step 2 – Match the metric**

MOIC $=$ (distributions $+$ remaining value)/capital invested. A 2.0x MOIC means the value doubled, whether it took 3 years or 10.

$\Rightarrow$ MOIC is the absolute measure that ignores holding periods.

**Interpretation**

That is its strength (absolute, comparable across fund sizes, harder to manipulate than the IRR) and its weakness (no time value of money, cannot be annualised, no benchmark).

**Why the other options are wrong**

**A is incorrect.** The time-zero IRR (MIRR) is still a rate driven by the timing of cash flows; it only changes the reinvestment assumption.

**B is incorrect.** The IRR is an annual rate, explicitly sensitive to holding periods and to the timing of cash flows.

**C is incorrect.** The PME benchmarks timed cash flows against an index: a relative, timing-dependent measure, not an absolute multiple.

**LO:** IM-11 c: Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.c

### Solution IM-11 c.2 — Answer: A

Bank Q489 • Schweser Module Quiz 93.2 Q1

**Step 1 – Rule out the multiples**

RVPI and MOIC are ratios of value to capital; they ignore when cash moves.

**Step 2 – Which timing-based metric can be inflated this way**

The IRR annualises the return over the time the LPs' money is out. Early distributions, or calls delayed by a credit line, shorten that time and raise the IRR without improving the economics (Section 2).

$\Rightarrow$ The IRR is the metric that uses timing and can be gamed through it.

**Interpretation**

The PME also uses timing (correction box (1)), but Schweser names the IRR as the metric these tricks inflate and calls the PME less prone to manipulation.

**Why the other options are wrong**

**B is incorrect.** RVPI has no time dimension; it is vulnerable to overstated NAVs, not to early distributions.

**C is incorrect.** PME is the benchmarked measure and is described as less prone to manipulation; the credit-line and early-distribution weakness is the IRR's.

**D is incorrect.** MOIC ignores holding periods entirely.

**LO:** IM-11 c: Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.c

### Solution IM-11 c.3 — Answer: C

New • built from SchweserNotes 2026 Reading 93

#### Step 1 – What PME is

The sum of index-discounted distributions (plus any discounted NAV) over the sum of index-discounted calls. Above 1.0, the fund beat the index.

#### Step 2 – Its advantages and disadvantages

Advantages: explicit benchmark, market context, neutralises market cycles, less prone to manipulation. Disadvantages: deployed capital only, sensitive to the benchmark, suitable benchmarks may not exist, most reliable for fully realised funds.
⇒ C lists the disadvantages correctly next to the advantage.

#### Interpretation

Ask what each metric adds. TVPI says the money came back; IRR says how fast; only PME says whether the index would have done better with the same cash on the same dates.

#### Why the other options are wrong

**A is incorrect.** Sibling swap: a PME above 1 means the fund beat the index. A fund with a positive IRR and a TVPI above 1.0x can still have a PME below 1, as in Workbook A28.1 (TVPI 1.42x, PME 0.96).
**B is incorrect.** All three metrics measure deployed capital only; none captures the cash drag on the uncalled commitment.
**D is incorrect.** Only the IRR annualises. PME is a ratio; it can be compared with 1.0, not with an IRR.

**LO:** IM-11 c: Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.c

### Solution IM-11 c.4 — Answer: B

Bank Q492 • New question written for this bank • re-filed from IM-11 d

#### Step 1 – Back out RVPI

$$\text{RVPI} = \text{TVPI} - \text{DPI} = 1.65 - 0.55 = 1.10\text{x}$$

#### Step 2 – How much of the value is the manager's mark

$$\frac{\text{RVPI}}{\text{TVPI}} = \frac{1.10}{1.65} = 66.7\%$$

⇒ Two-thirds of the reported value is unrealised NAV, determined by the manager.

#### Step 3 – Read the gap

DPI rests on realised cash; TVPI depends on the NAV estimate. A substantial gap between them may indicate reporting risk.

#### Interpretation

Applying the metrics is what earns the conclusion: the identity TVPI = DPI + RVPI turns two reported numbers into a measure of how much of the track record is still a mark. (In the bank this question sits under IM-11 d, where the DPI-to-TVPI gap is listed as the tool for reporting risk; the options were reordered by RVPI.)

#### Why the other options are wrong

**A is incorrect.** Takes the DPI figure as RVPI, and assigns the gap to selection risk, which is assessed by comparing a fund's TVPI with its category's pooled average.

**C is incorrect.** RVPI is right, but the fund is mostly *un*realised, and DPI, based on cash, is the least exposed to manipulation.
**D is incorrect.** Added instead of subtracting: $1.65 + 0.55 = 2.20$, then misread the result.

**LO:** IM-11 c: Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. IM-11 d: Explain how different risks can arise for investors in private markets and assess how these risks can be measured. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.c and 93.d

### Solution IM-11 c.5 — Answer: B

New • built from SchweserNotes 2026 Reading 93

#### Step 1 – The LPs' cash flows

The line, not the LPs, paid for the company at date 0. The LPs paid $200 \times 1.07 = 214$ at year 1 and received 330 at year 4.
$\Rightarrow$ From the LPs' side the money was out for three years, not four.

#### Step 2 – IRR

$$\text{IRR} = \left(\frac{330}{214}\right)^{1/3} - 1 = 1.54206^{1/3} - 1 = 15.5\%$$

#### Step 3 – TVPI

$$\text{TVPI} = \frac{330 + 0}{214} = 1.54\text{x} \qquad \text{(DPI too: nothing is left)}$$

#### Step 4 – Compare with no credit line

Paying 200 at date 0: IRR $= 1.65^{1/4} - 1 = 13.3\%$ and TVPI $= 1.65$x.
$\Rightarrow$ The line lifts the IRR by 2.2 points and lowers the multiple by 0.11x: the same deal looks faster but returns less per euro paid in.

#### Interpretation

This is the credit-line weakness of the IRR in numbers. The EUR 250 million commitment is a red herring: dividing by it $(330/250 = 1.32\text{x})$ is the committed-capital trap; the metrics use what was paid in.

#### Why the other options are wrong

**A is incorrect.** Measured the deal, not the LPs: 200 at date 0 and 330 at year 4 give 13.3% and 1.65x.
**C is incorrect.** Right IRR, but the TVPI divides by the line draw of 200 instead of the 214 the LPs paid in.
**D is incorrect.** Treated the line as free: LPs paying 200 at year 1 gives $(330/200)^{1/3} - 1 = 18.2\%$ and 1.65x.

**LO:** IM-11 c: Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.c

### Solution IM-11 c.6 — Answer: B

Workbook A28.1 • S5, S3 • self-generated

#### Step 1 – Paid-in capital and distributions

Paid-in $= 80 + 40 = 120$; total distributions $= 50 + 60 + 60 = 170$; residual value $= 0$.

$\Rightarrow$ The plan actually deployed CAD 120 million of its CAD 150 million commitment: the metrics are built on the 120.

**Step 2 – TVPI (equal to DPI here)**

DPI $= 170/120 = 1.4167$; RVPI $= 0/120 = 0$; TVPI $= 1.4167 + 0 = 1.42$x.

$\Rightarrow$ The plan got back CAD 1.42 per CAD 1 called. With no residual NAV, the multiple rests entirely on realised cash, so it is immune to valuation manipulation.

**Step 3 – Index discount factors $I_0/I_t$**

Year 0: $100/100 = 1.0000$; year 1: $100/125 = 0.8000$; year 2: $100/140 = 0.7143$; year 3: $100/160 = 0.6250$; year 4: $100/175 = 0.5714$.

**Step 4 – Index-discounted calls**

$80 \times 1.0000 + 40 \times 0.8000 = 80.00 + 32.00 = 112.00$.

**Step 5 – Index-discounted distributions**

$50 \times 0.7143 + 60 \times 0.6250 + 60 \times 0.5714 = 35.71 + 37.50 + 34.29 = 107.50$.

**Step 6 – PME**

$$\text{PME} = \frac{107.50}{112.00} = 0.96$$

$\Rightarrow$ The same cash invested in the index on the call dates and withdrawn on the distribution dates would have done better: the fund underperformed the public benchmark despite a 1.42x multiple.

**Interpretation**

B is correct. TVPI ignores time and benchmark; the plan's IRR on the tabled cash flows is 13.8%; the 17.5% headline is timing-sensitive (credit lines shorten the measured holding period) and neither IRR is risk adjusted or benchmarked. PME adds the benchmark: below 1, so the analyst should challenge the re-up case. A fully realised fund is exactly where PME is most reliable.

**Why the other options are wrong**

**A is incorrect.** Dropped term: discounts the distributions but leaves the calls undiscounted, $107.50/120 = 0.90$.

**C is incorrect.** Wrong-input twin: divides distributions by committed rather than called capital, $170/150 = 1.13$x; the metrics measure deployed capital only.

**D is incorrect.** Dropped term: discounts the calls but leaves the distributions undiscounted, $170/112.00 = 1.52$, which flips the conclusion to outperformance.

**LO:** IM-11 c: Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.c

IM-11 • Private Markets Investing • Vardeez: IMPORTANT

# IM-11 e

P1 CONCEPT

Describe the practice of co-investing in private markets and assess its advantages and disadvantages.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: shape S6 of section A28, the section reported from the August 2026 sitting (Map #49, 2 of 16 sheets). Vardeez flags co-investing, "benefits and risks", as a focus of this IMPORTANT reading.
**Vardeez:** co-investing: benefits and risks.

**Reported exam shapes**

- **S6** Private markets structures, co-investing and intermediated vehicles. Workbook A28.4 (worked as Question IM-11 b.6) tests co-investing in its correct option and in a polarity trap on procyclical deal flow.

**What the questions look like:** all conceptual. "Which is an advantage (or disadvantage) of co-investing compared with investing as an LP in the primary fund", "what does adverse selection mean here", or a scenario (a two-week deadline, a small team, a deal too big for the fund) and "which assessment is correct". Wrong options reverse a listed point: more legal protection, more diversification, less due diligence, countercyclical deal flow.

## 1. What co-investing is

Recap from IM-11 b: a primary fund commitment pools the LP's money with others and the GP picks the deals; a direct investment gives full control at the price of concentration. Co-investing sits between them.

- **Definition.** A private investor invests *directly in a specific deal alongside a private market fund*, typically **by invitation from the fund manager**.
- **Why the GP offers it.** The manager needs additional capital to complete a deal that **exceeds its usual investment limits**.
- **Who is invited.** Access is usually restricted to **large, established investors** with a strong relationship with the manager.
- **How fast.** Opportunities often arise at short notice, so due diligence and decisions are **accelerated**. Some managers formalise access through dedicated co-investment vehicles.
- **Where else it appears.** Co-investments are one of the "specialised investment structures" that motivate private markets investing (IM-11 a). In IM-11 d they help with tactical allocation: a fund commitment may take years to deploy, while co-investments (and secondary transactions) let an investor adjust its private market exposure in a more timely way.

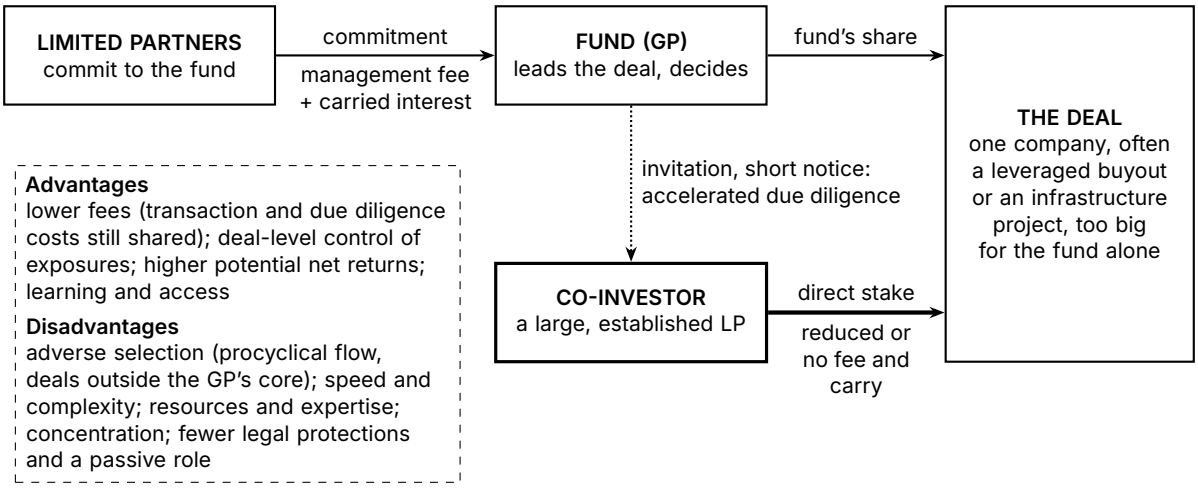



Figure 1. A co-investment. Solid arrows: capital; the heavy arrow is the co-investor's direct stake in the deal, taken alongside the fund. Dotted arrow: the GP's invitation, which sets the timetable and decides which deals are offered (the source of adverse selection). Dashed box: the advantages and disadvantages to assess.

## 2. Advantages

- **Lower fees.** Co-investors typically pay reduced, sometimes no, management fees or carried interest compared with primary fund investors. They still share in *transaction and due diligence costs*.
- **Greater control.** The investor targets specific deals, shaping its exposures and portfolio weights more precisely.
- **Potential for higher net returns.** Because fees are lower, co-investments often produce higher after-fee returns than investing through the primary fund.
- **Learning and access.** Hands-on experience, stronger relationships with GPs, and exposure to opportunities a fund commitment alone may not give.

## 3. Disadvantages

- **Adverse selection.** The deals offered may be less attractive or outside the manager's core expertise, and co-investment deal flow can be **procyclical**: opportunities often appear when valuations are already high. This is the structural one: the GP chooses which deals to offer.
- **Speed and complexity.** Accelerated due diligence and rapid decisions, with less preparation than for a primary fund commitment.
- **Resource and expertise requirements.** Effective co-investing needs significant internal resources, specialised expertise and the ability to evaluate deals quickly; investors without them often underperform.
- **Concentration and diversification risk.** Co-investments are typically larger, more concentrated positions than standard fund exposures, so the portfolio becomes more sensitive to deal-specific risk. Many involve leveraged buyouts or infrastructure projects, which raises the risk of unintended concentration.
- **Legal and governance risks.** Co-investors are usually **passive participants** with limited influence over deal structure, management decisions or exit timing, and they often accept fewer legal protections than LPs investing through the primary fund.

## 4. Assessing a co-investment programme

The objective's verb is "assess": weigh the two lists for a given investor.

- **Two kinds of control.** The advantage is control over *which* deals and *how much*: exposure and weights. Inside each deal the co-investor is passive, with fewer protections than a fund LP. Both statements are true at once.
- **The fee saving is real but conditional.** It turns into higher net returns only for investors with the resources to evaluate deals fast; the reading warns that investors lacking them often underperform.
- **Due diligence is compressed, not removed.** The co-investor does its own work on each deal, on the GP's timetable.
- **The timing works against the co-investor twice:** short deadlines, and deal flow that tends to arrive late in the cycle, when prices are high.
- **Portfolio role.** Co-investments complement fund commitments (targeted exposures, faster adjustment, cheaper access); they concentrate rather than diversify, so they do not replace them. Workbook A28.4's interpretation makes the same point for a first allocation.

| | LP in the primary fund | Co-investor in one deal |
|---|---|---|
| **Fees** | Management fee plus carried interest | Reduced or none; transaction and due diligence costs still shared |
| **Who picks the deal** | The GP | The investor accepts or declines deals the GP offers |
| **Influence in the company** | Through the fund only | Usually passive: little say on structure, management or exit timing |
| **Legal protections** | Those of an LP in the fund | Often fewer than the fund's LPs |
| **Diversification** | Many deals in one commitment | Large, concentrated, deal-specific positions |
| **Due diligence** | On the manager and strategy | On each deal, accelerated |
| **Access** | Any LP that commits | By invitation; large, established LPs |
| **Internal resources** | Moderate | Significant; without them, often underperformance |

**Exam traps**

- **"No costs".** Fees and carry are reduced or waived; transaction and due diligence costs are still shared.
- **Control read as governance.** Greater control means choosing deals and weights; inside the deal the co-investor is passive with fewer legal protections.
- **Due diligence reduced.** It is accelerated and done by the investor itself, not reduced or delegated.
- **More diversification.** Co-investments are larger, concentrated positions; diversification falls.
- **Countercyclical deal flow.** The flow can be procyclical, arriving when valuations are high; that worsens adverse selection.
- **Open access.** Co-investments come by invitation to large, established investors with strong GP relationships.
- **Adverse selection mislabelled.** It is about which deals are offered (less attractive, outside the GP's core), not about the GP's fees or the co-investor's own skill.

## Practice questions: IM-11 e

### Question IM-11 e.1

New • built from SchweserNotes 2026 Reading 93

The head of private markets at a Canadian pension plan, a large and long-standing LP of a buyout manager, receives an offer from the manager to co-invest in a carve-out acquisition. Which of the following statements best describes co-investing as it is typically practised?

**A.** Every LP in the primary fund has the right to co-invest pro rata, on the same timetable as the fund's own investment committee.

**B.** The plan would invest directly in the deal alongside the fund, often because the deal exceeds the fund's usual limits, and must decide quickly.

**C.** The plan would act as a general partner for this deal, taking over management decisions and exit timing in return for paying carried interest.

**D.** The plan would commit capital to the manager's next fund vintage, which gives it the option to see that fund's deals before other LPs.

### Question IM-11 e.2

Bank Q494 • Schweser Module Quiz 93.3 Q1

Which of the following is a key advantage of co-investing compared with investing as a limited partner in the primary fund?

**A.** Increased legal protections.

**B.** Reduced management fees.

**C.** Reduced need for due diligence.

**D.** Increased portfolio diversification.

### Question IM-11 e.3

New • built from SchweserNotes 2026 Reading 93

A board member of a Swiss foundation reads that co-investing offers "greater control", while the foundation's legal counsel warns about the "legal and governance risks" of co-investing. Which of the following statements correctly reconciles the two?

**A.** The control is over which deals to take and how much, shaping exposures; inside each deal the co-investor is usually passive, with fewer legal protections than the fund's LPs.

**B.** The control is over the company's management decisions and exit timing; the governance risk arises only if the GP refuses to give the co-investor a board seat.

**C.** Co-investors receive stronger legal protections than LPs in the primary fund, so the governance risk refers only to the exposure of the general partner.

**D.** The two cannot both hold: greater control is an advantage of co-investing, and the reading lists no legal or governance disadvantage for co-investors.

**Question IM-11 e.4** Bank Q495 • New question written for this bank

A fund manager invites a large limited partner to co-invest in a leveraged buyout that exceeds the fund's usual investment limit. Compared with investing only through the primary fund, which of the following is most likely a disadvantage for the co-investor?

**A.** The co-investor bears none of the transaction and due diligence costs, reducing its incentive to monitor.

**B.** The co-investment is typically a larger, more concentrated position, leaving the portfolio less diversified and more exposed to deal-specific risk.

**C.** The co-investor has less ability to shape its exposures and portfolio weights.

**D.** The co-investor pays higher management fees and carried interest than primary fund investors.

**Question IM-11 e.5** Bank Q496 • New question written for this bank

An investor with a small internal team is offered several co-investment opportunities by its general partners, each requiring a decision within two weeks. A consultant warns the investor about adverse selection. Which of the following best describes this concern?

**A.** Co-investors are given more legal protections than fund investors, which raises the cost of each deal.

**B.** Co-investment access is open to all investors on equal terms, so the best deals are quickly competed away.

**C.** Offered deals may be less attractive or outside the manager's core expertise, and deal flow tends to be procyclical, appearing when valuations are already high.

**D.** Co-investors always receive better deals than the fund, which leads to crowding out of the fund's limited partners.

**Question IM-11 e.6** New • built from SchweserNotes 2026 Reading 93

The CIO of a mid-sized university endowment, whose private markets team has three people, proposes a co-investment programme to cut the fees the endowment pays. Which of the following assessments of the proposal is most consistent with the reading?

**A.** The programme removes the costs of private equity investing, because co-investors pay no management fees, carried interest, transaction costs or due diligence costs.

**B.** Lower fees can raise net returns and build GP relationships, but co-investing needs significant internal resources and quick deal evaluation, and investors without them often underperform.

**C.** Because the GP has already done the due diligence on each deal, the endowment can rely on that work and keep its team at three people without any loss.

**D.** Co-investing gives the same diversification as the endowment's fund commitments at a lower cost, so the programme can replace those commitments entirely.

# Solutions • IM-11 e

## Solution IM-11 e.1

**Answer: B**

New • built from SchweserNotes 2026 Reading 93

### Step 1 – The practice

Co-investing is a direct investment in a specific deal alongside a private market fund, typically by invitation from the fund manager.

### Step 2 – Why and how

Managers offer co-investments when they need additional capital for a deal that exceeds their usual investment limits. The offers go to large, established investors with strong relationships, often at short notice, so due diligence and the decision are accelerated.

$\Rightarrow$ B describes the practice; the plan's size and long relationship are why it was invited.

### Interpretation

The invitation is the access and the problem: the GP picks the deals offered and sets the timetable.

### Why the other options are wrong

**A is incorrect.** Access is by invitation, on a compressed timetable.

**C is incorrect.** Role reversal: the co-investor stays a passive participant with reduced or no carry; the GP leads the deal.

**D is incorrect.** Definition swap: committing to the next fund vintage is a primary fund commitment, not a co-investment.

**LO:** IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.e

## Solution IM-11 e.2

**Answer: B**

Bank Q494 • Schweser Module Quiz 93.3 Q1

### Step 1 – List the advantages

Lower fees, greater control over exposures, potential for higher net returns, learning and access.

### Step 2 – Test the options

Only reduced management fees is on the list. Co-investors typically pay reduced, or no, management fees and carried interest on the deal.

$\Rightarrow$ B is the advantage; the other three reverse a disadvantage.

### Interpretation

The saving drives the higher potential net return, but co-investors still share transaction and due diligence costs.

### Why the other options are wrong

**A is incorrect.** Polarity: co-investors usually accept *fewer* legal protections than LPs in the primary fund.

**C is incorrect.** Due diligence is compressed and done by the co-investor on each deal, not reduced.

**D is incorrect.** Co-investments are larger, concentrated positions, so diversification falls.

**LO:** IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.e

### Solution IM-11 e.3 — Answer: A

New • built from SchweserNotes 2026 Reading 93

**Step 1 – What "greater control" means**

Co-investing lets investors target specific deals, "enabling more precise shaping of exposures and desired portfolio weights". It is control over the portfolio.

**Step 2 – What the governance risk means**

Co-investors are usually passive participants with limited influence over deal structure, management decisions or exit timing, and they often accept fewer legal protections than LPs in the primary fund.

$\Rightarrow$ Both hold at once, at different levels: A.

**Interpretation**

A board that hears "control" and expects a say in the company has confused co-investing with direct investment, where the investor owns and runs the position itself (IM-11 b).

**Why the other options are wrong**

**B is incorrect.** Places the control inside the company, where co-investors are passive; no board seat is promised.

**C is incorrect.** Polarity: co-investors have *fewer* protections than fund LPs.

**D is incorrect.** Absolute: legal and governance risks are one of the five listed disadvantages.

**LO:** IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.e

### Solution IM-11 e.4 — Answer: B

Bank Q495 • New question written for this bank

**Step 1 – Compare the exposures**

Through the fund, the LP holds a slice of many deals. A co-investment adds a large stake in one, here a leveraged buyout.

**Step 2 – Name the disadvantage**

Co-investments are typically larger and more concentrated than standard fund exposures, so the portfolio is less diversified and more exposed to deal-specific risk; leveraged buyouts and infrastructure projects raise the risk of unintended concentration.

$\Rightarrow$ B.

**Interpretation**

The same LBO is now held twice, through the fund and directly: the concentration is larger than the co-investment alone suggests.

**Why the other options are wrong**

**A is incorrect.** Co-investors still share transaction and due diligence costs.

**C is incorrect.** Polarity: co-investing gives *more* ability to shape exposures and weights.

**D is incorrect.** Polarity: fees and carry are reduced or waived, which is the main advantage.

**LO:** IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.e

### Solution IM-11 e.5 Answer: C

Bank Q496 • New question written for this bank

**Step 1 – What adverse selection means here**

The deals offered may be worse than average: less attractive than those the fund keeps, or outside the manager's core expertise.

**Step 2 – The timing element**

Co-investment deal flow can be procyclical: opportunities often appear when valuations are already high.

⇒ C states both parts.

**Interpretation**

The two-week deadline and the small team make it worse: the investor has the least time to spot a weak deal when the risk of being offered one is highest.

**Why the other options are wrong**

**A is incorrect.** Co-investors usually accept fewer legal protections, not more; and that is not adverse selection.

**B is incorrect.** Access is restricted to large, established investors, not open to all.

**D is incorrect.** The opposite of adverse selection, with an absolute qualifier the reading does not support.

**LO:** IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.e

### Solution IM-11 e.6 Answer: B

New • built from SchweserNotes 2026 Reading 93

**Step 1 – The case for**

Reduced or no management fees and carry, so higher potential net returns; experience and stronger GP relationships.

**Step 2 – The case against, for this investor**

Effective co-investing needs significant internal resources, specialised expertise and the ability to evaluate deals quickly; investors without them often underperform. A three-person team facing accelerated deadlines is exactly that risk.

⇒ B weighs both sides as the reading does.

**Interpretation**

An "assess" question rewards the balanced statement. Here the fee saving is real, but the endowment would need to add resources first, or keep co-investing small next to its fund commitments.

**Why the other options are wrong**

**A is incorrect.** Absolute: transaction and due diligence costs are still shared, and fees are reduced, not always zero.

**C is incorrect.** The co-investor must do its own accelerated due diligence; relying on the GP's work ignores adverse selection.

**D is incorrect.** Polarity: co-investments are concentrated positions; they complement fund commitments, they do not replace their diversification.

**LO:** IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.e

IM-11 • Private Markets Investing • Vardeez: IMPORTANT

# IM-11 f

P1 CONCEPT

Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: shape S6 of section A28, the section reported from the August 2026 sitting (Map #49, 2 of 16 sheets).
**Vardeez:** no separate pointer; the objective sits in an IMPORTANT reading, and Vardeez covers it in one benefits-and-challenges table.

**Reported exam shapes**

- **S6** Private markets structures, co-investing and intermediated vehicles. Workbook A28.4 (worked as Question IM-11 b.6) tests the intermediated fund through blind pool risk (a role trap: who decides which assets are bought) and the fee layer.

**What the questions look like:** all conceptual. "Which is a key benefit (challenge) of investing through a PE or private debt fund", or a scenario (an LP that needs cash, a GP that holds a company's debt and its equity, a GP investing where the LP would not have) and "which challenge does this illustrate". Wrong options name a challenge as a benefit (liquidity, transparency, no fees) or pick the neighbouring challenge (illiquidity for blind pool, transparency for conflicts).

## 1. Delegation in one picture

Recap from IM-11 b: the private fund is the most common structure, usually a closed-end limited partnership in which LPs commit capital and the GP decides. Investing "through an intermediated vehicle" means accepting that arrangement. Every benefit below is a service the manager provides; every challenge is a consequence of having delegated (Figure 1).

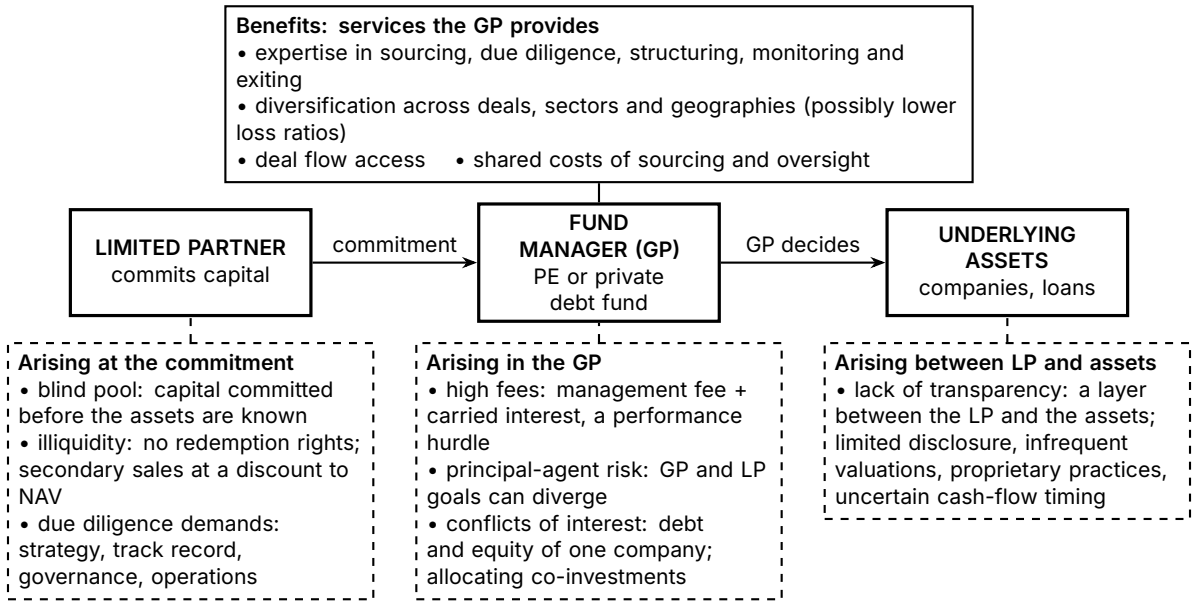



Figure 1. The intermediated vehicle. Solid box above: the four benefits, all supplied by the GP. Dashed boxes below: the seven challenges, placed where each arises. A wrong option usually moves a challenge to the benefit side (liquidity, transparency, no fees) or into the wrong dashed box.

## 2. Four benefits

- **Expertise.** Managers bring expertise in sourcing, due diligence, structuring, monitoring and exiting investments. Given how complex private markets are, this can reduce operational risk for investors who lack the skills.
- **Risk reduction.** Diversification is a key benefit: pooled capital reaches a wider range of deals, sectors and geographies. Professional management may also lead to lower loss ratios.
- **Deal flow access.** Managers often see attractive opportunities that most individual investors cannot reach, which matters more as private markets become more competitive and specialised.
- **Cost and resource sharing.** Pooling lets investors share the substantial costs of sourcing, analysis and ongoing oversight, costs that might otherwise keep an individual investor out of private markets altogether.

## 3. Seven challenges

- **High fees.** Professional management is expensive: a management fee and carried interest, which create a natural **performance hurdle** before the LP earns an attractive net return.
- **Principal-agent risk.** The GP's goals may not match the LPs', creating governance and performance concerns, including the potential for abuse or misconduct.
- **Lack of transparency.** Intermediation puts layers between investors and the assets, limiting their ability to monitor or influence them. Information asymmetry comes from limited disclosure, infrequent valuations, proprietary management practices and uncertainty about the timing of cash flows.
- **Blind pool risk.** Many PE and private debt funds are blind pools: investors commit before knowing which assets will be bought. Strategy guidelines exist, but the manager decides how the capital is deployed, which can produce exposures the investor would not have chosen.

- **Illiquidity risk.** Capital is committed for long periods with no redemption rights and limited secondary market options; an LP that needs cash may have to accept a significant discount to NAV.
- **Due diligence demands.** Investors must examine the fund's strategy, the manager's track record, governance structures and operational processes to reduce the risk of underperformance or misconduct.
- **Conflicts of interest.** For example, a manager whose funds hold both the equity and the debt of one company faces competing incentives between debtholders and equity holders; conflicts can also arise in allocating co-investment opportunities.

## 4. Explaining the trade-off

- **Four against seven, and the asymmetry is the point.** The investor buys the manager's services (expertise, diversification, deals, shared cost) and accepts the consequences of handing over the decisions.
- **Blind pool is its own risk.** It is about *not knowing what will be bought*. It is not illiquidity (you cannot get out) and not lack of transparency (you cannot see in), which are listed separately.
- **The responses sit in the rest of the reading.** Due diligence on the manager is the first defence. The structure can be changed: a managed account gives a large investor customisation and transparency; co-investments cut fees on chosen deals for an investor with the resources (IM-11 e); a fund of funds adds diversification but a second fee layer (IM-11 b). And the risks can be measured: agency risk through the fee structure (including carried interest), conflicts and the strength of governance; reporting risk through the gap between DPI and TVPI (IM-11 c and d).

| | What it means for the LP | Not to be confused with |
|---|---|---|
| **Expertise** | The GP sources, checks, structures, monitors and exits | "No due diligence needed" (the LP still vets the GP) |
| **Risk reduction** | Diversification; possibly lower loss ratios | The concentration of direct and co-investments |
| **Deal flow access** | Opportunities individual investors cannot reach | Adverse selection (a co-investing risk) |
| **Cost sharing** | Sourcing and oversight costs are pooled | "Lower fees" (fees are a challenge) |
| **High fees** | Management fee plus carry: a hurdle | Reinvestment risk (an IRR issue) |
| **Principal-agent** | GP and LP goals can diverge | Blind pool (what is bought, not why) |
| **Lack of transparency** | A layer between LP and assets | Illiquidity (getting out, not seeing in) |
| **Blind pool** | Commit first, the GP chooses later | Lack of transparency; illiquidity |
| **Illiquidity** | No redemption; secondary sale at a discount to NAV | Semiliquid funds' redemption windows (IM-10) |
| **Due diligence demands** | Strategy, track record, governance, operations | A benefit of intermediation |
| **Conflicts of interest** | Debt and equity of one firm; co-investment allocation | Settlement risk (IM-11 d) |

**Exam traps**

- **Liquidity, transparency or control as benefits.** All three are challenges of intermediation; the benefits are expertise, risk reduction, deal flow and cost sharing.
- **Fees eliminated or shared.** Costs of sourcing and oversight are shared; fees are paid on top, a management fee plus carry, and they set a hurdle.
- **Blind pool read as illiquidity or opacity.** Blind pool is committing before the assets exist; the GP decides within the guidelines.
- **Concentration as a challenge of funds.** Pooling reduces concentration; concentration belongs to direct investment and co-investing.
- **Carry as perfect alignment.** Carried interest is a fee and a hurdle; principal-agent risk remains, and agency risk is assessed through fees, conflicts and governance.
- **Due diligence delegated away.** Intermediation shifts due diligence from deals to the manager; it does not remove it.

## Practice questions: IM-11 f

### Question IM-11 f.1

Bank Q497 • Schweser Module Quiz 93.3 Q2

Which of the following is a key benefit of investing in private markets through intermediated vehicles such as private equity or private debt funds?

**A.** Access to expertise.

**B.** More access to liquidity.

**C.** Elimination of management fees.

**D.** Increased transparency and control.

### Question IM-11 f.2

New • built from SchweserNotes 2026 Reading 93

An investment consultant drafts a note for a family office that is considering its first commitment to a private debt fund. Which of the following statements in the note is most accurate?

**A.** Because the fund pools capital across many loans, the family office no longer needs to perform any due diligence of its own.

**B.** The fund's carried interest always aligns the general partner's interests with those of the limited partners, which offsets the cost of its fees.

**C.** Diversification and professional management may lower loss ratios, but the layer between the office and the loans limits its ability to monitor or influence them.

**D.** The fund's quarterly NAVs remove information asymmetry, because they are prepared by the manager that knows the loans best.

**Question IM-11 f.3** Bank Q498 • Schweser Module Quiz 93.3 Q3

Which of the following is a key challenge of investing in private markets through intermediated vehicles such as private equity or private debt funds?

**A.** Reinvestment risk.

**B.** Low information asymmetry.

**C.** Increased concentration risk.

**D.** Performance hurdle from fees.

**Question IM-11 f.4** New • built from SchweserNotes 2026 Reading 93

Two years ago a European pension fund committed to a buyout fund whose guidelines allow investments in mid-market industrial and healthcare companies across Europe. The GP has since invested most of the capital in healthcare companies in one country, an exposure the pension fund would not have chosen. Every investment is within the fund's guidelines. Which challenge of intermediated investing does this illustrate most directly?

**A.** Adverse selection.

**B.** Blind pool risk.

**C.** Illiquidity risk.

**D.** Lack of transparency.

**Question IM-11 f.5** New • built from SchweserNotes 2026 Reading 93

A private markets manager runs a buyout fund that owns the equity of a manufacturing company and a private debt fund that holds a loan to the same company. The company, now in difficulty, is negotiating a covenant waiver with its lenders. An LP in the debt fund asks which challenge of intermediated investing this situation raises. Which of the following answers is correct?

**A.** Blind pool risk, because the debt fund's LPs did not know in advance which loans the manager would make.

**B.** A conflict of interest, because a manager on both sides of the capital structure faces competing incentives between debtholders and equity holders.

**C.** Illiquidity risk, because the loan cannot be sold while the waiver is being negotiated.

**D.** Settlement risk, because the waiver has to be agreed and settled with a counterparty.

**Question IM-11 f.6** New • built from SchweserNotes 2026 Reading 93

An endowment committed to a private equity fund four years ago and now needs cash to cover a budget shortfall. Which of the following statements about its options is most accurate?

**A.** It can redeem its interest at the latest NAV at the fund's next quarterly redemption window.

**B.** The GP must buy back the interest at NAV, because LPs in closed-end funds hold a put right after the lockup period.

**C.** It has no redemption right and may have to sell its interest in a limited secondary market, possibly at a significant discount to NAV.

**D.** Illiquidity is not a concern, because the fund's diversification across deals ensures that some distributions arrive every quarter.

# Solutions • IM-11 f

## Solution IM-11 f.1

**Answer: A**

Bank Q497 • Schweser Module Quiz 93.3 Q2

### Step 1 – List the benefits

Expertise, risk reduction, deal flow access, cost and resource sharing.

### Step 2 – Test the options

Only access to expertise is on the list: managers bring skills in sourcing, due diligence, structuring, monitoring and exiting.
⇒ A. The other three are challenges stated the wrong way round.

### Interpretation

Expertise can reduce operational risk for investors who lack the skills.

### Why the other options are wrong

**B is incorrect.** Funds are illiquid: long commitments, no redemption rights, a limited secondary market at a discount to NAV.
**C is incorrect.** Intermediation adds fees (management fee plus carried interest).
**D is incorrect.** Intermediation adds a layer between investors and assets, reducing transparency and control.

**LO:** IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.f

## Solution IM-11 f.2

**Answer: C**

New • built from SchweserNotes 2026 Reading 93

### Step 1 – The benefit in C

Risk reduction: pooled capital is spread across deals, sectors and geographies, and professional management may lead to lower loss ratios.

### Step 2 – The challenge in C

Lack of transparency: layers between investors and assets limit monitoring and influence.
⇒ C pairs a real benefit with a real challenge.

### Interpretation

This is the trade the objective asks you to explain: the office gains the manager's services and loses direct sight and say.

### Why the other options are wrong

**A is incorrect.** Due diligence demands are a listed challenge: strategy, track record, governance and operations must still be examined.
**B is incorrect.** Absolute: carried interest is part of the fee hurdle; principal-agent risk remains.
**D is incorrect.** Infrequent valuations and limited disclosure cause information asymmetry; the manager's own NAV does not remove it.

**LO:** IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.f

### Solution IM-11 f.3 — Answer: D

Bank Q498 • Schweser Module Quiz 93.3 Q3

**Step 1 – List the challenges**

High fees, principal-agent risk, lack of transparency, blind pool risk, illiquidity, due diligence demands, conflicts of interest.

**Step 2 – Match the option**

Investors pay a management fee and carried interest on top of the underlying returns, so the fund must earn enough to cover them before the LP earns an attractive net return: a performance hurdle.

$\Rightarrow$ D.

**Interpretation**

The fee hurdle is why co-investing, with reduced or no fees, appeals to large LPs (IM-11 e).

**Why the other options are wrong**

**A is incorrect.** Reinvestment risk belongs to the IRR's reinvestment assumption (IM-11 c), not to intermediation.

**B is incorrect.** Polarity: intermediation *increases* information asymmetry.

**C is incorrect.** Polarity: pooling across many deals reduces concentration; concentration is a co-investing and direct-investment risk.

**LO:** IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.f

### Solution IM-11 f.4 — Answer: B

New • built from SchweserNotes 2026 Reading 93

**Step 1 – What happened**

The pension fund committed before the assets were known. The GP, acting within the strategy guidelines, chose investments that produced an exposure the LP would not have picked.

**Step 2 – Name it**

That is blind pool risk: investors commit capital before knowing exactly which assets will be acquired, and the manager ultimately decides how the capital is deployed.

$\Rightarrow$ B.

**Interpretation**

The pension fund can see what was bought and is not trying to sell: the risk is about who chooses, and when.

**Why the other options are wrong**

**A is incorrect.** Adverse selection is a co-investing risk: the GP offering weaker deals to co-investors.

**C is incorrect.** Illiquidity is about getting out (no redemption rights, a discount to NAV); the stem does not involve an exit.

**D is incorrect.** Lack of transparency is about seeing in; here the investments are known.

**LO:** IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.f

### Solution IM-11 f.5 — Answer: B

New • built from SchweserNotes 2026 Reading 93

**Step 1 – Where the manager stands**

Its buyout fund owns the equity and its debt fund holds the loan. In a waiver negotiation the lender wants protection and the shareholder wants flexibility.

**Step 2 – Name the challenge**

Schweser's own example of a conflict of interest: a fund manager that invests in both the equity and the debt of the same company faces competing incentives between debtholders and equity holders.

$\Rightarrow$ B. The debt fund's LPs depend on a manager that also represents the other side.

**Interpretation**

IM-11 d assesses this agency risk through fees, conflicts of interest and the strength of governance.

**Why the other options are wrong**

**A is incorrect.** A true feature of the fund, but blind pool risk is not what the waiver raises.

**C is incorrect.** The issue is the manager's incentives, not the saleability of the loan.

**D is incorrect.** Settlement risk is a counterparty risk in transactions (IM-11 d), not a conflict between the manager's funds.

**LO:** IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.f

### Solution IM-11 f.6 — Answer: C

New • built from SchweserNotes 2026 Reading 93

**Step 1 – The fund's terms**

A PE fund commits the LP's capital for a long period with no redemption rights.

**Step 2 – The way out**

Only the secondary market, which is limited; an LP that needs liquidity may have to accept a significant discount to NAV.

$\Rightarrow$ C.

**Interpretation**

A funding liquidity need has met an illiquid asset; IM-11 d measures this risk through the time until expected liquidity.

**Why the other options are wrong**

**A is incorrect.** Redemption windows belong to semiliquid evergreen funds (IM-10); a closed-end PE fund has none.

**B is incorrect.** No such put right exists; the GP has no obligation to buy back interests.

**D is incorrect.** Diversification is a benefit, but it does not make distributions regular or the interest liquid.

**LO:** IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. • **Reference:** Demaria et al., Asset Allocation and Private Markets (2021), Ch.3; SchweserNotes 2026 Book 5, Reading 93, LO 93.f

IM-14 • Madoff: A Riot of Red Flags • Vardeez: VERY IMPORTANT

# IM-14 b

P1 CONCEPT

Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk.

Workbook: A24 Hedge Funds, Survivorship and Madoff • Tier A • August 2026 recall

### WHAT IS TESTED HERE

**Why P1.** Tier A workbook topic: Madoff came up in the August 2026 sitting (1 question), and the A24 shapes were reported on 3 of 16 recall sheets (Map #34). Vardeez rates the reading Very Important: "Know this well and expect 1 question."
**Vardeez:** no pointer beyond the reading flag. His IM-14 b table groups the flags under returns and strategy, audit, governance and fee economics.

**Content authority:** SchweserNotes 2026 Reading 96 (Gregoriou and Lhabitant, 2009), used in place of the missing GARP 2026 book.

**Reported exam shapes**

- **S4** Madoff red flags as a case-to-lesson matching item against other failed-fund cases (Workbook A24.4, Question IM-14 b.10).

**What the questions look like:** all conceptual. Pick the warning sign from a list of BMIS features; explain why a feature signals fraud risk (fees, auditor, in-house custody, style analysis, options volume); sort flags into operational and investment; or match the Madoff case to its lesson among other hedge fund failures. The bank holds four questions here (Q545 fits IM-14 a better and is re-filed there), so six are new.

## 1. The case in one paragraph (history in IM-14 a)

Bernard L. Madoff Investment Securities (BMIS) was a genuine brokerage founded in 1960, and Madoff a former chairman of NASDAQ's board. Its advisory arm, set up in the late 1980s, managed about USD 17 billion by 2008 in 23 discretionary accounts, mostly *feeder funds*. It claimed a *split-strike conversion* strategy (long S&P 100 stocks, short out-of-the-money calls, long out-of-the-money puts) and reported 8% to 12% a year for about 17 years, with no down year and more than 95% of months positive. On 10 December 2008, unable to meet redemptions, Madoff told his family that the business was "a giant Ponzi scheme": the "returns" paid out had come from new investors' money. Losses were estimated at up to USD 50 billion.

"From regulators to investors, few asked the right questions": Madoff's standing deterred criticism, and end investors could examine only the feeder funds (Figure 1). The warning signs are **operational** (how the firm was organised and controlled) or **investment** (whether the stated strategy could have produced the returns).

## 2. Operational red flags: each removed an independent check

- **No independent service providers.** Execution, custody, administration and management, normally separate firms, were all in-house, "which made it far easier to manipulate performance results and potentially misappropriate assets".
- **Conflict of interest.** BMIS was broker-dealer and market maker for some of its own strategies: room for self-dealing, one less layer of oversight.
- **Questionable auditor.** USD 17 billion audited by Friehling & Horowitz: three people, 550 square

feet, one active accountant, and a firm that had told the AICPA it had no audit clients. The feeder funds' large auditors lent "an illusion of credibility" but relied on its work without verifying it.

- **Family in key roles.** Brother, sons, niece and nephew in senior posts; filings showed only one to five people running the advisory business. No independence, no segregation of duties.
- **Unusual fees.** No management or performance fee, only brokerage commissions: forgoing millions is uneconomic unless the manager is paid another way. It let the feeder funds charge high fees, which "likely encouraged feeder funds to overlook potential red flags".
- **Secrecy.** Feeders could not name Madoff or BMIS in their documents; AUM was not disclosed; visits and on-site due diligence were denied or restricted; questions on the strategy went unanswered.
- **Late SEC registration.** Only in 2006, when a rule change closed the loophole of counting each feeder fund as one "client"; BMIS stayed registered after the SEC reversed the rule. It gave a false sense of oversight.
- **Paper trade confirmations.** Tickets came by mail, with no electronic access and no third-party verification, so confirmations could be fabricated after the fact to match the reported results.

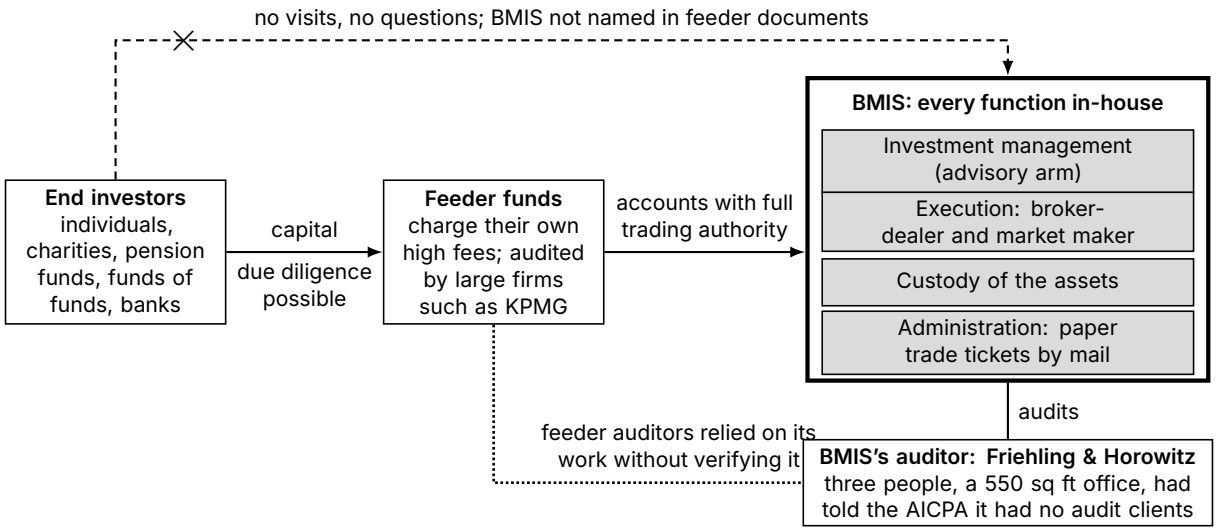



Figure 1. Who could check what. Solid arrows: money and authority. Dashed arrow with a cross: the check end investors could not make. Dotted line: reliance on BMIS's tiny auditor. Grey boxes in the heavy border: four functions normally run by separate firms, all inside BMIS.

## 3. Investment red flags: could the strategy earn these returns?

- **Black box.** A split-strike conversion "will inevitably produce some negative months"; Madoff reported almost none. Investors rationalised it (stock picking, smoothing with market-making profits, leverage, even illegal front running); no legitimate explanation fits. The long put limits the *size* of a loss, not its *frequency*: whenever the index falls by less than the distance to the put strike, the collar loses (Figure 2).
- **Questionable style exposure.** Style analysis of the returns showed exposures *opposite* to those of long stock, short call, long put: the trades were misrepresented or never executed.
- **Illogical Form 13F filings.** Small holdings in minor non-S&P 100 companies. Madoff's excuse, that he moved to cash at each quarter end, is implausible at USD 17 billion.

- **Impossible use of S&P 100 options.** These are less liquid than S&P 500 options; trades of the claimed size would have moved the market, and none did. "Over the counter" fails too: counterparties would have lost year after year, carried huge credit exposures and hedged, and none were ever identified.

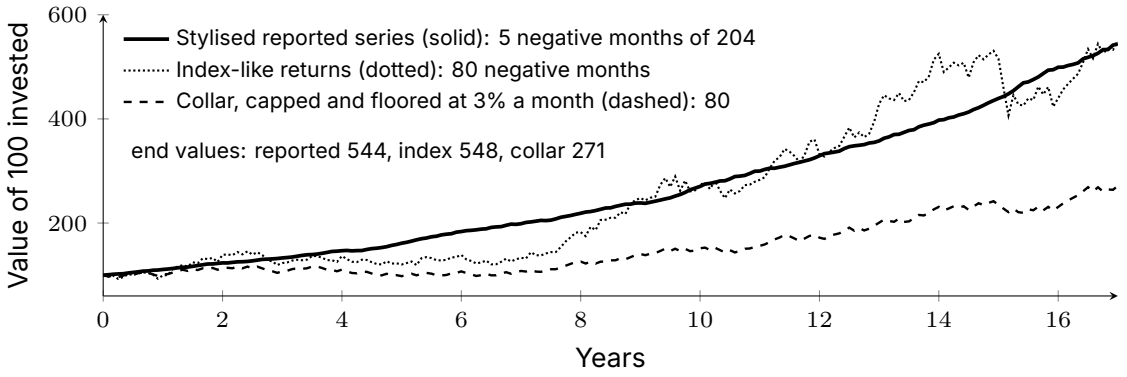



Figure 2. Simulated illustration, not fund data (inputs and seed in the verification script). Index-like months: mean 0.8%, standard deviation 4.33%. The collar clips each month to between $-3\%$ and $+3\%$ (premiums ignored): it caps the size of a loss, not its frequency, so it has as many negative months as the index. The stylised reported series (mean 0.83%, standard deviation 0.45%) has the feeder funds' profile: no down year, the index's end value, a tenth of its volatility.

## 4. Who saw it, and the lessons (shape S4)

The flags were visible from outside. Harry Markopolos repeatedly warned the SEC; his 2005 report, *The World's Largest Hedge Fund Is a Fraud*, argued the returns were impossible without front running or a Ponzi scheme, and was dismissed. A few banks refused to deal with BMIS, and a method by Brown, Goetzmann, Liang and Schwarz (2008) using only public Form ADV data would have flagged it. No investigation found the fraud: Madoff confessed (IM-14 a). Each flag is a due diligence item of IM-12: independent administration, custody and audit (operational, IM-12 e); fees against the cost of the business, related-party execution and too-smooth returns (IM-12 f); access to the decision makers (IM-12 c). GARP tests the lesson by swapping it between famous cases:

- **Madoff (BMIS), 2008:** in-house execution, custody and administration and a tiny auditor let impossibly smooth returns be fabricated.
- **LTCM, 1998:** assets that normally move independently become correlated in stress; monitor leverage and liquidity.
- **Lehman, MF Global, Refco:** cash at a failed prime broker or dealer may be recovered only in part (perhaps 30% to 40%).
- **Archegos, 2021:** leveraged total return swaps cost the bank counterparties about USD 10 billion.

**Correction to the source notes.** Vardeez says investors who asked detailed questions "were threatened with having their accounts returned". The reading, as SchweserNotes reports it, says only that Madoff refused questions and denied or restricted visits: answer from that (Bank Q544). Vardeez's four categories (returns, audit, governance, fees) also omit eight of the reading's twelve flags; use the card below.

## Red flags card: IM-14 b

| Red flag | The due diligence check that catches it |
|---|---|
| *Operational* | |
| Execution, custody and administration in-house | Independent administrator and custodian (IM-12 e) |
| Broker-dealer and market maker for its own strategies | Related-party execution (IM-12 f) |
| Three-person auditor for USD 17 billion | Auditor's size against the fund's; who audits the manager |
| Family in key roles; one to five advisory staff | Organisation chart, segregation of duties |
| Only brokerage commissions, no management or performance fee | Revenue against the cost of the business (IM-12 f) |
| No visits, no questions, BMIS not named | Access to the decision makers (IM-12 c) |
| SEC registration only in 2006 | Form ADV history |
| Paper confirmations by mail | Custodian statements, electronic access |
| *Investment* | |
| Almost no negative months | Count down months against the strategy |
| Style exposures opposite to the stated strategy | Returns-based style analysis |
| Form 13F: small holdings outside the S&P 100 | Read the public filings |
| Huge S&P 100 option volume, no counterparty | Market volume; name the counterparties |

## Exam traps

- **Fee polarity.** The flag was *no* management or performance fee, only commissions: not high fees, not zero fees.
- **Auditor role.** The problem was BMIS's own three-person auditor; the feeder funds' large auditors relied on it.
- **Smooth, not volatile.** Under 5% negative months was the flag, not unexplained large losses. Puts cap the size of a loss, not its frequency.
- **Style analysis.** Opposite exposures mean misrepresented or unexecuted trades, not market timing.
- **OTC excuse.** Over-the-counter trades need counterparties who would have lost and hedged; none existed.
- **Registration is not oversight.** The 2006 registration gave false comfort.
- **Case cross-mapping.** Correlations in stress: LTCM. Cash at a failed dealer: Lehman, MF Global, Refco. In-house custody and a tiny auditor: Madoff.

## Practice questions: IM-14 b

### Question IM-14 b.1

Bank Q546 • AnalystPrep Q.4854

Bernard Madoff told his family that his advisory business was "a giant Ponzi scheme". A trainee analyst at a fund of hedge funds is asked to define the term for an internal fraud-awareness course. Which of the following best describes a Ponzi scheme?

**A.** An investment firm forwards funds gathered from new investors to existing investors disguised as returns instead of investing the funds.

**B.** An investment firm gets into positions that are riskier than investors can tolerate.

**C.** An investment firm reports losses even though it has made substantial profits.

**D.** An investment firm buys securities other than those the investors have been promised as detailed in the prospectus.

### Question IM-14 b.2

Bank Q547 • Schweser Module Quiz 96.2 Q1

A due diligence analyst is preparing a case study of Bernard L. Madoff Investment Securities (BMIS) for new hires. The fee structure at BMIS was a warning sign to investors because Madoff's firm:

**A.** charged unusually high performance fees.

**B.** charged unusually high asset management fees.

**C.** charged only a brokerage commission on trades.

**D.** did not charge any fees, relying solely on investment returns.

### Question IM-14 b.3

Bank Q544 • Mock AP Mock 3 Q31 (rebuilt)

The investment committee of a pension fund asks its consultant which features of Bernard L. Madoff Investment Securities (BMIS) should have alerted investors before 2008. Which of the following was a warning sign that should have alerted investors to potential fraud at BMIS?

**A.** BMIS refused to answer questions about its investment strategy and restricted on-site due diligence.

**B.** BMIS reported occasional large negative monthly returns that were not explained by market moves.

**C.** BMIS used a large, well-known audit firm that was independent of its feeder funds' auditors.

**D.** BMIS charged high management and performance fees that exceeded those of comparable hedge funds.

**Question IM-14 b.4** New • built from SchweserNotes 2026 Reading 96

The operational due diligence team of a fund of hedge funds is reviewing why audits failed to protect investors in Madoff's feeder funds, which were themselves audited by large, well-known firms. Which statement best explains why the audit arrangements around BMIS should have been read as a fraud-risk indicator?

**A.** The feeder funds' auditors were too large to be independent of BMIS, so investors should have insisted that BMIS use a small specialist firm instead.

**B.** BMIS was audited by the same large firm as its feeder funds, so no auditor in the chain was independent of the feeder funds' managers.

**C.** BMIS's auditor issued a qualified opinion every year, which investors overlooked because the feeder funds' own opinions were unqualified.

**D.** BMIS's own auditor, a three-person firm that had told the AICPA it had no audit clients, could not credibly audit USD 17 billion, and the feeder funds' auditors relied on its work.

**Question IM-14 b.5** New • built from SchweserNotes 2026 Reading 96

A forensic accountant teaching a fraud-detection course asks which red flag at BMIS most directly explains how the firm could fabricate trade confirmations after the fact to match its reported performance. Which answer is most accurate?

**A.** The absence of management and performance fees, which left BMIS dependent on brokerage commissions.

**B.** Late registration with the SEC as an investment adviser, which came only in 2006.

**C.** Paper trade tickets sent by mail, combined with the absence of any third-party verification of trades.

**D.** Family members in senior positions, together with a very small advisory staff.

**Question IM-14 b.6** New • built from SchweserNotes 2026 Reading 96

A consultant sorts the warning signs at BMIS into operational red flags (how the firm was organised and controlled) and investment red flags (whether the reported returns could have come from the stated strategy). Which sorting is correct?

**A.** Late registration with the SEC as an investment adviser: investment red flag.

**B.** Form 13F filings showing small holdings in companies outside the S&P 100: investment red flag.

**C.** Reported returns with almost no negative months: operational red flag.

**D.** Paper trade confirmations sent by mail: investment red flag.

**Question IM-14 b.7** New • built from SchweserNotes 2026 Reading 96

A quantitative analyst at a fund of hedge funds runs a returns-based style analysis on a Madoff feeder fund. The fund states that it holds S&P 100 stocks, writes out-of-the-money S&P 100 calls and buys out-of-the-money S&P 100 puts. The analysis finds exposures opposite to those this combination would produce. Which conclusion is most appropriate?

**A.** The exposures show skilled market timing: the manager reverses the collar ahead of market turns, which explains the smooth returns.

**B.** The reported trades were either misrepresented or never executed, a fraud-risk indicator that calls for independent verification of positions.

**C.** Style analysis cannot be applied to option strategies, so the result says nothing about the fund and can be disregarded.

**D.** The result reflects the portfolio being moved into cash at each quarter end, so it raises no concern about the strategy.

**Question IM-14 b.8** New • built from SchweserNotes 2026 Reading 96

Asked why listed-market data showed no trace of BMIS's very large S&P 100 option trades, Madoff said the options were traded over the counter (OTC). A risk manager reviewing the case explains why this answer did not remove the red flag. Which explanation is most accurate?

**A.** OTC options must be reported in Form 13F filings, so the options would have appeared in BMIS's quarterly filings.

**B.** OTC index options are more liquid than listed S&P 500 options, so trades of that size would have been easy to observe.

**C.** OTC options are cleared through the exchange, so the trades would still have appeared in listed-market volume data.

**D.** The OTC counterparties would have lost money year after year, carried large credit exposures and needed to hedge, yet no counterparty was ever identified.

**Question IM-14 b.9** New • built from SchweserNotes 2026 Reading 96

After the collapse, a trustee of a pension fund that lost money through a feeder fund argues that the BMIS fraud could not have been detected before December 2008 without confidential information. Which response is most consistent with the reading?

**A.** The trustee is wrong: Markopolos warned the SEC that the returns were impossible without front running or a Ponzi scheme, some banks declined to deal with BMIS, and public Form ADV data would have flagged the firm.

**B.** The trustee is right: the fraud was uncovered only by an SEC investigation that relied on subpoenaed trade records unavailable to investors.

**C.** The trustee is right: BMIS's audited statements and its registration with the SEC gave investors no reason for concern.

**D.** The trustee is wrong, but only because the feeder funds' auditors had reported their doubts about BMIS to investors before 2008.

**Question IM-14 b.10** Workbook A24.4 • S4 • Aug-2026 • Map #34

A risk consultant is presenting failed hedge fund cases to the due diligence committee of a fund of hedge funds. Which of the following lessons is best illustrated by the case of Bernard L. Madoff Investment Securities (BMIS)?

**A.** Assets that normally behave independently can become highly correlated in extreme stress, so a fund's leverage and liquidity must be monitored closely.

**B.** Unusually high management and performance fees gave the manager an incentive to report smooth returns, which the feeder funds' due diligence failed to question.

**C.** Keeping execution, custody and administration in-house with a three-person auditor allowed returns with over 95% positive months to be fabricated undetected.

**D.** A fund can fail through no fault of its own when its prime broker goes bankrupt, because cash held as margin may be recovered only in part after litigation.

## Solutions • IM-14 b

### Solution IM-14 b.1 — Answer: A

Bank Q546 • AnalystPrep Q.4854

**Step 1 – The defining feature**

A Ponzi scheme pays "returns" to existing investors out of capital raised from new investors, instead of from investment profits. The money is not invested, or not invested as claimed.
⇒ A describes exactly that.

**Step 2 – Link to BMIS**

Madoff's reported returns were fabricated; withdrawals were paid from new subscriptions. The scheme survived while inflows exceeded redemptions and collapsed in late 2008 when they did not (IM-14 a).

**Interpretation**

This is why the red flags matter: a Ponzi scheme can only show smooth, steady returns, and it needs secrecy and the absence of independent custody to keep going. Every operational red flag at BMIS is a condition the scheme needed.

**Why the other options are wrong**

**B is incorrect.** Excessive risk-taking is a suitability or risk management failure, not recycled capital.
**C is incorrect.** The opposite of a Ponzi scheme, which overstates performance.
**D is incorrect.** Buying securities other than those promised is a mandate breach or style drift; the money is still invested.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant, Madoff: A Riot of Red Flags (EDHEC, 2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a and 96.b

### Solution IM-14 b.2 — Answer: C

Bank Q547 • Schweser Module Quiz 96.2 Q1

**Step 1 – What BMIS charged**

No management fee and no performance fee: BMIS collected only brokerage commissions on the trades it said it executed.
⇒ C is the fact.

**Step 2 – Why that is a red flag**

A manager with 17 years of steady 8% to 12% returns would normally earn millions in management and performance fees. Forgoing them is uneconomic unless the manager is paid some other way. It also let the feeder funds charge high fees of their own, which gave them a reason not to probe.

**Interpretation**

Read a fee structure as a business model question (IM-12 f): who earns what, and does it make sense? Here it did not, and the anomaly bought the silence of the intermediaries.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: BMIS charged no performance fee.
**B is incorrect.** Polarity flip: BMIS charged no asset management fee.
**D is incorrect.** Too absolute: BMIS did charge brokerage commissions, so it was not fee-free.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential

"red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.b (Unusual fee structure)

### Solution IM-14 b.3 — Answer: A

Bank Q544 • Mock AP Mock 3 Q31

#### Step 1 – The transparency red flag

Madoff refused to answer questions about his strategy, did not disclose AUM, barred feeder funds from naming BMIS, and denied or restricted on-site due diligence.
⇒ A is one of the reading's operational red flags.

#### Step 2 – Check the other three against the facts

B: BMIS reported almost no negative months, fewer than 5%. C: its auditor was a three-person firm, Friehling & Horowitz. D: it charged no management or performance fees.
⇒ B, C and D each reverse a fact.

#### Interpretation

Secrecy blocks verification. A manager who refuses the questions due diligence asks removes the only way an investor can test the claimed strategy, which is why restricted access is itself a fraud-risk indicator.

#### Why the other options are wrong

**B is incorrect.** Reversed: the warning sign was the near absence of negative months, not large losses.
**C is incorrect.** Reversed: the auditor was tiny; the large firms audited the feeder funds and relied on it.
**D is incorrect.** Reversed: BMIS charged only brokerage commissions, which was odd in its own way.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.b (Lack of transparency and restricted communication)

### Solution IM-14 b.4 — Answer: D

New • built from SchweserNotes 2026 Reading 96

#### Step 1 – Who audited BMIS

Friehling & Horowitz: three people in a 550 square foot office, one partner in his late 70s and largely inactive, one practising accountant, and a declaration to the AICPA that it had no audit clients. It audited a firm managing about USD 17 billion.
⇒ An auditor that small cannot give assurance on a firm that large.

#### Step 2 – Why the large feeder auditors did not help

Feeder funds were audited by major firms such as KPMG, which "lent an illusion of credibility". Those firms relied on Friehling & Horowitz's work without verifying it.
⇒ D states both halves.

#### Interpretation

The red flag is about capacity relative to the entity, and about the chain of reliance. An investor who sees a household-name auditor on a feeder fund's statements should still ask who audits the manager that holds the assets.

#### Why the other options are wrong

**A is incorrect.** Polarity flip: the risk came from the tiny auditor, not from the large ones.

**B is incorrect.** Wrong fact: BMIS and the feeder funds had different auditors; BMIS's was the three-person firm.
**C is incorrect.** No basis in the reading; the issue was the auditor's capacity and the reliance on it, not a qualified opinion.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.b (Questionable auditing firm)

### Solution IM-14 b.5 — Answer: C

New • built from SchweserNotes 2026 Reading 96

#### Step 1 – What fabrication needs

To fabricate a confirmation after the fact, the firm must control the record the client receives, and no outside party may hold a conflicting record.

#### Step 2 – Match the red flag

BMIS mailed paper trade tickets and gave no electronic account access. "This practice, combined with the lack of third-party verification, made it possible for the firm to fabricate trade confirmations after the fact to match reported performance results."
⇒ C is the mechanism.

#### Interpretation

A, B and D are genuine red flags, but none of them is the mechanism asked for. The question tests the "explain how" half of the LO: link each flag to what it makes possible.

#### Why the other options are wrong

**A is incorrect.** A real red flag (economic illogic), but fees do not create or alter trade records.
**B is incorrect.** A real red flag (false comfort of oversight), but registration does not explain how confirmations were fabricated.
**D is incorrect.** A real red flag (weak independence and oversight), but not the direct mechanism; the paper tickets were.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.b (Use of paper trade confirmations; Lack of independent service providers)

### Solution IM-14 b.6 — Answer: B

New • built from SchweserNotes 2026 Reading 96

#### Step 1 – The two lists

Operational: no independent service providers, broker-dealer conflict, questionable auditor, family in key roles, unusual fees, secrecy, late SEC registration, paper confirmations. Investment: black box (no negative months), style exposure, Form 13F filings, S&P 100 options volume.

#### Step 2 – Sort each option

Form 13F filings showing small non-S&P 100 holdings contradict the stated strategy: an investment red flag.
⇒ B is sorted correctly.

#### Interpretation

The test for "investment" is whether the flag says something about the returns or holdings against the strategy; "operational" flags describe how the firm was organised and controlled. Madoff's

excuse (moving to cash at each quarter end) was itself implausible at USD 17 billion.

### Why the other options are wrong

**A is incorrect.** Late SEC registration is an operational red flag.
**C is incorrect.** Returns with almost no negative months are the black-box investment red flag.
**D is incorrect.** Paper confirmations are an operational red flag.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.b (Operational red flags; Investment red flags)

## Solution IM-14 b.7 — Answer: B

New • built from SchweserNotes 2026 Reading 96

### Step 1 – What the stated position should show

Long S&P 100 stocks, short out-of-the-money calls and long out-of-the-money puts: positive exposure to the index between the strikes, with the upside capped and the downside floored.

### Step 2 – What the analysis found

Exposures opposite to those of the stated position. The reading draws the conclusion that the reported trades "were either misrepresented or never executed".
⇒ B: a fraud-risk indicator that calls for independent verification of the positions.

### Interpretation

Statistical analysis of published returns needs no confidential data. It was available to any investor holding the monthly return series.

### Why the other options are wrong

**A is incorrect.** A rationalisation, the kind investors used; no legitimate explanation fits the results.
**C is incorrect.** Style analysis applies to any return series; dismissing an inconvenient result is how the red flag was missed.
**D is incorrect.** The quarter-end cash claim was itself implausible (the Form 13F red flag) and does not explain reversed exposures.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.b (Questionable style exposure)

## Solution IM-14 b.8 — Answer: D

New • built from SchweserNotes 2026 Reading 96

### Step 1 – The listed-market problem

S&P 100 options are less liquid and less traded than S&P 500 options. With about USD 17 billion, BMIS's trades would have moved the market, and no such moves were seen.

### Step 2 – Why the OTC answer fails

OTC trades need counterparties. Those counterparties "would have suffered losses year after year, faced enormous credit exposures, and needed to hedge their positions, yet there is no evidence that any of this occurred". No counterparty was ever confirmed.
⇒ D: moving the trades OTC only moves the red flag; the trades were never executed.

### Interpretation

An excuse that creates new, checkable consequences is a test, and here every consequence was absent. Ask the manager to name the counterparties.

### Why the other options are wrong

**A is incorrect.** No basis in the reading: its Form 13F red flag is about the small non-S&P 100 stocks the filings showed. The OTC claim fails because of the missing counterparties.
**B is incorrect.** Reversed: S&P 100 options were the illiquid ones, which is the original red flag.
**C is incorrect.** OTC options are bilateral contracts, not exchange-cleared; that is why they did not show in listed volume.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.b (Impossible use of S&P 100 options)

## Solution IM-14 b.9

**Answer: A**

New • built from SchweserNotes 2026 Reading 96

### Step 1 – Who saw the signs

Harry Markopolos repeatedly warned the SEC; his 2005 report argued that the returns were impossible without front running or a Ponzi structure. A few banks declined to do business with BMIS.

### Step 2 – What public data could show

Using only public Form ADV information, the method of Brown, Goetzmann, Liang and Schwarz (2008) would have identified BMIS as a problem fund. The smooth return series was public too.
⇒ A: the fraud was detectable from outside.

### Interpretation

The reading's verdict is that due diligence was largely absent and "greed often overrode sound judgment". The lesson for investors is never to let attractive returns overshadow clear red flags.

### Why the other options are wrong

**B is incorrect.** Wrong: no investigation uncovered the fraud; Madoff confessed to his family on 10 December 2008.
**C is incorrect.** Wrong: the tiny auditor and the late, loophole-driven registration were themselves red flags.
**D is incorrect.** No basis: the feeder funds' auditors relied on BMIS's auditor without verifying its work.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (Fund collapse and confession) and LO 96.b

## Solution IM-14 b.10

**Answer: C**

Workbook A24.4 • S4 • Aug-2026

### Step 1 – Match the case to its lesson

BMIS performed execution, custody, administration and management itself (no independent service providers), was audited by Friehling & Horowitz, a three-person firm in a 550 square foot office that had told the AICPA it had no audit clients, and reported 8% to 12% a year with no down year and over 95% of months positive.
⇒ In-house functions made it easy to fabricate performance; independent verification of service providers would have exposed it. C is correct.

### Interpretation

Mirabile's operational due diligence lesson in one line: independent administration, audits and

top-tier service providers are "generally not negotiable", and "if it looks too good to be true, it probably is". Madoff was not found by investigators: he confessed to his family on 10 December 2008.

### Why the other options are wrong

**A is incorrect.** Case cross-mapping: this is the lesson of LTCM (1998): monitor liquidity and leverage, because assets that usually behave independently move together in stress.

**B is incorrect.** Polarity flip: BMIS charged no management or performance fee, only brokerage commissions. Forgoing fees was the red flag, and it left room for feeder funds to charge high fees of their own.

**D is incorrect.** Case cross-mapping: this is the lesson of Lehman, MF Global and Refco in Mirabile's counterparty discussion, where funds recovered perhaps 30 to 40 percent of cash held at the failed dealer.

**LO:** IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Gregoriou and Lhabitant (2009) (SchweserNotes 2026 Book 5, Reading 96); Mirabile, Ch.12 (GARP 2025 Ch.10); Stowell and Stowell, Ch.14 (SchweserNotes 2026 Book 5, Reading 91)

IM-17 • Illiquid Assets • Vardeez: VERY IMPORTANT

# IM-17 c

P1 CONCEPT

Assess the impact of biases on reported returns for illiquid assets.

Workbook: A24 Survivorship and Other Biases • Tier A • Map #34 D9 Illiquid Assets • Tier D • Map #48

**WHAT IS TESTED HERE**

**Why P1.** Tier A workbook topic: section A24 (August 2026 recall: hedge funds 2Q, Madoff 1Q) carries the survivorship-bias shape, reported on 3 of 16 recall sheets (Map #34: "survivorship bias questions"). The same biases come back in Tier D section D9 (Map #48, 2 of 16).
**Vardeez:** no pointer for this LO; the reading is rated VERY IMPORTANT, with LO 99.e and 99.f flagged Important.

**Reported exam shapes**

- **A24·S3** Survivorship bias against its siblings: selection, reporting, backfill. Map #34: "the bias sibling set ... is the discrimination GARP tests".
- **D9·S1** Biases in reported illiquid asset returns: survivorship, selection, infrequent trading, unsmoothing. The unsmoothing half is drilled in IM-17 d, where Workbook D9.1 is worked.

**What the questions look like:** all conceptual. A data problem is described (dead funds dropped, sales recorded only at high prices, appraisal-based values, voluntary reporting) and you must name the bias, its direction (return up or risk down) and its remedy. Some stems carry a table of returns and volatilities: the numbers are there to tempt you into trusting reported risk measures, not to be computed.

## 1. Background: what makes an asset illiquid

Shared background for the whole reading. IM-17 a and b are P4 and are not drilled, but the biases only make sense against them.

**Four characteristics of illiquid markets (IM-17 a).**

- **Most asset classes are illiquid**, at least to some degree: illiquid assets trade infrequently, in small amounts and with low turnover. Public equities trade within seconds (turnover above 100% a year); OTC equities usually within a day (turnover 25% to 35%); corporate bonds daily; municipal bonds about twice a year; real estate turns over about 5% (residential) and 7% (institutional) a year; art changes hands every 40 to 70 years; infrastructure commitments run 50 to 60 years (up to 99).
- **Illiquid markets are large.** In 2012 the US residential mortgage market (USD 16 trillion) and institutional real estate (USD 9 trillion) compared with about USD 17 trillion for NYSE and Nasdaq combined. Total illiquid wealth exceeds the wealth in liquid stock and bond markets.
- **Investors hold lots of illiquid assets:** about 90% of individuals' wealth excluding human capital (mainly the home); US high-net-worth investors hold about 10% in "treasure" assets such as art and jewellery (20% abroad); endowments about 25% (5% in the early 1990s) and pension funds about 20% (5% in 1995).
- **Liquidity dries up** even in normally liquid markets: repo and commercial paper froze in 2007 to 2009, as did MBS, structured credit and auction rate securities. Major liquidity crises have come at least once every ten years.

**Six market imperfections behind illiquidity (IM-17 b):** participation costs and clientele effects,

transaction costs, search frictions, asymmetric information (the lemons problem), price impact and funding constraints. The last four can stop a trade however much the investor is willing to pay.

## 2. "Reported illiquid asset returns are not returns"

Ang's verdict is that investors should be sceptical of the reported returns of illiquid assets. Three biases make the reported numbers flatter the asset class: together they **overstate expected returns and understate risk**. Each answers a different question about the data set:

- **Survivorship bias**: *which funds* are in the data?
- **Sample selection bias**: *when* is an asset's value observed?
- **Infrequent trading** (infrequent sampling): *how often* is a price observed?

## 3. Survivorship bias and its siblings (shape A24·S3)

**Mechanism.** Private equity, buyout and hedge funds have no obligation to report to database vendors. Poor performers stop reporting, some never start, and many eventually fail. Performance studies then keep only the funds that survived the whole period. Non-survivors have below-average returns and survivors above-average returns, but only the survivors are reported, so **reported returns are too high**. A study of smoking that sampled only smokers who lived to old age would make the same mistake.

**Size.** At least 1% to 2% a year for mutual funds, and more than 4% when defunct and live funds are compared. For illiquid asset managers these are lower bounds.

**Remedy.** "The only way to completely remove the effect of survivorship bias is to observe the entire population of funds", which is impossible in illiquid markets. Mutual fund databases now keep failed funds (Bodie, Kane and Marcus).

**The siblings GARP sets against it.**

- **Reporting bias.** Reporting is voluntary, so funds stop when returns are poor, or never start because the track record never becomes attractive; managers also "put the best face forward" by reporting only their better funds (self-selection). Ang treats reporting bias as part of survivorship bias, with the same remedy.
- **Sample selection bias** is about *when* values are recorded, not *who* is in the database (Section 4).
- **Backfill (instant history) bias.** When a fund joins a database, its incubation-period returns are loaded into its history. Failed incubations never join, so the early history of new funds is biased upward. It is not one of Ang's three biases; it comes from the hedge fund literature (Fung and Hsieh). Remedy: discard returns from before the fund joined.

## 4. Sample selection bias

**Mechanism.** Asset values and returns are recorded when they are high, because that is when assets are sold:

- houses and office buildings are sold when prices are high, and owners wait for a recovery before selling;
- buyout funds take companies public when stock prices are high, and venture capitalists sell when values are high;

- distressed portfolio companies are often not liquidated but left as "zombie" shells, so old data with no recent transaction cannot tell a live company from a zombie.

**Effect: both halves of the risk-return pair.** Fitting the security market line to the observed returns (Figure 1) gives

- **alpha too high**: a true alpha of zero shows up as positive;
- **beta too low**: the fitted line is *flatter*, because the low returns that would pin it down in bad markets are missing;
- **volatility too low**: only the upper part of the return distribution is seen.

Cochrane's venture capital estimate is the headline number: an alpha above 90% (log returns) falls to $-7\%$ after correcting for selection. The effect is smaller in real estate, whose volatility is lower and whose samples include the downturns of the early 1990s and early 2000s. **Remedy:** Heckman (1979) selection models, which let the threshold above which returns are observed vary over time.

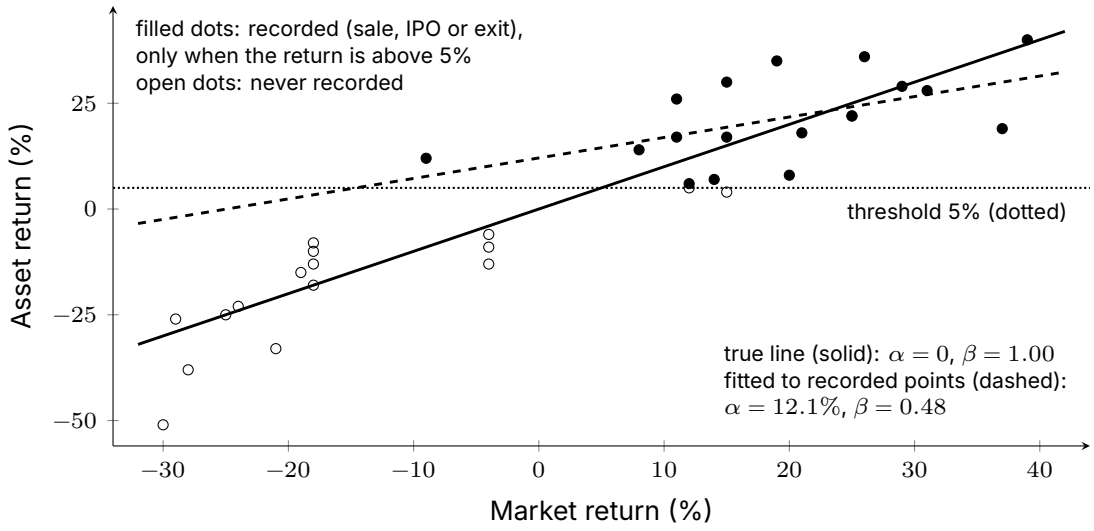



Figure 1. Sample selection bias, simulated. Every asset has a true alpha of zero and a beta of one (solid line; a regression on all 34 points gives $\alpha = -0.3\%$, $\beta = 1.00$). Only the 18 returns above 5% are recorded. The line fitted to them (dashed) has a positive alpha and a flatter slope, and their volatility is 10.2% against 23.3% for all points: alpha overstated, beta and volatility understated.

## 5. Infrequent trading

**Mechanism.** Illiquid assets are priced by appraisal or at rare transactions, so reported values are stale and reported returns are **smoothed**. The fingerprint of smoothing is positive autocorrelation in the reported returns.

**Effect.** Volatilities, correlations and betas computed from reported returns are **too low**; correlations with liquid assets are artificially low because volatility is muted. Even a liquid series looks calmer when sampled less often: in Ang's illustration the same prices give a volatility of 0.23 from quarterly data against 0.28 from daily data. The mean return is essentially unaffected: infrequent trading is a *risk* bias.

**Not a small-sample problem.** Ang calls it a "population" problem: a longer history of smoothed returns gives a more precise estimate of the wrong number.

**Remedy:** unsmoothing (Geltner, Ross and Zisler), which adds the noise back (IM-17 d). For the NCREIF real estate index, quarterly volatility rises from 2.25% to 6.26% and the correlation with the S&P 500 from 9.2% to 15.8%.

## 6. What the biases do to the numbers you use

- **Sharpe ratio: every bias pushes it up.** Survivorship and selection raise the numerator; infrequent trading and selection shrink the denominator (Figure 2).
- **Correlation and beta are understated,** so illiquid assets look like better diversifiers than they are, and a mean-variance optimiser fed reported statistics over-allocates to them.
- **Mixed comparisons mislead.** Statistics computed from market prices (an ETF) are reliable; statistics computed from reported private equity or real estate returns are biased toward high return and low risk. Ranking the two on reported return per unit of volatility favours the illiquid asset by construction.
- **One remedy per bias:** the full population for survivorship, Heckman models for selection, unsmoothing for infrequent trading. None of them fixes another bias.

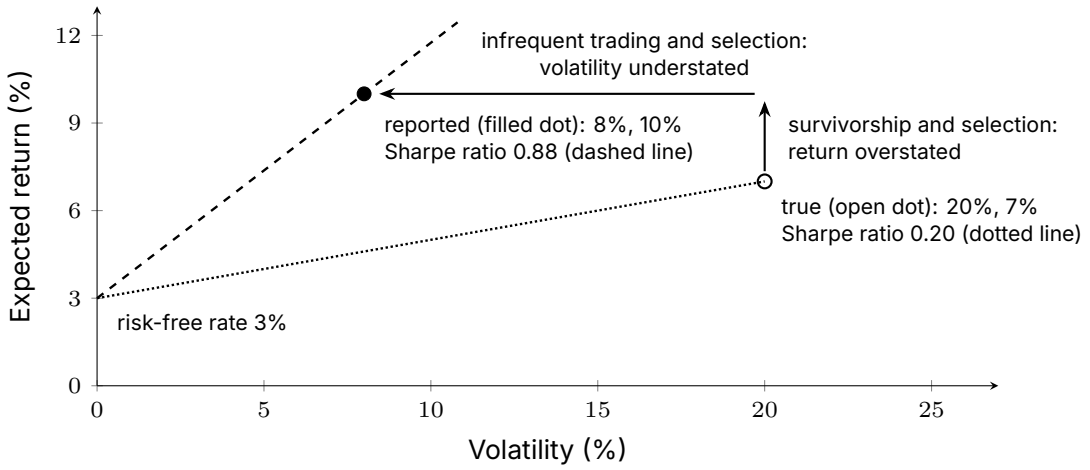



Figure 2. The three biases on one risk-return chart (illustrative numbers, risk-free rate 3%). The reported point sits up and to the left of the true point, so the slope from the risk-free rate, the Sharpe ratio, more than quadruples.

**Correction to the source notes.**

- Vardeez's box "Two biases inflate returns, one deflates risk" says survivorship and selection act on returns and infrequent trading on risk, "different halves of the risk-return pair". That is incomplete for selection bias: Schweser (following Ang) states that it produces "overestimated expected returns *and* underestimated risk as measured by beta and the standard deviation". Selection bias acts on both halves; only survivorship bias is purely a return bias and only infrequent trading purely a risk bias.
- Schweser writes that survivorship-adjusted returns "may be as much as 4% lower for illiquid asset markets", which reads like a ceiling. Ang (quoted in Workbook A24) gives at least 1% to 2% for mutual funds and more than 4% when defunct and live funds are compared, both *lower bounds* for illiquid asset managers. Treat 4% as a floor.
- Vardeez (IM-17 b) says search frictions and asymmetric information are "the two that cannot be priced away". Schweser lists four imperfections that can stop a trade "no matter how high the transaction cost": search frictions, asymmetric information, price impact and funding constraints.

**Comparison card: IM-17 c**

| Bias | What goes wrong in the data | Reported return | Reported risk | Remedy |
|---|---|---|---|---|
| **Survivorship (with reporting bias)** | Failed and non-reporting funds are missing | Too high (at least 1% to 2%; over 4%) | Not its main effect | Observe the entire population (impossible in practice) |
| **Sample selection** | Values recorded only when high (sales, IPOs, exits) | Alpha too high (VC: over 90% to $-7\%$) | Beta and volatility too low; SML flatter | Heckman selection model |
| **Infrequent trading** | Stale, appraisal-based prices; smoothed returns | Mean essentially unaffected | Volatility, correlation and beta too low | Unsmoothing (IM-17 d) |
| **Backfill (outside Ang's three)** | Incubation history added when a fund joins | Too high for new funds | Not its main effect | Drop pre-entry history |

**Exam traps**

- **Direction (polarity flip).** Every bias flatters the asset: returns up, risk down. Survivorship bias never makes reported returns look worse than true returns.
- **Sibling swap.** Dead funds dropped: survivorship. Values recorded only when high: sample selection. Stale or appraisal prices: infrequent trading. Incubation returns added on entry: backfill.
- **Reporting bias is wider than "stopped reporting".** Funds that never start reporting bias the database too.
- **Selection bias does two things.** Alpha up *and* beta and volatility down: the fitted SML is flatter, not steeper.
- **Remedy mismatch.** Unsmoothing fixes infrequent trading only; Heckman models address selection; only the full population removes survivorship bias.
- **"More data will fix it."** Infrequent trading is a population problem, not a small-sample problem.
- **Reported low volatility or low correlation read as low risk.** For appraisal-based assets they are artefacts of smoothing.
- **Invented biases.** "Turnover bias" and "availability bias" are not among Ang's three.

## Practice questions: IM-17 c

### Question IM-17 c.1

Bank Q606 • Schweser Module Quiz 99.1 Q3 (rebuilt)

Blue Sky Funds, a private equity fund, has suffered low returns for the last five years. As a result, the fund has decided to quit reporting returns. The fund did report returns each year for the previous ten years, when performance was strong. This problem of reporting leads to:

**A.** survivorship bias.

**B.** sample selection bias.

**C.** infrequent trading bias.

**D.** smoothing bias.

### Question IM-17 c.2

Bank Q600 • BT P2.T8.707.2 • BT Mock Q58 (edited)

Andrew Ang writes that "reported illiquid asset returns are not returns": investors overstate the expected returns and understate the risk of illiquid assets, and he attributes this to three key biases. Each of the following statements about biases in the reported returns of illiquid assets is accurate EXCEPT:

**A.** Survivorship bias can inflate reported returns by 4.0% or more.

**B.** Infrequent sampling (infrequent trading) artificially reduces risk and risk-related metrics such as volatility, correlation and beta.

**C.** Turnover bias decreases the typical time between transactions and tends to artificially increase the expected return by 5.0% or more.

**D.** Selection bias is a distortion of the sample that artificially increases (overestimates) alpha and artificially decreases (underestimates) beta.

### Question IM-17 c.3

Bank Q591 • AnalystPrep Q.2439 (rebuilt)

A consultant estimates the alpha and beta of the real estate sector using only the returns on buildings that were sold during the year. Owners in this market typically hold properties until values have recovered before selling. The consultant reports a beta of 0.65 and an alpha of 3.45%. Which of the following is the most likely implication of the way the data were collected?

**A.** Alpha is overestimated and beta is underestimated.

**B.** Alpha and beta are both overestimated.

**C.** Alpha is underestimated and beta is overestimated.

**D.** Alpha is correctly estimated but beta is underestimated.

**Question IM-17 c.4** Bank Q604 • Mock 1 Q60 (rebuilt)

An analyst at a pension fund is reviewing an allocation to illiquid alternative assets such as real estate and private equity. The analyst finds that the reported volatilities, betas and correlations with public equities of these assets are all very low, and is concerned that the fund will understate the true risk of the allocation. Which source of bias in the reported data best explains the analyst's concern?

**A.** Survivorship bias

**B.** Sample selection bias

**C.** Infrequent trading

**D.** Unsmoothing of reported returns

**Question IM-17 c.5** Bank Q596 • AnalystPrep Q.2582 (rebuilt, key corrected in bank)

Alexa Thompson is analysing a data set of active and passive funds. She notes two issues: (1) the data of some active funds that performed poorly and subsequently closed are not included in the database; (2) for the passive funds, the data set includes only funds that have outperformed their benchmark in the recent past. Which biases is Thompson most likely facing?

**A.** Survivorship bias in the active funds and selection bias in the passive funds

**B.** Selection bias in both the active and the passive funds

**C.** Survivorship bias in both the active and the passive funds

**D.** Infrequent trading in the active funds and survivorship bias in the passive funds

**Question IM-17 c.6** New • built from SchweserNotes 2026 Reading 99 • A24·S3

A risk analyst at a Japanese pension fund audits a vendor's alternative-asset database before it is used in the fund's asset allocation study. She records four practices:

| | |
|---|---|
| I | Funds that are liquidated during the sample period are deleted, together with their whole return history. |
| II | When a fund joins the database, the returns it earned during its incubation period are loaded into its history. |
| III | The property sub-index is computed only from buildings sold during each year; owners usually wait for prices to recover before selling. |
| IV | Quarterly property values come from annual appraisals carried forward between valuation dates. |

Which of the following correctly matches each practice to the bias it creates?

**A.** I survivorship; II backfill; III sample selection; IV infrequent trading

**B.** I survivorship; II sample selection; III backfill; IV infrequent trading

**C.** I sample selection; II backfill; III survivorship; IV infrequent trading

**D.** I survivorship; II backfill; III infrequent trading; IV sample selection

### Question IM-17 c.7

Bank Q602 • GARP PE2 Q11 • Mock 4 Q26

The chief investment officer (CIO) of a large university endowment fund is considering adding some illiquid assets to improve the performance of the university's investment portfolio. The CIO asks an investment manager to prepare a report discussing the characteristics of illiquid asset returns. The manager reviews the dynamics of illiquid assets and the determinants of their returns. Which of the following is a correct statement for the manager to include in the report?

**A.** Survivorship bias overstates reported returns of illiquid assets by reporting only the returns of those funds that have achieved a return above a required threshold.

**B.** Reporting bias overstates reported returns of illiquid assets because only the returns of those funds that continue to remain in business over a given time period are considered.

**C.** Survivorship bias and reporting bias can theoretically be eliminated by including the returns of the entire population of funds.

**D.** Infrequent trading, although considered a bias, can still generate sufficient data for accurate beta and correlation estimates.

### Question IM-17 c.8

Bank Q605 • Mock 3 Q30

A risk analyst at an investment bank is conducting performance analyses of hedge funds and real estate funds. The analyst notes the following two issues regarding the funds' annual performance data:

- Whenever a hedge fund stops reporting its performance, it is removed from the database of hedge funds.
- Assets owned by the real estate funds are valued only once a year due to infrequent trading.

Which of the following best describes the impacts of using the data with the issues above on the results of the performance analyses?

**A.** The average Sharpe ratio of hedge funds is understated and the average Sharpe ratio of real estate funds is overstated.

**B.** The average Sharpe ratio of hedge funds is overstated and the average Sharpe ratio of real estate funds is also overstated.

**C.** The average volatility of hedge funds is overstated and the average volatility of real estate funds is also overstated.

**D.** The average volatility of hedge funds is overstated and the average volatility of real estate funds is understated.

### Question IM-17 c.9

Bank Q603 • GARP PE2 Q51

An individual investor wants to invest USD 8 million in exchange-traded funds (ETFs) or private equity funds (PEFs). The investor obtains the previous year's returns for several ETFs and calculates summary statistics such as volatility and correlation based on these returns. The investor also reviews a database of reported returns and volatilities for several PEFs and then selects two potential investments in each asset class. Using these data, the investor generates the following information:

| Asset | Annual 1-year return | Volatility of returns |
|---|---|---|
| Broad equity market index ETF (ETF1) | 6.5% | 11.4% |
| Growth stock ETF (ETF2) | 8.3% | 13.6% |
| Private equity fund 1 (PEF1) | 7.4% | 12.3% |
| Private equity fund 2 (PEF2) | 10.2% | 11.1% |
| Correlation of returns, ETF1 and ETF2 | | 0.67 |
| Correlation of returns, ETF1 and PEF1 | | 0.25 |
| Correlation of returns, PEF1 and PEF2 | | 0.41 |

The investor evaluates this information while also considering the potential biases and uncertainties in the reported data. Which of the following conclusions is most appropriate for the investor to make?

**A.** The correlation of returns between PEF1 and ETF1 is more accurate than the correlation of returns between ETF1 and ETF2.

**B.** The entire USD 8 million should be allocated to PEF2 because it is clearly the superior investment from a return-to-risk perspective.

**C.** The number of assets in PEFs is typically higher than the number of assets in ETFs, which makes PEFs much less risky than ETFs over longer time horizons.

**D.** Summary statistics computed using reported returns of PEFs can be biased downward, which compromises the reliability of these risk measures.

### Question IM-17 c.10

Workbook A24.3 • S3 • Map #34

An investment consultant at a Canadian endowment is studying a commercial database of hedge fund and private equity returns. Funds report to the database voluntarily. Over the last ten years, the average annual return of funds still reporting today is 9.8%; including funds that stopped reporting or closed during the period, it is 7.1%. Which of the following statements should the consultant include in the report?

**A.** The 2.7-point gap reflects sample selection bias: returns are recorded only when underlying asset values are high, as when buyout companies are taken public in strong markets.

**B.** The 2.7-point gap reflects survivorship bias: poorly performing funds stop reporting and drop out, and only observing the entire population of funds removes the bias completely.

**C.** Sample selection bias in private equity data makes the fitted security market line steeper than the true line, so beta and volatility estimated from reported deals are overstated.

**D.** Reporting bias arises only when a fund that has been reporting stops after poor returns, so funds that never begin reporting cannot bias the database's average return.

# Solutions • IM-17 c

## Solution IM-17 c.1

**Answer: A**

Bank Q606 • Schweser Module Quiz 99.1 Q3

### Step 1 – What is missing from the database

Blue Sky's ten strong years are in the data; the five weak years are not, because the fund chose to stop reporting.
⇒ The sample loses poor outcomes because of *who* keeps reporting.

### Step 2 – Name the bias

Ang: "survivorship bias results from the tendency of poorly performing funds to stop reporting". Reporting bias, stopping or never starting, is part of it.
⇒ Survivorship bias: the database average is too high.

### Interpretation

If many funds behave like Blue Sky, the database average is built from the good years of the funds that kept reporting. Only observing the entire population of funds would remove the bias.

### Why the other options are wrong

**B is incorrect.** Sample selection bias is about *when* an asset's value is recorded: sales and exits happen when values are high. Here the fund's decision to stop reporting takes it out of the data, which is survivorship.
**C is incorrect.** Infrequent trading concerns stale, smoothed prices and biases risk estimates; it says nothing about which funds report.
**D is incorrect.** Smoothing is the symptom of infrequent trading; nothing in the stem is smoothed.

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Illiquid Asset Reported Returns Are Not Returns (Survivorship Bias); SchweserNotes 2026 Book 5, Reading 99, LO 99.c

## Solution IM-17 c.2

**Answer: C**

Bank Q600 • BT P2.T8.707.2 • BT Mock Q58

### Step 1 – List Ang's three biases

Survivorship bias, sample selection bias and infrequent trading (infrequent sampling).
⇒ "Turnover bias" is not among them.

### Step 2 – Check the other three statements

A: survivorship bias is at least 1% to 2% for mutual funds and more than 4% when defunct and live funds are compared, and larger for illiquid assets: accurate. B: smoothed returns understate volatility, correlation and beta: accurate. D: recording only high values raises alpha and flattens beta: accurate.

### Step 3 – Why C fails on its own logic

A shorter time between transactions would make prices *less* stale and reduce bias, not create a return premium.
⇒ C is the inaccurate statement.

### Interpretation

The bank's original option D read "Selection bias (aka, reporting bias)". Ang groups reporting bias with survivorship bias (funds that stop or never start reporting), so the label was removed here; the substance of D is accurate and the key is unchanged.

### Why the other options are wrong

**A is incorrect.** This statement is accurate: 4% is a floor for illiquid assets, not a ceiling (see the correction box).
**B is incorrect.** This statement is accurate: it is the infrequent trading bias in Ang's words.
**D is incorrect.** This statement is accurate: selection bias overstates alpha and understates beta (Figure 1).

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Illiquid Asset Reported Returns Are Not Returns; SchweserNotes 2026 Book 5, Reading 99, LO 99.c

### Solution IM-17 c.3 — Answer: A

Bank Q591 • AnalystPrep Q.2439

#### Step 1 – Identify the bias

Only buildings that were sold enter the data, and owners sell after values have recovered. Returns are observed only when values are high.
⇒ Sample selection bias.

#### Step 2 – Its effect on the regression

The low returns that would have pulled the fitted line down in weak markets are missing. The fitted intercept rises and the slope flattens (Figure 1).
⇒ The reported alpha of 3.45% is biased up; the reported beta of 0.65 is biased down.

#### Interpretation

The true alpha could well be zero. Selection bias flatters both halves of the risk-return pair: it overstates the return and understates systematic risk. The effect is smaller in real estate than in venture capital because real estate volatility is lower, and Heckman-type selection models are the correction.

#### Why the other options are wrong

**B is incorrect.** Beta is understated, not overstated: with the low observations missing, measured co-movement with the market falls.
**C is incorrect.** Reverses both effects.
**D is incorrect.** Alpha is biased too: recording only high prices inflates it.

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Selection Bias; SchweserNotes 2026 Book 5, Reading 99, LO 99.c (Sample Selection Bias)

### Solution IM-17 c.4 — Answer: C

Bank Q604 • Mock 1 Q60

#### Step 1 – Read the symptom

The concern is that volatility, beta and correlation are all too low: risk is understated. The stem says nothing about the level of returns.
⇒ Look for the bias that acts on risk measures across the board.

#### Step 2 – Match it

Real estate and private equity are valued by appraisal or at rare transactions, so reported returns are smoothed. Smoothing lowers volatility, beta and correlation with public equities together.
⇒ Infrequent trading.

#### Interpretation

The fix is unsmoothing (IM-17 d), which raises all three estimates and leaves the mean return essentially unchanged.

#### Why the other options are wrong

**A is incorrect.** Survivorship bias drops failed funds and overstates the average return; it does not explain low volatility, beta and correlation.
**B is incorrect.** Selection bias also flattens beta and lowers volatility, but its defining effect is an overstated alpha, and it does not explain why correlations with public equities are muted. Stale, appraisal-based values affecting every risk measure point to infrequent trading.
**D is incorrect.** Unsmoothing is the remedy for the bias, not a source of it.

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Infrequent Sampling; SchweserNotes 2026 Book 5, Reading 99, LO 99.c (Infrequent Trading)

### Solution IM-17 c.5 Answer: A

Bank Q596 • AnalystPrep Q.2582

#### Step 1 – Issue (1), the active funds

Poor performers that closed are missing from the database.
⇒ Survivorship bias.

#### Step 2 – Issue (2), the passive funds

Funds enter the sample only because they recently did well: the sample is chosen on high outcomes.
⇒ Sample selection bias.

#### Interpretation

Both biases overstate the average return, but the remedies differ: putting the dead funds back cures the first; modelling the selection rule (Heckman) addresses the second.

#### Why the other options are wrong

**B is incorrect.** Labels the missing closed funds as selection bias; the term for dropping failed funds is survivorship bias.
**C is incorrect.** The passive funds are chosen for recent outperformance, not for having survived or kept reporting.
**D is incorrect.** Infrequent trading concerns stale, smoothed prices, not missing funds; and the passive-fund issue is selection, not survivorship.

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Survivorship Bias and Selection Bias; SchweserNotes 2026 Book 5, Reading 99, LO 99.c

### Solution IM-17 c.6 Answer: A

New • built from SchweserNotes 2026 Reading 99 (backfill from GARP 2025 Ch.9, background)

#### Step 1 – Practice I

Liquidated funds disappear with their history, so only survivors remain.
⇒ Survivorship bias.

#### Step 2 – Practice II

Returns earned before the fund joined are added after the fact, and unsuccessful incubations never join.
⇒ Backfill (instant history) bias.

#### Step 3 – Practice III

Values are observed only at sales, and sales happen after prices recover.
⇒ Sample selection bias.

#### Step 4 – Practice IV

Annual appraisals carried forward make quarterly values stale and returns smooth.

⇒ Infrequent trading.

**Interpretation**

Practices I, II and III inflate reported returns; III and IV understate risk. Each needs its own remedy: restore the dead funds (I), discard pre-entry history (II), a Heckman selection model (III) and unsmoothing (IV). Backfill is not one of Ang's three biases, but it belongs to the sibling set the recall map says GARP tests.

**Why the other options are wrong**

**B is incorrect.** Swaps II and III. Backfill is about history loaded when a fund joins; selection is about values recorded only when high.

**C is incorrect.** Swaps I and III. Deleting dead funds is survivorship; recording values only at high prices is selection.

**D is incorrect.** Swaps III and IV. Carried-forward appraisals concern *how often* values are observed (infrequent trading), not *which* values are observed (selection).

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** SchweserNotes 2026 Book 5, Reading 99, LO 99.c; Ang, Asset Management (2014), Ch.13; Fung and Hsieh in GARP 2025 RMIM Ch.9, footnote 21 (backfill, background only); Workbook A24 revision notes

## Solution IM-17 c.7

**Answer: C**

Bank Q602 • GARP PE2 Q11

**Step 1 – Classify the two biases**

Survivorship and reporting bias are missing-data problems: funds that fail, stop reporting or never start are absent from the database.

**Step 2 – The remedy**

Ang: the only way to remove survivorship bias completely is to observe the entire population of funds.

⇒ In theory the full population eliminates both; in illiquid markets it cannot be observed in practice, hence "theoretically".

**Interpretation**

GARP builds A and B by swapping the definitions: the return-threshold story belongs to selection, and the "remain in business" story is survivorship.

**Why the other options are wrong**

**A is incorrect.** A return threshold above which results are reported describes selection bias, not survivorship.

**B is incorrect.** Keeping only funds that remain in business is survivorship bias; reporting bias concerns funds that stop or never start reporting.

**D is incorrect.** Infrequent trading smooths returns and biases beta and correlation downward. It is a population problem, so more data does not make the estimates accurate.

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Survivorship Bias; SchweserNotes 2026 Book 5, Reading 99, LO 99.c; GARP Practice Exam 2, Q11

## Solution IM-17 c.8

**Answer: B**

Bank Q605 • Mock 3 Q30

**Step 1 – Hedge fund data**

Funds that stop reporting are removed: survivorship bias. Average returns are too high.

⇒ The Sharpe ratio's numerator is inflated.

**Step 2 – Real estate data**

Annual valuation smooths returns: infrequent trading. Volatility is too low.

⇒ The Sharpe ratio's denominator is too small.

**Step 3 – Combine**

⇒ Both average Sharpe ratios are overstated.

**Interpretation**

The two biases work on different parts of the ratio but in the same direction (Figure 2). That is the general rule: every bias in reported illiquid returns flatters risk-adjusted performance.

**Why the other options are wrong**

**A is incorrect.** Removing poor performers raises average hedge fund returns, so the Sharpe ratio is overstated, not understated.

**C is incorrect.** Annual valuation understates real estate volatility; and survivorship bias does not overstate hedge fund volatility.

**D is incorrect.** Half right: real estate volatility is understated, but survivorship bias does not overstate hedge fund volatility; its main effect is on the return level.

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Survivorship Bias and Infrequent Sampling; SchweserNotes 2026 Book 5, Reading 99, LO 99.c

**Solution IM-17 c.9** **Answer: D**

Bank Q603 • GARP PE2 Q51

**Step 1 – Where each statistic comes from**

ETF statistics are computed from market prices. PEF statistics come from a database of reported returns.

**Step 2 – The biases in the PEF data**

Reported PEF returns are appraisal-based and smoothed, so volatility and correlations are biased down; the database adds survivorship and selection, so returns are biased up.

⇒ The PEF risk measures are not reliable.

**Step 3 – What the table tempts you to do**

Return per unit of volatility: ETF1 0.57, ETF2 0.61, PEF1 0.60, PEF2 0.92. PEF2 looks best only on biased inputs.

**Interpretation**

Unsmoothed volatility can be several times the reported figure (NCREIF: 2.25% to 6.26% a quarter). The low PEF correlations (0.25, 0.41) are equally suspect. The investor needs bias-adjusted inputs before comparing the two asset classes.

**Why the other options are wrong**

**A is incorrect.** Reversed: the correlation between ETF1 and ETF2 uses market prices and is the more accurate one; the correlation between PEF1 and ETF1 is depressed by smoothing.

**B is incorrect.** PEF2's 0.92 ratio rests on an understated volatility and an inflated return, and an all-in allocation to one fund ignores diversification.

**C is incorrect.** Private equity funds hold far fewer assets than broad ETFs, and the number of holdings does not cure smoothing: true PEF risk is higher than reported.

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Illiquid Asset Reported Returns Are Not Returns; SchweserNotes 2026 Book 5, Reading 99, LO 99.c; GARP Practice Exam 2, Q51

**Solution IM-17 c.10** **Answer: B**

Workbook A24.3 • S3

### Step 1 – Name the bias that the data isolate

The two averages differ only in whether dead and non-reporting funds are kept. Ang: "survivorship bias results from the tendency of poorly performing funds to stop reporting", and "the only way to completely remove the effect of survivorship bias is to observe the entire population of funds".

$\Rightarrow$ The 2.7-point gap $(9.8\% - 7.1\%)$ is survivorship bias, of the size Ang reports: at least 1% to 2% for mutual funds and above 4% when defunct and live funds are compared, both lower bounds for illiquid asset managers.

### Interpretation

Using the live-fund average would overstate expected returns by about 2.7 points a year. Bodie, Kane and Marcus give the same warning for mutual funds: limiting a sample to funds alive over the whole period creates survivorship bias, which is why most mutual fund databases now include failed funds. Stowell lists survivorship bias among the portfolio-level risks of hedge fund investing (IM-9 a).

### Why the other options are wrong

**A is incorrect.** Sibling swap: recording returns only when values are high is sample selection bias, a different mechanism; the gap here comes from dropping dead funds.

**C is incorrect.** Polarity flip: with selection bias the fitted SML is *flatter* than the true SML, so beta and volatility are underestimated while alpha and expected return are overestimated.

**D is incorrect.** Scope error: Ang states that reporting bias "also occurs when you don't start reporting your returns in the first place because your fund never achieves a sufficiently attractive track record".

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. IM-9 a: Describe different risks faced by investors in hedge funds. • **Reference:** Ang, Asset Management (2014), Ch.13, Survivorship Bias and Selection Bias; Bodie, Kane and Marcus, Ch.24, Selection Bias and Portfolio Evaluation; Stowell and Stowell, Ch.14 (SchweserNotes 2026 Book 5, Reading 91)

RISK MANAGEMENT AND INVESTMENT MANAGEMENT • PART

# P2

## Second priority

Tier B topics, or LOs Vardeez flags inside a Very Important chapter. 24 LOs from 9 readings.

IM-2 • Factors • Vardeez: UNTIERED

# IM-2 d

P2 CONCEPT

Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example.

Workbook: B4 Factor Theory, Fama-French and Style • Tier B • Map #18 (5/16)

### WHAT IS TESTED HERE

**Why P2.** Tier B workbook topic: B4 Factor Theory, Fama-French and Style was reported on 5 of 16 recall sheets (Map #18).
**Vardeez:** no pointer on this LO (the reading is untiered; his flags are on IM-2 c and IM-2 e).

**Reported exam shapes**

- **S1** SMB (small minus big) vs SML (security market line), with HML, in a Fama-French portfolio composition. One handwritten recall reads "Portfolio composition (SML, HML)": almost certainly SMB.
- **S2** Investing style (style, investment or dynamic factors) as a theory item.

**What the questions look like:** statements. How SMB and HML are built and what they sort on, what a positive or negative loading means, what the market's own loadings are, the static versus dynamic and style versus macro vocabulary, and which effect survived (value) and which did not (size). The calculations filed here in the bank (three- and four-factor expected returns, Bank Q74 from AP Mock 3 Q29, Q75 and Q76; an FX carry return, Q66) are left out under the CONCEPT tag.

## 1. Two kinds of factor (background for IM-2)

- **Macro (fundamental) factors:** economic growth, inflation, volatility, productivity, demographic and political risk. Bad outcomes define bad times for the average investor, but they are not usually traded directly, at least not in scale (volatility is the exception).
- **Investment (style) factors** are tradeable. **Static** factors, like the market, are simply held long to collect the premium. **Dynamic** factors "can only be exploited through constantly trading different types of securities": size, value, momentum.
- **Vocabulary (shape S2).** Ang uses "style factors", "investment factors" and "dynamic factors" interchangeably; practitioners say "smart beta" or "alternative beta". Ang prefers "dynamic factor" because beta strictly measures the exposure to a factor, not the factor: "we invest in factor portfolios, not betas".
- **Why they matter.** Dynamic factors can be implemented easily in a portfolio, which macro factors cannot. Many hedge funds and private equity investments are essentially bundles of dynamic factors, and macro factors are often embedded in the performance of investment factors.

## 2. The Fama-French (1993) model

$$E(r_i) = r_f + \beta_{i,\text{MKT}}\, E(r_m - r_f) + \beta_{i,\text{SMB}}\, E(\text{SMB}) + \beta_{i,\text{HML}}\, E(\text{HML})$$ (GARP Eq. 2.2)

- **SMB**: returns of **s**mall stocks **m**inus **b**ig stocks, sorted on **market capitalization**. It was designed to capture small firms' outperformance.

- **HML**: returns of **h**igh minus **l**ow **book-to-market** stocks. Book-to-market is book value divided by market capitalization, so high book-to-market stocks are value stocks (cheap relative to book) and low ones are growth stocks.
- **Factor-mimicking, long-short portfolios.** Each factor averages across many stocks (the CAPM idea of diversification) so that it captures only the size or value effect. Both take positions away from the market portfolio, and both have to be re-sorted and re-traded as firms change size and valuation: that is what makes them dynamic.
- **Credit.** Fama and French did not discover size (Banz, 1981; Reinganum, 1981) or value (Basu, 1977); they gave a parsimonious model that captures both.

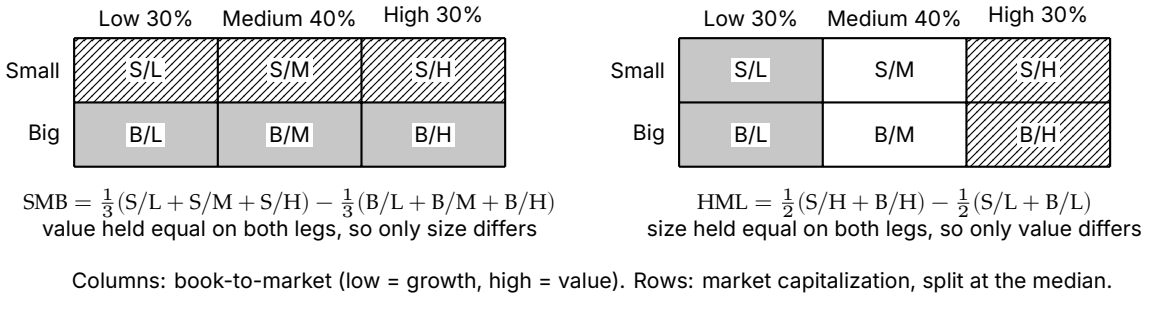



Figure 1. How the two Fama-French factors are built from six size and book-to-market portfolios. Hatched cells are the long leg, grey cells the short leg, white cells are not used. This 2 by 3 sort is the construction in Fama and French (1993); the GARP chapter describes the factors but not the sort.

## 3. How the loadings enter the expected return

- **Value stock (Ang's "Huffet").** Positive HML beta: relative to the CAPM, its expected return is adjusted *up* by $\beta_{i,\text{HML}} \times E(\text{HML})$, because HML is built to have a positive premium. **Growth stock ("Inron")**: negative HML beta, adjusted *down*.
- **Size** works the same way: a positive SMB loading means the asset moves with small stocks, a negative one with large stocks.
- **Centered at zero.** The SMB and HML betas are centered around zero: the market is size neutral and value neutral, and the average stock has only market exposure (every stock cannot be small). In the CAPM the average beta is one, the beta of the market. This is the factor version of "the average investor holds the market".
- **Reading a regression.** $r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + s\,\text{SMB}_t + h\,\text{HML}_t + \varepsilon_{it}$. Berkshire Hathaway (GARP Ch.3, IM-3) loads $s = -0.50$ (large) and $h = 0.38$ (value): Figure 2.
- **Constant betas are an assumption.** Exposures vary over time and tend to rise in bad times, which made risky assets' losses in the financial crisis larger than constant betas imply.
- **Extensions.** Momentum (WML) is added as a fourth factor (Carhart, 1997; IM-2 e). The same structure works in other asset classes: FX carry is a value factor, $\text{HML}_{FX}$, long high interest rate currencies minus low ones (GARP Eq. 2.3).

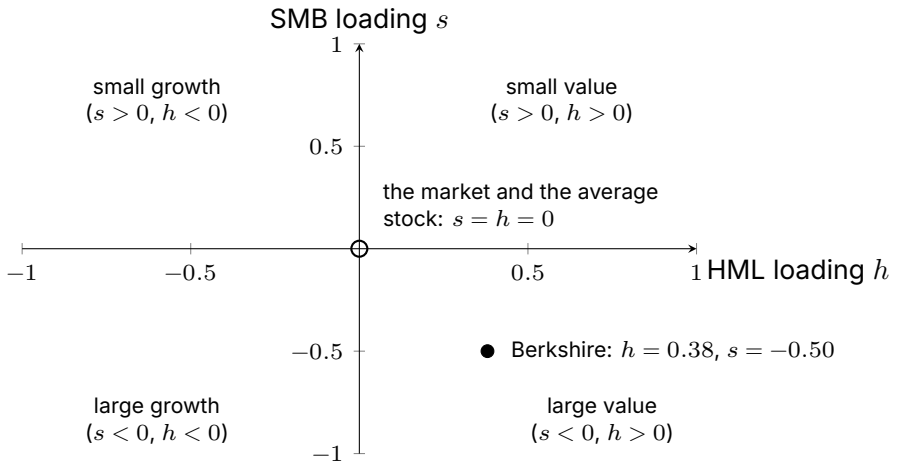



Figure 2. Reading Fama-French loadings. The open circle is the market (no size or value tilt); the filled circle is Berkshire Hathaway's regression from GARP Ch.3. The sign of each loading names the quadrant; the market factor's own beta is a separate, third axis.

## 4. What the two factors have earned

- **Size has disappeared.** Market-adjusted SMB compounded up to the early 1980s, just after Banz and Reinganum published, and has shown no significant premium since the mid-1980s, internationally too. Two readings: data mining (Fischer Black, 1993; the "file drawer problem") or rational investors bid small caps up until the effect was gone (Grossman-Stiglitz near-efficiency). Either way, Ang says size "does not deserve to be a systematic factor and should be removed from the Fama-French model". Small stocks still have higher raw returns and are less liquid, and value and momentum are stronger among them, but the unconstrained case for a small-cap tilt is not compelling: it suits only investors who want higher returns and cannot short the market or cannot lever.
- **Value is robust**: gains over fifty years, with losing spells (IM-2 e). The Fama-French model itself is silent on *why* value earns a premium; the CAPM, by contrast, pins down the market premium as $\bar{\gamma}\sigma_m^2$.

**Correction to the source notes.** SchweserNotes Reading 84 (LO 84.d and its Key Concepts) says the Fama-French model "explains asset returns based on three dynamic factors". Only SMB and HML are dynamic. GARP classes the market as a *static* factor, "which we simply go long to collect a risk premium"; dynamic factors are the ones that need constant trading of long and short positions.

| | Long minus short (sort variable) | Static or dynamic | Loading of the market | Record in GARP 2025 Ch.2 |
|---|---|---|---|---|
| **MKT** | Market portfolio minus T-bills | Static | 1 (average beta) | The CAPM premium |
| **SMB** | Small minus big (market capitalization) | Dynamic | 0 | No significant premium since the mid-1980s |
| **HML** | High minus low book-to-market | Dynamic | 0 | Robust for fifty years, with losing spells |
| **WML** | Past winners minus past losers (about six months of returns) | Dynamic | 0 | Largest returns; crash-prone (Carhart add-on, IM-2 e) |
| **SML** | Not a factor: the CAPM line $E(r_i) - r_f = \beta_i[E(r_m) - r_f]$ | n/a | n/a | A pricing relation, not a portfolio |

**Exam traps**

- **SMB versus SML (shape S1, sibling swap).** SMB is the size factor; the SML is the CAPM's beta pricing line. A "loading on the SML" does not exist, and the market factor stays in the regression.
- **Sort-variable swap.** SMB sorts on market capitalization, HML on book-to-market. "SMB captures high minus low book-to-market" is the classic corruption.
- **The market's loadings.** The market, and the average stock, has SMB and HML loadings of zero and a market beta of one. Value outperforming growth does not give the market a positive HML loading.
- **Sign of a loading.** Positive SMB means small-cap behavior (not large); negative HML means growth (not value). Read the factor's long leg.
- **Static or dynamic.** The market is static; SMB, HML and WML are dynamic. Dynamic factors are tradeable style factors, not macro variables whose levels change.
- **Robustness profiles swapped.** Size vanished after the mid-1980s; value has been robust; momentum earns the most but crashes.
- **WML in the three-factor model.** It is the Carhart (1997) fourth factor, not one of Fama and French's three.

## Practice questions: IM-2 d

### Question IM-2 d.1

Bank Q70 • AnalystPrep Q.3019

In the Fama-French three-factor model, in addition to the market factor and the HML factor, what is the third factor and what does it capture?

**A.** The momentum factor, WML, which captures the outperformance of past winners relative to past losers.

**B.** The SMB factor, which captures the outperformance of high book-to-market stocks relative to low book-to-market stocks.

**C.** The SMB factor, which captures the outperformance of small firms relative to larger firms.

**D.** The momentum factor, WML, which captures the outperformance of high-growth stocks relative to low-growth stocks.

### Question IM-2 d.2

Bank Q71 • AnalystPrep Q.4547 (rebuilt)

Which of the following is a macroeconomic (fundamental-based) risk factor rather than a tradeable dynamic (style) factor?

**A.** Size (SMB)

**B.** Economic growth

**C.** Value (HML)

**D.** Momentum (WML)

### Question IM-2 d.3

New • built from GARP Ch.2 (2.2, 2.4) • S2

A consultant presents a slide titled "Investing style" to the equity committee of a Gulf sovereign wealth fund, describing how style factors enter a multifactor model of asset returns. Which statement on the slide is correct?

**A.** Dynamic factors are macro factors, such as inflation and economic growth, that are called dynamic because their levels change over time.

**B.** Smart beta products invest in betas rather than in factor portfolios, which is why the term is more precise than "dynamic factor".

**C.** The market factor is a dynamic factor, because index funds must keep trading as market capitalizations change.

**D.** Style, investment and dynamic factors are interchangeable names for tradeable long-short factors such as size and value; unlike the static market factor, they can be harvested only by constantly trading.

### Question IM-2 d.4

Bank Q65 • BT answers P2.T8.703.2 • BT notes P2.T9.703.2 (rebuilt)

The Fama-French three-factor model is $E(r_i) = r_f + \beta_{i,\text{MKT}}E(r_m - r_f) + \beta_{i,\text{SMB}}E(\text{SMB}) + \beta_{i,\text{HML}}E(\text{HML})$. Which of the following statements about the model is TRUE?

**A.** A value investor has a positive HML beta, so the expected return is adjusted upward relative to the CAPM expected return.

**B.** Growth stocks have positive HML betas, so their expected returns are adjusted upward relative to the CAPM.

**C.** The average stock has an SMB beta of 1 and an HML beta of 1, as the average stock has a market beta of 1.

**D.** Value stocks are expected to outperform growth stocks especially during bad times for the economy, which is why they earn a premium.

### Question IM-2 d.5

Bank Q79 • AnalystPrep Q.2399 (re-filed from IM-2 e)

The Fama-French model with a momentum factor added is:

$$E(r_i) = r_f + \beta_{i,\text{MKT}}E(r_m - r_f) + \beta_{i,\text{SMB}}E(\text{SMB}) + \beta_{i,\text{HML}}E(\text{HML}) + \beta_{i,\text{WML}}E(\text{WML})$$

A fund manager buys stocks with increasing prices and sells stocks with decreasing prices. For stocks with increasing prices, $\beta_{i,\text{WML}}$ will be:

**A.** Positive

**B.** Negative

**C.** Zero

**D.** Equal to 1

### Question IM-2 d.6

Bank Q72 • GARP PE1 Q9 • Mock 3 Q71

A risk analyst regresses thirty years of weekly portfolio returns on the three Fama-French factors and obtains:

| Coefficient | Value |
|---|---|
| Alpha | 0.10 |
| Market coefficient | 0.52 |
| SMB coefficient | 0.18 |
| HML coefficient | −0.70 |

Assuming all coefficients are statistically significant, which statement is correct?

**A.** There is a positive correlation between portfolio return and the size factor, which indicates that the portfolio moves together with large-cap stocks.

**B.** There is a positive correlation between portfolio return and the value factor, which indicates that the portfolio moves together with growth stocks.

**C.** There is a negative correlation between portfolio return and the size factor, which indicates that the portfolio moves together with large-cap stocks.

**D.** There is a negative correlation between portfolio return and the value factor, which indicates that the portfolio moves together with growth stocks.

**Question IM-2 d.7** Workbook B4.1 • S1 • Map #18

A consultant to a Singapore pension fund is reviewing a junior analyst's research note on the Fama-French (1993) model before it goes to the investment committee. The note's section heading reads "Portfolio composition (SML, HML)", and the committee wants to know whether to add a size tilt to the fund's SGD-denominated equity mandate. Which of the following statements should the consultant accept as correct?

**A.** The SML in the heading is the security market line, so a fund's loading on it measures beta-adjusted exposure to the market risk premium and can replace the market factor in the regression.

**B.** Because value stocks beat growth stocks on average, the market portfolio itself carries a positive HML loading, so the average stock in the economy is tilted toward value.

**C.** The size factor is SMB, a long-short factor-mimicking portfolio long small caps and short large caps, and since the mid-1980s it has shown no significant premium after adjusting for beta.

**D.** Like the value premium, the size premium has stayed robust out of sample since Banz (1981), so an unconstrained investor should tilt toward small caps for higher risk-adjusted returns.

# Solutions • IM-2 d

## Solution IM-2 d.1

**Answer: C**

Bank Q70 • AnalystPrep Q.3019

### Step 1 – Name the three factors

Fama and French (1993): the CAPM market factor, SMB and HML.

### Step 2 – What SMB sorts on

SMB is small minus big by market capitalization: it captures the outperformance of small firms over large firms.

$\Rightarrow$ C.

### Interpretation

Each Fama-French factor is defined by its sort variable: market capitalization for SMB, book-to-market for HML. Swapping the two is the standard corruption (option B).

### Why the other options are wrong

**A is incorrect.** WML is a real factor, but it is Carhart's fourth factor, not one of the three.

**B is incorrect.** Right name, wrong definition: high minus low book-to-market is HML, already in the model.

**D is incorrect.** Wrong factor and wrong definition: WML sorts on past returns (winners minus losers), not on growth.

**LO:** IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.4 Dynamic Factors, Eq. 2.2; SchweserNotes 2026 Book 5, Reading 84, LO 84.d

## Solution IM-2 d.2

**Answer: B**

Bank Q71 • AnalystPrep Q.4547

### Step 1 – The two families

Macro (fundamental) factors are economy-wide: growth, inflation, volatility, productivity, demography, political risk. Investment (style) factors are tradeable portfolios.

### Step 2 – Classify the options

SMB, HML and WML are long-short, tradeable dynamic factors. Economic growth is a macro variable.

$\Rightarrow$ B.

### Interpretation

Macro factors are not usually traded directly in scale (volatility is the exception). That is the practical advantage of dynamic factors: they can be put into a portfolio. The two families are linked, since macro risks are often embedded in the returns of style factors.

### Why the other options are wrong

**A is incorrect.** SMB is long small caps, short large caps: a dynamic factor.

**C is incorrect.** HML is long value, short growth: a dynamic factor.

**D is incorrect.** WML is long past winners, short past losers: a dynamic factor.

**LO:** IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.2, 2.3 Macro Factors and 2.4 Dynamic Factors

### Solution IM-2 d.3 — Answer: D

New • built from GARP Ch.2 (2.2, 2.4)

**Step 1 – The vocabulary**

Ang uses "style factors", "investment factors" and "dynamic factors" interchangeably; practitioners add "smart beta" and "alternative beta".

**Step 2 – Static against dynamic**

Static factors, like the market, are simply held long to collect a premium. Dynamic factors can only be exploited by constantly trading different securities, as long-short portfolios that are re-sorted over time.

$\Rightarrow$ D.

**Interpretation**

This is shape S2, the "investing style" theory item. Dynamic factors are the investment-style factors that a multifactor model adds to the market; hedge funds and private equity are often bundles of them.

**Why the other options are wrong**

**A is incorrect.** Sibling swap: inflation and growth are macro factors; "dynamic" refers to how a tradeable factor must be traded, not to a variable whose level changes.

**B is incorrect.** Reverses Ang's point: beta measures the exposure to a factor, so investors invest in factor portfolios, not betas. That is why he prefers "dynamic factor".

**C is incorrect.** The market is GARP's example of a *static* factor; a capitalization-weighted index fund needs little trading to stay at market weights.

**LO:** IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.2 Value Investing (static and dynamic factors) and 2.4 Dynamic Factors, footnote 18

### Solution IM-2 d.4 — Answer: A

Bank Q65 • BT answers P2.T8.703.2

**Step 1 – The sign of HML**

HML is long high book-to-market (value) and short low book-to-market (growth); it is built to have a positive premium.

**Step 2 – A value tilt**

A value investor has $\beta_{i,\text{HML}} > 0$, so relative to the CAPM the expected return is raised by $\beta_{i,\text{HML}} E(\text{HML})$.

$\Rightarrow$ A.

**Step 3 – Check C**

SMB and HML betas are centered on zero; only the market beta averages one.

**Interpretation**

This is how a dynamic factor is *used* in the model: the loading on the factor scales the factor premium, which is added to the CAPM expected return. Growth stocks are pushed down by the same mechanism.

**Why the other options are wrong**

**B is incorrect.** Sign reversed: growth stocks have negative HML betas, which lower their expected returns.

**C is incorrect.** Scope error: carries the CAPM's average beta of one over to SMB and HML. The market is size neutral and value neutral.

**D is incorrect.** Reverses the rational story: value stocks do badly in bad times, and the premium

compensates for that.

**LO:** IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.4 Fama-French (1993) Model (Huffet and Inron)

### Solution IM-2 d.5 — Answer: A

Bank Q79 • AnalystPrep Q.2399

#### Step 1 – What WML is long

WML is the return on past winners minus past losers.

#### Step 2 – Rising-price stocks

They are past winners, the long leg, so their returns move with the factor.
$\Rightarrow$ Positive co-movement means a positive loading: A.

#### Interpretation

Filed in the bank under IM-2 e; it sits here because it shows a fourth dynamic factor at work in the multifactor model. As with SMB and HML, the WML beta is centered around zero: winners load positively and have their risk premiums adjusted up, losers load negatively, and the market, neither winner nor loser, has no momentum loading.

#### Why the other options are wrong

**B is incorrect.** A negative loading describes past losers, the short leg.
**C is incorrect.** Zero is the market's loading; a winner is on the long leg.
**D is incorrect.** A beta is a regression sensitivity; nothing fixes the WML loading at one.

**LO:** IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.5 Momentum, Eq. 2.4

### Solution IM-2 d.6 — Answer: D

Bank Q72 • GARP PE1 Q9

#### Step 1 – Size loading

SMB $= +0.18$: the portfolio moves with small minus big, so with small caps.

#### Step 2 – Value loading

HML $= -0.70$: the portfolio moves against high minus low book-to-market, so with growth stocks.
$\Rightarrow$ Only D gets both the sign and its meaning right.

#### Step 3 – What is not needed

The alpha $(0.10)$ and the market coefficient $(0.52)$ play no part in the size and value reading.

#### Interpretation

The portfolio is a small-cap growth tilt with a below-one market beta. Read each loading against the factor's long leg: SMB is long small, HML is long value.

#### Why the other options are wrong

**A is incorrect.** The sign is right but the meaning is reversed: a positive SMB loading means small-cap, not large-cap, co-movement.
**B is incorrect.** The HML loading is negative, not positive; the growth conclusion only follows from a negative loading.
**C is incorrect.** The SMB loading is positive, not negative.

**LO:** IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-

French model as an example. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.4 Fama-French (1993) Model; GARP Practice Exam 1, Q9

### Solution IM-2 d.7

**Answer: C**

Workbook B4.1 • S1

#### Step 1 – Identify the object in the heading

Fama-French adds two factors to the CAPM market factor: SMB (small minus big, by market capitalization) and HML (high minus low book-to-market). The SML is the CAPM pricing relationship $E(r_i) - r_f = \beta_i(E(r_m) - r_f)$, not a factor portfolio.

$\Rightarrow$ "SML" in the note is a transcription error for SMB; the committee should be reading about the size factor.

#### Step 2 – How SMB is built

SMB is USD 1 in small caps minus USD 1 in large caps: a long-short, factor-mimicking portfolio that diversifies across many stocks to isolate the size effect.

#### Step 3 – What the size factor has delivered

After Banz (1981) and Reinganum (1981), the compound returns of market-adjusted SMB peaked in the early 1980s; since the mid-1980s there has been no significant size premium, in the US or internationally.

$\Rightarrow$ The weak size effect means an asset owner should not tilt toward small stocks solely for higher risk-adjusted returns.

#### Interpretation

The consultant should correct the heading to SMB and advise the committee that a size tilt is hard to justify on risk-adjusted grounds alone; a case exists only for investors who want higher returns but cannot short the market or cannot lever.

#### Why the other options are wrong

**A is incorrect.** Sibling swap: substitutes the CAPM's security market line for the Fama-French size factor. The SML describes how beta is priced; it is not a factor with its own loading, and the market factor stays in the regression.

**B is incorrect.** Scope error: extends a property of value stocks to the market. In the Fama-French model the SMB and HML betas are centered around zero; the market is size neutral and value neutral.

**D is incorrect.** Sibling swap: gives size the robustness profile of value. Unlike size, the value premium is robust; the size effect has not been significant since the mid-1980s (data mining, or arbitraged away).

**LO:** IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example. IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.4 Dynamic Factors, Fama-French (1993) Model, Size Factor; Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lesson 5

IM-2 • Factors • Vardeez: UNTIERED

# IM-2 e

P2 CONCEPT

Compare value and momentum investment strategies, including their return and risk profiles.

Workbook: B4 Factor Theory, Fama-French and Style • Tier B • Map #18 (5/16) • VARDEEZ FOCUS Value vs momentum

**WHAT IS TESTED HERE**

**Why P2.** Tier B workbook topic (B4, Map #18, 5 of 16 sheets), and Vardeez flags it: "LO 84.E is imp which is value vs momentum strategies".
**Vardeez:** value vs momentum strategies.

**Reported exam shapes**

- **S3** Value vs momentum: return and risk profiles (Workbook B4.2, which also carries the S2 style theme and, in two options, the low-risk anomaly of IM-3 h and i).

**What the questions look like:** statements that set the two strategies side by side: the sort signal, the feedback direction, the size of the returns, the crash risk and skew, the correlation between them, and the economic stories behind each. Value's process and premium (IM-2 a) are P4, so the notes recap only what the comparison needs.

## 1. Value, in brief

- **The strategy.** Book-to-market = book value of equity / market capitalization. Value stocks have high ratios (low prices relative to book), growth stocks low ratios. The value-growth strategy (HML) is long value, short growth.
- **Return profile.** Robust: gains over fifty years, but with losing spells in the early-1990s recession, the late-1990s internet boom and 2007 to 2008 (and 2011). The book-to-market effect is a three- to six-month effect, unlike the five- to ten-year "deep value" of investors such as Buffett.
- **Why it pays (rational).** Value stocks move together and the common risk cannot be diversified away. Value firms are burdened with unproductive capital they cannot shed in bad times (high, asymmetric adjustment costs), and value betas rise in bad times. Value's bad times do not always match the economy's: it lost in the booming late 1990s.
- **Why it pays (behavioral).** Investors overextrapolate past growth (Lakonishok, Shleifer and Vishny, 1994), so growth stocks are too expensive and value stocks too cheap; or loss aversion with mental accounting makes beaten-down stocks look riskier (Barberis and Huang, 2001).
- **Everywhere.** Riding the yield curve in fixed income, the roll return in commodities, carry in currencies.

## 2. Momentum

- **The strategy.** Buy stocks that have gone up over the past six months or so (winners) and short those with the lowest returns (losers): **WML**, winners minus losers, also **UMD**, up minus down. Jegadeesh and Titman (1993), the same year as Fama and French; Carhart (1997) added it as a fourth factor (GARP Eq. 2.4). It is found in every asset class ("trend" investing; in commodities it is what commodity trading advisers do).

- **Return profile.** Momentum "blows size and value out of the water": from 1965 to 2011 its cumulated profits were an order of magnitude larger.
- **Risk profile.** Prone to periodic crashes; negatively skewed (skewness $-1.43$). Of the eleven largest momentum crashes, seven came in the Great Depression, one in 2001 and three in 2008: government bailouts put a floor under the loser financials (Citi, Bank of America, Goldman, Morgan Stanley) that momentum was short, and they skyrocketed. Crashes cluster when policymakers interrupt trends, in times of high volatility.
- **Why it pays.** The most widely cited theories are behavioral: delayed overreaction (conservatism; overconfidence with biased self-attribution) or underreaction (Hong and Stein's "news watchers"). Either way prices revert to fundamentals in the long run. The rational story (links to the business cycle, the state of the market, liquidity) is "still far from being fully fleshed out".

## 3. Side by side

**The same.** Both are **cross-sectional**: they compare one group of stocks against another at a point in time, not one stock over time; winners and losers, like value and growth, are relative. Both are long-short dynamic factors with loadings centered on zero, and both appear across asset classes.

**Different.** Value is a **negative feedback** strategy: prices that fall far enough make a stock cheap, value investors buy it, and their buying is **stabilizing**. Momentum is a **positive feedback** strategy: stocks that have risen attract more buying and rise further, which is **destabilizing** and subject to periodic crashes (Figure 3). Yet momentum is **not the opposite of value**: the correlation of HML with WML is only $-16\%$, so the two tilts are close to complementary. Many self-described growth investors, especially mutual funds, are really momentum investors, as pure growth underperforms value in the long run.

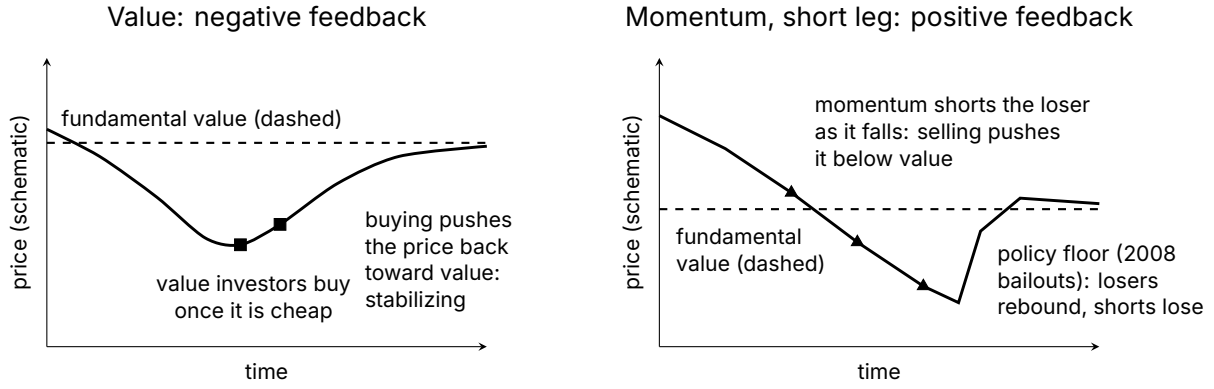



Figure 3. Schematic, shapes only. Solid lines are prices, dashed lines fundamental value. Squares mark value buying after a fall; triangles mark momentum selling into a fall. Value trades against the move and is stabilizing; momentum trades with the move, overshoots, and is exposed to sharp reversals when a trend is interrupted.

**Correction to the source notes.** SchweserNotes Reading 84 (LO 84.e) says value stocks "co-vary with growth stocks"; GARP says they "covary negatively with growth stocks". The same section adds that "an investor following a momentum strategy should still rebalance"; GARP does not say this. It separates the two: momentum is a cross-sectional strategy *within* an asset class, while rebalancing is done at the asset-class or strategy level. Bank Q82's key explanation leans on the Schweser line; its answer stands for GARP's reason (Solution IM-2 e.4).

| | Value (HML) | Momentum (WML or UMD) |
|---|---|---|
| **Sort signal** | Price relative to book value (book-to-market) | Past returns, about six months |
| **Long / short** | High book-to-market / low book-to-market | Past winners / past losers |
| **Feedback** | Negative: buys after falls, stabilizing | Positive: buys after rises, destabilizing |
| **Returns, 1965 to 2011** | Robust gains for fifty years | An order of magnitude larger than size or value |
| **Risk profile** | Losing spells: early 1990s, late-1990s boom, 2007 to 2008 | Periodic crashes; skewness $-1.43$; 7 of the 11 worst in the 1930s, 3 in 2008 |
| **Bad times** | May not match the economy's (the late-1990s boom) | Policy interruptions of trends, high volatility |
| **Explanations** | Rational and behavioral | Mostly behavioral (over- or underreaction) |
| **Shared** | Cross-sectional long-short; found across asset classes | Same; correlation with HML only $-16\%$, so not the opposite of value |

**Exam traps**

- **Feedback polarity flipped.** Value is negative feedback and stabilizing; momentum is positive feedback and destabilizing.
- **Momentum as the opposite of value.** Opposite only in feedback; the HML and WML correlation is $-16\%$, not strongly negative, so a model can hold both.
- **Cross-sectional for one only.** Both are cross-sectional; a distractor that makes momentum a single-stock time-series strategy claims a difference that is not there.
- **Robustness profiles swapped.** Size vanished; value is robust; momentum earns the most but crashes and is negatively skewed.
- **Crash cause.** Momentum's 2008 crashes came when bailouts lifted the losers it was short, not from its winners collapsing.
- **Explanation swap.** Overextrapolation of past growth is value's behavioral story; momentum's is over- or underreaction to news.
- **Equities only.** Both value and momentum appear in bonds, currencies and commodities as well.

## Practice questions: IM-2 e

### Question IM-2 e.1

Bank Q78 • AP Mock 3 Q13

A portfolio manager focused on buying stocks that have gone up in recent months and shorting stocks with the lowest returns during the same period is most likely pursuing which of these investment strategies?

**A.** Value investing.

**B.** Growth investing.

**C.** Momentum investing.

**D.** Income investing.

### Question IM-2 e.2

Bank Q83 • Schweser Module Quiz 84.3 Q1

Which of the following investment strategies stabilizes asset prices?

**A.** A value investment strategy.

**B.** A momentum investment strategy.

**C.** A size investment strategy.

**D.** Value, momentum, and size strategies all stabilize asset prices.

### Question IM-2 e.3

Bank Q81 • BT Mock Q70 • BT answers P2.T8.703.3 • BT notes P2.T9.703.3

Adding a momentum factor, WML (winners minus losers) or UMD (up minus down), to the Fama-French model produces a four-factor model. Over the period January 1965 to December 2011, which of the following is TRUE about the momentum factor?

**A.** Momentum is a negative feedback strategy which is inherently stabilizing.

**B.** The momentum factor is observed in equities but is NOT observed in bonds, commodities, and real estate.

**C.** Momentum investing by definition is an anti-value strategy; correlations between HML and WML are strongly negative.

**D.** The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value.

**Question IM-2 e.4** Bank Q82 • BT answers P2.T9.20.4.2 • BT notes P2.T9.20.4.2

Ang analyzed the historical performance of several macroeconomic (fundamental) factors and style (investment) factors. Style factors include value, momentum, and size. According to Ang's analysis, which of the following is TRUE about style factors?

**A.** Style factors do not persist because eventually, investors discover the factor.

**B.** Investing in momentum is the most profitable strategy among all style factors but momentum strategies exhibit negative skew.

**C.** The opposite of a momentum strategy is portfolio rebalancing; i.e., an investor who rebalances cannot invest in the momentum factor by definition.

**D.** The opposite of a momentum strategy is a value strategy; i.e., a factor model with both value and momentum contradicts itself or at least suffers from perfect multicollinearity.

**Question IM-2 e.5** New • built from GARP Ch.2 (2.5)

The investment committee of a Canadian foundation is deciding whether its value and momentum tilts rest on the same economic stories. The CIO summarizes the evidence in Ang's chapter. Which statement is correct?

**A.** Value has both rational stories (unproductive capital in bad times) and behavioral ones (overextrapolated growth), while momentum's most widely cited stories are behavioral (over- or underreaction to news).

**B.** Both premiums rest mainly on rational risk stories: value and momentum both lose money in recessions, and recessions define their bad times.

**C.** Momentum is explained by investors overextrapolating past growth rates into the future, while value is explained by investors underreacting to good news about the firm.

**D.** Because value is a negative feedback strategy, its premium cannot be explained by behavioral biases, which produce only positive feedback.

**Question IM-2 e.6** Workbook B4.2 • S2, S3, S4 • Map #18

The CIO of a Canadian university endowment is reviewing a consultant's proposal to add three investing-style tilts (value, momentum and low volatility) to the endowment's CAD 2.4 billion equity portfolio. Before the investment committee meets, the CIO checks four statements in the proposal. Which statement is correct?

**A.** Momentum is a negative feedback strategy that buys stocks after they have fallen, so like value it is inherently stabilizing and rarely suffers crashes.

**B.** The beta anomaly shows that high-beta stocks earn much lower average returns than low-beta stocks, so a long-short of beta quintiles captures it without leverage.

**C.** Value and momentum returns are only weakly negatively correlated, and momentum's largest crashes came when policymakers interrupted trends, as with bailouts of loser stocks in 2008.

**D.** Leverage constraints fully explain the low-risk anomaly, because constrained institutions overweight high-beta stocks, which both overprices high-beta and underprices low-beta stocks.

# Solutions • IM-2 e

## Solution IM-2 e.1

**Answer: C**

Bank Q78 • AP Mock 3 Q13

### Step 1 – Read the sort signal

The manager selects on past returns over a recent window, not on valuation or yield.

### Step 2 – Name the strategy

Long recent winners, short recent losers: momentum (WML), Jegadeesh and Titman's strategy.
⇒ C.

### Interpretation

The signal decides the strategy. Value sorts on price relative to fundamentals (book-to-market); momentum sorts on past price moves. A stock can be a value stock and a momentum loser at the same time.

### Why the other options are wrong

**A is incorrect.** Value sorts on book-to-market and buys stocks after they have fallen far enough to be cheap.

**B is incorrect.** Growth means low book-to-market stocks, a valuation sort, not a sort on recent returns.

**D is incorrect.** Income investing targets dividend or coupon yield.

**LO:** IM-2 e: Compare value and momentum investment strategies, including their return and risk profiles. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.5 Momentum Investing; SchweserNotes 2026 Book 5, Reading 84, LO 84.e

## Solution IM-2 e.2

**Answer: A**

Bank Q83 • Schweser Module Quiz 84.3 Q1

### Step 1 – Value's feedback

Stocks with declining prices eventually fall far enough to become value stocks; value investors then buy them. Buying after a fall pushes prices back toward fundamentals: negative feedback.

### Step 2 – Momentum's feedback

Stocks with high past returns attract momentum buyers and keep going up: positive feedback, which is destabilizing.
⇒ Only value stabilizes prices: A.

### Interpretation

This is the one sense in which value and momentum are opposites. The destabilizing side is why momentum is "subject to periodic crashes".

### Why the other options are wrong

**B is incorrect.** Momentum is positive feedback and destabilizing.

**C is incorrect.** GARP gives size no stabilizing feedback mechanism, and the size premium has largely disappeared since the mid-1980s.

**D is incorrect.** Momentum does not stabilize, so "all three" fails.

**LO:** IM-2 e: Compare value and momentum investment strategies, including their return and risk profiles. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.5 Momentum Investing

### Solution IM-2 e.3 — Answer: D

Bank Q81 • BT Mock Q70 • BT answers P2.T8.703.3

#### Step 1 – The return evidence

GARP's Figure 2.6 (1965 to 2011): "the cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on size and value". One dollar in WML peaked above USD 60.
$\Rightarrow$ D is true.

#### Step 2 – Test A to C

Momentum is positive feedback (A false); it is observed in international equities, commodities, government and corporate bonds, industries and real estate (B false); the HML and WML correlation is only $-16\%$ (C false).

#### Interpretation

The high return comes with crash risk: the same dollar fell below USD 30 in the 2007 to 2009 crisis, a drop of more than 50%.

#### Why the other options are wrong

**A is incorrect.** Reverses the feedback: value is negative feedback and stabilizing; momentum is positive feedback.
**B is incorrect.** Momentum appears in every asset class GARP lists.
**C is incorrect.** A correlation of $-16\%$ is weak; momentum is not simply anti-value.

**LO:** IM-2 e: Compare value and momentum investment strategies, including their return and risk profiles. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.5 Momentum Investing, Figure 2.6, Eq. 2.4

### Solution IM-2 e.4 — Answer: B

Bank Q82 • BT answers P2.T9.20.4.2

#### Step 1 – Return and risk of momentum

Momentum has the largest returns of the style factors, but it crashes periodically; its skewness is $-1.43$.
$\Rightarrow$ B is true.

#### Step 2 – Test the others against GARP

Only size disappeared after discovery; value has persisted (A false). Momentum and rebalancing are not opposites that exclude each other: momentum is cross-sectional within an asset class, while rebalancing is done at the asset-class or strategy level, so one investor can do both (C false). Value and momentum correlate at $-16\%$, far from the perfect collinearity D needs.

#### Interpretation

The bank's key justifies C's rejection with "a momentum investor should still rebalance", a Schweser line not in GARP (see the correction box). The answer is unchanged; GARP's reason is the level at which each operates.

#### Why the other options are wrong

**A is incorrect.** Overgeneralizes the size story: the value premium has persisted for fifty years.
**C is incorrect.** Rebalancing works across asset classes and strategies; momentum works within one. They do not rule each other out.
**D is incorrect.** Opposite only in feedback; a correlation of $-16\%$ is nowhere near perfect multicollinearity, and the four-factor model holds both.

**LO:** IM-2 e: Compare value and momentum investment strategies, including their return and risk profiles. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.4 Size Factor and 2.5 Momentum Investing, Characterizing Momentum Risk

### Solution IM-2 e.5 — Answer: A

New • built from GARP Ch.2 (2.5)

**Step 1 – Value's stories**

Rational: value stocks share undiversifiable risk; value firms are stuck with unproductive capital in bad times; firm investment risk. Behavioral: overextrapolation of past growth (Lakonishok, Shleifer and Vishny), loss aversion with mental accounting (Barberis and Huang). GARP: value has "a combination of rational and behavioral stories".

**Step 2 – Momentum's stories**

"The most widely cited theories are behavioral": delayed overreaction or underreaction to news. The rational story is "still far from being fully fleshed out".

$\Rightarrow$ A.

**Interpretation**

For the asset owner, Ang's test is the same for both: do I behave like the average investor, and can I bear the strategy's bad times? For momentum that means tolerating large drawdowns when policymakers interrupt trends.

**Why the other options are wrong**

**B is incorrect.** Momentum's rational story is incomplete, and value's bad times do not always line up with recessions: value lost in the booming late 1990s.

**C is incorrect.** Sibling swap: overextrapolation of growth is value's behavioral story; under- or overreaction to news is momentum's.

**D is incorrect.** Absolute claim: value has well-known behavioral explanations; the feedback direction describes the trading, not the source of the premium.

**LO:** IM-2 e: Compare value and momentum investment strategies, including their return and risk profiles. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.5 Rational and Behavioral Theories of the Value Premium; Momentum, Implications for Asset Owners; Ch.6 (GARP 2025 Ch.1), 1.7

### Solution IM-2 e.6 — Answer: C

Workbook B4.2 • S2, S3, S4

**Step 1 – Value vs momentum (IM-2 e)**

Value buys stocks whose prices have fallen far enough to look cheap: a negative feedback, stabilizing strategy. Momentum buys past winners and shorts past losers: a positive feedback, destabilizing strategy. The correlation of HML with WML is only $-16\%$.

$\Rightarrow$ Momentum is not the opposite of value, so the two tilts are close to complementary, not offsetting.

**Step 2 – Momentum's risk profile**

Momentum is negatively skewed $(-1.43)$. Of the eleven largest crashes, seven came in the Great Depression, one in 2001 and three in 2008, when bailouts put a floor under loser financials (Citi, Bank of America, Goldman, Morgan Stanley) that momentum was short.

$\Rightarrow$ The endowment must be able to tolerate large drawdowns concentrated in periods of policy intervention.

**Step 3 – Low-risk tilt (IM-3 h, i)**

The beta anomaly shows up in Sharpe ratios (0.9 falling to 0.4), not in average returns, which are roughly flat across beta quintiles; BAB therefore levers the low-beta leg to unit beta. Leverage constraints explain why high-beta stocks are overpriced but not why low-beta stocks are underpriced.

**Interpretation**

The CIO can accept the value and momentum diversification argument and the crash warning, but should ask the consultant to correct the description of the beta anomaly and to drop the claim that one explanation accounts for the whole risk anomaly.

### Why the other options are wrong

**A is incorrect.** Polarity flip: momentum is a positive feedback strategy that buys stocks after they have risen; positive feedback is destabilizing and subject to periodic crashes.

**B is incorrect.** Sibling swap: gives the beta anomaly the volatility anomaly's profile. High-beta stocks do not have low average returns; they have high volatility and low Sharpe ratios, so the BAB factor must lever the low-beta portfolio.

**D is incorrect.** Absolute claim: the leverage-constraint story explains only the overpricing of high-beta stocks, not the underpricing of low-beta stocks, and institutions in fact underweight high-risk stocks.

**LO:** IM-2 e: Compare value and momentum investment strategies, including their return and risk profiles. IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. IM-3 i: Describe potential explanations for the risk anomaly. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.5 Momentum, Characterizing Momentum Risk; Ch.10 (GARP 2025 Ch.3), Beta Anomaly, Betting against Beta, Explanations

IM-3 • Alpha (and the Low-Risk Anomaly) • Vardeez: VERY IMPORTANT

# IM-3 a

P2 CONCEPT

Describe and evaluate the low-risk anomaly of asset returns.

Workbook: B4 Factor Theory, Fama-French and Style • Tier B

**WHAT IS TESTED HERE**

**Why P2.** Tier B workbook topic: B4 (Map #18, 5 of 16 recall sheets) lists the low-risk anomaly as one of its reported shapes, next to Fama-French and style.
**Vardeez:** no pointer on this LO; the chapter is VERY IMPORTANT.

**Reported exam shapes**

- **S4 (B4)** The low-risk anomaly: volatility anomaly versus beta anomaly and their explanations. Workbook B4.2 tests it and is worked as Question IM-2 e.6.

This block covers what the anomaly is and how to judge it as a source of alpha. The volatility-versus-beta comparison is drilled in IM-3 h and the explanations in IM-3 i; both build on the background here.

**What the questions look like:** statement questions, no calculations. Which description of the anomaly is right, which portfolio illustrates it, and how a low-volatility strategy's alpha changes with the benchmark.

## 1. The anomaly

The CAPM says risk, measured by beta, is rewarded: the security market line slopes upward. The **low-risk anomaly** says the opposite. In GARP's words it is "a combination of three effects, with the third a consequence of the first two":

1. **Volatility** is negatively related to future returns (the volatility anomaly);
2. **Realized beta** is negatively related to future returns (the beta anomaly);
3. **Minimum variance portfolios** do better than the market.

Robin Greenwood (Harvard, a fellow adviser to Martingale): "the Mother of all inefficiencies may be standing right in front of us: the risk anomaly."

**Read "returns" carefully.** Sorted on past volatility, high-volatility stocks have "abysmally low" average returns. Sorted on past beta, average returns are roughly *flat*; the high-beta stocks lose on a *risk-adjusted* basis because they are more volatile (IM-3 h). Measured over the same period (contemporaneously), beta and returns are positively related, as the CAPM predicts. The summary line that "stocks with low betas and low volatilities have high returns" is shorthand for higher risk-adjusted returns than the CAPM allows.

## 2. A long history

- **Friend and Blume (1970)**, stocks 1960 to 1968: "risk-adjusted performance is dependent on risk. The relationship is inverse and highly significant."
- **Black, Jensen and Scholes (1972)**: beta and returns positively related, but "too flat" against the CAPM.
- **Haugen and Heins (1975)**, 1926 to 1971: portfolios "with lesser variance in monthly returns have experienced greater average returns than 'riskier' counterparts".

- **Fama and French (1992)**: "beta shows no power to explain average returns"; the point estimate was negative.
- These results were largely forgotten until **Ang, Hodrick, Xing and Zhang (2006)** found high-volatility stocks earn "abysmally low" returns, launching the modern "risk anomaly" literature.

## 3. Evaluating it as a source of alpha

GARP's case: GM Asset Management (which runs the GM pension fund) is pitched Martingale's low volatility strategy on the Russell 1000 universe.

- **Performance.** It beat the Russell 1000 by 1.50% a year over the full sample and by 1.83% a year after going live in 2008. Sharpe ratio $8.59/12.22 = 0.70$ against 0.45 for the index. Against a beta-adjusted benchmark (beta 0.73) its alpha rises to 3.44% a year (IM-3 b).
- **Against richer benchmarks.** Low-risk strategies "appear to have significant alpha relative to standard market capitalization benchmarks and sophisticated factor benchmarks" that add value-growth and momentum. After controlling for the Fama-French and momentum factors, the low-volatility factor still earns 0.42% a month ($t = 4.37$); betting against beta earns 0.33% ($t = 1.89$, borderline).
- **Alpha is all about the benchmark.** If the low volatility strategy itself were the benchmark, there would be no alpha: "alpha morphs into the benchmark (or beta, as some in industry like to call it)".
- **The catch for GM: tracking error.** Against the Russell 1000 the strategy's tracking error is 6.16%, above the sub-6% of the strategies GM usually considers, "because Martingale's product has a low beta". GM would have to offset that risk or persuade its investment committee to change the benchmark, a lengthy process. The same tracking-error limits help explain why the anomaly persists (IM-3 i). For GM the strategy also had a liability case: lower risk and a higher correlation with pension liabilities.
- **Robust and pervasive.** The effect appears in recessions and expansions, in each G7 market and across developed markets, and in Treasuries, corporate bonds, credit derivatives, commodities, currencies and options. Schweser's illustration: a U.S. minimum volatility ETF returned 68.75% over 2011 to 2016 against 65.27% for the S&P 500, with less risk.
- **Will it persist?** Ang hopes it disappears (low-risk investors would then earn large capital gains) but doubts it will: most institutions appear underweight low-risk stocks, and an effect this pervasive "requires a deep explanation".

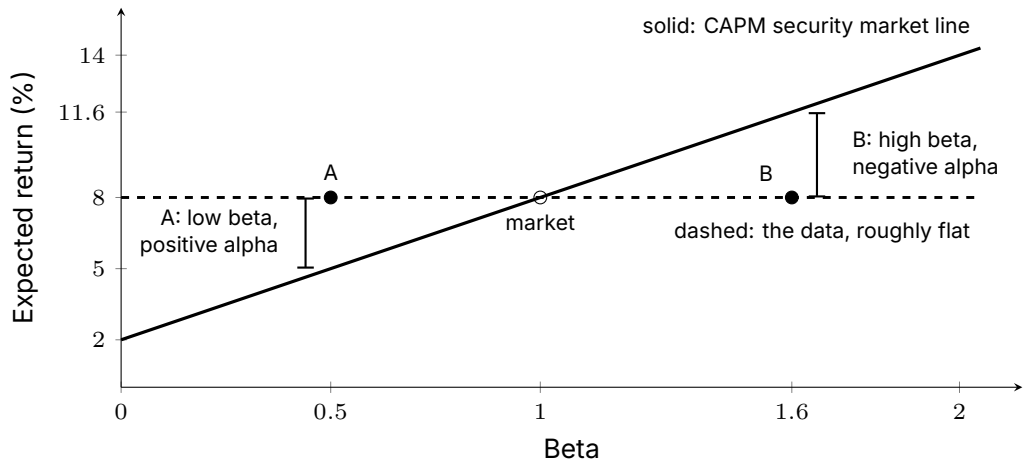



Figure 1. The low-risk anomaly in one picture (after GARP Figures 3.12 and 3.14; illustrative numbers: $r_f = 2\%$, market 8%). Against past betas, average returns are roughly flat (dashed) instead of rising along the CAPM line (solid). Low-beta stock A earns more than the CAPM requires (positive alpha, left bracket); high-beta stock B earns less (negative alpha, right bracket). IM-3 h and IM-3 i build on this figure.

| | CAPM prediction | Evidence (GARP Ch.3) |
|---|---|---|
| **Past volatility and future return** | No relation (only beta is priced) | Negative; highest-volatility quintile earns about 0.1% |
| **Past beta and future return** | Positive | Roughly flat raw returns; Sharpe ratio falls with beta |
| **Beta and return, same period** | Positive | Positive, as predicted (a "too flat" SML) |
| **Minimum variance portfolio** | Below the market's return | Beats the market |
| **Low-volatility strategy's alpha** | Zero | Positive and significant against cap-weighted and factor benchmarks |
| **Where it shows up** | Nowhere | U.S. and international stocks, bonds, credit, commodities, FX, options |

**Exam traps**

- **Raw versus risk-adjusted.** High-beta stocks do not have low raw returns; they have low risk-adjusted returns. High-volatility stocks do have low raw returns.
- **Lagged versus contemporaneous.** The anomaly is about risk measured *before* the return period. Over the same period, beta and returns line up positively.
- **Consequence, not a separate cause.** Minimum variance portfolios beating the market follows from the first two effects.
- **Naive benchmark.** A low-beta strategy measured against the raw index looks *worse* than against a beta-adjusted benchmark (1.50% against 3.44%).
- **Benchmark morphing.** Make the low-volatility strategy the benchmark and its alpha becomes zero.
- **Low risk means low tracking error?** No: a low-beta strategy has a *high* tracking error against a cap-weighted index.
- **Confined to small U.S. stocks?** No: the effect is pervasive across countries and asset classes.

## Practice questions: IM-3 a

### Question IM-3 a.1

Bank Q87 • Schweser Module Quiz 85.1 Q1

Which of the following statements is correct concerning the relationship between the low-risk anomaly and the capital asset pricing model (CAPM)?

**A.** The low-risk anomaly provides support for the CAPM.

**B.** The notion that the low-risk anomaly violates the CAPM has not been proven empirically.

**C.** The low-risk anomaly violates the CAPM and suggests that low-beta stocks will outperform high-beta stocks.

**D.** Both the CAPM and the low-risk anomaly point to a positive relationship between risk and reward.

### Question IM-3 a.2

Bank Q85 • AnalystPrep Q.4594 (rebuilt)

Which of the following is the most likely reason why low-beta stocks are attractive to investors?

**A.** They have historically delivered higher risk-adjusted returns (Sharpe ratios) than high-beta stocks, contrary to the CAPM's prediction that return rises with beta.

**B.** The CAPM predicts that low-beta stocks earn a higher expected return than the market to compensate for their lower diversification.

**C.** They are guaranteed to outperform the market in every period, including strongly rising markets.

**D.** They earn a larger risk premium because a low beta signals greater systematic risk.

### Question IM-3 a.3

Bank Q86 • BT P2.T9.20.5.1

Consider the following four portfolios:

I. Albert's portfolio has a high beta and generates high (strong) returns.

II. Betty's portfolio exhibits low volatility with high (strong) returns.

III. Chuck's portfolio exhibits low beta (systematic risk) but generates a high volatility (total risk).

IV. Donald's portfolio tracks the market's minimum variance portfolio with low tracking error and generates low (weak) returns.

Which of the portfolios best illustrates and confirms the low-risk anomaly?

**A.** Albert's portfolio

**B.** Betty's portfolio

**C.** Chuck's portfolio

**D.** Donald's portfolio

### Question IM-3 a.4

New • built from GARP Ch.3 (Ang), Section 3.5

A consultant is summarising the low-risk anomaly for a pension fund's investment committee, which knows the CAPM but not the empirical literature. Which of the following summaries is most accurate?

**A.** High-beta stocks earn lower average returns than low-beta stocks whether betas are measured before or during the return period, so the security market line slopes downward at every horizon.

**B.** Stocks with low past volatility or low past beta have earned higher risk-adjusted returns than the CAPM allows, and as a consequence minimum variance portfolios have outperformed the market.

**C.** The anomaly concerns idiosyncratic volatility only; once stocks are sorted on past beta instead, the positive risk-return relation of the CAPM is restored.

**D.** The anomaly is confined to small, illiquid U.S. stocks in the 1963 to 2011 sample, so it is most likely an artefact of data mining.

### Question IM-3 a.5

Bank Q84 • AP Mock 1 Q51

A pension fund portfolio manager is running a training session on low volatility investment strategies, using past performance data and explaining the benchmarks applied. Which assertion about low volatility strategies would be accurate for the manager to put forward?

**A.** These strategies typically demonstrate significant alphas when compared to conventional market capitalization benchmarks.

**B.** When evaluated against dynamic factors such as value or momentum, these strategies usually yield negative alphas.

**C.** Over the risk-free rate, these strategies tend to yield high alphas, but when compared to any other benchmark, the alphas are insignificant.

**D.** These strategies usually generate low alphas if the benchmark is risk-adjusted and high alphas otherwise.

### Question IM-3 a.6

Bank Q105 • BT P2.T9.20.5.3 • re-filed from IM-3 c

William evaluates a low-volatility fund against its naive benchmark, the Russell 1000. Over three years the fund's average return exceeded the Russell 1000 by $+0.28\%$ per year, which he tentatively calls the fund's "alpha". His colleague Emily argues the correct benchmark is a risk-adjusted portfolio of 65.0% Russell 1000 and 35.0% risk-free asset (the risk-free return is significantly less than the Russell 1000 return). If William re-evaluates performance over the same period using this risk-adjusted benchmark, which is the most likely outcome compared with his initial analysis?

**A.** The fund's alpha will increase.

**B.** The fund's beta will increase.

**C.** The fund's active return will decrease.

**D.** The fund's information ratio will decrease because its tracking error will increase.

**Question IM-3 a.7** New • built from GARP Ch.3 (Ang), Sections 3.2, 3.3 and 3.6

GM Asset Management is reviewing Martingale's low volatility strategy, which is built from the Russell 1000 universe and benchmarked to the Russell 1000. Over the full sample the strategy's Sharpe ratio was 0.70 against 0.45 for the index, its beta to the index was 0.73, and its tracking error against the index was 6.16%, while the other U.S. equity strategies GM considers typically have tracking errors below 6%. Which of the following conclusions is most appropriate?

**A.** Because the strategy is less volatile than the index, its 6.16% tracking error must come from stock-specific bets alone and would shrink in a longer sample.

**B.** The high tracking error comes largely from the beta of 0.73, so GM must offset that risk or justify a change of benchmark, and limits like this help the anomaly persist.

**C.** If GM adopted the low volatility strategy itself as the benchmark, the strategy's measured alpha would rise above the 3.44% a year it earns against the beta-adjusted benchmark.

**D.** Since the strategy kept beating the index after going live in 2008, its alpha is now statistically certain and the anomaly can be expected to persist indefinitely.

# Solutions • IM-3 a

## Solution IM-3 a.1

**Answer: C**

Bank Q87 • Schweser Module Quiz 85.1 Q1

### Step 1 – What the CAPM says

Expected return rises with beta: a positive risk-reward relation.

### Step 2 – What the anomaly says

Low-beta and low-volatility stocks have earned higher risk-adjusted returns than high-beta, high-volatility stocks: the relation runs the other way.

$\Rightarrow$ The anomaly violates the CAPM; C.

### Interpretation

Read "outperform" as *risk-adjusted*. Across beta quintiles raw average returns are roughly flat (about 15%, then 12.7% for the top quintile), while Sharpe ratios fall from 0.9 to 0.4. Across volatility quintiles raw returns themselves collapse at the top.

### Why the other options are wrong

**A is incorrect.** Polarity flip: the anomaly contradicts the CAPM.

**B is incorrect.** The violation is documented repeatedly, from Friend and Blume (1970) to Ang et al. (2006) and Frazzini and Pedersen.

**D is incorrect.** Only the CAPM implies a positive relation; the anomaly is a negative one.

**LO:** IM-3 a: Describe and evaluate the low-risk anomaly of asset returns. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5 Low Risk Anomaly; SchweserNotes 2026 Book 5, Reading 85, LO 85.a

## Solution IM-3 a.2

**Answer: A**

Bank Q85 • AnalystPrep Q.4594

### Step 1 – The CAPM benchmark

Expected return increases with beta, so low-beta stocks should earn *less* than the market.

### Step 2 – The evidence

Low-beta stocks earned about the same raw returns with less risk, so their Sharpe ratios were higher (0.9 for the lowest-beta quintile against 0.4 for the highest).

$\Rightarrow$ Better risk-adjusted performance, against the CAPM's prediction: A.

### Interpretation

This is the attraction GM saw in Martingale's product: a higher Sharpe ratio (0.70 against 0.45) with lower risk and a higher correlation with pension liabilities.

### Why the other options are wrong

**B is incorrect.** Reverses the CAPM: it predicts a *lower* expected return for a low beta.

**C is incorrect.** Absolute qualifier: low-beta stocks tend to lag in strong rising markets; nothing is guaranteed in every period.

**D is incorrect.** A low beta means less systematic risk and so, under the CAPM, a smaller premium.

**LO:** IM-3 a: Describe and evaluate the low-risk anomaly of asset returns. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Beta Anomaly (Figure 3.10)

### Solution IM-3 a.3 Answer: B

Bank Q86 • BT P2.T9.20.5.1

**Step 1 – What would illustrate the anomaly**

Low risk (volatility or beta) together with high returns: the reverse of the CAPM's trade-off.

**Step 2 – Check each portfolio**

Albert: high risk, high return, which is what the CAPM predicts. Betty: low volatility and high returns, the anomaly. Chuck: two risk measures and no return information. Donald: low risk and low return, the conventional trade-off.

$\Rightarrow$ Only Betty's portfolio contradicts the positive risk-return relation.

**Interpretation**

Donald's description is the opposite of the third effect in the anomaly: minimum variance portfolios have done *better* than the market, not worse.

**Why the other options are wrong**

**A is incorrect.** High risk with high return is the CAPM's prediction.

**C is incorrect.** Says nothing about returns, so it cannot illustrate a risk-return anomaly.

**D is incorrect.** Low risk with low return is the conventional relation; the evidence says minimum variance portfolios beat the market.

**LO:** IM-3 a: Describe and evaluate the low-risk anomaly of asset returns. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5 Low Risk Anomaly

### Solution IM-3 a.4 Answer: B

New • built from GARP Ch.3, Section 3.5

**Step 1 – The three effects**

GARP: volatility is negatively related to future returns; realized beta is negatively related to future returns; and, as a consequence, minimum variance portfolios do better than the market.

$\Rightarrow$ B states all three, with the right timing (past risk, future return) and the right yardstick (risk-adjusted).

**Step 2 – Test the others against the evidence**

Contemporaneous betas line up positively with returns; raw returns are roughly flat across past-beta quintiles; the effect appears in G7 markets, bonds, commodities, currencies and options, and the volatility anomaly is strongest in large, liquid stocks.

**Interpretation**

Two qualifiers carry most exam traps here: "past" (lagged) risk, and "risk-adjusted" returns for the beta sort.

**Why the other options are wrong**

**A is incorrect.** Absolute claim: over the same period beta and returns are positively related (as the CAPM predicts), and raw returns across past-beta quintiles are roughly flat, not falling.

**C is incorrect.** Sibling swap: the beta anomaly is the second of the three effects; sorting on beta does not restore the CAPM.

**D is incorrect.** The effect is pervasive across countries and asset classes, the best argument against data mining (IM-3 i).

**LO:** IM-3 a: Describe and evaluate the low-risk anomaly of asset returns. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5 Low Risk Anomaly (History; Volatility Anomaly; Beta Anomaly; Data Mining)

### Solution IM-3 a.5 — Answer: A

Bank Q84 • AP Mock 1 Q51

#### Step 1 – Against cap-weighted benchmarks

Martingale's strategy beat the Russell 1000 by 1.50% a year (1.83% live), and low-risk strategies "appear to have significant alpha relative to standard market capitalization benchmarks".
$\Rightarrow$ A is accurate.

#### Step 2 – Against richer benchmarks

The same sentence continues: "and sophisticated factor benchmarks that control for risk using dynamic value-growth and momentum factors". After the Fama-French and momentum factors, the volatility factor's alpha is 0.42% a month ($t = 4.37$) and betting against beta's 0.33% ($t = 1.89$).

#### Interpretation

Adjusting the benchmark for the strategy's low beta *raises* its alpha (1.50% to 3.44% for Martingale). The bank's explanation says the alpha "shrinks toward insignificance" against value and momentum factors; GARP's evidence says it stays positive and, for the volatility factor, highly significant.

#### Why the other options are wrong

**B is incorrect.** Polarity flip: the alphas against value and momentum benchmarks stay positive.
**C is incorrect.** Absolute claim: the alphas are significant against cap-weighted and factor benchmarks, not only against the risk-free rate.
**D is incorrect.** Reversed: a risk-adjusted benchmark gives a low-beta strategy a *higher* alpha than the raw index.

**LO:** IM-3 a: Describe and evaluate the low-risk anomaly of asset returns. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Betting-against-Beta and Volatility Factors; 3.6 GM Asset Management and Martingale Redux

### Solution IM-3 a.6 — Answer: A

Bank Q105 • BT P2.T9.20.5.3

#### Step 1 – What the naive benchmark assumes

The Russell 1000 alone assumes a beta of 1. Emily's 65/35 benchmark recognises a beta of about 0.65.

#### Step 2 – Effect on the hurdle

Because the risk-free rate is well below the index return, the 65/35 benchmark has a lower return. Illustration: index 8%, risk-free 2%: $0.65 \times 8\% + 0.35 \times 2\% = 5.9\%$ against 8%.
$\Rightarrow$ Over the same period, the fund's return over its benchmark rises (from 0.28% to about 2.4% in the illustration): alpha increases.

#### Interpretation

This is Martingale again: 1.50% against the raw index, 3.44% against the beta-adjusted one. The tracking error also falls, because the unhedged 35% market exposure no longer counts as active risk, so the IR rises on both counts.

#### Why the other options are wrong

**B is incorrect.** Beta is a property of the fund's returns; a new benchmark only recognises it.
**C is incorrect.** Active return over a lower-returning benchmark rises, not falls.
**D is incorrect.** Reversed: tracking error against the risk-matched benchmark is lower and the IR higher.

**LO:** IM-3 a: Describe and evaluate the low-risk anomaly of asset returns. IM-3 c: Explain the impact of benchmark choice on alpha and describe characteristics of an effective benchmark to measure alpha. • **Reference:** Ang, Asset

Management, Ch.10 (GARP 2025 Ch.3), Benchmarks Matter; 3.6 GM Asset Management and Martingale Redux

**Solution IM-3 a.7** **Answer: B**

New • built from GARP Ch.3, Sections 3.2, 3.3 and 3.6

**Step 1 – Where the tracking error comes from**

A beta of 0.73 means the strategy holds, in effect, only 73% of the index's market exposure. Against the full index, the missing 27% of market risk shows up as active risk. GARP: "The tracking error is high because Martingale's product has a low beta."

**Step 2 – GM's choices**

GARP: the equities group "would need to offset that risk or convince the investment committee that a change of benchmark was warranted".

**Step 3 – Link to the anomaly**

Tracking-error constraints against cap-weighted benchmarks stop institutions from buying low-risk stocks, one reason the anomaly is not arbitraged away (IM-3 i).

$\Rightarrow$ B.

**Interpretation**

Low volatility and low tracking error are different things: the less a strategy looks like the index, the higher its tracking error, even when its own volatility is lower.

**Why the other options are wrong**

**A is incorrect.** Wrong source: the low beta itself generates tracking error against a beta-1 index; it is not stock-specific risk that a longer sample would remove.

**C is incorrect.** Alpha morphs into the benchmark: with the strategy as its own benchmark, alpha is zero.

**D is incorrect.** Absolute claim: alpha is hard to detect statistically even over long samples, and Ang does not claim certainty about persistence.

**LO:** IM-3 a: Describe and evaluate the low-risk anomaly of asset returns. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.2 GM Asset Management and Martingale; Benchmarks Matter; 3.6 Redux

IM-3 • Alpha (and the Low-Risk Anomaly) • Vardeez: VERY IMPORTANT

# IM-3 e

P2 BOTH

Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark.

Workbook: B4 Factor Theory, Fama-French and Style • Tier B • Vardeez focus

### WHAT IS TESTED HERE

**Why P2.** Tier B workbook topic (B4, Map #18, 5 of 16 sheets), and Vardeez flags it in this VERY IMPORTANT chapter.
**Vardeez:** "Calculation of FAMA French Model using SMB, HML (Pls know this very well)".

**Reported exam shapes**

- **S1 (B4)** SMB (small minus big) versus SML (security market line), with HML, in a Fama-French portfolio composition. One recall sheet wrote "Portfolio composition (SML, HML)", almost certainly SMB. Workbook B4.1 tests the SMB/SML confusion and is worked as Question IM-2 d.7.

**What the questions look like:** a mix. Most give a factor regression table (coefficients, $t$-statistics, adjusted $R^2$) and ask what the loadings say about the fund's style, which dollar positions the mimicking portfolio holds, or whether the alpha is significant. The calculation form gives the loadings and the average factor returns and asks for the alpha against the multifactor benchmark.

## 1. A regression is a benchmark portfolio

IM-3 b showed the CAPM benchmark for Martingale: 27% T-bills plus 73% Russell 1000. The general point: rearrange the CAPM and **beta becomes a portfolio weight**. With $\beta_i = 1.3$,

$$E(r_i) - r_f = 1.3\,\big(E(r_m) - r_f\big) \quad \Longrightarrow \quad \underbrace{E(r_i)}_{\$1 \text{ in asset } i} = \underbrace{-\,0.3\,r_f + 1.3\,E(r_m)}_{\$1 \text{ in the benchmark}}$$

\$1 in asset $i$ is matched by borrowing \$0.30 at the T-bill rate and holding \$1.30 of the market. Alpha is any expected return in excess of this replicating portfolio: $E(r_i) = \alpha_i + E(r^{\text{bmk}})$. "Every time we run a factor regression, we are assuming that we can create a factor benchmark portfolio."

**Berkshire Hathaway, CAPM benchmark** (monthly, 1990 to May 2012): $\alpha = 0.72\%$ $(t = 2.02)$, $\beta = 0.51$ $(t = 6.51)$, adjusted $R^2 = 0.14$. Benchmark: $(1 - 0.51) = \$0.49$ in T-bills plus \$0.51 in the market; Buffett adds $0.72\% \times 12 = 8.6\%$ a year with about half the market's risk. This CAPM alpha is **Jensen's alpha**. A $t$-statistic above 2 is the usual 95% threshold; "most factor regressions do not produce significant alpha estimates", and CAPM $R^2$s for single stocks are usually below 10%.

## 2. Fama-French: adding size and value

Fama and French (1993) add two long-short factor-mimicking portfolios to the market:

$$\text{SMB} = \$1 \text{ in small caps} - \$1 \text{ in large caps}$$

$$\text{HML} = \$1 \text{ in value (high book-to-market)} - \$1 \text{ in growth (low book-to-market)}$$

$$\boxed{r_{it} - r_{ft} = \alpha + \beta\,(r_{mt} - r_{ft}) + s\,\text{SMB}_t + h\,\text{HML}_t + \varepsilon_{it}} \qquad \text{(Eq. 3.10)}$$

- **Size loading $s$:** zero for a medium-size stock; positive when it co-moves with small stocks; **negative when it moves with large stocks**.

- **Value loading $h$:** positive for a value orientation; negative for a growth orientation.
- **The market itself** is neither small nor big, neither value nor growth: zero $s$ and $h$.
- **SMB is not the SML (shape S1).** The security market line is the CAPM pricing line $E(r_i) - r_f = \beta_i(E(r_m) - r_f)$; it is not a factor and has no loading. SMB is a long-short portfolio whose loading is a dollar position in the benchmark.

## 3. Berkshire through the Fama-French lens

| | CAPM | $t$ | Fama-French | $t$ |
|---|---|---|---|---|
| Alpha (per month) | 0.72% | 2.02 | 0.65% | 1.96 |
| Market loading | 0.51 | 6.51 | 0.67 | 8.94 |
| SMB loading | | | −0.50 | −4.92 |
| HML loading | | | 0.38 | 3.52 |
| Adjusted $R^2$ | 0.14 | | 0.27 | |

- **The market beta moves** from 0.51 to 0.67: a sign that SMB and HML explain something (it would stay put if they explained nothing).
- $s = -0.50$**:** Berkshire acts like a *large* stock. Being large counts against Buffett's performance, because in the model large stocks tend to underperform small ones.
- $h = +0.38$**:** a strong value orientation. "A large, value investor. Duh, of course it is!"
- **Alpha** falls from 0.72% to 0.65% a month (8.6% to 7.8% a year, "nearly 1%"), still outsized and still significant at about $t = 2$; the fit improves, with adjusted $R^2$ from 14% to 27%.

**The mimicking portfolio**, per $1 in Berkshire: $(1 - 0.67) = \$0.33$ in T-bills + \$0.67 in the market − \$0.50 in small caps + \$0.50 in large caps + \$0.38 in value stocks − \$0.38 in growth stocks; Buffett adds 0.65% a month on top. Cash and market sum to \$1, and each long-short overlay nets to zero, so the benchmark still represents \$1 of capital (Figure 1).

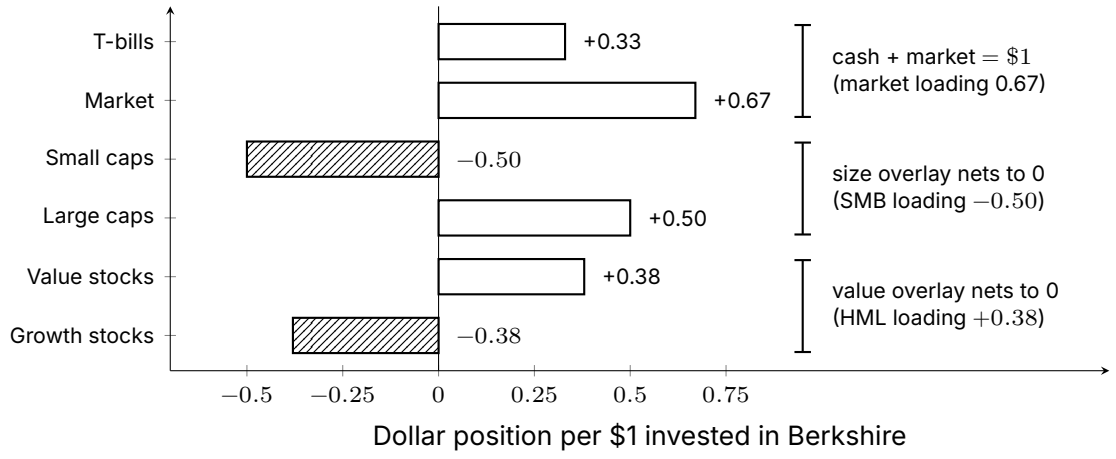



Figure 1. Berkshire Hathaway's Fama-French benchmark as a portfolio (GARP Ch.3 estimates). White bars are long positions, hatched bars short positions. A negative SMB loading is a short position in small caps funding a long position in large caps; a positive HML loading is long value, short growth.

## 4. Adding momentum

Carhart (1997) adds UMD, long stocks that have gone up and short stocks that have gone down (Eq. 3.11): $r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + s\,\text{SMB}_t + h\,\text{HML}_t + u\,\text{UMD}_t + \varepsilon_{it}$. For Berkshire: alpha 0.68% ($t = 2.05$), market 0.66, SMB $-0.50$, HML 0.36, UMD $-0.04$ ($t = -0.66$), adjusted $R^2$ 0.27. The momentum loading is near zero and insignificant (Buffett eschews momentum), the fit does not improve, and the alpha even rises slightly. Mimicking portfolio: \$0.34 in T-bills, \$0.66 in the market, the same $\mp$\$0.50 size overlay, a $\pm$\$0.36 value overlay, and $-$\$0.04 in past winners against $+$\$0.04 in past losers.

> **Correction to the source notes.** GARP prints the four-factor mimicking portfolio as "$(1 - 0.66) = \$0.37$ in T-bills". The expression is right and the figure is not: $1 - 0.66 = 0.34$. With a market loading of 0.66, \$0.34 in T-bills is what makes the benchmark \$1 of capital. Vardeez flags the same slip.

## 5. Measuring alpha against the multifactor benchmark

Because OLS fits through the means, the alpha is the fund's average excess return minus what the loadings earn on the average factor returns:

$$\boxed{\alpha = (\bar{r}_i - \bar{r}_f) - \left[\beta\,(\bar{r}_m - \bar{r}_f) + s\,\overline{\text{SMB}} + h\,\overline{\text{HML}}\right]}$$

**Worked example (monthly).** A fund's average excess return is 0.90%, with $\beta = 0.85$, $s = -0.30$, $h = 0.50$. The factors averaged $\bar{r}_m - \bar{r}_f = 0.55\%$, $\overline{\text{SMB}} = 0.20\%$, $\overline{\text{HML}} = 0.30\%$.

$$\text{Benchmark} = 0.85(0.55) + (-0.30)(0.20) + 0.50(0.30) = 0.4675 - 0.0600 + 0.1500 = 0.5575\%$$

$$\alpha = 0.90\% - 0.5575\% = 0.3425\% \text{ a month}$$

About 4.1% a year. Dropping the size and value terms would give $0.90 - 0.4675 = 0.43\%$: the fund's value tilt earns 0.15% a month that a market-only benchmark would book as skill, partly offset by its large-cap tilt. The benchmark is \$0.15 in T-bills and \$0.85 in the market, short \$0.30 small and long \$0.30 large, long \$0.50 value and short \$0.50 growth (Figure 2).

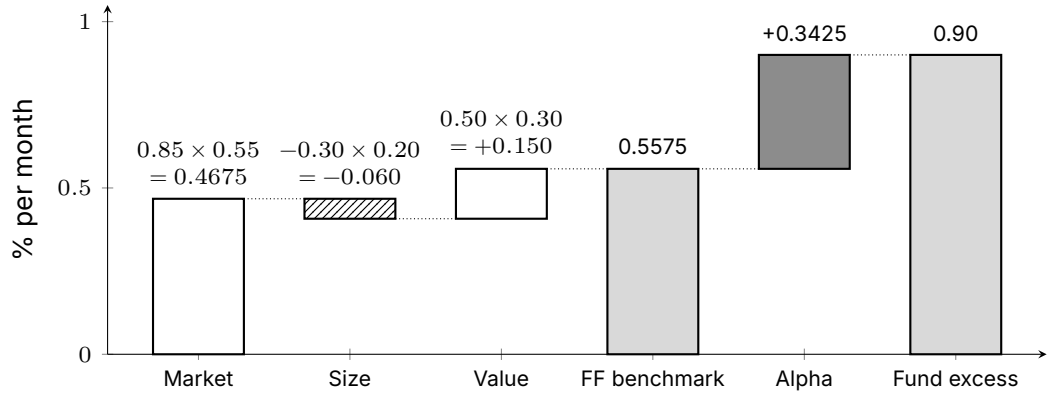



Figure 2. The worked example as a waterfall: each factor's contribution is its loading times the factor's average return. White bars add, the hatched bar (size) subtracts; the light grey bars are the Fama-French benchmark and the fund's average excess return, and the dark grey bar between them is alpha.

## 6. Benchmarks without a risk-free asset

Benchmark portfolios need not hold T-bills. Regress *gross* returns on tradeable asset returns and **constrain the loadings to sum to one**, so \$1 in the fund is matched by \$1 in the benchmark (Eq. 3.12). CalPERS, annual returns 1990 to 2011, against the S&P 500 and a long-term corporate bond index: alpha $-1.11\%$ ($t = -1.16$), bonds 0.32, stocks 0.68, adjusted $R^2$ 0.90. The fund behaves like a 32/68 bond-stock index portfolio; the negative alpha is not significant, so "we fail to reject the hypothesis that CalPERS adds value", but a fund with expense ratios of 0.50% to 0.80% could run that benchmark for close to zero. Direct real estate regressed on REITs, bonds and stocks shows no significant alpha either; stocks and bonds alone mimic it with 35% stocks and 65% bonds.

## 7. What a factor benchmark cannot do

- **Alpha depends on the factor list.** A positive alpha under one set of factors can turn negative under another. "Choosing the right set of factors is the most relevant issue for asset owners." This is the joint hypothesis problem: Hansen and Jagannathan (1997) show an ex-post benchmark with zero alpha can always be found.
- **Tradeability.** SMB and HML are not directly tradeable, and the loadings are assumed constant over the sample. Style analysis fixes both (IM-3 f).
- **Fit is not skill.** A high adjusted $R^2$ says the benchmark explains the fund's returns; whether the fund adds value is the alpha and its $t$-statistic.

**Formula card: IM-3 e**

| | |
|---|---|
| CAPM benchmark | $r^{\text{bmk}} = (1-\beta)\, r_f + \beta\, r_m$; $\beta = 1.3$: $-0.3\, r_f + 1.3\, r_m$ |
| Fama-French regression | $r_i - r_f = \alpha + \beta(r_m - r_f) + s\,\text{SMB} + h\,\text{HML}\,[+u\,\text{UMD}] + \varepsilon$ |
| Factor definitions | SMB = small − big; HML = value (high B/M) − growth (low B/M); UMD = winners − losers |
| Mimicking portfolio per \$1 | $(1-\beta)$ T-bills, $\beta$ market; $s$: $+s$ small, $-s$ large; $h$: $+h$ value, $-h$ growth |
| Alpha from averages | $\alpha = (\bar{r}_i - \bar{r}_f) - [\beta(\bar{r}_m - \bar{r}_f) + s\,\overline{\text{SMB}} + h\,\overline{\text{HML}}]$ |
| Significance | $t$ = coefficient/standard error; $\lvert t\rvert > 2$ about 95%; $\lvert t\rvert > 1.645$ about 90% |
| Gross-return benchmark | $r_i = \alpha + \sum_k \beta_k r_k + \varepsilon$ with $\sum_k \beta_k = 1$ (no T-bills needed) |
| Time | monthly alpha $\times 12$ for the annual figure |

**Exam traps**

- **SMB read as SML.** SMB is a size factor; the SML is the CAPM pricing line.
- **Sign of $s$.** Negative SMB loading means large-cap, not small-cap; it is short small caps, long large caps.
- **Sign of $h$.** Positive HML means value; negative means growth. HML is about value and growth, SMB about size: do not pair SMB with growth stocks.

- **T-bill leg.** $1 - \beta$ in T-bills: long when $\beta < 1$, a short (borrowing) when $\beta > 1$.
- **Alpha in the benchmark.** Alpha is earned on top of the benchmark; it is not one of its weights.
- **Raw market return.** Loadings multiply the market's *excess* return.
- **Factor terms dropped.** A market-only benchmark books the fund's size and value tilts as alpha.
- $R^2$ **as skill.** A high $R^2$ means a good benchmark fit, not value added.
- **Searching for factors.** Re-running regressions until the loadings are significant (or the alpha vanishes) is data mining.

## Practice questions: IM-3 e

### Question IM-3 e.1

Bank Q129 • Schweser Module Quiz 85.2 Q1

Why would an investor include multiple factors in a regression study?

I. To attempt to improve the adjusted $R^2$ measure.

II. To reduce the $t$-statistic values on the respective regression coefficients.

**A.** I only.

**B.** II only.

**C.** Both I and II.

**D.** Neither I nor II.

### Question IM-3 e.2

Bank Q130 • AP Mock 2 Q31

James is a portfolio manager of the ABC Fund. He is constructing a blended benchmark from the S&P 500 and a risk-free portfolio of T-bills. The CAPM regression of the fund's excess returns yields:

| Coefficient | Estimate | $t$-statistic |
|---|---|---|
| Alpha | 0.65% | 2.45 |
| MKT loading | 0.79 | 6.70 |
| Adjusted $R^2$ | 0.17 | |

Which of the following best describes the benchmark?

**A.** $0.65 + 0.79\,R_{\text{Tbill}} + 0.79\,R_{\text{SP500}}$

**B.** $0.65 + 0.21\,R_{\text{Tbill}} + 0.79\,R_{\text{SP500}}$

**C.** $0.21\,R_{\text{Tbill}} + 0.79\,R_{\text{SP500}}$

**D.** $0.79\,R_{\text{Tbill}} + 0.79\,R_{\text{SP500}}$

### Question IM-3 e.3

Bank Q128 • BT P2.T8.705.2

Below are Andrew Ang's regression results for the annual gross returns of the CalPERS pension fund against a passive portfolio of index funds in stocks and bonds. The returns are gross returns, not excess returns (they are NOT net of the risk-free rate).

| CalPERS gross annual returns, $R(p)$, versus benchmark portfolio | Coefficient | Standard error |
|---|---|---|
| Alpha, $\alpha$ | $-1.110\%$ | 0.957% |
| Bond loading, $\beta(B)$ | 0.320 | 0.0229 |
| Stock loading, $\beta(S)$ | 0.680 | 0.0487 |
| Adjusted $R^2$ | 0.900 | |

Which of the following statements about these regression results is TRUE?

**A.** The high adjusted $R^2$ validates a hypothesis that the CalPERS active fund managers do add value relative to the benchmark.

**B.** Because the factor loadings are not statistically significant, different factors should be repeatedly tested until a set is found that is significant.

**C.** This regression should be re-run with at least one additional factor because robust benchmark portfolios must include at least one risk-free asset.

**D.** It is acceptable to exclude risk-free assets and regress gross returns against the benchmark portfolio subject to the constraint that the factor loadings sum to one; here $\beta_B + \beta_S = 1.0$.

### Question IM-3 e.4

Bank Q127 • BT P2.T8.705.1 (rebuilt, key corrected)

Andrew Ang regressed Berkshire Hathaway's monthly excess returns (1990 to 2012) first on a CAPM benchmark and then on the Fama-French three-factor benchmark:

| CAPM | Coefficient | Std error | Fama-French | Coefficient | Std error |
|---|---|---|---|---|---|
| Alpha | 0.720% | 0.356% | Alpha | 0.650% | 0.332% |
| Beta | 0.510 | 0.0783 | MKT loading | 0.670 | 0.075 |
| | | | SMB loading | $-0.500$ | 0.102 |
| | | | HML loading | 0.380 | 0.108 |
| Adjusted $R^2$ | 0.140 | | Adjusted $R^2$ | 0.270 | |

All of the following statements about these results are true EXCEPT which one is false?

**A.** The SMB loading of $-0.50$ and HML loading of $+0.38$ imply Berkshire is a large-cap, value investor.

**B.** Berkshire's Fama-French alpha is significant at the 90% confidence level, and adding SMB and HML improves the fit of the model.

**C.** The rise in adjusted $R^2$ from 0.14 to 0.27 shows that SMB and HML add no explanatory power, so the CAPM benchmark is adequate.

**D.** The Fama-French benchmark implied by the estimates is USD 0.33 in T-bills + USD 0.67 in the market + (long USD 0.50 large caps, short USD 0.50 small caps) + (long USD 0.38 value, short USD 0.38 growth).

### Question IM-3 e.5

Bank Q126 • BT Mock Q5 • BT P2.T9.21.1.2

Richard ran a factor regression for his firm's Flagship Portfolio using the three-factor Fama-French specification (MKT, SMB = small minus big, HML = high minus low, i.e. value minus growth). Results are based on monthly excess returns, $R(p) - R_f$:

| Variable | Coefficient | Standard error |
|---|---|---|
| Alpha | 0.300% | 0.033% |
| MKT loading | 1.250 | 0.096 |
| SMB loading | 0.640 | 0.043 |
| HML loading | −0.920 | 0.066 |
| Adjusted $R^2$ | 0.290 | |

Each of the following statements about these results is TRUE EXCEPT which one is false?

**A.** The portfolio's style appears to be growth-oriented, small-cap.

**B.** The portfolio manager appears to be skillful; that is, the alpha is significant.

**C.** The low adjusted $R^2$ and low standard errors imply these factors are not appropriate to this portfolio.

**D.** The implied CAPM-type benchmark is leveraged: it borrows cash (T-bills) to invest more than 100% in the market portfolio.

### Question IM-3 e.6

New • built from GARP Ch.3 (Ang), Eq. 3.10

A risk analyst at a Canadian pension fund evaluates an external equity manager with the Fama-French three-factor model, using monthly data. Over the sample the fund's average monthly return was 1.15% and the average T-bill return 0.20%. The regression and the factors' average monthly returns are:

| Factor | Loading | Average factor return (% per month) |
|---|---|---|
| Market excess return, $r_m - r_f$ | 1.05 | 0.55 |
| SMB | 0.70 | 0.30 |
| HML | −0.25 | 0.32 |
| Adjusted $R^2$ | 0.88 | |
| Fund assets under management | CAD 2.4 billion | |

Which of the following is closest to the fund's monthly alpha measured against its Fama-French benchmark?

**A.** 0.03%

**B.** 0.08%

**C.** 0.24%

**D.** 0.37%

**Question IM-3 e.7** Bank Q122 • Mock 4 Q41 • GARP PE2 Q40 • S1

An individual investor decides to create a USD 1 million portfolio that mimics the strategy of Fund CRN, which has consistently generated high alphas. The investor regresses Fund CRN's monthly excess returns over the risk-free rate against the three Fama-French factors plus a momentum factor and obtains the following statistically significant estimates:

| Factor | Coefficient | $t$-statistic |
|---|---|---|
| Alpha | 0.08 | 2.10 |
| MKT loading | 0.55 | 6.72 |
| SMB loading | $-0.63$ | $-4.35$ |
| HML loading | 0.36 | 3.20 |
| UMD loading | $-0.07$ | $-2.77$ |
| Adjusted $R^2$ | 0.56 | |

Which of the following positions is a component of the mimicking portfolio?

**A.** A USD 7,000 long position in stocks showing positive momentum

**B.** A USD 360,000 long position in value stocks

**C.** A USD 450,000 short position in T-bills

**D.** A USD 630,000 short position in growth stocks

# Solutions • IM-3 e

## Solution IM-3 e.1

**Answer: A**

Bank Q129 • Schweser Module Quiz 85.2 Q1

### Step 1 – Statement I

Relevant extra factors can raise explanatory power. Berkshire: adjusted $R^2$ rises from 0.14 (CAPM) to 0.27 (Fama-French). True.

### Step 2 – Statement II

Nobody adds factors to *lower* $t$-statistics; a lower $t$ means less significance. The aim is a benchmark that represents the fund's style and explains more of its return. False.
$\Rightarrow$ I only.

### Interpretation

Adding factors also changes the alpha: whatever the new factors explain is no longer counted as skill (Berkshire: 0.72% to 0.65% a month).

### Why the other options are wrong

**B is incorrect.** Reverses the purpose: weaker significance is never a goal.
**C is incorrect.** Statement II is false.
**D is incorrect.** Statement I is true.

**LO:** IM-3 e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4 Factor Benchmarks, Size and Value-Growth Benchmarks; SchweserNotes 2026 Book 5, Reading 85, LO 85.e

## Solution IM-3 e.2

**Answer: C**

Bank Q130 • AP Mock 2 Q31

### Step 1 – Market weight

The market loading is the weight on the S&P 500: 0.79.

### Step 2 – T-bill weight

Weights must sum to one: $1 - 0.79 = 0.21$ in T-bills.

### Step 3 – Where alpha goes

Alpha (0.65% a month, $t = 2.45$) is the return *above* the benchmark, not a component of it.
$\Rightarrow$ Benchmark $= 0.21\, R_{\text{Tbill}} + 0.79\, R_{\text{SP500}}$.

### Interpretation

Same construction as Berkshire's $0.49/0.51$ and Martingale's $0.27/0.73$. The alpha is significant ($|t| > 2$), and the adjusted $R^2$ of 0.17 is typical of a CAPM regression.

### Why the other options are wrong

**A is incorrect.** Puts alpha in the benchmark and 0.79 on T-bills: weights of 1.58.
**B is incorrect.** Puts alpha inside the benchmark; the weights are right.
**D is incorrect.** Sets the T-bill weight equal to beta instead of $1 - \beta$: weights of 1.58.

**LO:** IM-3 e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, CAPM Benchmark, Eq. 3.7 to 3.9

### Solution IM-3 e.3 — Answer: D

Bank Q128 • BT P2.T8.705.2

**Step 1 – Is a risk-free asset needed?**

The dependent variable is the gross return, so the benchmark can be a fully invested stock-bond mix as long as the weights sum to 100%.

**Step 2 – Check the constraint**

$\beta_B + \beta_S = 0.320 + 0.680 = 1.000$: valid weights of 32% bonds and 68% stocks.

**Step 3 – Read the alpha**

$t = -1.110/0.957 = -1.16$: not significant; "we fail to reject the hypothesis that CalPERS adds value" (and the point estimate is negative).

$\Rightarrow$ D.

**Interpretation**

An adjusted $R^2$ of 0.90 means CalPERS behaves almost exactly like a cheap 32/68 index portfolio; with expense ratios of 0.50% to 0.80%, that is Ang's case for considering indexing.

**Why the other options are wrong**

**A is incorrect.** Confuses fit with skill: value added is the alpha, which is negative and insignificant.

**B is incorrect.** The loadings are highly significant ($t = 13.97$ and $13.96$), and searching factor sets until something is significant is data mining.

**C is incorrect.** A risk-free asset is not required with gross returns and loadings constrained to sum to one.

**LO:** IM-3 e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, Doing without Risk-Free Assets (CalPERS), Eq. 3.12

### Solution IM-3 e.4 — Answer: C

Bank Q127 • BT P2.T8.705.1

**Step 1 – A: the signs**

$s = -0.50$: co-moves with large stocks; $h = +0.38$: value. True.

**Step 2 – B: significance and fit**

Fama-French alpha $t = 0.650/0.332 = 1.96 > 1.645$, so significant at 90% (two-tailed); adjusted $R^2$ rises from 0.14 to 0.27. True.

**Step 3 – D: the mimicking portfolio**

Market 0.67, T-bills $1 - 0.67 = 0.33$, long large/short small \$0.50, long value/short growth \$0.38. True.

**Step 4 – C**

Adjusted $R^2$ nearly doubles, so SMB and HML clearly add explanatory power. C says the opposite.

$\Rightarrow$ C is the false statement.

**Interpretation**

The key was corrected in the bank once already and is confirmed here. At 95% the Fama-French alpha is borderline ($t = 1.96$, GARP's own figure), against $t = 2.02$ for the CAPM alpha: richer benchmarks make alpha harder to detect.

**Why the other options are wrong**

**A is incorrect.** True: large-cap, value.

**B is incorrect.** True: $t = 1.96$ clears the 90% threshold of 1.645, and the fit improves.
**D is incorrect.** True: the dollar version of the loadings.

**LO:** IM-3 e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, Size and Value-Growth Benchmarks, Eq. 3.10

### Solution IM-3 e.5 Answer: C

Bank Q126 • BT Mock Q5 • BT P2.T9.21.1.2

#### Step 1 – Style from the loadings

SMB $+0.64$ ($t = 0.64/0.043 = 14.9$): small-cap. HML $-0.92$ ($t = -13.9$): growth. A is true.

#### Step 2 – Alpha

$t = 0.300/0.033 = 9.1$, far above 2: significant. B is true.

#### Step 3 – Market loading

1.25 ($t = 13.0$): the benchmark holds \$1.25 of the market financed by $-$\$0.25 of T-bills, a levered position. D is true.

#### Step 4 – C

Small standard errors mean precisely estimated, significant loadings, which supports the factors' relevance; an adjusted $R^2$ of 0.29 is respectable for one portfolio.
$\Rightarrow$ C is the false statement.

#### Interpretation

Read every loading as a position: \$1.25 market, $-$\$0.25 T-bills, $+$\$0.64 small/$-$\$0.64 large, $-$\$0.92 value/$+$\$0.92 growth.

#### Why the other options are wrong

**A is incorrect.** True: positive SMB is small-cap, negative HML is growth.
**B is incorrect.** True: $t \approx 9$.
**D is incorrect.** True: $1 - 1.25 = -0.25$ in T-bills is borrowing.

**LO:** IM-3 e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, CAPM Benchmark (Eq. 3.7) and Size and Value-Growth Benchmarks

### Solution IM-3 e.6 Answer: C

New • built from GARP Ch.3, Eq. 3.10

#### Step 1 – Fund excess return

$1.15\% - 0.20\% = 0.95\%$ a month.

#### Step 2 – Return of the Fama-French benchmark (excess over T-bills)

$$1.05(0.55) + 0.70(0.30) + (-0.25)(0.32) = 0.5775 + 0.2100 - 0.0800 = 0.7075\%$$

$\Rightarrow$ The fund's market, small-cap and growth exposures earned 0.71% a month on their own.

#### Step 3 – Alpha

$$\alpha = 0.95\% - 0.7075\% = 0.2425\% \approx 0.24\% \text{ a month (about 2.9\% a year)}$$

#### Interpretation

The benchmark per \$1: borrow \$0.05 at the T-bill rate, \$1.05 in the market, long \$0.70 small and short \$0.70 large, short \$0.25 value and long \$0.25 growth. The adjusted $R^2$ and the fund's size

are red herrings.

### Why the other options are wrong

**A is incorrect.** Multiplied the market loading by the raw market return ($0.55 + 0.20 = 0.75\%$) instead of the excess return: $0.95 - (0.7875 + 0.21 - 0.08) = 0.03\%$.

**B is incorrect.** Treated the negative HML loading as a value tilt (sign flipped): $0.95 - (0.5775 + 0.21 + 0.08) = 0.08\%$.

**D is incorrect.** Dropped the SMB and HML terms, a market-only benchmark: $0.95 - 0.5775 = 0.37\%$, which books the small-cap premium as skill.

**LO:** IM-3 e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, Size and Value-Growth Benchmarks, Eq. 3.10

## Solution IM-3 e.7 Answer: B

Bank Q122 • Mock 4 Q41 • GARP PE2 Q40

### Step 1 – Each loading times USD 1 million is a dollar position

MKT 0.55: USD 550,000 long the market; T-bills: $(1 - 0.55) \times 1{,}000{,}000 =$ USD 450,000 *long*.

### Step 2 – The long-short overlays

SMB $-0.63$: short USD 630,000 small caps, long USD 630,000 large caps. HML $+0.36$: long USD 360,000 value, short USD 360,000 growth. UMD $-0.07$: short USD 70,000 past winners, long USD 70,000 past losers.

$\Rightarrow$ Only B matches.

### Interpretation

Alpha (0.08) is a return earned on top of the benchmark, not a position. The bank prints the SMB and UMD $t$-statistics as positive; they are shown here with the signs of their coefficients. This is the "portfolio composition" shape: SMB (size), not the SML.

### Why the other options are wrong

**A is incorrect.** Decimal slip and sign error: the momentum position is USD 70,000, and it is short winners, not long.

**C is incorrect.** Sign error: with a market loading below 1, the remaining USD 450,000 is invested in T-bills, not borrowed.

**D is incorrect.** Wrong factor: a $-0.63$ SMB loading is short small caps (long large caps); growth stocks belong to HML, where the short is USD 360,000.

**LO:** IM-3 e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, Size and Value-Growth Benchmarks and Adding Momentum, Eq. 3.10 and 3.11; GARP Practice Exam 2, Q40

IM-3 • Alpha (and the Low-Risk Anomaly) • Vardeez: VERY IMPORTANT

# IM-3 f

P2 CONCEPT

Explain how to use style analysis to handle time-varying factor exposures.

Workbook: none (Vardeez focus only); style analysis appears in the B4 revision notes • Tier B

**WHAT IS TESTED HERE**

**Why P2.** Vardeez names style analysis as the third focus of this VERY IMPORTANT chapter. No recall sheet reports it, so it sits in P2 on Vardeez's flag alone.
**Vardeez:** style analysis.

**Reported exam shapes:** none. The questions here are built from Vardeez's focus and GARP Ch.3 (Time-Varying Factor Exposures), checked with Schweser.

**What the questions look like:** statements. What style analysis fixes in a static Fama-French regression, how the time-varying weights are estimated and used, what they showed for four well-known funds, and what changes when short positions are allowed. Bodie, Kane and Marcus's version of Sharpe's style analysis (returns-based, $R^2$ as the share explained by style) is IM-7 e.

## 1. Two weaknesses of a static factor regression

The Fama-French regressions of IM-3 e estimate one set of loadings for the whole sample. Ang runs them for four funds on monthly data from January 2001 to December 2011:

| | LSVEX | FMAGX | GSCGX | BRK |
|---|---|---|---|---|
| Alpha (per month) | 0.00% | −0.27% | −0.14% | 0.22% |
| $t$-statistic | 0.01 | −2.23 | −1.33 | 0.57 |
| Market loading | 0.94 | 1.12 | 1.04 | 0.36 |
| HML loading | 0.51 | −0.05 | −0.17 | 0.34 |

LSVEX is LSV Value Equity, FMAGX Fidelity Magellan, GSCGX Goldman Sachs Capital Growth and BRK Berkshire Hathaway. Only Magellan's alpha is significant, and it is negative: $-0.27\% \times 12 = -3.24\%$ a year. Berkshire's alpha, significant from 1990, is not significant from 2001: "detecting statistical significance of outperformance is hard, even in samples of more than ten years". LSV is a value shop ($h = 0.51$, $t = 14.6$); Magellan is a levered play on the market ($\beta = 1.12$); none of the funds loads on momentum. Style analysis "seeks to rectify two potential shortcomings" of this analysis:

1. **The Fama-French portfolios are not tradeable**, so the benchmark is not something an asset owner can actually buy.
2. **The factor loadings may vary over time**, and a full-sample regression averages them away.

## 2. Style analysis: a benchmark that moves

William Sharpe (1992) introduced **style analysis**: "a factor benchmark where the factor exposures evolve through time". It replicates the fund by investing passively in low-cost index funds; the replicating weights are the **style weights**. Ang uses three SPDR ETFs: SPY (S&P 500), SPYV (S&P 500 Value) and SPYG (S&P 500 Growth).

$$r_{t+1} = \alpha_t + \beta_{\text{SPY},t}\,\text{SPY}_{t+1} + \beta_{\text{SPYV},t}\,\text{SPYV}_{t+1} + \beta_{\text{SPYG},t}\,\text{SPYG}_{t+1} + \varepsilon_{t+1}$$
$$\text{with} \quad \beta_{\text{SPY},t} + \beta_{\text{SPYV},t} + \beta_{\text{SPYG},t} = 1 \qquad \text{(Eq. 3.14)}$$

- **Tradeable factors.** The ETFs can be bought, unlike SMB and HML; other ETFs or index funds would do.
- **Weights sum to one**, so they form a replicating portfolio for $1 in the fund. In Ang's first version the weights are also **non-negative** (no shorting).
- **Timing.** The weights are estimated with information up to time $t$ (Ang uses the previous 60 months), and the first weights are estimated at January 2006. The fund's excess return over the next month is measured against the portfolio formed at $t$:

$$r^x_{t+1} = r_{t+1} - \left[\beta_{\text{SPY},t}\,\text{SPY}_{t+1} + \beta_{\text{SPYV},t}\,\text{SPYV}_{t+1} + \beta_{\text{SPYG},t}\,\text{SPYG}_{t+1}\right]$$

Each month the window rolls forward and the weights are re-estimated (Figure 1). Ang's question: "Can we find a robot that makes time-varying investments in SPY, SPYV, and SPYG that, together, match the returns of Buffett?"

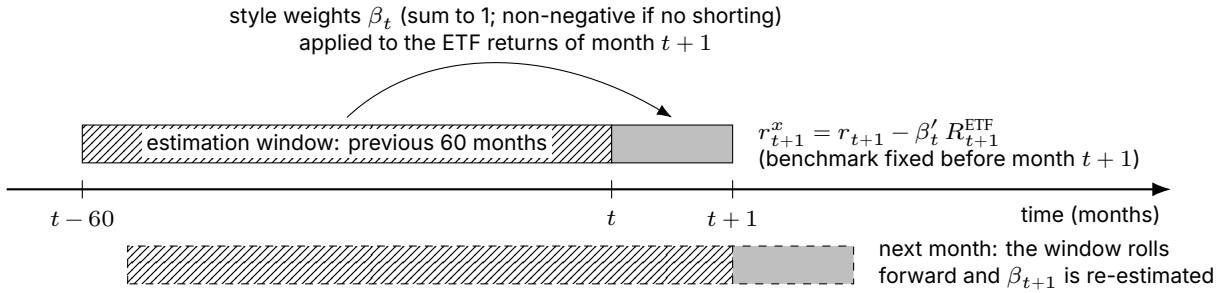



Figure 1. How the style benchmark moves. Weights are estimated on the hatched window (data up to $t$ only) and applied to the next month (grey); then the window rolls forward one month (dashed boxes). Using month $t+1$ data in the weights would be look-ahead.

## 3. What the style weights showed (no shorting, 2006 to 2012)

- **LSV:** "merely a combination of the market (SPY) and value (SPYV)"; cumulated excess returns about zero.
- **Fidelity Magellan:** starts in 2006 as a mix of all three ETFs and ends up all growth (SPYG); cumulated excess returns trend downward.
- **Goldman Sachs Capital Growth:** mostly market and growth throughout; excess returns about zero.
- **Berkshire Hathaway:** "strongly value" in 2006, switches to growth during the financial crisis, then back to strong value; the only fund with an upward trend in cumulated excess returns. A full-sample regression would have averaged these switches into one constant value loading (Figure 2).

## 4. Allowing short positions

Ang's second version lets the weights go negative and is written in excess returns (Eq. 3.15):

$$r_{t+1} - r_{f,t+1} = \alpha_t + \beta_{\text{SPY},t}\left(\text{SPY}_{t+1} - r_{f,t+1}\right) + h_t\left(\text{SPYV}_{t+1} - \text{SPYG}_{t+1}\right) + \varepsilon_{t+1}$$

- **SPYV − SPYG** is long value, short growth: the ETF analogue of HML. The regression is the "ETF version" of Fama-French *without the SMB factor*, with loadings that change over time.

- **Loadings:** LSV a positive $h$ (value); Magellan and the Goldman growth fund increasingly negative $h$ (growth); Berkshire negative $h$ in 2008 and 2009.
- **Alphas fall.** Allowing shorting "not surprisingly, reduces the alphas": Magellan's trend becomes more negative and Berkshire's positive excess returns shift down. The cumulated excess returns otherwise change little.

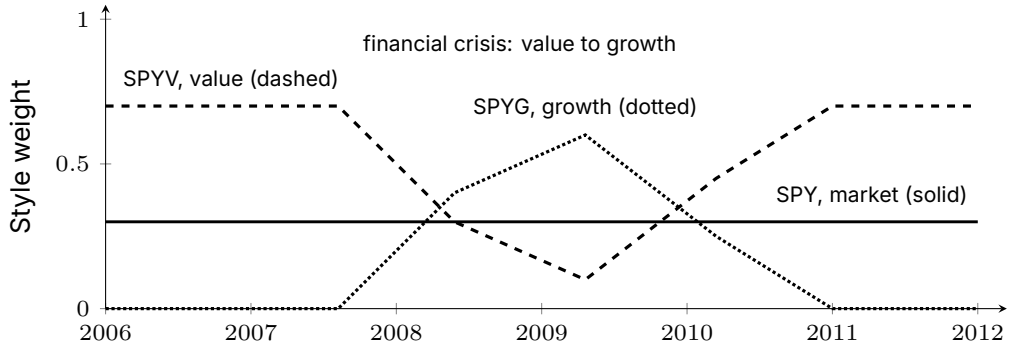



Figure 2. Schematic of the pattern GARP reports for Berkshire Hathaway (Figure 3.5, Panel A): strongly value in 2006, growth during the crisis, value again afterwards. The levels are illustrative, not estimates; only the pattern and the constraints (weights non-negative and summing to one every month) come from the reading.

## 5. The price: statistical inference

"The problems of statistical inference with time-varying portfolio benchmarks are serious." Significance is hard to detect even with constant benchmarks (Berkshire, 2001 to 2011), and estimated time-varying styles have even larger standard errors. Style analysis buys a more realistic, tradeable benchmark at the cost of noisier alpha estimates.

| | Static Fama-French regression | Style analysis, no shorting | Style analysis with shorting |
|---|---|---|---|
| **Factors** | MKT, SMB, HML (UMD): not tradeable | SPY, SPYV, SPYG: tradeable ETFs | SPY $-r_f$ and SPYV $-$ SPYG |
| **Loadings** | One set for the whole sample | Re-estimated each month, rolling 60 months | Re-estimated each month |
| **Constraints** | None | Sum to one, non-negative | Short positions allowed |
| **Shows style drift?** | No: averaged away | Yes | Yes |
| **Alphas** | Constant-benchmark alpha | Time-varying benchmark | Lower than with no shorting |
| **Main drawback** | Untradeable, static | Larger standard errors | Larger standard errors |

**Exam traps**

- **Look-ahead.** Weights for month $t+1$ use data up to $t$ only.
- **Sum to one, not "at least one".** The weights replicate \$1; leverage is not allowed for in the no-shorting version.
- **"Style analysis cannot short."** Ang runs it both ways; shorting is a choice, and it lowers the alphas.
- **Easier inference?** No: time-varying loadings mean larger standard errors.

- **Momentum is not the advantage.** UMD can be added to a static regression too; tradeability and time variation are what style analysis adds.
- **Sign of $h$ on SPYV − SPYG.** Negative means growth, as with HML.
- **SMB is absent** from Ang's ETF version.

## Practice questions: IM-3 f

### Question IM-3 f.1

New • built from GARP Ch.3 (Ang), Time-Varying Factor Exposures

A risk analyst at a fund of funds runs a static Fama-French regression (with a momentum factor) for four funds on monthly data from January 2001 to December 2011, with factor loadings held constant over the sample. Selected results:

| | LSVEX | FMAGX | GSCGX | BRK |
|---|---|---|---|---|
| Alpha (per month) | 0.00% | −0.27% | −0.14% | 0.22% |
| $t$-statistic of alpha | 0.01 | −2.23 | −1.33 | 0.57 |
| Market loading | 0.94 | 1.12 | 1.04 | 0.36 |
| HML loading | 0.51 | −0.05 | −0.17 | 0.34 |

The momentum loadings (not shown) are all small and insignificant. Which of the following conclusions is most appropriate?

**A.** Berkshire's alpha of 0.22% a month is statistically significant, which confirms the significant alpha found for 1990 to 2012.

**B.** Only Magellan's alpha is statistically significant, about $-3.2\%$ a year, and loadings held constant for 11 years may hide style changes that style analysis can track.

**C.** LSV's HML loading of 0.51 identifies a growth manager, and its zero alpha shows that its value tilt earns no premium.

**D.** Magellan's market loading of 1.12 shows that it is a momentum strategy, so adding a momentum factor would remove its negative alpha.

### Question IM-3 f.2

Bank Q135 • BT P2.T9.21.1.3

In contrast to factor regressions that assume constant factor weights, which of the following is the primary advantage of style analysis for determining the factor benchmark?

**A.** The factor benchmark uses actual tradeable funds.

**B.** The factor benchmark can include a momentum factor.

**C.** The factor benchmark naturally overcomes the beta anomaly.

**D.** Statistical inference is easier when the factor loadings vary over time.

**Question IM-3 f.3** New • built from GARP Ch.3 (Ang), Eq. 3.14

A consultant implements Ang's version of Sharpe's style analysis for a client's equity fund, using the SPDR S&P 500 (SPY), S&P 500 Value (SPYV) and S&P 500 Growth (SPYG) ETFs, with no short positions allowed. Which of the following describes how the benchmark return and the fund's excess return for month $t+1$ are obtained?

**A.** Style weights are estimated once over the full sample, constrained to sum to one, and every month's excess return is the fund return minus that fixed-weight ETF portfolio's return.

**B.** Style weights are estimated each month from the previous 60 months of data up to time $t$, constrained to be non-negative and to sum to one, and applied to the ETF returns of month $t+1$.

**C.** Style weights are estimated each month from data that include month $t+1$, so that the benchmark matches the fund's realised style exactly and the excess return is pure skill.

**D.** Style weights are estimated each month from the previous 60 months, constrained only to be non-negative, so that they can sum to more than one when the fund uses leverage.

**Question IM-3 f.4** Bank Q134 • BT P2.T8.706.1 (rebuilt)

A risk committee evaluates fund performance with a static three-factor Fama-French regression. A proposal would replace it with style analysis, illustrated by the following benchmark for next-period return, estimated each month using information up to time $t$:

$$R_{t+1} = \alpha_t + \beta_{\text{SPY},t} R^{\text{SPY}}_{t+1} + \beta_{\text{SPYV},t} R^{\text{SPYV}}_{t+1} + \beta_{\text{SPYG},t} R^{\text{SPYG}}_{t+1} + \varepsilon_{t+1}, \qquad \beta_{\text{SPY},t} + \beta_{\text{SPYV},t} + \beta_{\text{SPYG},t} = 1$$

where SPY, SPYV and SPYG are the SPDR S&P 500, S&P 500 Value and S&P 500 Growth ETFs. Each of the following statements about style analysis versus a static benchmark is true EXCEPT which is false?

**A.** A drawback of style analysis is that it cannot introduce short positions into the benchmark.

**B.** A drawback of style analysis is that statistical significance is harder to detect because of larger standard errors.

**C.** An advantage of style analysis is that factor loadings can vary over time, as shown by $\beta_{\text{SPY},t}$ rather than $\beta_{\text{SPY}}$.

**D.** An advantage of style analysis is that actual tradeable funds, such as SPDR ETFs, can be used as the factor benchmark.

**Question IM-3 f.5** New • built from GARP Ch.3 (Ang), Style Analysis with No Shorting

Using style analysis with no shorting, an analyst estimates rolling style weights on SPY, SPYV and SPYG for four funds from 2006 to 2012. Which of the following readings of the results is consistent with Ang's findings?

**A.** LSV Value Equity is replicated mostly by the growth ETF, and its cumulated excess returns trend strongly upward over the period.

**B.** Fidelity Magellan starts as a mix of all three ETFs and ends up almost entirely in growth, while its cumulated excess returns trend downward.

**C.** Berkshire Hathaway's weights stay fixed on the value ETF throughout, which is why a static Fama-French regression captures its style fully.

**D.** Goldman Sachs Capital Growth is replicated mostly by the value ETF, and it is the only fund whose cumulated excess returns trend upward.

**Question IM-3 f.6** New • built from GARP Ch.3 (Ang), Style Analysis with Shorting, Eq. 3.15

An analyst re-runs a style analysis allowing short positions. She regresses each fund's return over T-bills on the return of SPY over T-bills and on the return of SPYV minus SPYG, with the loadings re-estimated through time. Which of the following statements about this specification is correct?

**A.** The SPYV minus SPYG factor is a long-short position analogous to HML, so the regression is an ETF version of the Fama-French model without the size factor.

**B.** Allowing short positions typically raises the estimated alphas, because the benchmark can no longer fully match the fund's value or growth tilt.

**C.** A fund with an increasingly negative loading on SPYV minus SPYG is moving towards a value style, as LSV Value Equity did.

**D.** Because SPYV and SPYG are not tradeable, this specification gives up the main advantage of style analysis over Fama-French regressions.

## Solutions • IM-3 f

### Solution IM-3 f.1 — Answer: B

New • built from GARP Ch.3, Time-Varying Factor Exposures

**Step 1 – Which alphas are significant?**

Only Magellan has $|t| > 2$ ($t = -2.23$). Its alpha is $-0.27\% \times 12 = -3.24\%$ a year: investors lose money, significantly.

**Step 2 – What a static regression misses**

The loadings are averages over 11 years. Ang's style analysis on the same funds shows Magellan drifting to all growth and Berkshire switching from value to growth and back: changes a constant loading cannot show.

$\Rightarrow$ B.

**Interpretation**

These results motivate style analysis: the Fama-French factors are not tradeable and the loadings may vary over time. Berkshire's significant 1990s alpha disappears in this window, a reminder that alpha is hard to detect even over a decade.

**Why the other options are wrong**

**A is incorrect.** $t = 0.57$ is far below 2: not significant for 2001 to 2011.

**C is incorrect.** A positive HML loading of 0.51 is a strong *value* tilt; a zero alpha means the benchmark, value premium included, explains the returns, not that value earns nothing.

**D is incorrect.** A market loading above 1 is a levered market play; momentum would show in the UMD loading, which is insignificant.

**LO:** IM-3 f: Explain how to use style analysis to handle time-varying factor exposures. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, Time-Varying Factor Exposures

### Solution IM-3 f.2 — Answer: A

Bank Q135 • BT P2.T9.21.1.3

**Step 1 – The two fixes**

Style analysis replaces the untradeable Fama-French portfolios with tradeable funds (SPY, SPYV, SPYG) and lets the loadings change over time.

**Step 2 – Check the options**

Only A names one of those fixes.

**Interpretation**

Schweser: "Unlike Fama and French's untradeable SMB and HML indices, style analysis uses tradeable assets." A benchmark the asset owner can buy is the point of an alpha measured against it.

**Why the other options are wrong**

**B is incorrect.** A momentum factor (UMD) can be added to a static regression too.

**C is incorrect.** Style analysis says nothing about the beta anomaly, an unrelated empirical finding.

**D is incorrect.** Reversed: time-varying loadings have larger standard errors, so inference is harder.

**LO:** IM-3 f: Explain how to use style analysis to handle time-varying factor exposures. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.4, Time-Varying Factor Exposures; SchweserNotes 2026 Book 5, Reading 85, LO 85.f

### Solution IM-3 f.3 — Answer: B

New • built from GARP Ch.3, Eq. 3.14

**Step 1 – Estimation**

Weights are estimated with information up to $t$, using the previous 60 months, and re-estimated as the window rolls forward.

**Step 2 – Constraints**

They sum to one (a replicating portfolio of \$1) and, in the no-shorting version, are non-negative.

**Step 3 – Use**

$r^x_{t+1} = r_{t+1} - \beta'_t R^{\text{ETF}}_{t+1}$: the portfolio formed at $t$ is the benchmark for month $t+1$.
$\Rightarrow$ B.

**Interpretation**

The sum-to-one restriction is what makes the weights a benchmark portfolio; the timing is what makes it implementable: a "robot" could have held it.

**Why the other options are wrong**

**A is incorrect.** Fixed full-sample weights are the static benchmark that style analysis replaces.
**C is incorrect.** Look-ahead: using month $t+1$ data would fit the benchmark to the outcome.
**D is incorrect.** The weights must sum to one; they cannot replicate leverage in the no-shorting version.

**LO:** IM-3 f: Explain how to use style analysis to handle time-varying factor exposures. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Style Analysis with No Shorting, Eq. 3.14

### Solution IM-3 f.4 — Answer: A

Bank Q134 • BT P2.T8.706.1

**Step 1 – C and D: the advantages**

The time subscript on $\beta_{\text{SPY},t}$ is time variation (C true); SPDR ETFs are tradeable (D true).

**Step 2 – B: the cost**

Time-varying styles have larger standard errors, so significance is harder to detect (B true).

**Step 3 – A**

Ang runs style analysis both without shorting (non-negative weights) and with shorting (Figure 3.6, Eq. 3.15). Short positions are a modelling choice, not an impossibility.
$\Rightarrow$ A is the false statement.

**Interpretation**

The only restriction in the equation shown is that the loadings sum to one; nothing there rules out a negative weight. Allowing shorting lowers the measured alphas.

**Why the other options are wrong**

**B is incorrect.** True: larger standard errors.
**C is incorrect.** True: the defining feature.
**D is incorrect.** True: tradeable ETFs replace untradeable SMB and HML.

**LO:** IM-3 f: Explain how to use style analysis to handle time-varying factor exposures. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Style Analysis with No Shorting and with Shorting, Eq. 3.14 and 3.15

### Solution IM-3 f.5 — Answer: B

New • built from GARP Ch.3, Style Analysis with No Shorting

#### Step 1 – Recall the four style paths

LSV: market plus value. Magellan: all three ETFs, ending all growth. Goldman: market plus growth. Berkshire: value, then growth in the crisis, then value.

#### Step 2 – Recall the cumulated excess returns

LSV and Goldman about zero; Magellan trending down; Berkshire the only one trending up.
$\Rightarrow$ B.

#### Interpretation

Magellan's style drifted while it lost money against its own replicating portfolio. A constant benchmark would have measured it against a style it no longer had.

#### Why the other options are wrong

**A is incorrect.** LSV is replicated by market and *value*, and its excess returns are about zero.
**C is incorrect.** Berkshire's weights move from value to growth and back; a static regression averages that away.
**D is incorrect.** The Goldman growth fund is market plus *growth*, with excess returns about zero; Berkshire is the only upward trend.

**LO:** IM-3 f: Explain how to use style analysis to handle time-varying factor exposures. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Style Analysis with No Shorting (Figure 3.5)

### Solution IM-3 f.6 — Answer: A

New • built from GARP Ch.3, Eq. 3.15

#### Step 1 – The factors

SPY $-r_f$ is the market excess return; SPYV $-$ SPYG is long value, short growth, "analogous to the HML factor". There is no size factor.
$\Rightarrow$ A: the ETF version of Fama-French without SMB, with time-varying loadings.

#### Step 2 – Check the claims about shorting and signs

Allowing shorting *reduces* the alphas; a negative loading on SPYV $-$ SPYG is a growth tilt (Magellan, Goldman), while LSV's is positive.

#### Interpretation

The shorting version gives each fund a single value-growth loading $h_t$ that can be read like an HML loading, month by month.

#### Why the other options are wrong

**B is incorrect.** Polarity flip: allowing shorting lowers the alphas.
**C is incorrect.** Negative $h$ means growth; LSV had a positive $h$.
**D is incorrect.** SPYV and SPYG are tradeable ETFs; tradeability is kept.

**LO:** IM-3 f: Explain how to use style analysis to handle time-varying factor exposures. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), Style Analysis with Shorting, Eq. 3.15 (Figure 3.6)

IM-3 • Alpha (and the Low-Risk Anomaly) • Vardeez: VERY IMPORTANT

# IM-3 h

P2 CONCEPT

Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly.

Workbook: B4 Factor Theory, Fama-French and Style • Tier B

**WHAT IS TESTED HERE**

**Why P2.** Tier B workbook topic: B4 (Map #18, 5 of 16 sheets) reports the volatility-versus-beta comparison as shape S4.
**Vardeez:** no pointer on this LO; the chapter is VERY IMPORTANT. His summary: the volatility anomaly destroys the numerator of the Sharpe ratio, the beta anomaly inflates the denominator.

**Reported exam shapes**

- **S4 (B4)** The low-risk anomaly: volatility anomaly versus beta anomaly and their explanations. Workbook B4.2 tests it and is worked as Question IM-2 e.6.

The background (the three effects, the history, the flat security market line) is in IM-3 a; the explanations are IM-3 i.

**What the questions look like:** statements. What happens to average returns and Sharpe ratios across volatility and beta quintiles, lagged versus contemporaneous risk, and how the two anomalies are traded (BAB and VOL factors).

## 1. Two anomalies, one conclusion, different arithmetic

Both anomalies end with the Sharpe ratio falling as risk rises (IM-3 a, Figure 1). They get there by different routes:

- **Volatility anomaly:** high-volatility stocks have low *raw* returns. The numerator collapses.
- **Beta anomaly:** raw returns are roughly flat across betas; high-beta stocks are more volatile, so their Sharpe ratios are lower. The denominator grows.

A statement that "high-beta stocks earn low returns" describes the volatility anomaly and labels it the beta anomaly.

## 2. The volatility anomaly

**Theory says the opposite.** Under the CAPM and its multifactor extensions, a stock's own volatility should not matter; idiosyncratic volatility (tracking error against a risk-adjusted benchmark) "should definitely not have any relation to expected returns". Models with segmented markets (clienteles) or noise traders predict a *positive* relation, because investors must be paid to bear idiosyncratic risk.

**Evidence.** Ang, Hodrick, Xing and Zhang (2006) found high-volatility stocks earn "abysmally low" returns, robust to controls for size, value, leverage, liquidity risk, volume, turnover, bid-ask spreads, co-skewness, dispersion in analysts' forecasts and momentum. Aggregate volatility risk does not explain it, and the 2009 follow-up found it in each G7 country and across developed markets, after controls for private information, transaction costs, analyst coverage, institutional ownership and delay. Skewness does not explain it.

**Lagged volatility and future returns (GARP Figure 3.8).** U.S. stocks, quarterly rebalanced, September 1963 to December 2011, sorted into quintiles on idiosyncratic volatility (Fama-French residuals,

daily data over the past quarter; total volatility gives similar results), value-weighted within quintiles. Volatility rises across quintiles by construction. Average returns are **above 10% for the first three quintiles, 6.8% for quintile 4 and 0.1% for the highest-volatility quintile**. Raw Sharpe ratios (mean over standard deviation) **fall monotonically from 0.8 to 0.0**.

**Contemporaneous volatility (Figure 3.9).** Sorting at the end of each period on the volatility realised during it still gives a negative relation: the most volatile stocks lose money now (not forecastable) and in the future (forecastable). These portfolios are not tradeable.

## 3. The beta anomaly

**Earlier tests.** The 1970s tests of the CAPM found a positive but "too flat" relation between beta and returns (Black, Jensen and Scholes, 1972). Fama and French (1992): "beta shows no power to explain average returns"; their point estimate was negative.

**Lagged beta and future returns (Figure 3.10).** Quintiles on betas estimated over the previous quarter with daily returns, rebalanced quarterly, equal-weighted (to get the largest spreads). Average returns are **roughly flat: about 15% for the first four quintiles and 12.7% for quintile 5**. Raw Sharpe ratios **fall from 0.9 to 0.4**. "The beta anomaly is not that stocks with high betas have low returns: they don't. Stocks with high betas have high volatilities." (Implied volatilities: about $15/0.9 = 17\%$ for the lowest quintile, $12.7/0.4 = 32\%$ for the highest.)

**Pre- versus post-ranking betas.** Betas measured in the quarter *after* formation are much flatter across quintiles than the betas used to rank: betas are noisy. The post-ranking spread is still over 1.0.

**Contemporaneous beta (Figure 3.11).** The CAPM is a statement about betas and returns over the same period, and that relation is *positive*, as the CAPM predicts (though the line is "too flat" near the axis). The real mystery "is not so much that beta does not work; it is that we have such difficulty in predicting future betas, especially with past betas". Betas estimated from option prices (Buss and Vilkov, 2012) or from accounting data (Cosemans et al., 2012) predict future betas better and give a positive risk-return relation.

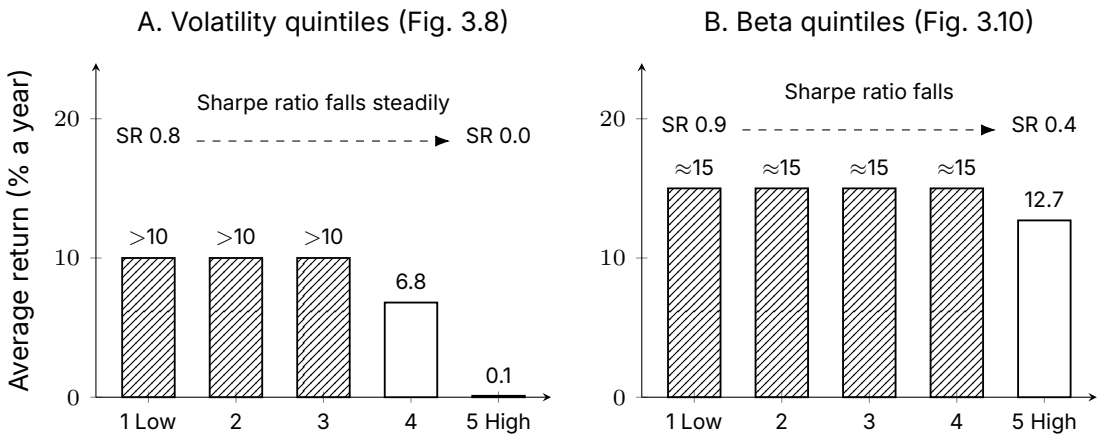



Figure 1. Average returns by quintile of past risk (GARP Ch.3). Hatched bars mark quintiles for which GARP reports only a bound or an approximate level ("above 10%", "around 15%"); the bar is drawn at that level. White bars are reported values. Only the end-point Sharpe ratios are reported; the dashed arrows show the direction. Panel A: the return itself collapses at the top. Panel B: returns are roughly flat and the Sharpe ratio falls because high-beta stocks are more volatile.

## 4. Trading the two anomalies: BAB and VOL

**Betting against beta** (Frazzini and Pedersen): differences in average returns across beta quintiles are tiny, so a plain long-short of the quintiles does not capture the anomaly; what is large is the difference in Sharpe ratios. BAB scales each leg to a beta of one:

$$\text{BAB}_{t+1} = \frac{r_{L,t+1} - r_f}{\beta_{L,t}} - \frac{r_{H,t+1} - r_f}{\beta_{H,t}} \qquad \text{VOL}_{t+1} = \sigma_{\text{target}} \left( \frac{r_{L,t+1} - r_f}{\sigma_{L,t}} - \frac{r_{H,t+1} - r_f}{\sigma_{H,t}} \right)$$

The long leg levers up the low-beta portfolio; the short leg combines the high-beta portfolio with T-bills (Figure 2). Only two beta portfolios can be used: the lowest quintile's pre-ranking beta is close to zero, and levering it to a beta of one would need almost infinite leverage. Ang's **VOL** factor scales the lowest and highest volatility quintiles to a target volatility ($\sigma_{\text{target}} = 15\%$); because expected returns, not only volatilities, differ across volatility quintiles, it can be traded without the risk-free asset.

| Fama-French + momentum regression, Oct 1963 to Dec 2011 | BAB | $t$ | VOL | $t$ |
|---|---|---|---|---|
| Alpha (per month) | 0.33% | 1.89 | 0.42% | 4.37 |
| Market loading | −0.17 | −4.13 | 0.87 | 38.8 |
| SMB loading | 0.29 | 5.20 | −0.63 | −20.3 |
| HML loading | 0.48 | 7.85 | 0.20 | 5.73 |
| UMD loading | 0.09 | 2.35 | 0.13 | 6.00 |

- **Distinct anomalies.** The correlation between BAB and VOL is only $-9\%$. VOL has higher cumulative returns and a slightly higher Sharpe ratio (0.6 against 0.5).
- **Significance.** BAB's alpha (about 4% a year) is borderline ($t = 1.89$, $p = 0.06$); VOL's (about 5% a year) is strongly significant.
- **Where they live.** Both tilt to value and momentum. BAB loads positively on SMB: the beta anomaly shows up more in *small* stocks. VOL loads negatively: the volatility anomaly is more pervasive in *large*, liquid, easier-to-trade stocks.
- **Which to do?** "This is not an either-or choice. You should do both."

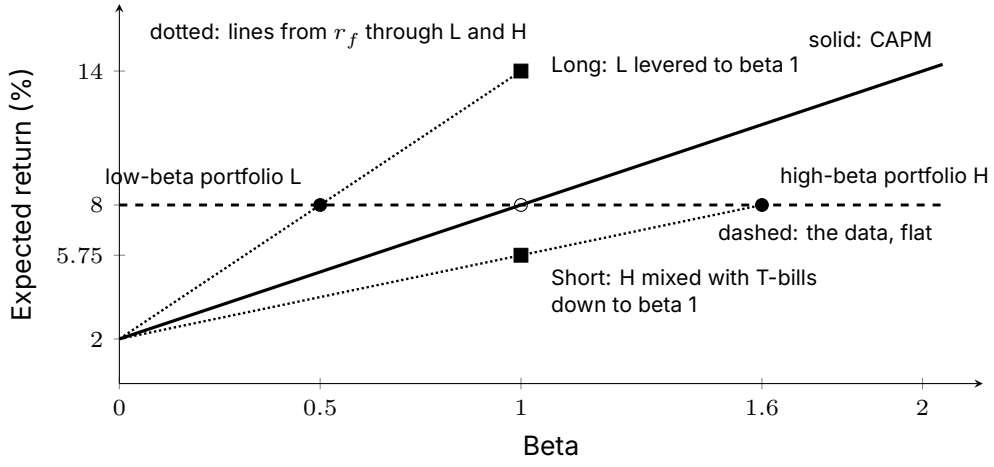



Figure 2. Building BAB (after GARP Figure 3.12; illustrative numbers: $r_f = 2\%$, flat data line at 8%, $\beta_L = 0.5$, $\beta_H = 1.6$). Levering L along its dotted line to a beta of 1 gives the Long point (14%); mixing H with T-bills down to a beta of 1 gives the Short point (5.75%). Long minus Short has zero beta and a positive expected return: it harvests the flatness of the data line.

**Correction to the source notes.** (1) SchweserNotes (LO 85.h) says that "as standard deviation increased, both the average returns and the Sharpe ratios decreased". GARP reports average returns above 10% for each of the three lowest-volatility quintiles, then 6.8% and 0.1%: the collapse is at the top end, and only the Sharpe ratio is reported to decline monotonically (0.8 to 0.0). Vardeez makes the same correction. (2) Vardeez and Schweser attach the September 1963 to December 2011 quintile results to Ang, Hodrick, Xing and Zhang (2006), and Schweser credits the beta-quintile evidence to the same paper. In GARP both are Ang's own sorts for the chapter, built "similar to Ang et al. (2006, 2009)"; a 2006 paper could not use data to 2011. The findings are unchanged; only the attribution is loose.

| | Volatility anomaly | Beta anomaly |
|---|---|---|
| **Sort on** | Past idiosyncratic (or total) volatility | Past beta (previous quarter, daily data) |
| **Average return, low to high** | Above 10% for Q1 to Q3, 6.8%, 0.1%: collapses | About 15% for Q1 to Q4, 12.7%: roughly flat |
| **Sharpe ratio, low to high** | 0.8 to 0.0 | 0.9 to 0.4 |
| **Driver of the falling Sharpe** | Numerator (return) | Denominator (volatility) |
| **Contemporaneous risk** | Still negative | Positive, as the CAPM predicts |
| **Key evidence** | Ang et al. (2006, 2009): robust to many controls, every G7 market | BJS (1972) "too flat"; Fama-French (1992) "no power"; noisy post-ranking betas |
| **Factor** | VOL: target volatility 15%, no risk-free asset needed | BAB: unit-beta legs, levered long |
| **Alpha after FF + UMD** | 0.42% a month, $t = 4.37$ | 0.33% a month, $t = 1.89$ (borderline) |
| **Size tilt** | Large, liquid stocks (SMB $-0.63$) | Small stocks (SMB $+0.29$) |

**Exam traps**

- **Merged anomalies.** High-beta stocks do not earn low raw returns; that is the volatility anomaly's signature.
- **"Both returns and Sharpe ratios fall steadily."** Only the Sharpe ratio is reported as monotonic; returns collapse at the top.
- **Sharpe direction.** Volatility is the Sharpe ratio's denominator; higher volatility with equal or lower return lowers the ratio.
- **Lagged versus contemporaneous beta.** Same-period beta and returns are positively related; the puzzle is predicting future betas.
- **Highly correlated?** BAB and VOL are nearly uncorrelated ($-9\%$): two distinct effects.
- **Size tilts swapped.** Beta anomaly in small stocks, volatility anomaly in large stocks.
- **BAB without leverage.** The long leg is levered; differences in quintile returns are too small to trade directly.
- **Option-implied betas.** They predict future betas *better* and restore a positive risk-return relation.

## Practice questions: IM-3 h

### Question IM-3 h.1

Bank Q139 • AnalystPrep Q.4630 (rebuilt)

Ang, Hodrick, Xing and Zhang (2006), and Ang's later sorts of U.S. stocks, form quintile portfolios by volatility (standard deviation). Which of the following statements best describes the volatility anomaly they document?

**A.** Average returns and Sharpe ratios rise as volatility rises, with the highest-volatility quintile performing best.

**B.** Sharpe ratios fall steadily as volatility rises and average returns collapse in the highest-volatility quintiles, so the lowest-volatility stocks perform best on a risk-adjusted basis.

**C.** Average returns fall as volatility rises, but Sharpe ratios rise because volatility is in the numerator.

**D.** Volatility is related to returns only contemporaneously, and lagged volatility has no relation to risk-adjusted returns.

### Question IM-3 h.2

Bank Q140 • BT P2.T8.704.2 (rebuilt)

Regarding the volatility anomaly and the beta anomaly, each of the following statements is consistent with the empirical evidence EXCEPT:

**A.** Across volatility quintiles, Sharpe ratios fell from about 0.8 to 0.0, and the highest-volatility quintile earned an average return of only about 0.1%.

**B.** Stocks with higher betas earned lower raw returns than stocks with lower betas, which is the essence of the beta anomaly.

**C.** Across beta quintiles, the Sharpe ratio fell from about 0.9 for the lowest-beta stocks to about 0.4 for the highest-beta stocks.

**D.** Higher-beta stocks tend to have higher standard deviations, which is the denominator of the Sharpe ratio.

### Question IM-3 h.3

Bank Q138 • BT notes P2.T9.704.2

The low-risk anomaly is a combination of three true effects. Which of the following statements is FALSE (and not technically included in the low-risk anomaly)?

**A.** Both contemporaneous and lagged volatility are inversely (negatively) related to returns.

**B.** Contemporaneous beta is inversely (negatively) related to raw returns.

**C.** Lagged beta is inversely (negatively) related to risk-adjusted returns.

**D.** Minimum variance portfolios do better than the market.

**Question IM-3 h.4** New • built from GARP Ch.3 (Ang), Beta Anomaly

A quantitative analyst replicates Ang's beta sorts. Each quarter she ranks stocks into quintiles on betas estimated from the previous quarter's daily returns (pre-ranking betas), then measures the portfolios' realised (post-ranking) betas and returns over the following quarter. She also forms portfolios at the end of each quarter on the betas realised during that quarter. Which of the following findings is consistent with the evidence in the reading?

**A.** Post-ranking betas spread as widely as pre-ranking betas, so past betas forecast future betas reliably and estimation noise cannot contribute to the anomaly.

**B.** Portfolios sorted on betas realised over the same quarter show a positive relation between beta and returns, so the puzzle lies largely in predicting future betas from past ones.

**C.** Portfolios sorted on pre-ranking betas show average returns falling steeply from the lowest to the highest quintile, which is what the CAPM predicts for lagged betas.

**D.** Betas estimated from option prices predict future betas worse than past-return betas do, and they confirm a negative relation between beta and returns.

**Question IM-3 h.5** Bank Q141 • BT P2.T9.21.2.3 (rebuilt)

An analyst compares four equity portfolios (risk-free rate 6.0%):

| Portfolio | Standard deviation | Beta |
|---|---|---|
| A | 8% | 0.6 |
| B | 11% | 1.3 |
| C | 15% | 1.0 |
| D | 24% | 1.6 |

Assuming the volatility anomaly and the beta anomaly hold, which portfolio is expected to have the highest Sharpe ratio?

**A.** Portfolio A

**B.** Portfolio B

**C.** Portfolio C

**D.** Portfolio D

**Question IM-3 h.6** New • built from GARP Ch.3 (Ang), Betting-against-Beta and Volatility Factors

The CIO of a sovereign fund is deciding whether to add a betting-against-beta (BAB) factor, a low-minus-high volatility (VOL) factor, or both to the fund's factor benchmark. Her analyst summarises Ang's evidence for October 1963 to December 2011. Which of the following statements in the summary is correct?

**A.** The BAB factor can be built from the lowest and highest of five beta quintiles without leverage, because average returns differ widely across the beta quintiles.

**B.** The BAB and VOL factors are highly correlated, so they capture a single anomaly and adding the second factor would add little.

**C.** After controlling for the Fama-French and momentum factors, the VOL alpha is the more significant, and the volatility anomaly is more pervasive in large, liquid stocks.

**D.** Both factors load negatively on SMB, so both anomalies are concentrated in large stocks, and neither factor is related to value or momentum.

## Solutions • IM-3 h

### Solution IM-3 h.1 — Answer: B

Bank Q139 • AnalystPrep Q.4630

**Step 1 – Average returns across volatility quintiles**

Above 10% for quintiles 1 to 3, 6.8% for quintile 4, 0.1% for the highest-volatility quintile: a collapse at the top.

**Step 2 – Sharpe ratios**

Raw Sharpe ratios fall monotonically from 0.8 to 0.0.

$\Rightarrow$ The least volatile stocks did best on a risk-adjusted basis; the most volatile did worst. B.

**Interpretation**

Option B is reworded from the bank's "average returns and Sharpe ratios fall as volatility rises" to match what GARP reports (see the correction box). The dates 1963 to 2011 are Ang's own sorts, built like the 2006 paper.

**Why the other options are wrong**

**A is incorrect.** States the CAPM-style positive relation, the reverse of the evidence.

**C is incorrect.** Volatility is the Sharpe ratio's denominator, not its numerator; lower returns with higher volatility push the ratio down.

**D is incorrect.** Both lagged (Figure 3.8) and contemporaneous (Figure 3.9) volatility are negatively related to returns.

**LO:** IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Volatility Anomaly (Figures 3.8 and 3.9); SchweserNotes 2026 Book 5, Reading 85, LO 85.h

### Solution IM-3 h.2 — Answer: B

Bank Q140 • BT P2.T8.704.2

**Step 1 – Test A, C and D**

A: the volatility anomaly's numbers (Sharpe 0.8 to 0.0; 0.1% top quintile). C: the beta anomaly's Sharpe ratios. D: the reason high-beta Sharpe ratios fall. All consistent.

**Step 2 – Test B**

Across beta quintiles average returns are roughly flat (about 15%, then 12.7%). "The beta anomaly is not that stocks with high betas have low returns: they don't."

$\Rightarrow$ B misstates the beta anomaly and is the exception.

**Interpretation**

B describes the volatility anomaly's mechanism (raw returns collapsing) under the beta anomaly's label: the classic merge.

**Why the other options are wrong**

**A is incorrect.** Consistent with GARP's Figure 3.8. (Option A is reworded from the bank's "both average returns and Sharpe ratios decreased" to match the reported numbers.)

**C is incorrect.** Consistent: 0.9 to 0.4.

**D is incorrect.** Consistent: high betas come with high volatility.

**LO:** IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Volatility Anomaly and Beta Anomaly (Figures 3.8 and 3.10)

### Solution IM-3 h.3 — Answer: B

Bank Q138 • BT notes P2.T9.704.2

**Step 1 – The three effects**

Volatility negatively related to returns (both lagged and contemporaneous: A true); realised past beta negatively related to risk-adjusted returns (C true); minimum variance portfolios beat the market (D true).

**Step 2 – Contemporaneous beta**

Over the same period, beta and returns are *positively* related, exactly as the CAPM predicts (Figure 3.11).
$\Rightarrow$ B is false.

**Interpretation**

The CAPM's prediction is about contemporaneous betas, and there it holds. The anomaly lives in *lagged* betas, because past betas forecast future betas poorly.

**Why the other options are wrong**

**A is incorrect.** True: Figures 3.8 and 3.9.
**C is incorrect.** True: Sharpe ratios fall from 0.9 to 0.4 across past-beta quintiles.
**D is incorrect.** True: the third effect, a consequence of the first two.

**LO:** IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Contemporaneous Beta and Returns (Figure 3.11)

### Solution IM-3 h.4 — Answer: B

New • built from GARP Ch.3, Beta Anomaly

**Step 1 – Pre- versus post-ranking betas**

Post-ranking betas are much flatter across quintiles than pre-ranking betas: "Betas are noisy!" (though the spread is still over 1.0).

**Step 2 – Contemporaneous sort**

Betas and returns measured over the same quarter are positively related, as the CAPM predicts.
$\Rightarrow$ The real mystery is predicting future betas from past ones: B.

**Interpretation**

Lagged-beta quintiles show roughly flat average returns (about 15%, then 12.7%), not falling ones. Better beta forecasts, from option prices (Buss and Vilkov) or accounting data (Cosemans et al.), give a positive risk-return relation.

**Why the other options are wrong**

**A is incorrect.** Post-ranking betas are much flatter: past betas are poor forecasts of future betas.
**C is incorrect.** Returns across lagged-beta quintiles are roughly flat; and the CAPM makes a contemporaneous prediction, not one about lagged betas.
**D is incorrect.** Option-implied betas predict future betas *better* and yield a *positive* relation.

**LO:** IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Lagged Beta and Future Returns; Contemporaneous Beta and Returns (Figures 3.10 and 3.11)

### Solution IM-3 h.5 Answer: A

Bank Q141 • BT P2.T9.21.2.3

**Step 1 – Volatility anomaly**

Sharpe ratios decline as volatility rises: A has the lowest standard deviation (8%).

**Step 2 – Beta anomaly**

Sharpe ratios decline as beta rises: A has the lowest beta (0.6).
$\Rightarrow$ A ranks best on both.

**Interpretation**

No calculation is needed: both anomalies point the same way. The risk-free rate is a red herring. Note B: lower volatility than C but a higher beta, so the two anomalies would rank B and C differently.

**Why the other options are wrong**

**B is incorrect.** Second-lowest volatility, but a high beta (1.3) that the beta anomaly penalises.
**C is incorrect.** Treats a beta of 1 (the market) as the efficient choice: CAPM reasoning the anomalies contradict.
**D is incorrect.** Highest volatility and beta: the lowest expected Sharpe ratio.

**LO:** IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Volatility Anomaly and Beta Anomaly

### Solution IM-3 h.6 Answer: C

New • built from GARP Ch.3, Betting-against-Beta and Volatility Factors

**Step 1 – Significance**

After the Fama-French and momentum factors: VOL alpha 0.42% a month, $t = 4.37$; BAB 0.33%, $t = 1.89$ (borderline, $p = 0.06$).

**Step 2 – Size tilts**

BAB's SMB loading is $+0.29$ (the beta anomaly shows more in small stocks); VOL's is $-0.63$ (the volatility anomaly is more pervasive in large, liquid stocks).
$\Rightarrow$ C.

**Interpretation**

With a correlation of $-9\%$ the two factors are distinct: Ang's advice is to do both, not to choose.

**Why the other options are wrong**

**A is incorrect.** Return differences across beta quintiles are tiny; BAB scales each leg to unit beta, so the low-beta leg is levered, and only two portfolios are feasible because the lowest quintile's beta is near zero.
**B is incorrect.** The correlation is $-9\%$: distinct anomalies.
**D is incorrect.** BAB loads positively on SMB, and both factors have significant value (HML) and momentum (UMD) tilts.

**LO:** IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Betting against Beta; Volatility Factor; Betting-against-Beta and Volatility Factors (Eq. 3.17 and 3.18)

IM-3 • Alpha (and the Low-Risk Anomaly) • Vardeez: VERY IMPORTANT

# IM-3 i

P2 CONCEPT

Describe potential explanations for the risk anomaly.

Workbook: B4 Factor Theory, Fama-French and Style • Tier B

**WHAT IS TESTED HERE**

**Why P2.** Tier B workbook topic: B4 (Map #18, 5 of 16 sheets) reports the low-risk anomaly "and their explanations" as shape S4.
**Vardeez:** no pointer on this LO; the chapter is VERY IMPORTANT. His trap: leverage constraints (investors who cannot borrow) and agency problems (managers who cannot short or deviate) are different explanations.

**Reported exam shapes**

- **S4 (B4)** The low-risk anomaly: volatility anomaly versus beta anomaly and their explanations. Workbook B4.2 (option D: "leverage constraints fully explain the low-risk anomaly") is worked as Question IM-2 e.6.

The anomaly itself is described in IM-3 a and the evidence in IM-3 h.

**What the questions look like:** statements. Which explanation is on Ang's list, what each one can and cannot explain, and the direction of each constraint (cannot borrow, cannot short, limited tracking error).

## 1. No single explanation

"We are still searching for a comprehensive explanation for the risk anomaly. In my opinion, the true explanation is a combination of all of the explanations [that] follow, plus potentially others being developed." Ang's four: data mining, leverage constraints, agency problems and preferences.

## 2. Data mining

Some papers raise data-mining concerns about the original Ang et al. (2006) results, and there is some sensitivity to portfolio weighting schemes and to illiquidity. But Chen et al. (2012) conclude that "idiosyncratic volatility is a common stock phenomenon", not a microstructure or liquidity bias. **The best argument against data mining is that the effect appears in many other contexts**: recessions and expansions, stable and volatile periods, international stock markets, Treasury bonds, corporate bonds across maturities and ratings, credit derivatives, commodities, foreign exchange and options. "Low risk is pervasive."

## 3. Leverage constraints

Many investors want more risk but cannot borrow (Black, 1972; Frazzini and Pedersen apply it to the beta anomaly). They do the next best thing and buy stocks with "built-in" leverage: high-beta stocks. Their demand bids high-beta stocks up until they are overpriced and deliver low returns, which flattens the security market line.

- **What it explains:** the overpricing of high-beta stocks, and why some low-beta assets have positive alphas.
- **What it does not explain:** the *underpricing* of low-beta stocks relative to the market, so it cannot explain why low-beta or low-volatility assets beat the market portfolio.

- **A prediction that fails:** leverage-constrained institutions should be drawn to high-risk stocks. In reality, stocks with high idiosyncratic volatility are held and traded mostly by retail investors.

## 4. Agency problems

"Many institutional managers can't or won't play the risk anomaly." The use of market-weighted benchmarks may itself cause the anomaly. Take the flat data line of IM-3 a, Figure 1: low-beta stock A has a positive CAPM alpha, high-beta stock B a negative one.

- **Unconstrained investors** buy A and sell B until both alphas disappear; in that world there is no anomaly.
- **A long-only manager with a tracking-error limit** against a cap-weighted index does not buy A: it would add significant tracking error. She cannot short B, so the best she can do is underweight it slightly, and the tracking-error limit caps even that. If the data line slopes slightly upward, she has an incentive to *buy* B, which could beat the market.
- Tracking error is a binding constraint for GM Asset Management and for most institutional asset owners. Changing the benchmark is the obvious fix, but it needs the investment committee and is slow.

**Buffett and low risk.** Frazzini, Kabiller and Pedersen (2012): Buffett's alpha from 1976 to 2011 is 12.5% a year against the Fama-French and momentum benchmark, 11.1% once BAB is added, and 7.0% once a quality factor is added too. Some of his prowess is selecting low-risk stocks; most comes from "ferreting out gems with high underlying quality", true skill unrelated to holding low-volatility stocks (Figure 1).

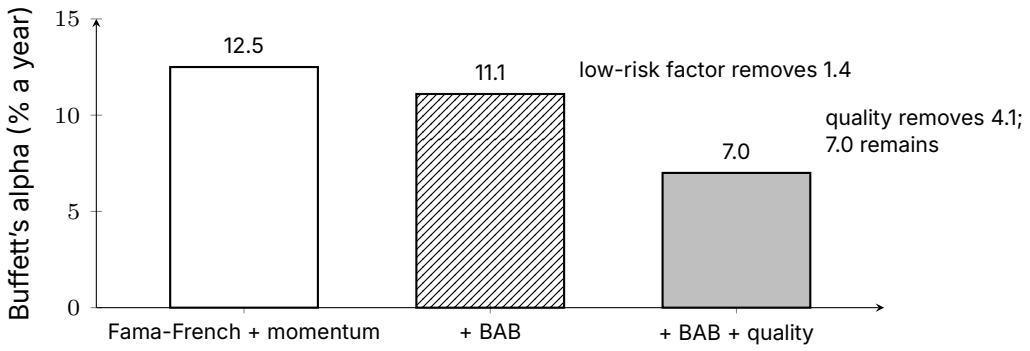



Figure 1. Berkshire Hathaway's alpha, 1976 to 2011, as factors are added to the benchmark (Frazzini, Kabiller and Pedersen, as reported in GARP Ch.3). White: Fama-French plus momentum; hatched: plus betting against beta; grey: plus quality. The low-risk factor explains a slice of Buffett's alpha (1.4 points) and the quality factor more (4.1 points). GARP's reading: some of his edge is selecting low-risk stocks, most of it is finding high-quality firms.

## 5. Preferences

- **"Hopes and dreams".** If asset owners simply prefer high-volatility, high-beta stocks, they bid them up until they earn low returns, and they shun safe, low-volatility, low-beta stocks, leaving them cheap with high returns.
- **Hou and Loh (2012)** test many explanations in three groups: lottery preferences; market frictions including illiquidity; and "other" (uncertainty, short-sale constraints, financial distress, investor inattention, growth options, earnings shocks). **Each explanation alone explains less than one**

**tenth** of the volatility anomaly. As a group, **lottery preferences explain close to half**; close to half remains unexplained.

- **Disagreement plus short-sale limits** (Hong and Sraer, 2012). When disagreement is low and everyone is long-only, the CAPM holds. When disagreement is high, some investors want to short and cannot, so high-beta stocks become overpriced; large enough disagreement makes the beta-return relation slope *downward*.

**Correction to the source notes.** (1) SchweserNotes (LO 85.i) says the leverage-constraint mechanism "works to lower the prices of low-beta stocks and, therefore, results in higher risk-adjusted returns". GARP says the opposite: the leverage-constraint story "does not explain the underpricing of low-beta stocks relative to the market, only the overpricing of high-beta stocks". Workbook B4.2 uses this as its distractor. (2) GARP's sentence that institutions "tend to underweight high-risk stocks" carries a footnote reporting that institutions hold 53.5% high-risk and 46.5% low-risk Russell 1000 stocks, which points the other way (and matches GARP's later remark that institutions appear underweight low-risk stocks). The safe exam reading is the part both support: high idiosyncratic-volatility stocks are held mostly by retail investors, not by leverage-constrained institutions.

| Explanation | Mechanism | Explains | Weakness / evidence |
|---|---|---|---|
| **Data mining** | Anomaly found by searching the data | Sensitivity to weighting and illiquidity | Effect is pervasive across periods, countries and asset classes |
| **Leverage constraints** | Investors who cannot borrow buy high-beta stocks for built-in leverage | Overpriced high-beta stocks; flat SML | Not the underpricing of low-beta stocks; high-IVOL stocks are retail-held |
| **Agency problems** | Cap-weighted benchmarks, tracking-error limits, no shorting | Why institutions do not buy A or short B | Fix (changing benchmarks) is slow |
| **Preferences** | "Hopes and dreams", lottery preferences bid up high-risk stocks | Close to half of the volatility anomaly (as a group) | Each story alone explains less than one tenth |
| **Disagreement + short-sale limits** | Optimists set prices when pessimists cannot short | Downward-sloping beta-return relation when disagreement is high | CAPM holds when disagreement is low |

**Exam traps**

- **Cannot borrow, not highly levered.** The leverage story is about investors who are leverage *constrained*.
- **Presence, not lack, of constraints.** It is the presence of short-sale and tracking-error constraints that stops institutions from arbitraging the anomaly.
- **Leverage constraints are not the whole story.** They explain overpriced high-beta stocks, not underpriced low-beta stocks.
- **Leverage versus agency.** Leverage: investors who cannot borrow. Agency: managers who cannot short or deviate from the benchmark.
- **Data mining direction.** Evidence across many markets argues *against* data mining.
- **Preference polarity.** Investors prefer (and overpay for) high-volatility stocks; they shun safe ones.

- **Hou and Loh scale.** Each explanation alone, less than one tenth; lottery preferences together, close to half.
- **Hong and Sraer condition.** The downward slope needs high disagreement *and* short-sale constraints.

## Practice questions: IM-3 i

### Question IM-3 i.1

Bank Q144 • Schweser Module Quiz 85.3 Q1

Which of the following characteristics is a potential explanation for the risk anomaly?

**A.** Investor preferences.

**B.** The presence of highly leveraged retail investors.

**C.** Lack of short selling constraints for institutional investors.

**D.** Lack of tracking error constraints for institutional investors.

### Question IM-3 i.2

Bank Q143 • BT P2.T8.706.3

Which of the following statements about the risk anomaly is TRUE?

**A.** The primary argument against the risk anomaly, with respect to either low volatility or low beta, is that economists have generally demonstrated that markets are efficient; i.e., the efficient market hypothesis (EMH) is true.

**B.** Due to time-varying reality, it is theoretically intractable (i.e., not possible) to create a reproducible benchmark for either the low beta or low volatility risk anomalies, such that empirical tests of the risk anomalies are not robust and the discussion remains "largely theoretical".

**C.** The risk anomaly might be explained by investors who are leverage constrained (i.e., who cannot borrow so instead bid up high beta stocks) and/or have an "agency problem" created by a need to minimize tracking error with the benchmark (e.g., they cannot short high volatility or high beta stocks).

**D.** The presence of the low-risk anomaly in several different contexts, including U.S. equities, international stock markets, Treasury bonds, corporate bonds, commodity, option and foreign exchange markets, is compelling evidence that "data mining" almost surely creates an illusion of a relationship between idiosyncratic return volatility (IVOL) and future returns.

**Question IM-3 i.3** New • built from GARP Ch.3 (Ang), Leverage Constraints

A research analyst writes that leverage constraints explain the whole low-risk anomaly. The head of research asks which part of the argument holds up against the evidence in the reading. Which of the following is correct?

**A.** Investors who cannot borrow buy high-beta stocks for their built-in leverage, which can explain why high-beta stocks are overpriced but not why low-beta and low-volatility stocks beat the market portfolio.

**B.** Investors who cannot borrow sell high-beta stocks short instead, which makes high-beta stocks underpriced and steepens the security market line relative to the CAPM.

**C.** The story is confirmed by holdings data: leverage-constrained institutions hold most of the stocks with high idiosyncratic volatility, which retail investors avoid.

**D.** Because constrained investors bid up high-beta stocks, the same mechanism fully accounts for the underpricing of low-beta stocks relative to the market portfolio.

**Question IM-3 i.4** New • built from GARP Ch.3 (Ang), Agency Problems (Figure 3.14)

A long-only equity manager is benchmarked to a market-capitalisation-weighted index and has a tight tracking-error limit. Her analyst identifies stock A, a low-beta stock with a positive CAPM alpha, and stock B, a high-beta stock with a negative CAPM alpha. Empirically, average returns are roughly flat across beta. Which of the following best describes how her constraints contribute to the risk anomaly?

**A.** She buys A heavily and shorts B, so her trading should eliminate the anomaly over time, just as an unconstrained investor's would.

**B.** Buying A adds substantial tracking error and she cannot short B, so at most she underweights B slightly; if the data line slopes up slightly, she may even prefer to hold B.

**C.** The tracking-error limit binds only on short positions, so she can overweight A as much as she wishes, and the anomaly persists only because of short-sale bans.

**D.** Because A has a positive alpha, holding more of it reduces her tracking error against the index, so tracking-error limits push institutions towards low-risk stocks.

**Question IM-3 i.5** New • built from GARP Ch.3 (Ang), Preferences

An analyst at a fund of funds is preparing a briefing on why the low-risk anomaly persists. Which of the following statements about preference-based explanations should she include?

**A.** Hou and Loh find that each individual explanation accounts for about half of the volatility anomaly, so any one of them is sufficient on its own.

**B.** Investors with "hopes and dreams" preferences shun high-volatility, high-beta stocks, which makes those stocks cheap and gives them high future returns.

**C.** Taken as a group, lottery preferences are the most promising explanation in Hou and Loh's study, accounting for close to half of the volatility anomaly, with close to half unexplained.

**D.** Hong and Sraer show that when investor disagreement is high and short selling is unrestricted, the relation between beta and returns turns downward sloping.

**Question IM-3 i.6** New • built from GARP Ch.3 (Ang), Agency Problems

A trustee argues that Berkshire Hathaway's long-run outperformance is mostly a low-risk anomaly trade. The CIO cites Frazzini, Kabiller and Pedersen (2012), who measure Buffett's alpha from 1976 to 2011 against several benchmarks: 12.5% a year against the Fama-French and momentum benchmark, 11.1% when a betting-against-beta (BAB) factor is added, and 7.0% when a factor for company quality is also added. Which conclusion is best supported?

**A.** Low-risk factors explain most of Buffett's alpha, since adding BAB removes most of the 12.5%.

**B.** Some of Buffett's outperformance comes from selecting low-risk stocks, but most of it comes from finding high-quality companies, which the reading treats as true skill.

**C.** Because 7.0% of alpha remains after both factors are added, the low-risk factor explains none of Buffett's performance.

**D.** Because adding BAB lowers the alpha, the beta anomaly must be concentrated in large stocks such as Berkshire's holdings rather than in small stocks.

## Solutions • IM-3 i

### Solution IM-3 i.1 — Answer: A

Bank Q144 • Schweser Module Quiz 85.3 Q1

**Step 1 – Ang's list**

Data mining, leverage constraints, agency problems (no shorting, tracking-error limits) and preferences.

**Step 2 – Check each option against the list**

Only A names one of them as stated. The other three reverse a constraint.
⇒ A.

**Interpretation**

A taste for high-volatility, high-beta stocks pushes their prices up and their future returns down.

**Why the other options are wrong**

**B is incorrect.** The story is about investors who are leverage *constrained* (cannot borrow), not highly levered ones.
**C is incorrect.** Polarity flip: it is the *presence* of short-selling constraints that stops institutions from exploiting the mispricing.
**D is incorrect.** Polarity flip: it is the *presence* of tracking-error constraints that keeps institutions away from low-risk stocks.

**LO:** IM-3 i: Describe potential explanations for the risk anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Explanations; SchweserNotes 2026 Book 5, Reading 85, LO 85.i

### Solution IM-3 i.2 — Answer: C

Bank Q143 • BT P2.T8.706.3

**Step 1 – Test C**

Leverage-constrained investors bid up high-beta stocks; institutions bound by tracking error and short-sale limits cannot arbitrage the mispricing. Both are on Ang's list.
⇒ C is true.

**Step 2 – Test the others**

A: no source shows markets are demonstrably efficient; the anomaly is a puzzle relative to the CAPM. B: the anomalies are documented with tradeable factors (BAB, VOL). D: pervasiveness argues *against* data mining.

**Interpretation**

Option C's "cannot short" should be read as being unable to short the expensive high-risk stocks (B in IM-3 a, Figure 1); the bank's original wording ("cannot short low volatility") had the side reversed and is corrected here.

**Why the other options are wrong**

**A is incorrect.** The efficient market hypothesis is not the main argument against the anomaly; Grossman and Stiglitz show perfectly efficient markets cannot exist.
**B is incorrect.** Benchmarks for both anomalies have been built (BAB, VOL) and tested; the evidence is empirical, not just theoretical.
**D is incorrect.** Reversed: evidence across many markets is "the best argument against data mining".

**LO:** IM-3 i: Describe potential explanations for the risk anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Explanations (Data Mining; Leverage Constraints; Agency Problems)

### Solution IM-3 i.3 Answer: A

New • built from GARP Ch.3, Leverage Constraints

#### Step 1 – The mechanism

Investors who cannot borrow hold high-beta stocks for their built-in leverage; demand bids them up until they are overpriced and earn low returns, flattening the SML.

#### Step 2 – Its limits

GARP: the story "does not explain the underpricing of low-beta stocks relative to the market, only the overpricing of high-beta stocks. Thus, it cannot explain why low-beta or low-volatility assets have higher returns than the market portfolio."
$\Rightarrow$ A.

#### Interpretation

It also predicts that leverage-constrained institutions should be drawn to high-risk stocks, while high idiosyncratic-volatility stocks are held mostly by retail investors. Workbook B4.2 uses "fully explain" as its trap.

#### Why the other options are wrong

**B is incorrect.** Investors who cannot borrow buy high-beta stocks; they do not short them, and the SML flattens rather than steepens.
**C is incorrect.** Reversed: high idiosyncratic-volatility stocks are predominantly held and traded by retail investors.
**D is incorrect.** Absolute claim: the mechanism explains overpricing of high-beta stocks, not underpricing of low-beta stocks (see the correction box).

**LO:** IM-3 i: Describe potential explanations for the risk anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Leverage Constraints

### Solution IM-3 i.4 Answer: B

New • built from GARP Ch.3, Agency Problems

#### Step 1 – The unconstrained benchmark case

Unconstrained investors buy A and sell B until both alphas vanish: no anomaly.

#### Step 2 – The constrained manager

Buying A, a low-beta stock, moves her away from the cap-weighted index and adds tracking error. She cannot short B, so at most she avoids or slightly underweights it, and the tracking-error limit caps that too. If the data line is slightly upward sloping, B might outperform the market, so she may even want to hold it.
$\Rightarrow$ B.

#### Interpretation

Market-weighted benchmarks plus tracking-error limits make it hard to bet on low risk or against high risk, so the mispricing survives. Changing the benchmark is the fix, but it is slow.

#### Why the other options are wrong

**A is incorrect.** That is the unconstrained investor of Step 1.
**C is incorrect.** The tracking-error limit also binds on overweighting A; it is not only a short-sale problem.
**D is incorrect.** Reversed: a low-beta stock adds tracking error against a cap-weighted index; alpha does not reduce tracking error.

**LO:** IM-3 i: Describe potential explanations for the risk anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Agency Problems (Figure 3.14)

### Solution IM-3 i.5 Answer: C

New • built from GARP Ch.3, Preferences

#### Step 1 – Hou and Loh (2012)

Individually, each explanation explains less than one tenth of the volatility anomaly. As groups, lottery preferences are the most promising: together they explain close to half; close to half remains unexplained.
$\Rightarrow$ C.

#### Step 2 – Check the other statements

"Hopes and dreams" investors *prefer* high-volatility stocks; Hong and Sraer need short-sale constraints for the downward slope.

#### Interpretation

Preference stories work through prices: overpaying for lottery-like, high-risk stocks lowers their future returns, while unloved safe stocks stay cheap.

#### Why the other options are wrong

**A is incorrect.** Each explanation alone explains less than one tenth.
**B is incorrect.** Polarity flip: these investors bid up high-volatility, high-beta stocks and shun safe ones.
**D is incorrect.** The downward slope arises when disagreement is high and pessimists *cannot* short.

**LO:** IM-3 i: Describe potential explanations for the risk anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Preferences

### Solution IM-3 i.6 Answer: B

New • built from GARP Ch.3, Agency Problems

#### Step 1 – What BAB removes

$12.5\% - 11.1\% = 1.4$ points: a slice, not most, of Buffett's alpha.

#### Step 2 – What quality removes

$11.1\% - 7.0\% = 4.1$ points more.

#### Step 3 – GARP's reading

"Some of Buffett's investing prowess is due to Buffett selecting stocks with low risk, but most of Buffett's investment prowess comes from ferreting out gems with high underlying quality: true skill that is unrelated to just holding low-volatility stocks."
$\Rightarrow$ B.

#### Interpretation

The example sits in GARP's agency-problems discussion: Berkshire is "well known for its ability to go against the crowd and avoid common agency issues", so it can harvest low-risk stocks that benchmarked institutions avoid.

#### Why the other options are wrong

**A is incorrect.** BAB removes only 1.4 of 12.5 points.
**C is incorrect.** Absolute claim: 1.4 points is not "none".
**D is incorrect.** Wrong conclusion and wrong fact: the beta anomaly shows up more in *small* stocks (BAB loads positively on SMB).

**LO:** IM-3 i: Describe potential explanations for the risk anomaly. • **Reference:** Ang, Asset Management, Ch.10 (GARP 2025 Ch.3), 3.5, Agency Problems

IM-4 • Portfolio Construction • Vardeez: VERY IMPORTANT

# IM-4 d

P2 CONCEPT

Explain the implications of transaction costs on portfolio construction.

Workbook: D28 Portfolio Construction and the No-Trade Region • Tier D • notes only

**WHAT IS TESTED HERE**

**Why P2.** Vardeez puts transaction costs first in his focus for IM-4, a VERY IMPORTANT reading. The recall workbook covers the reading in Tier D section D28 (notes only; Map #66, reported on 1 of 16 recall sheets).

**Vardeez:** transaction costs (with the no-trade region, IM-4 g, and the construction techniques, IM-4 h).

**Reported exam shapes**

- **S2** Transaction costs: amortisation over the holding period.

**What the questions look like:** statement questions. Why costs complicate construction (a cost paid at one point in time against alpha earned over a period), the amortisation rule and its direction, what costs do to the trade-off between alpha and active risk and to the precision needed in the alphas, how certain costs are compared with alphas, and what costs mean for revising the portfolio. The tag is CONCEPT, so the rule is tested in words; the numerical version is the no-trade region of IM-4 g.

**Shared background.** Section 1 is the background for the whole reading. IM-4 g and h below and IM-4 b and c (in P3) point back to it.

## 1. The reading on one page (shared background)

Grinold and Kahn define implementation as "the efficient translation of research into portfolios": "Good implementation can't help poor research, but poor implementation can foil good research." The standard objective is active return minus a penalty for active risk, the **value added**, and once costs enter, value added less transaction costs (the figure of merit, Eq. 4.10):

$$\text{VA} = \alpha_P - \lambda_A \psi_P^2 \qquad \text{maximise} \quad \text{VA} - \text{TC}$$

Here $\alpha_P$ is the portfolio's alpha (expected active return), $\psi_P$ its active risk (tracking error) and $\lambda_A$ the active risk aversion. Five inputs feed the problem (Figure 1). **Only the current portfolio is measured with near certainty.** The alphas are "often unreasonable and subject to hidden biases". Covariances and transaction cost estimates are noisy: "we hope that they are unbiased, but we know that they are not measured with certainty". Even the risk aversion is uncertain, and it must be made consistent with the manager's target active risk ($\lambda_A = \text{IR}/2\psi_P$, IM-4 e). Hence the theme of the reading: "Many of the procedures used in portfolio construction are, in fact, indirect methods of coping with noisy data." Grinold and Kahn prefer a direct attack: refine the alphas (IM-4 b, c), price the costs (IM-4 d, g) and choose the technique (IM-4 h).

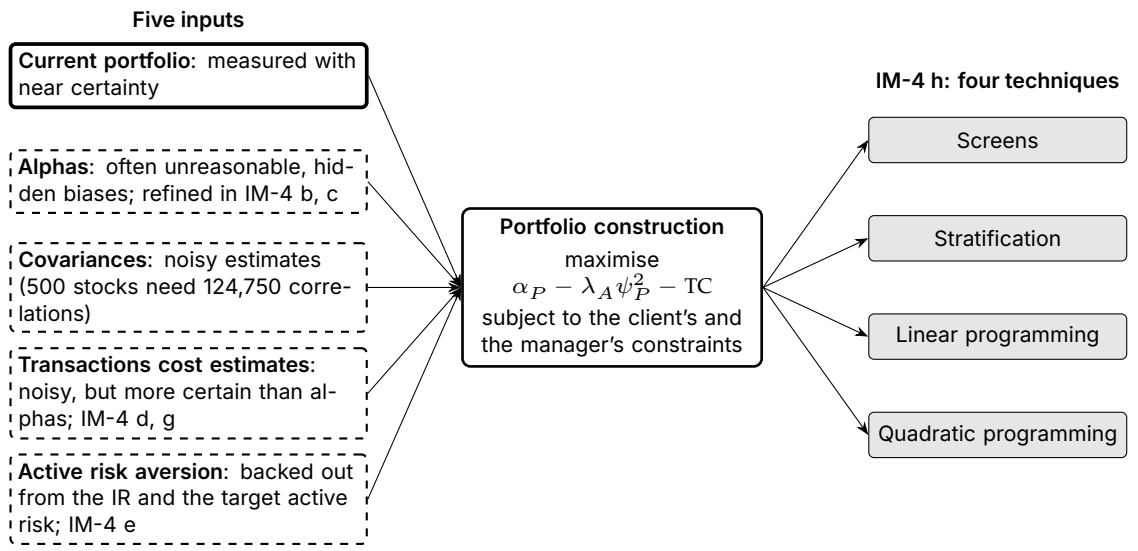



Figure 1. The five inputs and the four techniques of the reading. Thick solid border: the one input measured with near certainty. Dashed borders: estimated inputs, all subject to error. Grey boxes: the generic construction techniques of IM-4 h. Redrawn from the sketch in Vardeez's notes, with the GARP labels.

## 2. What transaction costs do to the problem

- **The third rubber chicken.** "Up to this point, the struggle has been between alpha and active risk. Any klutz can juggle two rubber chickens." Transaction costs, "the cost of moving from one portfolio to another", are the third. They are paid when the portfolio is built from cash, when it is rebalanced and when it is liquidated.
- **One dimension becomes two.** Without costs, a wrong scale in the alphas can be offset by turning one knob: raise or lower the active risk aversion. With costs there are two trade-offs, alpha against active risk and alpha against costs, and one knob cannot fix both. So "transactions costs force greater precision on our estimates of alpha": the scale must be right (IM-4 b).
- **A point in time against a period.** Costs are paid on the trade date; alpha and active risk are annual rates over the holding period. "We must amortize the transactions costs to compare them to the annual rate of gain from the alpha and the annual rate of loss from the active risk." The rate of amortisation depends on the anticipated holding period (Section 3).
- **More certain than alphas, not certain.** "It has been said that accurate estimation of transactions costs is just as important as accurate forecasts of exceptional return. That is an overstatement": returns are much less certain than costs, so perfect information about returns is worth much more than perfect information about costs. The remark "does point out the crucial role transactions costs play". Schweser: costs are uncertain, "although typically less so than alphas".
- **Revisions (IM-4 f, g).** With zero costs, revise whenever new information arrives. With costs, small changes in alpha are not worth a trade: the no-trade region of IM-4 g. A manager who underestimates costs, makes large changes in alpha estimates very frequently and revises daily will "churn the portfolio and suffer higher than expected transactions costs and lower than expected alpha"; "a crude but effective cure is to revise the portfolio less frequently". Short-horizon alphas are noisier, but "the transactions costs stay the same, whether we are reacting to signal or noise".
- **Elsewhere in the reading.** Costs make some dispersion across separate accounts optimal (IM-4 i), and the techniques of IM-4 h are judged on how they handle alpha, risk and costs together.

## 3. Amortisation over the holding period (shape S2)

$$\text{Annualised transactions cost} = \frac{\text{round-trip cost}}{\text{holding period in years}}$$

**GARP's example.** Perfect certainty, a zero risk-free rate, start and end in cash. Two stocks at USD 100, each costing USD 0.75 to buy or to sell, so a *round trip* costs USD 1.50 (1.5%). Stock 1 rises to USD 102 in six months and then stays there; the manager can find another stock like it every six months. Stock 2 rises to USD 108 over 24 months. Both earn an annual alpha of 4%.

| | Stock 1 and successors | Stock 2 |
|---|---|---|
| Holding period | 6 months (four in a row) | 24 months |
| Costs over two years (USD) | $0.75 + 1.50 + 1.50 + 1.50 + 0.75 = 6.00$ | $0.75 + 0.75 = 1.50$ |
| Gain on the shares (USD) | $4 \times 2 = 8.00$ | 8.00 |
| Profit over two years (USD) | 2.00 | 6.50 |
| Annual alpha | 4% | 4% |
| Annualised cost, round trip ÷ years | $1.5\%/0.5 = 3.00\%$ | $1.5\%/2 = 0.75\%$ |
| Annual return after costs | 1.00% | 3.25% |

"For a 6-month holding period, we double the round-trip transactions cost to get the annual transactions cost, and for a 24-month holding period, we halve the round-trip transactions cost." Schweser's version uses a 1% round trip: two 2% six-month trades net about $(2\% - 1\%) \times 2 = 2\%$ a year, a single 4% one-year trade nets $4\% - 1\% = 3\%$. Holding periods are uncertain and differ across holdings, so the horizon cannot be separated from the trade-off between alpha, risk and costs (IM-4 f). In the no-trade region, GARP "subsumes the amortization horizon into the costs", so the purchase and sales costs there are already annual rates (IM-4 g).

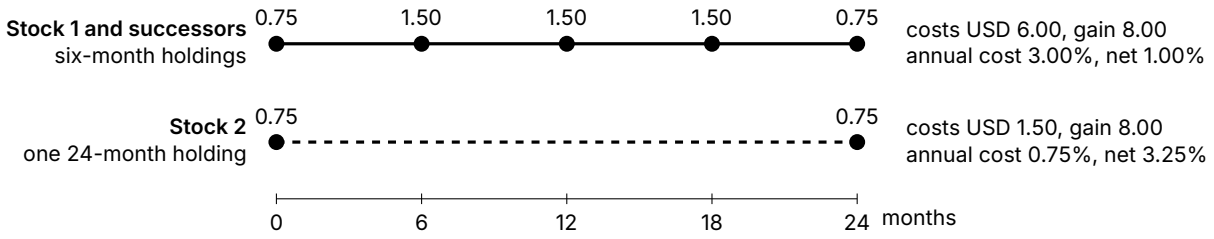



Figure 2. GARP's two-stock example. Dots are trade dates, labelled with the cost paid (USD, on a USD 100 position). The six-month series (solid line) pays a round trip every six months; the single holding (dashed line) pays one round trip in two years. Same 4% annual alpha, four times the annual cost.

| | Alphas | Transaction costs |
|---|---|---|
| **What it is** | Forecast of exceptional return, earned over the holding period | Cost of moving from one portfolio to another, paid on the trade date |
| **How certain** | Much less certain; "often unreasonable and subject to hidden biases" | More certain than alphas, but still a noisy estimate |
| **Value of perfect information** | Much higher | Lower: "an overstatement" to call them equally important |
| **How it is handled** | Refined: neutralise, scale, trim (IM-4 b, c) | Amortised: round trip ÷ holding period in years |
| **Effect on construction** | Alpha against risk alone is one knob ($\lambda_A$) | Adds a second trade-off; forces a precise alpha scale; creates a no-trade region (IM-4 g) |

**Exam traps**

- **Multiplying instead of dividing.** The annual cost is the round-trip cost *divided* by the holding period in years: a six-month holding doubles it, a 24-month holding halves it.
- **One-way versus round trip.** GARP's USD 0.75 is per trade; the round trip is USD 1.50. A position built from cash and sold later pays both legs.
- **Comparing unlike quantities.** A one-off cost set against an annual alpha without amortising it, or every cost amortised over one year whatever the holding period.
- **Polarity of certainty.** Costs are *more* certain than alphas. "Costs are uncertain while the benefits are relatively certain" is the reverse.
- **One knob.** Changing the risk aversion can absorb mis-scaled alphas only when there are no costs; with costs the alpha scale itself must be right.
- **Revision frequency.** Underestimating costs leads to trading too *often*, not too rarely. Zero costs mean revising on every piece of new information.

## Practice questions: IM-4 d

### Question IM-4 d.1

Bank Q161 • Schweser Module Quiz 86.1 Q3

A trainee on a quantitative equity desk asks why the optimizer cannot simply subtract the cost of each trade from the stock's annual alpha. Which of the following statements most correctly describes a consideration that complicates the incorporation of transaction costs into the portfolio construction process?

**A.** The transaction costs must be amortized over the horizon of the benefit from the trade.

**B.** The transaction costs are uncertain, while the benefits are relatively certain.

**C.** The transaction costs and the benefits always occur in two distinct time periods.

**D.** There are no complicating factors from the introduction of transaction costs.

### Question IM-4 d.2

Bank Q160 • AP Mock 1 Q61 • AnalystPrep Q.2446

A portfolio manager makes the following comments about transaction costs in portfolio construction:

I. The correct way to compare transaction costs with the annual rate of gain from alpha and the annual rate of loss from active risk is to amortize the transaction costs, with the rate of amortization depending on the anticipated holding period.

II. The annualized transaction cost is given by the round-trip cost divided by the holding period in years.

Which of the comments is/are correct?

**A.** Only I is correct.

**B.** Only II is correct.

**C.** Both I and II are correct.

**D.** Neither I nor II is correct.

**Question IM-4 d.3** New • built from GARP Ch.4, 4.4 • S2

An analyst presents Grinold and Kahn's two-stock example to an investment committee. Both stocks cost USD 100, and buying or selling either stock costs USD 0.75 per trade. Stock 1 rises to USD 102 over six months and then stays flat; the manager expects to find a similar stock every six months and to switch into it. Stock 2 rises to USD 108 over 24 months. Both positions earn an annual alpha of 4%, the risk-free rate is zero, and the manager starts and ends in cash. Which of the following statements is correct?

**A.** Because both positions earn the same 4% annual alpha, they deliver the same annual return after transaction costs.

**B.** The six-month series pays a round-trip cost every six months, so its annualized transaction cost is four times that of the single 24-month holding.

**C.** The 24-month holding carries the higher annualized cost, because the round-trip cost is multiplied by the longer holding period.

**D.** Transaction costs should be amortized over one year for every position, so the two positions carry the same annualized cost.

**Question IM-4 d.4** New • built from GARP Ch.4, 4.4 (and note 5)

A consultant tells a pension fund's investment committee: "Accurate estimation of transaction costs is just as important as accurate forecasts of exceptional return." Based on Grinold and Kahn, which of the following is the most appropriate assessment of the consultant's statement?

**A.** Correct: transaction costs are as uncertain as alphas, so perfect information about either input is equally valuable.

**B.** An understatement: transaction costs are the least certain input to portfolio construction, so they matter more than the alphas.

**C.** Irrelevant: transaction costs are paid once, when a position is opened, so they can be left out of portfolio construction.

**D.** An overstatement: returns are much less certain than transaction costs, so perfect information about returns is much more valuable, although costs still play a crucial role.

**Question IM-4 d.5** New • built from GARP Ch.4, 4.4

The CIO of a quantitative equity fund finds that the alphas sent to the optimizer have about twice their proper scale. Before transaction costs were added to the optimizer, the team offset the problem by raising the active risk aversion. The optimizer now includes transaction costs. Which of the following statements is correct?

**A.** Raising the risk aversion can restore the trade-off between alpha and active risk, but not the trade-off between alpha and transaction costs, so the alphas themselves must be rescaled.

**B.** Raising the risk aversion still corrects the problem fully, because transaction costs act as one more component of active risk.

**C.** The fund should lower the risk aversion, because transaction costs already penalize the large positions that overscaled alphas create.

**D.** The scale of the alphas no longer matters, because trades are now driven by transaction costs rather than by alphas.

**Question IM-4 d.6** Bank Q166 • bank-written (GARP style) • re-filed from IM-4 f

The risk committee of an asset manager is reviewing the rebalancing policy of a quantitative equity portfolio. The portfolio manager revises holdings using updated alpha forecasts, estimated active risk and estimated transaction costs, all of which are subject to estimation error. Which of the following statements about the manager's revision and rebalancing decisions is correct?

**A.** Underestimating transaction costs will cause the manager to trade too infrequently, so the portfolio will drift away from its optimal holdings.

**B.** Shortening the horizon over which alphas are forecast increases the certainty of those alphas, which justifies more frequent rebalancing.

**C.** Because alpha, risk and transaction cost estimates are uncertain, the manager should trade more aggressively to capture alpha before the forecasts decay.

**D.** If transaction costs were zero, the manager should revise the portfolio every time new information about alphas arrives.

## Solutions • IM-4 d

### Solution IM-4 d.1 — Answer: A

Bank Q161 • Schweser Module Quiz 86.1 Q3

**Step 1 – When the cost and the benefit arrive**

A trade's cost is paid once, on the trade date. Its benefit, the extra alpha, accrues as an annual rate over however long the position is held.

⇒ A one-off cost cannot be netted against an annual rate as it stands.

**Step 2 – What makes them comparable**

Spread the cost over the horizon of the benefit: amortise it. The annual cost is the round-trip cost divided by the holding period in years, so it depends on a holding period that is itself uncertain.

⇒ The complication is the amortisation, and it is why the horizon enters the trade-off (IM-4 f).

**Interpretation**

This is the reason the trainee's shortcut fails: subtracting the full cost from one year's alpha is right only for a one-year holding. A six-month position must bear twice the round trip per year; a two-year position half of it.

**Why the other options are wrong**

**B is incorrect.** Polarity flip: the alphas are the noisy input; transaction costs are comparatively well estimated ("returns are much less certain than the transactions costs").

**C is incorrect.** Absolute qualifier: the issue is not that costs and benefits "always" fall in distinct periods, but that a lump-sum cost meets a stream of benefits.

**D is incorrect.** Costs turn a one-dimensional problem into a two-dimensional one; that is a real complication.

**LO:** IM-4 d: Explain the implications of transaction costs on portfolio construction. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.4 Transactions Costs; SchweserNotes 2026 Book 5, Reading 86, LO 86.d

### Solution IM-4 d.2 — Answer: C

Bank Q160 • AP Mock 1 Q61

**Step 1 – Statement I: the principle**

Alpha and active risk are annual rates; costs are incurred at a point in time. GARP: "We must amortize the transactions costs to compare them to the annual rate of gain from the alpha and the annual rate of loss from the active risk. The rate of amortization will depend on the anticipated holding period."

⇒ I is correct.

**Step 2 – Statement II: the rule**

"The annualized transactions cost is the round-trip cost divided by the holding period in years." A 1% round trip held six months costs 2% a year; held two years, 0.5% a year.

⇒ II is correct.

**Interpretation**

I is the reason and II is the arithmetic that carries it out. GARP states both, in this order, in Section 4.4.

**Why the other options are wrong**

**A is incorrect.** Accepts the principle but rejects the rule that implements it.

**B is incorrect.** Accepts the rule but rejects the reason for it; the rule exists only because costs

must be amortised.
**D is incorrect.** Would hold only if a one-off cost could be set against an annual alpha without adjustment.

**LO:** IM-4 d: Explain the implications of transaction costs on portfolio construction. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.4 Transactions Costs

### Solution IM-4 d.3 — Answer: B

New • built from GARP Ch.4, 4.4

#### Step 1 – Count the round trips

The six-month series buys at 0 and sells at 24 months, and switches at 6, 12 and 18 months: four round trips in two years, costing USD 0.75, 1.50, 1.50, 1.50 and 0.75, a total of USD 6.00. The 24-month holding makes one round trip, USD 1.50.
$\Rightarrow$ Four times the cost for the same two-year period.

#### Step 2 – Apply the amortisation rule

Round trip 1.5%. Six-month holding: $1.5\%/0.5 = 3\%$ a year. 24-month holding: $1.5\%/2 = 0.75\%$ a year.
$\Rightarrow$ The ratio is $3/0.75 = 4$; dividing by the holding period, not multiplying, is what produces it.

#### Interpretation

Same annual alpha, very different net returns: 1% a year for the six-month series against 3.25% for the single holding. Short-horizon alpha has to clear a much higher annual cost hurdle, which is the bridge to the revision trade-offs of IM-4 f.

#### Why the other options are wrong

**A is incorrect.** Ignores the costs: equal alphas give equal returns only before costs (1.00% against 3.25% after).
**C is incorrect.** Multiplies by the holding period instead of dividing: the longer holding has the lower annual cost.
**D is incorrect.** Amortises every cost over one year; the rate of amortisation must follow the anticipated holding period.

**LO:** IM-4 d: Explain the implications of transaction costs on portfolio construction. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.4 Transactions Costs (the two-stock example)

### Solution IM-4 d.4 — Answer: D

New • built from GARP Ch.4, 4.4

#### Step 1 – Recall what GARP says about the claim

"It has been said that accurate estimation of transactions costs is just as important as accurate forecasts of exceptional return. That is an overstatement, but it does point out the crucial role transactions costs play."

#### Step 2 – The reason (note 5)

"Perfect information regarding returns is much more valuable than perfect information regarding transactions costs. The returns are much less certain than the transactions costs."
$\Rightarrow$ Overstatement, yet costs still matter: they add a second trade-off and force precision on the alphas.

#### Interpretation

The asymmetry in certainty is why costs are treated as an input to the optimiser ("we will assume that we know the cost for each anticipated trade"), while the alphas need a whole procedure of

refinement (IM-4 b, c).

### Why the other options are wrong

**A is incorrect.** Wrong on certainty: alphas are much less certain than costs, so perfect alpha information is worth more.

**B is incorrect.** Polarity flip: costs are not the least certain input; alphas are the least reliable.

**C is incorrect.** Costs are paid on construction, rebalancing and liquidation and must be amortised into the objective; they cannot be dropped.

**LO:** IM-4 d: Explain the implications of transaction costs on portfolio construction. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.4 Transactions Costs and note 5; SchweserNotes 2026 Book 5, Reading 86, LO 86.d

## Solution IM-4 d.5 Answer: A

New • built from GARP Ch.4, 4.4

### Step 1 – Without costs: one knob

"Finding the correct trade-off between alpha and active risk is a one-dimensional problem. By turning a single knob, we can find the right balance." Doubling the alpha scale and doubling $\lambda_A$ leave the optimal active positions unchanged.

### Step 2 – With costs: two trade-offs

"Transactions costs make this a two-dimensional problem." Alpha must now also be traded off against costs. Overscaled alphas make every trade look twice as valuable relative to its cost, and the risk aversion does not touch that comparison.

$\Rightarrow$ The manager overtrades unless the alphas themselves are rescaled. "We therefore must be precise in our choice of scale."

### Interpretation

This is one of the two implications Schweser lists for LO 86.d: costs "increase the importance of both precision in estimating alphas and of the choice of scale". Rescaling is the IM-4 b procedure: alpha scale = volatility $\times$ IC.

### Why the other options are wrong

**B is incorrect.** Costs are not a component of active risk; the alpha-against-cost comparison is a separate dimension.

**C is incorrect.** Polarity flip, and still the wrong tool: a lower risk aversion would enlarge positions further, and the costs do not correct the alpha scale.

**D is incorrect.** Costs make the scale *more* important, not irrelevant: "transactions costs force greater precision on our estimates of alpha".

**LO:** IM-4 d: Explain the implications of transaction costs on portfolio construction. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.4 Transactions Costs; SchweserNotes 2026 Book 5, Reading 86, LO 86.d

## Solution IM-4 d.6 Answer: D

Bank Q166 • bank-written (GARP style)

### Step 1 – The frictionless benchmark

"How often should you revise your portfolio? Whenever you receive new information. That's the short answer." With zero costs, acting on new information is free.

$\Rightarrow$ D is correct.

### Step 2 – Test the other statements against the reading

Costs, and the error in estimating them, are what create a reason not to trade. Underestimating costs makes trades look cheap, so the manager churns; uncertain inputs argue for less frequent revision, not more; shorter horizons make alphas noisier, while "the transactions costs stay the same, whether we are reacting to signal or noise".

### Interpretation

The implication of costs for revisions: a frictionless manager trades on every signal; a manager facing costs trades only when the expected gain clears the amortised cost, which is the no-trade region of IM-4 g. "A crude but effective cure is to revise the portfolio less frequently."

### Why the other options are wrong

**A is incorrect.** Polarity flip: underestimated costs make the manager trade too *often* ("churn the portfolio").

**B is incorrect.** Reverses the horizon effect: shorter-horizon alphas "contain larger amounts of noise".

**C is incorrect.** Uncertain inputs call for a more conservative policy, possibly less frequent revision as a safeguard.

**LO:** IM-4 d: Explain the implications of transaction costs on portfolio construction; IM-4 f: Describe portfolio revisions and rebalancing, and analyze the tradeoffs between alpha, risk, transaction costs, and time horizon. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.6 Portfolio Revisions

IM-4 • Portfolio Construction • Vardeez: VERY IMPORTANT

# IM-4 g

P2 BOTH

Determine the optimal no-trade region for rebalancing with transaction costs.

Workbook: D28 Portfolio Construction and the No-Trade Region • Tier D • notes only

**WHAT IS TESTED HERE**

**Why P2.** Vardeez puts the no-trade region in his focus for IM-4, a VERY IMPORTANT reading ("Transaction Costs and No-Trade Region"), and the workbook section D28 is named after it (Tier D, notes only, Map #66).
**Vardeez:** no-trade region.

**Reported exam shapes**

- **S3** The optimal no-trade region for MCVA and for alpha, and Leland's band around a target allocation.

**What the questions look like:** a mix. Numerical: given $\alpha_n$, $\lambda_A$, $\psi_P$, $\text{MCAR}_n$ and the purchase and sales costs, compute the MCVA and decide to buy, sell or hold; locate the band for alpha; find the net benefit of a trade or how far alpha must move before a trade pays. Concept: the band's polarity and asymmetry, what sets its width and what sets its location, and Leland's "trade back to the boundary" rule. The costs in the band are the amortised costs of IM-4 d (Section 3 there); the objective and inputs are in IM-4 d, Section 1.

## 1. Marginal contribution to value added

Start from value added, $\text{VA} = \alpha_P - \lambda_A \psi_P^2$. Raise the holding of stock $n$ a little, with an offsetting decrease in cash. Portfolio alpha rises at the rate $\alpha_n$ (cash has no alpha); active risk rises at the rate $\text{MCAR}_n$, so the penalty $\lambda_A \psi_P^2$ rises at $2\lambda_A \psi_P \, \text{MCAR}_n$:

$$\boxed{\text{MCVA}_n = \alpha_n - 2\,\lambda_A\,\psi_P\,\text{MCAR}_n} \qquad \text{(Eq. 4.7)}$$

- **MCVA** "shows how value added, as measured by risk-adjusted alpha, changes as the holding of the stock is increased, with an offsetting decrease in cash". **MCAR** "measures the rate at which active risk changes as we add more of stock $n$". Call $2\lambda_A\psi_P\text{MCAR}_n$ the *risk charge*.
- **Units.** GARP works in percent: $\psi_P$ in percent and $\lambda_A$ per percent ($\lambda_A = \text{IR}/2\psi_P$: IR 0.5 and 5% active risk give 0.05), so MCVA is in percent per year.
- **No costs.** At the optimum (no limits on holdings or cash, no other constraints) every $\text{MCVA}_n = 0$: this is the first-order condition $\alpha = 2\lambda_A V h_{PA}$ behind the modified alphas of IM-4 b.

## 2. The no-trade region for MCVA

With costs, an optimal portfolio cannot have a stock whose MCVA exceeds its purchase cost (buying would pay, net $\text{MCVA}_n - \text{PC}_n$) or falls below minus its sales cost (selling would pay, net $-\text{MCVA}_n - \text{SC}_n$). So

$$\boxed{-\text{SC}_n \;\le\; \text{MCVA}_n \;\le\; \text{PC}_n \quad\Longrightarrow\quad \text{do not trade}} \qquad \text{(Eq. 4.8)}$$

**GARP's numbers.** $\text{PC}_n = 0.50\%$, $\text{SC}_n = 0.75\%$. An MCVA of $+0.80\%$ exceeds the purchase cost: buying yields a net benefit of $0.80\% - 0.50\% = 0.30\%$. An MCVA of $-1.30\%$ is below $-0.75\%$: cutting the holding saves 1.30% at the margin for a cost of 0.75%, a net benefit of 0.55%.

- **Polarity.** Trade only when MCVA leaves the band. "As long as the alpha stays within that band, the portfolio will remain optimal, and we should not react to new information." The intuitive answer, that any new information warrants a trade, is the wrong one.
- **Width and asymmetry.** The bandwidth is $\text{PC}_n + \text{SC}_n = 1.25\%$. The band is not centred on zero when the costs differ: here a larger negative MCVA is tolerated than a positive one.
- **Just traded.** After a purchase, $\text{MCVA}_n = \text{PC}_n$: the stock sits at the top of the band. Any rise in alpha leads to further purchases, but "the alpha would have to decrease by 1.25 percent before we would consider selling". After a sale it sits at $-\text{SC}_n$.
- **Amortised costs.** "This analysis has simplified the problem by subsuming the amortization horizon into the costs SC and PC." MCVA is an annual rate, so $\text{PC}_n$ and $\text{SC}_n$ are annualised costs (IM-4 d).

## 3. The same band, written for alpha

Substitute Eq. 4.7 into Eq. 4.8:

$$\boxed{2\lambda_A\psi_P\text{MCAR}_n - \text{SC}_n \;\le\; \alpha_n \;\le\; 2\lambda_A\psi_P\text{MCAR}_n + \text{PC}_n} \qquad \text{(Eq. 4.9)}$$

- **Costs set the width; risk sets the location.** The alpha band is still $\text{PC}_n + \text{SC}_n$ wide. The risk charge moves it: a higher $\lambda_A$, $\psi_P$ or (positive) $\text{MCAR}_n$ shifts the whole band up, so a larger alpha is needed to buy and a sale is triggered at a higher alpha.
- **Example.** $\text{PC}_n = 0.50\%$, $\text{SC}_n = 0.75\%$, $\lambda_A = 0.05$, $\psi_P = 5\%$, $\text{MCAR}_n = 0.6$. Risk charge $2(0.05)(5)(0.6) = 0.30\%$. Alpha band: $0.30 - 0.75 = -0.45\%$ to $0.30 + 0.50 = 0.80\%$. Buy only if the new alpha exceeds 0.80%; sell only if it falls below $-0.45\%$. Doubling $\lambda_A$ doubles the charge to 0.60%: band $-0.15\%$ to $1.10\%$, still 1.25% wide.

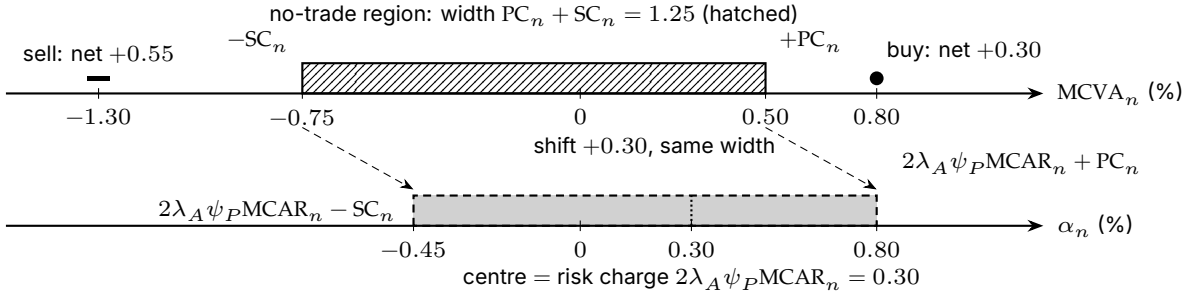



Figure 1. Top: the no-trade band for MCVA with GARP's costs (purchase 0.50%, sale 0.75%), hatched; the dot (buy) and the square (sell) are GARP's two cases. Bottom: the same band written for alpha in the Section 3 example, grey with a dashed border. It is the MCVA band moved right by the risk charge of 0.30 (dashed arrows); the dotted line marks its centre. Costs fix the width, risk fixes the location.

**Correction to the source notes.** SchweserNotes (LO 86.f/g) says "the size of the no-trade region is determined by transaction costs, risk aversion, alpha, and the riskiness of the assets", and its Module Quiz 86.2 Q1 keys that an increase in risk aversion or in MCAR "will increase the no-trade region for the alpha". In Eqs. 4.8 and 4.9 the width is $\text{PC}_n + \text{SC}_n$ whatever the risk inputs (GARP: "the bandwidth is the total of the sale plus purchase costs"); risk aversion, active risk and MCAR only move the alpha band. The sentence belongs to Leland's dynamic allocation problem (Section 4), where the size of the region does depend on risk aversion and on the assets' expected returns and risks. The bank's rebuilt version of the quiz item is Question IM-4 g.6.

## 4. Leland: a band around a target allocation

Leland (1996) solves the rebalancing problem around an optimal stock/bond mix, say 60/40, with linear costs and a utility that penalises active variance (relative to the target) and costs over time.

- The optimal strategy has a **no-trade region around 60/40**. Inside it, do nothing. Outside it, **trade back to the boundary, not to the target**.
- "Trading only to the boundary, not to the target allocation, cuts the turnover and transactions costs roughly in half, with effectively no change in risk over time."
- The **size** of the region "depends on the transactions costs, the risk aversion, and the expected return and risk of stocks and bonds"; changing it changes the turnover.
- The problem is simpler than active management: one-dimensional, a static target, no flow of new information. Its value is the rule it motivates: rebalance when the portfolio falls outside boundaries, not on the calendar (monthly or quarterly).
- **Information horizon.** Applying rules like Eq. 4.9 to one position over an indefinite future, the after-cost information ratio falls as the one-way cost rises and as the half-life of the signals shortens. Two effects: we pay the costs, and the costs make us less eager to trade: "we lose by intimidation".

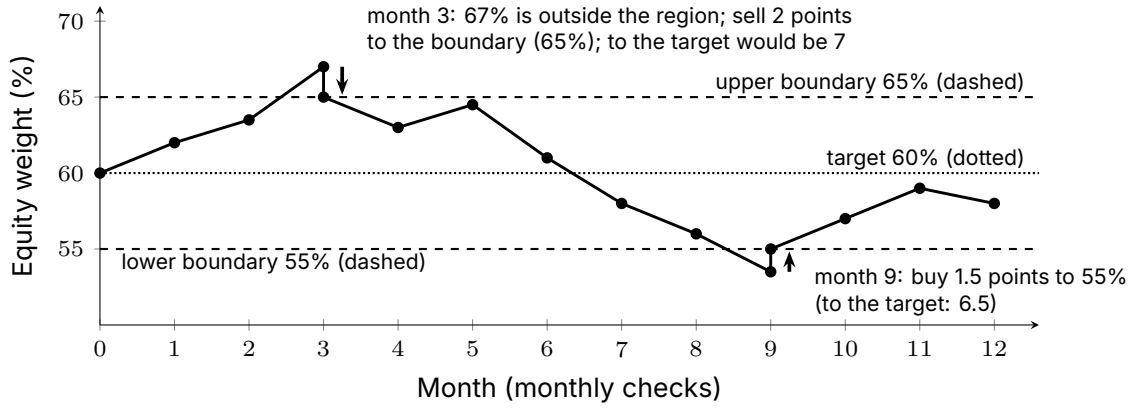



Figure 2. Leland's rule on an illustrative path (solid line with dots) around a 60/40 target with a 55% to 65% equity region (the region's size is assumed, not GARP's). Inside the dashed boundaries nothing is traded; outside, the thick arrows trade back to the nearer boundary. At the two crossings the rule trades 2 and 1.5 points; trading back to the target at the same dates would have traded 7 and 6.5.

**Formula card: IM-4 g**

| | |
|---|---|
| Value added | $\text{VA} = \alpha_P - \lambda_A \psi_P^2$; construction maximises VA − TC |
| Marginal value added | $\text{MCVA}_n = \alpha_n - 2\lambda_A \psi_P \, \text{MCAR}_n$ (holding up, cash down) |
| No-trade band, MCVA | $-\text{SC}_n \le \text{MCVA}_n \le \text{PC}_n$: hold; above $\text{PC}_n$ buy; below $-\text{SC}_n$ sell |
| No-trade band, alpha | $2\lambda_A \psi_P \text{MCAR}_n - \text{SC}_n \le \alpha_n \le 2\lambda_A \psi_P \text{MCAR}_n + \text{PC}_n$ |
| Width and location | width $\text{PC}_n + \text{SC}_n$ in both forms; the risk charge $2\lambda_A \psi_P \text{MCAR}_n$ sets the alpha band's centre |
| Net benefit of a trade | buy: $\text{MCVA}_n - \text{PC}_n$; sell: $-\text{MCVA}_n - \text{SC}_n$ |
| Costs in the band | annualised: round trip ÷ holding period in years (IM-4 d) |
| Risk aversion | $\lambda_A = \text{IR}/2\psi_P$ with $\psi_P$ in percent (IM-4 e) |
| Leland | outside the region, trade to the boundary, not the target: about half the turnover, same risk |

| | Stock-level band (Eqs. 4.8, 4.9) | Leland's allocation band |
|---|---|---|
| **Problem** | Many stocks, one period, new alphas arriving | One stock/bond mix over many periods, static target |
| **Test** | MCVA against $-\text{SC}_n$ and $\text{PC}_n$ | Current allocation against the region's boundaries |
| **Width set by** | Purchase plus sales cost only | Costs, risk aversion, expected returns and risks |
| **Outside the band** | Trade until MCVA is back at the edge of the band | Trade back to the boundary, not to the target |

**Exam traps**

- **Polarity.** Trade when MCVA is *outside* the band; new information that leaves MCVA inside it is ignored.
- **Swapped or symmetric ends.** The band is $[-\text{SC}_n, +\text{PC}_n]$: the purchase cost is the buy hurdle, the sales cost the sell hurdle. Using either cost on both sides, or swapping them, is the standard corruption.
- **Negative is not enough.** A negative MCVA justifies a sale only below $-\text{SC}_n$; a positive one justifies a purchase only above $\text{PC}_n$.
- **Gross for net.** The benefit of a purchase is $\text{MCVA}_n - \text{PC}_n$, not $\text{MCVA}_n$; of a sale, $|\text{MCVA}_n| - \text{SC}_n$.
- **Width versus location.** Risk aversion, active risk and MCAR shift the alpha band; they do not widen it. Doubling $\lambda_A$ doubles the risk charge.
- **Risk charge.** It is $2\lambda_A \psi_P \text{MCAR}_n$, subtracted from alpha: dropping the 2, or adding the charge, gives the wrong MCVA.
- **Unamortised costs.** The band compares annual rates; a round-trip cost must first be annualised over the holding period.
- **Leland.** Trade back to the boundary of the region, not to the 60/40 target, and on the boundary rule, not the calendar.

## Practice questions: IM-4 g

### Question IM-4 g.1

New • built from GARP Ch.4, 4.6 • S3

A risk analyst reviews the rebalancing rule of an equity optimizer. For each stock $n$, the optimizer compares the marginal contribution to value added, $\text{MCVA}_n = \alpha_n - 2\lambda_A\psi_P\,\text{MCAR}_n$, with the stock's purchase cost $\text{PC}_n$ and sales cost $\text{SC}_n$. Which of the following statements about the resulting no-trade region is correct?

**A.** The region runs from $-\text{SC}_n$ to $+\text{PC}_n$, so its width is $\text{PC}_n + \text{SC}_n$; the risk aversion and $\text{MCAR}_n$ shift the matching band for $\alpha_n$ but do not change its width.

**B.** The region is symmetric around zero, from $-\text{PC}_n$ to $+\text{PC}_n$, because only the cost of the next purchase has to be recovered.

**C.** When new information moves $\text{MCVA}_n$ to a point inside the region, the manager should still trade toward the new optimum that ignores costs.

**D.** A higher active risk aversion widens the no-trade band for $\alpha_n$, because it raises the penalty on active risk.

### Question IM-4 g.2

Bank Q167 • bank-written (GARP style) • re-filed from IM-4 f

A portfolio manager at a long-only equity fund receives a revised alpha forecast for a stock held in the portfolio. Rebalancing the position to reflect the new forecast is expected to add 0.9% per year of active return on that position. The round-trip transaction cost of the rebalancing trade is 0.6%, and the manager expects to hold the revised position for six months before the forecast is updated again. Ignoring the effect on active risk, which of the following is the best decision?

**A.** Rebalance, because the annualized transaction cost of 0.3% is well below the 0.9% expected annual benefit.

**B.** Rebalance, because the 0.9% expected annual benefit exceeds the 0.6% transaction cost.

**C.** Do not rebalance, because the annualized transaction cost of 1.2% exceeds the 0.9% expected annual benefit.

**D.** Do not rebalance, because any trade that incurs transaction costs reduces the portfolio's active return.

### Question IM-4 g.3

New • built from GARP Ch.4, 4.6 • S3

A quantitative equity manager's portfolio was optimal before this morning's alpha update. For every stock, the purchase cost is 0.50% and the sales cost is 0.75%, both amortized and expressed in percent per year. After the update, the optimizer reports:

| Stock | Active weight | MCVA (% per year) |
|---|---|---|
| P | $+1.2\%$ | $+0.65$ |
| Q | $-0.4\%$ | $-0.60$ |
| R | $+2.0\%$ | $-1.30$ |
| S | $0.0\%$ | $+0.95$ |

Which of the following correctly states the trades the manager should make and the net benefit of the most valuable trade?

**A.** Buy S only and sell R only; the most valuable trade is the sale of R, with a net benefit of 0.55%.

**B.** Buy P and S and sell R only; the most valuable trade is the sale of R, with a net benefit of 0.55%.

**C.** Buy S only and sell Q and R; the most valuable trade is the sale of R, with a net benefit of 0.80%.

**D.** Buy P and S and sell Q and R; the most valuable trade is the sale of R, with a net benefit of 1.30%.

### Question IM-4 g.4

Bank Q168 • GARP PE2 Q55 (rebuilt)

A portfolio manager holds stock QYZ and is deciding whether to buy more, sell, or hold after new information. The manager uses MCVA $= \alpha - 2\lambda_A\psi \cdot$ MCAR. All quantities are in percent per year:

| | | | |
|---|---|---|---|
| Alpha, $\alpha$ | 0.25 | Active risk aversion, $\lambda_A$ | 0.32 |
| Marginal contribution to active risk, MCAR | 0.17 | Purchase cost, PC | 0.30 |
| Active risk, $\psi$ | 2.3 | Sales cost, SC | 0.40 |

Which is the best action and rationale?

**A.** Purchase more of QYZ, because its MCVA is greater than the purchase cost.

**B.** Purchase more of QYZ, because its MCVA is greater than its MCAR.

**C.** Sell the position in QYZ, because its MCVA is negative.

**D.** Maintain the position in QYZ, because its MCVA is greater than the negative of the sales cost and below the purchase cost.

### Question IM-4 g.5

New • built from GARP Ch.4, 4.6 • S3

A portfolio manager has just bought stock K up to its optimal holding. For stock K the purchase cost is 0.40% and the sales cost is 0.60%, both amortized (percent per year). The stock's marginal contribution to active risk is 0.30, the manager's active risk aversion is 0.05 and the portfolio's active risk is 4%. Assuming the marginal contribution to active risk and the other inputs stay unchanged, by how much must stock K's alpha fall before the manager would consider selling it?

**A.** 0.40%

**B.** 0.60%

**C.** 1.00%

**D.** 1.12%

### Question IM-4 g.6

Bank Q169 • Schweser Module Quiz 86.2 Q1 (rebuilt, key corrected)

For a stock, the purchase cost is $\text{PC} = 0.30\%$ and the sales cost is $\text{SC} = 0.40\%$. Trading is not worthwhile when $-\text{SC} \leq \alpha - 2\lambda_A \psi \cdot \text{MCAR} \leq \text{PC}$. Initially $2\lambda_A \psi \cdot \text{MCAR} = 0.20\%$. If the manager's active risk aversion $\lambda_A$ doubles, holding all else equal, the no-trade region for the stock's alpha becomes:

**A.** $-0.40\%$ to $0.30\%$

**B.** $-0.20\%$ to $0.50\%$

**C.** $0.00\%$ to $0.70\%$

**D.** $0.10\%$ to $0.80\%$

### Question IM-4 g.7

New • built from GARP Ch.4, 4.6 (Leland) • S3

A pension fund's strategic allocation is 60% equities and 40% bonds. Following Leland's analysis of rebalancing with linear transaction costs, the fund's consultant has estimated a no-trade region of 55% to 65% equities around the target. After a strong equity rally, the portfolio has drifted to 68% equities. Which of the following actions is consistent with Leland's result?

**A.** Sell equities to restore the 60/40 target, because the target allocation is the optimal portfolio.

**B.** Wait for the next scheduled quarterly rebalancing date, then trade back to the 60/40 target.

**C.** Leave the portfolio at 68/32, because outside the no-trade region the cost of trading exceeds the benefit.

**D.** Sell equities until the allocation reaches 65/35, the boundary of the region, which roughly halves turnover and costs with effectively no change in risk over time.

## Solutions • IM-4 g

### Solution IM-4 g.1 — Answer: A

New • built from GARP Ch.4, 4.6

**Step 1 – Where the band comes from**

A purchase pays if $\text{MCVA}_n > \text{PC}_n$; a sale pays if $\text{MCVA}_n < -\text{SC}_n$. Between the two, no trade pays: $-\text{SC}_n \le \text{MCVA}_n \le \text{PC}_n$.
$\Rightarrow$ Width $\text{PC}_n + \text{SC}_n$ (1.25% in GARP's example), centred on zero only if the two costs are equal.

**Step 2 – Rewrite it for alpha**

$2\lambda_A \psi_P \text{MCAR}_n - \text{SC}_n \le \alpha_n \le 2\lambda_A \psi_P \text{MCAR}_n + \text{PC}_n$. The risk terms appear at both ends with the same sign.
$\Rightarrow$ They move the band; only the costs set its width.

**Interpretation**

The two halves of the correct statement are the two numbers you need in any band question: where the edges are (costs, plus the risk charge for alpha) and how wide the band is (costs only).

**Why the other options are wrong**

**B is incorrect.** The sales cost sets the lower edge: the band is $[-\text{SC}_n, +\text{PC}_n]$ and is symmetric only if $\text{SC}_n = \text{PC}_n$.
**C is incorrect.** Polarity flip: inside the region "we should not react to new information"; trading toward the cost-free optimum loses more in costs than it gains.
**D is incorrect.** Width versus location: a higher $\lambda_A$ raises the risk charge and shifts the alpha band up; the width stays $\text{PC}_n + \text{SC}_n$ (see the correction box in Section 3).

**LO:** IM-4 g: Determine the optimal no-trade region for rebalancing with transaction costs. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.6, Eqs. 4.7 to 4.9

### Solution IM-4 g.2 — Answer: C

Bank Q167 • bank-written (GARP style)

**Step 1 – Put the cost on the same annual basis as the benefit**

$$\text{Annualized cost} = \frac{0.6\%}{0.5} = 1.2\% \text{ per year}$$

$\Rightarrow$ With a six-month holding the round trip is effectively paid twice a year.

**Step 2 – Compare**

Benefit 0.9% per year $<$ cost 1.2% per year.
$\Rightarrow$ The trade loses about 0.3% a year on the position: stay inside the no-trade region and do not rebalance.

**Interpretation**

This is the no-trade test with the amortisation made explicit. GARP's band compares MCVA with PC and SC after "subsuming the amortization horizon into the costs"; here the stem gives a raw round trip, so it must be annualised first. With a one-year holding (cost 0.6%) the same trade would pay.

**Why the other options are wrong**

**A is incorrect.** Multiplied by the holding period instead of dividing: $0.6\% \times 0.5 = 0.3\%$.
**B is incorrect.** Compared an annual benefit with an unamortised one-off cost.

**D is incorrect.** Overgeneralises: a trade is worth making whenever its benefit exceeds its amortised cost.

**LO:** IM-4 g: Determine the optimal no-trade region for rebalancing with transaction costs; IM-4 d: Explain the implications of transaction costs on portfolio construction. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.4 and 4.6

### Solution IM-4 g.3 — Answer: B

New • built from GARP Ch.4, 4.6

#### Step 1 – Set up the band

$-\text{SC} \leq \text{MCVA} \leq \text{PC}$: from $-0.75\%$ to $+0.50\%$. Buy above 0.50%, sell below $-0.75\%$.

#### Step 2 – Test each stock

P: $0.65 > 0.50$, buy, net $0.65 - 0.50 = 0.15\%$. Q: $-0.75 \leq -0.60 \leq 0.50$, hold. R: $-1.30 < -0.75$, sell, net $1.30 - 0.75 = 0.55\%$. S: $0.95 > 0.50$, buy, net $0.95 - 0.50 = 0.45\%$.
$\Rightarrow$ Buy P and S, sell R, hold Q.

#### Step 3 – Most valuable trade

Net benefits 0.15% (P), 0.55% (R), 0.45% (S): the sale of R, 0.55%.

#### Interpretation

P and Q are the asymmetry tests. P's MCVA (0.65%) would sit inside a band built on the sales cost alone, and Q's ($-0.60\%$) would trigger a sale if the purchase cost were used on the downside. The active weights are a red herring: the decision depends only on MCVA and the two costs.

#### Why the other options are wrong

**A is incorrect.** Used the sales cost on both sides ($\pm 0.75\%$): P is wrongly held.
**C is incorrect.** Swapped the ends of the band ($-0.50\%$ to $+0.75\%$): P is held, Q is sold, and R's net benefit becomes $1.30 - 0.50 = 0.80\%$.
**D is incorrect.** Traded on the sign of MCVA with no band and reported R's gross 1.30% as its net benefit.

**LO:** IM-4 g: Determine the optimal no-trade region for rebalancing with transaction costs. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.6, Eq. 4.8 and the worked costs of 0.50% and 0.75%

### Solution IM-4 g.4 — Answer: D

Bank Q168 • GARP PE2 Q55

#### Step 1 – Risk charge

$$2\lambda_A \psi \cdot \text{MCAR} = 2(0.32)(2.3)(0.17) = 0.25024$$

#### Step 2 – MCVA

$$\text{MCVA} = 0.25 - 0.25024 = -0.00024 \approx 0$$

$\Rightarrow$ The stock's alpha is almost exactly used up by the active risk it adds at the margin.

#### Step 3 – Compare with the band

$-0.40 \leq -0.00024 \leq 0.30$: inside the no-trade region.
$\Rightarrow$ Hold: neither a purchase nor a sale covers its cost.

#### Interpretation

The question is built so that MCVA lands at zero, the middle of the band; any reading that trades on the sign of MCVA, or compares MCVA with the wrong quantity, gives a trade. In alpha terms

the band runs from $0.25024 - 0.40 = -0.15$ to $0.25024 + 0.30 = 0.55$, and the alpha of 0.25 is inside it.

**Why the other options are wrong**

**A is incorrect.** MCVA $(-0.00024)$ is far below the 0.30 purchase cost.
**B is incorrect.** MCAR is not a hurdle; the buy test is MCVA against the purchase cost.
**C is incorrect.** A negative MCVA justifies a sale only below $-0.40$, the negative of the sales cost.

**LO:** IM-4 g: Determine the optimal no-trade region for rebalancing with transaction costs. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.6, Eqs. 4.7 and 4.8; GARP Practice Exam 2, Q55

## Solution IM-4 g.5

**Answer: C**

New • built from GARP Ch.4, 4.6

**Step 1 – Where the stock sits now**

Just after a purchase up to the optimal holding, $\text{MCVA}_K = \text{PC}_K = +0.40\%$: the top edge of the band.

**Step 2 – Where selling starts**

A sale pays only once $\text{MCVA}_K < -\text{SC}_K = -0.60\%$.

**Step 3 – Distance**

With MCAR and the other inputs fixed, MCVA moves one-for-one with alpha, so alpha must fall by

$$\text{PC}_K + \text{SC}_K = 0.40\% + 0.60\% = 1.00\%$$

$\Rightarrow$ The full width of the band. GARP: "The alpha would have to decrease by 1.25 percent before we would consider selling the stock" with its costs of 0.50% and 0.75%.

**Interpretation**

The risk charge, $2(0.05)(4)(0.30) = 0.12\%$, is a red herring: it sits inside MCVA at both ends and cancels. The same logic after a sale: alpha must *rise* by PC + SC before the manager buys back.

**Why the other options are wrong**

**A is incorrect.** Treated MCVA turning negative as the sell signal: a fall of 0.40% only brings MCVA to zero, the middle of the band.
**B is incorrect.** Started the stock at zero instead of at the top of the band, so counted only the sales cost.
**D is incorrect.** Added the risk charge of 0.12% to the width; it does not change.

**LO:** IM-4 g: Determine the optimal no-trade region for rebalancing with transaction costs. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.6, Eq. 4.8

## Solution IM-4 g.6

**Answer: C**

Bank Q169 • Schweser Module Quiz 86.2 Q1 (rebuilt)

**Step 1 – Rearrange for alpha**

$$2\lambda_A \psi \,\text{MCAR} - \text{SC} \le \alpha \le 2\lambda_A \psi \,\text{MCAR} + \text{PC}$$

$\Rightarrow$ Initially: $0.20 - 0.40 = -0.20\%$ to $0.20 + 0.30 = 0.50\%$.

**Step 2 – Double the risk aversion**

The risk charge is linear in $\lambda_A$: $0.20\% \to 0.40\%$.

**Step 3 – New band**

Lower edge $0.40 - 0.40 = 0.00\%$; upper edge $0.40 + 0.30 = 0.70\%$.
$\Rightarrow$ Width $0.70\% =$ PC + SC, unchanged; the band has moved up by 0.20%.

### Interpretation

A more risk-averse manager needs a larger alpha before buying (0.70% instead of 0.50%) and sells sooner (below 0% instead of $-0.20\%$). The original Schweser item keyed this as a wider region; the bank's rebuilt version asks for the band itself, which settles it (correction box, Section 3).

### Why the other options are wrong

**A is incorrect.** Ignored the risk charge: $-$SC to $+$PC is the band for MCVA, not for alpha.
**B is incorrect.** The band before the change; the doubled risk aversion was not applied.
**D is incorrect.** Swapped the costs: $0.40 - 0.30 = 0.10\%$ to $0.40 + 0.40 = 0.80\%$.

**LO:** IM-4 g: Determine the optimal no-trade region for rebalancing with transaction costs. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.6, Eq. 4.9; SchweserNotes 2026 Book 5, Reading 86, LO 86.g

## Solution IM-4 g.7

**Answer: D**

New • built from GARP Ch.4, 4.6

### Step 1 – Is the portfolio outside the region?

68% equities is above the 65% boundary, so the no-trade rule no longer protects it: a trade is required.

### Step 2 – Trade to where?

Leland: outside the region, "the optimal strategy is to trade back to the boundary". Sell 3 points of equities to 65/35, not 8 points to 60/40.
$\Rightarrow$ "Trading only to the boundary, not to the target allocation, cuts the turnover and transactions costs roughly in half, with effectively no change in risk over time."

### Interpretation

Two features separate Leland's rule from ordinary rebalancing: the trigger is the boundary, not the calendar, and the destination is the boundary, not the target. The region's size depends on costs, risk aversion and the assets' expected returns and risks.

### Why the other options are wrong

**A is incorrect.** Trades to the target: about twice the turnover and cost for effectively no reduction in risk.
**B is incorrect.** Calendar rebalancing; Leland motivates rules driven by the boundaries, not by the passage of time.
**C is incorrect.** Polarity flip: it is inside the region that the costs exceed the benefit; outside it, a trade to the boundary pays.

**LO:** IM-4 g: Determine the optimal no-trade region for rebalancing with transaction costs. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.6 (Leland, 1996)

IM-4 • Portfolio Construction • Vardeez: VERY IMPORTANT

# IM-4 h

P2 CONCEPT

Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming.

Workbook: D28 Portfolio Construction and the No-Trade Region • Tier D • notes only

**WHAT IS TESTED HERE**

**Why P2.** Vardeez makes the construction techniques his second focus for IM-4, a VERY IMPORTANT reading, and the workbook (D28, Tier D) calls them "a ready-made four-statement audit". **Vardeez:** portfolio construction techniques.

**Reported exam shapes**

- **S4** The four techniques as a four-statement audit: screens, stratification, LP, QP.

**What the questions look like:** concept only. Name the technique from a description; pick the strength or the weakness that does not belong to a technique; and the audit itself, four statements, one per technique, where three either swap in a property of a sibling technique or turn a strength into a weakness. Background on the inputs and the objective: IM-4 d, Section 1.

## 1. The yardstick

"We are interested in high alpha, low active risk, and low transactions costs." The figure of merit is value added less costs, $\alpha_P - \lambda_A \psi_P^2 - \text{TC}$. Each technique is judged on how it handles those three elements. Four generic classes cover "the vast majority of institutional portfolio management applications" (monthly or quarterly rebalancing of up to 1,000 assets). They run from simple and robust to sophisticated and input-hungry: QP's strength and its weakness are the same property, it uses every input.

## 2. Screens

**Recipe:** rank by alpha, take the top 50 (say), equal- or capitalisation-weight them. **For rebalancing:** with alphas on 200 stocks, put the top 40 on a buy list, the next 60 on a hold list and the bottom 100 on a sell list; buy what is on the buy list and not held, sell what is held and on the sell list. The sizes of the lists regulate turnover.

- **Strengths.** "Beauty in simplicity": easy to understand, clear link from cause (list membership) to effect (portfolio membership), easy to computerise. **Robust**: it depends solely on ranking, so wild positive or negative alphas do not alter the result. Enhances alpha by concentrating in high-alpha stocks; strives for risk control through the number of stocks and the weighting; limits costs through the list sizes.
- **Weaknesses.** Ignores all information in the alphas apart from the rankings; no protection against biases in the alphas (if all utilities rank low, the portfolio holds no utilities); risk control is "fragmentary at best": screened portfolios have turned out "considerably more risky than their managers had imagined". Still very popular.

## 3. Stratification

"Glorified screening." Split the followed stocks into **exclusive categories** (10 sectors $\times$ 3 size groups $= 30$), screen by alpha within each, and weight each category at its **benchmark weight**.

- **Strengths.** Everything a screen offers, plus: it matches the benchmark on the chosen dimensions and ignores any biases in the alphas *across* categories; robust, somewhat transparent, easy to code; same turnover control.
- **Weaknesses.** Ignores some information; cannot slightly overweight one category and underweight another; often "little substantive research underlies the selection of the categories", so risk control is rudimentary; if important risk dimensions are left out, risk control fails.

## 4. Linear programming

"Space-age stratification." Characterise stocks along risk dimensions (industry, size, volatility, beta) that need **not** partition them into exclusive categories, and build a portfolio close to the benchmark on all of them. It can include explicit transaction costs, a turnover limit and upper and lower position limits; the objective is to maximise alpha less costs while staying close to the benchmark on the risk-control dimensions.

- **Strengths.** Takes all the information about alpha into account; controls risk by keeping the portfolio's characteristics close to the benchmark's.
- **Weaknesses.** Difficulty producing portfolios with a prespecified number of stocks; the risk-control characteristics must not work at cross purposes with the alphas (if the alphas sometimes favour small stocks and sometimes large ones, do not control risk on size).

## 5. Quadratic programming

"The ultimate in portfolio construction." QP explicitly considers all three elements of the figure of merit, alpha, risk and costs, and since it includes a linear program as a special case, it can carry every LP constraint. "This should be the best of all worlds. Alas, nothing is perfect."

- **Weakness: inputs.** A universe of 500 stocks needs 500 volatilities and $500 \times 499/2 = 124{,}750$ correlations (Schweser: 400 stocks, 79,800 covariances). More inputs mean more noise: "a fear of garbage in, garbage out".
- **Why noise hurts more here.** In ordinary risk estimation the errors largely cancel; in optimisation the optimiser "will take advantage of opportunities that appear in the noisy estimates of covariance but are not present in reality".
- **Figure 4.2 (cash versus market).** True market volatility 17%. Estimates within 1 point barely hurt; beyond about 3 points the loss becomes significant, especially for underestimates; an estimate of 12% gives *negative* value added (Figure 1).
- **The lessons.** Covariance errors lead to inefficient implementation, and it is vital to have good estimates: "Rather than abandon the attempt, try to do a good job." Given sufficiently accurate risk estimates, QP "most consistently achieves high value added".

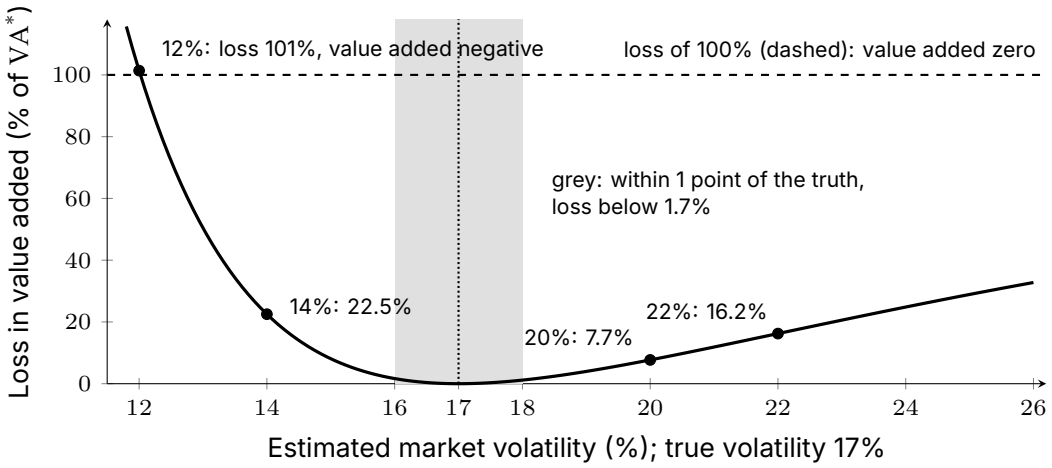



Figure 1. GARP's Figure 4.2 redrawn: the share of the optimal value added lost when a cash-versus-market QP uses an estimated volatility $\hat{\sigma}$ instead of the true 17% (solid curve; the dotted line marks the truth). The curve is $\left(1 - 17^2/\hat{\sigma}^2\right)^2$, rebuilt from the cash-versus-market problem (Eq. 4.11 is not reproduced in the text); it matches every reading GARP takes from the figure. Underestimates are punished far more than overestimates of the same size: 14% costs 22.5%, 20% costs 7.7%.

## 6. The test: same alphas, four methods

Muller fed identical alphas (actual next-year returns of the S&P 500 stocks plus noise, IC 0.1, so a target IR of $0.1\sqrt{500} = 2.24$) to two screens, a stratification (top $J$ stocks in each of 55 BARRA industries) and a QP, at three risk aversions and four dates, ignoring costs. Ex post information ratios:

| | Screen I (equal) | Screen II (cap) | Strat | QP |
|---|---|---|---|---|
| Average | 0.86 | 1.10 | 1.27 | **1.88** |
| Standard deviation | 0.27 | 0.79 | 0.89 | 0.40 |
| Maximum | 1.43 | 2.24 | **2.82** | 2.51 |
| Minimum | 0.50 | −0.53 | 0.33 | 0.98 |

QP led "to consistently the highest ex post information ratios"; a stratified portfolio had the single highest IR "but no consistency over time"; a screen had a negative IR in one period. None reached 2.24, partly because of the constraints imposed on the optimiser (constraints shrink the effective IC, IM-4 b).

**Correction to the source notes.** (1) Vardeez's table lists "ignores diversification" as a weakness of screens. GARP says the opposite about intent: a screen "strives for risk control by including a sufficient number of stocks ... and by weighting them to avoid concentration in any single stock"; its weakness is that this risk control is "fragmentary at best". (2) Vardeez gives linear programming's weakness as "may result in a portfolio very close to the benchmark, potentially missing alpha". That is not GARP's: staying close to the benchmark on the risk dimensions is how LP controls risk. GARP's two weaknesses are the difficulty of producing a prespecified number of stocks and risk-control dimensions that work at cross purposes with the alphas. (3) SchweserNotes says that "with 5% estimation error, value added may actually be negative". Only the underestimate does that (12% against a true 17%: loss 101%); a 5-point overestimate (22%) loses 16% of the value added.

| | How it works | Strengths | Weaknesses |
|---|---|---|---|
| **Screens** | Rank by alpha; top $N$; equal- or cap-weight; buy, hold and sell lists | Simple, transparent, easy to code; robust to wild alphas (ranks only); list sizes control turnover | Uses only the rankings; no protection against alpha biases; fragmentary risk control |
| **Stratification** | Exclusive categories (sector $\times$ size); screen within each; benchmark category weights | Matches the benchmark on the chosen categories; ignores alpha biases across categories; robust | Ignores some information; no slight over- or underweighting of categories; risk control only as good as the categories |
| **Linear programming** | Non-exclusive risk dimensions (industry, size, volatility, beta); maximise alpha less costs near the benchmark | Uses all alpha information; can include costs, turnover and position limits | Hard to hit a prespecified number of stocks; risk dimensions must not fight the alphas |
| **Quadratic programming** | Maximise alpha less a risk penalty less costs, with any LP constraints | Theoretically the best; explicit alpha, risk and costs; LP is a special case | Most inputs, so most noise; exploits covariance errors; underestimated risk can make value added negative |

**Exam traps**

- **Sibling swap: exclusive categories.** Mutually exclusive categories belong to stratification; LP characterises stocks along overlapping dimensions.
- **Screens and wild alphas.** Screens are robust to extreme alphas *because* they use only ranks; the weakness is the information thrown away, not sensitivity.
- **Stratification weights.** Categories are held at benchmark weights; stratification does not tilt toward low-risk categories.
- **QP and noise.** More inputs mean *more* noise and more chances of error, not less; QP considers alpha, risk *and* costs, not just two of them.
- **Special case direction.** LP is a special case of QP, not the reverse.
- **Which estimate hurts.** Underestimating volatility hurts value added far more than overestimating it.
- **Counting inputs.** $N(N-1)/2$ correlations: 124,750 for 500 stocks, 79,800 for 400; with the 400 variances, 80,200 distinct entries.

## Practice questions: IM-4 h

### Question IM-4 h.1

Bank Q173 • AnalystPrep Q.2745

A portfolio manager has a specific approach to building a portfolio. The manager first ranks stocks based on their alpha, then selects the top ten stocks from this ranking. The selected stocks are then used to create a portfolio in which each stock is given equal weight. Which portfolio construction technique is the manager most likely using?

**A.** Stratification

**B.** Linear programming

**C.** Quadratic programming

**D.** Screens

### Question IM-4 h.2

Bank Q171 • AnalystPrep Q.2452

A member of a fund's investment committee lists four possible shortcomings of the screening technique for portfolio construction. All of the following are shortcomings of screens, EXCEPT:

**A.** It is hard to code.

**B.** It ignores all information in the alphas apart from the rankings.

**C.** It does not protect against biases in the alphas.

**D.** It may result in more risky portfolios.

### Question IM-4 h.3

New • built from GARP Ch.4, 4.7 • S4

A consultant compares two proposals for a US equity mandate drawn from a followed list of 200 stocks. Proposal 1 sorts the stocks into 10 economic sectors and 3 size groups, screens by alpha within each of the 30 cells, and weights each cell at its benchmark weight. Proposal 2 characterizes every stock by industry, size, volatility and beta, and maximizes alpha less transaction costs while staying close to the benchmark on each of those characteristics. Which of the following statements is correct?

**A.** Proposal 2 is a stratification approach, because it requires the risk dimensions to split the stocks into mutually exclusive categories.

**B.** Proposal 2 is a linear program, and its main strength is that it produces portfolios with exactly the number of stocks the manager specifies.

**C.** Proposal 1 is a stratification approach: it is robust to alpha biases across categories, but it cannot slightly overweight one category and underweight another, and its risk control fails if an important dimension is left out.

**D.** Proposal 1 uses all of the information in the alphas, including their magnitudes, and departs from benchmark category weights wherever the alphas justify it.

**Question IM-4 h.4** Bank Q172 • AnalystPrep Q.2455

Quadratic programming is a type of mathematical optimization that can handle a range of constraints. Which of the following options does NOT accurately describe the characteristics or considerations of quadratic programming as a portfolio construction technique?

**A.** It includes the linear program as a special case.

**B.** It considers only two elements: risk and transaction costs.

**C.** It requires a large number of inputs.

**D.** The use of a large number of inputs increases the noise in the portfolio construction process.

**Question IM-4 h.5** New • built from GARP Ch.4, 4.7 (Figure 4.2) • S4

A quantitative manager tests a simple cash-versus-market quadratic program in which the true market volatility is 17%. Following Grinold and Kahn, the manager studies how much of the optimal value added is lost when the optimizer uses an estimated volatility instead of the true one. Which of the following conclusions is correct?

**A.** Overestimates of volatility are more damaging than underestimates of the same size, because the optimizer then takes positions that are too large.

**B.** Estimation errors in the covariance inputs cancel out in optimization, as they do in ordinary risk estimation, so their quality matters little.

**C.** Because covariance errors make implementation inefficient, the manager should abandon quadratic programming in favour of screens.

**D.** An estimate within about 1 percentage point of 17% costs little value added, but errors beyond about 3 points are costly, especially underestimates, and an estimate of 12% makes value added negative.

**Question IM-4 h.6** Bank Q170 • AP Mock 1 Q65

Which of the following statements regarding portfolio construction techniques is correct?

**A.** The stratification technique provides superior risk control by overweighting the categories with lower risks and underweighting the categories with higher risks.

**B.** The linear programming approach characterizes stocks along dimensions of risk, and requires that these dimensions distinctly and exclusively partition the stocks.

**C.** Although screening ignores all information in alphas except the rankings, it limits transaction costs through judicious choice of the buy, sell and hold lists.

**D.** Quadratic programming requires many more inputs than other portfolio construction techniques and as a result minimizes both noise and the potential for poor calibration.

**Question IM-4 h.7** Bank Q174 • GARP PE1 Q23 • Mock 3 Q57

The risk committee of an equity mutual fund is reviewing a portfolio construction technique proposed by a new portfolio manager who has recently been allocated capital to manage. The fund typically grants its portfolio managers flexibility in selecting and implementing portfolio construction procedures but requires that any methodology adopted fulfill key risk control objectives set by the firm. Which of the following is correct regarding a portfolio construction technique and its approach to risk control?

**A.** The screening technique ranks stocks by risk-adjusted alpha but does not apply any additional risk control measures such as weighting the selected stocks by their relative capitalization.

**B.** The quadratic programming technique takes into account additional risk parameters compared to other major portfolio construction techniques but also requires more inputs, which leads to more noise.

**C.** The stratification technique splits the list of stocks into categories and maintains risk control by overweighting the categories with lower risks and underweighting the categories with higher risks.

**D.** The linear programming technique focuses on the pair-wise correlations of stocks rather than characterizing each stock along multiple dimensions of risk such as size, industry, volatility or beta.

## Solutions • IM-4 h

### Solution IM-4 h.1 — Answer: D

Bank Q173 • AnalystPrep Q.2745

**Step 1 – Match the recipe**

GARP's screen: "1. Rank the stocks by alpha. 2. Choose the first 50 stocks (for example). 3. Equal-weight (or capitalization-weight) the stocks." The manager does exactly this with ten stocks.

**Step 2 – Rule out the others**

No categories, no benchmark matching on risk characteristics, no risk or cost model.
⇒ A screen.

**Interpretation**

Rank-select-weight is the signature of a screen; once categories appear it becomes stratification, once risk characteristics are matched it becomes LP, and once a covariance matrix enters it is QP. A ten-stock screen also shows the weakness: risk control comes only from the number of names and the weighting.

**Why the other options are wrong**

**A is incorrect.** Stratification would first split the stocks into exclusive categories and weight each at its benchmark weight.
**B is incorrect.** LP keeps the portfolio close to the benchmark on risk characteristics; nothing here does that.
**C is incorrect.** QP trades off alpha, risk and costs with a covariance matrix; none is used.

**LO:** IM-4 h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.7 Screens

### Solution IM-4 h.2 — Answer: A

Bank Q171 • AnalystPrep Q.2452

**Step 1 – List the screen's weaknesses in GARP**

"They ignore all information in the alphas apart from the rankings. They do not protect against biases in the alphas. … Risk control is fragmentary at best", with screened portfolios "considerably more risky than their managers had imagined".
⇒ B, C and D are genuine shortcomings.

**Step 2 – The odd one out**

"The screen is easy to computerize; it might be that mythical computer project that can be completed in two days!"
⇒ Ease of coding is a strength, so "hard to code" is not a shortcoming.

**Interpretation**

EXCEPT questions on screens hinge on the pair simple and robust (strengths) against information loss and weak risk control (weaknesses).

**Why the other options are wrong**

**B is incorrect.** A genuine weakness: magnitudes are discarded.
**C is incorrect.** A genuine weakness: if all utilities rank low, the screen holds no utilities.
**D is incorrect.** A genuine weakness: fragmentary risk control can leave the portfolio riskier than intended.

**LO:** IM-4 h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.7 Screens

### Solution IM-4 h.3 — Answer: C

New • built from GARP Ch.4, 4.7

#### Step 1 – Name the proposals

Proposal 1 uses exclusive cells (10 sectors $\times$ 3 sizes), screens within each and holds cells at benchmark weights: stratification, "glorified screening". Proposal 2 uses overlapping risk characteristics and maximises alpha less costs near the benchmark: linear programming, "space-age stratification".

#### Step 2 – Check the properties claimed

Stratification "ignores any biases in the alphas across categories" but "does not consider slightly over-weighting one category and underweighting another", and "if some important risk dimensions are excluded, risk control will fail".
$\Rightarrow$ C is correct.

#### Interpretation

The two proposals differ in exactly the way GARP separates them: exclusive categories against non-exclusive dimensions. Both match the benchmark on risk; LP uses all the alpha information, stratification only the rankings inside each cell.

#### Why the other options are wrong

**A is incorrect.** Sibling swap: LP "does not require that these dimensions distinctly and exclusively partition the stocks"; that requirement belongs to stratification.
**B is incorrect.** Polarity flip: "the linear program has difficulty producing portfolios with a prespecified number of stocks".
**D is incorrect.** Stratification ignores some information (it screens on ranks within cells) and holds categories at benchmark weights.

**LO:** IM-4 h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.7 Stratification and Linear Programming

### Solution IM-4 h.4 — Answer: B

Bank Q172 • AnalystPrep Q.2455

#### Step 1 – What QP optimises

"The quadratic program explicitly considers each of the three elements in our figure of merit: alpha, risk, and transactions costs."
$\Rightarrow$ "Only two elements: risk and transaction costs" leaves out alpha, the point of the exercise.

#### Step 2 – Check the other statements

QP "includes a linear program as a special case"; it "requires a great many more inputs"; and "more inputs mean more noise".
$\Rightarrow$ A, C and D are accurate.

#### Interpretation

QP's strength (every input used) is also its weakness (every input's error used). That double edge is what the audit questions test.

#### Why the other options are wrong

**A is incorrect.** Accurate: LP is a special case of QP, so QP can carry every LP constraint.
**C is incorrect.** Accurate: 500 stocks need 500 volatilities and 124,750 correlations.
**D is incorrect.** Accurate: more inputs bring more estimation noise.

**LO:** IM-4 h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.7 Quadratic Programming

### Solution IM-4 h.5 — Answer: D

New • built from GARP Ch.4, 4.7 (Figure 4.2)

#### Step 1 – Read the loss curve

GARP: "market volatility estimates within 1 percent of the true market volatility will not hurt value added very much, but as estimation error begins to exceed 3 percent, the effect on value added becomes significant, especially if the error is an underestimate of volatility. In fact, an underestimate of 12 percent market volatility ... leads to a negative value added."

#### Step 2 – Check with the rebuilt curve

Loss $= \left(1 - 17^2/\hat{\sigma}^2\right)^2$: 1.7% at 16%, 1.2% at 18%, 22.5% at 14%, 7.7% at 20%, 101% at 12%.
$\Rightarrow$ D is correct.

#### Interpretation

An underestimate makes the market look safer than it is, so the optimiser takes positions that are too large; the loss grows much faster than for an overestimate. The two lessons: covariance errors cause inefficient implementation, and good covariance estimates are vital.

#### Why the other options are wrong

**A is incorrect.** Polarity flip: underestimates do more damage, and it is underestimates that make positions too large.
**B is incorrect.** Errors largely cancel in ordinary risk estimation but are "an obstacle in optimization", because the optimiser exploits them.
**C is incorrect.** "Rather than abandon the attempt, try to do a good job": the lesson is better estimates, not a cruder technique.

**LO:** IM-4 h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.7 Quadratic Programming, Eq. 4.11 and Figure 4.2

### Solution IM-4 h.6 — Answer: C

Bank Q170 • AP Mock 1 Q65

#### Step 1 – Audit each statement against its technique

Screens: "They ignore all information in the alphas apart from the rankings" and "transactions costs are limited by controlling turnover through judicious choice of the size of the buy, sell, and hold lists".
$\Rightarrow$ C is a correct strength-and-weakness pair.

#### Step 2 – The three corrupted statements

Stratification holds categories at benchmark weights (A); exclusive partitions are stratification's, not LP's (B); QP's many inputs add noise (D).

#### Interpretation

A four-statement audit (shape S4): one statement per technique, three of them corrupted by a sibling swap or a polarity flip. Work through the techniques one at a time against the comparison

table.

### Why the other options are wrong

**A is incorrect.** Stratification matches the benchmark's category weights; it does not tilt toward low-risk categories.

**B is incorrect.** Sibling swap: LP "does not require that these dimensions distinctly and exclusively partition the stocks".

**D is incorrect.** Polarity flip: "more inputs mean more noise".

**LO:** IM-4 h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.7

## Solution IM-4 h.7

**Answer: B**

Bank Q174 • GARP PE1 Q23

### Step 1 – QP

QP models risk through the full covariance matrix, more risk parameters than any other technique, and needs far more inputs (500 volatilities and 124,750 correlations for 500 stocks), which brings more noise.

$\Rightarrow$ B is correct.

### Step 2 – The other three

Screens rank by alpha and may be equal- or capitalisation-weighted; stratification holds categories at benchmark weights; LP characterises stocks along size, industry, volatility and beta, while pairwise correlations belong to QP.

### Interpretation

GARP's version of the audit, with a risk-control frame. The correct statement is QP's double edge, and two of the distractors are borrowed from other techniques (correlations from QP into LP; risk tilting that no technique uses into stratification).

### Why the other options are wrong

**A is incorrect.** Screens rank by alpha, not risk-adjusted alpha, and can capitalisation-weight the selected stocks (Screen II in Muller's test).

**C is incorrect.** Stratification sets category weights equal to benchmark weights; it does not overweight low-risk categories.

**D is incorrect.** Sibling swap: pairwise correlations are QP's input; LP characterises each stock along dimensions of risk.

**LO:** IM-4 h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.7 and 4.8; GARP Practice Exam 1, Q23

IM-6 • VaR and Risk Budgeting in Investment Management • Vardeez: VERY IMPORTANT

# IM-6 e

P2 BOTH

Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk.

Workbook: C4 Policy-Mix VaR and Risk Budgeting • Tier C • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** Vardeez flags it (surplus at risk calculation) in a VERY IMPORTANT chapter. The workbook section C4 is Tier C (Map #35, reported on 4 of 16 recall sheets: "then VaR was asking to be calculated by policy mix method").
**Vardeez:** surplus at risk calculation.

**Reported exam shapes**

- **C4·S1** VaR calculated by the policy-mix method: policy weights on benchmark returns, diversified.
- **C4·S2** The six-way sibling set: absolute, relative, policy-mix, active-management, funding and sponsor risk.

**What the questions look like:** half are statement questions on the six risks, where the wrong options swap siblings (cash-flow against economic risk, policy-mix against active risk, defined benefit against defined contribution). The other half are calculations: policy-mix VaR, relative (tracking-error) VaR, the change in surplus from asset and liability durations, and surplus at risk from asset and liability amounts, growth rates, volatilities and their correlation. The calculations sit inside the Vardeez pointer and shape S1, so they are in scope for this BOTH LO.

## 1. Six risks, three questions

Risk is easy to define for a bank trader: the loss on the marked-to-market position. An investment manager can perceive risk differently. Jorion's six risks are best kept apart by three questions (Figure 1, after Vardeez):

1. **Against what is the loss measured?** Zero (absolute risk) or a benchmark (relative risk).
2. **Whose decision caused it?** The policy mix chosen by the fund (policy-mix risk) or the managers' deviations from it (active-management risk). Both are parts of the fund's absolute (asset) risk.
3. **Whose problem is it?** The fund's surplus over its liabilities (funding risk) or the firm that stands behind the fund (sponsor risk).

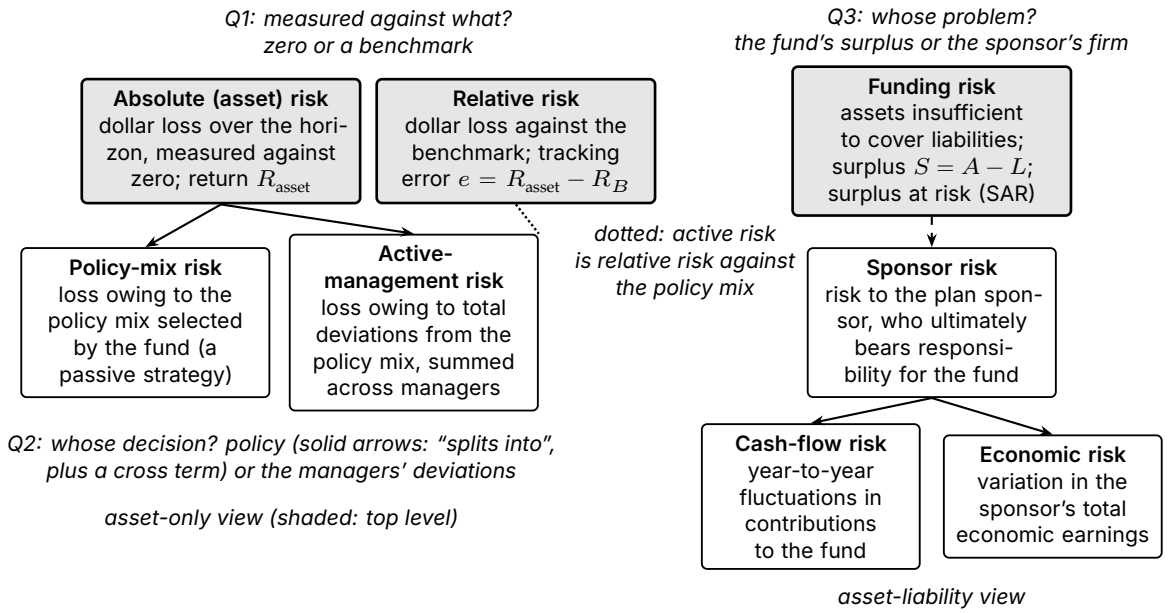



Figure 1. The six risks as a tree (shaded boxes are the top level). Solid arrows split a risk into its parts; the dashed arrow extends funding risk to the plan sponsor; the dotted line links active-management risk to relative risk measured against the policy benchmark.

## 2. Absolute and relative risk

- **Absolute risk** is "the risk of a dollar loss over the horizon"; the usual definition in a trading environment, sometimes called **asset risk**. The relevant return is $R_{\text{asset}}$ and, with normality, VaR $= \alpha\sigma W$.
- **Relative risk** is "the risk of a dollar loss in a fund relative to its benchmark", measured as the dollar difference between the fund return and a like amount invested in the benchmark. It is the relevant risk for a manager with a mandate to beat a benchmark. The relevant return is Jorion's tracking error $e = R_{\text{asset}} - R_B$, and with normality $\boxed{\text{VaR} = \alpha W \sigma_e}$.
- **Notation trap.** Jorion's "tracking error" is the return difference itself; its volatility $\sigma_e$ (TEV) is what most other readings, and most stems, call tracking error. Read which one the stem gives: a level or a standard deviation.
- **Example.** A USD 500 million equity mandate with 15% volatility and a 2% TEV has a 95% absolute VaR of $1.645 \times 0.15 \times 500 = 123.4$ million but a relative VaR of only $1.645 \times 0.02 \times 500 = 16.5$ million.

## 3. Policy-mix and active-management risk (shape C4·S1)

A fund that allocates to a pool of active managers can split its absolute performance into a policy part and an active part (Eq. 6.1 and 6.2):

$$R_{\text{asset}} = \sum_i w_i R_i = \underbrace{\sum_i w_i^p R_i^b}_{\text{policy mix}} + \underbrace{\sum_i \left(w_i R_i - w_i^p R_i^b\right)}_{\text{active management}}$$

where $R_i^b$ is the benchmark return of fund $i$ and $w_i^p$ its policy weight. Hence:

- **Policy-mix risk** is "the risk of a dollar loss owing to the policy mix selected by the fund". The policy mix can be implemented with passive funds, so this is the risk of a passive strategy. **Policy-mix VaR** uses the *policy* weights and the *benchmark* volatilities and correlations, and it is diversified:

$$\text{VaR}_{\text{policy}} = \alpha W \sqrt{\sum_i \sum_j w_i^p w_j^p \sigma_i \sigma_j \rho_{ij}}$$

  Example: policy mix 60% stocks (benchmark volatility 15%), 40% bonds (5%), correlation 0.2, USD 2,000 million, $\alpha = 2.33$: $\sigma = \sqrt{0.0081 + 0.0004 + 2(0.6)(0.4)(0.2)(0.15)(0.05)} = 9.60\%$, so $\text{VaR}_{\text{policy}} = 2.33 \times 0.0960 \times 2{,}000 =$ USD 447 million, against an undiversified 513.

- **Active-management risk** is "the risk of a dollar loss owing to the total deviations from the policy mix": the sum of profits or losses across all managers relative to their benchmarks, so it diversifies across managers with different styles. It also includes temporary deviations of the current asset mix from the policy mix ($w_i \neq w_i^p$). Current weights and the managers' TEVs therefore belong here, not in the policy-mix VaR.

- **OTPPB** (annual, 99%, % of fund value): policy-mix VaR 19.6%, active-management VaR 1.6%, asset VaR 19.3%. Three lessons. (1) Most of the risk is policy mix, as Brinson et al. (1986) found for performance. (2) Active VaR is small: diversification across styles and managers, or assets held in indexed funds. (3) The two do *not* add up: total VaR comes from the policy-mix VaR, the active VaR and a cross-product term. Here the implied correlation is $(19.3^2 - 19.6^2 - 1.6^2)/(2 \times 19.6 \times 1.6) = -0.23$, so "active managers could take greater deviations from their benchmark without affecting the plan's total VaR".

## 4. Funding risk and surplus at risk (Vardeez pointer)

For a defined-benefit plan the fixed payments to retirees make asset volatility alone the wrong measure: risk belongs in an asset/liability (ALM) framework. **Funding risk** is "the risk that the value of assets will not be sufficient to cover the liabilities of the fund". The variable is the surplus $S = A - L$, with $\Delta S = \Delta A - \Delta L$ and (Eq. 6.3)

$$R_S = \frac{\Delta S}{A} = R_{\text{asset}} - R_{\text{liabilities}} \frac{L}{A}$$

- **Liabilities behave like a short long-term bond** when they are mainly nominal payments (accumulated benefit obligations discounted at a market rate). "Decreases in interest rates, while beneficial for equities on the asset side, can increase even more the value of liabilities." Example: assets 1,000 (duration 5), liabilities 900 (duration 12), rates fall 1%: $\Delta A = +50$, $\Delta L = +108$, so the surplus *falls* by 58. Inflation-indexed liabilities behave like inflation-protected bonds.

- **Minimum-risk position:** immunization, with asset duration equal to liability duration. It may be infeasible (not enough long bonds) and carries an opportunity cost if other assets earn more.

- **Who bears it:** in a defined-benefit plan the shortfall "will have to be made up by the fund's owner"; "defined-contribution plans put the risk on the employees". Funding risk is the true long-term risk to the owner; it is also called **surplus at risk (SAR)**.

- **PERF (GARP).** Assets 1,000, liabilities 900, surplus 100. Expected surplus return, scaled by assets, 5%: growth of 50 to an expected surplus of 150. Surplus volatility 9.4% gives a 99% VaR of $2.33 \times 9.4\% \approx 22\%$, or 220. Hence $150 - 220 = -70$: a 1% chance that the surplus becomes a deficit of 70 or more within a year (Figure 2). If acceptable, risk budgeting allocates the SAR of 220 (IM-6 h).

- **Horizon equivalence (OTPPB):** since $(1.28/2.33) \times \sqrt{4} = 1.10$ is close to 1, a 99% VaR over one year is roughly a 90% VaR over four years (a loss expected in one of ten four-year periods).

**Building the surplus distribution from assets and liabilities.** When the stem gives amounts, growth rates, volatilities and a correlation:

$$E[S_1] = A(1+R_A) - L(1+R_L), \qquad \sigma_S = \sqrt{(A\sigma_A)^2 + (L\sigma_L)^2 - 2\rho\,(A\sigma_A)(L\sigma_L)}$$

The minus sign matters: assets and liabilities that move together offset. Three numbers can be asked, so read the reference point:

| Quantity | Measured from | Formula |
|---|---|---|
| Surplus VaR (relative SaR) | expected surplus | $z\sigma_S$ |
| Absolute SaR | current surplus | $z\sigma_S - E[\Delta S]$ |
| Worst-case surplus (deficit if negative) | zero | $E[S_1] - z\sigma_S$ |

Example (SchweserNotes): $A = 200$ (growth 4%, volatility 10%), $L = 180$ (3%, 7%), $\rho = 0.4$, $z = 1.65$. $E[\Delta S] = 8 - 5.4 = 2.6$; $\sigma_S = \sqrt{20^2 + 12.6^2 - 2(0.4)(20)(12.6)} = 18.90$; surplus VaR $1.65 \times 18.90 = 31.18$; absolute SaR 28.58; worst-case surplus $20 + 2.6 - 31.18 = -8.58$, a deficit of about 8.6.

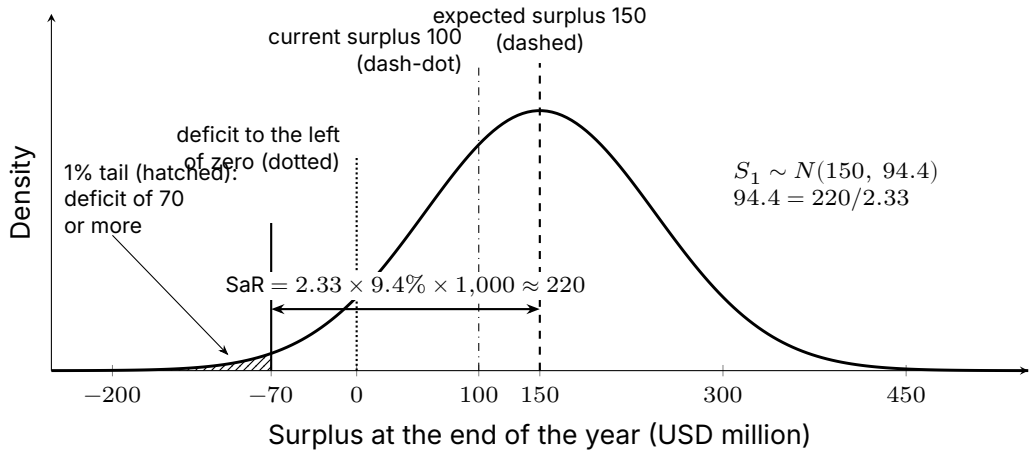



Figure 2. GARP's PERF example. The surplus is expected to grow from 100 (dash-dot) to 150 (dashed). The 99% surplus VaR of 220 is measured from the expected surplus, so the 1% cut-off (solid line) sits at $150 - 220 = -70$.

## 5. Sponsor risk

Surplus risk extends to the owner of the fund, the **plan sponsor**, "who ultimately bears responsibility for the pension fund". Two measures:

- **Cash-flow risk**: "the risk of year-to-year fluctuations in contributions to the pension fund". A sponsor that can absorb greater variation in funding costs can adopt a more volatile risk profile.
- **Economic risk**: "the risk of variation in total economic earnings of the plan sponsor". Surplus falls matter less if they occur when the firm's operating profits are high. A fund heavy in stocks has the opposite effect: a downturn cuts the assets just when the company is least able to contribute.

From the sponsor's viewpoint risk is measured not only by the assets or even the surplus but by the effect on the economic value of the firm, so pension management should be integrated with the sponsor's financial goals: enterprisewide risk management. Sponsor risk is the only one of the six that reaches outside the fund.

**Formula card: IM-6 e**

| | |
|---|---|
| Absolute VaR | $\alpha\sigma_{\text{asset}}W$, return $R_{\text{asset}}$ |
| Relative VaR | $\alpha\sigma_e W$, $e = R_{\text{asset}} - R_B$ ($\sigma_e$ = TEV) |
| Return decomposition (6.2) | $R_{\text{asset}} = \sum w_i^p R_i^b + \sum(w_i R_i - w_i^p R_i^b)$ |
| Policy-mix VaR | $\alpha W\sqrt{\sum_i \sum_j w_i^p w_j^p \sigma_i \sigma_j \rho_{ij}}$ (policy weights, benchmark risks) |
| Combining the two | $\text{VaR}^2_{\text{asset}} = \text{VaR}^2_{\text{policy}} + \text{VaR}^2_{\text{active}} + 2\rho\,\text{VaR}_{\text{policy}}\text{VaR}_{\text{active}}$ |
| Surplus return (6.3) | $R_S = R_{\text{asset}} - R_{\text{liabilities}}(L/A)$; $\Delta S = \Delta A - \Delta L$ |
| Duration effect | $\Delta A \approx -D_A A\Delta y$, $\Delta L \approx -D_L L\Delta y$ |
| Surplus distribution | $E[S_1] = A(1+R_A) - L(1+R_L)$; $\sigma_S^2 = (A\sigma_A)^2 + (L\sigma_L)^2 - 2\rho(A\sigma_A)(L\sigma_L)$ |
| Surplus at risk | from expected surplus $z\sigma_S$; worst-case surplus $E[S_1] - z\sigma_S$ |

| | Loss of what, against what | Driver | Measure |
|---|---|---|---|
| **Absolute** | Fund value against zero | All asset moves | $\alpha\sigma W$ (asset VaR) |
| **Relative** | Fund against its benchmark | Deviations from benchmark | $\alpha\sigma_e W$ (TEV) |
| **Policy-mix** | Passive policy portfolio | Strategic asset allocation | Policy-mix VaR |
| **Active mgmt** | Fund against policy mix | Managers' deviations, mix drift | Active VaR |
| **Funding** | Assets against liabilities | Asset-liability mismatch | Surplus VaR, SAR |
| **Sponsor** | Sponsor's contributions and earnings | Surplus and the firm's business | Cash-flow, economic risk |

**Exam traps**

- **Policy weights versus current weights (wrong input).** Policy-mix VaR uses policy weights on benchmark returns; the current mix's drift from policy belongs to active-management risk.
- **Additivity (absolute claim).** Policy-mix VaR plus active VaR is not the asset VaR (OTPPB: 19.6% and 1.6% give 19.3%).
- **Cash-flow versus economic risk (sibling swap).** Cash-flow risk is the fluctuation in contributions; economic risk is the variation in the sponsor's total economic earnings.
- **Who bears funding risk (role).** The owner (sponsor) in a defined-benefit plan; the employees in a defined-contribution plan.
- **Falling rates help the surplus (polarity flip).** Long-duration nominal liabilities usually gain more than the assets, so the surplus falls.
- **Sign of the covariance term.** $\sigma_S^2$ subtracts $2\rho(A\sigma_A)(L\sigma_L)$; adding it overstates surplus risk.
- **Reference point for SaR.** From the expected surplus, from the current surplus, or from zero: read which number the stem wants, and use the expected (not the current) surplus for the worst case.

- **Tracking error level or volatility (notation).** Jorion's tracking error is the return difference; its volatility is the TEV.

## Practice questions: IM-6 e

### Question IM-6 e.1

Bank Q311 • BT L2.T5.65.1

A risk analyst at a pension fund reviews an equity mandate with a value of USD 10 million that is benchmarked against an index. The mandate's return volatility is 10.0%, the benchmark's volatility is 5.0%, and the mandate's active risk (tracking error volatility) is 8.0%; all are normally distributed. What is the 95% tracking error VaR (relative VaR) of the mandate? Use $z = 1.645$.

**A.** USD 0.82 million

**B.** USD 1.32 million

**C.** USD 1.65 million

**D.** USD 1.96 million

### Question IM-6 e.2

Bank Q299 • BT P2.T8.13.1 (rebuilt)

A pension fund invests in only two asset classes, equities and fixed income. Its policy mix is 50% equities and 50% fixed income. Over the last period the benchmark return was 7.0% for equities and 5.0% for fixed income. The fund did not deviate from its policy mix (50% and 50%). The equity managers returned +4.0% and the fixed-income managers returned +10.0%. What is the fund's return attributable to active management?

**A.** $-1.5\%$

**B.** $0.0\%$

**C.** $+1.0\%$

**D.** $+2.5\%$

### Question IM-6 e.3

Workbook C4.1 • S1, S2, S5 • Map #35

A risk analyst at an Australian superannuation fund with AUD 8,000 million of assets is asked by the investment committee for the fund's one-year VaR at the 99% confidence level, calculated by the policy-mix method. Returns are assumed normal and the analyst uses $\alpha = 2.33$.

| Asset class | Policy weight | Current weight | Benchmark volatility | Managers' TEV |
|---|---|---|---|---|
| Domestic equities | 50% | 45% | 15% | 3.0% |
| Global equities | 20% | 20% | 17% | 4.0% |
| Fixed income | 30% | 35% | 6% | 1.5% |

Benchmark correlations: domestic/global equities 0.60; domestic equities/fixed income 0.10; global equities/fixed income 0.05. Which of the following is closest to the policy-mix VaR?

**A.** AUD 1,571 million

**B.** AUD 1,794 million

**C.** AUD 1,910 million

**D.** AUD 2,367 million

### Question IM-6 e.4

Bank Q286 • AnalystPrep Q.2485 (rebuilt)

A risk analyst at a large pension plan reviews the plan's annual VaR decomposition (same confidence level and horizon, as a percentage of assets):

| Source of risk | VaR |
|---|---|
| Policy-mix VaR | 19.6% |
| Active-management VaR | 1.6% |
| Total (asset) VaR | 19.3% |

Treating total VaR as the combination of the policy-mix VaR and the active-management VaR, the implied correlation between the two components is closest to:

**A.** $-0.45$

**B.** $-0.23$

**C.** $-0.03$

**D.** $+0.23$

### Question IM-6 e.5

Bank Q308 • Mock 1 Q71 (rebuilt)

A pension fund has plan assets and liabilities currently valued at USD 150 million and USD 120 million, respectively. The assets are fully invested in equities, while the liabilities resemble fixed-income obligations with an average duration of 14. Over the next year, the plan assets fall 30% and yields fall by 2 percentage points. Which of the following is the change in the surplus?

**A.** −USD 78.6 million

**B.** −USD 48.6 million

**C.** −USD 11.4 million

**D.** USD 78.6 million

### Question IM-6 e.6

Bank Q283 • AP Mock 1 Q66 (rebuilt)

The risk manager of an Indian pension fund reports the following (amounts in INR million):

| Item | Assets | Liabilities |
|---|---|---|
| Amount | 500 | 410 |
| Expected growth per year | 8% | 9% |
| Annual volatility of growth | 15% | 8% |

Annual asset returns and liability growth are jointly normal with a correlation of 0.7. Using a one-sided 95% confidence level ($z = 1.645$), the worst-case surplus value at the end of one year is closest to:

**A.** INR $-41.6$ million

**B.** INR $-18.8$ million

**C.** INR $-0.8$ million

**D.** INR 93.9 million

### Question IM-6 e.7

Bank Q306 • GARP PE2 Q72

A risk manager at a defined-benefit pension fund for a large automobile manufacturer is training newly hired analysts on the fund's risk measurement and management practices. The manager describes some of the types of risk that are most important for defined-benefit pension funds and provides examples of how each type of risk can be mitigated. Which of the following statements would be most appropriate for the manager to make?

**A.** Relative risk is mitigated when investment decisions made by portfolio managers decrease the fund's tracking error.

**B.** Cash-flow risk is minimized when the marginal VaR of all assets held by the fund is equal.

**C.** Funding risk is mitigated when the duration of the fund's liabilities is decreasing as interest rates are also decreasing.

**D.** Sponsor risk is eliminated when the policy mix selected by the fund generates the expected returns.

**Question IM-6 e.8** Workbook C4.3 • S2 • Map #35

The CRO of a Canadian defined-benefit teachers' pension plan is preparing a report for the board's risk committee, which has asked for a clear statement of how the plan's risk types differ. Which of the following statements is correct to include in the report?

**A.** Funding risk is measured on the surplus; because nominal pension liabilities behave like a short position in a long-term bond, a fall in interest rates can reduce the surplus even in a year when the plan's equities rise.

**B.** Sponsor risk has two parts: cash-flow risk, the variation in the total economic earnings of the plan sponsor, and economic risk, the year-to-year fluctuation in contributions to the fund.

**C.** Policy-mix risk and active-management risk are additive, so the plan's total asset VaR always equals the policy-mix VaR plus the active-management VaR reported for its managers.

**D.** Funding risk is borne by the plan sponsor in a defined-contribution plan, which must make up any shortfall of the fund's assets relative to the benefits promised to retirees.

## Solutions • IM-6 e

### Solution IM-6 e.1

**Answer: B**

Bank Q311 • BT L2.T5.65.1

**Step 1 – Choose the volatility**

Relative VaR is the VaR of the excess return over the benchmark, so the relevant volatility is the tracking error volatility, 8.0%.

**Step 2 – Relative VaR**

$$\text{VaR} = \alpha W \sigma_e = 1.645 \times 10 \times 0.08 = \text{USD } 1.316 \text{ million}$$

$\Rightarrow$ There is a 5% chance the mandate underperforms its benchmark by more than about USD 1.32 million.

**Interpretation**

The 10% and 5% volatilities are red herrings for a relative VaR; together with the 8% TEV they imply a correlation of 0.61 between the mandate and the index. The absolute VaR, $1.645 \times 0.10 \times 10 = 1.65$, answers a different question (loss against zero).

**Why the other options are wrong**

**A is incorrect.** Used the benchmark's volatility: $1.645 \times 0.05 \times 10 = 0.82$.
**C is incorrect.** Used the mandate's total volatility: $1.645 \times 0.10 \times 10 = 1.65$. That is absolute VaR.
**D is incorrect.** Total volatility with the two-tailed $z = 1.96$: $1.96 \times 0.10 \times 10 = 1.96$.

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2 Absolute and Relative Risks

### Solution IM-6 e.2

**Answer: C**

Bank Q299 • BT P2.T8.13.1

**Step 1 – Fund return**

$$R_{\text{asset}} = 0.5 \times 4.0\% + 0.5 \times 10.0\% = 7.0\%$$

**Step 2 – Policy-mix return (Eq. 6.2, first term)**

$$\sum w_i^p R_i^b = 0.5 \times 7.0\% + 0.5 \times 5.0\% = 6.0\%$$

**Step 3 – Active-management return**

$$7.0\% - 6.0\% = +1.0\% = 0.5(4-7) + 0.5(10-5) = -1.5\% + 2.5\%$$

$\Rightarrow$ The managers added 1.0% in total: the equity managers lost 1.5% of fund value against their benchmark and the fixed-income managers gained 2.5%.

**Interpretation**

Holding the policy weights does not remove active-management risk: the managers' deviations from their benchmarks remain. With weights equal to policy, the whole active return comes from the managers; if the weights had drifted, part of it would come from the mix.

**Why the other options are wrong**

**A is incorrect.** Counted only the equity managers: $0.5 \times (4\% - 7\%) = -1.5\%$.
**B is incorrect.** Assumed that sticking to the policy mix means no active return.
**D is incorrect.** Counted only the fixed-income managers: $0.5 \times (10\% - 5\%) = +2.5\%$.

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2, Eq. 6.1 and 6.2

### Solution IM-6 e.3 — Answer: C

Workbook C4.1 • S1, S2, S5

#### Step 1 – Choose the inputs

Policy-mix risk is "the risk of a dollar loss owing to the policy mix selected by the fund": the first term of Eq. 6.2, $\sum_i w_i^p R_i^b$. So the weights are the policy weights and the volatilities are the benchmark volatilities. The current weights and the managers' TEVs describe active-management risk and are not inputs here.

#### Step 2 – Variance of the policy-mix return

$$\begin{aligned}\sigma_p^2 &= (0.50 \times 0.15)^2 + (0.20 \times 0.17)^2 + (0.30 \times 0.06)^2 \\ &+ 2(0.50)(0.20)(0.60)(0.15)(0.17) + 2(0.50)(0.30)(0.10)(0.15)(0.06) \\ &+ 2(0.20)(0.30)(0.05)(0.17)(0.06) \\ &= 0.005625 + 0.001156 + 0.000324 + 0.003060 + 0.000270 + 0.000061 = 0.010496 \\ \sigma_p &= \sqrt{0.010496} = 10.25\%\end{aligned}$$

$\Rightarrow$ A passive implementation of the policy mix would have an annual volatility of about 10.25%.

#### Step 3 – Policy-mix VaR

$$\text{VaR}_{\text{policy}} = \alpha \sigma_p W = 2.33 \times 0.10245 \times 8{,}000 = 1{,}909.7$$

$\Rightarrow$ There is a 1% chance that the chosen policy mix alone loses more than about AUD 1,910 million over a year.

#### Interpretation

The individual risk budgets are $2.33 \times 0.15 \times 0.50 \times 8{,}000 = 1{,}398$, $2.33 \times 0.17 \times 0.20 \times 8{,}000 = 634$ and $2.33 \times 0.06 \times 0.30 \times 8{,}000 = 336$; their sum, 2,367, is the undiversified VaR, and diversification removes about 458. The managers' TEVs would feed a separate active-management VaR; as at OTPPB, the fund's asset VaR need not equal the sum of the two, so the committee should not add them.

#### Why the other options are wrong

**A is incorrect.** Ignored the correlations (treated benchmark returns as uncorrelated): $\sqrt{0.005625 + 0.001156 + 0.000324} = 8.43\%$, $2.33 \times 0.0843 \times 8{,}000 = 1{,}571$.
**B is incorrect.** Used the current weights (45/20/35) instead of the policy weights: volatility 9.62%, VaR $2.33 \times 0.0962 \times 8{,}000 = 1{,}794$. The tilt away from policy is a deviation from the policy mix, which Jorion places in active-management risk.
**D is incorrect.** Summed the individual risk budgets, $1{,}398 + 634 + 336 = 2{,}367$: the undiversified VaR, as if all correlations were 1.

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2 Eq. 6.1, 6.2 and 6.5

### Solution IM-6 e.4 Answer: B

Bank Q286 • AnalystPrep Q.2485

**Step 1 – Combine the two VaRs like two positions**

$$\mathrm{VaR}_T^2 = \mathrm{VaR}_P^2 + \mathrm{VaR}_A^2 + 2\rho\,\mathrm{VaR}_P\mathrm{VaR}_A$$

**Step 2 – Solve for $\rho$**

$$19.3^2 - 19.6^2 - 1.6^2 = 372.49 - 384.16 - 2.56 = -14.23, \qquad 2 \times 19.6 \times 1.6 = 62.72$$

$$\rho = -14.23/62.72 = -0.227 \approx -0.23$$

$\Rightarrow$ The active deviations are slightly negatively correlated with the policy mix.

**Interpretation**

These are the OTPPB figures in GARP. Total VaR (19.3%) is below the simple sum (21.2%) and even below the policy-mix VaR alone (19.6%): the active bets diversify the policy risk. Jorion's conclusion is that the managers could then take greater deviations from their benchmarks without raising the plan's total VaR.

**Why the other options are wrong**

**A is incorrect.** Dropped the factor 2 in the cross term: $-14.23/31.36 = -0.45$.
**C is incorrect.** Worked with the VaRs instead of their squares: $(19.3 - 19.6 - 1.6)/62.72 = -0.03$.
**D is incorrect.** Sign error: a total below the root-sum-of-squares value (19.67%) needs a negative correlation.

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2 Policy Mix and Active Management Risk (OTPPB decomposition)

### Solution IM-6 e.5 Answer: A

Bank Q308 • Mock 1 Q71

**Step 1 – Change in assets**

$$\Delta A = 150 \times (-0.30) = -45.0$$

**Step 2 – Change in liabilities (duration approximation)**

$$\Delta L \approx -D_L L \Delta y = -14 \times 120 \times (-0.02) = +33.6$$

$\Rightarrow$ Falling yields raise the present value of the liabilities: they behave like a short position in a long-term bond.

**Step 3 – Change in surplus**

$$\Delta S = \Delta A - \Delta L = -45.0 - 33.6 = -78.6$$

$\Rightarrow$ The surplus moves from $+30$ to $105 - 153.6 = -48.6$: a fall of USD 78.6 million.

**Interpretation**

Both legs hurt at once: equities fall and the liabilities rise. This is the funding-risk scenario Jorion describes, in which lower rates increase the liabilities by more than any gain on the asset side.

**Why the other options are wrong**

**B is incorrect.** Reported the new surplus level ($30 - 78.6 = -48.6$), not the change.
**C is incorrect.** Treated the liability increase as a gain: $-45.0 + 33.6 = -11.4$.
**D is incorrect.** Right size, wrong sign: the surplus falls.

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2 Funding Risk, Eq. 6.3

## Solution IM-6 e.6

**Answer: C**

Bank Q283 • AP Mock 1 Q66

### Step 1 – Expected surplus at the end of the year

$$E[S_1] = 500(1.08) - 410(1.09) = 540 - 446.9 = 93.1$$

### Step 2 – Volatility of the surplus

$$A\sigma_A = 500 \times 15\% = 75, \qquad L\sigma_L = 410 \times 8\% = 32.8$$

$$\sigma_S = \sqrt{75^2 + 32.8^2 - 2(0.7)(75)(32.8)} = \sqrt{5{,}625 + 1{,}075.8 - 3{,}444} = \sqrt{3{,}256.8} = 57.07$$

$\Rightarrow$ The positive correlation lets the liabilities offset part of the asset risk.

### Step 3 – Worst-case surplus at 95%

$$E[S_1] - z\sigma_S = 93.1 - 1.645 \times 57.07 = 93.1 - 93.88 = -0.78 \approx -0.8$$

$\Rightarrow$ With 5% probability the plan ends the year in a small deficit of about INR 0.8 million or worse.

### Interpretation

Read the reference point. The surplus VaR, measured from the expected surplus, is $z\sigma_S = 93.9$; the worst-case surplus subtracts it from the expected surplus of 93.1, not from the current surplus of 90.

### Why the other options are wrong

**A is incorrect.** Ignored the correlation: $\sigma_S = \sqrt{75^2 + 32.8^2} = 81.9$, so $93.1 - 1.645 \times 81.9 = -41.6$.
**B is incorrect.** Used the two-sided $z = 1.96$: $93.1 - 1.96 \times 57.07 = -18.8$.
**D is incorrect.** Reported only $z\sigma_S = 93.9$, the surplus VaR, without subtracting it from the expected surplus.

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2 Funding Risk, Eq. 6.3 and the PERF example; SchweserNotes 2026 Book 5, Reading 88, LO 88.e

## Solution IM-6 e.7

**Answer: A**

Bank Q306 • GARP PE2 Q72

### Step 1 – Match each risk to its definition

Relative risk: loss against the benchmark, measured by the tracking error. Cash-flow risk: fluctuations in contributions. Funding risk: assets falling short of liabilities. Sponsor risk: cash-flow and economic risk of the sponsor.

### Step 2 – Test each mitigation

Lowering tracking error lowers the volatility of the return against the benchmark, which is relative risk.
⇒ A is the appropriate statement.

### Interpretation

Each wrong option attaches a real tool or event to the wrong risk. Equal marginal VaRs belong to the minimum-risk portfolio (IM-5 d); falling rates hurt a defined-benefit plan's surplus; and no policy mix "generates" its expected return with certainty.

### Why the other options are wrong

**B is incorrect.** Equal marginal VaRs is the condition for a minimum-risk portfolio; it says nothing about the year-to-year variation in the sponsor's contributions.
**C is incorrect.** Falling rates raise the value of nominal liabilities, often by more than the assets, so funding risk worsens; a shorter liability duration in that scenario is not a mitigation the fund controls.
**D is incorrect.** Sponsor risk is never eliminated: returns are uncertain, and the risk also depends on the sponsor's own earnings.

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2 What Are the Risks?; GARP Practice Exam 2, Q72

## Solution IM-6 e.8 — Answer: A

Workbook C4.3 • S2

### Step 1 – Why A is correct

Funding risk is "the risk that the value of assets will not be sufficient to cover the liabilities of the fund"; the relevant variable is the surplus $S = A - L$, with $R_S = R_{\text{asset}} - R_{\text{liabilities}}(L/A)$. Jorion: when liabilities consist mainly of nominal payments, "their value in general will behave like a short position in a long-term bond. Thus decreases in interest rates, while beneficial for equities on the asset side, can increase even more the value of liabilities, thereby negatively affecting the surplus."
⇒ The board should judge the plan on the surplus, not on asset returns alone.

### Interpretation

The minimum-risk position is an immunized portfolio matching asset and liability durations, at an opportunity cost.

### Why the other options are wrong

**B is incorrect.** Sibling swap: the labels are reversed. Cash-flow risk is "the risk of year-to-year fluctuations in contributions"; economic risk is "the risk of variation in total economic earnings of the plan sponsor".
**C is incorrect.** Absolute claim: total VaR comes from the policy-mix VaR, the active-management VaR and a cross-product term. At OTPPB, 19.6% and 1.6% gave an asset VaR of 19.3% because of a slightly negative correlation.
**D is incorrect.** Role misassignment: a defined-benefit plan's shortfall "will have to be made up by the fund's owner"; "defined-contribution plans put the risk on the employees".

**LO:** IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. • **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.2 (Funding Risk, Sponsor Risk, Policy Mix and Active Management Risk)

IM-6 • VaR and Risk Budgeting in Investment Management • Vardeez: VERY IMPORTANT

# IM-6 h

P2 BOTH

Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.

Workbook: C4 Policy-Mix VaR and Risk Budgeting • Tier C • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** Vardeez flags risk budgeting calculations (his "LO 88.h") in a VERY IMPORTANT chapter; the workbook section C4 is Tier C (Map #35, 4 of 16), with "3-4 on portfolio return computation" and "IR/2 active risk numerical" reported.
**Vardeez:** risk budgeting calculations.

**Reported exam shapes**

- **C4·S3** Portfolio return computation: the expected excess return of a pool of managers.
- **C4·S4** IR and active risk numerical: IR from expected excess return and TEV; the portfolio IR.
- **C4·S5** Risk budgets across asset classes and across active managers.

**What the questions look like:** mostly numerical. Turn a volatility target into a fund VaR budget, convert weights into individual risk budgets, split a class budget between correlated managers, combine independent managers' TEVs, and allocate across managers in proportion to their information ratios with the slack in the index. One or two ask you to describe the top-down process. Background on the policy mix and on SAR is in IM-6 e.

## 1. The risk budgeting process

**Risk budgeting** is "the process of allocating and managing risk using a top-down approach to different aspects of the investment process". It is "equivalent to a top-down allocation of economic risk capital starting from the asset classes down to the choice of the active manager and even to the level of individual securities", and it gives "a dynamic comparison of current risk profiles with prespecified risk budgets". What is allocated is *risk*, not capital: a manager with less money but a riskier mandate can receive the larger budget.

1. **Set the total.** Translate the board's risk appetite into a fund VaR: a volatility target ($\text{VaR} = \alpha\sigma W$) or, for a pension fund, the SAR of IM-6 e (PERF "allocates the SAR of $220 million to different aspects of the investment process").
2. **Allocate across asset classes** (Section 2), then **across managers within each class** (Section 3).
3. **Monitor.** Managers within their budgets "can execute new transactions without requiring approval of senior management": no micromanaging. A breach is flagged and examined (IM-6 f).

## 2. Budgeting across asset classes

PERF's board sets a total volatility of 10%: VaR $= 2.33 \times 0.10 \times 1{,}000 =$ USD 233 million at 99%. The optimal weights (from Table 6.2's returns, volatilities and correlations) are then converted into **risk budgets, or individual VaRs**, $\alpha\sigma_i w_i W$:

| Asset class | Weight | Volatility | Principal | Risk budget |
|---|---|---|---|---|
| U.S. stocks | 52.5% | 15.1% | 525 | 184 |
| Non-U.S. stocks | 10.4% | 16.7% | 104 | 41 |
| U.S. bonds | 12.2% | 7.3% | 122 | 21 |
| Non-U.S. bonds | 24.9% | 10.9% | 249 | 63 |
| Sum of budgets (undiversified VaR) | | | | 308 |
| Fund (diversified) VaR | 100% | 10.0% | 1,000 | 233 |

"The risk budgets sum to $308 million, which is the undiversified VaR. The actual fund VaR is lower, at $233 million, owing to diversification effects." (From the rounded weights printed, the budgets are 184.7, 40.5, 20.8 and 63.2; the table's inputs give a fund volatility of 10.05%.)

**The next level: managers inside a class.** PERF gives its USD 525 million of U.S. stocks to two equally good managers, USD 262.5 million each, with volatility 16% and correlation 0.78. Each gets $2.33 \times 0.16 \times 262.5 =$ USD 98 million, and together

$$\sqrt{98^2 + 98^2 + 2(0.78)(98)(98)} = 98\sqrt{3.56} \approx \text{USD } 184 \text{ million},$$

the class budget. Two equal budgets $V$ with correlation $\rho$ combine to $V\sqrt{2+2\rho}$, so a class budget $V_T$ splits into $V = V_T/\sqrt{2+2\rho}$ each.

**Is a proposed addition inside the budget?** (Vardeez, SchweserNotes.) A manager holds USD 500 million of W (volatility 10%) and may add USD 500 million of X (9%, correlation 0.7 with W) or Y (12%, correlation 0) under a 99% VaR limit of USD 200 million. On the new USD 1,000 million: with X, $\sigma = 8.76\%$ and VaR 204 (over); with Y, $\sigma = 7.81\%$ and VaR 182 (inside). Stand-alone VaRs (X 104.9, Y 139.8) would have picked the wrong asset.

> **Correction to the source notes.** Bank Q338 (GARP PE2 Q20, Mock 4 Q21) was rebuilt with the risk budget of each asset class "defined as the amount of fund VaR attributable to it (taking diversification into account)" and keyed C, the fund VaR of EUR 11.63 million. GARP's own text defines the risk budgets as the classes' individual VaRs, which "sum to ... the undiversified VaR". Under that definition the sum of the four budgets is EUR 22.10 million, and the fund VaR is the smaller, diversified figure. Question IM-6 h.2 uses the stem without the bank's added definition and is keyed to GARP's definition (D).

## 3. Budgeting across active managers (shapes C4·S3 and S4)

Active managers are evaluated on their **tracking error**, which Jorion defines as the active return minus the benchmark return. With $\mu_i$ its expected value and $\omega_i$ its volatility (TEV), the **information ratio** is (Eq. 6.4)

$$\text{IR}_i = \mu_i/\omega_i \qquad \text{(Grinold and Kahn: an IR of 0.50 is "good", top quartile).}$$

A fund puts a fraction $x_i$ with manager $i$ and the rest in the index, whose TEV is zero. With the managers' active returns **independent**:

$$\mu_P = \sum_i x_i\mu_i \;\; (6.5), \qquad \omega_P = \sqrt{\textstyle\sum_i x_i^2\omega_i^2} \;\; (6.6)$$

$$\boxed{x_i\omega_i = \frac{\mathrm{IR}_i}{\mathrm{IR}_P}\,\omega_P} \quad (6.7), \qquad \mathrm{IR}_P = \sqrt{\textstyle\sum_i \mathrm{IR}_i^2}$$

Maximizing the fund's IR for a fixed TEV gives Eq. 6.7: **relative risk budgets $x_i\omega_i$ are proportional to the information ratios**. The weights need not sum to 1; the remainder goes to the index. The fund's expected excess return is $\mu_P = \mathrm{IR}_P\,\omega_P$.

**GARP Table 6.4.** PERF allocates USD 525 million to two managers, each with TEV 6%, IRs 0.60 and 0.40, for a fund TEV of 4% (a 99% risk budget of USD 48.9 million).

$$\mathrm{IR}_P = \sqrt{0.60^2 + 0.40^2} = 0.7211, \qquad x_1 = \frac{0.60}{0.7211} \times \frac{4\%}{6\%} = 55.5\%$$

$$x_2 = \frac{0.40}{0.7211} \times \frac{4\%}{6\%} = 37.0\%, \qquad \text{index} = 100\% - 55.5\% - 37.0\% = 7.5\%$$

| | TEV | IR | Weight | Principal | Excess return $x_i\mu_i$ | Relative risk budget |
|---|---|---|---|---|---|---|
| Manager 1 | 6.0% | 0.60 | 55% | 291 | $0.555 \times 3.6\% = 2.0\%$ | 40.6 |
| Manager 2 | 6.0% | 0.40 | 37% | 194 | $0.370 \times 2.4\% = 0.9\%$ | 27.1 |
| Index | 0.0% | 0.00 | 8% | 40 | 0.0% | 0.0 |
| Fund | 4.0% | 0.72 | 100% | 525 | 2.9% | 48.9 |

The managers' expected excess returns are $\mu_i = \mathrm{IR}_i\omega_i$ (3.6% and 2.4%). The fund expects 2.9%, an IR of $2.9/4.0 = 0.72$: higher than either manager's, because the independent active returns diversify. The relative risk budgets are $\alpha x_i\omega_i W$ (GARP's table uses $\alpha = 2.326$; with 2.33 the first is 40.7). Their ratio, $40.6/27.1 = 1.5$, equals the ratio of the IRs, $0.60/0.40$. With equal IRs the fund IR is $\mathrm{IR}\sqrt{N}$, the **law of active management**; in practice it is hard to find many truly unrelated strategies. Jorion adds that budgeting on tracking error alone is "second-best because it fails to control total risk".

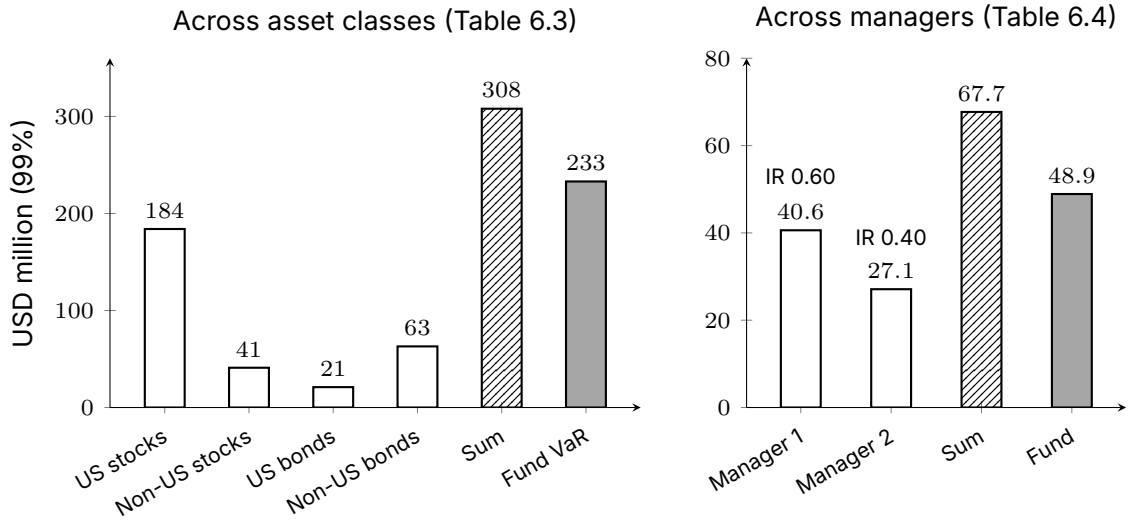



Figure 3. Left: individual risk budgets (white) sum to the undiversified VaR of 308 (hatched); the fund's diversified VaR is 233 (grey). Right: relative risk budgets (white) in proportion to the information ratios, 40.6 : 27.1 = 0.60 : 0.40. Their sum (hatched, 67.7) exceeds the fund's 4% TEV budget of 48.9 (grey) because independent active bets diversify.

## Formula card: IM-6 h

| | |
|---|---|
| Fund VaR budget | $\alpha\sigma_{\text{target}}W$ (or the SAR of IM-6 e) |
| Asset-class risk budget | $\alpha\sigma_i w_i W$ (individual VaR); $\sum$ budgets = undiversified VaR $\geq$ fund VaR |
| Two managers, equal budgets $V$ | combined $V\sqrt{2+2\rho}$; so $V = V_T/\sqrt{2+2\rho}$ |
| Information ratio (6.4) | $\text{IR}_i = \mu_i/\omega_i$ ($\mu$ expected tracking error, $\omega$ TEV) |
| Fund excess return (6.5) | $\mu_P = \sum_i x_i\mu_i = \text{IR}_P\,\omega_P$ |
| Fund TEV, independent (6.6) | $\omega_P = \sqrt{\sum_i x_i^2\omega_i^2}$ |
| Optimal budgets (6.7) | $x_i\omega_i = (\text{IR}_i/\text{IR}_P)\,\omega_P$, $x_i = (\text{IR}_i/\text{IR}_P)(\omega_P/\omega_i)$; index $= 1 - \sum x_i$ |
| Fund IR, independent | $\text{IR}_P = \sqrt{\sum_i \text{IR}_i^2}$; equal IRs: $\text{IR}\sqrt{N}$ |
| Relative risk budget (money) | $\alpha x_i\omega_i W$ |

## Exam traps

- **Sum of budgets as the fund VaR (sibling swap).** Individual risk budgets add to the undiversified VaR (308), not to the fund VaR (233).
- **Splitting a budget by 2 (dropped correlation).** Two equal budgets combine as $V\sqrt{2+2\rho}$: divide by 2 only if $\rho = 1$, by $\sqrt{2}$ only if $\rho = 0$.
- **Portfolio IR as a sum or average (absolute claim).** With independent managers it is $\sqrt{\sum \text{IR}_i^2}$, which exceeds every manager's IR.
- **Weights forced to 100% (scope).** The optimal manager weights usually sum to less than 1; the slack in the index is what hits the TEV target exactly.
- **Budgets proportional to money or to expected return (wrong input).** Relative risk budgets $x_i\omega_i$ are proportional to IRs; the weights $x_i$ are proportional to $\text{IR}_i/\omega_i$.
- **Inverted ratios in Eq. 6.7.** $x_i = (\text{IR}_i/\text{IR}_P)(\omega_P/\omega_i)$: the manager's own TEV is in the denominator.
- **Combining TEVs linearly (dropped diversification).** Independent active risks add in squares: $\sqrt{\sum(x_i\omega_i)^2}$.

## Practice questions: IM-6 h

### Question IM-6 h.1

Bank Q270 • BT P2.T8.408.3 (rebuilt) • re-filed from IM-6 a

Which of the following statements best describes risk budgeting as applied by a large investment fund?

**A.** A top-down process in which the total acceptable portfolio VaR is set first and then allocated to asset classes and managers, taking diversification effects into account

**B.** A bottom-up process in which stand-alone VaRs are computed for each position and simply added to obtain the fund's risk level

**C.** A process equivalent to market value allocation, in which capital is divided among asset classes in proportion to their weights in the benchmark

**D.** A top-down process in which the total VaR budget is divided so that the stand-alone VaRs of the asset classes sum exactly to the fund's total VaR limit

### Question IM-6 h.2

Bank Q338 • GARP PE2 Q20 • Mock 4 Q21 (rebuilt; key changed, see the correction box)

An investment fund uses risk budgeting with delta-normal VaR at the 99% confidence level ($z = 2.326$). The fund has EUR 100 million of principal and maintains a total portfolio volatility of 5% (annual). The fund allocates this risk across four asset classes, converting its allocation into a risk budget for each class. Information on the asset classes:

| Asset class | Weight | Volatility |
|---|---|---|
| European stocks | 40% | 9.25% |
| Non-European stocks | 12% | 11.91% |
| European bonds | 22% | 5.76% |
| Non-European bonds | 26% | 11.94% |

The pairwise correlations between the asset classes are all between $-0.07$ and $+0.02$. What is the sum of the risk budgets allocated to the four asset classes?

**A.** EUR 5.00 million

**B.** EUR 8.23 million

**C.** EUR 11.63 million

**D.** EUR 22.10 million

### Question IM-6 h.3

Bank Q326 • AP Mock 1 Q62

The BC&C pension fund has 60% (about USD 21 billion) of its assets invested in equities. It measures the absolute risk of the equity allocation with a 95% one-year VaR of USD 3.4545 billion, assuming a normal distribution and 10% annual volatility. The fund wishes to allocate this risk to two equity managers, each with the same VaR budget. The correlation between the managers' returns is 0.5. The VaR budget for each manager is closest to:

**A.** USD 1.0731 billion

**B.** USD 1.9945 billion

**C.** USD 2.4427 billion

**D.** USD 3.4545 billion

### Question IM-6 h.4

Bank Q339 • Mock 3 Q15

A pension fund allocates its active risk budget across three external managers, each running an independent strategy benchmarked to the same index. The fund's chief investment officer gathers the following data:

| Manager | Allocation | Tracking error |
|---|---|---|
| Manager A | 50% | 4.0% |
| Manager B | 30% | 6.0% |
| Manager C | 20% | 3.0% |

Assuming the managers' active returns are uncorrelated with one another, what is the tracking error of the fund as a whole?

**A.** 2.76%

**B.** 4.33%

**C.** 4.40%

**D.** 7.81%

### Question IM-6 h.5

Bank Q327 • AP Mock 3 Q5

A pension fund wants to allocate USD 400 million across a pool of active managers so as to maximize the information ratio of the fund, subject to an overall tracking error volatility (TEV) of 4%. Any balance not allocated to active managers is placed in an index fund with zero tracking error.

| Item | TEV | Information ratio |
|---|---|---|
| Manager 1 | 8.0% | 0.48 |
| Manager 2 | 6.0% | 0.36 |
| Index fund | 0.0% | 0.00 |

Assuming the managers' excess returns are independent of each other, the optimal allocation to the index fund is equal to:

**A.** USD 0

**B.** USD 80 million

**C.** USD 160 million

**D.** USD 320 million

### Question IM-6 h.6

Workbook C4.2 • S3, S4, S5 • Map #35

A senior analyst at a Dutch pension fund is setting risk budgets for the EUR 600 million domestic equity sleeve. The sleeve can be split between two active managers and a passive index fund (TEV 0%). The investment committee wants to maximize the sleeve's information ratio subject to an overall tracking-error volatility of 3%. The managers' active returns are assumed independent of each other.

- Manager X: expected excess return over the benchmark 2.5%; TEV 5%
- Manager Y: expected excess return over the benchmark 2.4%; TEV 8%

Using the optimal relative risk budgets, which of the following is closest to the weight allocated to Manager Y and the expected excess return of the sleeve?

**A.** Weight to Y 14.1%; expected excess return 1.28%

**B.** Weight to Y 19.3%; expected excess return 1.75%

**C.** Weight to Y 19.3%; expected excess return 1.20%

**D.** Weight to Y 37.5%; expected excess return 2.46%

### Question IM-6 h.7

Bank Q337 • BT P2.T9.21.8.3 (rebuilt)

A pension fund sub-allocates assets to two active managers whose excess returns are independent. It maximizes the portfolio information ratio subject to an overall TEV of 3.0%; any remainder is invested in the benchmark index. The proposed allocation is:

| Holding | TEV | IR | Weight | Relative risk budget (weight × TEV) |
|---|---|---|---|---|
| Manager 1 | 5.0% | 0.63 | 36.0% | 1.80% |
| Manager 2 | 7.0% | 0.84 | 34.3% | 2.40% |
| Index | 0.0% | 0.00 | 29.7% | 0.00% |
| Portfolio | 3.0% | 1.05 | 100.0% | |

Is this optimization correct (best answer)?

**A.** No, the index allocation should be zero; otherwise the information ratio is not maximized

**B.** No, the overall TEV of 3.0% cannot be less than each manager's TEV (5.0% and 7.0%)

**C.** Yes, because each weight satisfies the optimal condition, so that relative risk budgets are proportional to the information ratios

**D.** Yes, because the index is allocated a relative risk budget of zero

# Solutions • IM-6 h

## Solution IM-6 h.1

**Answer: A**

Bank Q270 • BT P2.T8.408.3

### Step 1 – Direction of the process

Risk budgeting is top-down: the fund first sets the total risk it will accept (a VaR or SAR), then allocates it to asset classes, managers and even securities.

### Step 2 – What is allocated, and how it adds up

Risk, not capital, is allocated. Unless the classes are perfectly correlated, the stand-alone VaRs add to more than the fund VaR, so the allocation must recognize diversification.

$\Rightarrow$ Only A states both points.

### Interpretation

GARP: risk budgeting is "equivalent to a top-down allocation of economic risk capital". PERF's budgets illustrate the second point: individual budgets of 308 sit under a fund VaR of 233.

### Why the other options are wrong

**B is incorrect.** Bottom-up aggregation that also ignores diversification: the reverse of the process.

**C is incorrect.** Market value allocation: risk budgeting allocates risk, which differs from allocating capital.

**D is incorrect.** Top-down is right, but the stand-alone VaRs need not sum to the fund limit: they sum to the undiversified VaR, which exceeds the fund VaR.

**LO:** IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.

• **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), introduction and 6.5 Risk Budgeting

## Solution IM-6 h.2

**Answer: D**

Bank Q338 • GARP PE2 Q20 • Mock 4 Q21

### Step 1 – Risk budget of each class (Jorion, Table 6.3: individual VaR)

$$\alpha\,\sigma_i w_i W: \quad 2.326 \times 0.0925 \times 40 = 8.61, \quad 2.326 \times 0.1191 \times 12 = 3.32$$
$$2.326 \times 0.0576 \times 22 = 2.95, \quad 2.326 \times 0.1194 \times 26 = 7.22$$

### Step 2 – Sum of the budgets

$$8.61 + 3.32 + 2.95 + 7.22 = \text{EUR } 22.10 \text{ million}$$

$\Rightarrow$ This is the undiversified VaR, the sum of the classes' stand-alone VaRs.

### Step 3 – Compare with the fund VaR

$$\text{VaR}_{\text{fund}} = 2.326 \times 0.05 \times 100 = \text{EUR } 11.63 \text{ million}$$

$\Rightarrow$ The fund VaR is about half the sum because the near-zero correlations diversify the classes away.

### Interpretation

This mirrors GARP's PERF table, where budgets of 184, 41, 21 and 63 sum to the undiversified 308 against a fund VaR of 233. The correlations are not needed for the budgets; they only explain why the 5% target volatility is far below the 9.5% weighted-average volatility (zero correlations would give 5.19%). See the correction box in Section 2 for the bank's different key.

**Why the other options are wrong**

**A is incorrect.** Forgot the $z$-value: $\sigma_P W = 0.05 \times 100 = 5.00$.
**B is incorrect.** Fund VaR with the 95% $z$ of 1.645: $1.645 \times 0.05 \times 100 = 8.23$.
**C is incorrect.** The fund's diversified VaR, $2.326 \times 0.05 \times 100 = 11.63$: the total the budgets are set against, not the sum of the budgets.

**LO:** IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.
• **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.5 Budgeting across Asset Classes, Table 6.3; GARP Practice Exam 2, Q20

## Solution IM-6 h.3

**Answer: B**

Bank Q326 • AP Mock 1 Q62

### Step 1 – Check the total

$$1.645 \times 0.10 \times 21 = \text{USD } 3.4545 \text{ billion}$$

### Step 2 – Two equal budgets $x$ with $\rho = 0.5$

$$\text{VaR}^2 = x^2 + x^2 + 2(0.5)x^2 = 3x^2 \quad \Rightarrow \quad x = \frac{3.4545}{\sqrt{3}} = \text{USD } 1.9945 \text{ billion}$$

$\Rightarrow$ Each manager receives more than half the total, because their imperfect correlation diversifies.

### Interpretation

The general split is $V_T/\sqrt{2+2\rho}$: at $\rho = 1$ each budget is half the total (1.727); at $\rho = 0$ it is $V_T/\sqrt{2}$. PERF's two U.S. equity managers (98 each, $\rho = 0.78$, class budget 184) are the same calculation in reverse.

### Why the other options are wrong

**A is incorrect.** Forgot to square the total VaR: solved $3x^2 = 3.4545$, so $x = \sqrt{3.4545/3} = 1.0731$. It is even below the perfect-correlation budget of 1.727, which is impossible.
**C is incorrect.** Assumed zero correlation: $3.4545/\sqrt{2} = 2.4427$.
**D is incorrect.** The total VaR, not divided between the managers.

**LO:** IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.
• **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.5 Budgeting across Asset Classes (two U.S. stock managers)

## Solution IM-6 h.4

**Answer: A**

Bank Q339 • Mock 3 Q15

### Step 1 – Weighted tracking errors

$$0.50 \times 4.0\% = 2.00\%, \qquad 0.30 \times 6.0\% = 1.80\%, \qquad 0.20 \times 3.0\% = 0.60\%$$

### Step 2 – Uncorrelated active returns: variances add (Eq. 6.6)

$$\omega_P = \sqrt{2.00^2 + 1.80^2 + 0.60^2} = \sqrt{4.00 + 3.24 + 0.36} = \sqrt{7.60} = 2.76\%$$

$\Rightarrow$ The fund's tracking error is below each manager's, because independent active bets diversify.

### Interpretation

This diversification is what lifts the fund's IR above the managers' IRs in Table 6.4. With perfectly correlated managers the TEVs would add linearly (4.40%).

### Why the other options are wrong

**B is incorrect.** Simple average of the three tracking errors, $(4+6+3)/3 = 4.33\%$: ignores both the weights and the diversification.
**C is incorrect.** Added the weighted tracking errors, $2.00 + 1.80 + 0.60 = 4.40\%$: the perfect-correlation case.
**D is incorrect.** Combined the unweighted tracking errors in quadrature, $\sqrt{16+36+9} = 7.81\%$: forgot the allocations.

**LO:** IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.
• **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.5 Budgeting across Active Managers, Eq. 6.6

## Solution IM-6 h.5

**Answer: B**

Bank Q327 • AP Mock 3 Q5

### Step 1 – Portfolio IR

$$\text{IR}_P = \sqrt{0.48^2 + 0.36^2} = \sqrt{0.36} = 0.60$$

### Step 2 – Optimal weights (Eq. 6.7)

$$x_1 = \frac{0.48}{0.60} \times \frac{4\%}{8\%} = 40\%, \qquad x_2 = \frac{0.36}{0.60} \times \frac{4\%}{6\%} = 40\%$$

$\Rightarrow$ Check: $\sqrt{(0.4 \times 8\%)^2 + (0.4 \times 6\%)^2} = \sqrt{3.2^2 + 2.4^2} = 4.0\%$, the TEV target.

### Step 3 – Index allocation

$$1 - 0.40 - 0.40 = 20\%, \qquad 0.20 \times 400 = \text{USD } 80 \text{ million}$$

### Interpretation

The relative risk budgets are $3.2\%$ and $2.4\%$, in the ratio $0.48 : 0.36$ of the IRs. The index holds the slack that brings the fund's TEV down to exactly 4%.

### Why the other options are wrong

**A is incorrect.** Assumed the managers absorb all the capital; the optimal weights sum to only 80%.
**C is incorrect.** The allocation to each manager $(0.40 \times 400)$, not to the index.
**D is incorrect.** The total allocated to the two managers $(0.80 \times 400)$.

**LO:** IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.
• **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.5, Eq. 6.6 and 6.7, Table 6.4

## Solution IM-6 h.6

**Answer: B**

Workbook C4.2 • S3, S4, S5

### Step 1 – Managers' information ratios

$$\text{IR}_X = \frac{\mu_X}{\omega_X} = \frac{2.5\%}{5\%} = 0.50, \qquad \text{IR}_Y = \frac{2.4\%}{8\%} = 0.30$$

$\Rightarrow$ X earns 0.50 units of excess return per unit of active risk ("good", top quartile); Y earns 0.30.

### Step 2 – Portfolio IR with independent managers

$$\text{IR}_P = \sqrt{0.50^2 + 0.30^2} = \sqrt{0.34} = 0.5831$$

**Step 3 – Relative risk budgets and weights (Eq. 6.7)**

$$x_Y\omega_Y = \frac{\text{IR}_Y}{\text{IR}_P}\omega_P = \frac{0.30}{0.5831}(3\%) = 1.5435\% \;\Rightarrow\; x_Y = \frac{1.5435\%}{8\%} = 19.29\%$$

$$x_X\omega_X = \frac{0.50}{0.5831}(3\%) = 2.5725\% \;\Rightarrow\; x_X = \frac{2.5725\%}{5\%} = 51.45\%$$

$$\text{index} = 100\% - 51.45\% - 19.29\% = 29.26\%$$

$\Rightarrow$ Check: $\sqrt{2.5725^2 + 1.5435^2} = 3.00\%$, the TEV target. The 29.3% left in the index is the slack needed to hit exactly 3%.

**Step 4 – Expected excess return of the sleeve**

$$\mu_P = x_X\mu_X + x_Y\mu_Y = 0.5145(2.5\%) + 0.1929(2.4\%) = 1.286\% + 0.463\% = 1.749\%$$

$\Rightarrow$ Equivalently $\text{IR}_P \times \omega_P = 0.5831 \times 3\% = 1.749\%$: the sleeve's IR of 0.58 exceeds both managers', because independent active bets diversify.

**Interpretation**

In money terms the 99% relative risk budgets are $2.33 \times 2.5725\% \times 600 =$ EUR 36.0 million for X and $2.33 \times 1.5435\% \times 600 =$ EUR 21.6 million for Y; the sleeve's budget is $2.33 \times 3\% \times 600 =$ EUR 41.9 million, smaller than their sum because of diversification. Within those budgets the managers trade without seeking approval.

**Why the other options are wrong**

**A is incorrect.** Took $\text{IR}_P$ as the sum $0.50 + 0.30 = 0.80$, as if active returns were perfectly correlated: $x_Y = (0.30/0.80)(3\%)/8\% = 14.1\%$, $x_X = 37.5\%$, $\mu_P = 0.375(2.5\%) + 0.141(2.4\%) = 1.28\%$.

**C is incorrect.** Correct weights, but computed the sleeve's excess return as the average manager IR times the TEV: $0.40 \times 3\% = 1.20\%$, ignoring the diversification of independent active bets.

**D is incorrect.** Split the whole sleeve in proportion to the IRs with nothing in the index: $x_Y = 0.30/0.80 = 37.5\%$, $\mu_P = 0.625(2.5\%) + 0.375(2.4\%) = 2.46\%$. This ignores the TEV constraint: the sleeve's TEV would be $\sqrt{3.125^2 + 3.0^2} = 4.33\%$.

**LO:** IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.

• **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.5 Eq. 6.4 to 6.7, Table 6.4

**Solution IM-6 h.7** **Answer: C**

Bank Q337 • BT P2.T9.21.8.3

**Step 1 – Portfolio IR**

$$\text{IR}_P = \sqrt{0.63^2 + 0.84^2} = \sqrt{0.3969 + 0.7056} = 1.05$$

**Step 2 – Recompute the optimal weights (Eq. 6.7)**

$$x_1 = \frac{0.63}{1.05} \times \frac{3.0}{5.0} = 36.0\%, \qquad x_2 = \frac{0.84}{1.05} \times \frac{3.0}{7.0} = 34.3\%, \qquad \text{index} = 29.7\%$$

$\Rightarrow$ The proposed weights are the optimal ones.

**Step 3 – Check proportionality and the TEV**

Relative risk budgets $1.80\% : 2.40\% = 0.75 = 0.63 : 0.84$, the ratio of the IRs; $\sqrt{1.80^2 + 2.40^2} = 3.0\%$, the TEV constraint.

### Interpretation

The fund expects $\text{IR}_P\,\omega_P = 1.05 \times 3\% = 3.15\%$ of excess return. A fund TEV below each manager's TEV is normal: less than 100% sits with the managers, and their independent bets diversify.

### Why the other options are wrong

**A is incorrect.** The weights need not sum to 1; the index (IR 0) holds the slack that scales total active risk down to the 3.0% constraint.

**B is incorrect.** With weights below 100% and independent managers, the fund TEV is naturally below each manager's TEV; this is not a flaw.

**D is incorrect.** The index always has a zero relative risk budget; that alone proves nothing. Optimality is shown by the proportionality of the risk budgets to the IRs.

**LO:** IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.

• **Reference:** Jorion, Value at Risk 3e, Ch.17 (GARP 2025 Ch.6), 6.5, Eq. 6.6 and 6.7

IM-7 • Portfolio Performance Evaluation • Vardeez: VERY IMPORTANT

# IM-7 a

P2 BOTH

Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses.

Workbook: VARDEEZ FOCUS Time-weighted vs dollar-weighted returns • no workbook question (A18 revision notes only)

**WHAT IS TESTED HERE**

**Why P2.** Vardeez flags it in a Very Important chapter. There is no workbook question and no reported exam shape; the A18 revision notes cover it in one bullet.
**Vardeez:** time-weighted vs dollar-weighted returns.

**Reported exam shapes:** none reported. The block is built from Vardeez's focus, GARP Ch.8 (Bodie, Kane and Marcus) and SchweserNotes Reading 89. TYPE is BOTH by the user's decision, so the two calculations stay in.

**What the questions look like:** a two- or three-period share history (buy, collect dividends, buy or sell more, sell out). You compute the time-weighted return (the geometric mean of the sub-period holding period returns), the dollar-weighted return (the IRR of the investor's cash flows), or both. The wrong options are the other measure, the arithmetic mean, a cumulative figure that was never annualised, or a cash flow with a dividend left out. One question asks which measure fits which use.

## 1. Two ways to average the same history

Over one period with no interim cash flows there is only one return:

$$r = \frac{\text{total proceeds}}{\text{initial investment}} - 1 = \frac{\text{income} + \text{capital gain}}{\text{initial investment}}$$

A share bought at \$50 that pays a \$2 dividend and is sold at \$53 earns $(2+3)/50 = 10\%$. The same number solves the discounted cash flow problem $50 = (2+53)/(1+r)$. The trouble starts when money is added or withdrawn during the measurement period: the two framings then give two different averages.

- **Time-weighted return (TWR)** is the geometric average of the sub-period returns:

$$\boxed{(1+r_G)^n = (1+r_1)(1+r_2)\cdots(1+r_n)}$$

  Each period's return has the same weight, however much money was invested in it, which is why GARP calls the geometric average a time-weighted average. It is not affected by the size or timing of external cash flows.

- **Dollar-weighted return (DWR)** is the internal rate of return: the rate that equates the present value of everything the investor paid in with the present value of everything received.

$$\boxed{\sum_t \frac{\text{outlay}_t}{(1+r_D)^t} = \sum_t \frac{\text{proceeds}_t}{(1+r_D)^t}}$$

  Periods with more money at work have more influence on it: it is "dollar weighted".

### Computing the time-weighted return (Schweser's three steps)

1. Value the portfolio just before each external cash flow (deposit, withdrawal, or a dividend paid out and not reinvested), and form sub-periods that start and end on those dates. Calendar valuations with no cash flow are not needed.
2. Compute the holding period return of each sub-period on the money actually invested in it.
3. Chain-link: multiply the $(1 + \text{HPR}_t)$ terms. If the whole horizon is longer than one year, take the $n$-th root for the annual figure.

The arithmetic average of the same HPRs is always at least as large as the geometric average, and the gap grows with volatility. It is a frequent distractor.

## 2. GARP's worked case

A share costs \$50 and pays a \$2 dividend at the end of each year. At the end of year 1 the price is \$53 and a second share is bought at \$53. At the end of year 2 both shares pay \$2 and are sold at \$54 each.

**Time-weighted.** Work per share; the second purchase does not enter.

$$r_1 = \frac{53 + 2 - 50}{50} = 10.00\%, \qquad r_2 = \frac{54 + 2 - 53}{53} = 5.66\%$$

$$r_G = (1.10 \times 1.0566)^{1/2} - 1 = 7.81\%$$

**Dollar-weighted.** The investor pays \$50 at $t = 0$; at $t = 1$ receives \$2 and pays \$53 (net $-\$51$); at $t = 2$ receives \$4 of dividends plus \$108 of sale proceeds ($+\$112$):

$$50 + \frac{53}{1+r} = \frac{2}{1+r} + \frac{112}{(1+r)^2} \quad \Longrightarrow \quad r_D = 7.117\%$$

**Why the DWR is lower.** Year 2 earned the lower return (5.66%) and had twice as much money at work (\$106 against \$50). A rough money-weighted average of the two year returns, $(50 \times 10\% + 106 \times 5.66\%)/156 = 7.05\%$, lands close to the IRR and shows the mechanism.

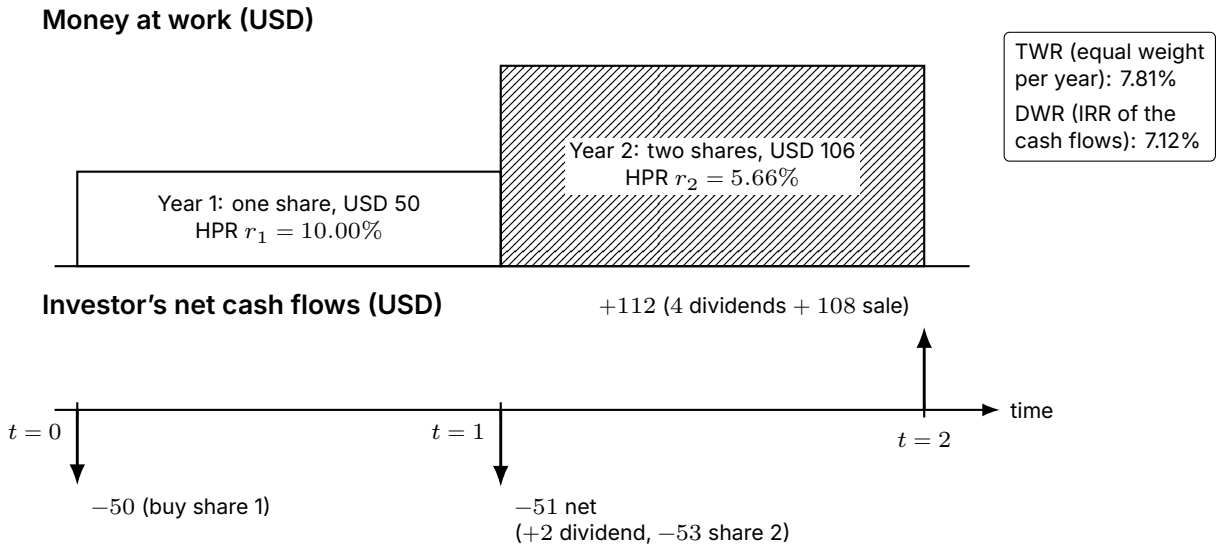



Figure 1. GARP's two-year case. Top: money at work in each year (white bar, one share; hatched bar, two shares) with that year's holding period return. Bottom: the investor's net cash flows (arrows down: paid in; arrow up: received). The time-weighted return gives both years equal weight; the IRR leans towards year 2, which had more money and the lower return.

### Solving a two-period IRR without a calculator

Put $x = 1/(1+r)$ and the IRR condition becomes a quadratic in $x$. For GARP's case, $-50 - 51x + 112x^2 = 0$:

$$x = \frac{51 + \sqrt{51^2 + 4(112)(50)}}{2(112)} = \frac{51 + 158.12}{224} = 0.93356 \quad \Longrightarrow \quad r = \frac{1}{x} - 1 = 7.117\%$$

On a calculator enter CF0 $= -50$, C01 $= -51$, C02 $= +112$ and solve for IRR. In the exam it is often faster to plug each option into the net present value and keep the one that gives zero.

## 3. Which one comes out higher

The DWR is pulled towards the returns of the periods with the most money in them:

| Case | Cash flows | TWR | DWR |
|---|---|---|---|
| GARP: second share bought before the weaker year | $-50, -51, +112$ | 7.81% | 7.12% |
| Schweser: buy at 100, add a share at 120, sell both at 130, \$2 dividends | $-100, -118, +264$ | 15.84% | 13.86% |
| Second share bought after a fall: 40, then 36, then 45 (no dividends) | $-40, -36, +90$ | 6.07% | 11.60% |

- **Money added before a weak period, or withdrawn before a strong one:** DWR below TWR (first two rows).
- **Money added before a strong period, or withdrawn after one:** DWR above TWR (last row: the second share earns the 25% rebound while the first year lost 10%).
- **No interim cash flows:** the two coincide.

## 4. Appropriate uses

- **Evaluating a manager: time-weighted.** Clients decide when money comes in and goes out, and the manager does not control that. A measure driven by the timing of contributions would reward or punish the manager for the client's decisions. The TWR isolates the decisions the manager actually made. It is the industry's preferred measure, and comparison universes rank funds on (usually time-weighted) returns.
- **Measuring what the investor earned: dollar-weighted.** The IRR describes the return on the money actually invested. It is the right measure when the person being evaluated controls the flows: Schweser's example is a private investor with complete control over the money flowing into and out of the account. An investor who adds money before good periods earns a DWR above the TWR.
- **Vardeez's rule of thumb, with its reason.** The source notes compress this to "overall performance of a manager: time-weighted". Carry the reason with the rule: judge the manager on the measure that is not driven by flows the manager does not control, and reach for the dollar-weighted figure when the question is about the money, not the manager.

**Formula card: IM-7 a**

| | |
|---|---|
| Holding period return | $\text{HPR}_t = \left(V_{\text{end}} + \text{income} - V_{\text{begin}}\right)/V_{\text{begin}}$, on the money invested in that sub-period |
| Time-weighted (geometric) | $r_G = \left[\prod_{t=1}^{n}(1+\text{HPR}_t)\right]^{1/n} - 1$; sub-periods break at each external cash flow |
| Arithmetic time-weighted | $\bar{r} = \frac{1}{n}\sum \text{HPR}_t \geq r_G$ (not the geometric figure) |
| Dollar-weighted (IRR) | $\sum_t \text{CF}_t/(1+r_D)^t = 0$, with purchases negative, dividends and sales positive |
| Two-period IRR | $x = 1/(1+r)$: $\text{CF}_0 + \text{CF}_1 x + \text{CF}_2 x^2 = 0$, take the positive root |
| Net the flows | a dividend received and a purchase made on the same date enter as one net flow |

| | Time-weighted | Dollar-weighted |
|---|---|---|
| **Also called** | Geometric average return | Internal rate of return (money-weighted) |
| **Weights** | Each period equally | Each period by the money at work in it |
| **External cash flows** | No effect | Drive the result |
| **Data needed** | Portfolio value at every cash flow date | Only the cash flows and the end value |
| **Use to evaluate** | The manager (who does not control the flows) | The investor's own result (who does) |
| **Higher when** | Money arrives before weak periods | Money arrives before strong periods |

**Exam traps**

- **Swapped labels.** The IRR is the dollar-weighted return; the geometric mean is the time-weighted return. GARP's options often pair the right two numbers with the wrong labels.
- **Arithmetic for geometric.** $(r_1 + r_2)/2$ is not the time-weighted return the reading uses. It is always at least as large.
- **Cumulative, not annual.** $(1+r_1)(1+r_2) - 1$ is the two-year return. Take the root.
- **Lost dividends.** A dividend received on the day a share is bought must be netted into that date's flow; dropping it understates the IRR.
- **Wrong base in a sub-period.** A sub-period return uses the money invested at its start: two shares at \$53 is \$106, not \$53.
- **Single-period shortcut.** Treating a year with a mid-year dividend paid out as one period ignores the change in base; break the period at the cash flow.
- **Wrong measure for the manager.** A manager is judged on the time-weighted return because the client controls the flows.

## Practice questions: IM-7 a

### Question IM-7 a.1

Bank Q353 • Mock 2 Q12 (rebuilt)

A performance analyst at a wealth manager is documenting how the firm computes annual time-weighted returns for client accounts. The time-weighted rate of return measures compound growth over a specified time horizon. Which of the following actions is NOT required to compute the annual time-weighted return for an investment?

**A.** Value the portfolio at every month-end, regardless of whether any cash flows occurred.

**B.** Compute the product of (1 + holding period return) across the sub-periods to obtain the total return for the entire measurement period.

**C.** Form sub-periods that correspond to the dates of deposits and withdrawals.

**D.** Compute the holding period return of the portfolio for each sub-period.

### Question IM-7 a.2

Bank Q351 • Mock 1 Q58

An investor purchases a security for USD 100. Six months later the security is priced at USD 108 and pays a dividend of USD 4, which the investor does not reinvest. At the end of twelve months the security is priced at USD 112. Assuming dividends are not reinvested, what is the time-weighted rate of return on this investment over the 12-month holding period?

**A.** 12.00%

**B.** 15.70%

**C.** 16.00%

**D.** 16.15%

### Question IM-7 a.3

Bank Q343 • AP Mock 3 Q19 (rebuilt)

David Parker purchased 500 shares of Yamacha Petroleum at USD 60 per share on 1 January 2016, and another 500 shares at USD 75 on 1 January 2017. The stock paid a dividend of USD 5 per share on 31 December 2016 and again on 31 December 2017. On 31 December 2017 he sold all his shares at USD 90 each. The time-weighted rate of return per year on the investment is closest to:

**A.** 26.67%

**B.** 29.96%

**C.** 33.33%

**D.** 68.89%

### Question IM-7 a.4

Bank Q342 • AP Mock 2 Q18 (rebuilt)

You buy one share of a stock at USD 40 now. The stock pays an annual dividend of USD 4 per share. At the end of year 1, after receiving the dividend, you buy another share at USD 42. At the end of year 2, you receive the dividend on both shares and sell both shares at USD 43 each. The dollar-weighted rate of return on the investment is closest to:

**A.** 9.5%

**B.** 13.0%

**C.** 13.4%

**D.** 15.0%

### Question IM-7 a.5

Bank Q352 • Mock 1 Q59

An investor purchases one share of a security for USD 100. At the end of Year 1 the share is worth USD 110, and the investor purchases a second share at that price. At the end of Year 2 each share is worth USD 99 and the investor sells both. There are no dividends. What is the dollar-weighted rate of return on this investment over the holding period?

**A.** $-5.71\%$

**B.** $-3.92\%$

**C.** $-1.00\%$

**D.** $0.00\%$

### Question IM-7 a.6

Bank Q349 • BT P2.T8.15.3 (rebuilt)

Shares of XYZ Corp pay a USD 2 dividend at the end of every year on December 31. An investor buys two shares on January 1 at USD 20 each, sells one share for USD 22 exactly one year later (the next January 1), and sells the second share one year after that for USD 19. What are, respectively, the dollar-weighted and time-weighted annual rates of return on the 2-year investment?

**A.** 7.0% (dollar) and 11.9% (time)

**B.** 8.8% (dollar) and 7.0% (time)

**C.** 11.9% (dollar) and 7.0% (time)

**D.** 11.9% (dollar) and 7.7% (time)

### Question IM-7 a.7

Bank Q350 • BT P2.T9.21.10.1 (rebuilt)

Bertha buys one share at Time 0 for USD 100. At Time 1 the price has fallen to USD 80 and the share has paid a USD 5.00 dividend; she buys a second share at USD 80. At Time 2 she receives USD 8.00 in dividends on two shares, the price is USD 92, and she buys a third share. At Time 3 the price is USD 106; she receives USD 13.80 in dividends on three shares and sells all three shares for USD 318.00.

| Time | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Share price (USD) | 100.00 | 80.00 | 92.00 | 106.00 |
| Dividend per share (USD) | | 5.00 | 4.00 | 4.60 |
| Net cash flow (USD) | −100.00 | −75.00 | −84.00 | +331.80 |

What are, respectively, the time-weighted and dollar-weighted annual returns over the three-year period?

**A.** 7.0% (time) and 7.9% (dollar)

**B.** 7.0% (time) and 12.5% (dollar)

**C.** 8.4% (time) and 12.5% (dollar)

**D.** 12.5% (time) and 7.0% (dollar)

### Question IM-7 a.8

New • built from GARP Ch.8 and SchweserNotes 2026 Reading 89

The investment committee of a Canadian university endowment reviews the external equity manager who runs a CAD 400 million account. Contributions and withdrawals are decided by the endowment's treasurer, not by the manager. Over the year the account returned $+5\%$, $+4\%$, $-8\%$ and $+5.5\%$ in the four quarters, and the treasurer added CAD 200 million at the end of the second quarter. The account's time-weighted return for the year is 6.0% and its dollar-weighted return is 3.6%. Which of the following statements is correct?

**A.** The manager should be judged on the 3.6% dollar-weighted return, because it reflects the money that was actually invested in the weak third quarter.

**B.** The manager should be judged on the 6.0% time-weighted return; the gap arises because the treasurer's contribution put more money to work in the weak quarter.

**C.** The time-weighted return is the IRR of the account's cash flows, so the 6.0% already reflects the timing of the treasurer's contribution.

**D.** The gap shows that the manager timed the market poorly in the third quarter, a weakness that the time-weighted return understates.

# Solutions • IM-7 a

## Solution IM-7 a.1

**Answer: A**

Bank Q353 • Mock 2 Q12

### Step 1 – List what the time-weighted return needs

Break the horizon at each external cash flow, value the portfolio at those dates, compute each sub-period HPR, and chain-link the HPRs.

$\Rightarrow$ B, C and D are each one of the required steps.

### Step 2 – Test option A

Sub-periods are set by deposits and withdrawals, not by the calendar. A month-end with no cash flow adds a valuation date but changes nothing in the chain.

$\Rightarrow$ Valuing at every month-end regardless of cash flows is not required.

### Interpretation

The point of breaking at cash flows is that each sub-period return is then computed on a fixed amount of money, so the size of the flows drops out. That is what makes the measure time-weighted.

### Why the other options are wrong

**B is incorrect.** Required: geometric linking of $(1 + \text{HPR})$ terms is the final step (then the $n$-th root if the horizon exceeds a year).

**C is incorrect.** Required: the sub-periods must start and end on the cash flow dates.

**D is incorrect.** Required: each sub-period HPR is the building block of the chain.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** SchweserNotes 2026 Book 5, Reading 89, LO 89.a (steps 1 to 3); Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1 Average Rates of Return

## Solution IM-7 a.2

**Answer: D**

Bank Q351 • Mock 1 Q58

### Step 1 – Split the year at the dividend date

The \$4 dividend is paid out and not reinvested, so it is an external cash flow at six months: two sub-periods.

### Step 2 – Sub-period returns

$$\text{HPR}_1 = \frac{108 + 4 - 100}{100} = 12.00\%, \qquad \text{HPR}_2 = \frac{112 - 108}{108} = 3.70\%$$

$\Rightarrow$ The second half starts from the \$108 still invested, not from \$100.

### Step 3 – Chain-link

$$\text{TWR} = 1.12 \times 1.037037 - 1 = 16.15\%$$

### Interpretation

The horizon is exactly one year, so no root is needed. The answer differs from the one-period figure (16.00%) only because the base changes at the cash flow date.

### Why the other options are wrong

**A is incorrect.** Stopped after the first sub-period: 12.00%.

**B is incorrect.** Added the two sub-period returns, $12.00\% + 3.70\% = 15.70\%$, instead of compounding them.

**C is incorrect.** Treated the year as one period: $(112 + 4 - 100)/100 = 16.00\%$, ignoring the mid-year cash flow.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1; SchweserNotes 2026 Book 5, Reading 89, LO 89.a

### Solution IM-7 a.3 Answer: B

Bank Q343 • AP Mock 3 Q19

#### Step 1 – Year 1 return per share

$$\text{HPR}_1 = \frac{75 - 60 + 5}{60} = 33.33\%$$

#### Step 2 – Year 2 return per share

$$\text{HPR}_2 = \frac{90 - 75 + 5}{75} = 26.67\%$$

$\Rightarrow$ The 500 extra shares bought at \$75 do not enter: the time-weighted return works per share.

#### Step 3 – Geometric mean, annualised

$$r_G = \sqrt{1.3333 \times 1.2667} - 1 = \sqrt{1.6889} - 1 = 29.96\%$$

#### Interpretation

The share counts are a red herring for the time-weighted return. They matter only for the dollar-weighted return, which would lean towards year 2 because twice as much money was invested in it.

#### Why the other options are wrong

**A is incorrect.** Reported the second-year return only: 26.67%.
**C is incorrect.** Reported the first-year return only: 33.33%.
**D is incorrect.** Gave the cumulative two-year return, $1.6889 - 1 = 68.89\%$, without taking the square root.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1 Time-Weighted Returns versus Dollar-Weighted Returns

### Solution IM-7 a.4 Answer: B

Bank Q342 • AP Mock 2 Q18

#### Step 1 – The investor's net cash flows

$t = 0$: $-40$. $t = 1$: $+4$ dividend $-42$ purchase $= -38$. $t = 2$: $2 \times 4$ dividends $+ 2 \times 43$ sales $= +94$.

#### Step 2 – Solve for the IRR

$$-40 - \frac{38}{1+r} + \frac{94}{(1+r)^2} = 0$$

With $x = 1/(1+r)$: $94x^2 - 38x - 40 = 0$, so $x = \dfrac{38 + \sqrt{38^2 + 4(94)(40)}}{188} = 0.88505$ and $r = 12.99\%$.

$\Rightarrow$ The dollar-weighted return is about 13.0%.

**Interpretation**

Year 1 earned 15.0% on \$40 and year 2 earned 11.9% on \$84, so the IRR sits below the time-weighted 13.4%: the weaker year had more money in it, as in GARP's own case.

**Why the other options are wrong**

**A is incorrect.** Left out the \$4 dividend at $t = 1$ (flows $-40$, $-42$, $+94$): IRR 9.5%.
**C is incorrect.** Time-weighted return: $\text{HPR}_1 = (42 - 40 + 4)/40 = 15.0\%$, $\text{HPR}_2 = (43 + 4 - 42)/42 = 11.9\%$, geometric mean 13.4%.
**D is incorrect.** First-year holding period return only: 15.0%.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1; SchweserNotes 2026 Book 5, Reading 89, LO 89.a (dollar-weighted example)

## Solution IM-7 a.5 — Answer: B

Bank Q352 • Mock 1 Q59

**Step 1 – Cash flows**

$t = 0$: $-100$; $t = 1$: $-110$; $t = 2$: $+2 \times 99 = +198$.

**Step 2 – Solve for the IRR**

$$-100 - \frac{110}{1+r} + \frac{198}{(1+r)^2} = 0 \quad \Rightarrow \quad 198x^2 - 110x - 100 = 0,\ x = 1.04080,\ r = -3.92\%$$

**Step 3 – Compare with the time-weighted return**

$(1.10)(0.90) - 1 = -1.00\%$ over two years: per year $\sqrt{0.99} - 1 = -0.50\%$.
$\Rightarrow$ The dollar-weighted return is worse because the second, larger investment was made just before the 10% fall.

**Interpretation**

The time-weighted return describes the security; the dollar-weighted return describes this investor's result, which suffered from adding money at the top.

**Why the other options are wrong**

**A is incorrect.** Simple loss over total money invested, $(198 - 210)/210 = -5.71\%$, with no discounting and no annualisation.
**C is incorrect.** The two-year time-weighted return, $1.10 \times 0.90 - 1 = -1.00\%$: the other measure.
**D is incorrect.** Arithmetic average of $+10\%$ and $-10\%$: 0.00%.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1

## Solution IM-7 a.6 — Answer: C

Bank Q349 • BT P2.T8.15.3 (GARP Concept Check 8.1)

**Step 1 – Dollar-weighted: cash flows**

$t = 0$: buy two shares, $-40$. $t = 1$: dividends on two shares plus the sale of one, $4 + 22 = +26$.

$t = 2$: dividend plus the sale of the other, $2 + 19 = +21$.

$$-40 + \frac{26}{1+r} + \frac{21}{(1+r)^2} = 0 \quad \Rightarrow \quad r_D = 11.91\%$$

### Step 2 – Time-weighted: per-share returns

$$r_1 = \frac{2 + (22 - 20)}{20} = 20.00\%, \qquad r_2 = \frac{2 + (19 - 22)}{22} = -4.55\%$$

$$r_G = [(1.20)(0.9545)]^{1/2} - 1 = 7.03\%$$

### Interpretation

Here the dollar-weighted return is the higher one: two shares were held through the strong first year and only one through the losing second year. This is GARP's Concept Check 8.1; GARP also reports the arithmetic time-weighted average, 7.73%.

### Why the other options are wrong

**A is incorrect.** The two measures swapped.

**B is incorrect.** Total profit over the initial investment, $(47 - 40)/40 = 17.5\%$, halved to 8.8%: no discounting, no account of timing.

**D is incorrect.** Right dollar-weighted figure, but the arithmetic mean $(20\% - 4.55\%)/2 = 7.7\%$ in place of the geometric mean.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1, Concept Check 8.1 and its solution

## Solution IM-7 a.7

**Answer: B**

Bank Q350 • BT P2.T9.21.10.1

### Step 1 – Per-share holding period returns

$$\text{HPR}_1 = \frac{80 + 5 - 100}{100} = -15.00\%, \qquad \text{HPR}_2 = \frac{92 + 4 - 80}{80} = 20.00\%$$

$$\text{HPR}_3 = \frac{106 + 4.60 - 92}{92} = 20.22\%$$

### Step 2 – Time-weighted (geometric) return

$$[(0.85)(1.20)(1.2022)]^{1/3} - 1 = 1.22622^{1/3} - 1 = 7.03\%$$

### Step 3 – Dollar-weighted return

$$-100 - \frac{75}{1+r} - \frac{84}{(1+r)^2} + \frac{331.80}{(1+r)^3} = 0 \quad \Rightarrow \quad r_D = 12.50\%$$

$\Rightarrow$ Plugging in $r = 12.5\%$: $-100 - 66.67 - 66.37 + 233.03 \approx 0$.

### Interpretation

Bertha bought after the price had fallen, so the extra money earned the two 20% years and missed the $-15\%$ year. Her dollar-weighted return beats the security's time-weighted return by about 5.5 points.

### Why the other options are wrong

**A is incorrect.** Right time-weighted figure, but the IRR ignores the dividends (flows $-100$, $-80$, $-92$, $+318$): 7.9%.
**C is incorrect.** Arithmetic mean of the HPRs, $(-15 + 20 + 20.22)/3 = 8.4\%$, in place of the geometric mean.
**D is incorrect.** The two measures swapped.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1

### Solution IM-7 a.8 — Answer: B

New • built from GARP Ch.8 and SchweserNotes 2026 Reading 89

#### Step 1 – Check the two numbers

Time-weighted: $1.05 \times 1.04 \times 0.92 \times 1.055 - 1 = 5.99\%$. The account is worth CAD 436.8 million at mid-year, CAD 636.8 million after the contribution and CAD 618.1 million at year-end. Dollar-weighted: solve $400(1 + r) + 200(1 + r)^{0.5} = 618.08$, giving $r_D = 3.62\%$.

#### Step 2 – Explain the gap

The CAD 200 million arrived just before the $-8\%$ quarter, so more money was at work in the worst quarter. The IRR weights that quarter more heavily; the time-weighted return does not.

#### Step 3 – Choose the measure for the manager

The treasurer, not the manager, chose the size and timing of the contribution.
$\Rightarrow$ The manager is judged on the time-weighted 6.0%.

#### Interpretation

The 3.6% is still a true and useful number: it is what the endowment earned on the money it committed, and it reflects the treasurer's timing decision. It is the wrong number for judging the manager.

#### Why the other options are wrong

**A is incorrect.** Sibling swap: the dollar-weighted return mixes in the treasurer's timing, which the manager did not control.
**C is incorrect.** Definition swap: the IRR is the dollar-weighted return; the time-weighted return is built to ignore the flows.
**D is incorrect.** Wrong person: the timing that hurt the result was the treasurer's contribution. The manager's own decisions are what the 6.0% captures.

**LO:** IM-7 a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.1 and Adjusting Returns for Risk (comparison universes); SchweserNotes 2026 Book 5, Reading 89, LO 89.a

IM-7 • Portfolio Performance Evaluation • Vardeez: VERY IMPORTANT

# IM-7 e

P2 CONCEPT

Describe style analysis.

Workbook: VARDEEZ FOCUS Style analysis • no workbook question (A18 revision notes only)

### WHAT IS TESTED HERE

**Why P2.** Vardeez flags it in a Very Important chapter. He calls style analysis "the last LO" because his source notes run the objectives out of order; in the 2026 official order it is IM-7 e. There is no workbook question and no reported exam shape.
**Vardeez:** style analysis.

**Reported exam shapes:** none reported. Built from Vardeez's focus, GARP Ch.8, Section 8.2, and SchweserNotes Reading 89.

**What the questions look like:** CONCEPT. What the regression uses, which constraints it imposes, what the slopes, the $R^2$ and the intercept mean, and what benchmark the method measures performance against. The bank's one numerical item (a return difference against the style benchmark) is left out under the TYPE rule; the same idea is tested in words in Questions IM-7 e.1 and e.6.

## 1. From the index model to style analysis

The index model regression already describes a manager's style: high-beta portfolios are *cyclical* or *aggressive*, low-beta portfolios *defensive*. A multifactor model describes exposures to several factors, and each exposure is an implicit asset allocation decision. William Sharpe (1992) turned this into a tool: regress the fund's returns on the returns of indexes that represent a range of asset classes ("styles"),

$$r_t = a + \sum_i b_i R_{i,t} + e_t, \qquad b_i \geq 0, \qquad \sum_i b_i = 1$$

- **Regressors:** tradable asset-class indexes (bills, bonds, large-cap, small-cap, growth, value and so on), chosen to be exhaustive and mutually exclusive.
- **Constraints:** funds are barred from short positions, so the slopes are constrained to be zero or positive and to sum to 100%. The slopes then read as a complete asset allocation: the fund's *effective* (implicit) weights. A slope of zero is allowed; a negative slope, a slope above 1, or slopes that do not add to 1 break the method.
- **Style benchmark:** the fitted combination $\sum b_i R_{i,t}$ is a passive portfolio with the same style as the fund. Performance is measured against it, not against a broad index or a peer-group median.
- **$R^2$:** the percentage of return *variability* attributable to style choice (asset allocation). $1 - R^2$ is attributable to security selection within asset classes and to timing that shows up as periodic changes in allocation.
- **Intercept:** the average return from security selection. The residuals around the style benchmark track the manager's selection success period by period.

## 2. GARP's case: Fidelity Magellan under Peter Lynch

Monthly returns, October 1986 to September 1991, regressed on seven style portfolios (six stock indexes and T-bills). Only three slopes are positive (Figure 1), and together they explain 97.5% of the variance of Magellan's returns; the residual variability is 2.5%.

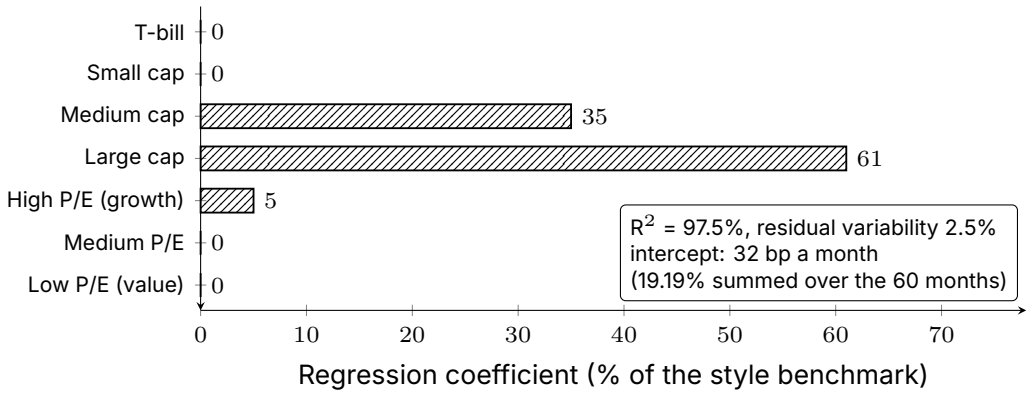



Figure 1. GARP Table 8.4 redrawn: Magellan's style weights (hatched bars). The constrained regression puts zero weight on four of the seven style portfolios. The printed weights add to 101 because of rounding.

- **The misreading GARP warns against.** A high $R^2$ is often read as evidence that security selection does not matter. That ignores the intercept. The $R^2$ can be 100% and the intercept still positive, because consistently superior stock selection shows up as a constant, not as unexplained variance. Magellan's intercept was 32 basis points a month, a cumulative abnormal return of 19.19% over the five years (about $0.32\% \times 60$, summed rather than compounded).
- **Two benchmarks, one picture.** GARP's Figure 8.4 plots Magellan's cumulative return difference against the style benchmark and against the SML (CAPM) benchmark; both climb steadily over the period.
- **Magellan is the exception.** Across 636 mutual funds (1985 to 1989), the average residuals are bell-shaped around a slightly negative mean of $-0.074\%$ a month.
- **Why asset allocation dominates.** A well-known study found that most of the variation in returns of 82 mutual funds could be explained by their allocation to bills, bonds and stocks. The exact interpretation is disputed, but there is wide agreement that asset allocation explains a high proportion of the variation across funds.

SchweserNotes quotes 97.3% style and 2.7% selection for Magellan: that is Sharpe's own study on January 1985 to December 1989. GARP's 97.5% is the textbook authors' re-estimate on a later window. The lesson is the same.

## 3. Where it fits

- **Same machinery, two jobs.** In IM-3 f (Ang), style analysis handles factor exposures that change over time. Here it finds the right benchmark in the first place. One technique, two objectives.
- **Variance, not return.** Style analysis splits the *variability* of returns ($R^2$ against $1 - R^2$) and measures the average selection return (the intercept). Performance attribution (IM-7 i) splits the *return* in one period into the contributions of allocation and selection decisions.
- **Inferring the style.** The slopes reveal the style a manager actually followed, which can be compared with the stated strategy: weights concentrated in large-cap and growth indexes mean a large-cap growth manager, whatever the prospectus says.

| | Index model (Jensen's alpha, IM-7 b) | Sharpe's style analysis |
|---|---|---|
| **Regressors** | Market index excess return | Several tradable asset-class (style) index returns |
| **Constraints on slopes** | None | Each $\geq 0$, sum to 100% (no shorting) |
| **Slopes mean** | Beta: systematic exposure | Effective allocation to each style |
| **Benchmark implied** | The SML: $r_f + \beta(r_M - r_f)$ | A passive portfolio of the same style |
| **Intercept** | Alpha against the CAPM | Average return from security selection |
| **$R^2$** | Share explained by the market | Share of variability due to style choice |
| $1 - R^2$ | Nonsystematic risk | Selection within classes plus timing |

**Exam traps**

- **$R^2$ and $1 - R^2$ swapped (polarity).** $R^2$ is the style (asset allocation) share; $1 - R^2$ is selection and timing.
- **High $R^2$ means no selection skill (absolute claim).** The intercept measures average selection return and can be positive at any $R^2$, even 100%.
- **Unconstrained slopes.** Negative slopes or slopes above 1 do not belong in Sharpe's method; a zero slope and a nonzero intercept do.
- **A slope as a share of variability.** A 23% weight on T-bills is an effective allocation, not the share of variability explained by anything.
- **Wrong benchmark (sibling swap).** Style analysis measures performance against a passive portfolio of the same style, not a broad market index, a single index fund, or the average of active peers (that is the comparison universe of IM-7 b).
- **Holdings for returns.** It is a returns-based method: the weights are implied by the return history, not read from the fund's holdings, and they describe average effective exposure, not the holdings at any date.
- **Non-tradable factors.** The regressors are tradable index returns, which is what makes the fitted combination a benchmark an investor could actually hold.

## Practice questions: IM-7 e

### Question IM-7 e.1

Bank Q399 • Schweser Module Quiz 89.3 Q1

An investment consultant uses Sharpe's style analysis to evaluate the performance of an active portfolio manager. The style analysis measures the manager's performance relative to:

**A.** a passive benchmark of the same style.

**B.** broad-based market indices.

**C.** the performance of an equity index fund.

**D.** an average of similar actively managed investment funds.

### Question IM-7 e.2

Bank Q394 • Mock 2 Q74

A pension fund manager wants to understand the underlying factors driving an actively managed equity portfolio's returns and to compare them with the fund's stated strategy. Which approach would be most appropriate to decompose the portfolio's returns into style and factor exposures?

**A.** Conduct a returns-based style analysis using a multi-factor regression model.

**B.** Perform a holdings-based analysis using the individual security characteristics.

**C.** Apply a Sharpe ratio analysis to compare risk-adjusted returns across different time periods.

**D.** Implement a tracking error minimization technique to identify style drift.

### Question IM-7 e.3

Bank Q397 • BT P2.T8.20.2

To conduct a style analysis on a mutual fund that cannot short securities, an analyst regresses the fund's returns on five regressors: T-bills, a small capitalization index, a large capitalization index, a growth index and a value index. The regression returned is

$$R(t) = 0 \times \text{TBILL} - 0.2 \times \text{SMALL} + 1.4 \times \text{LARGE} + 0.3 \times \text{GROWTH} + 0.7 \times \text{VALUE} + 1.57$$

A colleague criticizes the style analysis with the following statements:

I. You should not have a zero coefficient $(0 \times \text{TBILL})$.

II. You should not have a nonzero intercept $(1.57)$.

III. You should not have a negative coefficient $(-0.2 \times \text{SMALL})$.

IV. You should not have a coefficient that exceeds one $(1.4 \times \text{LARGE})$.

V. The sum of the coefficients should not differ from 1.0 $(0 - 0.2 + 1.4 + 0.3 + 0.7 = 2.2)$.

Which of the criticisms is (are) correct?

**A.** None are correct; the regression is fine.

**B.** I and II only.

**C.** III, IV and V only.

**D.** All are correct.

**Question IM-7 e.4** Bank Q398 • GARP PE1 Q64 • Mock 3 Q17 (rebuilt)

A group of junior risk analysts at a wealth management firm is discussing whether style analysis would be an appropriate approach to assess a mutual fund's investment decisions. Which of the following statements about Sharpe's style analysis is correct?

**A.** Style analysis shows that the contribution of security selection to the variation in a fund's returns is usually much higher than the contribution of asset allocation.

**B.** The regression slopes are unconstrained and may be negative, so that they capture short positions in the asset class indices.

**C.** Returns are regressed on relevant asset class benchmark indices, and the R-squared measures the percentage of return variability attributable to style choice.

**D.** Returns are regressed on non-tradable factors, and the intercept of the regression measures the average return attributable to style choice.

**Question IM-7 e.5** Bank Q395 • AnalystPrep Q.2566 (rebuilt)

A fund manager runs a Sharpe style analysis on his fund. He regresses the fund's returns on four style portfolios and obtains the following results (slopes constrained to be nonnegative and to sum to 100%):

| Style portfolio | Regression weight |
|---|---|
| T-bills | 23% |
| Small cap | 20% |
| High P/E | 28% |
| Large cap | 29% |
| $R^2$ | 94% |

Which of the following statements is correct?

**A.** 94% of the variability in the fund's returns is attributable to security selection and market timing.

**B.** 6% of the variability in the fund's returns is attributable to style (asset allocation) choice.

**C.** 6% of the variability in the fund's returns is attributable to security selection and market timing.

**D.** 23% of the variability in the fund's returns is attributable to security selection and market timing.

**Question IM-7 e.6** New • built from GARP Ch.8, Section 8.2

A consultant at a Hong Kong wealth manager runs Sharpe's style analysis on an actively managed equity fund using 60 months of returns and seven style indexes. The constrained regression gives nonzero slopes of 0.58 on a large-cap index, 0.37 on a mid-cap index and 0.05 on a growth index, an $R^2$ of 97%, and an intercept of 0.28% per month. Which of the following conclusions is correct?

**A.** Because the $R^2$ is 97%, security selection contributed almost nothing to the fund's average return over the period.

**B.** The intercept measures the fund's average monthly return from security selection, and it can be positive even when the $R^2$ is close to 100%.

**C.** The 3% of variability that the styles leave unexplained is the fund's average return from security selection, about 3% a year.

**D.** The slopes show that the fund held 58% of its assets in large-cap stocks at every month-end of the sample period.

## Solutions • IM-7 e

### Solution IM-7 e.1 — Answer: A

Bank Q399 • Schweser Module Quiz 89.3 Q1

**Step 1 – What the regression builds**

The constrained slopes are effective weights on asset-class indexes, so the fitted combination is a passive portfolio with the fund's own style.

**Step 2 – What performance is measured against**

The fund's return in excess of that style benchmark is the manager's selection contribution.
⇒ Performance is measured relative to a passive benchmark of the same style.

**Interpretation**

This is why GARP says style analysis measures "performance relative to similar-style funds": the benchmark is customised to the fund. A fund that beats the S&P 500 only because it was tilted to a style that did well gets no credit.

**Why the other options are wrong**

**B is incorrect.** A broad market index ignores the fund's style tilts; that is the CAPM (SML) comparison.
**C is incorrect.** A single equity index fund is a fixed benchmark that need not match the fund's style mix.
**D is incorrect.** The average of active peers is the comparison universe of IM-7 b, which is not an investable benchmark.

**LO:** IM-7 e: Describe style analysis. • **Reference:** SchweserNotes 2026 Book 5, Reading 89, LO 89.e; Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.2 Style Analysis and Summary point 4

### Solution IM-7 e.2 — Answer: A

Bank Q394 • Mock 2 Q74

**Step 1 – Name the method that decomposes returns**

Sharpe's returns-based style analysis regresses the portfolio's returns on the returns of style (asset-class) indexes.

**Step 2 – Read the output against the stated strategy**

The constrained slopes are the portfolio's effective style exposures; comparing them with the stated strategy reveals any mismatch (style drift).
⇒ Only option A describes a regression that decomposes the returns into style exposures.

**Interpretation**

A "multi-factor regression model" here means the style regression with tradable index returns as the factors and the nonnegativity and sum-to-one constraints.

**Why the other options are wrong**

**B is incorrect.** A holdings-based analysis describes what the fund holds at given dates; it does not decompose the return history into exposures, and it is not Sharpe's method.
**C is incorrect.** The Sharpe ratio is a single risk-adjusted number; it says nothing about which styles drove the returns.
**D is incorrect.** Tracking error minimisation is a portfolio-construction tool, not a way to estimate exposures from realised returns.

**LO:** IM-7 e: Describe style analysis. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.2 Style Analysis; SchweserNotes 2026 Book 5, Reading 89, LO 89.e (step 3)

### Solution IM-7 e.3 — Answer: C

Bank Q397 • BT P2.T8.20.2

**Step 1 – State Sharpe's constraints**

The fund cannot short, so each slope must be nonnegative and the slopes must sum to 100%.

**Step 2 – Test each criticism**

I: a zero slope simply means no T-bill exposure; it is allowed, so I is wrong. II: the intercept is the average return not explained by style (selection); it is allowed, so II is wrong. III: $-0.2$ breaks nonnegativity: correct. IV: with nonnegative slopes summing to 1, no slope can exceed 1: correct. V: $0 - 0.2 + 1.4 + 0.3 + 0.7 = 2.2 \neq 1$: correct.
$\Rightarrow$ III, IV and V only.

**Interpretation**

An unconstrained regression can produce these numbers, but then the slopes no longer describe a portfolio the fund could hold, and the fitted line is no longer a style benchmark.

**Why the other options are wrong**

**A is incorrect.** Ignores that three slopes break the nonnegativity and sum-to-one constraints.
**B is incorrect.** Treats a zero slope and a nonzero intercept as errors; both are acceptable, and the intercept is the selection return.
**D is incorrect.** Wrongly includes I and II.

**LO:** IM-7 e: Describe style analysis. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.2 Style Analysis

### Solution IM-7 e.4 — Answer: C

Bank Q398 • GARP PE1 Q64 • Mock 3 Q17

**Step 1 – The regressors and the $R^2$**

Fund returns are regressed on asset-class indexes, and the $R^2$ is the percentage of return variability attributable to style choice ($1 - R^2$ to selection).
$\Rightarrow$ C is correct.

**Step 2 – Check the other three claims**

The slopes are constrained (nonnegative, summing to 100%); the regressors are tradable index returns; the intercept is the average return from security selection, not from style. The evidence runs the other way from A: asset allocation explains most of the variation (Magellan: 97.5% in GARP's regression, 97.3% in Sharpe's original study).

**Interpretation**

The question tests three properties at once: tradable regressors, constrained slopes, and the meaning of $R^2$ against the intercept.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: style (asset allocation) usually explains most of the variation; selection explains little.
**B is incorrect.** Sharpe's slopes are constrained to be nonnegative and to sum to 100%, because the funds cannot short.
**D is incorrect.** The regressors are tradable indexes, and the intercept measures selection, not style.

**LO:** IM-7 e: Describe style analysis. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.2 Style Analysis, Table 8.4; GARP Practice Exam 1, Q64

### Solution IM-7 e.5 — Answer: C

Bank Q395 • AnalystPrep Q.2566

**Step 1 – Check the constraints**

$23\% + 20\% + 28\% + 29\% = 100\%$, all nonnegative: a valid Sharpe style regression.

**Step 2 – Read the $R^2$**

Style (asset allocation) explains $R^2 = 94\%$ of the variability of returns.

**Step 3 – The remainder**

$1 - R^2 = 6\%$ is attributable to security selection and timing.
$\Rightarrow$ C states this.

**Interpretation**

The weights describe *how* the fund is allocated across styles; the $R^2$ describes *how much* of the return variability that allocation explains. They answer different questions.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: 94% is the style share, not the selection share.
**B is incorrect.** Polarity flip the other way: 6% is the selection share; style is 94%.
**D is incorrect.** Took a regression weight (T-bills, 23%) as a share of variability; weights are effective allocations.

**LO:** IM-7 e: Describe style analysis. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.2 Style Analysis; SchweserNotes 2026 Book 5, Reading 89, LO 89.e (step 2)

### Solution IM-7 e.6 — Answer: B

New • built from GARP Ch.8, Section 8.2

**Step 1 – What the $R^2$ says**

97% of the month-to-month *variability* of returns moves with the style benchmark. It says nothing about the *average level* of the unexplained part.

**Step 2 – What the intercept says**

The intercept is the average return from security selection relative to the style benchmark: 0.28% a month here, about 3.4% a year before compounding.
$\Rightarrow$ A fund can have a very high $R^2$ and a positive intercept at the same time, as Magellan did (97.5% and 32 basis points a month).

**Interpretation**

GARP warns that a high $R^2$ is often read as evidence against the importance of security selection, and that this reading "misses the important role of the intercept". A manager who beats the style benchmark by a steady amount every month adds return without adding unexplained variance.

**Why the other options are wrong**

**A is incorrect.** The misreading GARP names: a high $R^2$ does not rule out a positive average selection return.
**C is incorrect.** Confuses a share of variance with a return: $1 - R^2 = 3\%$ is the variability left unexplained, not 3% of return.
**D is incorrect.** Absolute claim: the slopes are effective average exposures implied by returns, not the fund's holdings at each month-end.

**LO:** IM-7 e: Describe style analysis. • **Reference:** Bodie, Kane and Marcus, Investments 12e, Ch.24 (GARP 2025 Ch.8), 8.2 Style Analysis, Table 8.4 and Figure 8.4

IM-14 • Madoff: A Riot of Red Flags • Vardeez: VERY IMPORTANT

# IM-14 a

P2 CONCEPT

Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance.

Workbook: VARDEEZ FOCUS History, strategy and performance • A24.4 touches this LO (worked as Question IM-14 b.10)

**WHAT IS TESTED HERE**

**Why P2.** Vardeez flags it in a Very Important chapter ("Know this well and expect 1 question"). No workbook question is assigned to it; Workbook A24.4 (worked as Question IM-14 b.10) touches it, and the A24 revision notes cover it in one bullet.
**Vardeez pointer:** none beyond the reading flag.

**Content authority:** SchweserNotes 2026 Reading 96 (Gregoriou and Lhabitant, 2009), used in place of the missing GARP 2026 book. The red flags are in IM-14 b (P1), which prints earlier in this book.

**Reported exam shapes:** none reported.

**What the questions look like:** conceptual. How BMIS began and grew (brokerage first, advisory business later); how the advisory business reached investors (feeder funds); the legs of the split-strike conversion and how its payoff behaves; the reported returns against the S&P 100; and how the fraud came to light and why it collapsed. The bank holds four questions; Q545 is re-filed here from IM-14 b and three are new.

## 1. Two businesses under one roof

**The brokerage (genuine).** Founded in 1960, Bernard L. Madoff Investment Securities (BMIS) quoted bid and ask prices, executed over-the-counter (OTC) trades and adopted electronic trading early. Madoff used NYSE Rule 390, which let NYSE-listed stocks trade away from the exchange floor (where NYSE members could not make markets in them), joined the struggling Cincinnati Stock Exchange and made it the first fully electronic US exchange. BMIS earned the bid-ask spread, then quoted in fractions, and paid brokers USD 0.01 a share for their orders: controversial *payment for order flow* that won it over 5% of trading volume by 1989 and about 10% by 2008. Madoff became chairman of NASDAQ's board. That reputation later lent the advisory business credibility and deterred questions.

**The advisory business (the fraud).** Set up in the late 1980s, as brokerage margins shrank, *in the same building*, which blurred the two and created conflicts that complicated oversight. BMIS was not strictly a hedge fund: its 2008 Form ADV showed 23 discretionary accounts, about USD 17 billion of assets and USD 700 million of equity capital. Most accounts were **feeder funds** that gave BMIS full trading authority, so end investors were the feeders' clients and could perform due diligence only on the feeders; Madoff allowed no visits, questions or direct contact. Access was sold as exclusive, for the wealthy and well connected.

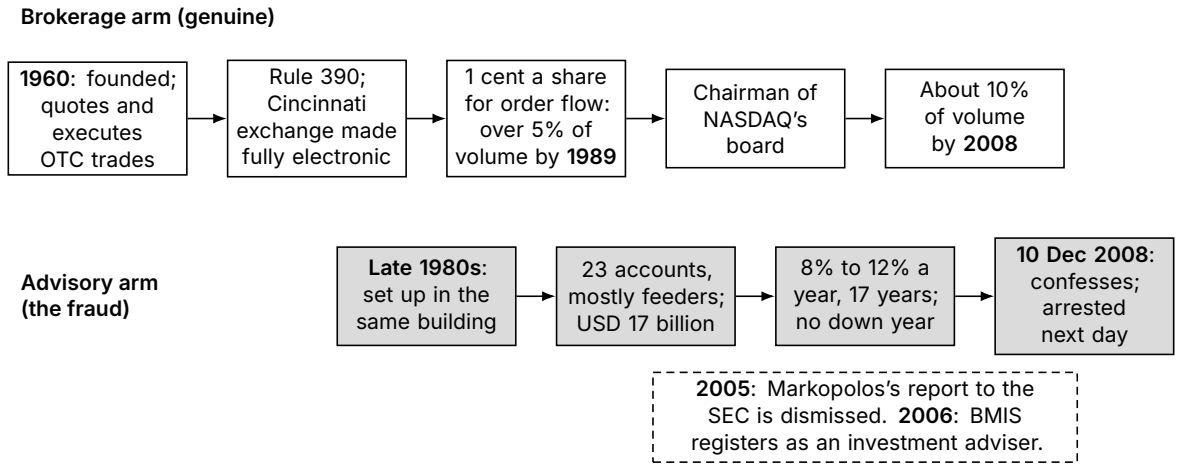



Figure 1. BMIS in two lanes, in time order (not to scale). Plain boxes: the genuine brokerage. Grey boxes: the advisory business, run from the same building, that became the fraud. Dashed box: two late events.

## 2. The reported strategy: split-strike conversion

Madoff said he traded a portfolio of stocks and options designed to mirror the S&P 100 Index:

1. Buy a basket of stocks highly correlated with the S&P 100.
2. Write (sell) out-of-the-money S&P 100 calls, with notional roughly equal to the stock basket.
3. Buy out-of-the-money S&P 100 puts, with notional roughly equal to the stock basket.

The short call puts a **ceiling** on the return: above its strike, gains on the stocks are given up. The long put sets a **floor**: below its strike, further losses on the stocks are offset. The package is a **collar**, a protective put plus a covered call, which option traders also call a bull spread, or a "vacation trade" because it needs little attention until the options near expiry (Figure 2). Madoff claimed the portfolio sat in **cash most of the time**, with each position held over short horizons, **typically less than 30 days**.

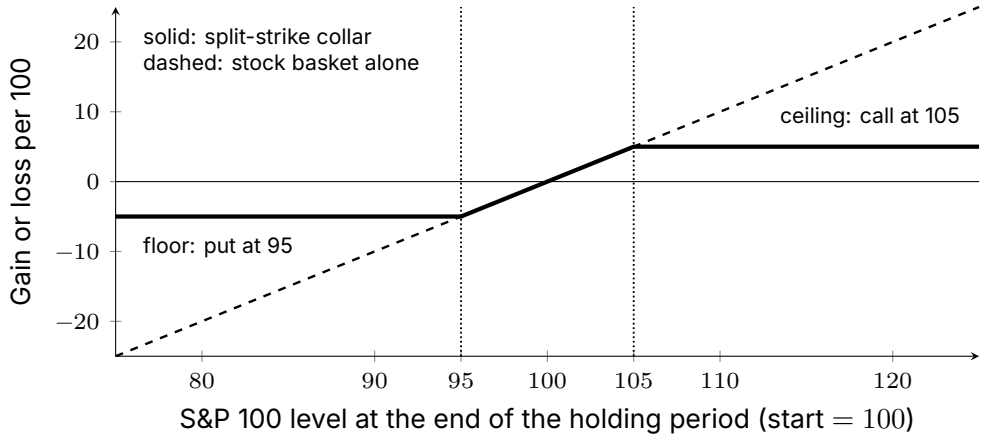



Figure 2. Payoff of the split-strike conversion at the end of a holding period, with illustrative strikes 5% either side of the starting level and option premiums ignored. The solid line is flat below the put strike and above the call strike (dotted verticals); the dashed line is the stock basket alone. The collar limits how much a bad period can lose, not how often a period is bad (IM-14 b, Figure 2).

## 3. The reported performance

The feeder funds reported steady returns of **8% to 12% a year** for about **17 years**, with **no down year** and **more than 95% of months positive**. The returns were **comparable to those of the S&P 100**, but their volatility was **dramatically lower**. The approach was "widely regarded as one of the most effective in the industry". As a red flag (IM-14 b): a collar still loses in every month the index falls (it only caps how much), so index-like returns with almost no losing months are impossible.

## 4. Collapse and confession

By late 2008 redemptions in the market turmoil outran new money, and Madoff could not raise capital fast enough. On **10 December 2008** he told his sons, wife and brother that the advisory business was "a giant Ponzi scheme", "all a big lie"; **his sons reported him** that evening, and he was arrested the next day and charged with securities fraud. He was sentenced to 150 years. Losses were estimated at up to USD 50 billion, spread across individuals, charities, pension funds, hedge funds, funds of funds and banks. He was *not* caught by an investigation, and did not confess out of remorse. A Ponzi scheme pays withdrawals from new money, so, like the subprime boom, it lasts only while prices rise and inflows cover redemptions.

**Correction to the source notes.** Vardeez says the reported returns had "near-zero correlation to equity markets". The reading, as SchweserNotes reports it, gives no correlation: returns comparable to the S&P 100's with far lower volatility (the key of Bank Q542). Answer from that. Vardeez's BLMIS is Schweser's BMIS.

**Summary card: IM-14 a**

| | Brokerage arm | Advisory arm |
|---|---|---|
| **Began** | 1960 | Late 1980s, same building |
| **Business** | OTC trades; market making in NYSE-listed stocks | 23 discretionary accounts (2008), mostly feeder funds; about USD 17 billion |
| **Earned from** | Bid-ask spread; paid 1 cent a share for order flow | Commissions only; no management or performance fee |
| **Standing** | Over 5% of volume (1989), about 10% (2008); NASDAQ chairman | No visits or questions; adviser registration only in 2006 |
| **Reported strategy** | Split-strike conversion: long S&P 100-correlated stocks, short OTM S&P 100 calls, long OTM S&P 100 puts (a collar). Cash most of the time; positions held under 30 days. | |
| **Reported performance** | 8% to 12% a year for about 17 years; no down year; more than 95% of months positive; returns like the S&P 100's with far lower volatility. | |

**Exam traps**

- **Founding.** BMIS began in 1960 as a genuine brokerage, not as an advisory business, a clearinghouse or a Cincinnati Stock Exchange member. The fraud was in the advisory arm (late 1980s).
- **Legs reversed.** Calls were *sold* (the ceiling) and puts *bought* (the floor). There was no long call.
- **Level versus volatility.** Returns comparable to the S&P 100, not higher; volatility far lower. Cash most of the time, positions under 30 days.

- **Discovery.** A confession to his family on 10 December 2008; his sons then reported him. Not the SEC, not Markopolos, and the sons did not find it on their own.
- **Why it collapsed.** Redemptions outran new money in the 2008 turmoil; there was no reported losing year.

## Practice questions: IM-14 a

### Question IM-14 a.1

Bank Q540 • Schweser Module Quiz 96.1 Q1

A trainee at a fund of hedge funds is reviewing the history of Bernard L. Madoff Investment Securities, LLC (BMIS) for a case study. BMIS began operations in 1960:

**A.** as a brokerage business.

**B.** as a derivatives clearinghouse.

**C.** as an investment advisory business.

**D.** conducting electronic trading on the Cincinnati Stock Exchange (CSE).

### Question IM-14 a.2

New • built from SchweserNotes 2026 Reading 96

A compliance trainer at a broker-dealer uses Madoff's career to show how a strong reputation can discourage questions. Which statement about the history of BMIS's brokerage business is correct?

**A.** Madoff used NYSE Rule 390 to become a specialist on the NYSE floor, which gave BMIS its competitive edge in listed stocks.

**B.** The brokerage was a front from the start: it never executed client orders and existed only to hold the advisory clients' money.

**C.** BMIS paid brokers about one cent a share to route orders to it, which helped it handle more than 5% of trading volume by 1989.

**D.** BMIS earned most of its brokerage profits from management and performance fees charged to the brokers that sent it orders.

### Question IM-14 a.3

New • built from SchweserNotes 2026 Reading 96

In 2007 a due diligence analyst at a European fund of hedge funds tries to evaluate BMIS's advisory business directly and finds that she cannot. Which description of how the business was organised best explains why?

**A.** BMIS ran a single offshore hedge fund in limited partnership form, and its offering memorandum was available only to existing limited partners.

**B.** The advisory business was run from a separate city by independent staff, so the brokerage could not release any of its records.

**C.** BMIS managed thousands of small retail accounts directly, so no single investor was large enough to demand an on-site visit.

**D.** Most of its 23 discretionary accounts were feeder funds, so end investors were clients of the feeders, and Madoff allowed no visits or questions.

**Question IM-14 a.4** Bank Q541 • Schweser Module Quiz 96.1 Q2

A derivatives analyst reconstructs the strategy Madoff described to his investors. Which of the following statements about Madoff's split-strike conversion strategy is incorrect?

**A.** Part of the strategy involved selling an out-of-the-money call option.

**B.** A floor value was created by purchasing an out-of-the-money put option.

**C.** A long position in a portfolio of stocks highly correlated with the S&P 100 was required.

**D.** Gains on the portfolio were supplemented by gains on the long, out-of-the-money call option position.

**Question IM-14 a.5** New • built from SchweserNotes 2026 Reading 96

A risk analyst tests the split-strike conversion strategy as Madoff described it, with out-of-the-money options struck either side of the index level at the start of each holding period. In one period the S&P 100 falls 15%; in another it rises 12%. Ignoring option premiums, which statement is correct?

**A.** The position loses about 15% in the falling period, because the puts were sold, and gains the full 12% in the rising period.

**B.** The loss in the falling period is limited to about the distance to the put strike, and the gain in the rising period is capped at about the distance to the call strike.

**C.** The position gains in both periods, because the two option legs make the strategy market-neutral whatever the index does.

**D.** The loss in the falling period is limited by the short calls, and the gain in the rising period is increased by the long puts.

**Question IM-14 a.6** Bank Q542 • Schweser Module Quiz 96.1 Q3

A consultant compares the reported returns of Madoff's feeder funds with the S&P 100 over the life of the funds. When comparing the reported returns of Madoff's feeder funds with the S&P 100, those returns were:

**A.** significantly higher than the S&P 100, and more volatile.

**B.** significantly higher than the S&P 100, and much less volatile.

**C.** approximately the same as the S&P 100, but with much lower volatility.

**D.** significantly lower than the S&P 100, but as stable as short-term Treasury bills.

**Question IM-14 a.7** Bank Q543 • Schweser Module Quiz 96.2 Q2

A member of a pension fund's investment committee asks how the Madoff fraud was finally discovered. The securities fraud at BMIS came to light when:

**A.** Harry Markopolos sent a 17-page letter to the SEC detailing the red flags at BMIS.

**B.** the SEC, after years of investigation, finally gathered enough evidence to indict Madoff.

**C.** Madoff confessed to his family that his investment advisory business was a giant Ponzi scheme.

**D.** Madoff's sons discovered evidence of the fraud on their own and immediately contacted authorities.

**Question IM-14 a.8** Bank Q545 • AnalystPrep Q.4849 (rebuilt) • re-filed from IM-14 b

Bernard Madoff was found guilty of running a Ponzi scheme, which collapsed in December 2008. Why is a Ponzi scheme not sustainable indefinitely?

**A.** Payouts to existing investors are funded by new investors' capital rather than real profits, so the scheme fails once inflows fall short of redemptions.

**B.** Regulators automatically detect the scheme once reported returns exceed those of the S&P 100.

**C.** The scheme's investment strategy eventually becomes unprofitable as option premiums rise.

**D.** Feeder funds eventually raise their fees, which erodes returns to the underlying investors.

# Solutions • IM-14 a

## Solution IM-14 a.1

**Answer: A**

Bank Q540 • Schweser Module Quiz 96.1 Q1

### Step 1 – What BMIS was in 1960

A brokerage that quoted bid and ask prices and executed OTC trades for clients.
⇒ A.

### Step 2 – Date the other options

The Cincinnati Stock Exchange work came later, after Madoff spotted NYSE Rule 390. The advisory business was set up only in the late 1980s, when brokerage margins were shrinking. BMIS was never a clearinghouse.

### Interpretation

The order matters for the story: decades of a genuine, innovative brokerage built the reputation that the advisory business later traded on.

### Why the other options are wrong

**B is incorrect.** BMIS never ran a derivatives clearinghouse; the options appear only in the strategy it later claimed.
**C is incorrect.** Anachronism: the advisory business, which became the fraud, dates from the late 1980s.
**D is incorrect.** The CSE electronic trading came after Rule 390, well after 1960.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant, Madoff: A Riot of Red Flags (EDHEC, 2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (BMIS Background and Growth)

## Solution IM-14 a.2

**Answer: C**

New • built from SchweserNotes 2026 Reading 96

### Step 1 – How the brokerage grew

BMIS made markets in NYSE-listed stocks off the exchange, earning the bid-ask spread, and paid brokers USD 0.01 a share to route orders to it. This payment for order flow, controversial in the industry, gave it more than 5% of trading volume by 1989 and about 10% by 2008.
⇒ C is correct.

### Step 2 – Rule 390

Rule 390 allowed NYSE-listed stocks to trade away from the floor and its specialists. NYSE members could not make markets in those stocks off the exchange, so the edge went to a non-member: Madoff joined the Cincinnati Stock Exchange instead.

### Interpretation

The brokerage was genuine and successful, and Madoff chaired NASDAQ's board. That reputation is what made investors and regulators reluctant to question the advisory business.

### Why the other options are wrong

**A is incorrect.** Reversed: the opening was to trade listed stocks *away* from the floor and its specialists, not to become one.
**B is incorrect.** The brokerage was a real business; the fraud was in the advisory arm.
**D is incorrect.** Sibling swap: management and performance fees belong to asset management; the brokerage earned spreads.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (BMIS Background and Growth)

### Solution IM-14 a.3 — Answer: D

New • built from SchweserNotes 2026 Reading 96

#### Step 1 – How the advisory business was organised

Its 2008 Form ADV listed 23 discretionary accounts with about USD 17 billion. Most were feeder funds that opened brokerage accounts at BMIS and gave it full trading authority; the feeders were marketed to investors worldwide.

#### Step 2 – What that meant for due diligence

End investors were clients of the feeder funds, not of BMIS, so they could examine only the feeders. Madoff did not let investors visit, ask questions or deal with him directly.
⇒ D explains why the analyst cannot reach BMIS.

#### Interpretation

The reading calls this distinction "central to understanding the Madoff story": investors had to rely on the feeder funds to do the due diligence they could not do themselves.

#### Why the other options are wrong

**A is incorrect.** BMIS was not a hedge fund in limited partnership form; it managed discretionary accounts.
**B is incorrect.** Reversed: the advisory firm was in the same building as the brokerage, which blurred the two.
**C is incorrect.** A small number of large accounts, mostly feeder funds, not thousands of retail accounts.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (BMIS Background and Growth)

### Solution IM-14 a.4 — Answer: D

Bank Q541 • Schweser Module Quiz 96.1 Q2

#### Step 1 – The three legs

Long a basket of stocks highly correlated with the S&P 100; write out-of-the-money S&P 100 calls; buy out-of-the-money S&P 100 puts.
⇒ A, B and C each describe a leg correctly.

#### Step 2 – Test D

The calls were *sold*, so there was no long call to add gains. The short call caps the upside; the long put sets the floor.
⇒ D is the incorrect statement.

#### Interpretation

Remember the collar as a ceiling and a floor: the premium from the call helps pay for the put, and the price is giving up the gains above the call strike.

#### Why the other options are wrong

**A is incorrect.** Correct: selling the out-of-the-money call is step 2.
**B is incorrect.** Correct: the long out-of-the-money put creates the floor.
**C is incorrect.** Correct: the stock basket is the base of the position.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (Reported Investment Strategy and Performance)

### Solution IM-14 a.5 Answer: B

New • built from SchweserNotes 2026 Reading 96

**Step 1 – The falling period**

The index ends below the put strike. The long put pays the difference, so losses on the stocks below the strike are offset: the position loses only about the distance from the start to the put strike (the floor in Figure 2).

**Step 2 – The rising period**

The index ends above the call strike. The short call costs the difference, so gains above the strike are given up: the position gains only about the distance to the call strike (the ceiling).
$\Rightarrow$ B describes both outcomes.

**Interpretation**

This is the payoff the investors were promised. Note what it does *not* do: in a period when the index falls by less than the distance to the put strike, the collar loses the whole fall. That is why almost no losing months was impossible (IM-14 b).

**Why the other options are wrong**

**A is incorrect.** Polarity flip: the puts were bought, not sold, and the short calls cap the 12% gain.
**C is incorrect.** Absolute claim: a collar is long the market between the strikes, so it loses when the index falls.
**D is incorrect.** Legs swapped: the long puts limit losses and the short calls cap gains.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (Reported Investment Strategy and Performance)

### Solution IM-14 a.6 Answer: C

Bank Q542 • Schweser Module Quiz 96.1 Q3

**Step 1 – The level of the returns**

Steady 8% to 12% a year, "comparable to those of the S&P 100".

**Step 2 – The volatility**

No down year in about 17 years and more than 95% of months positive: the volatility was dramatically lower than the index's.
$\Rightarrow$ C: about the same return, much lower volatility.

**Interpretation**

This combination is the core of the fraud's appeal and of its implausibility: equity-like returns with almost bond-like smoothness, from a strategy that is long equities between the strikes.

**Why the other options are wrong**

**A is incorrect.** Wrong on both counts: not significantly higher, and far less volatile.
**B is incorrect.** Right on volatility, wrong on level: returns were comparable, not significantly higher.
**D is incorrect.** Returns of 8% to 12% were neither far below the index nor Treasury-bill-like.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO

96.a (Reported Investment Strategy and Performance)

### Solution IM-14 a.7 Answer: C

Bank Q543 • Schweser Module Quiz 96.2 Q2

**Step 1 – How it ended**

With redemptions he could not meet, Madoff told his sons, wife and brother on 10 December 2008 that the advisory business was "a giant Ponzi scheme".

**Step 2 – What followed**

His sons reported him that evening; he was arrested the next day.
$\Rightarrow$ C: a confession, not an investigation.

**Interpretation**

The reading stresses that Madoff "was not discovered through an investigation", despite Markopolos's warnings to the SEC. The trigger was a funding crisis, not detection.

**Why the other options are wrong**

**A is incorrect.** Markopolos did warn the SEC, but his warnings were dismissed and did not expose the fraud.
**B is incorrect.** No SEC investigation uncovered the scheme.
**D is incorrect.** The sons reported him only after he confessed to them; they did not discover it on their own.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (Fund Collapse and Confession); Schweser files this quiz item under LO 96.b

### Solution IM-14 a.8 Answer: A

Bank Q545 • AnalystPrep Q.4849

**Step 1 – How a Ponzi scheme pays**

"Returns" to earlier investors come out of money raised from later investors; no real profits are earned.

**Step 2 – Why it must end**

The promised balances grow while the real assets do not, so the scheme needs inflows to keep exceeding withdrawals. When new money slows or redemptions spike, as in the 2008 turmoil, it cannot pay and collapses.
$\Rightarrow$ A.

**Interpretation**

This is why the BMIS collapse came when it did: the market crisis brought a wave of redemptions, and Madoff could not raise capital fast enough. Had prices kept rising, the scheme might have run longer.

**Why the other options are wrong**

**B is incorrect.** Regulators did not detect Madoff despite warnings; there is no automatic detection.
**C is incorrect.** No real strategy was being executed, so its profitability is irrelevant.
**D is incorrect.** Feeder fees did not cause the collapse; the cash shortfall did.

**LO:** IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. • **Reference:** Gregoriou and Lhabitant (2009); SchweserNotes 2026 Book 5, Reading 96, LO 96.a (Fund Collapse and Confession); bank LO 96.b, re-filed

IM-15 • Market-Driven Scenarios: An Approach for Plausible Scenario Construction • Vardeez: VERY IMPORTANT

# IM-15 b

P2 CONCEPT

Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios.

Workbook: C9 Stress-Testing Practices and Scenario Design • Tier C • Map #38

**WHAT IS TESTED HERE**

**Why P2.** A Tier C workbook topic (C9, Map #38: "2 to 3 questions on stress testing related: appropriate practices", 3 of 16 recall sheets), lifted to P2 because Vardeez flags it in a VERY IMPORTANT chapter.
**Vardeez:** Mahalanobis distance, scenario / volatility / correlation z-scores. The LO verb is *explain*: no need to get hung up on formulas.

**Reported exam shapes**

- **S2** (C9) Market-driven scenarios: plausible scenario construction, z-scores and covariance choice. Workbook C9.2 is the last question of this block.

**What the questions look like:** statement questions. Which of the four measures does what (the sibling swap among them is the standard trap), which way a change moves each measure (doubling every shock, flipping one sign against the correlation, adding policy variables), and how the choice of covariance matrix changes the reading. No question needs a formula to be evaluated. This block also carries the shared background for IM-15 (Section 1), which IM-15 c and IM-15 a build on.

## 1. The conditional stress testing framework

The market-driven scenario (MDS) approach is a **conditional stress test**. Of the $M$ risk factors that drive a portfolio, the experts shock only a small subset of $K$ **policy variables**: major traded instruments such as the S&P 500, bond yields, or factor-mimicking portfolios (momentum, growth). The shocks to the other $M - K$ **nonpolicy factors** are *implied* from the policy shocks by multivariate regression. Policy and implied shocks together form the **perturbation vector**: the set of simultaneous shocks across every risk factor.

$$\hat{r}_j = \sum_{k=1}^{K} \beta_{jk} r_k \quad (j = K+1, \ldots, M) \qquad \Longleftrightarrow \qquad \hat{r}_N = \Sigma_{NP} \Sigma_{PP}^{-1} r_P$$

- **The betas are the covariance matrix.** Estimating the regression coefficients is equivalent to estimating the covariance matrix of the $M$ factor returns. The matrix form on the right is the standard conditional-mean result ($\Sigma_{PP}$ among the policy variables, $\Sigma_{NP}$ between nonpolicy and policy factors). Change the matrix and every implied shock changes (Section 5).
- **Keep $K$ small.** Each policy shock is a judgment, so a small set of policy variables minimizes the arbitrariness of the perturbation vector (and, IM-15 c, limits multicollinearity).
- **P&L.** If asset returns are linear in the factor shocks, the portfolio return is $\sum_m x_m r_m$, exposures times shocks. Positions with embedded optionality (nonlinear payoffs) are **fully revalued** using the full perturbation vector.

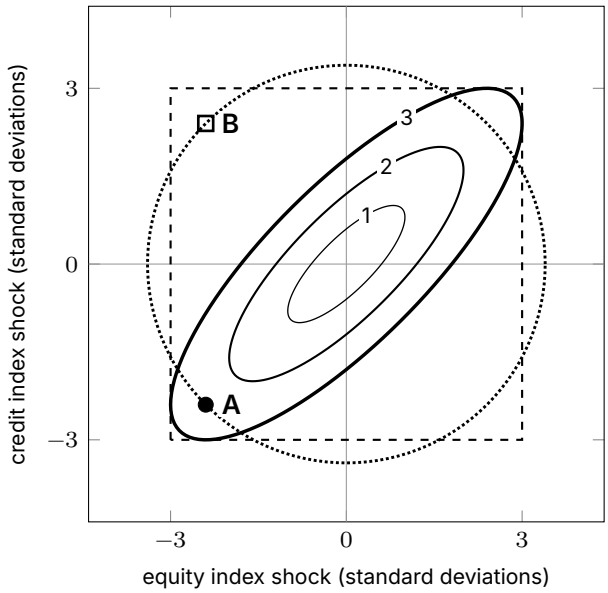
Every tool in this LO answers one question about the policy shock vector $r$ that the experts propose: *how far from history is it, and is it coherent?*

## 2. Mahalanobis distance: is the shock vector an outlier?

$$\boxed{\text{MD} = \sqrt{(r-\mu)'\,\Sigma^{-1}\,(r-\mu)}}$$

where $r$ is the shock vector, $\mu$ the mean vector and $\Sigma^{-1}$ the inverse covariance matrix of the policy variables.

- An **implausible** shock is one highly inconsistent with historical data: an outlier. The MD is the most common statistic for spotting *multivariate* outliers.
- It counts how many standard deviations the point lies from the centre of the distribution, *taking the shape of the data into account*. It is a multivariate generalization of a z-score: with one variable it is $|r-\mu|/\sigma$.
- **Shape matters** (Figure 1). Take two policy variables whose returns have a correlation of $+0.8$. Point A (both fall 2.4 standard deviations) has MD $= 2.53$; point B (one falls 2.4, the other rises 2.4) has MD $= 7.59$. Both pass a univariate 3-sigma screen. Only the MD sees that B cuts across the correlation.




**Solid ellipses:** equal Mahalanobis distance (1, 2, 3) when the two returns have correlation $+0.8$. The ellipses lie along the diagonal: falling together is ordinary.

**Dashed square:** a univariate screen, each shock within 3 standard deviations on its own.

**Dotted circle:** equal distance with the correlation ignored. A and B both lie on it (3.39), so they have the *same volatility z-score*.

**A** (filled circle): both fall 2.4$\sigma$. MD 2.53, inside the MD $= 3$ ellipse: with the grain, low correlation z-score.

**B** (open square): equities fall 2.4$\sigma$, credit rallies 2.4$\sigma$. MD 7.59, far outside every ellipse: against the grain, high correlation z-score.

Figure 1. Why the univariate view is not enough (Vardeez's figure, redrawn with numbers). A and B have identical factor z-scores and an identical zero-correlation distance, so they pass the same univariate screen. Only the Mahalanobis distance, and the scenario and correlation z-scores built on it, separate the plausible scenario from the implausible one. These are the inputs of Question IM-15 b.4.

## 3. Scenario z-score: severity on a common scale

- **The problem with the raw MD.** It grows with the number of policy variables $n$, roughly with $\sqrt{n}$. Nine uncorrelated one-sigma shocks give MD $= \sqrt{9} = 3$, although no single move is unusual (under normality the expected MD$^2$ is $n$).
- **The fix.** The **scenario z-score** rescales the MD so that it does not increase with $\sqrt{n}$. It states the **severity** of a scenario in standard deviations. **Lower means more plausible**; higher means more severe and less plausible. Scenario z-scores tend to rise during periods of market stress.
- **Risk ruler.** A spectrum of past events, from most to least plausible, each with its scenario z-score under the chosen covariance matrix. Many were outliers that looked implausible beforehand but

still happened. To size a hypothetical event, match its perceived likelihood to a historical event and read off the z-score: a higher z-score marks a more severe, less plausible stress.

- **Relative, not absolute, probability.** Scenario z-scores measure the *relative* probability of events, assuming returns follow an elliptical distribution. An absolute probability needs one more assumption: a **parametric** approach (assume a distribution for the z-scores) or a **nonparametric** approach (an empirical distribution of z-scores over a historical horizon).

## 4. Volatility and correlation z-scores: size versus shape

Kinlaw and Turkington (2014) split the scenario z-score into two parts:

- **Volatility z-score: the magnitude.** The scenario z-score recomputed assuming *all correlations between the shocks are zero*. It depends only on how large each shock is against its own volatility, and it rises when the shocks get bigger.
- **Correlation z-score: the consistency.** The scenario z-score normalized for the magnitude of the shocks. It can also be written directly in terms of the correlation matrix ($\Lambda$) and the individual factor z-scores. It depends on the **signs and relative magnitudes** of the factor z-scores, **not their absolute size**: doubling every shock leaves it unchanged. Close to zero: the shocks are consistent with the covariance matrix. High and positive (above about 1.0): the matrix may not capture the co-movement the experts are assuming.

**Reading the three together.** A high scenario z-score with a high volatility z-score and a low correlation z-score is a big but coherent scenario: the severity comes from the size of the moves. A moderate volatility z-score with a high correlation z-score is a scenario of ordinary-sized moves in an unusual combination, like point B in Figure 1: the implausibility comes from the shape.

## 5. Which covariance matrix? (shape S2)

Every z-score standardizes the shock vector by a covariance matrix, so the choice of dates and parameters can change the result materially. Three common choices:

| | |
|---|---|
| **Slow data** | Historical matrix with a long lookback, such as five years of weekly, equally weighted data before the event. |
| **Fast data** | Historical matrix with a short lookback, such as one year of weekly, equally weighted data before the event. |
| **Recent data** | The current matrix, which may reflect post-event dynamics. |

- Each choice changes how surprising a given scenario looks, so each produces a **different risk ruler** for the same set of events.
- **The rule: match the matrix to the scenario.** For a market crash, use data from periods with similar crashes, because correlations among risky assets typically spike during crises. Relying only on "normal" data would likely **understate** the severity of the stress.
- **Why calm data mislead twice.** Calm-period volatilities make the policy shocks look more extreme (higher z-scores), so a sound crash scenario can be dismissed as implausible. Calm-period correlations make the implied nonpolicy shocks too small, so the stress P&L understates the contagion.

> **Correction to the source notes.** Vardeez's table defines the volatility z-score as $(\sigma_{\text{scen}} - \mu_\sigma)/\sigma_\sigma$ and the correlation z-score as $(\rho_{\text{scen}} - \mu_\rho)/\sigma_\rho$, z-scores of a volatility *level* and a correlation *level*, and calls the sce-

nario z-score a univariate $(x-\mu)/\sigma$ ("three univariate measures and one multivariate one"). The reading defines them differently (SchweserNotes, after Kinlaw and Turkington): the scenario z-score is the Mahalanobis distance rescaled for the number of policy variables, so it is multivariate; the volatility z-score is that score recomputed with zero correlations (the size of the shocks); the correlation z-score normalizes it for that size (consistency with the covariance matrix). His trap line "z-scores give severity; Mahalanobis distance gives likelihood" follows from his definitions. In the reading, the MD and the scenario z-score measure the same distance, read both as severity (standard deviations) and as plausibility (lower is more plausible). His A-versus-B figure is right and is redrawn as Figure 1. The reading's definitions are used throughout, as in Workbook C9.2.

| | What it captures | Uses correlations? | Double every shock | Read it as |
|---|---|---|---|---|
| **Mahalanobis distance** | Distance of the shock vector from the historical centre, in standard deviations, given the shape of the data | Yes ($\Sigma^{-1}$) | Rises | Outlier screen; grows with $\sqrt{n}$ |
| **Scenario z-score** | The MD rescaled for the number of policy variables | Yes | Rises | Severity; lower is more plausible; placed on the risk ruler |
| **Volatility z-score** | Magnitude of the shocks | No: set to zero | Rises | How big the moves are |
| **Correlation z-score** | Consistency of the shock pattern with the covariance matrix | Yes, through signs and relative sizes | Unchanged | Near 0 consistent; above about 1.0 a warning |

**Exam traps**

- **Sibling swap among the z-scores.** Scenario z-score = severity (lower is more plausible); volatility z-score = magnitude with zero correlations; correlation z-score = consistency with the covariance matrix. Workbook C9.2's option A gives the scenario z-score the correlation z-score's job.
- **Univariate thinking.** The same number of standard deviations on each variable does not give the same scenario z-score: a sign that fights the correlation raises the MD (Figure 1).
- **Raw MD across different $n$.** The MD grows with $\sqrt{n}$; compare scenarios on scenario z-scores.
- **Doubling the shocks.** The correlation z-score is unchanged; the MD, the scenario z-score and the volatility z-score all rise.
- **Polarity of the ruler.** A higher scenario z-score means less plausible and more severe, not "more likely" or "already priced in".
- **Relative versus absolute.** A z-score is a number of standard deviations, not a probability; an absolute probability needs a parametric or nonparametric step.
- **Calm data for a crash.** Normal-period data understate crisis correlations and so understate the stress; they do not overstate it.
- **Passing the screen ends nothing.** What counts as "too extreme" is still a judgment.

## Practice questions: IM-15 b

### Question IM-15 b.1

Bank Q551 • Schweser Module Quiz 97.1 Q2 (rebuilt)

In the market-driven scenario (MDS) approach, experts propose a vector of shocks to several policy variables. A risk manager wants a single multivariate statistic that accounts for the correlations among the policy variables and measures how many standard deviations the shock vector lies from the historical centre, so that implausible outliers can be flagged. Before any adjustment for the number of policy variables, which statistic should the manager use?

**A.** Scenario z-score

**B.** Volatility z-score

**C.** Correlation z-score

**D.** Mahalanobis distance

### Question IM-15 b.2

New • built from SchweserNotes 2026 Reading 97

A risk analyst at a multi-asset manager screens two proposed market-driven scenarios with the Mahalanobis distance, using the same covariance matrix. Scenario 1 shocks two policy variables. Scenario 2 shocks nine policy variables that are close to uncorrelated, each by one standard deviation, so its Mahalanobis distance is about 3.0, the same as Scenario 1's. The analyst concludes that the two scenarios are equally severe and equally plausible. Which of the following is the most appropriate response?

**A.** The conclusion is correct, because the inverse covariance matrix in the Mahalanobis distance already adjusts for the number of policy variables.

**B.** The conclusion is flawed: the distance tends to grow with the square root of the number of policy variables, so the scenarios should be compared on scenario z-scores, which would rank Scenario 2 as less severe.

**C.** The conclusion is flawed: Scenario 2 should be ranked as more severe, because a scenario that shocks more variables at once is always less plausible.

**D.** The conclusion is correct, because a Mahalanobis distance of 3.0 describes a three-standard-deviation scenario whatever the number of variables shocked.

**Question IM-15 b.3** Bank Q553 • New, written for the bank (GARP style)

A risk committee compares two hypothetical scenarios using a risk ruler built from historical events. Scenario X has a scenario z-score of 2.6 and Scenario Y has a scenario z-score of 4.1. A committee member asks the risk manager how to interpret and use these results. Which of the following statements is most accurate?

**A.** Scenario X is more plausible than Scenario Y, and the z-scores measure relative probability under an elliptical distribution; absolute probabilities require a parametric or nonparametric distributional assumption.

**B.** Scenario Y is more plausible than Scenario X, and the z-scores directly equal the probabilities of occurrence of the scenarios.

**C.** Scenario X is more plausible than Scenario Y, and the z-scores can be converted into absolute probabilities without any additional assumptions.

**D.** Scenario Y is more plausible than Scenario X, because a higher scenario z-score indicates a larger shock that the market has already priced in.

**Question IM-15 b.4** New • built from SchweserNotes 2026 Reading 97

A risk manager at a US credit fund tests two candidate scenarios on the same two policy variables, a US equity index and a US high-yield credit index, whose returns have a correlation of $+0.8$ in the chosen covariance matrix. Each scenario moves each index by 2.4 standard deviations. In Scenario A both indices fall. In Scenario B the equity index falls and the high-yield credit index rallies. Which of the following statements is correct?

**A.** The two scenarios have the same scenario z-score, because each shock is the same number of standard deviations in both scenarios.

**B.** The two scenarios have the same volatility z-score, but Scenario B has the higher scenario z-score and the higher correlation z-score.

**C.** Scenario A has the higher scenario z-score, because two indices falling together produce a larger combined loss than offsetting moves.

**D.** The two scenarios have the same correlation z-score, because the correlation z-score does not depend on the absolute size of the shocks.

**Question IM-15 b.5** Bank Q552 • New, written for the bank (GARP style)

A risk manager reviews a market-driven scenario in which experts have shocked three policy variables. The scenario z-score is 3.4. After the experts double the size of every shock while preserving all signs and relative magnitudes, the risk manager recomputes the statistics. Which of the following is correct?

**A.** The scenario z-score is unchanged because it is normalized for the number of policy variables.

**B.** The correlation z-score doubles because it is the scenario z-score divided by the volatility z-score.

**C.** The volatility z-score is unchanged because it isolates the consistency of the shocks with the covariance structure.

**D.** The correlation z-score is unchanged because it depends on the signs and relative magnitudes of the factor z-scores rather than their absolute size.

**Question IM-15 b.6** New • built from SchweserNotes 2026 Reading 97

A risk team is building a risk ruler of historical stress events for its market-driven scenario program. The same event receives a scenario z-score of 6.8 when the covariance matrix uses five years of weekly data before the event, 4.1 when it uses one year of weekly data before the event, and 3.2 when it uses the current covariance matrix. A committee member asks what the differences mean. Which of the following responses is most appropriate?

**A.** Only the five-year figure is valid, because a longer equally weighted window always gives the most reliable scenario z-score.

**B.** The event's shocks must have been mismeasured, because an event has a single scenario z-score whatever covariance matrix is used.

**C.** Each matrix makes the same event look more or less surprising and gives a different risk ruler, so the matrix should match the conditions of the scenario being modeled.

**D.** The current matrix should always be used, because the most recent data always describe the conditions under which the next stress will occur.

**Question IM-15 b.7** Bank Q559 • New, written for the bank (GARP style)

A firm's risk team uses the Mahalanobis distance and scenario z-scores to screen proposed policy shocks for a market-driven scenario. A newly hired analyst argues that once a shock vector passes the statistical screen, no further judgment is needed. Which of the following best describes why this view is incorrect?

**A.** Plausibility testing is statistical in nature, but it still requires judgment about what is reasonable versus too extreme, and implied nonpolicy shocks may be unrealistic if the covariance matrix is poorly chosen.

**B.** The Mahalanobis distance can be computed only after the portfolio P&L has been estimated, so it cannot screen shocks beforehand.

**C.** The statistical screen is unnecessary because the scenario's plausibility is determined entirely by the catalyst chosen when the event is defined.

**D.** Scenario z-scores are valid only for historical events, so they cannot be used to assess hypothetical shocks.

**Question IM-15 b.8** Workbook C9.2 • S2, S3 • Map #38

A portfolio risk manager at a London asset manager is drafting internal guidance on building plausible stress scenarios, drawing on the market-driven scenario (MDS) approach and on lessons from the supervisory bank stress tests. Which of the following statements should the manager keep in the guidance?

**A.** The scenario z-score measures how consistent a set of policy shocks is with the chosen covariance matrix, so a value close to zero signals a coherent scenario.

**B.** The set of policy variables should be kept large, so that the regression that implies the non-policy factor shocks leaves less room for arbitrariness.

**C.** The 2009 SCAP achieved coherence with a rich scenario of more than 70 trading-book risk factors and eight macro factors for each participating country.

**D.** For a market crash scenario, the covariance matrix should be estimated from periods with similar crashes, since correlations among risky assets spike in crises.

## Solutions • IM-15 b

### Solution IM-15 b.1 — Answer: D

Bank Q551 • Schweser Module Quiz 97.1 Q2

**Step 1 – List what the manager needs**

One statistic, multivariate, that uses the correlations among the policy variables, measures distance from the historical centre in standard deviations, and comes *before* any adjustment for the number of variables.

**Step 2 – Match it**

$\text{MD} = \sqrt{(r-\mu)'\Sigma^{-1}(r-\mu)}$. The inverse covariance matrix brings in the correlations, and the result is a multivariate generalization of a z-score. It is the most common statistic for identifying multivariate outliers.

$\Rightarrow$ The Mahalanobis distance.

**Interpretation**

The scenario z-score is built from the MD by rescaling for the number of policy variables. The phrase "before any adjustment for the number of policy variables" is what separates the two.

**Why the other options are wrong**

**A is incorrect.** The scenario z-score is the adjusted measure: the MD rescaled so that it does not grow with $\sqrt{n}$. The stem asks for the measure before that adjustment.

**B is incorrect.** The volatility z-score sets every correlation to zero, so it ignores exactly what the manager wants included. It isolates the size of the shocks.

**C is incorrect.** The correlation z-score normalizes the size away and shows whether the shocks fit the covariance structure. It is not a distance from the centre.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.b, Mahalanobis Distance; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.5

### Solution IM-15 b.2 — Answer: B

New • built from SchweserNotes 2026 Reading 97

**Step 1 – Why the two distances match**

With nine uncorrelated shocks of one standard deviation each, $\text{MD}^2 = 1^2 + \cdots + 1^2 = 9$ and $\text{MD} = 3$.

$\Rightarrow$ Scenario 2 reaches 3.0 by adding up nine ordinary moves, not through any single unusual one.

**Step 2 – The scaling problem**

The MD tends to grow with the square root of the number of policy variables: under normality its expected square equals $n$. Raw distances therefore penalize the scenario with more variables.

**Step 3 – Use the scenario z-score**

The scenario z-score rescales the MD so that it does not increase with $\sqrt{n}$, and expresses severity in standard deviations. On that scale Scenario 2 is the less severe, more plausible scenario.

**Interpretation**

Rescaling is what lets scenarios built on different numbers of policy variables, and the historical events on a risk ruler, sit on one scale. Lower scenario z-score means more plausible.

**Why the other options are wrong**

**A is incorrect.** The inverse covariance matrix adjusts for each variable's volatility and for the

correlations, not for how many variables there are. That is why a separate rescaling exists.
**C is incorrect.** Polarity flip and absolute claim. More variables do not by themselves make a scenario less plausible; here every move is ordinary, so Scenario 2 is the milder one.
**D is incorrect.** Absolute claim. A distance of 3.0 means very different things for two variables and for nine; ignoring the $\sqrt{n}$ effect is the error the scenario z-score was introduced to fix.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.b, Scenario Z-Score; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.5

### Solution IM-15 b.3 — Answer: A

Bank Q553 • New, written for the bank

#### Step 1 – Order the scenarios

A lower scenario z-score means a more plausible scenario. X (2.6) is more plausible than Y (4.1).

#### Step 2 – What kind of probability

Scenario z-scores measure the *relative* probability of events under the assumption that returns follow an elliptical distribution.

#### Step 3 – Getting to an absolute probability

One more assumption is needed: a parametric approach (a probability distribution for the z-scores) or a nonparametric approach (an empirical distribution of z-scores over a historical horizon).
$\Rightarrow$ A is the only option right on both the ranking and the probability statement.

#### Interpretation

On a risk ruler, X would sit nearer the plausible end and Y nearer the extreme end. The committee can say "Y is less likely than X" without further work; "Y has a 1% chance" needs the extra step.

#### Why the other options are wrong

**B is incorrect.** Reverses the ranking, and a z-score is a number of standard deviations, not a probability.
**C is incorrect.** Right ranking, but absolute claim: converting a z-score into an absolute probability needs a parametric or nonparametric assumption.
**D is incorrect.** Reverses the ranking. A higher z-score marks a more severe, less plausible scenario; "already priced in" is the Step 1 criterion of IM-15 c, applied out of place.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.b, Scenario Z-Score and Figure 97.1 (risk ruler)

### Solution IM-15 b.4 — Answer: B

New • built from SchweserNotes 2026 Reading 97

#### Step 1 – Volatility z-score (correlations set to zero)

Both scenarios have two shocks of 2.4 standard deviations, so the zero-correlation distance is $\sqrt{2.4^2 + 2.4^2} = 3.39$ in both.
$\Rightarrow$ Same magnitude, same volatility z-score.

#### Step 2 – Scenario z-score (correlation $+0.8$ included)

$$\text{MD}_A = \sqrt{\frac{2.4^2 + 2.4^2 - 2(0.8)(2.4)(2.4)}{1 - 0.8^2}} = \sqrt{\frac{2.304}{0.36}} = 2.53$$

$$\text{MD}_B = \sqrt{\frac{2.4^2 + 2.4^2 + 2(0.8)(2.4)(2.4)}{1 - 0.8^2}} = \sqrt{\frac{20.736}{0.36}} = 7.59$$

$\Rightarrow$ A moves with the correlation; B cuts across it. B's scenario z-score is far higher.

### Step 3 – Correlation z-score

It normalizes for magnitude and depends on the signs and relative sizes of the factor z-scores. The sizes are identical and only B's sign pattern contradicts the $+0.8$ correlation, so B's correlation z-score is high and A's is low.

### Interpretation

These are points A and B of Figure 1. A univariate screen, or the volatility z-score alone, would treat the two scenarios as equally extreme. The figures in Step 2 are there only to show the direction: the question is answered by the definitions.

### Why the other options are wrong

**A is incorrect.** Univariate thinking: equal factor z-scores do not give equal scenario z-scores once the correlation enters ($2.53$ against $7.59$).

**C is incorrect.** Confuses portfolio loss with statistical plausibility. The z-scores measure distance from history, not P&L, and A is the scenario that sits with the grain.

**D is incorrect.** A true fact applied wrongly: the correlation z-score is independent of the absolute size, but it does depend on the signs, and B flips one.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.b, Volatility Z-Score and Correlation Z-Score (Kinlaw and Turkington, 2014)

## Solution IM-15 b.5 Answer: D

Bank Q552 • New, written for the bank

### Step 1 – Sort the measures by what they depend on

Volatility z-score: the magnitude of the shocks (correlations set to zero). Scenario z-score: severity in standard deviations, so it also depends on magnitude. Correlation z-score: the scenario z-score normalized for magnitude, so it depends only on signs and relative sizes.

### Step 2 – Apply the doubling

Signs and relative magnitudes are preserved, so the correlation z-score is unchanged. The scenario z-score and the volatility z-score both rise with the size of the shocks.

$\Rightarrow$ D.

### Interpretation

Doubling every shock makes a scenario more severe but no less coherent. The correlation z-score answers "is the pattern believable?", not "are the moves big?".

### Why the other options are wrong

**A is incorrect.** The normalization is for the number of policy variables, not for the size of the shocks: severity rises when the shocks double.

**B is incorrect.** The magnitude effect is what the correlation z-score removes, so a pure rescaling leaves it where it was.

**C is incorrect.** Sibling swap: the volatility z-score is the magnitude component and rises with the shocks; consistency with the covariance structure is the correlation z-score's job.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.b, Correlation Z-Score

### Solution IM-15 b.6 — Answer: C

New • built from SchweserNotes 2026 Reading 97

#### Step 1 – What a z-score is measured against

Every scenario z-score standardizes the shock vector by a covariance matrix. Change the matrix and the yardstick changes. Slow data (five years weekly before the event), fast data (one year weekly before the event) and recent data (the current matrix, which may reflect post-event dynamics) are the three common choices.

#### Step 2 – Read the numbers

The z-score falls from 6.8 to 4.1 to 3.2. The shorter window and the current matrix carry more volatility (or correlations closer to the event's own pattern) than the five-year window, so the same moves are fewer standard deviations.

$\Rightarrow$ Each matrix produces a different risk ruler for the same events.

#### Step 3 – Which one to use

The appropriate matrix depends on the scenario being modeled: data from similar crashes for a crash scenario; recent data for Brexit, which had no comparable precedent (IM-15 c).

#### Interpretation

None of the three numbers is "the" z-score of the event. The committee should ask which window describes the conditions of the scenario under review, and keep the choice consistent across the ruler.

#### Why the other options are wrong

**A is incorrect.** Absolute claim: a long equally weighted window is not always best. For a crash it dilutes the crisis correlations and understates the stress.

**B is incorrect.** A scenario z-score is not a property of the event alone; it depends on the covariance matrix used to standardize the shocks.

**D is incorrect.** Absolute claim: recent data suit some scenarios (Brexit) and not others, and the current matrix may reflect post-event dynamics.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.b, Covariance Matrix Selection and Figures 97.2 to 97.4

### Solution IM-15 b.7 — Answer: A

Bank Q559 • New, written for the bank

#### Step 1 – What the screen does

The Mahalanobis distance and the scenario z-score are applied to the proposed policy shock vector to flag outliers and to gauge severity and plausibility.

#### Step 2 – What it cannot do

There is no objective threshold separating "reasonable" from "too extreme": that line is a judgment. And the implied nonpolicy shocks come from the covariance structure, so a poorly chosen matrix can make them unrealistic even when the policy shocks pass the screen.

$\Rightarrow$ A.

#### Interpretation

The statistics discipline the judgment; they do not replace it. The reading lists this among the challenges of implementing the approach (IM-15 d).

### Why the other options are wrong

**B is incorrect.** The distance is computed on the shock vector before any P&L is estimated; screening shocks beforehand is its purpose.

**C is incorrect.** A specific catalyst helps define the event (IM-15 c, Step 1); it does not replace the statistical check of the shocks.

**D is incorrect.** Scenario z-scores are applied to hypothetical events; the risk ruler exists precisely to scale them against past events.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.b and LO 97.d (fifth challenge)

## Solution IM-15 b.8 Answer: D

Workbook C9.2 • S2, S3

### Step 1 – Covariance choice (IM-15 b, c)

Z-scores and implied nonpolicy shocks both depend on the covariance matrix. "If you are modeling a market crash, you should use data or a covariance matrix from periods that experienced similar crashes, because correlations among risky assets typically spike during market crises. Relying only on 'normal' data would likely understate the severity of the stress."

### Step 2 – Map the other statements to their true owners

Consistency with the covariance structure belongs to the correlation z-score; a small policy set is the MDS rule; the 70-plus factor scenario belongs to EBA 2011, while SCAP used GDP, unemployment and house prices plus a historical market shock.

### Interpretation

D is correct. Its supervisory counterpart in ORR-17 is the same lesson from the other side: a historical shock gives loose coherence but "does not test for anything new", so the data window must match the stress being modelled.

### Why the other options are wrong

**A is incorrect.** Sibling swap: the scenario z-score measures severity (lower means more plausible). The correlation z-score normalizes for magnitude and shows consistency with the covariance matrix: near zero consistent, above 1.0 a warning.

**B is incorrect.** Polarity flip: "keeping the number of policy variables small helps minimize arbitrariness in the resulting perturbation vector"; a large set invites multicollinearity (do not add a variable with adjusted $R^2$ of 90% or more).

**C is incorrect.** Sibling swap (case cross-mapping): over 70 trading-book factors and eight macro factors for each of 21 countries describe the 2011 EBA test. The 2009 SCAP state space had only three dimensions with a historical market shock from June to December 2008.

**LO:** IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. ORR-17 b: Explain challenges in designing stress test scenarios, including the problem of coherence in modeling risk factors. • **Reference:** Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.5 (via SchweserNotes 2026 Book 5, Reading 97, LO 97.b and 97.c); Schuermann, Stress Testing Banks, Section 17.3

IM-15 • Market-Driven Scenarios: An Approach for Plausible Scenario Construction • Vardeez: VERY IMPORTANT

# IM-15 c

P2 CONCEPT

Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided.

Workbook: C9 Stress-Testing Practices and Scenario Design • Tier C • Map #38

### WHAT IS TESTED HERE

**Why P2.** A Tier C workbook topic (C9, Map #38), lifted to P2 because Vardeez flags it in a VERY IMPORTANT chapter.
**Vardeez:** know how to construct a market-driven scenario.

**Reported exam shapes**

- **S2** (C9) Market-driven scenarios: plausible scenario construction, z-scores and covariance choice. Workbook C9.2 touches this LO; it is worked as Question IM-15 b.8 (Question IM-15 b.8).

**What the questions look like:** statement questions on the five steps in order, the rule behind each step (Step 1 criteria, the 90% adjusted $R^2$ test, the two calibration routes, the data window), and the facts of the Brexit example. The construction is what is tested, not the arithmetic: a bank item that only multiplied the 60/40 weights by two shocks is left out under the CONCEPT rule. The conditional framework, the z-scores and the covariance choice are in IM-15 b; this block recaps them only where a step uses them.

## 1. Five guidelines that reduce subjectivity

- Define hypothetical scenarios **precisely** enough to translate clearly into shocks on market factors.
- Choose key risk factors (policy variables) that are **not highly correlated** with one another.
- Keep them to a **small, straightforward set** that still captures the key market dynamics.
- Use **scenario z-scores** to evaluate the plausibility, magnitude and relative sizing of the shocks.
- Build the covariance matrix from a **data sample consistent with the nature of the scenario's shocks**.

## 2. The five steps

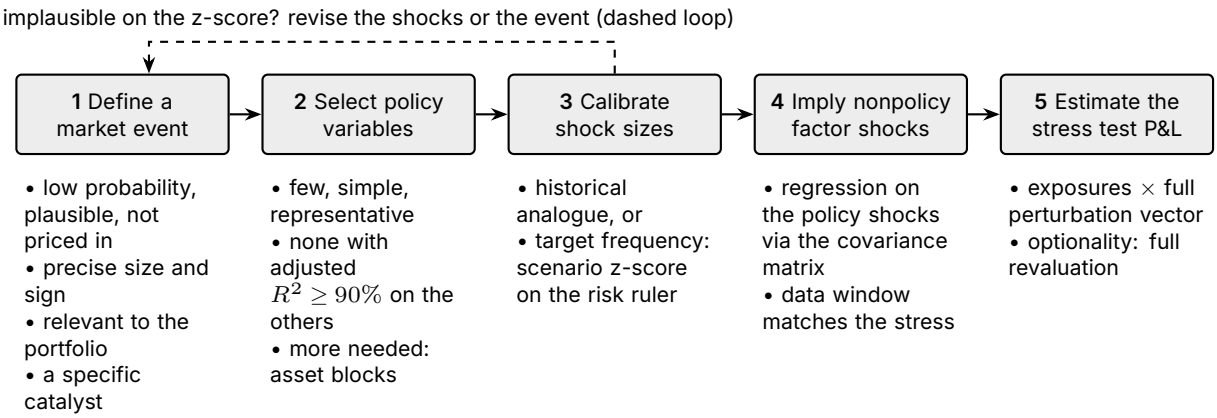



Figure 1. The five steps of the MDS process. Solid arrows show the order; the dashed loop is the iteration of IM-15 a, in which the policy shocks are checked for plausibility and refined until a collective version is agreed.

- **Step 1: Define a market event.** Two levels of uncertainty: whether the event will occur, and how markets will react if it does. Four criteria reduce it. (i) *Low probability but plausible*: known risks are already priced in, so look for one that is unlikely, possible and not yet in prices. (ii) *Precise*: the expected market impact, size and direction, is not open to interpretation. (iii) *Relevant*: value-versus-growth shocks mean little to a portfolio holding only US Treasuries. (iv) *A specific catalyst*: because the event is unlikely, name what could plausibly trigger it.
- **Step 2: Select policy variables.** Simple yet representative of the market dynamics, and few, to minimize multicollinearity (highly correlated variables carry overlapping, redundant information). Do not add a policy variable with an **adjusted $R^2$ of 90% or more** against the existing ones. If a larger set is unavoidable, use **asset blocks**: regress different groups of implied factors on different subsets of the policy variables.
- **Step 3: Calibrate shock sizes.** Two routes: (1) find a historical period that resembles the scenario, or (2) set a **target frequency** from a subjective view of its likelihood, the likelihood of a scenario at least as extreme as the one proposed, measured by the scenario z-score (the risk ruler of IM-15 b).
- **Step 4: Generate the nonpolicy shocks.** They are implied from the policy shocks through the covariance matrix, so the choice of matrix is critical. The window should reflect conditions like the scenario's: to model the end of quantitative easing on the short end of the curve, use a period when short rates were more volatile; for geopolitical risk, a period of political uncertainty.
- **Step 5: Estimate the stress test P&L.** Apply the full set of shocks to the portfolio's factor exposures. Embedded optionality is fully revalued, not approximated linearly.

## 3. The example: Brexit, 2016

- **The event (Step 1).** A low-probability UK referendum in June 2016 on leaving the European Union. By February 2016 scenarios existed for both outcomes. The "remain" camp appeared to hold an early 10% lead, and the vote to leave surprised markets. Any portfolio exposed to the euro, European or British sovereign debt, or European or British equities needed a view.
- **Three scenarios across the spectrum.** *Remain*; *Soft Brexit* (leave, trade relations with Europe largely intact); *Adverse Brexit* (leave, trade relations deteriorate significantly). The reading follows Soft Brexit, for which the post-exit models considered were the single market (Norway), a

customs union (Turkey) and "single-market-lite" (Switzerland). Consensus: risk-off sentiment and a dovish Bank of England.

- **Policy variables (Step 2).** UK-focused: the FTSE All Share index, the 10-year gilt yield and sterling. Care was taken to separate Soft Brexit from an ordinary downturn, since it would hit the banking sector and housing demand in ways a typical shock would not. Shocks to factors such as Italian sovereign debt and the Mexican peso were kept **deliberately small**, to dampen contagion outside the UK and stop them overwhelming the covariance matrix. Policy shocks were generally kept **within one asset class** to avoid spurious correlations.
- **Sizing (Step 3).** Direct historical comparisons gave limited guidance: sterling's 1992 exit from the Exchange Rate Mechanism was loosely related but was about currency policy. On the risk ruler (Figure 2), Remain was more likely than Soft Brexit, and Adverse Brexit far less likely with a much more severe impact. Asset class experts recommended no significant changes. The most debated variable was rates: would gilts be a safe haven or would investors leave UK fixed income? Soft Brexit assumed a **safe haven**; Adverse Brexit assumed **rising yields**.
- **Scenario z-score delta.** For each policy shock, the change in the scenario z-score if that shock were moved from its stressed value to its expected value conditional on the others. It was close to zero for every shock **except rates**: apart from rates, the shocks were largely consistent with one another.
- **Covariance matrix (Step 4).** No truly comparable historical event existed, so a matrix based on **recent data** was used.
- **P&L (Step 5).** Run through BlackRock's Aladdin for a portfolio of 60% FTSE 100 and 40% iBoxx GBP bonds: a sizable equity loss, partly offset by gains on the bonds.
- **The point of the exercise.** Not to quantify tail losses or make managers target outcomes, but to integrate different perspectives in a structured way. Quantifying returns under scenarios of varying plausibility makes managers reassess their strategic convictions and widens awareness of alternative outcomes.

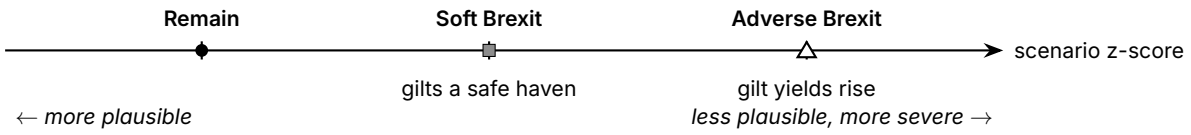



Figure 2. The Brexit scenarios on the risk ruler (after the reading's Figure 97.6). Schematic: the reading gives the order of the three scenarios, not their values, so the spacing carries no information. Markers: filled circle Remain, grey square Soft Brexit, open triangle Adverse Brexit.

**Correction to the source notes.** Vardeez lists four steps: define the narrative; shock the core assets (calibrated from historical analogues, option-implied moves or expert judgement); test plausibility; propagate and revalue. The reading has *five*: define a market event; select policy variables; calibrate shock sizes to the policy variables; generate nonpolicy factor shocks from the policy shocks; estimate the portfolio stress test P&L. Vardeez merges selection with calibration, makes plausibility testing a step of its own (in the reading it runs through calibration, via the scenario z-score, and through the iterative review), and merges P&L into propagation. The reading's two calibration routes are a historical analogue or a target frequency. Learn the five steps as listed here; Question IM-15 c.1 tests the order.

| Step | Rule to remember | In the Brexit example |
|---|---|---|
| **1 Event** | Low probability, plausible, not priced in; precise; relevant; catalyst | Leave vote; Remain, Soft and Adverse Brexit; risk-off and a dovish Bank of England |
| **2 Policy variables** | Few and simple; no addition with adjusted $R^2 \geq 90\%$; asset blocks; one asset class per shock set | FTSE All Share, 10-year gilt yield, sterling; small ITL and MXN shocks to dampen contagion |
| **3 Calibration** | Historical analogue or target frequency (scenario z-score, risk ruler) | 1992 ERM exit of limited help; ruler order Remain, Soft, Adverse; rates most debated; z-score delta near zero except rates |
| **4 Nonpolicy shocks** | Implied through the covariance matrix; window matches the stress | Recent data: no comparable precedent |
| **5 P&L** | Exposures times shocks; full revaluation of optionality | 60/40 FTSE 100 and iBoxx GBP: equity loss partly offset by bond gains |

**Exam traps**

- **Order.** The policy shocks are calibrated (Step 3) before the nonpolicy shocks are implied (Step 4); the P&L comes last. Implying shocks before the policy shocks exist is impossible.
- **Step 1 failures.** A risk the market already prices (a widely expected rate cut) fails "low probability"; value-versus-growth for a Treasury-only fund fails "relevant"; "markets fall at some point" fails "precise" and has no catalyst.
- **More variables is not better (polarity).** High adjusted $R^2$ means redundancy and multicollinearity, not stability; the remedy for needing more factors is asset blocks.
- **Calibration routes.** A historical analogue is one of two routes, not a requirement; Brexit had no good one.
- **The data window.** It must match the stress (crash: crisis data; geopolitics: political uncertainty; Brexit: recent data). Length alone is not the test.
- **Brexit facts swapped.** Gilts a safe haven in Soft Brexit, yields up in Adverse Brexit; Italian and Mexican shocks kept small, not large; shocks kept within asset classes.
- **Scenario z-score delta.** Near zero means consistent with the other shocks; the exception, rates, is the least consistent, not the most.
- **Purpose.** MDS is not a tail-loss measure and does not set loss targets.

## Practice questions: IM-15 c

### Question IM-15 c.1

New • built from SchweserNotes 2026 Reading 97

A newly hired risk analyst at a London asset manager is documenting the firm's market-driven scenario (MDS) process. Which of the following lists the five steps in the correct order?

**A.** Select policy variables; define a market event; generate nonpolicy factor shocks; calibrate shock sizes; estimate the portfolio stress test P&L.

**B.** Define a market event; select policy variables; calibrate shock sizes to the policy variables; generate nonpolicy factor shocks; estimate the portfolio stress test P&L.

**C.** Define a market event; calibrate shock sizes to all risk factors; select policy variables from the largest shocks; estimate the portfolio stress test P&L; generate nonpolicy factor shocks.

**D.** Define a market event; select policy variables; generate nonpolicy factor shocks; calibrate shock sizes to the implied shocks; estimate the portfolio stress test P&L.

### Question IM-15 c.2

New • built from SchweserNotes 2026 Reading 97

The risk team of a US fund that holds only US Treasury securities is asked to propose the market event for its next market-driven scenario. Which of the following proposals best meets the criteria for defining a market event?

**A.** A 25 basis point cut in the federal funds rate at the next policy meeting, which futures markets price with a 95% probability.

**B.** A sharp rotation from growth stocks into value stocks, triggered by a surprise in technology sector earnings.

**C.** A 120 basis point rise in 10-year Treasury yields within one week, triggered by a failed Treasury auction amid rising fiscal concerns.

**D.** A significant deterioration in global financial markets at some point over the next two years, from causes still to be identified.

### Question IM-15 c.3

Bank Q554 • Schweser Module Quiz 97.2 Q1

Which of the following statements is *not* recommended when developing a market-driven scenario (MDS) analysis?

**A.** Specifying hypothetical scenarios precisely so they can be translated into market shocks.

**B.** Using scenario z-scores to assess the plausibility, magnitude, and relative sizing of shocks.

**C.** Selecting a covariance matrix sample that reflects market conditions consistent with the scenario.

**D.** Including a large number of highly correlated key risk factors to maximize coverage of different risks.

**Question IM-15 c.4** Bank Q555 • New, written for the bank (GARP style)

A risk manager is selecting policy variables for a market-driven scenario on the European equity market. The existing policy variables are a European equity index and the 10-year Bund yield. An analyst proposes adding a European financials sector index; regressing it on the existing policy variables gives an adjusted $R^2$ of 93%. The team also wants several additional currency and credit factors included. Which of the following is the most appropriate action?

**A.** Add the financials index, because a high adjusted $R^2$ shows it is well explained by the existing variables and so improves the stability of the perturbation vector.

**B.** Do not add the financials index, because its adjusted $R^2$ is at or above 90%; if more variables are needed, group implied factors into asset blocks regressed on different subsets of policy variables.

**C.** Add the financials index, because higher multicollinearity improves coverage of the key market dynamics.

**D.** Do not add the financials index, and replace the Bund yield with a larger number of highly correlated rate factors to compensate.

**Question IM-15 c.5** New • built from SchweserNotes 2026 Reading 97

In February 2016 a risk team is sizing the policy shocks for a Soft Brexit scenario. The closest historical episode, sterling's 1992 exit from the Exchange Rate Mechanism, was driven mainly by currency policy and did not capture the broader economic effects of leaving the European Union. Which of the following approaches to calibrating the shock sizes is most consistent with the MDS process?

**A.** Copy the 1992 shocks exactly, since a historical episode is the only objective basis on which shock sizes can be set.

**B.** Set the shocks to a target frequency based on the team's view of the scenario's likelihood, measured by the scenario z-score against the risk ruler.

**C.** Size every policy shock at the same number of standard deviations, so that the correlation z-score of the scenario is zero by construction.

**D.** Choose the largest shocks the covariance matrix will accept, since the purpose of the scenario is to quantify the worst-case tail loss.

**Question IM-15 c.6** New • built from SchweserNotes 2026 Reading 97

A risk manager is reviewing how the market-driven scenario approach was applied to the 2016 UK referendum on leaving the European Union. Which of the following statements about the Soft Brexit scenario is correct?

**A.** Policy shocks were spread across asset classes, because mixing equity, rate and currency shocks in one block gives the most realistic correlations.

**B.** Gilts were assumed to sell off, because investors were expected to exit UK fixed income in both of the leave scenarios.

**C.** The covariance matrix was estimated over the 1992 ERM crisis, because it was the closest historical analogue to a vote to leave.

**D.** The policy variables focused on UK markets, and shocks to Italian sovereign debt and the Mexican peso were kept small to dampen contagion.

**Question IM-15 c.7** New • built from SchweserNotes 2026 Reading 97

A risk committee reviews a table of Soft Brexit policy shocks with a column headed "scenario z-score delta": the change in the scenario z-score if a policy shock were moved from its stressed value to its expected value conditional on the other shocks. The delta is close to zero for every policy variable except the 10-year gilt yield. Which of the following conclusions is most appropriate?

**A.** The rate shock is the least consistent with the other shocks under the chosen covariance matrix, so the safe-haven assumption for gilts deserves the closest challenge.

**B.** The rate shock is the most consistent with the other shocks, since a delta away from zero shows that the covariance matrix explains it well.

**C.** All shocks except the rate shock should be removed from the scenario, since a delta near zero shows that they add no information to it.

**D.** The scenario z-score of the full scenario is close to zero, so the scenario is too mild to be of use as a stress test of the portfolio.

## Solutions • IM-15 c

### Solution IM-15 c.1 — Answer: B

New • built from SchweserNotes 2026 Reading 97

**Step 1 – Start from what each step needs**

Policy variables can be chosen only for a defined event; shocks can be sized only once the policy variables exist; nonpolicy shocks are implied *from* the sized policy shocks; the P&L needs the full perturbation vector.

**Step 2 – Read off the order**

(1) Define a market event; (2) select policy variables; (3) calibrate shock sizes to the policy variables; (4) generate nonpolicy factor shocks from the policy shocks; (5) estimate the portfolio stress test P&L.

$\Rightarrow$ B.

**Interpretation**

Each step consumes the output of the one before, so the order follows from the logic. Vardeez's four-step version (see the correction box) is a common source of a wrong "count" or a wrong order.

**Why the other options are wrong**

**A is incorrect.** Selects policy variables before the event they are meant to represent is defined, and implies nonpolicy shocks before the policy shocks are sized.

**C is incorrect.** Calibrates shocks to all risk factors directly, which defeats the conditional design (the experts shock only the few policy variables), and computes the P&L before the perturbation vector is complete.

**D is incorrect.** Implies nonpolicy shocks before the policy shocks are sized, then calibrates to the implied shocks: the dependency runs the other way.

**LO:** IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.c, Steps for Defining and Developing MDSs

### Solution IM-15 c.2 — Answer: C

New • built from SchweserNotes 2026 Reading 97

**Step 1 – The four Step 1 criteria**

Low probability but plausible, and not already priced in; defined precisely (size and direction); relevant to the portfolio; a specific catalyst.

**Step 2 – Test each proposal**

A: priced at 95%, so it is a known risk, not a low-probability one. B: equity style shocks are irrelevant to a Treasury-only portfolio (the reading's own example). C: precise size and direction, directly relevant, unlikely but possible, with a named catalyst. D: no size, no direction, no catalyst.

$\Rightarrow$ C meets all four criteria.

**Interpretation**

The criteria reduce the two uncertainties of a scenario: whether the event happens, and how markets react if it does. A precise, relevant event with a catalyst also gives the team something it can monitor.

**Why the other options are wrong**

**A is incorrect.** Fails "low probability": a risk the market already prices teaches nothing new.
**B is incorrect.** Fails "relevant": shocks to value versus growth would have little relevance for a portfolio invested solely in US Treasuries.
**D is incorrect.** Fails "precise" and "catalyst": the expected impact is open to interpretation and nothing triggers it.

**LO:** IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.c, Step 1: Define a market event

### Solution IM-15 c.3 — Answer: D

Bank Q554 • Schweser Module Quiz 97.2 Q1

#### Step 1 – Check each statement against the guidelines and steps

A is Step 1 (precise definition); B is the guideline on scenario z-scores (used in Step 3); C is the guideline on the covariance sample (Step 4).

#### Step 2 – Find the one that breaks a rule

D asks for many highly correlated key risk factors. The guidelines say the opposite: a small, straightforward set of factors that are not highly correlated, to avoid multicollinearity and an unstable perturbation vector.
$\Rightarrow$ D is not recommended.

#### Interpretation

Highly correlated policy variables give redundant information and make the implied shocks unstable. Coverage comes from the regression that implies the nonpolicy shocks, not from adding policy variables.

#### Why the other options are wrong

**A is incorrect.** Recommended: scenarios must translate clearly into market shocks.
**B is incorrect.** Recommended: scenario z-scores gauge plausibility, magnitude and relative sizing.
**C is incorrect.** Recommended: a sample unlike the scenario would give misleading implied shocks.

**LO:** IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.c, Guidelines for Reducing Subjectivity in MDSs

### Solution IM-15 c.4 — Answer: B

Bank Q555 • New, written for the bank

#### Step 1 – Apply the Step 2 test

Do not add a policy variable with an adjusted $R^2$ of 90% or more against the existing policy variables. At 93%, the financials index is largely explained by the equity index and the Bund yield.
$\Rightarrow$ It is redundant and would add multicollinearity.

#### Step 2 – Handle the wish for more factors

If a larger set is required, use asset blocks: regress different groups of implied factors on different subsets of the policy variables.

#### Interpretation

A high $R^2$ is a reason to leave a variable out, because the existing policy variables already carry

its information. The financials index can still be shocked: as a nonpolicy factor it will be implied by the regression.

### Why the other options are wrong

**A is incorrect.** Polarity flip: a high $R^2$ signals redundancy and instability, not stability.
**C is incorrect.** Multicollinearity adds overlapping information and reduces reliability; it does not improve coverage.
**D is incorrect.** Right on the index, but adding many highly correlated rate factors recreates the multicollinearity the rule is meant to avoid.

**LO:** IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.c, Step 2: Select policy variables

## Solution IM-15 c.5 Answer: B

New • built from SchweserNotes 2026 Reading 97

### Step 1 – The two calibration routes

(1) Identify a historical period that resembles the scenario; (2) set a target frequency from a subjective view of the scenario's likelihood, the likelihood of a scenario at least as extreme, measured by the scenario z-score.

### Step 2 – Which route fits here

The stem rules out route (1): the 1992 episode was about currency policy and the reading notes that direct historical comparisons offered limited guidance for Brexit.
$\Rightarrow$ Route (2), using the risk ruler to place the scenario against historical events.

### Interpretation

The risk ruler turns "how likely do we think this is?" into a number of standard deviations. Soft Brexit sat between Remain (more likely) and Adverse Brexit (far less likely).

### Why the other options are wrong

**A is incorrect.** Absolute claim: an analogue is one of two routes, and the stem says this one does not resemble the scenario.
**C is incorrect.** Equal sizing does not make a scenario coherent: the correlation z-score depends on the signs and relative sizes against the correlations, so nothing is "zero by construction".
**D is incorrect.** Purpose error: MDS aims at plausible scenarios, not at the largest acceptable shock, and it is not designed to quantify tail losses.

**LO:** IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.c, Step 3 and the Brexit example (Figure 97.6)

## Solution IM-15 c.6 Answer: D

New • built from SchweserNotes 2026 Reading 97

### Step 1 – Recall the policy variables

The Soft Brexit policy variables focused on UK markets: the FTSE All Share index, the 10-year gilt yield and sterling.

### Step 2 – Recall the design choices

A few shocks, such as those to Italian sovereign debt and the Mexican peso, were kept small on purpose, to dampen contagion outside the UK and stop those factors overwhelming the covariance matrix. Shocks were generally restricted to the same asset class.

⇒ D is correct.

### Interpretation

Each design choice protects the covariance step: small peripheral shocks and single-asset-class blocks keep the implied shocks from being driven by spurious relationships.

### Why the other options are wrong

**A is incorrect.** Reverses the rule: shocks were restricted to the same asset class to avoid spurious correlations.

**B is incorrect.** Scenario swap: gilts were a safe haven in Soft Brexit; rising yields were the Adverse Brexit assumption.

**C is incorrect.** No comparable event existed; the 1992 ERM exit was only loosely related, and a matrix based on recent data was used.

**LO:** IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.c, Applying the MDS Approach: Brexit Example (Figures 97.5 and 97.7)

## Solution IM-15 c.7

**Answer: A**

New • built from SchweserNotes 2026 Reading 97

### Step 1 – What the delta measures

How much the scenario z-score changes when one policy shock is moved to the value the covariance matrix would expect given the other shocks. Near zero: that shock already sits where the other shocks imply.

### Step 2 – Read the table

Every delta is near zero except rates, so the shocks are largely consistent with one another, with the exception of rates.

⇒ The gilt rally is the assumption that departs most from what the matrix implies.

### Step 3 – Link to the example

Rates were also the most debated variable: safe haven or exit from UK fixed income. The team kept the safe-haven assumption for Soft Brexit, but this is where the judgment is concentrated and must be defended.

### Interpretation

A large delta does not prove the assumption wrong; it marks where the scenario leans hardest on expert judgment rather than on history.

### Why the other options are wrong

**B is incorrect.** Polarity flip: a delta near zero signals consistency; the exception is the least consistent shock.

**C is incorrect.** Near zero means consistent, not redundant: those shocks define the scenario and drive its P&L.

**D is incorrect.** Confuses the delta (a change in the z-score) with the level of the scenario z-score, which the table does not report.

**LO:** IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.c, Figure 97.7 (Expanded Soft Brexit Policy Variable Selection)

IM-16 • Liquidity Risk Management • Vardeez: VERY IMPORTANT

# IM-16 a

P2 CONCEPT

Explain best practices for managing liquidity risk in an investment fund.

Workbook: none • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** No workbook section covers IM-16, but Vardeez rates the reading VERY IMPORTANT ("LO 98.b is imp and rest all Learning Objectives, pls know it"), so all five of its LOs sit in P2. **Vardeez:** no pointer for this LO beyond "know it".

**Reported exam shapes:** none reported. The block is built from Vardeez's notes, checked against SchweserNotes 2026 Reading 98 (the content authority for this Schweser-only reading).

**What the questions look like:** statement questions. What fund liquidity risk is and where it comes from; the four elements of the framework and how they sit in the three lines of defense; and the core objective, which is to meet redemptions without leaving the remaining investors with an illiquid fund. A bank calculation filed under this LO (a position limit set as a share of average value traded) tests IM-16 c and is worked there. This block also carries the shared vocabulary of IM-16: fund liquidity risk, redemptions, and the three investor groups.

## 1. Fund liquidity risk

In this reading "liquidity" means **fund liquidity risk**: the risk that a fund may be unable to obtain enough liquidity to meet investor **redemptions**. It is particularly complex for hedge funds, because of the assets they hold and because redemptions must be met without disadvantaging the shareholders who stay.

- **Where it comes from.** It can arise as a **byproduct** of other problems, such as poor fund performance or reputational problems involving the managers: these trigger the redemptions that the fund then has to fund.
- **The response.** A robust, **top-down governance** structure, supported by a strong liquidity risk management framework that is clearly defined and **operates independently** within the fund.

## 2. The framework: four elements, three lines of defense

1. **Liquidity risk measurement.** Identify, assess and quantify the key sources of liquidity risk (for example, how quickly each asset can be sold).
2. **Liquidity risk management.** Monitor the fund's liquidity profile and evaluate redemption risks against established limits.
3. **Portfolio manager liquidity risk awareness.** Risk managers review and discuss liquidity risks with the portfolio managers regularly, and escalate material concerns to senior decision-makers (the CRO or CIO) in a timely manner.
4. **Redemption toolkit.** Know the backup liquidity tools available and be ready to activate extraordinary measures when necessary (IM-16 d).

The framework sits inside the **three lines of defense**. *First line:* the portfolio managers, who must stay aware of liquidity risk and hold primary responsibility for managing it. *Second line:* an independent

and robust risk management function. *Third line:* a periodic audit that checks whether the first two lines follow the established policies.

## 3. The core objective: do not strand the remaining investors

A comprehensive liquidity risk management system **measures the liquidity of the fund's assets**, **analyzes redemption patterns**, and has a **process that ensures enough cash is available to meet expected redemptions in a timely manner**. Its critical goal is to avoid creating liquidity for departing investors in a way that leaves the remaining shareholders with an illiquid portfolio: the fund must balance meeting redemptions against preserving a sound strategy and structure for those who stay. The variables that drive liquidity are **market conditions** (normal or stressed), **position sizes**, **transaction costs** and potential **market impact**.

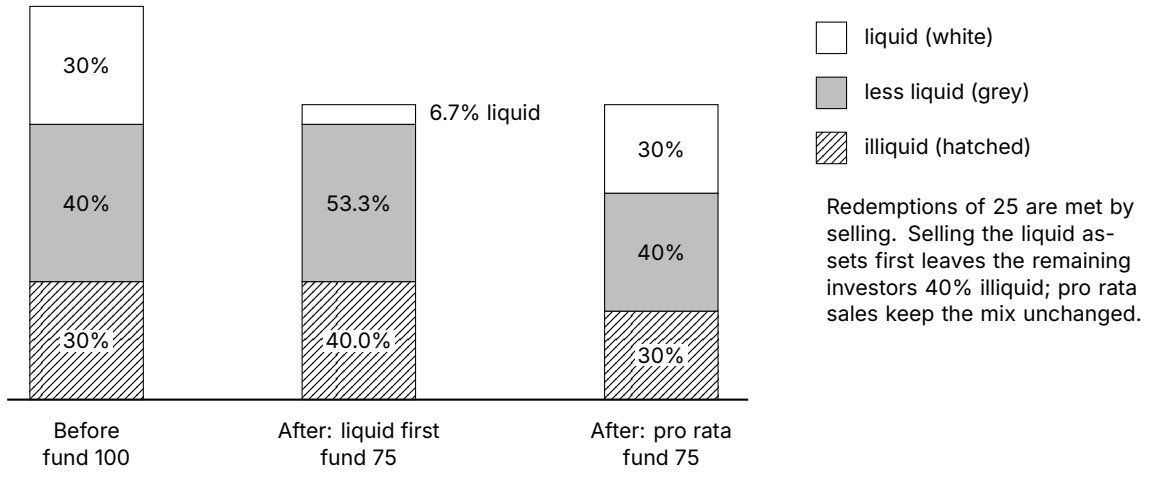



Figure 1. The same 25% redemption met two ways (illustrative numbers). Liquid-first selling is cheapest today but hands the illiquid assets to the investors who stay; that is the outcome the framework is designed to avoid. The redemption waterfall of IM-16 e uses pro rata (risk-constant) sales once its cash layer is used up.

**Correction to the source notes.** Vardeez lists the four elements as liquidity risk measurement, liquidity risk management, *portfolio construction* and *governance*. The reading's four elements (per SchweserNotes) are measurement, management, **portfolio manager liquidity risk awareness** and the **redemption toolkit**. Governance is the top-down structure the framework sits in, not one of its elements, and portfolio construction is not on the list. The SchweserNotes list is used here and in the questions.

| Element | What it does | What a breakdown looks like |
|---|---|---|
| **Measurement** | Identifies, assesses and quantifies the key sources of liquidity risk | Asset liquidity never estimated under stressed conditions |
| **Management** | Monitors the liquidity profile; compares redemption risk with limits | No limits, or limits that never trigger action |
| **PM awareness** | Regular risk-PM discussion; timely escalation to CRO or CIO | Reports show market depth tightening but nobody discusses or escalates it |
| **Redemption toolkit** | Backup liquidity tools; readiness to use extraordinary measures | Tools exist on paper but have never been tested (IM-16 d) |
| **Three lines of defense** | 1: portfolio managers (primary responsibility); 2: independent risk management; 3: periodic audit | Risk management treated as the owner of liquidity risk; audit doing daily monitoring |

**Exam traps**

- **Definition.** Fund liquidity risk is about meeting redemptions. It is not a fall in asset values, and it can arise as a byproduct of performance or reputational problems.
- **The four elements.** Measurement, management, PM awareness, redemption toolkit. "Portfolio construction" and "governance" are not on the list.
- **Measurement versus management (sibling swap).** Measurement quantifies the sources of risk; management monitors the profile against limits.
- **Lines of defense.** Portfolio managers are the first line and own the risk; independent risk management is the second; audit is the third.
- **Top-down and independent.** Not a bottom-up set of limits chosen by each manager.
- **Liquid-first selling.** Cheap today, unfair tomorrow: it leaves the remaining investors with the illiquid assets.
- **Scope.** The system covers assets *and* redemption patterns, in normal *and* stressed markets.

## Practice questions: IM-16 a

### Question IM-16 a.1

New • built from SchweserNotes 2026 Reading 98

The board of a multi-strategy hedge fund asks its new chief risk officer to explain fund liquidity risk after a reputational problem involving one of the fund's star managers. Which of the following statements is most consistent with best practice?

**A.** It is the risk of being unable to obtain enough liquidity to meet redemptions; it can arise as a byproduct of poor performance or reputational problems, and is best met with top-down governance.

**B.** It is the risk that the fund's holdings fall in value as bid-ask spreads widen; it does not depend on investor behavior, so a reputational problem leaves it unchanged.

**C.** It arises only on the asset side of the fund, so a fund whose holdings are all liquid has no need for a liquidity risk management framework.

**D.** It is best managed bottom-up, with each portfolio manager setting liquidity limits for his or her own book and no separate fund-wide framework.

**Question IM-16 a.2** New • built from SchweserNotes 2026 Reading 98

The chief risk officer of a credit fund is writing job descriptions for two teams within the fund's liquidity risk framework. Which of the following correctly distinguishes the liquidity risk measurement element from the liquidity risk management element?

**A.** Measurement monitors the liquidity profile against limits; management identifies and quantifies the sources of liquidity risk.

**B.** Measurement escalates concerns to the CIO; management maintains the fund's backup credit lines and other extraordinary tools.

**C.** Measurement identifies, assesses and quantifies the key sources of liquidity risk; management monitors the liquidity profile and evaluates redemption risks against limits.

**D.** Measurement and management are the same element, because quantifying liquidity risk automatically keeps it within the fund's limits.

**Question IM-16 a.3** Bank Q561 • Schweser Module Quiz 98.1 Q1

Within a robust liquidity risk management framework, which of the following situations most clearly indicates a governance or communication breakdown rather than a measurement or toolkit weakness?

**A.** The fund's redemption toolkit includes swing pricing mechanisms for stressed markets.

**B.** Stress testing results trigger predefined limits and prompt partial portfolio rebalancing.

**C.** The risk function evaluates asset liquidity profiles under normal and stressed conditions.

**D.** Daily liquidity reports show tightening market depth, but no follow-up discussion occurs between risk management and portfolio management.

**Question IM-16 a.4** New • built from SchweserNotes 2026 Reading 98

An internal auditor is mapping a mutual fund's liquidity risk framework to the three lines of defense model. Which of the following assignments is correct?

**A.** First line: the independent risk management function, which holds primary responsibility for liquidity risk; second line: portfolio managers; third line: periodic audit.

**B.** First line: portfolio managers, who hold primary responsibility for liquidity risk; second line: an independent risk management function; third line: periodic audit.

**C.** First line: periodic audit; second line: portfolio managers; third line: the board, which sets the limits and monitors the fund's liquidity profile each day.

**D.** First line: portfolio managers; second line: periodic audit; third line: the independent risk management function, which checks that audit follows policy.

**Question IM-16 a.5** New • built from SchweserNotes 2026 Reading 98

A risk manager is designing a comprehensive liquidity risk management system for an open-ended credit fund. Which of the following designs is most consistent with best practice?

**A.** Measure the liquidity of the fund's assets under normal conditions only, since stressed conditions are covered by the fund's extraordinary measures.

**B.** Measure the liquidity of the fund's assets and leave the analysis of redemption patterns to distributors, since investor behavior is outside the fund's control.

**C.** Measure the liquidity of the fund's assets, analyze redemption patterns, and keep a process that ensures enough cash to meet expected redemptions in a timely manner.

**D.** Set liquidity limits once at launch from position sizes alone, since market conditions and transaction costs do not affect the liquidity of a diversified fund.

**Question IM-16 a.6** New • built from SchweserNotes 2026 Reading 98

A bond fund receives redemption requests equal to 20% of its assets during a stressed market. To minimize transaction costs, a portfolio manager proposes meeting the whole amount by selling the fund's most liquid holdings. Which of the following concerns is most consistent with liquidity risk management best practice?

**A.** None: minimizing the transaction costs borne by redeeming investors is the overriding objective of the liquidity framework.

**B.** Selling liquid holdings first leaves the remaining shareholders with a less liquid portfolio, so the fund must balance meeting redemptions with preserving its strategy for those who stay.

**C.** Selling liquid holdings first raises transaction costs, because liquid assets carry wider bid-ask spreads than illiquid assets in stressed markets.

**D.** Selling liquid holdings first is prohibited, because funds must always meet redemptions by borrowing before they sell any assets.

## Solutions • IM-16 a

### Solution IM-16 a.1 — Answer: A

New • built from SchweserNotes 2026 Reading 98

**Step 1 – Define the risk**

Fund liquidity risk is the risk that a fund may be unable to obtain enough liquidity to meet investor redemptions.

**Step 2 – Where it comes from**

It can arise as a byproduct of other issues, such as poor fund performance or reputational problems involving managers: exactly the board's situation.

**Step 3 – The best-practice response**

A robust, top-down governance structure supported by a clearly defined liquidity risk framework that operates independently within the fund.

$\Rightarrow$ A.

**Interpretation**

The star manager's problem is a redemption trigger. The board should expect the risk to show up on the liability side first, and the framework should already have measured how fast the assets can be turned into cash.

**Why the other options are wrong**

**B is incorrect.** Wrong risk: a fall in asset values is market risk. Fund liquidity risk is about meeting redemptions, and reputational problems drive redemptions.

**C is incorrect.** Absolute claim: redemptions can still exceed what can be sold without stranding the remaining investors, and the framework also covers redemption patterns.

**D is incorrect.** Polarity flip: best practice is top-down governance with an independent framework, not bottom-up limits.

**LO:** IM-16 a: Explain best practices for managing liquidity risk in an investment fund. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.a; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.7

### Solution IM-16 a.2 — Answer: C

New • built from SchweserNotes 2026 Reading 98

**Step 1 – Measurement**

Risk managers identify, assess and quantify the key sources of liquidity risk.

**Step 2 – Management**

Risk managers monitor the fund's liquidity profile and evaluate redemption risks relative to established limits.

$\Rightarrow$ C.

**Interpretation**

Measurement produces the numbers; management acts on them against limits. The other two elements are portfolio manager awareness (discussion and escalation) and the redemption toolkit (backup tools).

**Why the other options are wrong**

**A is incorrect.** Sibling swap: the two definitions are reversed.

**B is incorrect.** Describes the other two elements: escalation is portfolio manager awareness; backup credit lines belong to the redemption toolkit.

**D is incorrect.** Absolute claim: a measured risk is not a managed one; monitoring against limits

is a separate element.

**LO:** IM-16 a: Explain best practices for managing liquidity risk in an investment fund. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.a, the four key elements

### Solution IM-16 a.3 — Answer: D

Bank Q561 • Schweser Module Quiz 98.1 Q1

#### Step 1 – The four elements

Measurement, management (limits), portfolio manager awareness (regular discussion and escalation) and the redemption toolkit.

#### Step 2 – Classify each situation

A: a toolkit tool in place. B: limits that trigger action. C: measurement under normal and stressed conditions. D: a known deterioration that nobody discusses or escalates.
⇒ D is a failure of the portfolio manager awareness element: governance and communication.

#### Interpretation

The data in D were good. The failure is that the information never reached the people who hold primary responsibility, which is why the framework includes escalation to the CRO or CIO.

#### Why the other options are wrong

**A is incorrect.** Having swing pricing in the toolkit is a strength, not a breakdown.
**B is incorrect.** Stress results triggering limits and rebalancing is the management element working as designed.
**C is incorrect.** Assessing liquidity under normal and stressed conditions is sound measurement.

**LO:** IM-16 a: Explain best practices for managing liquidity risk in an investment fund. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.a

### Solution IM-16 a.4 — Answer: B

New • built from SchweserNotes 2026 Reading 98

#### Step 1 – First line

The portfolio managers: they must remain consistently aware of liquidity risk and hold primary responsibility for managing it.

#### Step 2 – Second and third lines

Second: an independent and robust risk management function. Third: a periodic audit that verifies whether the first two lines adhere to established policies.
⇒ B.

#### Interpretation

Ownership sits with the people who take the positions; independence sits with risk management; assurance sits with audit.

#### Why the other options are wrong

**A is incorrect.** Swaps the first two lines: risk management is the independent second line, not the owner.
**C is incorrect.** Audit is periodic assurance, not the first line, and the board does not do daily monitoring.
**D is incorrect.** Swaps the second and third lines: audit checks the first two lines, not the other way round.

**LO:** IM-16 a: Explain best practices for managing liquidity risk in an investment fund. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.a, three lines of defense

### Solution IM-16 a.5 Answer: C

New • built from SchweserNotes 2026 Reading 98

**Step 1 – The three parts of a comprehensive system**

Measure the liquidity of the assets held; analyze redemption patterns; establish a process that ensures sufficient cash is available to meet expected redemptions in a timely manner.

**Step 2 – The variables it must reflect**

Market conditions (normal versus stressed), position sizes, transaction costs and potential market impact.

⇒ Only C covers both sides of the problem.

**Interpretation**

Liquidity risk is a mismatch between how fast assets can be sold and how fast investors can leave, so a system that sees only one side cannot measure it.

**Why the other options are wrong**

**A is incorrect.** Scope error: stressed conditions are where liquidity matters most; extraordinary measures are a last resort, not a substitute for measurement.

**B is incorrect.** Scope error: analyzing redemption patterns is part of the system.

**D is incorrect.** Absolute claim: market conditions and transaction costs are among the key variables, and limits must be monitored, not set once.

**LO:** IM-16 a: Explain best practices for managing liquidity risk in an investment fund. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.a

### Solution IM-16 a.6 Answer: B

New • built from SchweserNotes 2026 Reading 98

**Step 1 – The critical goal**

Avoid creating liquidity for departing investors in a way that leaves the remaining shareholders with an illiquid portfolio.

**Step 2 – Apply it**

Selling the most liquid 20% first meets today's redemptions cheaply but raises the illiquid share of the fund for everyone who stays (Figure 1).

⇒ The fund must balance meeting redemptions with preserving a sound strategy and structure for those who remain.

**Interpretation**

This is the fairness theme that IM-16 e makes explicit: redeeming, subscribing and remaining investors should all be treated fairly, which is why the waterfall's second layer sells pro rata or risk-constant.

**Why the other options are wrong**

**A is incorrect.** Ignores the remaining investors: transaction costs matter, but they are not the overriding objective.

**C is incorrect.** Polarity flip: liquid assets are cheaper to sell; the problem is what is left behind, not the cost.

**D is incorrect.** Absolute claim: borrowing is the last layer of the waterfall, not a requirement before any sale.

**LO:** IM-16 a: Explain best practices for managing liquidity risk in an investment fund. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.a

IM-16 • Liquidity Risk Management • Vardeez: VERY IMPORTANT

# IM-16 b

P2 CONCEPT

Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling.

Workbook: none • Vardeez focus: the key LO of IM-16

**WHAT IS TESTED HERE**

**Why P2.** Vardeez flags this as *the* important LO of a VERY IMPORTANT reading ("LO 98.b is imp"). No workbook section covers IM-16.
**Vardeez:** flagged as the key LO.

**Reported exam shapes:** none reported; built from Vardeez's focus and SchweserNotes 2026 Reading 98.

**What the questions look like:** statement questions on three themes. (1) Market structure: the post-crisis shift from price takers to price makers and what it did to liquidity. (2) Why bonds are harder than equities: the number of securities, sparse trading and the valuation of complex structures. (3) Data: TRACE's volume caps and what the transparency initiatives have and have not fixed. Numbers to carry: USD 1 million (high yield) and USD 5 million (investment grade) caps; uncapped data with a six-month delay; MiFID II's one-month lag for less liquid bonds.

## 1. After the crisis: from price takers to price makers

Low interest rates after the Global Financial Crisis encouraged heavy corporate borrowing, including a surge in nonstandard issues. Market behavior changed at the same time:

- **Before the GFC: price takers.** Asset managers asked several dealers for quotes in the OTC market and took the best one.
- **After the GFC: price makers.** Managers set the price at which they are willing to trade and then look for counterparties at that level.
- **Consequences.** Thinner secondary markets, with **lower turnover**, **wider bid-ask spreads** and a **larger price impact** for big trades. Managers now break large trades into smaller orders.
- **Immediacy costs more.** The cost of liquidity rises as the need for immediacy rises, and the effect is amplified for complex assets such as structured or specialized products.

## 2. Why bonds are harder than equities

- **Equities:** a manageable number of securities that trade frequently on public exchanges, so price discovery is strong and prices are transparent.
- **Fixed income:** fragmented, with an enormous number of securities (municipal bonds alone account for roughly one million CUSIPs). Only a small fraction trade on any given day, with activity concentrated in **large and recently issued** offerings. Price discovery is limited and uneven, so market depth and transaction costs are hard to estimate.
- **Valuation, by analogy.** A retired couple's mortgage-free house, owned for 30 years, is valued from the sale prices of comparable homes, but there is never a perfect comparable. A rental house with an established income stream is easier, because the income gives a second basis. Bonds work the same way, and the difficulty grows rapidly as the structure becomes more complex.

## 3. The data problem

FINRA's **TRACE** (Trade Reporting and Compliance Engine) gives daily bond prices and volumes, but reported volumes are **capped**. Exact sizes appear only below the cap: high-yield trades above **USD 1 million** and investment-grade trades above **USD 5 million** are shown only as "above the cap". Whether a trade was USD 6 million or USD 60 million matters greatly for liquidity (Figure 1). Models infer the true volume from historical trading patterns, which can be inaccurate when future conditions differ from the past. The higher cap applies to the more liquid category.

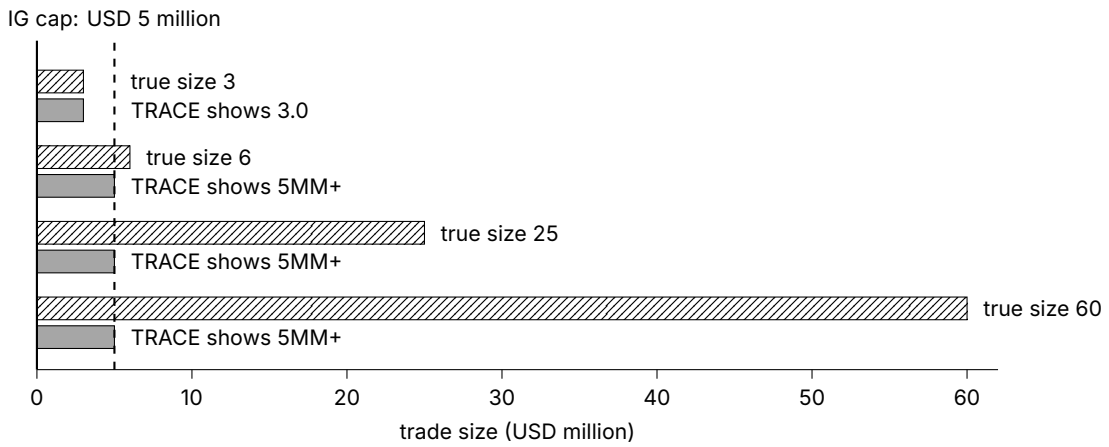



Figure 1. Four investment-grade trades (illustrative). Hatched bars: true size. Grey bars: what TRACE reports. Three very different trades, USD 6, 25 and 60 million, all look the same, so a model must infer their sizes. For high-yield bonds the cap is lower, USD 1 million.

**Transparency initiatives**, and what each leaves open:

| Initiative | What it gives | What is still missing |
|---|---|---|
| **TRACE, uncapped (as of 2025)** | Uncapped trading volume at the CUSIP level | Released with a six-month delay; the reported volumes are still capped |
| **MiFID II transparency** | Intraday price and volume snapshots for liquid bonds | Less liquid bonds released with a one-month lag |
| **MiFID II cost reporting** | Asset managers report realized transaction costs for European funds | Costs after the fact, not market depth |
| **Consolidated tape (proposed)** | One standardized feed of prices and volumes from all venues, for equities and bonds | Under consideration by the European Commission, not yet in place |

The pattern is Vardeez's point: the cap is lifted only at the price of a six-month delay, so the data become accurate when they stop being timely. The initiatives have widened access but have not resolved the problem of measuring and modelling bond liquidity.

| | Equities | Fixed income |
|---|---|---|
| **Number of securities** | Manageable | Enormous (about one million municipal CUSIPs alone) |
| **Where they trade** | Public exchanges, frequently | OTC; a small fraction on any day, mostly large recent issues |
| **Price discovery** | Strong, transparent | Limited and uneven |
| **Manager's role since the GFC** | n/a | Price maker: thinner markets, wider spreads, more impact |
| **Valuation** | Observable prices | Comparables, never perfect; harder as complexity grows |
| **Volume data** | Exchange-reported | TRACE caps (USD 1m HY, 5m IG); uncapped only after six months |

**Exam traps**

- **Price taker or price maker (polarity).** Managers moved *from* taking dealer quotes *to* setting prices, and liquidity got worse: thinner markets, wider spreads, more impact.
- **Which market is fragmented.** Fixed income has far more securities, trades less often and is mostly OTC; equities trade on exchanges.
- **The caps.** USD 1 million for high yield, USD 5 million for investment grade; both segments are capped. Capped trades are reported as "above the cap", not dropped.
- **Inference from history.** Filling in capped volumes from past patterns fails when conditions change.
- **The delays.** Uncapped TRACE: six months. MiFID II: intraday for liquid bonds, one-month lag for less liquid ones. Swapping the two categories is the trap.
- **Consolidated tape.** Proposed, not operating.
- **Immediacy and complexity.** Faster sales cost more, and more so for complex securities.

## Practice questions: IM-16 b

### Question IM-16 b.1

Bank Q562 • Schweser Module Quiz 98.1 Q2

Which of the following statements best explains why managing liquidity risk in fixed-income securities is often more difficult than in equity markets?

**A.** Fixed-income markets have fewer securities but trade more frequently with smaller bid-ask spreads.

**B.** Equity markets rely primarily on OTC transactions, reducing price transparency relative to fixed income.

**C.** Asset managers in fixed-income markets typically act as price takers, improving overall market liquidity.

**D.** Fixed-income markets contain a very large number of securities, many of which trade infrequently, making price discovery and valuation more difficult.

**Question IM-16 b.2** New • built from SchweserNotes 2026 Reading 98

A risk analyst explains to a fund's board why the bonds in its structured credit sleeve are harder to value and to liquidate than its large, recently issued investment-grade corporate bonds. Which of the following explanations is most accurate?

**A.** As with valuing a long-held house from sales of comparable homes, there is never a perfect comparable, and the difficulty grows rapidly as the structure of the bond becomes more complex.

**B.** Structured bonds trade on public exchanges with continuous quotes, but their prices are more volatile, which makes them harder to value than corporate bonds.

**C.** Trading in fixed-income markets is spread evenly across all outstanding issues, so the structured bonds should be as liquid as the corporates once dealers quote them.

**D.** Complex structures are easier to value than plain bonds, because their many cash-flow features give more data points with which to fit a pricing model.

**Question IM-16 b.3** Bank Q564 • New, written for the bank (GARP style)

A chief investment officer notes that since the Global Financial Crisis her bond fund's traders have increasingly set the price at which they are willing to transact and then sought counterparties at that level, rather than accepting the best of several dealer quotes. Which of the following is the most likely consequence for liquidity risk management in fixed-income portfolios?

**A.** Price discovery has improved because managers now observe quotes from several dealers before every trade.

**B.** Secondary markets have become deeper, with higher turnover and narrower bid-ask spreads, so large orders can be executed at once.

**C.** Secondary markets have become thinner, with lower turnover and wider bid-ask spreads, so managers often break large trades into smaller orders to limit market impact.

**D.** The cost of liquidity has become independent of the need for immediacy because managers control the transaction price.

**Question IM-16 b.4** New • built from SchweserNotes 2026 Reading 98

A portfolio manager must sell USD 80 million of a seasoned, thinly traded structured bond. She can complete the sale within one day or work the order over two weeks. Which of the following statements about the cost of the two plans is most consistent with the post-crisis bond market?

**A.** The one-day sale should be cheaper, because selling at once avoids paying the bid-ask spread on many small trades.

**B.** The one-day sale should be more expensive, because the cost of liquidity rises with the need for immediacy and the effect is amplified for complex securities.

**C.** The two plans should cost about the same, because transaction costs consist only of fixed costs such as commissions and bid-ask spreads.

**D.** The two-week plan should be more expensive, because managers now act as price makers and dealers reward large single trades with tighter prices.

**Question IM-16 b.5** Bank Q563 • New, written for the bank (GARP style)

A risk analyst at a bond fund builds a days-to-liquidate model using FINRA TRACE data. The analyst notes that investment-grade trades above $5 million and high-yield trades above $1 million are reported only as exceeding the cap. Which of the following is the most accurate description of the resulting data-modeling challenge?

**A.** Capped volumes cause TRACE to understate the number of trades but not their size, so the average trade size can be used without adjustment.

**B.** The true size of the largest trades is unknown, so models must infer volume from historical patterns, which may be inaccurate if future conditions differ from past trends.

**C.** Because the cap applies only to high-yield bonds, investment-grade volumes can be taken directly from TRACE without inference.

**D.** The cap removes the need to model price impact because large trades are excluded from the reported data entirely.

**Question IM-16 b.6** New • built from SchweserNotes 2026 Reading 98

A European bond fund's risk team is reviewing whether recent transparency initiatives have resolved its fixed-income liquidity data problems. Which of the following statements is correct?

**A.** FINRA has removed the caps on TRACE trade volumes, so the exact sizes of large trades are now available in real time.

**B.** MiFID II provides intraday price and volume data for less liquid bonds and releases the data for liquid bonds with a one-month lag.

**C.** A consolidated tape combining price and volume data for bonds and equities across all European venues is already in operation.

**D.** FINRA provides uncapped CUSIP-level volumes with a six-month delay, and MiFID II requires asset managers to report realized transaction costs for European funds.

# Solutions • IM-16 b

## Solution IM-16 b.1

**Answer: D**

Bank Q562 • Schweser Module Quiz 98.1 Q2

### Step 1 – Equities

A manageable number of securities trading frequently on public exchanges, so price discovery is strong.

### Step 2 – Fixed income

Fragmented: an enormous number of securities (about one million municipal CUSIPs alone), only a small fraction trading on any day.

$\Rightarrow$ Sparse trading limits observable prices, so valuation, market depth and transaction costs are harder to estimate: D.

### Interpretation

The difficulty is structural, not a matter of volatility: there are simply too many bonds and too few trades for prices to be discovered continuously.

### Why the other options are wrong

**A is incorrect.** Reverses the facts: fixed income has more securities, trades less often and has wider spreads.

**B is incorrect.** Reverses the markets: fixed income is mainly OTC; equities trade on exchanges.

**C is incorrect.** Managers have moved from price takers to price makers, and this has thinned markets rather than improving them.

**LO:** IM-16 b: Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.b; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.7

## Solution IM-16 b.2

**Answer: A**

New • built from SchweserNotes 2026 Reading 98

### Step 1 – The reading's analogy

A long-held, mortgage-free house is valued from comparable sales, and no comparable is perfect. A rental with an established income stream is easier to value, because the income adds a second basis.

### Step 2 – Apply it to bonds

"Fixed-income valuation works in a similar way. The difficulty in valuation grows rapidly as the structure of the fixed-income product becomes more complex." Trading also concentrates in large, recently issued offerings, so the corporates have the better price discovery.

$\Rightarrow$ A.

### Interpretation

Complexity hurts twice: it makes comparables poorer and it raises the cost of immediacy when the bonds must be sold.

### Why the other options are wrong

**B is incorrect.** Structured bonds trade OTC, not on exchanges, and the problem is sparse prices rather than volatile ones.

**C is incorrect.** Trading is concentrated in large, recent issues, not spread evenly; a dealer quote is not executed liquidity.

**D is incorrect.** Polarity flip: complexity makes valuation harder.

**LO:** IM-16 b: Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.b

### Solution IM-16 b.3 — Answer: C

Bank Q564 • New, written for the bank

#### Step 1 – Name the behavior

Setting a price and then seeking counterparties is the post-GFC "price maker" behavior; accepting the best of several dealer quotes was the pre-GFC "price taker" behavior.

#### Step 2 – Its consequence

The shift contributed to thinner secondary markets: lower turnover, wider bid-ask spreads and a larger price impact for big trades. Managers respond by splitting large trades.
⇒ C.

#### Interpretation

The manager gains control of the price but loses immediacy: the trade waits for a counterparty, and liquidity becomes more expensive when it must be fast.

#### Why the other options are wrong

**A is incorrect.** Describes the older price-taker practice, which has declined.
**B is incorrect.** Reverses the effect: markets became thinner, not deeper.
**D is incorrect.** The cost of liquidity rises with the need for immediacy; setting one's own price does not change that.

**LO:** IM-16 b: Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.b

### Solution IM-16 b.4 — Answer: B

New • built from SchweserNotes 2026 Reading 98

#### Step 1 – The market the manager faces

Thin secondary markets with a large price impact for big trades; managers split large trades into smaller orders to reduce impact.

#### Step 2 – Immediacy and complexity

"The cost of liquidity rises as the need for immediacy increases, and this effect is further amplified when dealing with more complex assets." A seasoned, thinly traded structured bond is the worst case.
⇒ The one-day sale should cost more: B.

#### Interpretation

The trade-off is time against cost. The two-week plan carries the risk that prices move against the fund while it waits, which is what t-cost models balance (IM-16 c).

#### Why the other options are wrong

**A is incorrect.** Ignores market impact, which dominates for a large block in a thin market.
**C is incorrect.** Transaction costs include market impact as well as fixed costs, and impact depends on how fast the position is sold.
**D is incorrect.** Being a price maker means searching for counterparties; it does not earn better prices for large blocks.

**LO:** IM-16 b: Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.b and LO 98.c (transaction costs)

### Solution IM-16 b.5 Answer: B

Bank Q563 • New, written for the bank

**Step 1 – What TRACE shows**

Exact volume only below the cap; larger trades are flagged as above it (USD 5 million investment grade, USD 1 million high yield).

**Step 2 – Why it matters**

A USD 6 million trade and a USD 60 million trade look the same, so the true volume is ambiguous (Figure 1).

**Step 3 – The modeling consequence**

Models infer volume from historical trading patterns, which can be inaccurate when future conditions differ from past ones.

⇒ B.

**Interpretation**

The cap biases volume estimates downward exactly for the large trades that matter most when a fund must sell in size.

**Why the other options are wrong**

**A is incorrect.** The trades are counted; it is their size that is hidden, so average trade size is understated.

**C is incorrect.** Both segments are capped; investment grade at USD 5 million.

**D is incorrect.** Capped trades are reported as above the cap, not excluded, and price impact must still be modeled.

**LO:** IM-16 b: Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.b, data availability

### Solution IM-16 b.6 Answer: D

New • built from SchweserNotes 2026 Reading 98

**Step 1 – TRACE**

As of 2025, FINRA still applies caps to reported trade volumes, but now provides uncapped trading volume at the CUSIP level with a six-month delay.

**Step 2 – MiFID II**

Intraday price and volume snapshots for liquid bonds; data for less liquid bonds with a one-month lag; asset managers must report realized transaction costs for European funds.

**Step 3 – Consolidated tape**

Under consideration by the European Commission as an amendment to MiFID II.

⇒ D is the only statement that is right in every part.

**Interpretation**

The initiatives widen access but leave the core trade-off: the most complete data arrive late, so liquidity models still work with incomplete, delayed information.

**Why the other options are wrong**

**A is incorrect.** The caps still apply to reported volumes; uncapped data arrive only after six months.

**B is incorrect.** Swaps the categories: intraday data are for liquid bonds, the one-month lag for less liquid ones.

**C is incorrect.** The consolidated tape is proposed, not in operation.

**LO:** IM-16 b: Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.b, recent initiatives

IM-16 • Liquidity Risk Management • Vardeez: VERY IMPORTANT

# IM-16 c

P2 BOTH

Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds.

Workbook: none • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** An LO of a VERY IMPORTANT reading (Vardeez: "know it"); no workbook section. The LO says *describe and apply*, so it is the one IM-16 LO with calculations.
**Vardeez:** no pointer beyond "know it".

**Reported exam shapes:** none reported. The bank's GARP-tagged item (Mock 3) is a days-to-liquidate ranking, and it is in the set below.

**What the questions look like:** a mix. Calculations: days to liquidate (one position, a ranking, a position limit), the expected volume of an infrequently traded bond, implementation shortfall against the right benchmark, and a dollar-weighted cost estimate. Concepts: what each approach estimates, latent liquidity, why ADV is conservative when volume is volatile, and why the previous close is the wrong benchmark. Stems carry red herrings (market values, previous closes, tail estimates).

## 1. Days to liquidate

$$\text{Days to liquidate} = \frac{\text{trade size}}{\text{participation rate} \times \text{ADV}}$$

- **Trade size.** Estimated from historical redemption patterns, portfolio leverage, derivatives exposures and other factors, case by case.
- **Participation rate.** The share of the security's ADV that can be sold without significant market impact. Inherently hard to estimate. Actual volume is usually lower than the volume the market could support; the gap is **latent liquidity**.
- **ADV** (average daily volume). Average volume over a window (20 or 60 days, say), from the firm's own data, exchanges or vendors; it can use modeled inputs such as bonds outstanding or bid-ask spreads, and when data are thin, asset characteristics, sector turnover or traders' experience.
- **Worked example.** USD 30 million of a bond with ADV of USD 8 million, 25% participation: $30/(0.25 \times 8) = 15$ days. Keep the units consistent: shares with shares, value with value.
- **Reading a limit the other way.** A rule that caps a holding at a fraction of average value traded caps it at one day's permitted sales: maximum position = participation × average value traded (Question IM-16 c.3).

## 2. Infrequently traded bonds

Filling the gaps in a sporadically traded bond's history with past data is unreliable, so the problem is split in two: (1) the **probability that a trade occurs**, and (2) the **volume if it does**. A **random forest regression** (a machine learning method) suits it, because it handles nonlinearity, missing data and large sets of explanatory variables. With expected volume zero on no-trade days:

$$\boxed{\text{E[daily volume]} = \Pr(\text{trade}) \times \text{E[volume} \mid \text{trade]}}$$

- **Other data.** History is a fair guide for frequently traded bonds and breaks down for sporadic ones. Analysts then use **comparable bonds**, and **dealer inventory** (posted bids and asks) to infer potential tradable volume. Both overstate what can actually be executed: quoted liquidity exceeds executed liquidity.
- **A range, not a point.** Latent liquidity can be estimated with tail measures such as the **90th percentile of the ADV distribution**. Liquidity then lies in a range between the observed ADV and the optimistic tail estimate.
- **Volume variance.** Large swings in daily volume produce large swings in estimated volume. When variance is high, **ADV is a conservative measure** of liquidity (Figure 1).
- **Worked example (Figure 1's bond).** It trades on 35% of days, averaging USD 1.14 million when it trades: expected volume $0.35 \times 1.14 =$ USD 0.40 million a day. A USD 3 million position at 25% participation needs $3/(0.25 \times 0.40) = 30$ days; at the optimistic 90th-percentile ADV of USD 0.61 million, $3/(0.25 \times 0.61) = 19.7$ days. The estimate is a range of about 20 to 30 days.

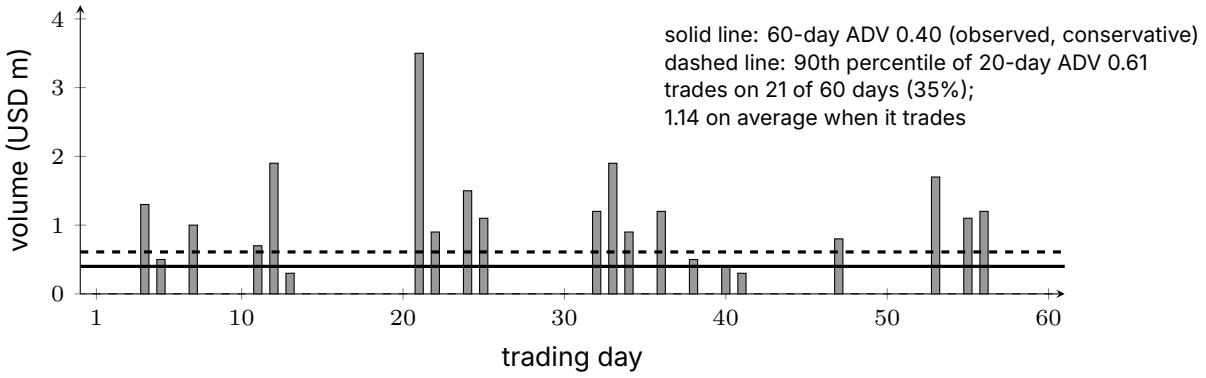



Figure 1. An infrequently traded corporate bond over 60 days (illustrative data, after the reading's Figure 98.1). Grey bars: daily volume; most days have none. Expected volume $= 0.35 \times 1.14 = 0.40$, the observed ADV (solid line). The 90th percentile of the rolling 20-day ADV (dashed line) is the optimistic end of the liquidity range.

## 3. Transaction cost (t-cost) models for corporate bonds

- **Why they matter.** T-costs are part of best-execution verification. Liquidating quickly on a pro rata basis can move prices; holding more liquidity to avoid that constrains returns. The model weighs the time needed to liquidate against the market impact.
- **Components. Fixed costs** (commissions, bid-ask spreads) plus **market impact costs** (how the size of the position affects the execution price). The bid-ask spread can be normalized by the midprice of benchmark quotes, but comparisons across bonds weaken when they trade infrequently. The model uses intraday benchmark prices, empirical transaction data and bond-specific attributes, with regression for some components: its standard form is a fixed-cost term plus a market-impact term.

- **Price is the key input** and is uncertain in fixed income, so estimates are noisy; measurement error matters most when it is large relative to the cost being measured. Standardized pricing data should help.

**Implementation shortfall** is the measure of market impact: the difference between the traded (execution) price and the **order-arrival price or another pre-trade benchmark**. For a sale,

$$\text{IS} = \frac{P_{\text{benchmark}} - P_{\text{execution}}}{P_{\text{benchmark}}} \qquad \text{cost} = (P_{\text{benchmark}} - P_{\text{execution}}) \times \frac{\text{par}}{100}$$

(for a purchase, execution minus benchmark).

- **Not the previous close.** Using yesterday's closing price puts every market move between the close and the execution into the measured cost, a sizable return component that is not the trade's doing. An **intraday, pre-trade benchmark** is best practice: it reflects prevailing conditions and is less open to manipulation (Figure 2).
- **Worked example.** Sell USD 10 million par; previous close 101.20, arrival price 100.80, average execution 100.45. Implementation shortfall $= 0.35$ points, or $0.35/100.80 = 34.7$ bp, USD 35,000. Against the previous close it would be 0.75 points (74.1 bp, USD 75,000): the extra USD 40,000 is the market's fall before the order arrived.
- **Estimating from many trades.** Weight observations by the **dollar amount traded**, because the many small odd-lot trades would otherwise bias the estimate. Security-level data are best; with little data, use a cohort of similar bonds (built on option-adjusted spread, duration, yield or bid-ask spread). Machine learning (clustering, tree-based models, neural networks) helps judge similarity.

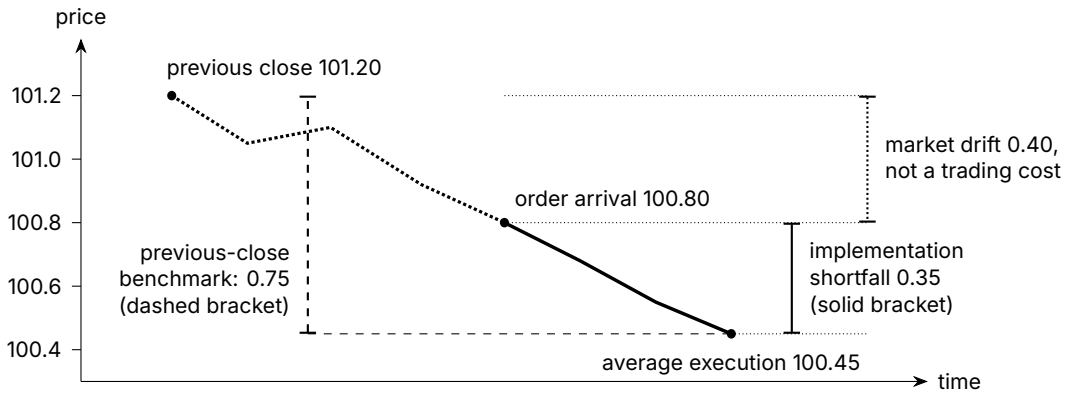



Figure 2. Choosing the benchmark for a sale (the notes' worked example). Dotted path: the market falls overnight and in the morning before the order arrives. Solid path: the price falls further while the order is worked. Only the solid part, 0.35 points, is the trade's market impact; the previous-close benchmark adds the 0.40 points of market drift.

**Formula card: IM-16 c**

| | |
|---|---|
| Days to liquidate | trade size/(participation rate $\times$ ADV), in consistent units |
| One-day position limit | maximum position = participation rate $\times$ ADV (or $\times$ average value traded) |
| Infrequently traded bond | $\mathrm{E}[\text{volume}] = \Pr(\text{trade}) \times \mathrm{E}[\text{volume} \mid \text{trade}]$; zero on no-trade days |
| Liquidity range | observed ADV (conservative) up to a tail estimate such as the 90th percentile of ADV (optimistic) |
| T-cost | fixed costs (commissions, bid-ask spread) + market impact cost |
| Implementation shortfall (sale) | $(P_{\text{arrival}} - P_{\text{exec}})/P_{\text{arrival}}$; cost $= (P_{\text{arrival}} - P_{\text{exec}}) \times \text{par}/100$; purchase: $P_{\text{exec}} - P_{\text{arrival}}$ |
| Benchmark | order-arrival or other intraday pre-trade price; not the previous close, not a post-trade price |
| Cohort cost estimate | dollar-weighted: $\sum_k(\text{size}_k \times \text{cost}_k)/\sum_k \text{size}_k$ |

| | Days to liquidate | Infrequently traded bonds | T-cost model |
|---|---|---|---|
| **Question answered** | How long to unwind a position without major impact | How much volume to expect when trades are sporadic | How much it costs to trade, and the market impact |
| **Inputs** | Trade size, participation rate, ADV | Trade probability, volume if traded; comparables; dealer inventory | Intraday benchmark prices, transaction data, bond attributes |
| **Method** | Ratio | Random forest regression; tail measures for latent liquidity | Regression; implementation shortfall; dollar weighting |
| **Weak spot** | Participation rate and latent liquidity hard to estimate | Quoted liquidity exceeds executed liquidity | Noisy prices; benchmark choice |

**Exam traps**

- **Participation slips.** 15% entered as 1.5% (ten times too many days); the participation rate ignored (shares ÷ ADV); the ratio inverted (capacity ÷ position).
- **Red-herring values.** Market value does not enter days to liquidate when shares and ADV are given; ranking by value or by share count alone picks the wrong security.
- **Conditional volume used as ADV.** For a sporadic bond, multiply by the probability of a trade; using the volume-when-traded understates the days.
- **Tail used as expected.** The 90th-percentile ADV is the optimistic end of the range, not the expected volume.
- **Polarity on ADV.** When volume variance is high, ADV is *conservative* (it understates liquidity). Quoted liquidity *exceeds* executed liquidity.
- **Wrong benchmark.** The previous close adds the market's move; a post-trade price hides part of the impact. Use the order-arrival or intraday pre-trade price.
- **Unweighted averages.** Many small odd-lot trades bias an equally weighted cost upward;

weight by dollar amount traded.

## Practice questions: IM-16 c

### Question IM-16 c.1

Bank Q568 • AnalystPrep Q.2516, Q.2518, Q.2534 (rebuilt)

A risk management unit at an investment fund wants to gauge how long it would take to unwind a large position during a period of heavy redemption requests. It decides to use the days-to-liquidate approach to model asset liquidity. Which of the following best describes what this approach estimates?

**A.** The expected change in the asset's price over the liquidation horizon, based on recent volatility.

**B.** The exact amount of cash the fund must hold to meet expected client redemptions.

**C.** The number of days needed to unwind a position, given the size of the trade, an assumed market participation rate, and the asset's average daily volume.

**D.** The credit-quality deterioration of the asset's issuer over the liquidation horizon.

### Question IM-16 c.2

Bank Q573 • AnalystPrep Q.2751 (rebuilt)

A risk analyst at an equity fund uses the days-to-liquidate approach to estimate how long it would take to exit a position, assuming that no more than 15% of the daily volume in the security will be sold.

| | |
|---|---|
| Number of shares held | 2,000,000 |
| Average daily volume of the security (shares) | 750,000 |

Which of the following is closest to the number of days needed to liquidate the position?

**A.** 0.056

**B.** 2.7

**C.** 17.8

**D.** 177.8

### Question IM-16 c.3

Bank Q560 • AnalystPrep Q.3228 (rebuilt)

XinWin pension fund has a liquidity risk policy that limits each equity holding to a maximum of 45% of the stock's 30-day average value traded. The fund's size is $7 billion. What is the maximum allocation (as a percentage of the fund) that the fund can hold in a stock whose 30-day average value traded is $390 million?

**A.** 1.25%

**B.** 2.51%

**C.** 5.57%

**D.** 25.07%

### Question IM-16 c.4

Bank Q565 • Mock 3 Q56

A hedge fund manager is concerned about a potential increase in investor redemptions and wants to assess the effects of such an event on the fund's liquidity. The manager asks a junior analyst to estimate the average number of days required to liquidate certain securities in the fund. The analyst uses the following information for four securities:

| Security | Market value in fund (CNY m) | Shares in fund | Average daily trading volume | Maximum share of daily volume sold |
|---|---|---|---|---|
| A | 93.00 | 500,000 | 522,000 | 22% |
| B | 173.04 | 420,000 | 1,328,000 | 12% |
| C | 58.88 | 256,000 | 710,000 | 18% |
| D | 28.80 | 640,000 | 848,000 | 20% |

Which of the four securities is expected to take the longest to liquidate?

**A.** Security A

**B.** Security B

**C.** Security C

**D.** Security D

### Question IM-16 c.5

Bank Q571 • AnalystPrep Q.2533 (rebuilt)

A risk analyst adapts the days-to-liquidate approach for an infrequently traded corporate bond by using historical data to fill in missing volume observations, assuming expected volume on no-trade days is zero. Which of the following correctly describes how the infrequent-trading problem is broken down?

**A.** Estimating the bid-ask spread and then the market impact cost of the trade.

**B.** Estimating the probability that a trade will occur and then predicting the trade volume if a trade does occur.

**C.** Estimating the coupon rate and then the residual maturity of the bond.

**D.** Estimating the participation rate and then the portfolio weight of the bond.

### Question IM-16 c.6

New • built from SchweserNotes 2026 Reading 98

A risk analyst at a high-yield fund estimates the liquidation horizon of a USD 6 million position in a corporate bond that trades only on some days. Her random forest model gives the following, with expected volume on no-trade days equal to zero:

| | |
|---|---|
| Probability that the bond trades on a given day | 30% |
| Expected volume on days when it trades | USD 2.5 million |
| 90th percentile of the bond's ADV distribution | USD 1.8 million |
| Maximum participation rate under the fund's policy | 20% |

Which of the following is closest to the expected number of days needed to liquidate the position?

**A.** 8.0

**B.** 12.0

**C.** 16.7

**D.** 40.0

### Question IM-16 c.7

Bank Q566 • Schweser Module Quiz 98.2 Q1

Which of the following statements best describes a key challenge in modeling the liquidity risk of infrequently traded fixed-income securities using the days-to-liquidate approach?

**A.** Random forest regression cannot handle large feature sets or missing data.

**B.** Trade frequency and volume follow a linear pattern that basic regression can capture.

**C.** Bid prices perfectly represent true liquidity conditions, making volume variance irrelevant.

**D.** Relying on recent trade volumes can bias estimates because latent liquidity requires additional inference.

### Question IM-16 c.8

Bank Q567 • Schweser Module Quiz 98.2 Q2

When estimating transaction costs using t-cost modeling, why is an intraday, pre-trade benchmark price typically preferred over the previous end-of-day price? The intraday, pre-trade benchmark price:

**A.** reduces benchmark-related noise.

**B.** ensures that market impact is zero.

**C.** allows bid-ask spreads to be ignored.

**D.** eliminates the need to measure implementation shortfall.

**Question IM-16 c.9** New • built from SchweserNotes 2026 Reading 98

A corporate bond trader at a US asset manager sells USD 25 million par of a BBB-rated corporate bond to meet redemptions. The fund's transaction cost analysis uses implementation shortfall to measure market impact. The following prices (per 100 of par) are recorded:

| | |
|---|---|
| Previous day's closing price | 99.10 |
| Price at order arrival (intraday pre-trade benchmark) | 98.40 |
| Average execution price | 97.95 |
| Closing price on the trade date | 98.10 |

Which of the following is closest to the market impact cost of the sale?

**A.** USD 37,500

**B.** USD 112,500

**C.** USD 175,000

**D.** USD 287,500

**Question IM-16 c.10** New • built from SchweserNotes 2026 Reading 98

A transaction cost analyst at a bond fund estimates the cost of trading a cohort of similar corporate bonds from the following recent trades. Each cost is the implementation shortfall against an intraday pre-trade benchmark.

| Trade size (USD million) | 0.05 | 0.10 | 0.20 | 0.50 | 2.00 | 8.00 |
|---|---|---|---|---|---|---|
| Cost (basis points) | 90 | 70 | 55 | 40 | 22 | 12 |

Following best practice for estimating transaction costs, which of the following is closest to the cohort's estimated cost?

**A.** 14.0 bp

**B.** 16.8 bp

**C.** 30.4 bp

**D.** 48.2 bp

# Solutions • IM-16 c

## Solution IM-16 c.1

**Answer: C**

Bank Q568 • AnalystPrep Q.2516, Q.2518, Q.2534

### Step 1 – Write the approach

Days to liquidate = size of the intended trade / (participation rate × ADV).

### Step 2 – Read the output

The participation rate is the share of ADV that can be sold without significant impact, so the result is the time, in days, to unwind the position in an orderly way.
⇒ C.

### Interpretation

It is an asset-liquidity measure. Combined with a redemption estimate (IM-16 e), it tells the fund whether it can meet outflows on time.

### Why the other options are wrong

**A is incorrect.** Confuses a time-to-exit measure with a market-risk forecast.
**B is incorrect.** Confuses asset liquidity with funding needs: the output is days, not a cash amount.
**D is incorrect.** Confuses liquidity with credit risk.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Days-to-Liquidate Approach; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.7

## Solution IM-16 c.2

**Answer: C**

Bank Q573 • AnalystPrep Q.2751

### Step 1 – Daily sales allowed

$0.15 \times 750{,}000 = 112{,}500$ shares a day.

### Step 2 – Days to liquidate

$$\frac{2{,}000{,}000}{112{,}500} = 17.78 \text{ days}$$

⇒ About 17.8 trading days to exit without exceeding the participation limit.

### Interpretation

The position is 2.7 days of total market volume, but the fund may take only 15% of each day's volume, so the exit takes more than six times as long.

### Why the other options are wrong

**A is incorrect.** Inverted the ratio: $112{,}500/2{,}000{,}000 = 0.056$.
**B is incorrect.** Ignored the participation limit: $2{,}000{,}000/750{,}000 = 2.7$.
**D is incorrect.** Entered 15% as 1.5%: $2{,}000{,}000/(0.015 \times 750{,}000) = 177.8$.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Days-to-Liquidate Approach

### Solution IM-16 c.3

**Answer: B**

Bank Q560 • AnalystPrep Q.3228

#### Step 1 – Maximum position in dollars

$0.45 \times \$390\text{m} = \$175.5\text{m}$.

$\Rightarrow$ The limit is one day's permitted sales at a 45% participation rate.

#### Step 2 – As a share of the fund

$$\frac{175.5}{7{,}000} = 2.51\%$$

#### Interpretation

A position limit tied to average value traded is the days-to-liquidate formula solved for the trade size with the horizon set to one day.

#### Why the other options are wrong

**A is incorrect.** Halved the limit, using 22.5% of value traded: $87.75/7{,}000 = 1.25\%$.

**C is incorrect.** Ignored the 45% factor: $390/7{,}000 = 5.57\%$.

**D is incorrect.** Unit slip, dividing by \$700 million instead of \$7,000 million: $175.5/700 = 25.07\%$.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c (participation rate and ADV); filed in the bank under LO 98.a

### Solution IM-16 c.4

**Answer: A**

Bank Q565 • Mock 3 Q56

#### Step 1 – Days for each security

Days = shares / (maximum share of volume $\times$ ADV); market value is not needed.

| Security | Daily shares allowed | Days |
|---|---|---|
| A | $0.22 \times 522{,}000 = 114{,}840$ | $500{,}000/114{,}840 = 4.35$ |
| B | $0.12 \times 1{,}328{,}000 = 159{,}360$ | $420{,}000/159{,}360 = 2.64$ |
| C | $0.18 \times 710{,}000 = 127{,}800$ | $256{,}000/127{,}800 = 2.00$ |
| D | $0.20 \times 848{,}000 = 169{,}600$ | $640{,}000/169{,}600 = 3.77$ |

#### Step 2 – Rank

A takes longest, 4.35 days.

#### Interpretation

The market value column is a red herring: days to liquidate is a ratio of shares to tradable shares. A holds almost a full day's total market volume.

#### Why the other options are wrong

**B is incorrect.** Ranked by market value (CNY 173.04 million, the largest); B needs only 2.64 days.

**C is incorrect.** The quickest, 2.00 days: chosen by ranking in the wrong direction.

**D is incorrect.** Ranked by share count (640,000, the largest) without dividing by tradable volume: 3.77 days.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Days-to-Liquidate Approach

### Solution IM-16 c.5 — Answer: B

Bank Q571 • AnalystPrep Q.2533

**Step 1 – Why the usual approach fails**

The bond trades sporadically, so observed ADV is unreliable.

**Step 2 – The decomposition**

(1) The probability that a trade occurs; (2) the trade volume if it does. With zero volume on no-trade days, $\mathrm{E}[\text{volume}] = \Pr(\text{trade}) \times \mathrm{E}[\text{volume} \mid \text{trade}]$; a random forest regression can model both parts.

$\Rightarrow$ B.

**Interpretation**

Splitting the problem lets the model learn separately what makes a bond trade at all and what makes a trade large.

**Why the other options are wrong**

**A is incorrect.** Describes the t-cost model (fixed costs and market impact), not volume modeling.

**C is incorrect.** Coupon and maturity may be explanatory variables, not the two components.

**D is incorrect.** Participation rate is an input to days to liquidate; portfolio weight is unrelated.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Modeling Liquidity for Infrequently Traded Bonds

### Solution IM-16 c.6 — Answer: D

New • built from SchweserNotes 2026 Reading 98

**Step 1 – Expected daily volume**

$$\mathrm{E}[\text{volume}] = 0.30 \times 2.5 = \text{USD } 0.75 \text{ million}$$

$\Rightarrow$ The bond trades on fewer than one day in three, so its expected volume is well below its volume when it trades.

**Step 2 – Days to liquidate**

$$\frac{6}{0.20 \times 0.75} = \frac{6}{0.15} = 40 \text{ days}$$

**Interpretation**

The 90th-percentile ADV is a red herring for the expected horizon: it is the optimistic end of the liquidity range ($6/(0.20 \times 1.8) = 16.7$ days). The analyst can report a range of about 17 to 40 days, with 40 as the expected case.

**Why the other options are wrong**

**A is incorrect.** Dropped the participation rate: $6/0.75 = 8.0$.

**B is incorrect.** Used the volume when the bond trades as its ADV, ignoring the 30% trade probability: $6/(0.20 \times 2.5) = 12.0$.

**C is incorrect.** Used the optimistic tail estimate as the expected ADV: $6/(0.20 \times 1.8) = 16.7$.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Modeling Liquidity for Infrequently Traded Bonds

### Solution IM-16 c.7 Answer: D

Bank Q566 • Schweser Module Quiz 98.2 Q1

**Step 1 – Observed versus potential volume**

For sporadically traded bonds, actual volume is lower than what the market could support; the gap is latent liquidity.

**Step 2 – The consequence**

Recent volumes alone are a biased input, and latent liquidity has to be inferred: from comparable bonds, dealer inventory, or tail measures such as the 90th-percentile ADV.

$\Rightarrow$ D.

**Interpretation**

High volume variance adds noise: large swings in daily volume give large swings in estimates, and ADV becomes a conservative measure.

**Why the other options are wrong**

**A is incorrect.** Reverses the reading: random forests are chosen because they handle missing data, nonlinearity and many explanatory variables.

**B is incorrect.** The relationships are nonlinear, which is why basic linear regression falls short.

**C is incorrect.** Absolute claim: posted bids overstate executable liquidity, and volume variance matters.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c

### Solution IM-16 c.8 Answer: A

Bank Q567 • Schweser Module Quiz 98.2 Q2

**Step 1 – What implementation shortfall measures**

Executed price minus a pre-trade benchmark: the market impact of the trade.

**Step 2 – What the previous close adds**

Every market move between yesterday's close and the execution enters the measured cost: a sizable return component unrelated to the trade.

**Step 3 – Why intraday is better**

An intraday, pre-trade benchmark reflects prevailing conditions, strips out that noise and is less open to manipulation.

$\Rightarrow$ A.

**Interpretation**

Figure 2 shows the size of the effect: 0.40 points of the 0.75-point previous-close shortfall is market drift.

**Why the other options are wrong**

**B is incorrect.** No benchmark makes impact zero; impact is what the shortfall measures.

**C is incorrect.** The bid-ask spread is a fixed-cost component and stays in the model.

**D is incorrect.** The benchmark is an input to implementation shortfall, which remains the measure of impact.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Transaction Cost Modeling for Corporate Bonds

### Solution IM-16 c.9

**Answer: B**

New • built from SchweserNotes 2026 Reading 98

**Step 1 – Choose the benchmark**

The order-arrival price, 98.40, is the intraday pre-trade benchmark.

**Step 2 – Implementation shortfall**

$$98.40 - 97.95 = 0.45 \text{ points} \qquad \frac{0.45}{98.40} = 45.7 \text{ bp}$$

**Step 3 – In dollars**

$$0.45 \times \frac{25{,}000{,}000}{100} = \text{USD } 112{,}500$$

**Interpretation**

The previous close (99.10) and the trade-date close (98.10) are red herrings. The 0.70-point fall from the close to the order's arrival was the market's move, not the trade's impact.

**Why the other options are wrong**

**A is incorrect.** Used the post-trade closing price: $(98.10 - 97.95) \times 250{,}000 = 37{,}500$. A post-trade benchmark hides the impact that has already reverted.

**C is incorrect.** Measured the market's drift before the order: $(99.10 - 98.40) \times 250{,}000 = 175{,}000$.

**D is incorrect.** Used the previous close as the benchmark: $(99.10 - 97.95) \times 250{,}000 = 287{,}500$, which includes the drift.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Transaction Cost Modeling for Corporate Bonds (implementation shortfall)

### Solution IM-16 c.10

**Answer: B**

New • built from SchweserNotes 2026 Reading 98

**Step 1 – The rule**

When estimating transaction costs, weight observations by the dollar amount traded, to offset the large number of small odd-lot trades.

**Step 2 – Dollar-weighted cost**

$$\sum \text{size} \times \text{cost} = 0.05(90) + 0.10(70) + 0.20(55) + 0.50(40) + 2.00(22) + 8.00(12) = 182.5$$

$$\frac{182.5}{10.85} = 16.8 \text{ bp}$$

**Interpretation**

The four trades of USD 0.5 million or less are two-thirds of the observations but under 8% of the dollars traded. An equally weighted average lets them dominate and nearly triples the estimate.

**Why the other options are wrong**

**A is incorrect.** Dropped the small trades instead of weighting them: $(44 + 96)/10 = 14.0$.

**C is incorrect.** Divided the dollar-weighted sum by the number of trades instead of the total size: $182.5/6 = 30.4$.

**D is incorrect.** Equally weighted average: $289/6 = 48.2$, the bias the rule exists to remove.

**LO:** IM-16 c: Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.c, Transaction Cost Modeling for Corporate Bonds (regression weighting)

IM-16 • Liquidity Risk Management • Vardeez: VERY IMPORTANT

# IM-16 d

P2 CONCEPT

Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed.

Workbook: none • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** An LO of a VERY IMPORTANT reading (Vardeez: "know it"); no workbook section. **Vardeez:** no pointer. His notes cover only temporary borrowing and "break glass" testing; the other four measures, and the US versus European (UCITS) rules for each, come from Schweser-Notes 2026 Reading 98.

**Reported exam shapes:** none reported.

**What the questions look like:** statement questions. Which measures a given fund may use (a US retail fund versus a UCITS fund is the standard sibling swap), how each measure works, and when it is used. A bank item that only computed a gated payout (10% of assets) is left out under the CONCEPT rule; gates are tested by what they do and where they are allowed.

## 1. Plan first, break glass second

The most effective way to handle unexpected redemptions is **advance planning**. Some jurisdictions also permit **extraordinary measures** when ordinary liquidity tools prove insufficient. They are rarely invoked but may be needed at short notice, so legal, operations, risk management and trading teams should run **"break glass" testing** to make sure each measure can be activated quickly. A tool that has never been exercised is not a tool.

## 2. The five measures

- **Temporary borrowing.** A short-term buffer that gives the fund time to sell less liquid assets in an orderly way: overdrafts, repurchase agreements (repos), bank credit facilities, interfund lending. *Permitted for US retail funds; not for European UCITS funds.*
- **In-kind redemptions.** Specific assets are transferred directly to the redeeming investor, so the fund need not sell them. *Permitted by regulation* (US and Europe), but practical only for certain asset types and mostly for **large institutional investors**, because it relies on specialized electronic transfer systems.
- **Gates.** Allow partial redemptions up to a predetermined threshold, such as 10% of fund assets, slowing or preventing a run by limiting the pace of outflows. *Permitted for UCITS funds; not for US-based funds.*
- **Price mechanisms.** Swing pricing and purchase redemption values (PRVs) shift transaction costs to the investors who subscribe or redeem. **Swing pricing** adjusts the fund's NAV when net flows exceed a threshold, so trading costs fall on the transacting investors rather than on those who remain; raising the cost of a rapid exit reduces the incentive to run. *Swing pricing: permitted and widely used for UCITS funds; the SEC does not currently allow it for US mutual funds* (under consideration in a rulemaking planned for 2025). *PRVs: allowed for US collective trust funds* (*CTFs*) *under OCC oversight.*
- **Suspension.** A complete halt to redemptions: the **most extreme** measure. It can be in investors'

long-term interest in severe market stress, but it creates significant operational and reputational problems. *Not permitted for US mutual funds without explicit SEC approval.*

| Measure | How it works; when it is used | United States | Europe (UCITS) |
|---|---|---|---|
| **Temporary borrowing** | Overdraft, repo, credit facility, interfund lending; buys time for orderly sales | Permitted for retail funds | Not permitted |
| **In-kind redemption** | Assets handed to the redeeming investor; certain asset types, large institutions | Permitted (limited in practice) | Permitted (limited in practice) |
| **Gate** | Partial redemptions up to a threshold (e.g. 10% of assets); slows a run | Not permitted | Permitted |
| **Swing pricing** | NAV adjusted when net flows pass a threshold; transacting investors bear the costs | Not for mutual funds (SEC, under consideration) | Permitted, widely used |
| **PRV** | Purchase and redemption values shift costs to transacting investors | US collective trust funds (OCC) | n/a |
| **Suspension** | Full halt; most extreme; severe stress only | Mutual funds only with explicit SEC approval | n/a in the reading |

**Exam traps**

- **Regime swap.** Borrowing: US yes, UCITS no. Gates and swing pricing: UCITS yes, US no. Swapping them is the standard trap.
- **PRV versus swing pricing.** PRVs are for US collective trust funds under the OCC; swing pricing is the UCITS tool.
- **Suspension.** The most extreme measure; for US mutual funds only with explicit SEC approval (not "never", not "at the fund's discretion").
- **Gate versus suspension.** A gate allows partial redemptions up to a threshold; a suspension halts them all.
- **In-kind.** Permitted, but practical only for certain assets and large institutional investors, not for retail.
- **Who pays.** Price mechanisms move costs *to* the transacting investors and *away from* those who remain.
- **Extraordinary means rare.** Advance planning comes first; break-glass testing makes the rare tools usable.

## Practice questions: IM-16 d

### Question IM-16 d.1

New • built from SchweserNotes 2026 Reading 98

A fund's board asks why its legal, operations, risk management and trading teams spend time on "break glass" testing of extraordinary liquidity measures that the fund has never used. Which of the following responses is most accurate?

**A.** Extraordinary measures are the fund's first response to any redemption request, so they must be exercised every day.

**B.** Testing is needed only by the legal team, because the measures are regulatory permissions rather than operational processes.

**C.** Advance planning is the most effective defense, but extraordinary measures, though rarely invoked, may be needed at short notice, so each must be ready to activate quickly.

**D.** Once a regulator has permitted a measure it can be activated immediately without preparation, so testing adds little value.

### Question IM-16 d.2

Bank Q575 • Schweser Module Quiz 98.3 Q1 (rebuilt)

A U.S.-based retail fund needs an extraordinary liquidity measure to meet unexpected redemption requests when its ordinary liquidity tools are insufficient. Which of the following measures is generally available to it?

**A.** Gates

**B.** Suspension

**C.** Swing pricing

**D.** Temporary borrowing

### Question IM-16 d.3

New • built from SchweserNotes 2026 Reading 98

A US bond mutual fund receives a redemption request for 12% of its assets from a single large pension plan during a stressed market. The fund wants to avoid selling bonds into the market. Which of the following statements about meeting the request with an in-kind redemption is correct?

**A.** It is not permitted for US funds, which can meet a request of this size only by gating or by suspending redemptions.

**B.** It transfers specific bonds directly to the pension plan and avoids a sale; it is permitted, but practical mainly for certain assets and large institutional investors.

**C.** It is the preferred tool for retail investors, because small accounts can easily receive individual bonds through standard brokerage systems.

**D.** It shifts the transaction costs of the redemption onto the remaining shareholders by adjusting the fund's net asset value.

**Question IM-16 d.4** Bank Q576 • New, written for the bank (GARP style)

A risk manager at a European UCITS fund is reviewing the fund's extraordinary liquidity measures ahead of a break-glass test. Which of the following statements is correct?

**A.** Gates are not permitted for UCITS funds, whereas they are available to U.S. mutual funds.

**B.** Temporary borrowing and gates are both permitted for UCITS funds, but swing pricing is not.

**C.** Swing pricing and temporary borrowing are both permitted for UCITS funds, but gates are not.

**D.** Gates and swing pricing are both permitted for UCITS funds, but temporary borrowing is not.

**Question IM-16 d.5** New • built from SchweserNotes 2026 Reading 98

A risk manager compares the price-based mechanisms available to funds in different regimes. Which of the following statements is correct?

**A.** Swing pricing adjusts the NAV when net flows pass a threshold, so transacting investors bear the trading costs; it is common in UCITS funds, and PRVs serve US collective trust funds.

**B.** Swing pricing passes the trading costs of large redemptions to the remaining shareholders, which increases the incentive for investors to run.

**C.** Swing pricing is widely used by US mutual funds under SEC rules, while UCITS funds rely on purchase redemption values instead.

**D.** Price mechanisms limit redemptions to a fixed share of fund assets, such as 10%, until market conditions recover.

**Question IM-16 d.6** New • built from SchweserNotes 2026 Reading 98

A European UCITS fund and a US mutual fund both face a surge of redemption requests during a severe market dislocation. Which of the following statements about gates and suspensions is correct?

**A.** Both funds may impose a gate that limits redemptions to a set share of fund assets, such as 10%, without regulatory approval.

**B.** A suspension allows partial redemptions up to a threshold, whereas a gate halts all redemptions and is the most extreme measure.

**C.** The UCITS fund may use a gate to allow partial redemptions up to a threshold, while the US mutual fund may suspend redemptions only with explicit SEC approval.

**D.** The US mutual fund may suspend redemptions at its own discretion, because suspension is the least disruptive of the extraordinary measures.

## Solutions • IM-16 d

### Solution IM-16 d.1 — Answer: C

New • built from SchweserNotes 2026 Reading 98

**Step 1 – The order of defenses**

Advance planning is the most effective way to manage unexpected redemptions; extraordinary measures apply when ordinary tools are insufficient.

**Step 2 – Why test them**

They are rarely invoked but may be needed on short notice, so legal, operations, risk management and trading teams conduct break-glass testing to be able to activate each one quickly.
⇒ C.

**Interpretation**

The test is cross-functional because each measure touches legal permissions, operational systems (transfer of assets in kind, NAV adjustments) and trading.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: extraordinary measures are rare, not the first response.
**B is incorrect.** Scope error: operations, risk and trading teams are part of the testing.
**D is incorrect.** A permission is not readiness; the measures may be needed at short notice.

**LO:** IM-16 d: Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.d; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.7

### Solution IM-16 d.2 — Answer: D

Bank Q575 • Schweser Module Quiz 98.3 Q1

**Step 1 – Check each measure for a US retail fund**

Temporary borrowing (overdrafts, repos, bank credit lines, interfund lending): permitted for US retail funds. Gates: UCITS only. Swing pricing: UCITS; the SEC does not currently allow it for mutual funds. Suspension: not permitted for US mutual funds without explicit SEC approval.
⇒ Only temporary borrowing is generally available: D.

**Interpretation**

The US toolkit is borrowing (and in-kind redemptions where practical); the European toolkit leans on gates and swing pricing.

**Why the other options are wrong**

**A is incorrect.** Gates belong to the UCITS regime, not to US funds.
**B is incorrect.** Suspension needs explicit SEC approval, so it is not generally available.
**C is incorrect.** Swing pricing is a UCITS tool; US collective trust funds use PRVs instead.

**LO:** IM-16 d: Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.d

### Solution IM-16 d.3 — Answer: B

New • built from SchweserNotes 2026 Reading 98

**Step 1 – What an in-kind redemption is**

Specific assets are transferred directly to the redeeming investor, so the fund does not have to sell

them.

### Step 2 – When it is allowed and practical

It is permitted by regulation, but used only for certain asset types and mostly with large institutional investors, because it relies on specialized electronic transfer systems. A pension plan redeeming 12% is the textbook case.

⇒ B.

### Interpretation

It solves the fund's problem without a fire sale, but only because the redeeming investor can receive and hold the bonds.

### Why the other options are wrong

**A is incorrect.** In-kind redemptions are permitted; and gates are not allowed for US funds.

**C is incorrect.** Reverses the practice: it suits large institutions, not retail accounts.

**D is incorrect.** Describes swing pricing, and even that moves costs onto transacting investors, not remaining ones.

**LO:** IM-16 d: Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.d, in-kind redemptions

## Solution IM-16 d.4 Answer: D

Bank Q576 • New, written for the bank

### Step 1 – Borrowing

Permitted for US retail funds, not for UCITS funds.

### Step 2 – Gates and swing pricing

Gates are permitted for UCITS funds (not US funds); swing pricing is permitted and widely used for UCITS funds.

⇒ Gates and swing pricing yes, borrowing no: D.

### Interpretation

For this UCITS fund, the break-glass test should cover the gate and the swing-pricing mechanics, not a credit facility.

### Why the other options are wrong

**A is incorrect.** Reverses the regimes.

**B is incorrect.** Allows borrowing and denies swing pricing: both wrong for UCITS.

**C is incorrect.** Applies the US pattern (borrowing yes, gates no) to a UCITS fund.

**LO:** IM-16 d: Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.d

## Solution IM-16 d.5 Answer: A

New • built from SchweserNotes 2026 Reading 98

### Step 1 – How swing pricing works

The NAV is adjusted when net flows exceed a threshold, so the costs of trading are absorbed by the transacting investors rather than the remaining shareholders. Raising the cost of rapid exits reduces the incentive to run.

### Step 2 – Where each mechanism applies

Swing pricing is permitted and widely used for UCITS funds; the SEC does not currently allow

it for mutual funds. PRVs are allowed for US collective trust funds under OCC oversight.
⇒ A.

### Interpretation

Price mechanisms do not stop redemptions; they make the redeeming investor pay for the liquidity he consumes.

### Why the other options are wrong

**B is incorrect.** Polarity flip: costs move to the transacting investors, and the incentive to run falls.
**C is incorrect.** Reverses the regimes: swing pricing is the UCITS tool; PRVs are for US collective trust funds.
**D is incorrect.** Describes a gate, not a price mechanism.

**LO:** IM-16 d: Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.d, price mechanisms

## Solution IM-16 d.6 Answer: C

New • built from SchweserNotes 2026 Reading 98

### Step 1 – Gates

Partial redemptions up to a predetermined threshold (such as 10% of fund assets); they slow or prevent a run. Permitted for UCITS funds, not for US-based funds.

### Step 2 – Suspension

A complete halt to redemptions, the most extreme measure, sometimes in investors' long-term interest in severe stress. Not permitted for US mutual funds without explicit SEC approval.
⇒ C.

### Interpretation

The two funds reach for different tools in the same crisis: the UCITS fund can gate; the US fund cannot gate and needs the SEC to suspend.

### Why the other options are wrong

**A is incorrect.** Gates are not allowed for US funds.
**B is incorrect.** Swaps the definitions: the gate is partial, the suspension total and most extreme.
**D is incorrect.** Suspension is the most disruptive measure and needs explicit SEC approval.

**LO:** IM-16 d: Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.d, gates and suspension

IM-16 • Liquidity Risk Management • Vardeez: VERY IMPORTANT

# IM-16 e

P2 CONCEPT

Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests.

Workbook: none • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** An LO of a VERY IMPORTANT reading (Vardeez: "know it"); no workbook section. **Vardeez:** no pointer.

**Reported exam shapes:** none reported.

**What the questions look like:** statement questions on the objective (fairness to three groups of investors), what drives large outflows, the layers of the redemption waterfall in order, how stress testing extends the model, the historical redemption-at-risk (HRaR) model and its limits, and which optimization constraint handles which concern. A bank item that only subtracted the waterfall layers from a redemption is left out under the CONCEPT rule; the waterfall is tested by its order and its rules.

## 1. What drives redemptions, and the objective

Unexpected or unusually large outflows can come from macroeconomic developments, poor fund performance, damage to the manager's reputation, severe underperformance by **peer funds** (contagion), and fund-specific features such as investor profiles and **concentration**. In normal markets redemption demand is typically low before a strategy matures; stressed markets can produce **large and highly correlated** redemptions. The objective is to treat all three groups of investors fairly and equitably: those **redeeming**, those **subscribing** and those **remaining** in the fund (recall IM-16 a, Figure 1).

## 2. The redemption waterfall

US '40 Act funds typically meet redemptions through a **waterfall**: a sequence of layers, moving to the next layer when the current one is not enough (Figure 1).

- **Layer 1: cash and highly liquid assets.** Used for three purposes: meeting routine or unplanned redemption "noise" at minimum transaction cost; supporting a strategy that uses a derivatives overlay for return enhancement; or responding strategically to market conditions.
- **Layer 2: pro rata or risk-constant sales.** When requests exceed Layer 1, sell less liquid holdings so that the remaining portfolio keeps approximately the same risk profile as before.
- **Layer 3: short-term borrowing,** the final option: reverse repos (a borrowing, from the fund's side), overdraft facilities, lines of credit. '40 Act funds may borrow no more than **33.3% of total fund assets**.

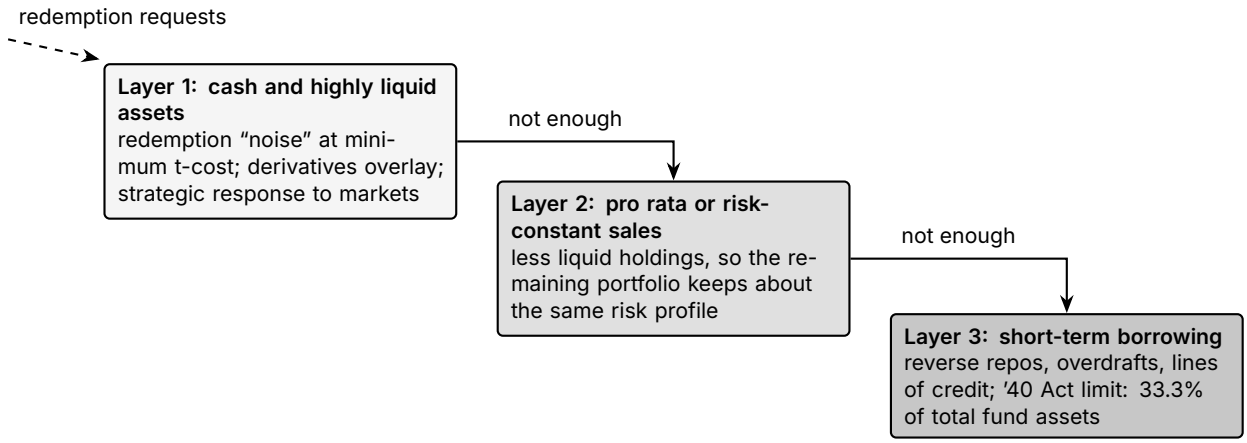



Figure 1. The redemption waterfall of a US '40 Act fund. Each layer is used only when the one above cannot meet the requests; shading darkens as the layers become more costly or more constrained.

**Stress testing** is a natural extension of redemption modeling: apply stress scenarios to fund assets, redemption volumes, ADV assumptions and transaction costs, and consider investor concentration, relevant historical redemption events and the effect of any gates in place.

## 3. Redemption-at-risk

- **HRaR.** Estimate worst-case outflows from the fund's historical flows, for example the **99th percentile of five-day redemptions** over the life of the fund. The result is the historical redemption-at-risk (HRaR) and gives managers stress signals.
- **Three limits of relying on history.** (a) New or fast-growing funds may not have experienced meaningful outflows. (b) Redemptions can be driven by idiosyncratic investor behavior that cannot be modeled reliably. (c) Managers may not see who is redeeming when a third party, such as a custodian, sits between the fund and the end investor.
- **Two kinds of drivers.** Large outflows usually combine **systemic** factors (sector-wide disruptions, macroeconomic shocks) and **idiosyncratic** factors (investor behavior, peer fund performance, discretionary cash needs, events such as insider-trading scandals, regulatory sanctions or reputational damage).
- **Machine learning** has been explored because it handles many factors and nonlinear relationships; it can estimate an **upper bound** on likely redemptions over a horizon. Data availability and quality remain significant obstacles.

## 4. Liquidity optimization

Liquidity management can be framed as an optimization: balance **liquidity**, **transaction costs** and **portfolio risk** while meeting redemption obligations. Securities $x_i$ have market value $m_i$ and notional $n_i$; the manager sells a pro rata fraction $p_{i,t}$ on day $t$. The first step is to find the optimal pro rata fraction; the objective is then maximized subject to constraints:

| Constraint | What it enforces | Example |
|---|---|---|
| **Redemption** | Total sales meet the required liquidation amount $v$ over a set period | USD 200 million within two days |
| **ADV** | Sales of security $i$ (ADV $v_i$, sector $j$) stay within the sector's participation rate $r_j$ of its ADV | No more than $r_j v_i$ a day |
| **Sector capacity** | Total sales in a sector $S_j$ stay within its capacity $c_{j,t}$: simultaneous sales of similar bonds can exhaust the market's ability to absorb supply | A sell-off in one bond sector |
| **Time lag** | Some assets cannot be sold before $d_i$ days | IPO lockups |
| **Risk** | The portfolio stays below a risk threshold $R$ after the sales; the sum runs over all securities in the benchmark and the portfolio, with benchmark weights $b_i = 0$ for absolute return funds | Index fund tracking error |

Other constraints (cash use, tracking error, leverage limits, compliance) can be added. The whole optimization is ultimately constrained by the **quality and availability of its input data**: every constraint needs a number that IM-16 b has just shown is hard to observe. (Vardeez lists four constraints; the reading also has the sector capacity constraint.)

| | Redemption waterfall | Redemption-at-risk (HRaR) | Liquidity optimization |
|---|---|---|---|
| **Question answered** | In what order to raise cash | How large outflows could be | What to sell, how much, on which day |
| **Key rule or input** | Cash, then pro rata or risk-constant sales, then borrowing ($\leq 33.3\%$ of assets) | 99th-percentile, five-day redemption over the fund's life | Redemption, ADV, sector capacity, time-lag and risk constraints |
| **Main limitation** | Layers can be exhausted in stress; stress test them | Short histories, idiosyncratic behavior, intermediaries hide investors | Input data quality and availability |

**Exam traps**

- **Fairness runs three ways.** Redeeming, subscribing and remaining investors; the remaining group is the one an ad hoc response forgets.
- **Waterfall order.** Cash and highly liquid assets, then pro rata or risk-constant sales, then borrowing as the final layer. Not "borrow first", not "sell the least liquid first".
- **The borrowing limit.** 33.3% of total fund assets for '40 Act funds (not 3.3%, not 50%). Gates are a UCITS tool, not a '40 Act waterfall layer.
- **HRaR's numbers.** 99th percentile, five-day, over the life of the fund.
- **History is not enough (least-likely trap).** New funds, idiosyncratic investors and intermedi-

aries all weaken HRaR; "history is reliable in all environments" is the false statement.

- **Drivers.** Systemic *and* idiosyncratic; peer underperformance can spread by contagion.
- **Constraint matching.** Lockup: time lag. Sector absorption: sector capacity. Tracking error: risk. Daily volume: ADV.
- **Machine learning.** Estimates an upper bound with many nonlinear factors; it still needs good data.

## Practice questions: IM-16 e

### Question IM-16 e.1

New • built from SchweserNotes 2026 Reading 98

During a sector-wide selloff, several peer funds suffer severe underperformance and a bond fund starts to receive unusually large redemption requests. The CIO asks what the overriding objective of the fund's redemption management should be. Which of the following responses is most consistent with best practice?

**A.** Meet the redemptions at the lowest possible transaction cost, since redeeming investors are the only group affected by how the fund raises cash.

**B.** Treat redeeming, subscribing and remaining investors fairly, recognizing that peer underperformance can spread by contagion and that stressed markets bring large, correlated redemptions.

**C.** Delay all redemptions until markets normalize, since the investors who remain in the fund take priority over those who redeem.

**D.** Disregard the peer funds' performance, since large outflows are driven only by the fund's own performance and reputation.

### Question IM-16 e.2

New • built from SchweserNotes 2026 Reading 98

A US '40 Act bond fund receives redemption requests that exceed its cash and highly liquid assets. Which of the following describes the next steps under a typical redemption waterfall?

**A.** Draw on short-term borrowing first, up to 50% of total fund assets, and sell less liquid holdings only if borrowing capacity is exhausted.

**B.** Sell the least liquid holdings first, so that the investors who remain keep the fund's most liquid assets.

**C.** Make pro rata or risk-constant sales of less liquid holdings so the portfolio keeps about the same risk profile, then use short-term borrowing as the final layer, within 33.3% of total fund assets.

**D.** Impose a gate limiting redemptions to 10% of fund assets, since '40 Act funds must gate redemptions before selling any assets.

**Question IM-16 e.3** New • built from SchweserNotes 2026 Reading 98

A risk manager at a US bond fund is extending the fund's redemption model with stress testing. Which of the following approaches is most consistent with best practice?

**A.** Apply stress scenarios to fund assets, redemption volumes, ADV assumptions and transaction costs, and consider investor concentration, past redemption events and the effect of any gates.

**B.** Stress only the fund's asset prices, since redemption volumes depend on investors and cannot meaningfully be stressed.

**C.** Replace the redemption model with stress testing, since historical redemption data lose all value once stress scenarios are run.

**D.** Stress transaction costs only, since ADV assumptions and redemption volumes stay stable when markets are stressed.

**Question IM-16 e.4** Bank Q578 • Schweser Module Quiz 98.3 Q2

An analyst is applying the historical redemption-at-risk (HRaR) model to estimate potential redemption demand. Which of the following statements is *least likely* to represent a key challenge of this approach?

**A.** Some funds may lack sufficient redemption history because they are new or have been growing rapidly.

**B.** Limited transparency between asset managers and end investors can reduce the quality of available data.

**C.** Investor withdrawal behavior may be driven by idiosyncratic factors that are difficult to model statistically.

**D.** Historical data provides a stable and reliable indicator of future redemption pressures across all market environments.

**Question IM-16 e.5** New • built from SchweserNotes 2026 Reading 98

A fund's risk team wants to improve its historical redemption-at-risk (HRaR) model, which uses the 99th-percentile five-day redemption over the fund's life. Which of the following statements is most accurate?

**A.** Large outflows are driven only by systemic factors, so a model that uses market returns alone captures redemption risk fully.

**B.** Large outflows usually mix systemic and idiosyncratic factors, so machine learning can help estimate an upper bound on redemptions, though data quality and availability remain obstacles.

**C.** Machine learning removes the need for fund flow data, because it infers redemptions directly from macroeconomic variables.

**D.** The HRaR model is most reliable for new and fast-growing funds, because their recent inflows give a clean record of investor behavior.

**Question IM-16 e.6** Bank Q580 • New, written for the bank (GARP style)

A fund manager builds a liquidity optimization model to meet redemptions. The fund holds a position subject to an IPO lockup, trades several bonds in a sector where simultaneous sales could exhaust market demand, and must keep tracking error below a specified level. Which of the following correctly matches each concern with the model constraint that addresses it?

**A.** Lockup: ADV constraint; sector demand: risk constraint; tracking error: market sector capacity constraint.

**B.** Lockup: risk constraint; sector demand: time-lag constraint; tracking error: ADV constraint.

**C.** Lockup: time-lag constraint; sector demand: market sector capacity constraint; tracking error: risk constraint.

**D.** Lockup: market sector capacity constraint; sector demand: ADV participation constraint only; tracking error: time-lag constraint.

# Solutions • IM-16 e

## Solution IM-16 e.1

**Answer: B**

New • built from SchweserNotes 2026 Reading 98

### Step 1 – The drivers at work

Severe underperformance by peer funds can create contagion, and stressed markets can produce large and highly correlated redemption requests.

### Step 2 – The objective

Manage redemptions so that all three groups, those redeeming, those subscribing and those remaining, are treated fairly and equitably.

$\Rightarrow$ B.

### Interpretation

Fairness is what rules out the easy answers: the cheapest exit for leavers (A) and an indefinite delay that protects stayers (C) each favor one group.

### Why the other options are wrong

**A is incorrect.** Ignores the subscribing and remaining investors, who bear the cost of how cash is raised.

**C is incorrect.** Favors the remaining investors at the expense of the redeeming ones; an indefinite delay is not the objective.

**D is incorrect.** Absolute claim: peer underperformance and macro shocks also drive outflows.

**LO:** IM-16 e: Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.e; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.7

## Solution IM-16 e.2

**Answer: C**

New • built from SchweserNotes 2026 Reading 98

### Step 1 – Layer 2

When requests exceed cash and highly liquid assets, sell less liquid holdings pro rata or risk-constant, so the remaining portfolio keeps approximately the same risk profile.

### Step 2 – Layer 3

Short-term borrowing (reverse repos, overdrafts, lines of credit) is the final option, within the '40 Act limit of 33.3% of total fund assets.

$\Rightarrow$ C.

### Interpretation

The order protects the remaining investors: cash first, then sales that keep the portfolio's shape, and leverage only at the end.

### Why the other options are wrong

**A is incorrect.** Wrong order and wrong limit: borrowing is the last layer, capped at 33.3%.

**B is incorrect.** Selling the least liquid first is not the waterfall's rule; Layer 2 keeps the risk profile constant.

**D is incorrect.** Gates are a UCITS tool and are not permitted for US funds (IM-16 d).

**LO:** IM-16 e: Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.e, redemption waterfall

### Solution IM-16 e.3 — Answer: A

New • built from SchweserNotes 2026 Reading 98

**Step 1 – What to stress**

Fund assets, redemption volumes, ADV assumptions and transaction costs.

**Step 2 – What else to consider**

Investor concentration, relevant historical redemption events and the potential effects of any gates in place.

⇒ A.

**Interpretation**

In a stress, both sides move together: redemptions rise while ADV falls and transaction costs widen, so a stress test that moves only one input understates the problem.

**Why the other options are wrong**

**B is incorrect.** Scope error: redemption volumes are a core stress input.

**C is incorrect.** Stress testing extends redemption modeling; it does not replace it.

**D is incorrect.** Polarity flip: ADV and redemption volumes are exactly what change in stressed markets.

**LO:** IM-16 e: Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.e, stress testing

### Solution IM-16 e.4 — Answer: D

Bank Q578 • Schweser Module Quiz 98.3 Q2

**Step 1 – What HRaR does**

Estimates worst-case outflows from historical fund flows, such as the 99th-percentile, five-day redemption over the life of the fund.

**Step 2 – The challenges the reading lists**

Short or benign histories (new or fast-growing funds); idiosyncratic investor behavior; poor visibility into end investors when a custodian or other third party intervenes. A, B and C restate them.

⇒ D, which claims history is reliable everywhere, is not a challenge but a false claim.

**Interpretation**

"Least likely" questions reward reading the list of limitations closely: three options will be true limitations, and the odd one out overstates the model.

**Why the other options are wrong**

**A is incorrect.** A stated challenge: no meaningful outflow history.

**B is incorrect.** A stated challenge: intermediaries hide who is redeeming.

**C is incorrect.** A stated challenge: idiosyncratic behavior is hard to model.

**LO:** IM-16 e: Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.e, Modeling Redemption-at-Risk

### Solution IM-16 e.5 Answer: B

New • built from SchweserNotes 2026 Reading 98

**Step 1 – What drives large outflows**

Usually a combination of systemic factors (sector-wide disruptions, macroeconomic shocks) and idiosyncratic factors (investor behavior, peer performance, cash needs, event-driven risks).

**Step 2 – What machine learning adds**

It can handle many factors and nonlinear relationships and estimate an upper bound on likely redemptions over a horizon. Data availability and quality remain significant obstacles.
$\Rightarrow$ B.

**Interpretation**

Machine learning widens the set of drivers the model can use; it does not cure the HRaR's data problems.

**Why the other options are wrong**

**A is incorrect.** Absolute claim: idiosyncratic factors matter too.
**C is incorrect.** Machine learning still needs flow data; data availability is its main obstacle.
**D is incorrect.** Polarity flip: new and fast-growing funds are where HRaR is weakest, because they lack meaningful outflows.

**LO:** IM-16 e: Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.e, Modeling Redemption-at-Risk

### Solution IM-16 e.6 Answer: C

Bank Q580 • New, written for the bank

**Step 1 – Lockup**

A fixed time restriction is handled by a time-lag constraint of $d_i$ days.

**Step 2 – Sector demand**

Large simultaneous sales of similar bonds that could exhaust the market's absorption are handled by the sector capacity constraint $c_{j,t}$.

**Step 3 – Tracking error**

Handled by the risk constraint with threshold $R$.
$\Rightarrow$ C.

**Interpretation**

The ADV constraint limits sales of each security; the sector capacity constraint limits the sector as a whole. Both can bind at once in a stressed sector.

**Why the other options are wrong**

**A is incorrect.** The ADV constraint caps daily sales, not lockups, and the risk constraint does not address absorption.
**B is incorrect.** Puts the lockup in the risk constraint and tracking error in the ADV constraint.
**D is incorrect.** ADV limits alone ignore sector absorption, and a time lag does not control tracking error.

**LO:** IM-16 e: Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests. • **Reference:** SchweserNotes 2026 Book 5, Reading 98, LO 98.e, Liquidity Optimization to Meet Redemption Requests

IM-17 • Illiquid Assets • Vardeez: VERY IMPORTANT

# IM-17 e

P2 CONCEPT

Compare illiquidity risk premiums across and within asset categories.

Workbook: D9 Illiquid Assets • Tier D • Map #48 • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** Vardeez flags LO 99.e as Important inside a VERY IMPORTANT reading, and the Tier D workbook section D9 (Map #48, 2 of 16 recall sheets) carries the shape.
**Vardeez:** no pointer beyond the Important flag.

**Reported exam shape**

- **D9·S2** Illiquidity risk premiums across versus within asset classes, and the four ways to harvest them. Workbook D9.2 (worked as Question IM-17 f.6) tests it together with shape S3.

**What the questions look like:** statement selection. Which way round is it, across or within? Which reason undermines the across-class chart? What is the evidence in Treasuries, corporate bonds, equities and secondary markets? Which of the four harvesting routes is easiest and has the greatest impact? No calculations.

## 1. Across asset classes: the conventional view

Ilmanen's chart (*Expected Returns*, 2011) plots average returns from 1990 to 2009 against his estimate of each asset class's illiquidity (Figure 1). It suggests a positive link: venture capital, the least liquid, earned about 16% to 17%; buyouts about 13%; timber close to 12%; hedge funds a little over 12%; real estate, about as liquid as hedge funds, nearly 8%; equities a bit over 4%; and cash, the most liquid, a little over 4%. Two caveats come with it: the illiquidity positions are Ilmanen's own estimates, and the classes overlap (some private equity funds are more liquid than some hedge funds or infrastructure investments).

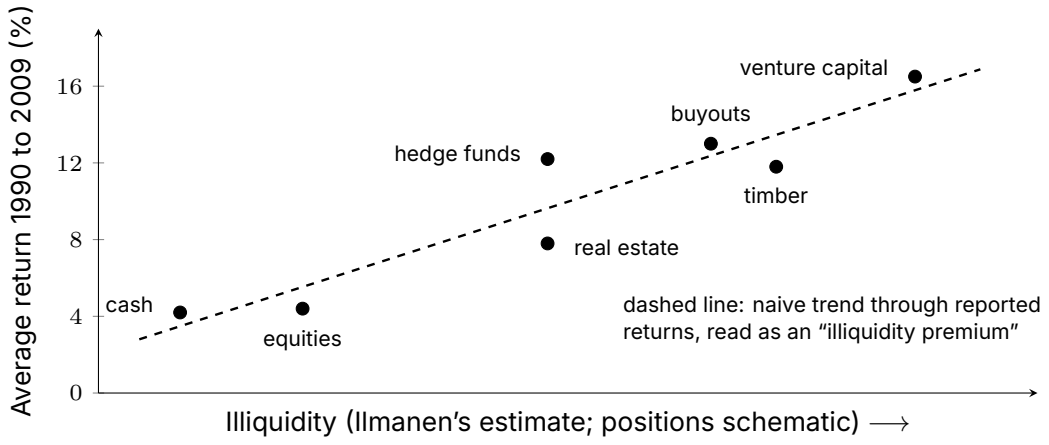



Figure 1. The across-class picture, redrawn from the returns Schweser quotes for Ilmanen's chart. The vertical positions are reported returns; the horizontal positions only rank illiquidity. The upward slope is what Ang argues is not a reliable premium (Section 2).

## 2. Why the across-class premium is doubtful

Ang's verdict: "there do not seem to be significant illiquidity risk premiums across classes". Four reasons:

1. **Illiquidity biases.** Reported returns of illiquid classes are too high, and their risks and correlations too low (survivorship, selection, infrequent trading: IM-17 c).
2. **Other risks.** Private equity, buyouts and timber carry significant risks beyond illiquidity; after adjusting for them they are much less attractive. One study finds most investors would have done better in the S&P 500 than in a private equity portfolio.
3. **No market index.** Private equity, hedge fund and real estate indexes are not investable, so nobody earns the index return. NCREIF holds thousands of properties; an investor holding a few faces large idiosyncratic risk.
4. **Manager skill cannot be separated from factor risk.** With no cheap, tradeable index fund, there is no way to split the passive return from the alpha of the manager.

## 3. Within asset classes: large premiums

Holding the asset class fixed holds most other risks roughly constant, so the comparison isolates illiquidity. Here the evidence is strong:

| Market | Evidence |
|---|---|
| **US Treasuries** | Newly issued on-the-run bills are more liquid and yield *less* than seasoned off-the-run bills (the on-the-run/off-the-run spread). In 2007 to 2009 T-bonds traded more than 5% cheaper than T-notes of the same maturity, in one of the most liquid markets in the world. |
| **Corporate bonds** | Wider bid-ask spreads and infrequent trading mean higher yields. Illiquidity explains 7% of the variation in investment-grade yields and 22% in junk yields; a 1 basis point rise in the bid-ask spread goes with a rise of more than 2 basis points in the yield spread. |
| **Equities** | Less liquid stocks earn more. Measures: bid-ask spreads, volume, turnover, volume signed by buyer or seller initiation, the Amihud ratio (absolute return to dollar volume), price impact, informed-trading (adverse selection) measures, quote size and depth, trade frequency, the number of *zero* returns, and return autocorrelations (stale prices). There are also illiquidity risk betas. Premiums of 1% to 8% depending on the measure, now diminished; about 1% for listed stocks against about 20% for OTC stocks. |
| **Hedge funds, PE, real estate** | Ang finds within-class premiums here too: hedge funds with lock-ups and withdrawal restrictions earn more than those without. |

**Why within but not across?** There is no formal theory. Investors may overpay for illiquid asset *classes*, chasing the illusion of high returns; institutions manage asset classes in silos, so capital moves slowly between them; and limits to arbitrage and institutional constraints keep across-class mispricing alive.

## 4. Secondary markets for private equity and hedge funds

- **Secondary buyouts.** PE funds sell portfolio companies to each other (about 15% of buyout deals in 2005). This lets a fund exit a *deal*, may return some cash to LPs and helps value portfolio companies, but it does not let an LP exit the *fund*.
- **LP stakes.** The market is immature, small and opaque. In the 1990s buyers were called "vultures" and bought at discounts of 30% to 50%; discounts fell below 20% in the early 2000s and widened again in 2007 to 2009. Harvard, whose endowment fell more than USD 8 billion (22%) from July to October 2008, faced discounts of about 50% when it tried to sell private equity stakes.

- **Hedge fund stakes.** Investors can redeem at set dates, so secondary discounts are much smaller: 6% to 8% on average in the crisis, with some closed funds trading at a premium.
- **Who can harvest it.** Large owners such as sovereign wealth funds and pension funds can supply liquidity, buying stakes at discounts.

## 5. Four ways to harvest illiquidity premiums

| Route | How |
|---|---|
| **1. Passive allocation to illiquid asset classes** | Hold a slice of, say, real estate. Exposed to every doubt in Section 2. |
| **2. Liquidity security selection** | Choose the less liquid assets within an asset class: the within-class premium. |
| **3. Market making** | Buy at a discount from sellers who need to trade quickly and sell at a premium to those who demand the asset. Dimensional Fund Advisors does this in small caps, avoiding adverse selection by dealing with counterparties who disclose information; large asset owners can do it too. |
| **4. Dynamic factor strategies at the aggregate portfolio level** | Long illiquid, short liquid, and rebalance countercyclically: buy when others want to sell, sell when others want to buy. **The easiest to implement, with the greatest effect on portfolio returns.** |

**Correction to the source notes.** Vardeez lists three reasons the across-class premium may not be real (illiquidity biases, additional risks, no market index). Schweser, and Ang as summarised in Workbook D9, give a fourth: investors must rely on manager skill, because without investable indexes factor risk cannot be separated from the manager's talent. Learn all four.

**Comparison card: IM-17 e**

| | Across asset classes | Within asset classes |
|---|---|---|
| **Raw evidence** | Ilmanen's chart: VC about 16% to 17%, cash and equities about 4% | Off-the-run over on-the-run Treasuries; illiquid corporates and stocks; HF lock-ups |
| **Verdict (Ang)** | No significant premium once data are corrected | Large premiums |
| **Why** | Biases, other risks, no investable index, skill not separable from factor risk | Same class holds other risks roughly fixed |
| **Possible explanation** | Investors overpay for illiquid classes; silos | Limits to arbitrage keep the gap open |
| **Harvest by** | Passive allocation (route 1) | Security selection (2), market making (3), dynamic rebalancing (4: easiest, greatest impact) |

**Exam traps**

- **Across versus within (polarity flip).** Premiums are large *within* classes and doubtful *across* them, not the reverse.
- **Raw chart read as a premium.** The upward slope uses reported, biased returns of non-investable indexes.
- **On-the-run polarity.** On-the-run Treasuries are the more liquid ones and yield less.

- **Number swaps.** Illiquidity explains 7% (investment grade) and 22% (junk) of yield variation; 1 bp of bid-ask goes with more than 2 bp of yield spread; equities 1% (listed) against about 20% (OTC).
- **Zero, not positive.** The illiquidity measure is the number of zero returns.
- **Deal versus fund.** Secondary buyouts let a fund exit a deal, not an LP exit the fund.
- **Ranking of the four routes.** Dynamic factor strategies (countercyclical rebalancing), not passive allocation, are the easiest with the greatest impact.
- **Discount sizes.** Hedge fund stakes trade at much smaller discounts than PE stakes, because investors can redeem.

## Practice questions: IM-17 e

### Question IM-17 e.1

Bank Q614 • Schweser Module Quiz 99.2 Q1

Which of the following variables is NOT an illiquidity factor that affects equity returns?

**A.** Measures of adverse selection (informed trading)

**B.** The number of recorded positive returns

**C.** Turnover

**D.** Volume

### Question IM-17 e.2

Bank Q611 • BT Mock Q64 • BT P2.T8.708.1 (rebuilt)

Antti Ilmanen's scatterplot (as cited by Andrew Ang) plots long-run average return against an index of illiquidity across asset classes, 1990 to 2009. Venture capital is the least liquid and has the highest average return (about 16% to 17%), while cash and public equities are the most liquid, with returns of roughly 4%. In regard to the illiquidity risk premium, which of the following statements is TRUE?

**A.** In general, illiquid asset classes offer high risk-adjusted returns.

**B.** The scatterplot demonstrates that large illiquidity risk premiums exist ACROSS asset classes.

**C.** Large illiquidity risk premiums exist WITHIN many asset classes, but are not demonstrated ACROSS asset classes.

**D.** Illiquid equities earn the same returns as liquid equities, and illiquid bonds earn the same returns as liquid bonds.

**Question IM-17 e.3** New • built from SchweserNotes 2026 Reading 99

The CIO of a Gulf sovereign wealth fund shows the board Ilmanen's chart, on which venture capital, the least liquid asset class, earned about 16% to 17% a year from 1990 to 2009, against a little over 4% for cash and public equities. The fund's head of risk is asked to explain why Ang doubts that the chart shows an illiquidity risk premium across asset classes. Which of the following points is correct for the head of risk to make?

**A.** The illiquid asset class indexes on the chart, such as NCREIF, are investable, so the chart understates what an investor could have earned from them.

**B.** Reported returns of the illiquid classes are overstated and their risks understated by data biases, and these classes carry significant risks other than illiquidity.

**C.** Within asset classes, less liquid securities earn lower returns than more liquid ones, so illiquidity cannot command a premium anywhere.

**D.** Venture capital returns can be split cleanly into factor returns and manager alpha, and the factor part alone explains the whole return gap.

**Question IM-17 e.4** New • built from SchweserNotes 2026 Reading 99

A fixed-income strategist at a German asset manager is preparing a note on illiquidity premiums within the bond and equity markets. Which of the following statements is correct for the note?

**A.** Newly issued on-the-run Treasury bills are less liquid than seasoned off-the-run bills and therefore trade at higher yields.

**B.** Illiquidity explains a larger share of the variation in investment-grade corporate bond yields than in high-yield bond yields.

**C.** A 1 basis point increase in a corporate bond's bid-ask spread is associated with an increase of more than 2 basis points in its yield spread.

**D.** Estimated illiquidity premiums in equities have risen over time and are now similar for listed stocks and OTC stocks.

**Question IM-17 e.5** Bank Q613 • GARP PE1 Q24 • Mock 4 Q28

The chief investment officer (CIO) of a large university endowment fund is considering whether to add illiquid assets to the university's investment portfolio. The CIO asks an investment analyst to review illiquidity risk premiums across and within asset categories and to prepare a report. Which of the following statements is correct for the analyst to include in the report?

**A.** Corporate bonds that trade less frequently or have larger bid-ask spreads have lower returns than more liquid corporate bonds.

**B.** Expected returns of illiquid assets can be overstated due to measurement biases.

**C.** US Treasury instruments are the only assets that do not exhibit an illiquidity risk premium.

**D.** Hedge funds that do not place restrictions on withdrawals exhibit higher returns.

**Question IM-17 e.6** New • built from SchweserNotes 2026 Reading 99

An investment officer at an Australian superannuation fund is considering supplying liquidity in the secondary markets for private equity and hedge fund stakes. Which of the following statements about these markets is correct?

**A.** Secondary buyouts, in which private equity funds sell portfolio companies to each other, allow limited partners to exit the private equity fund itself.

**B.** Because hedge fund investors can usually redeem at set dates, secondary discounts on hedge fund stakes have been much smaller than on private equity stakes.

**C.** Discounts on private equity secondaries have stayed close to 20% through the cycle, including during the 2007 to 2009 financial crisis.

**D.** Buying stakes from distressed sellers at a discount is a passive allocation to illiquid asset classes, the route Ang ranks as having the greatest impact.

**Question IM-17 e.7** Bank Q615 • Schweser Module Quiz 99.2 Q2

Rick Fairview, a general partner and portfolio manager with Fairview Funds, is considering ways to profit from illiquidity risk premiums. Which of the following strategies is most likely to produce the greatest impact on portfolio performance?

**A.** Acting as a market maker for individual securities.

**B.** Choosing the most illiquid assets within an asset class, even if the asset class is generally considered liquid.

**C.** Allocating a portion of the portfolio to illiquid asset classes.

**D.** Using dynamic factor strategies at the aggregate portfolio level.

# Solutions • IM-17 e

## Solution IM-17 e.1

**Answer: B**

Bank Q614 • Schweser Module Quiz 99.2 Q1

### Step 1 – Recall the equity illiquidity measures

Bid-ask spreads, volume, turnover, signed volume, the Amihud ratio, price impact, informed-trading (adverse selection) measures, quote size and depth, trade frequency, the number of zero returns and return autocorrelations.

### Step 2 – Find the impostor

Adverse selection, turnover and volume are on the list. The return-count measure is the number of *zero* returns: an illiquid stock often does not trade, so its price does not change.
⇒ "Positive returns" carry no information about liquidity.

### Interpretation

Zero returns and return autocorrelations both measure stale prices, the same symptom that smoothing produces in appraisal-based indexes (IM-17 d).

### Why the other options are wrong

**A is incorrect.** Informed trading (adverse selection) is a recognised liquidity cost and measure.
**C is incorrect.** Low turnover signals illiquidity.
**D is incorrect.** Trading volume is a core liquidity measure.

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. • **Reference:** Ang, Asset Management (2014), Ch.13, Illiquidity Risk Premiums Within Asset Classes (equities); SchweserNotes 2026 Book 5, Reading 99, LO 99.e

## Solution IM-17 e.2

**Answer: C**

Bank Q611 • BT Mock Q64 • BT P2.T8.708.1

### Step 1 – Across classes

The raw chart slopes up, but it uses biased reported returns of non-investable indexes, ignores other risks and cannot separate manager skill from factor returns.
⇒ Once these are allowed for, a significant across-class premium is not demonstrated.

### Step 2 – Within classes

Off-the-run Treasuries, illiquid corporate bonds, low-liquidity stocks and hedge funds with lock-ups all earn more than their liquid counterparts.
⇒ Large premiums within classes.

### Interpretation

Within-class comparisons hold most other risks roughly fixed, so they isolate illiquidity; the across-class chart mixes illiquidity with everything else that differs between asset classes.

### Why the other options are wrong

**A is incorrect.** After correcting for biases and other risks, illiquid classes look much less attractive; studies show no higher risk-adjusted returns.
**B is incorrect.** Reads the unadjusted chart at face value.
**D is incorrect.** Denies the within-class premium, which is where the evidence is strongest.

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. • **Reference:** Ang, Asset Management (2014), Ch.13, Illiquidity Risk Premiums; SchweserNotes 2026 Book 5, Reading 99, LO 99.e

### Solution IM-17 e.3 — Answer: B

New • built from SchweserNotes 2026 Reading 99

**Step 1 – List Ang's four reasons**

Illiquidity biases (returns overstated, risk understated); other risks beyond illiquidity; no investable market index; manager skill cannot be separated from factor risk.

**Step 2 – Match the options**

B states the first two reasons.

⇒ B is correct.

**Interpretation**

After adjusting for risk, one study finds most investors would have done better in the S&P 500 than in a private equity portfolio. The across-class slope is mostly bias and other risk, not an illiquidity premium.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: the indexes are *not* investable, so no investor earns the index return; that is one of Ang's reasons for doubt.

**C is incorrect.** Polarity flip: within asset classes less liquid securities earn *higher* returns; that is where the premium is large.

**D is incorrect.** The fourth reason says the opposite: without tradeable indexes, factor returns cannot be separated from manager alpha.

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. • **Reference:** Ang, Asset Management (2014), Ch.13, Illiquidity Risk Premiums Across Asset Classes; SchweserNotes 2026 Book 5, Reading 99, LO 99.e

### Solution IM-17 e.4 — Answer: C

New • built from SchweserNotes 2026 Reading 99

**Step 1 – Treasuries**

On-the-run bills are the more liquid ones and have *lower* yields than off-the-run bills.

⇒ A is reversed.

**Step 2 – Corporate bonds**

Illiquidity explains 7% of the variation in investment-grade yields and 22% in junk yields, so B is reversed. A 1 bp rise in the bid-ask spread goes with a rise of more than 2 bp in the yield spread.

⇒ C is correct.

**Step 3 – Equities**

Equity illiquidity premiums have diminished over time, and they differ sharply: about 1% for listed stocks against about 20% for OTC stocks.

⇒ D is wrong on both counts.

**Interpretation**

Within-class premiums show up even in the most liquid market in the world: during 2007 to 2009 T-bonds traded more than 5% cheaper than T-notes of the same maturity.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: on-the-run bills are more liquid and yield less.

**B is incorrect.** Number swap: illiquidity matters more for junk bonds (22%) than for investment grade (7%).

**D is incorrect.** Premiums have declined, and OTC stocks carry a much larger premium than listed stocks.

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. • **Reference:** Ang, Asset Man-

agement (2014), Ch.13, Illiquidity Risk Premiums Within Asset Classes; SchweserNotes 2026 Book 5, Reading 99, LO 99.e

### Solution IM-17 e.5 — Answer: B

Bank Q613 • GARP PE1 Q24

#### Step 1 – Test B

Survivorship, selection and infrequent trading biases make reported returns of illiquid assets too high and their risks too low.
⇒ B is correct, and it is the first reason the across-class premium is doubtful.

#### Step 2 – Test the within-class statements

Less liquid corporate bonds have higher yields and returns (A reversed). Treasuries show an on-the-run/off-the-run spread, so they do have illiquidity effects (C false). Ang reports that hedge funds *with* lock-ups and withdrawal restrictions earn higher returns (D reversed).

#### Interpretation

GARP mixes one across-class point (B) with three within-class points, each with its direction flipped or overstated.

#### Why the other options are wrong

**A is incorrect.** Polarity flip: less liquid bonds earn more.
**C is incorrect.** Absolute claim: even the Treasury market shows illiquidity premiums.
**D is incorrect.** Polarity flip: it is the funds that restrict withdrawals that earn more.

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. • **Reference:** Ang, Asset Management (2014), Ch.13, Illiquidity Risk Premiums; SchweserNotes 2026 Book 5, Reading 99, LO 99.c and 99.e; GARP Practice Exam 1, Q24

### Solution IM-17 e.6 — Answer: B

New • built from SchweserNotes 2026 Reading 99

#### Step 1 – Secondary buyouts

PE funds trade portfolio companies with each other (about 15% of buyout deals in 2005). That lets a fund exit a deal, not an LP exit the fund.
⇒ A is wrong.

#### Step 2 – LP stakes in PE funds

Discounts were 30% to 50% in the 1990s, fell below 20% in the early 2000s and widened again in the crisis; Harvard faced about 50% in 2008.
⇒ C is wrong.

#### Step 3 – Hedge fund stakes

Investors can redeem at set dates, so secondary discounts averaged 6% to 8% in the crisis, and some closed funds traded at a premium.
⇒ B is correct.

#### Interpretation

Buying stakes at a discount from forced sellers is market making (route 3). Large owners such as sovereign wealth funds and pension funds are natural suppliers of this liquidity.

#### Why the other options are wrong

**A is incorrect.** Scope error: deal, not fund.
**C is incorrect.** Discounts were far from stable; they widened sharply in the crisis.
**D is incorrect.** Sibling swap: buying from distressed sellers is market making, not passive allo-

cation; and the route with the greatest impact is dynamic factor strategies.

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. • **Reference:** Ang, Asset Management (2014), Ch.13, Secondary Markets for Private Equity and Hedge Funds; SchweserNotes 2026 Book 5, Reading 99, LO 99.e

### Solution IM-17 e.7

**Answer: D**

Bank Q615 • Schweser Module Quiz 99.2 Q2

#### Step 1 – Name the four routes

Passive allocation to illiquid asset classes (C), liquidity security selection (B), market making (A), and dynamic factor strategies at the aggregate portfolio level (D).

#### Step 2 – Apply Ang's ranking

Dynamic factor strategies (long illiquid, short liquid, rebalanced countercyclically) are "the easiest to implement and can have the greatest effect on portfolio returns".
$\Rightarrow$ D.

#### Interpretation

Rebalancing is the simplest way to provide liquidity: as long as the investor buys when others want to sell and sells when others want to buy, the strategy is countercyclical and earns the liquidity-provision premium.

#### Why the other options are wrong

**A is incorrect.** Market making is a valid route but needs scale and careful choice of counterparties; it is not ranked highest.
**B is incorrect.** Security selection earns the within-class premium but has less overall impact.
**C is incorrect.** Passive allocation is static and relies on the across-class premium Ang doubts; "simple" does not mean "passive".

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. • **Reference:** Ang, Asset Management (2014), Ch.13, Harvesting Illiquidity Premiums; SchweserNotes 2026 Book 5, Reading 99, LO 99.e

IM-17 • Illiquid Assets • Vardeez: VERY IMPORTANT

# IM-17 f

P2 CONCEPT

Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets.

Workbook: D9 Illiquid Assets • Tier D • Map #48 • Vardeez focus

**WHAT IS TESTED HERE**

**Why P2.** Vardeez flags LO 99.f as Important inside a VERY IMPORTANT reading, and the Tier D workbook section D9 (Map #48, 2 of 16 recall sheets) carries the shape.
**Vardeez:** no pointer beyond the Important flag.

**Reported exam shape**

- **D9·S3** Portfolio allocation with illiquid assets: optimal holdings, hurdle rates, rebalancing point. Workbook D9.2 (Question IM-17 f.6) also carries shape S2 from IM-17 e.

**What the questions look like:** statements on the direction of the Ang, Papanikolaou and Westerfield results (rarer liquidity events, lower holdings, higher hurdle rates), where to rebalance, why transaction-cost models fall short, and the cautions on illiquid assets; plus case items (an endowment with a near-term cash need, Harvard in 2008). The model's table values can be quoted in an option, but nothing is calculated.

## 1. Two features of illiquidity that portfolio models must handle

Personal circumstances drive the allocation (horizon, the lack of tradeable indexes, the need to hire and monitor talented managers), but portfolio choice models with illiquid assets must capture two features:

1. **long horizons between trades** (infrequent trading), and
2. **large transaction costs**.

The point of departure is simple: one cannot "eat" illiquid assets. Only liquid assets can be consumed, so illiquidity changes the whole portfolio decision.

## 2. Transaction-cost models, and what they miss

With proportional transaction costs (Constantinides), the investor does not rebalance continuously. There is a **no-trade region** around the target weight, and the investor trades only when the holding drifts outside it. The region can be wide: with 10% costs and 35% volatility, roughly private equity, the target is 25% and the investor trades only outside a band of 0% to 50% (Ang, as summarised in Workbook D9).

**What they miss.** These models assume an asset can always be traded if the investor pays the cost. In private equity, infrastructure, real estate and timber that is not true: a buyer may not be found quickly or at all, a counterparty must do due diligence, which takes time, and may walk away at the end of it. In stress, even liquid markets freeze and buyers disappear at any price.

## 3. Infrequent trading: the Ang, Papanikolaou and Westerfield model

The illiquid asset can be traded only when a **liquidity event** arrives, at a Poisson rate $\lambda$; the average interval between events is $1/\lambda$ (ten years: $\lambda = 1/10$).

| Average time between liquidity events | continuous | 6 months | 1 year | 2 years | 5 years | 10 years |
|---|---|---|---|---|---|---|
| Optimal holding of the risky illiquid asset | 59% | 44% | 37% | 24% | 11% | 5% |
| Required illiquidity premium (hurdle rate) | 0 | 0.7% | 0.9% | 2.0% | 4.3% | 6.0% |

**Consequences for the investor.**

- **Lower optimal holdings.** The rarer the liquidity event, the smaller the allocation: 59% with continuous trading, 5% if the asset can be sold once a decade on average.
- **Higher hurdle rates.** The investor must demand an illiquidity premium, larger the less often the asset trades (6.0% at ten years, a certainty-equivalent hurdle rate). Schweser cites a study putting the private equity premium needed for illiquidity at about 6% over public markets. Agency conflicts, capital calls and asset-liability mismatches push true hurdle rates higher still, and most illiquid assets do not clear them.
- **Allocations drift between trades.** The investor must wait for a liquidity event, so the holding can wander from too high to too low relative to the optimum (Figure 3).
- **Rebalance below the long-run average.** Illiquid wealth is right-skewed between events, so the optimal trading point is *lower* than the long-run average holding.
- **Consume less.** The investor cannot hedge falling values of an asset that cannot be sold, so consumes less than an investor holding only liquid assets.
- **No illiquidity arbitrage.** An arbitrage needs continuous trading, so the model never produces infinite long and short positions.
- **More risk-averse toward everything.** Illiquidity induces time-varying, endogenous risk aversion toward both liquid and illiquid assets.

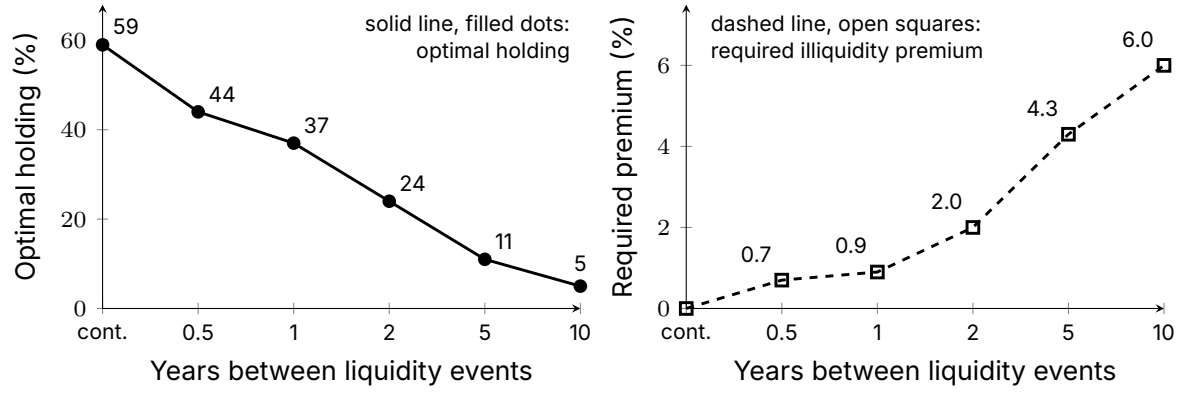



Figure 2. Ang, Papanikolaou and Westerfield (as reported in Ang, Ch.13, and Workbook D9). As liquidity events become rarer, the optimal holding falls (left, solid) and the premium needed to hold the asset rises (right, dashed). "cont." is continuous trading.

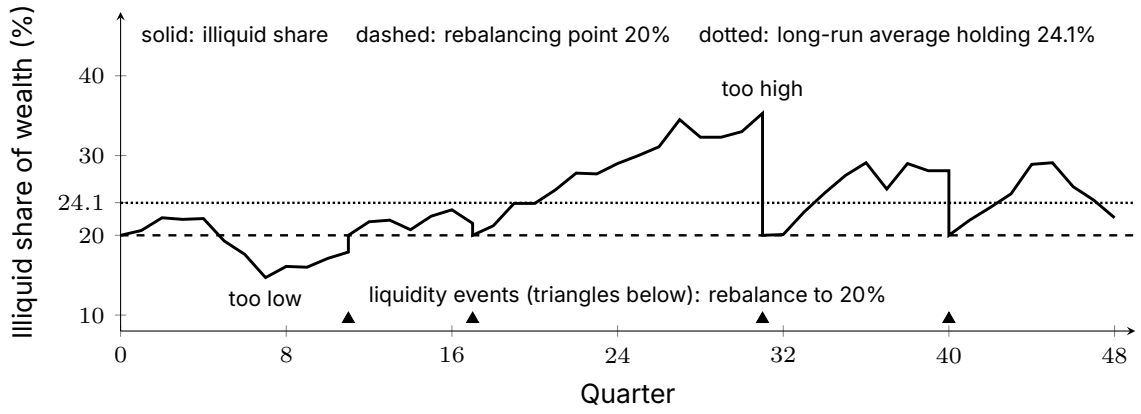



Figure 3. A simulated illiquid holding (schematic). Between liquidity events the share drifts with relative returns, from too low (quarter 7) to too high (quarter 31); at each event it is reset to the rebalancing point. Because the illiquid share drifts upward and is right-skewed, the long-run average holding (24.1%) sits above the rebalancing point (20%).

## 4. Liquidating Harvard

Harvard Management Company held about 55% in hedge funds, private equity and real assets. From July to October 2008 the endowment fell 22% (more than USD 8 billion). The university relies on it for operating funds; the money and commercial paper markets it would normally use had frozen, and buyers of a USD 1.5 billion private equity stake asked discounts of about 50%. Harvard borrowed instead, issuing USD 2.5 billion of bonds. Its "worst failing" was asset-liability management: its liabilities had a shorter duration than its assets.

## 5. Should illiquid assets be in the portfolio?

**Points against (the cautions).**

1. Studies show illiquid assets do not deliver higher risk-adjusted returns.
2. Agency problems: investors depend on the skill of external managers who are hard to monitor.
3. In many institutions illiquid assets are managed separately from the rest of the portfolio.
4. High idiosyncratic risk: there is no "market" portfolio of illiquid assets, and few investors can hold thousands of properties.

**Why they still attract skilled investors.** Illiquid markets are less efficient than stock and bond markets; information asymmetries are large; high transaction costs keep many investors out; and manager skill matters, with alpha widely dispersed. Endowments such as Harvard, Yale and Stanford have the skill and resources to find, evaluate and monitor these opportunities; less skilled investors can lose big. Size and patience are necessary, but not sufficient: the case rests on the word "skilled".

**Correction to the source notes.**

- Vardeez's "six cautions" list "illiquid asset markets are less efficient, with information asymmetries and high transaction costs deterring many investors" as a caution, and leave out Schweser's fourth caution, high idiosyncratic risk (no market portfolio of illiquid assets). In Schweser, inefficiency, information asymmetry, high transaction costs and dispersed alpha are the reasons illiquid assets are *compelling* for a skilled investor: opportunities, not cautions.
- Vardeez's "the optimal allocation to illiquid assets should be lower than for liquid ones" is loose. The

result is that the optimal holding falls the less frequently liquidity events are expected, relative to the same asset traded continuously (59% falling to 5%).

**Comparison card: IM-17 f**

| | Asset traded continuously | Asset traded only at liquidity events |
|---|---|---|
| **Optimal holding** | Mean-variance weight (59% in the model) | Lower, and lower still as events get rarer (5% at ten years) |
| **Required premium** | None for illiquidity | Rises with the wait (0.7% at six months to 6.0% at ten years) |
| **Rebalancing** | Continuous, or at the edges of a no-trade band if trading is costly | Only when an event arrives; reset below the long-run average holding |
| **Hedging and consumption** | Can sell to hedge | Cannot hedge; consumes less |
| **Arbitrage** | Possible | None: no infinite positions |
| **Risk aversion** | Constant | Higher, toward liquid and illiquid assets alike |
| **Benchmark and skill** | Cheap index separates beta from alpha | No investable index; manager skill decisive |

**Exam traps**

- **Direction (polarity flip).** Rarer liquidity events mean a *lower* optimal holding and a *higher* hurdle rate.
- **Rebalancing point.** Rebalance to a level *below* the long-run average holding, not above it.
- **"Pay the cost and you can always trade."** That is the assumption transaction-cost models make and illiquid markets break.
- **Consumption.** Illiquid-asset investors consume *less*, because they cannot hedge.
- **No illiquidity arbitrage.** Arbitrage needs continuous trading.
- **Risk aversion rises toward both** liquid and illiquid assets, not only the illiquid one.
- **Skill.** There is no tradeable index to separate factor risk from alpha, so selecting skilled managers is critical; size and patience alone are not enough.
- **Reported diversification.** Low reported correlations of illiquid assets are partly smoothing (IM-17 c); and a known near-term cash need argues against illiquid assets whatever their diversification.

## Practice questions: IM-17 f

### Question IM-17 f.1

Bank Q616 • AnalystPrep Q.2426

Jude Wang, a portfolio manager, has been asked to manage a university endowment fund. The head of the university informs her that the university plans to withdraw a substantial amount of the fund within the next two years to establish a financial laboratory, and that the board wants transaction costs kept as low as possible. A colleague, Jane Jones, suggests including private equity and real estate in the fund because these asset classes have low correlation with stocks and bonds, offer better diversification and would make it easier to generate excess return. Which of the following does NOT support the idea of including private equity and real estate in the portfolio?

**A.** It would provide a better diversification effect to the portfolio.

**B.** It would help Wang to generate an excess return.

**C.** The university needs a substantial amount of funds within the next two years for the laboratory.

**D.** It would improve the portfolio because of the assets' low correlation with stocks and bonds.

### Question IM-17 f.2

New • built from SchweserNotes 2026 Reading 99

A quantitative strategist at a Swiss foundation extends the foundation's asset allocation model to include private real estate by adding proportional transaction costs, in the spirit of Constantinides. A member of the investment committee asks what this change does and does not capture. Which of the following statements is correct?

**A.** With transaction costs, the model rebalances continuously to the target weight, but the target weight itself falls in proportion to the cost.

**B.** With transaction costs, the model leaves a wide no-trade region around the target weight and trades only when the holding moves outside it.

**C.** Because the model lets the foundation trade whenever it pays the cost, it fully captures the risk that no buyer can be found for a property in a crisis.

**D.** Adding transaction costs raises the optimal real estate allocation, because a property held for many years spreads its transaction costs thinly.

**Question IM-17 f.3** New • built from SchweserNotes 2026 Reading 99

A risk analyst at a US public pension plan explains to the trustees how an investor should manage an illiquid asset that can be traded only when a liquidity event arrives, as in the model of Ang, Papanikolaou and Westerfield. Which of the following statements is correct?

**A.** Between liquidity events the illiquid allocation drifts away from the optimum, and when an event arrives the investor should rebalance to a holding below the long-run average holding.

**B.** Because the illiquid asset cannot be sold, the investor can afford to consume more each period than an investor who holds only liquid assets.

**C.** If two illiquid assets are mispriced relative to each other, the investor should take very large offsetting long and short positions to capture the arbitrage.

**D.** Illiquidity makes the investor more risk-averse toward the illiquid asset but more risk-tolerant toward liquid assets, which are used to raise overall risk.

**Question IM-17 f.4** Bank Q617 • BT P2.T8.708.3

You are consulting to a large endowment fund that is determining its asset allocation budget, including the target allocation to, and hurdle rates for, illiquid assets. Four staff members make the assertions below. According to Ang, each of these is true EXCEPT which is probably false?

**A.** Phillip says: "The longer the time between liquidity events for an asset (i.e., the less liquid the asset), the LOWER its optimal allocation in the portfolio."

**B.** Debra says: "The longer we need to wait to exit an investment (i.e., the later the arrival of liquidity events), the HIGHER should be our illiquidity hurdle rate."

**C.** Peter says: "Because of the nature of factor risk and idiosyncratic risk in illiquid markets, it is really important to assign (or delegate to) genuinely skilled investors for the illiquid allocation."

**D.** Mary says: "It is not very important to identify skill in the illiquid asset class because we can easily benchmark against tradeable indexes, which lets us separate factor risk from manager skill (alpha)."

**Question IM-17 f.5** New • built from SchweserNotes 2026 Reading 99

A trustee of an Indian university endowment cites Harvard's experience in 2008 while the investment committee debates raising private equity and real assets to half of the portfolio. Which of the following lessons is consistent with Ang's analysis?

**A.** Harvard's losses show that every endowment should avoid illiquid assets, because studies find their risk-adjusted returns are negative for all types of investor.

**B.** Harvard's main failing was asset-liability management: its liabilities were shorter in duration than its assets, and illiquid assets cannot be consumed.

**C.** Harvard's problem arose because liquid assets were too large a share of its portfolio, so selling them would have depressed their prices.

**D.** A large endowment with a long horizon can rely on its size and patience alone to earn the illiquidity premium, so manager skill is secondary.

**Question IM-17 f.6** Workbook D9.2 • S2, S3 • Map #48

The CIO of a sovereign wealth fund proposes raising private equity from 10% to 35% of the portfolio, arguing that the fund's size and long horizon let it harvest a large illiquidity premium. A risk consultant is asked to reply using Ang's analysis of illiquid assets. Which of the following statements should the consultant include?

**A.** After correcting for data biases, illiquidity risk premiums are large across asset classes, while within asset classes the evidence for higher returns on less liquid securities is weak.

**B.** Of the four ways to harvest illiquidity premiums, a passive allocation to illiquid asset classes is the simplest to implement and has the greatest impact on portfolio returns.

**C.** In the Ang, Papanikolaou and Westerfield model, if liquidity events arrive on average once every ten years, the optimal holding falls to about 5% and the required illiquidity premium is about 6%.

**D.** Because illiquid wealth is right-skewed between liquidity events, the fund should rebalance its private equity holding to a level above its long-run average holding when trading is possible.

## Solutions • IM-17 f

### Solution IM-17 f.1 — Answer: C

Bank Q616 • AnalystPrep Q.2426

**Step 1 – The arguments for**

Low correlation with stocks and bonds, better diversification and the chance of alpha in less efficient markets: A, B and D all support the allocation.

**Step 2 – The constraints**

A large withdrawal is due within two years and the board wants low transaction costs. Illiquid assets cannot be "eaten": raising cash from them means searching for buyers, waiting through due diligence and accepting discounts.

$\Rightarrow$ The near-term cash need argues against the allocation: C.

**Interpretation**

The diversification arguments are themselves overstated, because the low reported correlations of private equity and real estate are partly an artefact of smoothing (IM-17 c), but they are still arguments *for*, which is what the question asks about.

**Why the other options are wrong**

**A is incorrect.** Diversification is a stated benefit, so it supports inclusion.

**B is incorrect.** Less efficient markets with dispersed manager skill offer alpha opportunities, which supports inclusion.

**D is incorrect.** Low correlation is the standard rationale for the allocation.

**LO:** IM-17 f: Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Portfolio Choice with Illiquid Assets; SchweserNotes 2026 Book 5, Reading 99, LO 99.f

### Solution IM-17 f.2 — Answer: B

New • built from SchweserNotes 2026 Reading 99

**Step 1 – What transaction costs do**

With proportional costs, rebalancing every period is too expensive. The optimal policy leaves a no-trade region around the target and trades only at its edges; with 10% costs and 35% volatility the region runs from 0% to 50% around a 25% target.

$\Rightarrow$ B is correct.

**Step 2 – What the model misses**

It assumes the asset can always be sold by paying the cost. For real estate, private equity, infrastructure and timber a buyer may not exist, due diligence takes time and a buyer may walk away; in a crisis even liquid markets freeze.

$\Rightarrow$ C claims exactly what the model cannot capture.

**Interpretation**

That gap is why Ang turns to infrequent-trading models (liquidity events), which produce much lower optimal holdings and higher hurdle rates than transaction-cost models.

**Why the other options are wrong**

**A is incorrect.** Continuous rebalancing is what transaction costs rule out.

**C is incorrect.** The model's key weakness: paying the cost does not guarantee a counterparty.

**D is incorrect.** Costs make an asset less attractive, not more; nothing in the model raises the allocation.

**LO:** IM-17 f: Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Asset Allocation with Transaction Costs (Constantinides), as summarised in Workbook D9; SchweserNotes 2026 Book 5, Reading 99, LO 99.f

### Solution IM-17 f.3 Answer: A

New • built from SchweserNotes 2026 Reading 99

#### Step 1 – Between events

The investor cannot trade, so the allocation drifts with returns and can move from too high to too low relative to the optimum (Figure 3).

#### Step 2 – At an event

Illiquid wealth is right-skewed, so the optimal trading point is below the long-run average holding.

⇒ A is correct.

#### Step 3 – The other behaviours

Unable to hedge, the investor consumes less; arbitrage needs continuous trading, so there are no infinite positions; and illiquidity raises risk aversion toward both liquid and illiquid assets.

#### Interpretation

Rebalancing below the average is the counterpart of drift: the investor resets low, knowing the holding will tend to rise again before the next chance to trade.

#### Why the other options are wrong

**B is incorrect.** Polarity flip: illiquid-asset investors consume less, because they cannot hedge declines.

**C is incorrect.** There is no illiquidity arbitrage: arbitrage requires continuous trading.

**D is incorrect.** Illiquidity makes the investor more risk-averse toward liquid and illiquid assets alike.

**LO:** IM-17 f: Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Asset Allocation with Infrequent Trading (Ang, Papanikolaou and Westerfield); SchweserNotes 2026 Book 5, Reading 99, LO 99.f

### Solution IM-17 f.4 Answer: D

Bank Q617 • BT P2.T8.708.3

#### Step 1 – Phillip and Debra (A and B)

Rarer liquidity events lower the optimal holding (59% continuous, 5% at ten years) and raise the required premium (0.7% at six months to 6.0% at ten years).

⇒ Both true.

#### Step 2 – Peter (C)

There is no market portfolio of illiquid assets, idiosyncratic risk is high and alpha is widely dispersed, so skilled managers are essential.

⇒ True.

#### Step 3 – Mary (D)

Illiquid asset classes have no tradeable indexes (NCREIF is not investable), so factor risk cannot be separated from manager skill.

⇒ False: the answer.

**Interpretation**

Because beta and alpha cannot be split, the allocation to illiquid assets is as much a bet on manager selection as on the asset class.

**Why the other options are wrong**

**A is incorrect.** True: less frequent liquidity events lower the optimal allocation.
**B is incorrect.** True: rarer trading requires a larger illiquidity premium.
**C is incorrect.** True: high idiosyncratic risk, agency problems and dispersed alpha make skill decisive.

**LO:** IM-17 f: Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Portfolio Choice with Illiquid Assets; SchweserNotes 2026 Book 5, Reading 99, LO 99.e and 99.f

## Solution IM-17 f.5 — Answer: B

New • built from SchweserNotes 2026 Reading 99

**Step 1 – What happened**

About 55% of Harvard's endowment was in hedge funds, private equity and real assets. It fell 22% from July to October 2008, the university needed cash for operations, and buyers of its private equity stakes asked discounts of about 50%.

**Step 2 – The lesson**

Ang calls asset-liability management Harvard's worst failing: its liabilities were shorter in duration than its assets, and only liquid assets can be consumed.
⇒ B is correct.

**Interpretation**

For the committee the question is not whether illiquid assets are good or bad, but whether the endowment's spending needs and its skill in selecting managers support a 50% allocation.

**Why the other options are wrong**

**A is incorrect.** Absolute claim: studies find no *higher* risk-adjusted returns on average, not negative returns for everyone; skilled endowments can profit.
**C is incorrect.** Reversed: the problem was too little liquidity, not too much.
**D is incorrect.** Size and patience are necessary but not sufficient; the case for illiquid assets rests on skill.

**LO:** IM-17 f: Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets. • **Reference:** Ang, Asset Management (2014), Ch.13, Liquidating Harvard, as summarised in Workbook D9; SchweserNotes 2026 Book 5, Reading 99, LO 99.e and 99.f

## Solution IM-17 f.6 — Answer: C

Workbook D9.2 • S2, S3

**Step 1 – Read the model's table**

Table 19.2: with continuous trading the optimal risky holding is 59%; when the average interval between liquidity events is ten years it falls to 5% (Panel A), and the investor should demand a 6.0% illiquidity premium, defined as a certainty-equivalent hurdle rate (Panel B).
⇒ C is correct. Ang adds that, given agency conflicts, capital calls and asset-liability mismatches, true hurdle rates are even higher, and most illiquid assets do not beat them.

**Interpretation**

A 35% target is far above what portfolio choice models with illiquidity risk support. "Size and pa-

tience are necessary but not sufficient": the premium Harvard earned came from skill in selecting managers, not from an asset class premium.

### Why the other options are wrong

**A is incorrect.** Polarity flip (across versus within). Ang: "there do not seem to be significant illiquidity risk premiums across classes", but "there are large illiquidity risk premiums within asset classes" (Treasuries, corporate bonds, equities, hedge funds, PE, real estate).

**B is incorrect.** Sibling swap. Of the four routes (passive allocation, liquidity security selection, market making, dynamic rebalancing), it is dynamic rebalancing at the aggregate level that "is simplest to implement and has the greatest impact on portfolio returns".

**D is incorrect.** Polarity flip. Because illiquid holdings are right-skewed, the average holding exceeds the optimal rebalanced holding: "the optimal trading point of illiquid assets is lower than the long-run average holding".

**LO:** IM-17 e: Compare illiquidity risk premiums across and within asset categories. IM-17 f: Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets. • **Reference:** Ang, Asset Management, Ch.13 (LTR source text Ch.19), 19.5 and 19.6, Table 19.2

RISK MANAGEMENT AND INVESTMENT MANAGEMENT • PART

# P3

## Third priority

Tier C and D topics, or LOs Vardeez flags in other chapters. 12 LOs from 7 readings.

IM-1 • Factor Theory • Vardeez: IMPORTANT

# IM-1 b

P3 CONCEPT

Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM.

Workbook: none assigned (B4.1 cross-referenced, worked as Question IM-2 d.7) • VARDEEZ FOCUS CAPM assumptions and limitations

**WHAT IS TESTED HERE**

**Why P3.** Vardeez flags it in a reading he rates Important, not Very Important. The reading's workbook section, B4 (Tier B, Map #18, 5 of 16 sheets), reports no shape on the CAPM itself: its shapes sit under IM-2 d, IM-2 e and IM-3.

**Vardeez:** CAPM assumptions and limitations. This block takes the assumptions and how the CAPM prices factor risk; the limitations (each assumption relaxed) are IM-1 c.

**Reported exam shapes:** none. Built from Vardeez's focus, GARP 2025 Ch.1 (Ang) and SchweserNotes Reading 83. Workbook B4.1 (worked as Question IM-2 d.7) turns on telling the security market line from the Fama-French SMB factor; the SML half is set out here.

**What the questions look like:** statement questions. Which item is (or is least likely) a CAPM assumption, what each assumption does in the model, what the CAPM counts as risk (beta, not the asset's own volatility), and which CAPM idea survives in practice. The bank's CAPM expected-return plug-ins (Bank Q10 and Q15 to Q18) are left out under the CONCEPT tag; the SML is in the notes in case a one-line use appears.

## 1. The idea behind IM-1 and IM-2: factor risk premiums

Ang's analogy is that **factors are to assets what nutrients are to food**. It is recapped here (IM-1 a is P4) because every LO of IM-1 and IM-2 leans on it.

- **Factors matter, not assets.** Look through the asset-class label to the factor content, as you read a food label for the nutrients.
- **Assets are bundles of factors.** Equities and government bonds are close to factors themselves; corporate bonds, hedge funds and private equity bundle equity, volatility, interest rate and default risk.
- **Different investors need different factors.** A bankruptcy lawyer can bear low GDP growth that would cost another investor his job.
- **The one difference: nutrients are good, factor risks are bad.** Each factor defines its own set of *bad times*. Investors exposed to losses in those bad times are compensated by risk premiums in good times.

The CAPM is the first and simplest theory of factor risk: **one factor**, the market portfolio, and **one definition of bad times**, low market returns.

## 2. The CAPM: one factor, priced through beta

Treynor (1961), Sharpe (1964), Lintner (1965) and Mossin (1966) built it on the principle of diversification and the mean-variance utility introduced by Markowitz in 1952. Its revolution: **the risk of an asset is not how it behaves in isolation but how it moves with the market**. Before the CAPM, risk was the asset's own volatility; the CAPM made that irrelevant.

$$E(r_i) - r_f = \beta_i[E(r_m) - r_f], \qquad \beta_i = \frac{\text{cov}(r_i, r_m)}{\text{var}(r_m)} = \frac{\rho_{i,m}\,\sigma_i}{\sigma_m}$$

$$E(r_m) - r_f = \bar{\gamma}\,\sigma_m^2$$

The first is the security market line (SML, GARP Eq. 1.2); the second pins down the market risk premium on the capital market line (GARP Eq. 1.1). Here $\bar{\gamma}$ is the risk aversion of the "average" investor, each investor weighted by wealth, and $\sigma_m^2$ is the variance of the market. Ang's verdict on the model is two-sided. Empirically it is "a spectacular failure": premiums depend on more than beta and on more than one factor. Yet it is the workhorse (75% of finance professors advocate it and 75% of CFOs use it in capital budgeting), because its intuition holds: factor exposures determine premiums, and premiums compensate for losses in bad times.

## 3. The seven assumptions

GARP lists seven; the table at the end of the notes gives each one's job in the model and the wording a distractor uses to corrupt it.

1. Investors have only financial wealth.
2. Investors have mean-variance utility.
3. Investors have a single-period investment horizon.
4. Investors have homogeneous expectations.
5. No taxes or transactions costs.
6. Individual investors are price takers.
7. Information is costless and available to all investors.

- **The two that carry the punchline.** Mean-variance utility makes variance the measure of risk and diversification the investor's main tool. Homogeneous expectations make everyone see the same means, volatilities and correlations, so everyone holds the *same* mean-variance efficient (MVE) portfolio, in different amounts. In equilibrium that portfolio must be the market (IM-1 c).
- **Assumption 5 is "no taxes or transactions costs".** Schweser and Vardeez's source notes shorten it to "frictionless". A distractor that says "uniform" or "low" taxes and costs is not the assumption. Once it is dropped, Ang treats taxes as a systematic factor in their own right.
- **Not all seven matter equally.** The single-period horizon is "a minor assumption" for the theory (Merton's dynamic version has the CAPM holding period by period); homogeneous expectations matter mainly in combination with short-sale limits (IM-1 c).

## 4. How factor risk is addressed in the CAPM

- **One factor: the market portfolio**, each stock held in proportion to its market capitalization (an index fund). It is tradeable, and it is *systematic*: every risky asset's premium is set only by its exposure to it.

- **Bad times are low (or negative) market returns.** The premium is a reward for how an asset pays off in those times.
- **Exposure is beta.** An asset that loses when the market loses has a high beta; its gains in good times do not cancel the losses in bad times, so it must offer a high expected return. An asset that pays off when the market tanks has a low beta, is valuable, and needs little compensation. With a low enough beta the expected return is *below* the risk-free rate: a negative premium. Gold, or sometimes government bonds, are the usual examples of low or negative beta assets.
- **Idiosyncratic risk earns nothing.** It is diversified away in the market portfolio ("the part that carries no nutritional value"). Two assets with the same beta have the same expected return whatever their own volatility; $\sigma_i$ enters only through $\rho_{i,m}\sigma_i/\sigma_m$.
- **Who escapes the factor.** Only the infinitely risk-averse investor, who holds nothing but the risk-free asset.

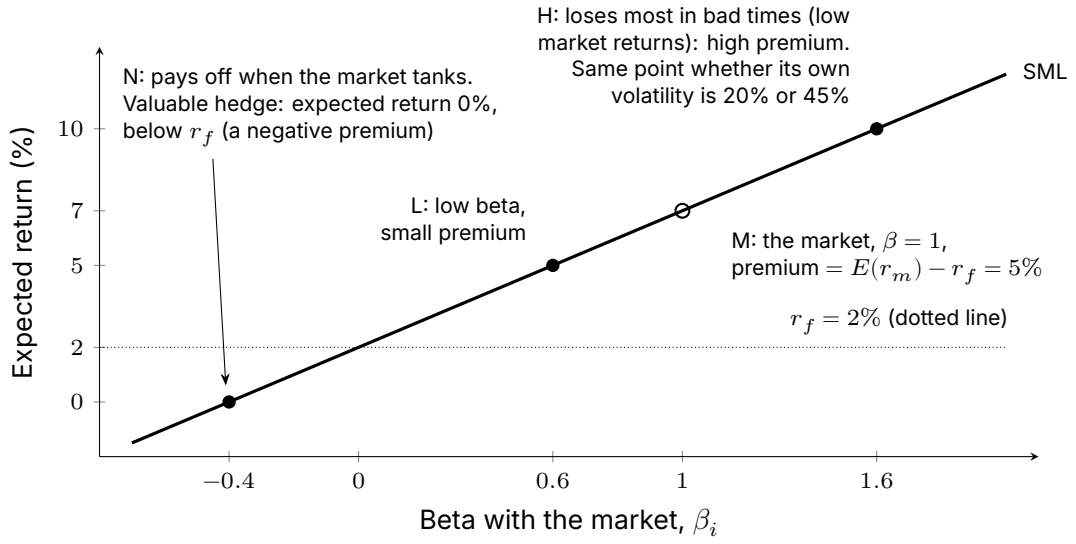



Figure 1. The security market line (solid) with an illustrative $r_f = 2\%$ and market premium 5%. Filled circles are assets; the open circle is the market. Height above the dotted $r_f$ line is the risk premium, set only by beta: own volatility does not move a point along the line.

| Assumption (GARP wording) | What it does in the model | Distractor wording that is *not* the assumption |
|---|---|---|
| 1. Only financial wealth | Portfolio choice ignores income and liabilities | Investors hedge labor income or real liabilities |
| 2. Mean-variance utility | Variance is risk; gains and losses are treated symmetrically | Investors dislike losses more than equal gains |
| 3. Single-period horizon | No rebalancing; the market is simply held | Long (multi-period) horizon |
| 4. Homogeneous expectations | Everyone holds the same MVE portfolio, so it becomes the market | Heterogeneous or differing beliefs |
| 5. No taxes or transactions costs | Frictionless trading; taxes are not a factor | "Uniform" or "low" taxes and costs |
| 6. Price takers | No trade moves prices | Large investors are price setters |
| 7. Information costless and available to all | Prices reflect what everyone knows | Information is costly or unevenly spread |

**Exam traps**

- **Own volatility as CAPM risk (sibling swap).** The CAPM measures risk by beta. A volatile asset with a low beta earns a low premium; total volatility is the pre-CAPM view.
- **Corrupted assumption wording.** "Uniform taxes", "multi-period horizon", "heterogeneous expectations", "price setters" and "costly information" are the real world, not the assumptions.
- **SML, CML or SMB?** The SML prices every asset against beta; the CML is the capital allocation line every investor shares in equilibrium, against volatility (IM-1 c); SMB is the Fama-French size factor (Workbook B4.1, worked as Question IM-2 d.7). A "loading on the SML" does not exist.
- **A negative premium is not an error.** An asset that pays off when the market crashes can have an expected return below $r_f$.
- **Rejected, but not useless.** The CAPM fails empirically, but its intuition (premiums compensate for bad-times losses; risk is factor exposure) is what multifactor models keep (IM-1 d).

## Practice questions: IM-1 b

**Question IM-1 b.1** Bank Q8 • AnalystPrep Q.2375 (rebuilt)

Which of the following is an assumption of the CAPM?

**A.** Investors have heterogeneous expectations about asset returns.

**B.** Investors face taxes and transaction costs that differ across securities.

**C.** Large investors are price setters whose trades move the market.

**D.** Investors have mean-variance utility.

**Question IM-1 b.2** Bank Q11 • AnalystPrep Q.4586

Which of the following is LEAST likely an assumption of the CAPM?

**A.** Investors have a long-period (multi-period) investment horizon.

**B.** Investors have identical expectations.

**C.** Investors trade without transaction costs or taxes.

**D.** Investors can access all information at the same time.

**Question IM-1 b.3** New • built from GARP Ch.1

A risk analyst at a Singapore insurer is reviewing a consultant's training slide that explains what each assumption of the CAPM contributes to the model. Which of the following statements on the slide is correct?

**A.** The CAPM assumes taxes and transaction costs are uniform across securities, so they shift every asset's required return by the same amount.

**B.** The single-period horizon is the assumption the CAPM cannot do without, because the model fails in every period once investors have long horizons.

**C.** Homogeneous expectations ensure that every investor holds the same mean-variance efficient portfolio, which in equilibrium becomes the market portfolio.

**D.** Mean-variance utility means investors dislike losses more than they like gains of the same size, which is why only market risk is priced.

**Question IM-1 b.4** Bank Q14 • AnalystPrep Q.3018 (rebuilt)

Which of the following best describes assets with a high positive beta?

**A.** Assets with lower volatility than the market, which tend to move slowly in response to market changes.

**B.** Assets that tend to go up when the market goes up and go down when the market goes down, by more than the market does.

**C.** Assets that tend to go down when the market goes up and up when the market goes down.

**D.** Assets with high volatility that is driven by firm-specific news and is unrelated to market moves.

**Question IM-1 b.5** Bank Q12 • AnalystPrep Q.4591 (rebuilt)

Under the CAPM the risk-free rate is 3% and the market risk premium is 6%. Asset X has a market beta of 0.5 and a total volatility of 40%. Asset Y has a market beta of 1.5 and a total volatility of 20%. Which statement is correct?

**A.** X has the higher expected return because its total volatility is higher.

**B.** Y has the higher expected return because it has greater exposure to the market factor.

**C.** X and Y have the same expected return because both earn the same market risk premium.

**D.** X has the higher expected return because its idiosyncratic risk is rewarded.

**Question IM-1 b.6** Bank Q9 • BT notes P2.T9.700.3

Andrew Ang catalogs both the failures and successes of the CAPM, where a success is an idea the CAPM gets right. Each of the following is an assumption or implication of the CAPM, but only one is true in practice and therefore useful. Which one is a genuine success?

**A.** Information is costless and available to all investors.

**B.** Risk is factor exposure: an asset's risk is measured by its exposure to the factor.

**C.** Investors have mean-variance utility.

**D.** Investors have homogeneous expectations.

## Solutions • IM-1 b

### Solution IM-1 b.1 — Answer: D

Bank Q8 • AnalystPrep Q.2375

**Step 1 – List the seven assumptions**

Only financial wealth; mean-variance utility; single-period horizon; homogeneous expectations; no taxes or transactions costs; price takers; costless information available to all.

**Step 2 – Test each option against the list**

A contradicts homogeneous expectations, B contradicts "no taxes or transactions costs", C contradicts price taking. D is assumption 2.

$\Rightarrow$ Only D is an assumption; the other three are the real-world departures that IM-1 c turns into shortcomings.

**Interpretation**

Mean-variance utility is what makes variance the measure of risk and diversification the investor's main tool. When it is relaxed (investors dislike losses more than they like gains), downside risk starts to earn its own premium.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: the CAPM assumes homogeneous (identical) expectations. Heterogeneous beliefs are the departure, and they matter once short sales are restricted.

**B is incorrect.** Corrupted assumption 5: the CAPM assumes no taxes or transactions costs at all. Costs that differ across securities are a real-world friction.

**C is incorrect.** Polarity flip: the CAPM assumes every investor is a price taker. Large investors who move prices are a departure.

**LO:** IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.6 Failures of the CAPM; SchweserNotes 2026 Book 5, Reading 83, LO 83.c

### Solution IM-1 b.2 — Answer: A

Bank Q11 • AnalystPrep Q.4586

**Step 1 – The horizon assumption**

The CAPM assumes a *single-period* investment horizon (assumption 3).

$\Rightarrow$ A states its opposite, so it is the least likely assumption.

**Step 2 – Check the other three**

B is homogeneous expectations (assumption 4), C is no taxes or transactions costs (5), and D restates costless information available to all investors (7).

**Interpretation**

Ang calls the single period "a minor assumption" for the theory: Merton's dynamic extension has the CAPM holding in each period. It bites in portfolio choice instead: the optimal strategy for a long-term investor is to rebalance, while the average investor, who holds the market, does not.

**Why the other options are wrong**

**B is incorrect.** A genuine assumption (homogeneous expectations); picking it misreads the LEAST likely format.

**C is incorrect.** A genuine assumption: no taxes or transactions costs.

**D is incorrect.** A genuine assumption: information is costless and available to all investors at

once.

**LO:** IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.6 Failures of the CAPM, assumption 3; SchweserNotes 2026 Book 5, Reading 83, LO 83.c

### Solution IM-1 b.3 — Answer: C

New • built from GARP Ch.1

#### Step 1 – What homogeneous expectations buy

If all investors share the same means, volatilities and correlations, they all find the same mean-variance efficient (maximum Sharpe ratio) portfolio. Risk aversion only sets how much of it each holds against the risk-free asset.

#### Step 2 – From one MVE portfolio to the market

In equilibrium, demand equals supply. If everyone holds the same risky portfolio, that portfolio must contain every asset in proportion to its market value.
$\Rightarrow$ The MVE portfolio becomes the market factor: this is GARP's chain, so C is correct.

#### Interpretation

By itself, Ang says, homogeneous expectations is "not important": a version of the CAPM holds with beliefs averaged across investors. Combined with short-sale limits, heterogeneous beliefs do produce deviations (IM-1 c).

#### Why the other options are wrong

**A is incorrect.** Corrupted wording: the assumption is *no* taxes or transactions costs, not uniform ones.

**B is incorrect.** Overstatement: the single period is a minor assumption; Merton's dynamic CAPM holds in each single period.

**D is incorrect.** Polarity flip and wrong mechanism: mean-variance utility treats gains and losses symmetrically (loss aversion is what breaks it), and only market risk is priced because idiosyncratic risk diversifies away.

**LO:** IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM (Equilibrium) and 1.6 Failures of the CAPM, assumptions 3 to 5

### Solution IM-1 b.4 — Answer: B

Bank Q14 • AnalystPrep Q.3018

#### Step 1 – What beta measures

$\beta_i = \text{cov}(r_i, r_m)/\text{var}(r_m)$: the sensitivity of the asset's return to the market's.

#### Step 2 – High and positive

A high positive beta ($\beta > 1$) means the asset moves with the market and by more than the market.
$\Rightarrow$ It loses most exactly when the market loses, the CAPM's bad times, so it must carry a high premium.

#### Interpretation

The size of the asset's own swings is not the point. A stock can be very volatile with a beta near zero if its moves are firm-specific (option D); the CAPM gives that volatility no premium.

#### Why the other options are wrong

**A is incorrect.** Describes a low-beta asset (below 1), which moves less than the market.

**C is incorrect.** Describes a negative-beta asset, which moves against the market and acts as a hedge.
**D is incorrect.** Describes idiosyncratic volatility: unrelated to market moves means a beta near zero, not a high one.

**LO:** IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lessons 5 and 6, Eq. 1.2

### Solution IM-1 b.5 — Answer: B

Bank Q12 • AnalystPrep Q.4591

#### Step 1 – Apply the SML to each asset

$$E(r_X) = 3\% + 0.5 \times 6\% = 6.0\% \qquad E(r_Y) = 3\% + 1.5 \times 6\% = 12.0\%$$

$\Rightarrow$ Y's expected return is double X's, although Y's volatility is half of X's.

#### Step 2 – Why total volatility does not enter

Most of X's 40% volatility is idiosyncratic and is diversified away in the market portfolio. The CAPM rewards only exposure to the single factor, which beta measures.

#### Interpretation

The total volatilities are the red herring. They could matter only through $\beta_i = \rho_{i,m}\sigma_i/\sigma_m$, and the betas given already contain them.

#### Why the other options are wrong

**A is incorrect.** Uses total volatility as the measure of risk: the pre-CAPM view the model replaced.
**C is incorrect.** The premium per unit of beta is the same for both, but their exposures differ, so their expected returns differ.
**D is incorrect.** Idiosyncratic risk is diversifiable and carries no premium in the CAPM.

**LO:** IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lessons 1 and 5, Eq. 1.2; SchweserNotes 2026 Book 5, Reading 83, LO 83.b

### Solution IM-1 b.6 — Answer: B

Bank Q9 • BT notes P2.T9.700.3

#### Step 1 – Test A, C and D against the real world

Information is costly and not available to all; investors treat losses and gains asymmetrically; beliefs are heterogeneous.
$\Rightarrow$ All three are assumptions that fail in practice.

#### Step 2 – Test B

"Risk is factor exposure" is CAPM Lesson 5. It survives the CAPM's empirical failure and carries straight into multifactor models, where risk is measured by factor betas.

#### Interpretation

Ang calls the CAPM "a spectacular failure" in the data, yet its intuition holds: factors determine premiums, and premiums compensate for losses in bad times. The lessons survive; the assumptions do not.

#### Why the other options are wrong

**A is incorrect.** Assumption 7 fails: information is costly and some of it is unavailable to many investors. This is the root of the Grossman-Stiglitz near-efficient market (IM-1 f).
**C is incorrect.** Assumption 2 fails: investors are more distressed by losses than pleased by gains, so downside risk carries a premium.
**D is incorrect.** Assumption 4 fails: investors have heterogeneous expectations.

**LO:** IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM (Lesson 5) and 1.6 Failures of the CAPM

IM-1 • Factor Theory • Vardeez: IMPORTANT

# IM-1 c

P3 CONCEPT

Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM.

Workbook: none assigned • VARDEEZ FOCUS CAPM limitations

**WHAT IS TESTED HERE**

**Why P3.** Vardeez flags it ("CAPM limitations") in a reading he rates Important; no workbook shape was reported on it.
**Vardeez:** CAPM limitations. The LO also asks for the four implications that come before them: equilibrium and optimal holdings, exposure to factor risk, and diversification.

**Reported exam shapes:** none. Built from Vardeez's focus, GARP 2025 Ch.1 and SchweserNotes Reading 83. The SML, the market premium $\bar{\gamma}\sigma_m^2$ and the seven assumptions are in IM-1 b.

**What the questions look like:** statements on the six CAPM lessons, on equilibrium (same MVE portfolio, different amounts), on CML versus SML, on beta as a measure of the lack of diversification, and on which deviation comes from which broken assumption. The bank's calculations filed here ($\bar{\gamma}\sigma_m^2$ plug-ins, SML mispricing, a tax gross-up: Bank Q19 to Q21, Q23, Q26, Q27) are left out under the CONCEPT tag; the direction of $\bar{\gamma}\sigma_m^2$ is tested in words (Question IM-1 c.4).

## 1. Equilibrium and optimal holdings

Recap from IM-1 b: one factor (the market), risk is beta, and the seven assumptions. The holdings come out of the mean-variance problem (Figure 2):

- Each investor mixes the risk-free asset with the **mean-variance efficient (MVE) portfolio**, the tangency portfolio with the highest Sharpe ratio, along a capital allocation line (CAL).
- Homogeneous expectations: everyone finds the *same* MVE portfolio and holds it in different quantities.
- **Equilibrium** (demand equals supply). Suppose every investor's MVE portfolio gave zero weight to stock AA. Nobody would hold it, so AA is overpriced and its expected return too low. Its expected payoff is fixed, so its price falls and its expected return rises until investors want exactly the shares outstanding. Hence the MVE portfolio becomes **the market portfolio, at capitalization weights**, and the CAL every investor shares is called the **capital market line (CML)**.
- **Lesson 2: each investor has his own optimal exposure to factor risk**, a different point on the CML. **Lesson 3: the average investor holds the market**: the tangency point (100% in the market) is the average investor, and the market's risk aversion is his (wealth-weighted) risk aversion. Only the infinitely risk-averse investor holds no market at all.

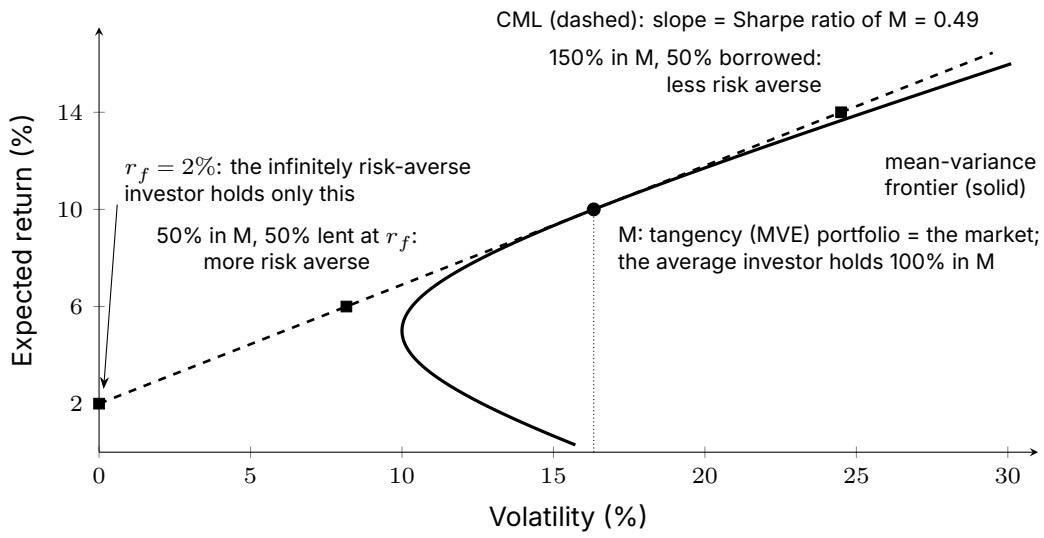



Figure 2. GARP's Figure 1.1 with illustrative numbers. Every investor holds the same risky portfolio M and differs only in where he sits on the dashed CML (Lesson 2); the average investor sits at M itself (Lesson 3). Squares are investors, the filled circle is the market.

## 2. Exposure to factor risk: Lessons 4, 5 and 6

- **Lesson 4: the factor risk premium has an economic story.** On the CML, $E(r_m) - r_f = \bar{\gamma}\,\sigma_m^2$. The premium rises when the average investor becomes more risk averse *and* when the market becomes more volatile: two separate levers, both upward. Because it is proportional to *variance*, doubling volatility quadruples it (with $\bar{\gamma} = 3$: $\sigma_m = 15\%$ gives 6.75%, $\sigma_m = 30\%$ gives 27%). A higher expected return on an unchanged expected payoff means a lower price today: in 2008 to 2009 volatility skyrocketed, equity prices nosedived, and realized returns in 2009 and 2010 were indeed high. The premium reflects investors' preferences and the economy; it will not disappear because clever hedge funds trade it, since **systematic risk cannot be arbitraged away**.
- **Lesson 5: risk is factor exposure.** The SML: the higher the beta, the higher the expected return, if the factor premium is positive.
- **Lesson 6: assets paying off in bad times have low risk premiums.** If the payoff is high when the market is low, the asset is valuable and investors may even pay to hold it (an expected return below $r_f$).

## 3. Treatment of diversification benefits

- **Lesson 1: don't hold an individual asset, hold the factor.** A stock carries market risk (rewarded) and idiosyncratic risk (not rewarded). Diversification offsets one asset's losses with another's gains and raises the Sharpe ratio; investors keep diversifying until they hold the most diversified portfolio possible, the market.
- **Beta measures the lack of diversification potential.** $\beta_i = \rho_{i,m}\sigma_i/\sigma_m$: the lower the correlation with the portfolio already held, the larger the diversification benefit. High beta means low diversification benefit, so the asset must pay a high expected return; low beta means a large benefit and a low required return.
- **Beta is the right measure only after diversifying.** It tells a holder of the market how much an asset adds to his risk. A concentrated holder still carries idiosyncratic risk that beta ignores; for

him the relevant measure is the asset's covariance with the portfolio he actually holds (its beta to that portfolio, as in IM-5 a).

**Correction to the source notes.** SchweserNotes Reading 83 (LO 83.c, Lesson 5) writes that "beta looks at idiosyncratic risk and the lack of diversification". Beta measures *systematic* risk, the exposure to the market factor; it is a measure of the lack of diversification potential with respect to the market, and idiosyncratic risk is exactly what it leaves out. The same paragraph says gold and government bonds "had negative expected returns" in the 2007 to 2009 crisis. GARP says only that assets with low enough betas have negative expected returns, giving gold and sometimes government bonds as examples of low or negative beta assets; government bonds were among the few assets that did well in 2008.

## 4. Shortcomings: relax each assumption

| Assumption relaxed | What appears (GARP 2025 Ch.1, Section 1.6) |
|---|---|
| 1. Only financial wealth | Income and real liabilities matter, and income falls when growth is low: inflation and growth become factors. Labor income (human capital) risk: Jagannathan and Wang (1996) found large improvements in the CAPM once it is included. |
| 2. Mean-variance utility | Losses hurt more than gains please: stocks with more downside risk earn more (Ang, Chen and Xing); higher-moment (skewness, kurtosis) premiums. |
| 3. Single period | Minor by itself (Merton: the CAPM holds in each period). But long-horizon investors should rebalance, and the average investor does not. |
| 4. Homogeneous expectations | By itself not important (a CAPM holds with averaged beliefs). With short-sale limits, optimists set prices (Miller, 1977, a rational setting): beliefs become a systematic factor. |
| 5. No taxes or transactions costs | Taxes are a systematic factor; costs vary across securities, so deviations are larger in illiquid markets (liquidity premiums). |
| 6. Price takers | Informed investors' large trades move prices; markets become illiquid with trading frictions. |
| 7. Costless information | Information itself can be a factor; deviations are strongest in small, illiquid stocks where information spreads poorly. |

**Summary.** When the assumptions fail, additional premiums appear: macro factors, asymmetric treatment of risk, illiquidity and transactions costs, and taxes. Failures of the CAPM should be most apparent in **illiquid, inefficient markets**, and the perfect-information assumption is one reason economists no longer believe markets are efficient in the CAPM's form (IM-1 f).

**Correction to the source notes.** Vardeez writes that "the seven assumptions above are the shortcomings: each one, when relaxed, generates an additional risk premium that the CAPM cannot price." GARP says otherwise for two of them: the single-period horizon is "an inconsequential assumption for the CAPM theory", and homogeneous expectations "by itself" is "not important"; heterogeneous beliefs produce deviations only in combination with frictions such as short-sale limits. Keep the summary at GARP's level: violations of the assumptions produce additional premiums, most visibly in illiquid, inefficient markets.

| | Capital market line (CML) | Security market line (SML) |
|---|---|---|
| **Plots** | Expected return against volatility $\sigma$ | Expected return against beta $\beta$ |
| **What lies on it** | Efficient portfolios: mixes of $r_f$ and the market | Every asset and portfolio, in equilibrium |
| **Slope** | Sharpe ratio of the market, $(E(r_m) - r_f)/\sigma_m$ | Market risk premium, $E(r_m) - r_f = \bar{\gamma}\sigma_m^2$ |
| **Below the line** | An inefficient portfolio | An overpriced asset (negative alpha) |
| **Where it comes from** | The CAL, shared by all investors in equilibrium | Beta pricing of each asset (Lesson 5) |

**Exam traps**

- **Different risky portfolios for different investors.** Under the CAPM all hold the same MVE portfolio, the market; risk aversion changes only the amount (Lesson 2).
- **The exception.** Only the infinitely risk-*averse* investor holds no market portfolio. An "infinitely risk-tolerant" investor still holds the market, levered.
- **One lever of $\bar{\gamma}\sigma_m^2$ only.** Higher risk aversion and higher market variance both raise the premium; it is proportional to variance, not volatility.
- **Arbitrage removes the premium.** Systematic risk cannot be diversified or arbitraged away, so the equilibrium premium persists.
- **Valuable means low beta.** High-beta assets lose in bad times and earn high premiums; the valuable hedges are low-beta, low-premium assets (Lesson 6 reversed is the classic distractor).
- **CML versus SML.** The CML uses volatility and holds only efficient portfolios; the SML uses beta and prices every asset.
- **Beta measures idiosyncratic risk.** It measures systematic risk, and it is the right measure only for a diversified holder.
- **Labels are not assumptions.** "A developing economy" is not a CAPM assumption; insider trading (information) and a shallow market (price takers) are the violated ones.

## Practice questions: IM-1 c

### Question IM-1 c.1

Bank Q29 • AnalystPrep Q.3017 (rebuilt)

Which of the following best explains, under the CAPM, why asset owners should hold the market portfolio rather than individual stocks?

**A.** The market portfolio earns a larger risk premium than any individual stock for the same exposure to the market factor.

**B.** Individual stocks carry idiosyncratic risk as well as market factor risk; the market portfolio diversifies away the uncompensated idiosyncratic risk while retaining the market risk premium.

**C.** Individual stocks have no exposure to the market factor, so they earn no risk premium.

**D.** Under the CAPM only a few investors hold the market portfolio, and the rest hold the stocks with the highest expected returns.

**Question IM-1 c.2** Bank Q30 • BT answers P2.T8.700.2 • BT notes P2.T9.700.2 (rebuilt)

Which of the following statements about the CAPM in equilibrium is correct?

**A.** The relevant risk of an asset is its own volatility, since risk is evaluated in isolation.

**B.** Investors with different risk aversion hold different risky portfolios, because they have different expected returns.

**C.** Arbitrageurs eventually eliminate the market factor risk premium, since it can be hedged away.

**D.** All investors hold the same risky mean-variance efficient portfolio, the market portfolio, and differ only in how much they hold relative to the risk-free asset.

**Question IM-1 c.3** Bank Q22 • AnalystPrep Q.2388

Which statement is NOT true regarding the capital market line (CML)?

**A.** The risk measure for the CML is standard deviation.

**B.** The CML is the line from the risk-free rate through the market portfolio.

**C.** The CML is the best attainable capital allocation line.

**D.** The CML is also called the security market line.

**Question IM-1 c.4** New • built from GARP Ch.1, Eq. 1.1

During a market crisis, the volatility of the market portfolio rises sharply while the risk aversion of the average investor is unchanged. The CIO of a Brazilian pension fund asks the risk team what the CAPM implies for the equity market. Which of the following statements is correct?

**A.** The expected market return rises and equity prices fall at the same time, because the market risk premium is proportional to the variance of the market.

**B.** The market risk premium is unchanged, because it depends only on the risk aversion of the average investor.

**C.** The market risk premium falls, because investors move out of equities into the risk-free asset.

**D.** The market risk premium rises in proportion to market volatility, so a doubling of volatility doubles the premium.

**Question IM-1 c.5** Bank Q28 • AnalystPrep Q.2467

A portfolio manager holds a portfolio which consists only of IT stocks. Borrowing some knowledge of the CAPM, he argues that since $\beta$ measures the contribution of each security to total portfolio risk, he has to include securities with low $\beta$ in his portfolio in order to reduce his portfolio risk. Is the manager's assertion correct?

**A.** Yes, because $\beta$ measures the contribution of one security to total portfolio risk.

**B.** No, because the portfolio held by the manager is not well diversified, so $\beta$ is not an appropriate risk parameter.

**C.** No, because $\beta$ measures non-systematic risk.

**D.** Yes, because $\beta$ measures systematic risk.

**Question IM-1 c.6** Bank Q32 • BT answers P2.T9.20.1.3 • BT notes P2.T9.20.1.3 (rebuilt)

A study group is reviewing the lessons of the capital asset pricing model (CAPM), which Ang describes as a failure in practice but whose basic intuition still holds. Each member states a lesson:

I. Allison: allocating to asset-class labels such as hedge funds is not the optimal way to build a portfolio, because what is rewarded is exposure to underlying factors, not the asset label.

II. Betty: each investor has their own optimal exposure to the factor risk, just as in the CAPM each investor holds a different mix of the risk-free asset and the market portfolio.

III. Chris: in equilibrium the market factor's risk premium cannot be arbitraged away, because systematic risk cannot be diversified, so the premium persists.

IV. Donald: an asset's risk is measured by its factor beta (its exposure to the factor), not by its stand-alone volatility.

V. Eric: assets with high CAPM betas are valuable because they pay off well in bad times, so they have lower risk premiums and lower expected returns.

Which of the statements are TRUE?

**A.** All five statements are true.

**B.** Allison and Betty are incorrect; the other statements are true.

**C.** Eric is incorrect; the other statements are true.

**D.** Chris and Donald are incorrect; the other statements are true.

**Question IM-1 c.7** Bank Q25 • AnalystPrep Q.2382 (rebuilt)

A fund manager in a developed market builds well-diversified portfolios using ten years of historical data and mean-variance utility, consistent with the CAPM framework. Which of the following best describes a shortcoming of the CAPM's assumption of mean-variance utility?

**A.** It ignores covariances between assets, so the benefit of diversification cannot be measured.

**B.** It assumes investors are price setters whose trades move asset prices.

**C.** It treats gains and losses symmetrically, whereas investors dislike losses more than they like equal-sized gains.

**D.** It assumes investors have a multi-period horizon and rebalance continuously.

**Question IM-1 c.8** Bank Q24 • AnalystPrep Q.2377

Country A, a developing economy, has recently witnessed the government's renewed interest and effort in reforming the financial sector. A recent report published by a non-governmental organization (NGO) indicates the presence of rampant insider trading. The report also points to the lack of depth in the financial market, which poses significant problems for investors and market participants. However, the presence of natural resources in the country makes it an attractive destination for foreign investors. A fund manager investing in Country A uses the CAPM to compute expected returns. Which of the following options is (are) correct?

I. Deviations from the CAPM will be observed due to insider trading.

II. Deviations from the CAPM will be observed due to the developing economy.

III. Deviations from the CAPM will be observed due to a lack of market depth.

IV. Deviations from the CAPM will be observed due to the country's underdeveloped corporate sector.

**A.** I only

**B.** I and II

**C.** I and III

**D.** III and IV

## Solutions • IM-1 c

### Solution IM-1 c.1 — Answer: B

Bank Q29 • AnalystPrep Q.3017

**Step 1 – Split a stock's risk**

Each stock's return carries exposure to the market factor, which earns the premium (the "nutrient"), plus idiosyncratic risk, which earns nothing.

**Step 2 – Diversify**

Holding many stocks that are not perfectly correlated diversifies the idiosyncratic part away and keeps the market premium, so the Sharpe ratio rises. Investors keep diversifying until they hold the most diversified portfolio possible, the market.

$\Rightarrow$ CAPM Lesson 1: don't hold an individual asset, hold the factor.

**Interpretation**

This is the CAPM's treatment of diversification benefits: diversification is free, so the model pays nothing for the risk it removes.

**Why the other options are wrong**

**A is incorrect.** Under the SML the premium depends only on the exposure; the same beta earns the same premium whether it comes from one stock or from the market.

**C is incorrect.** Individual stocks do have market exposure (a beta) and earn a premium for it; the problem is the extra, unrewarded idiosyncratic risk.

**D is incorrect.** In the CAPM every investor holds the market portfolio, in different amounts (Lessons 2 and 3).

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lesson 1; SchweserNotes 2026 Book 5, Reading 83, LO 83.c

### Solution IM-1 c.2 — Answer: D

Bank Q30 • BT answers P2.T8.700.2

**Step 1 – Same beliefs, same risky portfolio**

With homogeneous expectations, every investor identifies the same mean-variance efficient portfolio.

**Step 2 – Equilibrium makes it the market**

Demand must equal supply, so the portfolio everyone holds is the market at capitalization weights.

**Step 3 – Risk aversion sets the amount**

Investors differ only in the mix of the risk-free asset and the market, their position on the CML (Lesson 2).

$\Rightarrow$ D is the CAPM equilibrium.

**Interpretation**

The only investor who holds none of the market is the infinitely risk-averse one, who holds only the risk-free asset. Everyone else holds the same risky portfolio.

**Why the other options are wrong**

**A is incorrect.** The CAPM made own volatility irrelevant: risk is covariance with the market (beta).

**B is incorrect.** Different risky portfolios would require different beliefs. Expectations are homo-

geneous, so only the proportions differ.
**C is incorrect.** Systematic risk cannot be diversified or arbitraged away; the equilibrium premium persists (Lesson 4).

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM (Equilibrium; Lessons 2 to 4, Figure 1.1)

### Solution IM-1 c.3 — Answer: D

Bank Q22 • AnalystPrep Q.2388

#### Step 1 – What the CML is

In equilibrium every investor's CAL runs from $r_f$ through the same MVE portfolio, the market; that shared line is the CML. Its risk measure is standard deviation, and since the market has the highest Sharpe ratio it is the best attainable CAL.
$\Rightarrow$ A, B and C are true.

#### Step 2 – What the SML is

The SML plots expected return against *beta* and prices every individual asset: $E(r_i) - r_f = \beta_i[E(r_m) - r_f]$.
$\Rightarrow$ D is false: the CML and the SML are different lines.

#### Interpretation

The CML holds only efficient portfolios, so an individual stock lies below it. The same stock lies *on* the SML if it is fairly priced. Confusing the two lines is the sibling swap.

#### Why the other options are wrong

**A is incorrect.** True: the CML's horizontal axis is standard deviation.
**B is incorrect.** True: it starts at the risk-free rate and passes through the market portfolio.
**C is incorrect.** True: it is the CAL with the maximum Sharpe ratio.

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lessons 4 and 5, Eq. 1.1 and 1.2

### Solution IM-1 c.4 — Answer: A

New • built from GARP Ch.1, Eq. 1.1

#### Step 1 – Write the CAPM market premium

$$E(r_m) - r_f = \bar{\gamma}\,\sigma_m^2$$

#### Step 2 – Hold $\bar{\gamma}$, raise $\sigma_m$

The premium rises with the *square* of market volatility.
$\Rightarrow$ The expected return on the market rises.

#### Step 3 – From expected return to price

The expected payoff is unchanged, so a higher required return means a lower price today. GARP's example: in 2008 to 2009 volatility skyrocketed and equity prices nosedived; realized returns in 2009 and 2010 were high.

#### Interpretation

Lesson 4: the premium has an economic story, investors' aversion to variance. Risk aversion and market variance are two separate levers, and both push the premium up. With $\bar{\gamma} = 3$, a rise in volatility from 15% to 30% takes the premium from 6.75% to 27%: four times, not two.

### Why the other options are wrong

**B is incorrect.** Dropped term: the premium depends on market variance as well as on risk aversion.

**C is incorrect.** Polarity flip: more variance raises the premium the average investor demands to hold the market.

**D is incorrect.** Volatility for variance: the premium is proportional to $\sigma_m^2$, so doubling $\sigma_m$ quadruples it.

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lesson 4, Eq. 1.1

## Solution IM-1 c.5 — Answer: B

Bank Q28 • AnalystPrep Q.2467

### Step 1 – What CAPM beta measures

$\beta_i = \text{cov}(r_i, r_m)/\text{var}(r_m)$: exposure to the market factor, systematic risk. GARP: beta is "a measure of the lack of diversification potential" with respect to the market.

### Step 2 – For whom it is the right measure

For an investor who already holds the diversified market: his idiosyncratic risk is gone, so what a stock adds to his risk is its beta.

### Step 3 – The IT-only portfolio

A single-sector book still carries large sector and idiosyncratic risk that market betas do not see. A stock's contribution to *this* portfolio's risk is its covariance with the IT portfolio, not its market beta.

$\Rightarrow$ The manager's premise fails; the first step to cut risk is to diversify across sectors.

### Interpretation

The CAPM's treatment of diversification assumes the diversifying has already been done. Market beta prices risk for the diversified investor; it is not a general measure of a concentrated portfolio's risk.

### Why the other options are wrong

**A is incorrect.** CAPM beta measures a security's contribution to the risk of the *market* portfolio, not to the total risk of a concentrated one.

**C is incorrect.** Beta measures systematic, not non-systematic, risk (see the correction box in Section 3); the conclusion is right but the reason is wrong.

**D is incorrect.** The premise is right, beta is systematic risk, but it does not make beta the right measure for an undiversified portfolio.

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lessons 1 and 5

## Solution IM-1 c.6 — Answer: C

Bank Q32 • BT answers P2.T9.20.1.3

### Step 1 – Match I to IV to the lessons

I is Lesson 1 and factor theory (look through labels to factors). II is Lesson 2 (own optimal exposure). III is Lesson 4 (the equilibrium premium cannot be arbitraged away). IV is Lesson 5 (risk is factor exposure).

⇒ All four are true.

**Step 2 – Test V against Lessons 5 and 6**

A high beta means the asset loses when the market is low, in bad times. It is risky and earns a *higher* premium. The valuable assets that pay off in bad times are the *low*-beta ones.
⇒ Only Eric is wrong: C.

**Interpretation**

Eric has taken Lesson 6 (valuable assets have low premiums) and attached it to the wrong end of the beta scale. Reversing the sign of Lesson 6 is the standard distractor in this reading.

**Why the other options are wrong**

**A is incorrect.** Treats V as true, putting high beta and "valuable" together.
**B is incorrect.** Rejects Lessons 1 and 2, which are true, and misses that V is false.
**D is incorrect.** Rejects III and IV, which are true, and misses that V is false.

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.4 CAPM, Lessons 1 to 6

## Solution IM-1 c.7 — Answer: C

Bank Q25 • AnalystPrep Q.2382

**Step 1 – What mean-variance utility assumes**

Investors care only about mean and variance. Variance penalizes upside and downside deviations equally.

**Step 2 – What real investors do**

They are more distressed by losses than pleased by equal gains. Relaxing the assumption, stocks with more downside risk should earn higher returns (Ang, Chen and Xing), and higher moments such as skewness carry premiums.
⇒ The shortcoming is the symmetric treatment of risk: C.

**Interpretation**

Each shortcoming is a CAPM assumption that fails, and each failure points to a premium the CAPM cannot price; here, a premium for downside risk.

**Why the other options are wrong**

**A is incorrect.** Mean-variance analysis is built on covariances; that is how it measures diversification.
**B is incorrect.** Price taking is a separate assumption (6); mean-variance utility says nothing about price impact.
**D is incorrect.** The CAPM assumes a single period; a multi-period horizon is the relaxation, not the assumption.

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.6 Failures of the CAPM, assumption 2

## Solution IM-1 c.8 — Answer: C

Bank Q24 • AnalystPrep Q.2377

**Step 1 – Map each statement to an assumption**

I: insider trading means information is neither costless nor available to all investors (assumption 7). III: a shallow market means large trades move prices, so investors are not price takers

(assumption 6).
⇒ Both produce deviations from the CAPM.

### Step 2 – Test II and IV

"A developing economy" and "an underdeveloped corporate sector" are labels, not CAPM assumptions. They may coincide with violations, but they are not violations themselves.
⇒ I and III: C.

### Interpretation

GARP expects CAPM failures to be most apparent in illiquid, inefficient markets. The stem gives both: poor information (insider trading) and illiquidity (no depth).

### Why the other options are wrong

**A is incorrect.** Omits III: thin markets violate the price-taker assumption.
**B is incorrect.** Includes II, a label rather than a violated assumption, and omits III.
**D is incorrect.** Includes IV, a label, and omits I, the information violation.

**LO:** IM-1 c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.6 Failures of the CAPM, assumptions 6 and 7

IM-1 • Factor Theory • Vardeez: IMPORTANT

# IM-1 d

P3 CONCEPT

Describe multifactor models and compare and contrast multifactor models to the CAPM.

Workbook: none assigned • VARDEEZ FOCUS CAPM vs multifactor models

**WHAT IS TESTED HERE**

**Why P3.** Vardeez flags it ("CAPM vs multifactor models") in a reading he rates Important; no workbook shape was reported on it.
**Vardeez:** CAPM vs multifactor models. The LO's "compare and contrast" is GARP's six-lesson table set side by side.

**Reported exam shapes:** none. Built from Vardeez's focus, GARP 2025 Ch.1 and SchweserNotes Reading 83. The bank holds only three questions here, so two are re-filed from this reading (Bank Q7 from IM-1 a, Bank Q36 from IM-1 e) and one is new.

**What the questions look like:** statements. What a multifactor model adds to the CAPM (several definitions of bad times, several betas), which lessons carry over unchanged and which change, what can serve as a factor, and how the 2008 to 2009 crisis reads through a multifactor lens.

## 1. What a multifactor model is

- **Bad times are defined more broadly than low market returns.** Each factor provides its own definition of bad times: high inflation, low economic growth, value stocks lagging growth stocks, volatility spikes.
- **The first one was the arbitrage pricing theory (APT)** of Ross (1976). It is called "arbitrage" because the factors cannot be arbitraged or diversified away, just like the CAPM's market factor; in equilibrium investors must be compensated for bearing each of them.
- **Several betas.** The risk premium is a multi-beta relation (GARP Eq. 1.9):

$$\boxed{E(r_i) = r_f + \beta_{i,1}E(f_1) + \beta_{i,2}E(f_2) + \cdots + \beta_{i,K}E(f_K)}$$

  where $\beta_{i,k}$ is the asset's beta to factor $k$ and $E(f_k)$ is that factor's premium. The CAPM is the case $K = 1$, $f_1 = r_m - r_f$.
- **Kinds of factor.** Macro factors (inflation, growth, volatility, productivity, demography); tradeable investment-style factors (the market, value-growth, momentum: IM-2 d); and purely statistical factors (principal components), which Ang sets aside because they lack economic content.
- **The pricing-kernel view (IM-1 e, P4).** The stochastic discount factor $m$ is one index of all bad times. The CAPM is the special case $m = a + b\,r_m$ with $b < 0$: linear, one factor. A multifactor $m = m(f_1, \ldots, f_K)$ can be nonlinear, and it requires only no arbitrage; the CAPM needs its full set of onerous assumptions.
- **The $64,000 question: what is a bad time?** For the average investor, a time when an extra $1 is very valuable (high marginal utility): after losing a job, when consumption or GDP growth is low.

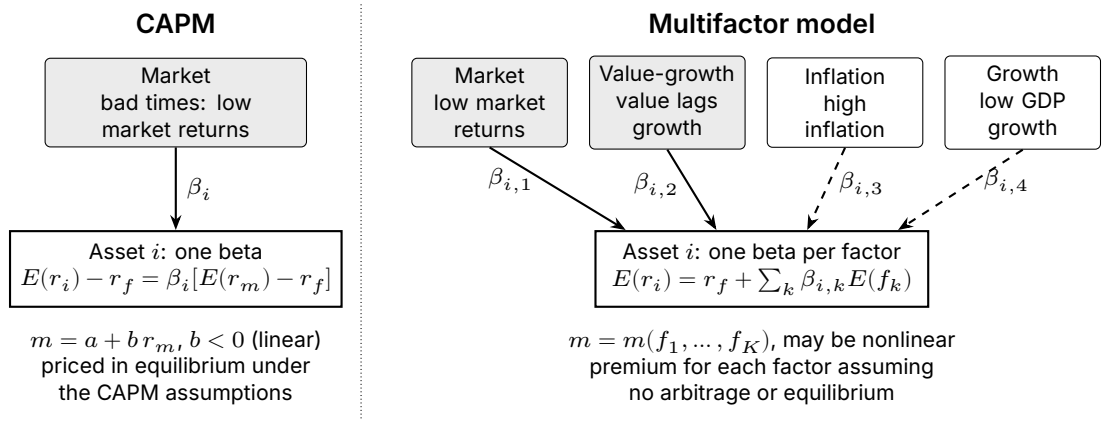



Figure 3. One definition of bad times against several. Grey boxes are tradeable investment factors (solid arrows); white boxes are macro factors (dashed arrows). Each arrow is a beta, and each factor carries its own premium.

## 2. Compare and contrast: the six lessons

GARP's answer to "compare and contrast" is a table: the intuition of all six CAPM lessons holds in a multifactor world, except that each factor now defines its own bad times. Read the table row by row; the words that change are the testable part.

| | CAPM (market factor) | Multifactor models |
|---|---|---|
| **Lesson 1** | Diversification works. The market diversifies away idiosyncratic risk. | Diversification works. The tradeable version of a factor diversifies away idiosyncratic risk. |
| **Lesson 2** | Each investor has her own optimal exposure of the market portfolio. | Each investor has her own optimal exposure of each factor risk. |
| **Lesson 3** | The average investor holds the market. | **The average investor holds the market** (identical). |
| **Lesson 4** | The market factor is priced in equilibrium under the CAPM assumptions. | Risk premiums exist for each factor assuming no arbitrage or equilibrium. |
| **Lesson 5** | Risk of an asset is measured by the CAPM beta. | Risk of an asset is measured by its factor exposures (factor betas). |
| **Lesson 6** | Assets paying off in bad times when the market return is low are attractive and have low risk premiums. | Assets paying off in bad times are attractive and have low risk premiums. |
| **Bad times** | Low market returns only | One definition per factor |
| **Pricing kernel** | $m = a + b\,r_m$, linear, $b < 0$ | $m = m(f_1, \ldots, f_K)$, may be nonlinear |

**Correction to the source notes.** Vardeez's comparison says "five of the six lessons survive essentially unchanged; only the wording of Lessons 1, 4 and 5 shifts", and his trap summary repeats that Lessons 1, 4 and 5 "are the ones that change wording". In GARP's table only **Lesson 3** is word for word the same. Lesson 2 also changes ("the market portfolio" becomes "each factor risk") and Lesson 6 drops "when the market return is low". He also writes that a multifactor premium "needs only no arbitrage"; GARP's Lesson 4 reads "assuming no arbitrage *or* equilibrium": either suffices. What survives unchanged is the intuition of all six, not the wording of five.

## 3. The 2008 to 2009 crisis through a multifactor lens

Nearly every risky asset class fell together in 2008 (GARP Table 1.1): US large caps $-37.0\%$, emerging market equities $-53.2\%$, US high yield $-26.3\%$, public REITs $-37.7\%$, equity hedge funds $-20.6\%$, commodities $-35.7\%$. Only cash (T-bills $+1.3\%$) and high-grade bonds went up: global government bonds $+10.9\%$, the core US bond index $+5.2\%$; GARP singles out safe-haven sovereign bonds, especially long-term Treasuries.

- **Not the death of diversification.** The common losses are what a multifactor model predicts when many asset classes carry the same factor exposures: very high volatility, low growth, uncertainty about policy, liquidity drying up. The commonality is evidence *for* multifactor models.
- **Labels mislead.** Diversifying across asset-class labels is not diversifying across factors. Factor exposures and correlations also rise in bad times, all the more reason to know the true factor drivers.

**Exam traps**

- **Lesson 3 changed (sibling).** The average investor holds the market in both models; it is the lesson candidates most often assume must have changed.
- **Diversification abandoned.** It still works: the tradeable factor portfolio diversifies idiosyncratic risk away, and idiosyncratic risk is priced in neither model.
- **Equilibrium in both (scope).** The CAPM prices the market in equilibrium under its assumptions; a multifactor premium exists assuming no arbitrage *or* equilibrium.
- **"Low market returns" as the multifactor definition of bad times.** That is the CAPM's definition only.
- **"Arbitrage" read backwards.** In the APT the factors *cannot* be arbitraged away; that is why they carry premiums.
- **Firm traits as factors.** A factor is a systematic source of bad times shared across assets; a firm's strategy or reputation is not.
- **2008 killed diversification.** It showed common factor exposures behind different labels, which is the multifactor story.

## Practice questions: IM-1 d

**Question IM-1 d.1** Bank Q33 • AnalystPrep Q.2376 (rebuilt)

Which of the following statements about multifactor models, relative to the CAPM, is CORRECT?

**A.** They treat the market portfolio as the only source of bad times, so an asset has a single beta.

**B.** An asset's risk is measured by its exposures (betas) to several factors, each of which carries a risk premium.

**C.** They abandon diversification, because idiosyncratic risk is priced when several factors are present.

**D.** They imply that the average investor does not hold the market portfolio.

**Question IM-1 d.2** Bank Q7 • BT Mock Q80 • BT answers P2.T9.20.1.1 (re-filed from IM-1 a)

Janice is preparing to conduct factor regressions to discover the factor loadings in her firm's equity portfolio. She will begin by regressing portfolio returns on three explanatory variables: the market factor (MKT) plus two additional factors. Each of the following pairs of factors is a good candidate for the two additional factors EXCEPT which pair is NOT a good candidate?

**A.** Quality and momentum (style factors)

**B.** Strategy and reputation (style factors)

**C.** Economic growth and inflation (macro factors)

**D.** Real interest rates and liquidity (macro factors)

**Question IM-1 d.3** Bank Q34 • AnalystPrep Q.4589 (rebuilt)

A member of a university endowment's investment committee asks how the lessons of the CAPM change when the endowment moves to a multifactor model. Which of the following statements correctly contrasts the two?

**A.** Under the CAPM the average investor holds the market portfolio, but in a multifactor model the average investor tilts toward the factors with the highest premiums.

**B.** Under the CAPM diversification removes idiosyncratic risk, but in a multifactor model idiosyncratic risk is priced because no single portfolio can diversify it across all factors.

**C.** Under the CAPM an asset's risk is its market beta, while in a multifactor model an asset's risk is its total volatility, since several factors drive it.

**D.** Under the CAPM the market factor is priced in equilibrium under the CAPM assumptions, while in a multifactor model a risk premium exists for each factor assuming no arbitrage or equilibrium.

**Question IM-1 d.4** Bank Q35 • BT answers P2.T8.701.3 • BT notes P2.T9.701.3 (rebuilt)

Which of the following is a lesson shared by the CAPM and multifactor models?

**A.** Bad times are defined only as periods of low market returns.

**B.** Valuable assets, which pay off in bad times, have low risk premiums.

**C.** An asset's risk is measured by a single market beta.

**D.** Idiosyncratic risk is rewarded with a risk premium.

**Question IM-1 d.5** Bank Q36 • AnalystPrep Q.2381 (rebuilt; re-filed from IM-1 e)

Fund managers in Country A estimate asset correlations from the last 10 years of calm-period data and use mean-variance utility. Their well-diversified portfolios fell sharply together in the most recent crisis. Which approach is best suited to modeling how assets behave in bad times?

**A.** The CAPM, with the market return as the only factor and a constant beta for each asset.

**B.** A single-index model estimated on the last 10 years of data.

**C.** A pricing kernel (SDF) model, in which $m$ indexes bad times across several factors and may depend nonlinearly on them.

**D.** A traditional discount-rate model that discounts expected cash flows at a fixed rate.

**Question IM-1 d.6** New • built from GARP Ch.1 (1.2, 1.5, 1.8)

After the 2008 to 2009 financial crisis, a pension trustee argues that diversification is dead: nearly every risky asset class the fund held, from equities and high-yield bonds to hedge funds, real estate and commodities, lost money at the same time. The fund's risk manager responds using Ang's factor theory. Which response is most appropriate?

**A.** The common losses are consistent with a multifactor model: many asset classes carried exposure to the same factors, such as volatility, growth and liquidity, so diversifying across asset-class labels did not diversify across factors.

**B.** The trustee is right: when correlations rise in a crisis, multifactor models break down, and only the CAPM's single market factor remains a valid description of risk.

**C.** The losses show that the asset classes had no exposure to systematic factors, since factor risks would have been diversified away within each asset class.

**D.** Under the arbitrage pricing theory, common factor losses would have been arbitraged away, so the losses must reflect behavioral mispricing rather than factor risk.

## Solutions • IM-1 d

### Solution IM-1 d.1

**Answer: B**

Bank Q33 • AnalystPrep Q.2376

#### Step 1 – The CAPM baseline

One factor, the market; bad times are low market returns; risk is one beta.

#### Step 2 – What a multifactor model adds

Bad times are defined by several factors (low growth, high inflation, value lagging growth, high volatility). Each factor that cannot be diversified away earns its own premium, so $E(r_i) = r_f + \sum_k \beta_{i,k} E(f_k)$.
$\Rightarrow$ An asset has several betas, each rewarded: B.

#### Interpretation

Everything else carries over from the CAPM: diversification still works and the average investor still holds the market. Only the definition of bad times widens.

#### Why the other options are wrong

**A is incorrect.** Describes the CAPM, not a multifactor model.
**C is incorrect.** Diversification still works: the tradeable factor portfolios remove idiosyncratic risk, which stays unpriced.
**D is incorrect.** Lesson 3 is identical in both models: the average investor holds the market.

**LO:** IM-1 d: Describe multifactor models and compare and contrast multifactor models to the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.5 Multifactor Models, Eq. 1.9 and Multifactor Model Lessons; SchweserNotes 2026 Book 5, Reading 83, LO 83.d

### Solution IM-1 d.2

**Answer: B**

Bank Q7 • BT Mock Q80 • BT answers P2.T9.20.1.1

#### Step 1 – What can be a factor

A factor is a systematic source of bad times shared across assets, with a premium that cannot be diversified away. Two families qualify: macro factors (growth, inflation, real rates, liquidity) and tradeable style factors (value, momentum, size, quality, low volatility).
$\Rightarrow$ Pairs A, C and D qualify.

#### Step 2 – Test strategy and reputation

These are firm-specific characteristics. They are diversifiable, define no common bad time, and cannot be held as a long-short factor portfolio.
$\Rightarrow$ B is the pair that is not a factor.

#### Interpretation

Filed in the bank under IM-1 a; it sits here because it asks what goes into a multifactor model. A multifactor regression adds factors that each define a different kind of bad time; it does not add firm traits.

#### Why the other options are wrong

**A is incorrect.** Quality and momentum are style factors implemented as long-short portfolios.
**C is incorrect.** Growth and inflation are the classic macro factors (GARP Eq. 1.9 uses them as $f_1$ and $f_2$).
**D is incorrect.** Real rates and liquidity are macro factors whose bad times (rate spikes, liquidity crises) are rewarded.

**LO:** IM-1 d: Describe multifactor models and compare and contrast multifactor models to the CAPM. • **Reference:**

Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.3 Factor Theory and 1.5 Multifactor Models

### Solution IM-1 d.3 — Answer: D

Bank Q34 • AnalystPrep Q.4589

#### Step 1 – Line up Lesson 4

CAPM: "the market factor is priced in equilibrium under the CAPM assumptions". Multifactor: "risk premiums exist for each factor assuming no arbitrage or equilibrium".
⇒ D states the contrast exactly.

#### Step 2 – Check the other lessons

Lesson 3 is identical (A is false), Lesson 1 survives unchanged in substance (B is false), and Lesson 5 moves from one beta to several factor betas, never to total volatility (C is false).

#### Interpretation

The weaker requirement in the multifactor column is why Ross named his model arbitrage pricing, and why the pricing kernel needs only no arbitrage. The CAPM needs its full list of assumptions to price even one factor.

#### Why the other options are wrong

**A is incorrect.** Lesson 3 holds in both models: the average investor holds the market; tilts are for investors who differ from the average.
**B is incorrect.** Lesson 1 holds in both: the tradeable version of each factor diversifies away idiosyncratic risk, which is priced in neither model.
**C is incorrect.** Lesson 5 becomes "risk is measured by factor betas", not total volatility.

**LO:** IM-1 d: Describe multifactor models and compare and contrast multifactor models to the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.5 Multifactor Model Lessons; SchweserNotes 2026 Book 5, Reading 83, LO 83.d

### Solution IM-1 d.4 — Answer: B

Bank Q35 • BT answers P2.T8.701.3

#### Step 1 – List the shared lessons

Diversification works; each investor has an optimal exposure; the average investor holds the market; factor risk is rewarded; risk is measured by betas; assets paying off in bad times have low premiums.

#### Step 2 – Lesson 6 in both columns

Assets that pay off in bad times are attractive and carry low premiums, whichever factor defines the bad times. The multifactor version only drops "when the market return is low".
⇒ B is shared.

#### Interpretation

The CAPM-only items are the narrow ones: one definition of bad times and one beta. The trap is to read a CAPM-only statement as general.

#### Why the other options are wrong

**A is incorrect.** True only in the CAPM; multifactor models also define bad times by low growth, high inflation or other factors.
**C is incorrect.** True only in the CAPM; multifactor models use several factor betas.
**D is incorrect.** False in both: idiosyncratic risk is diversified away and unpriced.

**LO:** IM-1 d: Describe multifactor models and compare and contrast multifactor models to the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.5 Multifactor Model Lessons

### Solution IM-1 d.5 — Answer: C

Bank Q36 • AnalystPrep Q.2381

**Step 1 – What went wrong**

Correlations from calm periods and a linear one-factor model miss how assets move together in bad times.

**Step 2 – What the SDF offers**

The pricing kernel $m$ is a single index of bad times across many factors and states of nature. The CAPM is the special case $m = a + b\, r_m$ $(b < 0)$: linear and one factor. The general $m(f_1, \dots, f_K)$ can be nonlinear ("the world is nonlinear").

**Step 3 – How it values a payoff**

$P_i = E[m \times \text{payoff}_i]$: payoffs in bad states, where $m$ is high, are weighted more heavily than with one fixed discount rate.

$\Rightarrow$ C.

**Interpretation**

Filed in the bank under IM-1 e; it sits here because it contrasts a multifactor kernel with the CAPM. The multifactor model widens the definition of bad times; the SDF is the notation that carries it.

**Why the other options are wrong**

**A is incorrect.** The CAPM is the linear, single-factor special case; it cannot capture several, nonlinear definitions of bad times.

**B is incorrect.** A single-index model on calm data shares the CAPM's limitation and adds a data-window problem.

**D is incorrect.** A fixed discount rate ignores that a payoff in a bad state is worth more than the same payoff in a good one.

**LO:** IM-1 d: Describe multifactor models and compare and contrast multifactor models to the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.5 Pricing Kernels, Eq. 1.3 to 1.6

### Solution IM-1 d.6 — Answer: A

New • built from GARP Ch.1 (1.2, 1.5, 1.8)

**Step 1 – The facts of 2008**

US large caps $-37.0\%$, emerging equities $-53.2\%$, high yield $-26.3\%$, equity hedge funds $-20.6\%$, commodities $-35.7\%$; only cash and high-grade, mainly government, bonds rose.

**Step 2 – Read them through factor theory**

Assets are bundles of factors. Volatility was very high, growth was low, policy uncertainty was large and liquidity dried up: factors that many asset classes shared. Common exposures produce common losses.

$\Rightarrow$ The commonality is evidence for a multifactor model of risk, not against diversification: A.

**Interpretation**

Ang's conclusion: diversification is not dead, but asset-class labels lulled investors into believing they were diversified when their factor exposures overlapped. Factor exposures and correlations also rise in bad times.

**Why the other options are wrong**

**B is incorrect.** Reverses the lesson: a single market factor cannot explain why so many classes fell together for different reasons; multiple shared factors can.

**C is incorrect.** Polarity flip: systematic factor risk is exactly what cannot be diversified away within an asset class.

**D is incorrect.** Reads "arbitrage" backwards: in the APT the factors cannot be arbitraged away, and their premiums compensate for losses like these.

**LO:** IM-1 d: Describe multifactor models and compare and contrast multifactor models to the CAPM. • **Reference:** Ang, Asset Management, Ch.6 (GARP 2025 Ch.1), 1.2 and 1.8 The 2008-2009 Financial Crisis (Table 1.1), 1.5 Multifactor Models

IM-2 • Factors • Vardeez: UNTIERED

# IM-2 c

P3 CONCEPT

Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk.

Workbook: none assigned • VARDEEZ FOCUS his "LO 84.c" (= IM-2 c)

**WHAT IS TESTED HERE**

**Why P3.** Vardeez flags it ("LO 84.C is Imp", Schweser numbering) in a reading he leaves untiered. No workbook shape was reported; B4's revision notes cover it under "Managing volatility risk".

**Vardeez:** his "LO 84.c". The notes follow his split: buying protection, selling protection, and the challenge.

**Reported exam shapes:** none. Built from Vardeez's focus, GARP 2025 Ch.2 (Ang) and SchweserNotes Reading 84. The bank holds four questions here; two are near-duplicates (Bank Q61 of Q63, Bank Q64 of the re-filed GARP item), so one GARP practice-exam question is re-filed from IM-1 a (Bank Q5) and three are new.

**What the questions look like:** statements on the two ways to cut volatility exposure (hold more bonds, buy protection) and their limits, why protection costs money (volatility has a negative price of risk), the crash profile of selling volatility, short-volatility exposure hidden in other strategies (carry, hedge funds, rebalancing), and why the volatility and expected-return relation is hard to pin down.

## 1. Volatility as a factor (recap of IM-2 b)

Volatility is one of the three most important macro factors, measured by the VIX index of S&P 500 option volatility. GARP's monthly correlations, March 1986 to December 2011:

| | VIX changes | Stock returns | Bond returns |
|---|---|---|---|
| VIX changes | 1.00 | −0.39 | 0.12 |
| Stock returns | −0.39 | 1.00 | −0.01 |
| Bond returns | 0.12 | −0.01 | 1.00 |

- **Stocks fall when volatility rises**, through two channels: the *leverage effect* (equity falls, debt is roughly constant, leverage and equity volatility rise) and a *discount-rate* channel (higher volatility raises the required return, so prices fall today). Large stocks averaged $-4.6\%$ in volatile times against $24.9\%$ in stable times (overall mean $11.3\%$).
- **Volatility is a factor with a negative price of risk.** Investors are so averse to it that, on average, they pay to avoid volatility risk rather than being paid to take it on. Many assets and strategies lose when volatility spikes: currency strategies especially, and hedge funds, which in aggregate sell volatility.

## 2. Two ways to mitigate volatility risk

1. **Hold more of the less volatile asset: bonds.** "Investors allergic to volatility could increase their holdings of bonds, but bonds do not always pay off during highly volatile periods." The correlation of bond returns with VIX changes is only $0.12$, so bonds give "some but not much respite". In 2008 to 2009 risk-free bonds did very well; in the turbulence of the late 1970s and early 1980s bonds did terribly, as did equities.
2. **Buy volatility protection** in derivatives markets: out-of-the-money puts, which pay off in high-volatility periods (these coincide with large market falls), or other contracts such as volatility swaps. Protection is not free: VIX implied volatility averages about 2 to 3 percentage points above realized volatility, so options are expensive on average and the buyer pays the volatility premium.

## 3. The other side: who sells protection

- **Collecting the premium** requires selling volatility, especially out-of-the-money puts. "Selling volatility is like selling insurance": high and steady payoffs in stable times, then, once every decade or so, a huge crash.
- **The record.** A Merrill Lynch volatility premium (swap) index rose steadily to 2007 with small blips (1998, 2001 to 2002, summer 2007), then lost close to 70% between September and November 2008. Its skewness is $-8.26$ over the whole sample but only $-0.37$ on data ending December 2007: before the crisis, selling volatility "looked like easy money". Volatility in the 1930s was even higher, and stayed high far longer.
- **Who should sell.** Only investors who can tolerate large losses in periods of very high volatility, which coincide with losses on most risky assets.
- **Rebalancing is a short volatility strategy.** A rebalancer earns a long-run volatility premium; the average investor, who holds the market and does not rebalance, is long volatility risk and gives that premium up. Two differences from selling options: rebalancing trades physical assets, not option (risk-neutral) volatility, so it does not earn the extra derivatives premium and its losses are potentially less steep; and pure volatility trading can be delta-hedged to take no view on expected returns.

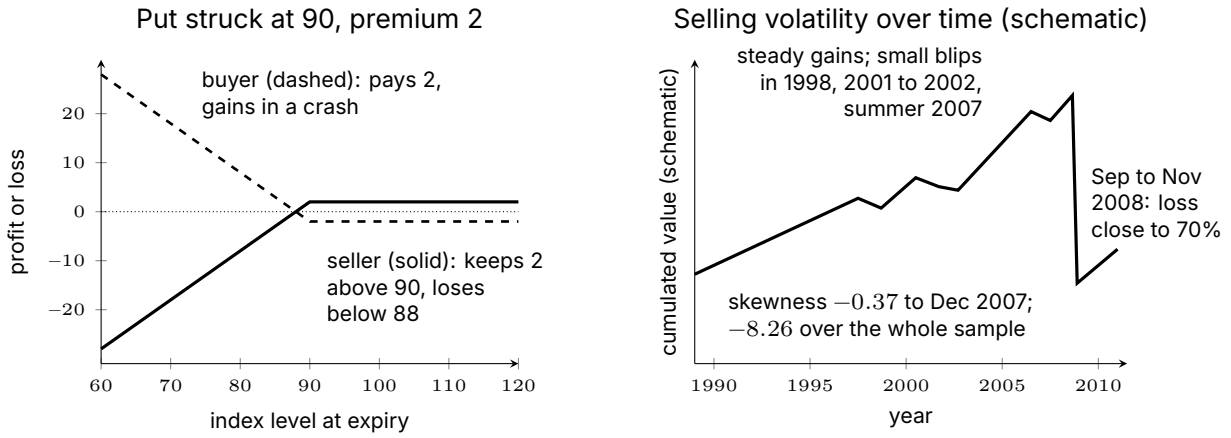



Figure 1. Left: profit or loss at expiry of a put on an index at 100, struck at 90 for a premium of 2 (illustrative); the seller is the solid line, the buyer the dashed line. Right: the shape of GARP's Figure 2.3 (a volatility premium index), drawn schematically with GARP's facts marked; the path is not data.

## 4. The challenges of managing volatility risk

- **The relation is hard to pin down.** "Constructing valuation models with volatility risk can be tricky because the relation between volatility and expected returns is time varying and switches signs." In $E(r_m) - r_f = \bar{\gamma}\sigma_m^2$, $\bar{\gamma}$ is positive in theory, but published estimates are positive, negative and zero; Glosten, Jagannathan and Runkle (1993) contains all three. What is undisputed: when volatility jumps, assets lose. (Keep this apart from the *contemporaneous* relation, which is reliably negative: $-0.39$.)
- **The cheap hedge is unreliable.** Bonds help only weakly and not in every high-volatility episode.
- **The reliable hedge is expensive.** Puts and volatility swaps cost the volatility premium in normal times.
- **Short-volatility exposure hides inside other positions**: currency carry, hedge funds, a rebalancing policy.
- **Short samples mislead.** A track record that ends before a crash shows little skew $(-0.37)$ for a strategy whose true skew is $-8.26$.
- **What the VIX measures.** It captures jump and skewness risk as well as volatility, but mainly volatility and the volatility risk premium; uncertainty risk is a separate factor in recent research, though highly correlated with volatility risk.

**Correction to the source notes.** SchweserNotes Reading 84 (LO 84.c) says that in the recent financial crisis volatility "caused risky assets, bonds and stocks included, to fall simultaneously", and its Key Concepts give 2007 to 2009 as the example of less volatile assets doing poorly. GARP says the opposite for 2008 to 2009: "risk-free bonds did very well". The episode when bonds did terribly along with equities was the late 1970s and early 1980s. Separately, Vardeez's figure caption calls the bond correlation of $0.12$ "a fifth of the magnitude" of the stock correlation of $-0.39$; it is about a third $(0.12/0.39 = 0.31)$.

| Position | Volatility exposure | Normal times | When volatility spikes | The catch |
|---|---|---|---|---|
| **Hold more bonds** | Reduced | Lower long-run return | Bonds rise weakly $(0.12)$; did well in 2008 to 2009, badly around 1980 | Partial, unreliable hedge |
| **Buy protection (OTM puts, volatility swaps)** | Long | Small steady losses: the premium | Large payoff | Pays the volatility premium |
| **Sell protection (sell OTM puts)** | Short | High, steady income | Huge losses (close to 70%, Sep to Nov 2008) | Skewness $-8.26$; only for those who can bear it |
| **Rebalance to fixed weights** | Short (implicitly) | Earns a long-run volatility premium | Losses, potentially less steep than option selling | No derivatives premium |
| **Hold the market, never rebalance** | Long | Gives up the volatility premium | Market losses | The average investor's position |

**Exam traps**

- **Direction (polarity).** Dislike volatility: buy protection. Can bear it: sell protection. The seller earns steadily and loses rarely but hugely, not the reverse.

- **Bonds as a full hedge (magnitude).** The bond correlation with VIX changes is $0.12$, not large; and the stock and bond correlation is $-0.01$, not strongly negative.
- **Which crisis hurt bonds?** Risk-free bonds did very well in 2008 to 2009; bonds and equities both did terribly in the late 1970s and early 1980s.
- **Sign of the volatility premium.** Implied volatility is above realized, so options are expensive and the seller collects the premium; volatility has a negative price of risk.
- **Two relations confused.** Contemporaneous stock returns and volatility changes: reliably negative. Volatility and *expected* returns ($\bar{\gamma}$): unstable, switching sign. $\bar{\gamma}$ is positive in theory.
- **Rebalancing mislabeled.** It is short volatility, not long; and it is not the same as selling options.
- **Volatility not tradeable.** It is the one macro factor that is traded directly in scale.

## Practice questions: IM-2 c

### Question IM-2 c.1

Bank Q5 • GARP PE1 Q79 (re-filed from IM-1 a)

A group of junior analysts is attending a training session on how risk factors are used in investment management. During the training, several macroeconomic factors as well as investment style factors are described and their effects on investors in different situations are evaluated. Which of the following is correct according to the theory of factor risk?

**A.** Investors in an asset that is likely to have a high payoff during adverse market conditions usually earn high risk premiums on that asset.

**B.** Investors who invest in the same assets are exposed to the same set of risk factors and have the same level of optimal exposure.

**C.** Investors are affected negatively when there is an unexpected decline in economic growth, regardless of the type of their investments.

**D.** Investors in an asset with a large exposure to volatility risk can hedge against this risk by purchasing out-of-the-money put options.

### Question IM-2 c.2

New • built from GARP Ch.2 (2.3, Volatility)

The CIO of a Japanese life insurer wants to cut the equity portfolio's exposure to volatility shocks and proposes simply raising the allocation to government bonds. A risk analyst checks the proposal against Ang's evidence for March 1986 to December 2011:

| | |
|---|---|
| Correlation of monthly VIX changes with stock returns | $-0.39$ |
| Correlation of monthly VIX changes with bond returns | $0.12$ |
| Correlation of stock returns with bond returns | $-0.01$ |

Which assessment is most appropriate?

**A.** Bonds are a reliable volatility hedge: their returns rise almost one for one with VIX changes, so the higher allocation offsets equity losses in volatility spikes.

**B.** Bonds add to the volatility exposure rather than reduce it, because bond returns are strongly negatively correlated with VIX changes, as stock returns are.

**C.** Bonds give some but not much protection: they rise only weakly when volatility jumps, and while risk-free bonds did very well in 2008 to 2009, they did terribly with equities around 1980.

**D.** Bonds are a reliable hedge because stock and bond returns are strongly negatively correlated, so bonds rise whenever stocks fall.

**Question IM-2 c.3** Bank Q63 • BT answers P2.T8.703.1 (rebuilt)

In the relationship $E(r_m) - r_f = \gamma \times \sigma_m^2$, $\gamma$ is the risk aversion of the average investor and $\sigma_m^2$ is the variance of the market return. Which of the following statements about volatility risk and premiums is most accurate?

**A.** The coefficient $\gamma$ must be negative in theory because investors dislike volatility, so a positive estimate in the data indicates a misspecified model.

**B.** Because realized volatility is on average 2% to 3% below VIX-implied volatility, selling volatility protection earns a premium, but the seller is exposed to large losses in a crash.

**C.** Selling out-of-the-money put options is a low-risk strategy because the payoffs are stable over long samples that include the 2007 to 2009 crisis.

**D.** Volatility carries a positive premium, so an investor earns it simply by holding a long position in equities.

**Question IM-2 c.4** Bank Q62 • GARP PE2 Q77 • Mock 4 Q78

A risk analyst at a bank is asked to prepare a report that tracks the relationship between volatility and asset performance. The analyst assesses the performance of various asset classes using empirical evidence over the last three decades and compares returns on those asset classes with changes in market volatility. Which of the following would be a correct statement for the analyst to include in the report?

**A.** Currency strategies such as currency carry trades tend to perform poorly during periods of high volatility.

**B.** When volatility is rising, all assets are either positively or negatively affected, with the exception of risk-free bonds.

**C.** Whether the relationship between stock returns and volatility is positive or negative depends on the phase of the business cycle.

**D.** When volatility is rising, stock returns tend to increase but bond returns tend to decrease.

**Question IM-2 c.5** New • built from GARP Ch.2 (2.3, Volatility)

A Dutch pension fund rebalances its 60/40 equity and bond mix back to target every quarter. A trustee asks how this policy relates to volatility risk. According to Ang, which response is correct?

**A.** Rebalancing is a long volatility strategy, because selling assets after they have risen works like buying protection against a crash.

**B.** Rebalancing carries no volatility exposure, because restoring fixed weights is neutral to how volatile markets are.

**C.** Rebalancing is equivalent to selling out-of-the-money puts, because both trade option-implied volatility and earn the same premium.

**D.** Rebalancing is a short volatility strategy that earns a long-run volatility premium, but its crash losses are potentially less steep than selling options because it trades physical assets.

**Question IM-2 c.6** New • built from GARP Ch.2 (2.3, Eq. 2.1)

A quantitative analyst at a Swiss asset manager wants to build a valuation model that prices volatility risk through the CAPM relation $E(r_m) - r_f = \bar{\gamma}\sigma_m^2$, where $\bar{\gamma}$ is the risk aversion of the average investor. Which of the following describes the main challenge the analyst faces?

**A.** The relation between volatility and expected returns is time varying and switches sign: estimates of $\bar{\gamma}$ are positive, negative and zero, sometimes within a single study.

**B.** The relation is reliably positive in the data, but it is too small to estimate without very long samples of returns.

**C.** Stock returns and volatility changes move together in some decades and against each other in others, so the contemporaneous relation cannot be signed.

**D.** Volatility cannot be traded, unlike the other macro factors, so no market price for volatility risk can be observed.

# Solutions • IM-2 c

## Solution IM-2 c.1

**Answer: D**

Bank Q5 • GARP PE1 Q79

### Step 1 – What hedges volatility risk

Periods of high volatility coincide with large market falls. Assets that pay off in those periods, such as out-of-the-money puts, hedge volatility risk.
$\Rightarrow$ D is the method GARP names.

### Step 2 – Test A to C against factor theory

A reverses CAPM Lesson 6: an asset that pays off in bad times is valuable and earns a *low* premium. B confuses identical exposures with identical optimal exposures, which differ across investors. C ignores investors who gain in low growth (government bonds rise; a bankruptcy lawyer's income rises).

### Interpretation

Filed in the bank under IM-1 a; it sits here because its answer is the core mitigation method. The hedge costs money: volatility has a negative price of risk, so the buyer of puts pays the volatility premium in normal times.

### Why the other options are wrong

**A is incorrect.** Polarity flip: a high payoff in bad times makes the asset a hedge, which earns a low, possibly negative, premium.
**B is incorrect.** The same assets carry the same exposures, but each investor's optimal exposure depends on his own risk aversion and circumstances.
**C is incorrect.** Absolute claim: some holdings, such as government bonds, do well when growth falls.

**LO:** IM-2 c: Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.3 Volatility; Ch.6 (GARP 2025 Ch.1), 1.3 and 1.4; GARP Practice Exam 1, Q79

## Solution IM-2 c.2

**Answer: C**

New • built from GARP Ch.2 (2.3, Volatility)

### Step 1 – Read the correlations

Bond returns correlate $+0.12$ with VIX changes: they rise a little when volatility jumps, with under a third of the magnitude of the stock correlation ($0.12/0.39 = 0.31$). The stock and bond correlation, $-0.01$, is essentially zero.
$\Rightarrow$ Bonds offer "some but not much respite".

### Step 2 – Check history

In 2008 to 2009 risk-free bonds did very well. In the late 1970s and early 1980s bonds did terribly, as did equities.
$\Rightarrow$ The protection is partial and not reliable in every volatile period: C.

### Interpretation

Holding more bonds is the cheap method; buying protection with puts or volatility swaps is the reliable one, but it costs the volatility premium. That trade-off is the assessment the LO asks for.

### Why the other options are wrong

**A is incorrect.** Magnitude error: $0.12$ is a weak correlation, nothing like one for one.
**B is incorrect.** Polarity flip: the bond correlation with VIX changes is positive ($+0.12$), not

negative like the stock correlation.
**D is incorrect.** Misread table: the stock and bond correlation is $-0.01$, not strongly negative.

**LO:** IM-2 c: Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.3 Macro Factors, Volatility (VIX correlation table and Figure 2.2)

### Solution IM-2 c.3 Answer: B

Bank Q63 • BT answers P2.T8.703.1

#### Step 1 – The sign of $\gamma$

In theory $\gamma$ is positive: a more risk-averse average investor demands a higher premium per unit of variance. The estimates in the data are positive, negative or zero.

#### Step 2 – Who collects the volatility premium

VIX implied volatility is on average 2 to 3 percentage points above realized volatility, so options are expensive and the seller of protection collects the premium. Volatility has a negative price of risk.

#### Step 3 – The seller's risk

Like selling insurance: steady income in normal times, then losses close to 70% from September to November 2008; skewness $-8.26$ with the crisis against $-0.37$ without it.
$\Rightarrow$ Only B states both the premium and the crash risk correctly.

#### Interpretation

GARP's conclusion: only an investor who can tolerate large losses in high-volatility periods should sell volatility protection.

#### Why the other options are wrong

**A is incorrect.** Reverses theory and data: $\gamma$ is positive in theory; it is the empirical estimates whose sign varies.
**C is incorrect.** Ignores the crisis: payoffs look stable only in samples that end before 2008.
**D is incorrect.** Confuses an asset premium with the volatility premium. Being long equities is being long volatility risk, which has a negative price; the premium is collected by selling volatility.

**LO:** IM-2 c: Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.3 Volatility (Figure 2.3, Eq. 2.1); SchweserNotes 2026 Book 5, Reading 84, LO 84.c

### Solution IM-2 c.4 Answer: A

Bank Q62 • GARP PE2 Q77 • Mock 4 Q78

#### Step 1 – What the evidence says

Volatility is negatively linked to the returns of many assets and strategies, and "currency strategies fare especially poorly in times of high volatility". Hedge funds, in aggregate, sell volatility.
$\Rightarrow$ A is a documented finding.

#### Step 2 – Test B to D

Stocks fall when volatility rises (correlation $-0.39$), and bonds rise a little ($+0.12$), so D is reversed. The contemporaneous relation between stock returns and volatility is reliably negative; it does not flip with the business cycle (C). B is an unsupported blanket claim.

#### Interpretation

For managing volatility risk this is the hidden-exposure challenge: a carry trade or a hedge fund can be short volatility without saying so. GARP's "switches signs" remark is about volatility

and *expected* returns (the estimate of $\gamma$), not about contemporaneous returns, which is what C misuses.

### Why the other options are wrong

**B is incorrect.** Not a documented finding, and the exception it names is not in the evidence.
**C is incorrect.** Confuses the two relations: contemporaneous returns move against volatility; it is the volatility and expected-return trade-off whose sign is unstable, and GARP does not tie it to the business-cycle phase.
**D is incorrect.** Reverses both signs: stocks fall and bonds rise slightly when volatility rises.

**LO:** IM-2 c: Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.3 Volatility; GARP Practice Exam 2, Q77

## Solution IM-2 c.5 — Answer: D

New • built from GARP Ch.2 (2.3, Volatility)

### Step 1 – Rebalancing's exposure

Rebalancing buys what has fallen and sells what has risen. GARP: "rebalancing as a portfolio strategy is actually a short volatility strategy" that reaps a long-run volatility risk premium; the investor who does not rebalance, the average investor holding the market, is long volatility risk and loses that premium.

### Step 2 – How it differs from selling options

Rebalancing trades physical assets, not risk-neutral (option) volatility, so it does not earn the additional derivatives premium, and its losses are potentially much less steep than those from trading volatility in derivatives markets.
$\Rightarrow$ D.

### Interpretation

For the trustee, the fund already carries some short-volatility exposure through its policy. That is one of the challenges of managing volatility risk: it hides in positions that are not labeled as volatility trades.

### Why the other options are wrong

**A is incorrect.** Polarity flip: rebalancing is short, not long, volatility; it loses in sustained crashes as it keeps buying the falling asset.
**B is incorrect.** Fixed weights do not make it neutral; the buy-low, sell-high pattern is a short volatility position.
**C is incorrect.** Sibling swap: selling options trades option-implied volatility and earns an extra premium that rebalancing does not.

**LO:** IM-2 c: Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.3 Volatility (rebalancing as a short volatility strategy)

## Solution IM-2 c.6 — Answer: A

New • built from GARP Ch.2 (2.3, Eq. 2.1)

### Step 1 – What theory says

$\bar{\gamma}$, the average investor's risk aversion, is positive in theory: more variance should raise the expected market return.

### Step 2 – What the data say

Estimates are positive, negative or zero; Glosten, Jagannathan and Runkle (1993) contains all three in the same paper, and theory shows the risk-return relation can be negative and change over time.
⇒ The relation "is time varying and switches signs and is thus very hard to pin down": A.

### Interpretation

What is undisputed is the contemporaneous fact: when volatility rises sharply, assets lose. That is why GARP's practical advice is about who should sell protection, not about a precise price of volatility risk.

### Why the other options are wrong

**B is incorrect.** The estimates are not reliably positive; their sign is unstable.
**C is incorrect.** Confuses the two relations: contemporaneous stock returns and volatility changes are reliably negative $(-0.39)$.
**D is incorrect.** Volatility is the exception among macro factors: it is traded directly, through options and volatility swaps.

**LO:** IM-2 c: Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk. • **Reference:** Ang, Asset Management, Ch.7 (GARP 2025 Ch.2), 2.3 Volatility, Eq. 2.1 and 2.4 Dynamic Factors

IM-4 • Portfolio Construction • Vardeez: VERY IMPORTANT

# IM-4 b

P3 CONCEPT

Evaluate the motivation for and the methods used for refining alphas in the implementation process.

Workbook: D28 Portfolio Construction and the No-Trade Region • Tier D • notes only

**WHAT IS TESTED HERE**

**Why P3.** Tier D workbook topic (D28, Map #66, reported on 1 of 16 recall sheets); Vardeez's focus for IM-4 does not name it.
**Vardeez:** no pointer for this LO.

**Reported exam shapes**

- **S1** Alpha refinement: scaling, trimming, and neutralisation (benchmark, cash, risk-factor, industry). Scaling and trimming are drilled here; neutralisation in IM-4 c.

**What the questions look like:** statement questions on why alphas are refined (constraints and noisy alphas; refining and then optimising without constraints reproduces any constrained process and shows what the constraints cost) and on what scaling and trimming do. One GARP-tagged mock question (Question IM-4 b.6) tests the alpha scale with numbers. It is kept outside the CONCEPT tag because it is the way the mocks test scaling.
**Background:** the five inputs and the objective $\alpha_P - \lambda_A \psi_P^2$ are in IM-4 d, Section 1 (P2).

## 1. Why refine the alphas (the motivation)

"Active management should be easy with the right alphas. Sometimes it isn't." Two reasons:

- **Constraints.** Clients impose some (no short sales, limited cash, restricted coverage for liquidity or self-dealing); managers add their own (sector or industry neutrality, limits on single positions, no bet on the benchmark). "These limits can make the portfolio less efficient, but they are hard to avoid."
- **Noisy inputs.** Alphas are "often unreasonable and subject to hidden biases". "Implementation schemes are, in part, safeguards against poor research."

**Another way to reach the same portfolio: adjust the inputs.** Any constrained optimum $h_{PA}$ can be reproduced by a simple *unconstrained* mean/variance optimisation run on modified alphas:

$$\alpha' = 2\,\lambda'_A\,V\,h_{PA} \qquad \text{with} \qquad \lambda'_A = \frac{\text{IR}}{2\psi_P} \qquad \text{(Eqs. 4.1, 4.2)}$$

**GARP's example (Table 4.1).** Twenty Major Market Index stocks, random alphas, $\lambda_A = 0.0833$. The unconstrained optimum sells American Express and Coca-Cola short and puts almost 18% (17.95%) in 3M, against a benchmark weight of 9.23%. Add two constraints, no short sales and no holding more than 5% above its benchmark weight: American Express and Coca-Cola drop to zero and 3M to exactly 14.23%. The modified alphas that deliver this portfolio without constraints are "pulled in toward zero": their standard deviation falls from 2.00% to 0.57%.

**Correction to the source notes.** GARP (Ch.4, 4.3) concludes that the constraints "effectively shrank the IC by 62 percent". With the standard deviations it quotes, recomputed from Table 4.1 as 2.000% and 0.574%, the shrinkage is $1-0.57/2.00 = 71.5\%$. The 62% is $1-0.57/1.50$: the modified alphas measured against the 1.5% scale of the text's own illustration (IC 0.05, residual risk 30%), not against the original alphas. Keep the point (constraints cut the effective IC sharply); do not rely on the 62%.

- "We can replace any portfolio construction process, regardless of its sophistication, by a process that first refines the alphas and then uses a simple unconstrained mean/variance optimization to determine the active positions."
- This "is not an argument against complicated implementation schemes. It simply focuses our attention on a reason for the complexity." If the scheme is a safeguard against unreasonable inputs, attack the inputs directly: preprocess the alphas, or design the procedure openly as an "input moderator".
- **What it buys.** Both investor and manager constraints are handled; attention goes to the alpha inputs and to what each constraint does to returns (Schweser); and the cost of the constraints is shown "explicitly, rather than hiding it under a rug of optimizer constraints". It does not make the cost go away: the refined alphas deliver the same, less efficient, constrained portfolio.

## 2. Scale the alphas

Alphas have a natural structure and therefore a natural scale:

$$\boxed{\alpha = \text{volatility} \times \text{IC} \times \text{score}} \quad \Longrightarrow \quad \boxed{\text{Std}\{\alpha\} = \text{volatility} \times \text{IC}}$$

Volatility (residual risk) and the IC are roughly constant across the set; the score has mean 0 and standard deviation 1. So the alphas should have mean 0 and a standard deviation, the **scale**, of volatility $\times$ IC. GARP: IC 0.05 and residual risk 30% give a scale of 1.5%: about two-thirds of alphas between $-1.5\%$ and $+1.5\%$, about 5% beyond $\pm 3.0\%$ (Figure 1). Schweser: IC 0.10 and 30% give 3%.

- "If the alphas input to portfolio construction do not have the proper scale, then rescale them."
- **Alternative (note 3).** Compute the information ratio the alphas imply, $\text{IR}_0 = \sqrt{\alpha^T V^{-1} \alpha}$, and rescale them by $\text{IR}/\text{IR}_0$ to match the manager's ex ante IR.
- The scale matters more once costs are in the optimiser: risk aversion can no longer absorb a misscaled set (IM-4 d).

## 3. Trim alpha outliers

"Very large positive or negative alphas can have undue influence."

- **Standard trimming.** Examine every alpha larger in magnitude than about **three times the scale**. Those that rest on questionable data are set to **zero**; the genuine ones are **pulled in to three times the scale**.
- **The extreme version.** Force the alphas into a normal distribution with benchmark alpha 0 and the required scale (each stock's cumulative benchmark weight is mapped to a normal quantile). Extreme "because it typically utilizes only the ranking information in the alphas and ignores the size of the alphas". Afterwards, "you must recheck benchmark neutrality and scaling".

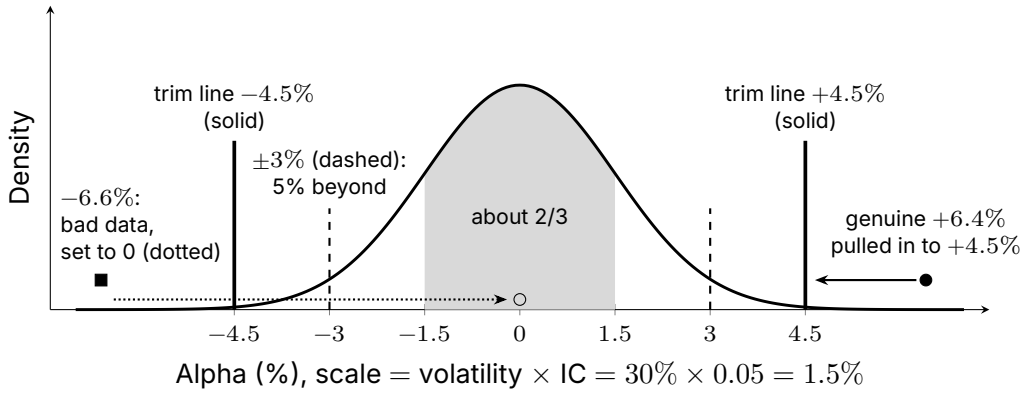



Figure 1. Alphas at GARP's illustrative scale of 1.5% (solid curve). About two-thirds lie in the grey band of $\pm 1$ scale and about 5% beyond the dashed lines at $\pm 2$ scale. Alphas beyond the solid trim lines at $\pm 3$ scale are examined: a genuine one (dot) is pulled in to 4.5%; one resting on questionable data (square) is set to zero.

## 4. The order of the steps

GARP presents scaling and trimming first "because of their simplicity", but "when we implement alpha analysis, we impose scaling and trimming as the final step in the process". Neutralisation (IM-4 c) comes first, because it can change the scale; and after the extreme trimming, neutrality and scale are rechecked.

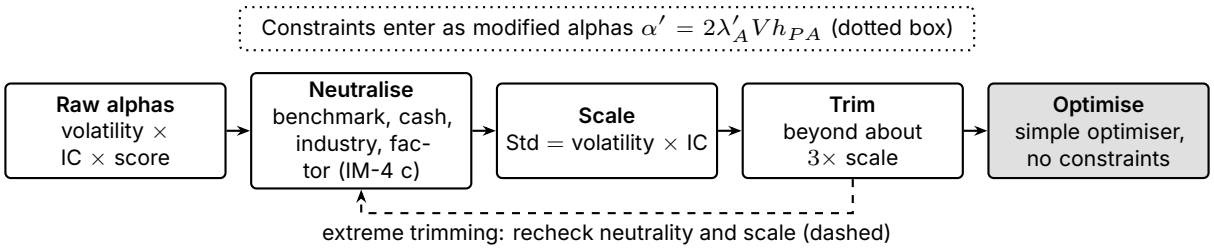



Figure 2. Alpha analysis in the order GARP implements it: neutralisation first, scaling and trimming last, then a simple optimisation (grey box). The dashed loop is the recheck required after forcing alphas into a normal distribution.

| | Scaling | Trimming | Neutralisation (IM-4 c) |
|---|---|---|---|
| **Fixes** | Alphas of the wrong size overall | A few extreme alphas with undue influence | Biases and unwanted bets in the alphas |
| **How** | Rescale to Std = volatility × IC (or match the ex ante IR) | Zero if questionable data; pull genuine ones in to 3× scale | Recentre so a chosen exposure has zero alpha |
| **Uses** | The size of the alphas | The size, case by case (the extreme version: ranks only) | The alphas and the exposures to be neutralised |
| **When** | Last, with trimming | Last, with scaling | First |

**Exam traps**

- **Order of operations.** Refine the alphas *then* optimise; optimising first and refining afterwards is the distractor.
- **Refining does not remove the cost.** Modified alphas reproduce the constrained portfolio; they

reveal what the constraints cost (the shrunken IC), they do not recover it.

- **Scale.** Std$\{\alpha\}$ = volatility × IC; the score has standard deviation 1 and drops out. Omitting the IC (or using the wrong manager's IC) is the numeric trap.
- **Trimming is selective.** Only alphas resting on questionable data go to zero; genuine outliers are capped at about three times the scale.
- **Forcing to normal.** The extreme version keeps only the rankings and must be followed by a recheck of neutrality and scale.
- **Sibling swap.** Scaling sets the size, trimming handles outliers, neutralisation removes bets; "interpolation" and "stratification" are not alpha refinements.

## Practice questions: IM-4 b

### Question IM-4 b.1

Bank Q146 • AnalystPrep Q.2441 (rebuilt) • re-filed from IM-4 a

At an investment committee meeting, a newly hired portfolio manager says: "Active management should be easy with the right alphas." Which of the following is the most appropriate assessment of this comment?

**A.** Correct: constraints change the weights but do not make the portfolio any less efficient, so the right alphas guarantee the optimal result.

**B.** Correct: refining the alphas for each constraint before optimizing fully eliminates the cost of the constraints, so the constrained portfolio is as efficient as the unconstrained one.

**C.** Incorrect: real portfolios are built under constraints (for example no short sales, cash and position limits), which make the constrained portfolio less efficient than the unconstrained optimum even when the alphas are right.

**D.** Correct: because most active managers do not short and hold little cash, their portfolios are more efficient than unconstrained portfolios.

### Question IM-4 b.2

Bank Q153 • Schweser Module Quiz 86.1 Q2

A client's investment policy prohibits short sales, and the portfolio manager imposes her own limit on active positions. The risk team reviews how the optimization process handles these restrictions. Which of the following is correct with respect to adjusting the optimal portfolio for portfolio constraints?

**A.** By refining the alphas and then optimizing, it is possible to include constraints of both the investor and the manager.

**B.** No reliable method exists.

**C.** By refining the alphas and then optimizing, it is possible to include constraints of the investor, but not of the manager.

**D.** By optimizing and then refining the alphas, it is possible to include constraints of both the investor and the manager.

### Question IM-4 b.3

Bank Q152 • BT P2.T9.21.3.3

Kelly is constructing a portfolio for which her initial unconstrained optimization generates two outcomes that violate her fund's style policy: it implies several short positions, and overweights in three positions that would exceed her self-imposed concentration limit (positions cannot exceed benchmark holdings by more than 10.0%). Kelly refines the alphas by scaling them and then trimming the extreme alphas. What is the most likely reason for Kelly to refine the alphas?

**A.** Performance: the refined alphas generate higher information ratios.

**B.** Precision: the active risk aversion coefficient is more accurately calibrated.

**C.** Sophistication: the refined alphas are more conducive to quadratic programming.

**D.** Simplification: to be able to size the active positions with a simple unconstrained mean-variance optimization.

### Question IM-4 b.4

Bank Q149 • AnalystPrep Q.2444 • AnalystPrep Q.2744

A junior analyst at a quantitative equity manager is documenting the alpha-analysis step that sits between the research team's raw forecasts and the optimizer. Which of the following is NOT a procedure used to refine alphas?

**A.** Trimming

**B.** Interpolation

**C.** Scaling

**D.** Neutralization

### Question IM-4 b.5

Bank Q150 • AnalystPrep Q.2457 (rebuilt)

The head of quantitative research asks a new analyst to explain one step of the firm's alpha-refinement routine. In refining alphas for portfolio construction, what is the primary purpose of scaling the alphas?

**A.** To reduce, or set to zero, extremely large alphas that may stem from questionable data.

**B.** To remove unintended bets, such as a beta or industry tilt, from the alphas.

**C.** To give the alphas the appropriate scale, with a standard deviation approximately equal to residual volatility $\times$ information coefficient.

**D.** To adjust each alpha for the transaction costs of establishing the position.

### Question IM-4 b.6

Bank Q147 • Mock 2 Q62 • kept outside the CONCEPT tag

A quantitative analyst regresses the returns of 500 stocks against the S&P 500. The resulting pool of 500 alphas has a residual risk of 18% and an information coefficient of 10%. Assuming the alphas are normally distributed with a mean of 0%, roughly how many stocks have an alpha greater than 3.6% or less than $-3.6\%$?

**A.** 5

**B.** 25

**C.** 250

**D.** 475

**Question IM-4 b.7** New • built from GARP Ch.4, 4.3 • S1

A quantitative analyst reviews a set of 600 alphas for a US equity portfolio. The alphas have a mean of zero and a scale of 1.5% (information coefficient 0.05, residual risk 30%), and a handful of stocks have alphas larger than 4.5% in magnitude. Following Grinold and Kahn's procedure for trimming alpha outliers, which of the following is correct?

**A.** Every alpha beyond three times the scale should be set to zero, because any such alpha is an outlier.

**B.** Forcing the alphas into a normal distribution with the required scale is the milder alternative, because it keeps the size of each alpha.

**C.** Trimming should be carried out before neutralization, so that the scale does not need to be checked again afterwards.

**D.** Alphas found to rest on questionable data should be set to zero, and the remaining genuine alphas should be pulled in to $\pm 4.5\%$.

## Solutions • IM-4 b

### Solution IM-4 b.1 — Answer: C

Bank Q146 • AnalystPrep Q.2441

**Step 1 – What GARP says about the comment**

"Active management should be easy with the right alphas. Sometimes it isn't. Most active managers construct portfolios subject to certain constraints, agreed upon with the client." Those limits "can make the portfolio less efficient, but they are hard to avoid".

⇒ The comment is incorrect: constraints shrink the feasible set.

**Step 2 – Where refining fits**

Refining the alphas (modified alphas $\alpha' = 2\lambda'_A V h_{PA}$) lets a simple unconstrained optimiser deliver the *constrained* portfolio. The efficiency loss is still there; it now shows up as alphas pulled toward zero.

**Interpretation**

This is the motivation half of the LO. Constraints are the reason refinement exists, and the refined alphas make the cost of the constraints visible (in GARP's example the alpha standard deviation fell from 2.00% to 0.57%).

**Why the other options are wrong**

**A is incorrect.** A restricted feasible set cannot produce a more efficient optimum than an unrestricted one.

**B is incorrect.** Refining imposes the constraint through the inputs; it does not remove its cost.

**D is incorrect.** Reverses the effect: no-short and cash limits are constraints and reduce efficiency.

**LO:** IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.2 Alphas and Portfolio Construction

### Solution IM-4 b.2 — Answer: A

Bank Q153 • Schweser Module Quiz 86.1 Q2

**Step 1 – The method**

Find the alphas under which an unconstrained optimiser would choose the constrained portfolio (for the no-short client, alphas that produce non-negative weights), then optimise.

⇒ Refine first, optimise second.

**Step 2 – Scope**

The investor's restriction (no short sales) and the manager's own limit (active positions) are both just constraints on the holdings, so both can be built into the alphas.

**Interpretation**

"We can replace any portfolio construction process, regardless of its sophistication, by a process that first refines the alphas and then uses a simple unconstrained mean/variance optimization."

**Why the other options are wrong**

**B is incorrect.** A reliable method exists: refine the alphas.

**C is incorrect.** Manager constraints (position limits, sector neutrality) are handled the same way as the client's.

**D is incorrect.** Wrong order: the alphas must be refined before the optimisation.

**LO:** IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.2, Eqs. 4.1 and 4.2;

SchweserNotes 2026 Book 5, Reading 86, LO 86.b

### Solution IM-4 b.3 Answer: D

Bank Q152 • BT P2.T9.21.3.3

**Step 1 – What the violations would otherwise need**

Short positions and concentration breaches would normally be fixed inside a constrained optimisation.

**Step 2 – What refining does instead**

Scaling and trimming pull the alphas in so that a simple unconstrained mean/variance optimisation produces positions inside the policy.

$\Rightarrow$ The motive is simplification: the constraints move into the inputs.

**Interpretation**

Schweser's key concept: refining "can achieve the same goal as a constrained optimization approach, but has the advantage of focusing on the alpha inputs and the effects of individual constraints on portfolio returns".

**Why the other options are wrong**

**A is incorrect.** Refining to satisfy constraints pulls the alphas toward zero and usually lowers the effective IC and IR (Section 1); it does not raise them.

**B is incorrect.** Refinement works on the alphas, not on the risk aversion, which is backed out from the IR and the target active risk (IM-4 e).

**C is incorrect.** The aim is to avoid a complex constrained program, not to feed one.

**LO:** IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.2 and 4.3; SchweserNotes 2026 Book 5, Reading 86, LO 86.b

### Solution IM-4 b.4 Answer: B

Bank Q149 • AnalystPrep Q.2444

**Step 1 – The refinements in GARP**

Section 4.3, Alpha Analysis: scale the alphas, trim alpha outliers, and neutralisation (benchmark, cash, industry and risk-factor).

**Step 2 – The odd one out**

Interpolation estimates values between known points; it plays no part in alpha analysis.

$\Rightarrow$ B.

**Interpretation**

The three refinements answer three different defects: wrong overall size (scaling), a few extreme values (trimming) and hidden bets (neutralisation).

**Why the other options are wrong**

**A is incorrect.** Trimming is a refinement: alphas beyond about three times the scale are examined and zeroed or capped.

**C is incorrect.** Scaling is a refinement: the alphas are rescaled to volatility $\times$ IC.

**D is incorrect.** Neutralisation is a refinement: it removes biases and undesirable bets.

**LO:** IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Alpha Analysis

### Solution IM-4 b.5 — Answer: C

Bank Q150 • AnalystPrep Q.2457

**Step 1 – The structure of an alpha**

$\alpha$ = volatility × IC × score, with the score of mean 0 and standard deviation 1.
$\Rightarrow$ The alphas should have mean 0 and standard deviation (scale) volatility × IC.

**Step 2 – What scaling does**

"If the alphas input to portfolio construction do not have the proper scale, then rescale them." For example, IC 0.10 and residual risk 30% give a scale of 3%.

**Interpretation**

Scaling matters because the optimiser takes the alphas at face value: a set twice too large produces bets twice too large, and with transaction costs the risk aversion can no longer compensate (IM-4 d).

**Why the other options are wrong**

**A is incorrect.** Describes trimming.
**B is incorrect.** Describes neutralisation.
**D is incorrect.** Costs are handled in the optimisation and the no-trade region (IM-4 g), not by scaling the alphas.

**LO:** IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. •
**Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Scale the Alphas

### Solution IM-4 b.6 — Answer: B

Bank Q147 • Mock 2 Q62

**Step 1 – Scale of the alphas**

$$\text{Std}\{\alpha\} = \text{residual risk} \times \text{IC} = 18\% \times 0.10 = 1.8\%$$

**Step 2 – Express the cut-off in standard deviations**

$$3.6\%/1.8\% = 2 \text{ standard deviations}$$

**Step 3 – Count the tails**

$$2 \times (1 - N(2)) = 2 \times 0.02275 = 4.55\%, \qquad 500 \times 4.55\% = 22.75 \approx 25$$

$\Rightarrow$ About 23 stocks lie beyond $\pm 3.6\%$; 25 is the closest option.

**Interpretation**

This is GARP's "natural scale" statement in numbers: about two-thirds of alphas within one scale, about 5% beyond two. The trimming line, three times the scale ($\pm 5.4\%$), would catch only about 1.35 stocks (0.27%) if the alphas were truly normal; many more beyond it is a sign of bad data or a wrong scale. This numeric question is kept although the LO is CONCEPT, because it is the way the mock tests scaling.

**Why the other options are wrong**

**A is incorrect.** Too few: 5 stocks is 1% of the pool, the tail beyond about $\pm 2.58$ standard deviations, not beyond $\pm 2$.
**C is incorrect.** Half the pool: the stocks on one side of the mean, not beyond two standard deviations.
**D is incorrect.** The complement: the stocks inside $\pm 2$ standard deviations ($500 \times 95.45\% = 477$).

**LO:** IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Scale the Alphas

### Solution IM-4 b.7 — Answer: D

New • built from GARP Ch.4, 4.3

#### Step 1 – Which alphas to examine

Three times the scale: $3 \times 1.5\% = 4.5\%$. Every alpha beyond $\pm 4.5\%$ is examined, not deleted automatically.

#### Step 2 – What to do with each

"Some of these alphas depend upon questionable data and should be ignored (set to zero), while others may appear genuine. Pull in these remaining genuine alphas to three times scale in magnitude."

$\Rightarrow$ Questionable: 0. Genuine: $\pm 4.5\%$.

#### Interpretation

Trimming limits the "undue influence" of a few extreme forecasts without throwing away the information in genuine ones (Figure 1). It is applied with scaling as the last refinement, after neutralisation.

#### Why the other options are wrong

**A is incorrect.** Absolute qualifier: only the alphas resting on questionable data are set to zero; genuine ones are capped.

**B is incorrect.** Polarity flip: forcing alphas into a normal distribution is the *extreme* approach, because it uses only the rankings and ignores the size of the alphas.

**C is incorrect.** Wrong order: scaling and trimming are "the final step in the process", after neutralisation, and the extreme version requires neutrality and scale to be rechecked.

**LO:** IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Trim Alpha Outliers and note 2

IM-4 • Portfolio Construction • Vardeez: VERY IMPORTANT

# IM-4 c

P3 CONCEPT

Describe neutralization and the different approaches used for refining alphas to be neutral.

Workbook: D28 Portfolio Construction and the No-Trade Region • Tier D • notes only

**WHAT IS TESTED HERE**

**Why P3.** Tier D workbook topic (D28, Map #66); Vardeez's focus for IM-4 does not name it.
**Vardeez:** no pointer for this LO.

**Reported exam shapes**

- **S1** Alpha refinement: scaling, trimming, and neutralisation (benchmark, cash, risk-factor, industry). The neutralisation part is drilled here; scaling and trimming in IM-4 b.

**What the questions look like:** statement questions. Identify the refinement from a description; say what each kind of neutralisation sets to zero; recognise the industry recipe; and apply the principles (decide in advance, not uniquely defined, cash and benchmark neutrality together). The bank's calculation items (benchmark-neutral alpha $\alpha_n - \beta_n \alpha_B$, industry recentring) fall outside the CONCEPT tag and are not drilled; the formulas are in the notes so that the statements make sense.
**Background:** IM-4 b, Section 1 (why alphas are refined) and IM-4 d, Section 1 (the inputs).

## 1. What neutralisation is

"Beyond scaling and trimming, we can remove biases or undesirable bets from our alphas. We call this process neutralization." It has implications for both the alphas and the portfolio. Two principles frame it:

- **Not uniquely defined.** "We can achieve even benchmark neutrality in more than one way": many different portfolios can hedge out an active beta.
- **Decide in advance.** "We should consider a priori how to neutralize our alphas. The choices will include benchmark, cash, industry, and factor neutralization. Do our alphas contain any information distinguishing one industry from another? If not, then industry-neutralize. The a priori approach works better than simply trying all possibilities and choosing the best performer."

## 2. Benchmark-neutral and cash-neutral alphas

- **Benchmark-neutral: the benchmark has zero alpha.** "By definition, the benchmark portfolio has 0 alpha." If the initial alphas imply a benchmark alpha $\alpha_B$, recentre them:

$$\alpha'_n = \alpha_n - \beta_n \, \alpha_B \qquad \Longrightarrow \qquad \alpha'_B = \sum_n h_{B,n} \alpha'_n = 0$$

  From the portfolio side, "the optimal portfolio will have a beta of 1": no bet on the benchmark, so no benchmark timing. Schweser adds that this equals putting a beta constraint in the optimisation.
- **GARP's Table 4.2.** The modified alphas of IM-4 b implied a benchmark alpha of only 1.6 basis points, so recentring barely moved them (American Express, $\beta = 1.21$: $-1.14\%$ to $-1.16\%$). That small change fits the observation that the optimal portfolio before neutralising already had a beta very close to 1.

- **Cash-neutral: no active cash.** "The alphas will not lead to any active cash position." Alphas can be both cash- and benchmark-neutral.

> **Correction to the source notes.** SchweserNotes (LO 86.c) says that under benchmark neutralisation "we say the portfolio alpha of the active portfolio is zero". It is the *benchmark's* alpha that is set to zero (GARP: "Benchmark neutralization means that the benchmark has 0 alpha"); the managed portfolio keeps its active alpha and ends with a beta of 1, so it makes no bet on the benchmark. Schweser's own worked example ($0.5\% - 0.013\% \times 1.2 = 0.48\%$) is right.

## 3. Risk-factor-neutral and industry-neutral alphas

A multiple-factor model splits return along several dimensions, and the manager labels each one either a **source of risk** or a **source of value added**. A risk factor is, by that definition, one the manager cannot forecast, so "he or she should neutralize the alphas against the risk factors". The neutralised alphas keep only information on the factors the manager can forecast, plus stock-specific information; "once neutralized, the alphas of the risk factors will be 0".

- **Industry recipe.** "Calculate the (capitalization-weighted) alpha for each industry, then subtract the industry average alpha from each alpha in that industry." The portfolio then carries no active industry bets (Figure 1). The same idea removes a size tilt.
- **Effect.** Matching the active portfolio's factor exposures to the benchmark's reduces active risk (Schweser).

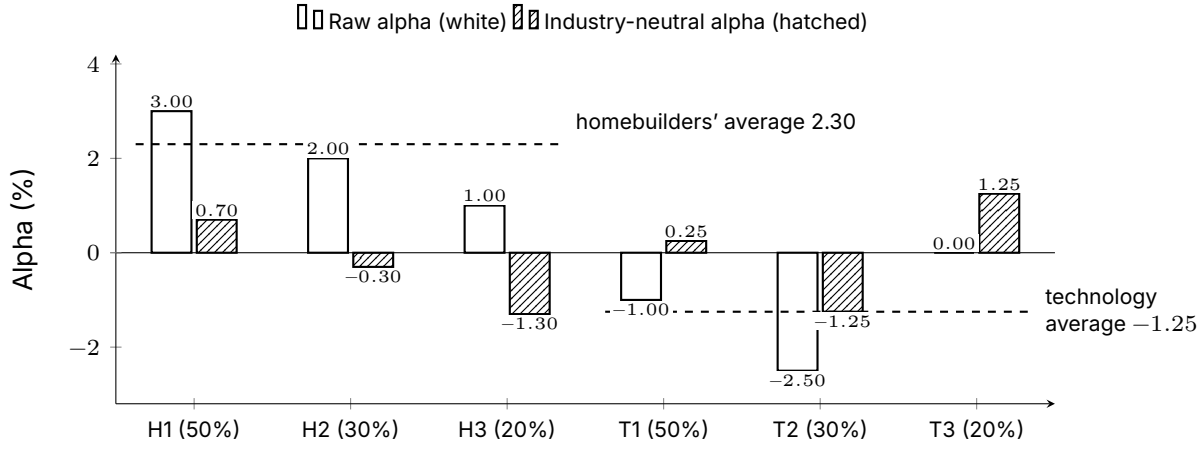



Figure 1. Industry neutralisation on an illustrative value signal that favours homebuilders (H) over technology (T). Benchmark weights within each industry in brackets. Dashed lines: capitalisation-weighted industry averages of the raw alphas (white bars). Subtracting them gives the hatched bars: each industry's weighted alpha is now zero, and the ranking inside each industry is unchanged.

| | Sets to zero | How | Portfolio result |
|---|---|---|---|
| **Benchmark-neutral** | The benchmark's alpha | $\alpha'_n = \alpha_n - \beta_n \alpha_B$ | Beta of 1: no benchmark timing |
| **Cash-neutral** | The active cash position the alphas imply | Adjust the alphas so the optimum holds no active cash | Fully invested relative to the benchmark; combinable with benchmark-neutral |
| **Risk-factor-neutral** | The alphas of factors the manager cannot forecast | Neutralise against those factors; keep forecastable factors and specific alpha | No active bets on those factors; lower active risk |
| **Industry-neutral** | Each industry's cap-weighted alpha | Subtract the industry's cap-weighted average from each alpha in it | No active industry bets |

**Exam traps**

- **Three different zeros.** Benchmark-neutral zeroes the benchmark's alpha, cash-neutral the active cash position, risk-factor-neutral the factors' alphas. Swapping the definitions is the standard distractor.
- **The managed portfolio is not set to zero alpha.** Benchmark neutralisation leaves the active alpha and gives the portfolio a beta of 1 (not 0).
- **Polarity of factor neutralisation.** Neutralise against the factors the manager *cannot* forecast; keep the ones he can.
- **Cap-weighted, not equal-weighted.** The industry average is capitalisation-weighted.
- **A priori, not by backtest.** Choose the neutralisation in advance from what the alphas contain; trying all and keeping the best performer is the corrupted version.
- **Not unique.** Many portfolios hedge out an active beta; cash and benchmark neutrality can be imposed together.
- **Sibling swap.** Neutralisation removes bets, trimming caps outliers, scaling sets the size; stratification is a construction technique (IM-4 h), not an alpha refinement.

## Practice questions: IM-4 c

### Question IM-4 c.1

Bank Q158 • GARP PE2 Q63

A portfolio manager is constructing an equity portfolio that will include several equities from multiple industries. The manager aims to achieve mean/variance optimization and asks a group of analysts to perform alpha analysis and to estimate the other portfolio inputs. As part of the alpha analysis, the analysts adjust alphas based on the portfolio's constraints and exclude very large positive and negative alphas from their calculations. Additionally, the analysts plan to remove biases or undesirable bets from the alphas. Which of the following procedures should the analysts apply to execute this plan?

**A.** Scaling

**B.** Trimming

**C.** Stratification

**D.** Neutralization

**Question IM-4 c.2** New • built from GARP Ch.4, 4.3 • S1

A portfolio manager benchmarked to a large-cap index finds that her refined alphas imply a benchmark alpha of 1.6 basis points. She wants the alphas to be benchmark-neutral. Which of the following statements about benchmark neutralization is correct?

**A.** It sets the alpha of the managed portfolio to zero, so the optimizer stops taking active positions.

**B.** It subtracts each stock's beta times the benchmark alpha from the stock's alpha, so that the benchmark has zero alpha and the optimal portfolio has a beta of 1.

**C.** It removes the active cash position implied by the alphas, so that the portfolio stays fully invested.

**D.** It makes the optimal portfolio's beta equal to zero, so that the portfolio carries no market risk.

**Question IM-4 c.3** New • built from GARP Ch.4, 4.3 • S1

A manager's multiple-factor model splits stock returns into industry, size and value factors plus a stock-specific component. Her research shows skill in forecasting the value factor and stock-specific returns, but no ability to forecast the industry or size factors. How should she neutralize her alphas?

**A.** Neutralize the alphas against the industry and size factors, so that the refined alphas carry only value-factor and stock-specific information.

**B.** Neutralize the alphas against the value factor, because the factor she forecasts is the one most likely to contain biases.

**C.** Neutralize the alphas against all three factors, so that the refined alphas carry only stock-specific information.

**D.** First forecast the industry and size factor returns, because neutralization requires a forecast for every factor in the model.

**Question IM-4 c.4** Bank Q157 • BT P2.T9.21.4.2

Robert is a value manager who picks among stocks with a relatively low EV/EBIT. However, he observes that these ratios are biased by industry; for example, homebuilders have much lower ratios than technology companies. How can he neutralize his alphas so that they are not biased by industry, that is, how can he achieve alphas that are risk-factor-neutral?

**A.** If industry is a risk factor, he should forecast the risk factors.

**B.** He can run a three-step screen: rank stocks by alpha, choose the first $X$ stocks, and equal-weight them.

**C.** He can subtract each industry's capitalization-weighted average alpha from his initial alphas.

**D.** His best approach is quadratic programming with the maximum number of inputs.

### Question IM-4 c.5

Bank Q159 • Mock 4 Q64

A quantitative team produces raw alpha forecasts for a universe of stocks. Before the forecasts are passed to the optimizer, the team applies two adjustments. First, it removes any unintended systematic exposures, for example a persistent tilt toward small-capitalization stocks, that were not part of the intended strategy. Second, it caps the most extreme positive and negative alpha values at a defined number of standard deviations from the mean. Which procedures has the team applied, in that order?

**A.** Neutralization, followed by trimming.

**B.** Scaling, followed by neutralization.

**C.** Trimming, followed by scaling.

**D.** Trimming, followed by neutralization.

### Question IM-4 c.6

New • built from GARP Ch.4, 4.3 • S1

The investment committee of a multi-strategy manager reviews the alpha-neutralization policy proposed by its quantitative team. Which of the following proposals is consistent with Grinold and Kahn's guidance?

**A.** Backtest the strategy with benchmark, cash, industry and factor neutralization in turn, and adopt whichever produced the best performance.

**B.** Choose either cash neutrality or benchmark neutrality, since alphas cannot be both cash- and benchmark-neutral at the same time.

**C.** Because the alphas contain no information that distinguishes one industry from another, make them industry-neutral.

**D.** Treat neutralization as uniquely defined, so that any two teams that benchmark-neutralize the same alphas will hold the same hedged portfolio.

## Solutions • IM-4 c

### Solution IM-4 c.1 — Answer: D

Bank Q158 • GARP PE2 Q63

**Step 1 – Identify what has already been done**

Adjusting alphas for the constraints is the modified-alpha step (IM-4 b); excluding very large alphas is trimming.

**Step 2 – Name the remaining plan**

"Remove biases or undesirable bets from our alphas. We call this process neutralization."
$\Rightarrow$ Neutralisation (benchmark, cash, industry or risk-factor).

**Interpretation**

GARP builds the stem as a checklist of alpha analysis so that the three refinements can be told apart: trimming handles size outliers, scaling the overall size, neutralisation the bets.

**Why the other options are wrong**

**A is incorrect.** Scaling sets the standard deviation of the alphas to volatility $\times$ IC; it does not remove a bias.
**B is incorrect.** Trimming deals with extreme values, and the stem says it has already been done.
**C is incorrect.** Stratification is a portfolio construction technique (IM-4 h), not an alpha refinement.

**LO:** IM-4 c: Describe neutralization and the different approaches used for refining alphas to be neutral. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Neutralization; GARP Practice Exam 2, Q63

### Solution IM-4 c.2 — Answer: B

New • built from GARP Ch.4, 4.3

**Step 1 – What is set to zero**

"Benchmark neutralization means that the benchmark has 0 alpha. If our initial alphas imply an alpha for the benchmark, the neutralization process recenters the alphas to remove the benchmark alpha": $\alpha'_n = \alpha_n - \beta_n \alpha_B$.

**Step 2 – What it means for the portfolio**

"From the portfolio perspective, benchmark neutralization means that the optimal portfolio will have a beta of 1, i.e., the portfolio will not make any bet on the benchmark."
$\Rightarrow$ B.

**Interpretation**

With a benchmark alpha of only 1.6 basis points, as in GARP's Table 4.2, the recentring barely moves the alphas, a sign that the optimal portfolio already had a beta close to 1.

**Why the other options are wrong**

**A is incorrect.** The benchmark's alpha, not the managed portfolio's, is set to zero; the portfolio keeps its active bets on individual stocks.
**C is incorrect.** Sibling swap: that is cash neutralisation.
**D is incorrect.** Polarity: the beta becomes 1 (no active beta), not 0 (no market exposure).

**LO:** IM-4 c: Describe neutralization and the different approaches used for refining alphas to be neutral. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Benchmark- and Cash-Neutral Alphas, Table 4.2

### Solution IM-4 c.3 Answer: A

New • built from GARP Ch.4, 4.3

#### Step 1 – Label each factor

Value: she can forecast it, a source of value added. Industry and size: she cannot, so they are sources of risk.

#### Step 2 – Apply the rule

A manager "does not have any ability to forecast the risk factors. Therefore, he or she should neutralize the alphas against the risk factors. The neutralized alphas will include only information on the factors the manager can forecast, along with specific asset information."
⇒ Neutralise against industry and size; keep value and the specific part.

#### Interpretation

Neutralisation removes the bets a manager has no reason to take; it should not remove the ones her research supports. Once neutralised, the industry and size alphas are zero.

#### Why the other options are wrong

**B is incorrect.** Polarity flip: the factor she can forecast is the one to keep.
**C is incorrect.** Over-neutralises: removing the value factor throws away the skill she has.
**D is incorrect.** Neutralisation needs no forecast of the factors; it exists because the manager has none.

**LO:** IM-4 c: Describe neutralization and the different approaches used for refining alphas to be neutral. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Risk-Factor-Neutral Alphas

### Solution IM-4 c.4 Answer: C

Bank Q157 • BT P2.T9.21.4.2

#### Step 1 – Name the problem

The EV/EBIT signal carries an unintended industry bet: it favours homebuilders over technology for reasons Robert does not claim to forecast.

#### Step 2 – GARP's industry recipe

"Calculate the (capitalization-weighted) alpha for each industry, then subtract the industry average alpha from each alpha in that industry."
⇒ Each industry's weighted alpha becomes zero, so the portfolio takes no active industry bets, and the ranking within each industry is kept (Figure 1).

#### Interpretation

Industry is treated as a risk factor: Robert's information distinguishes stocks within an industry, not one industry from another, which is the a priori test GARP gives for industry neutralisation.

#### Why the other options are wrong

**A is incorrect.** Polarity flip: risk factors are neutralised precisely because the manager cannot forecast them.
**B is incorrect.** A screen is a construction technique and does nothing to remove the bias; it would load up on homebuilders.
**D is incorrect.** QP is a construction technique; feeding it biased alphas, with more inputs, does not remove the bias.

**LO:** IM-4 c: Describe neutralization and the different approaches used for refining alphas to be neutral. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Risk-Factor-Neutral Alphas

### Solution IM-4 c.5 Answer: A

Bank Q159 • Mock 4 Q64

**Step 1 – First adjustment**

Removing an unintended systematic exposure, a small-cap tilt, is risk-factor neutralisation.

**Step 2 – Second adjustment**

Capping extreme alphas at a set number of standard deviations is trimming (GARP: about three times the scale).

⇒ Neutralisation, then trimming, which is also the order GARP implements (scaling and trimming last).

**Interpretation**

A mock favourite: two descriptions, four orderings. Read each adjustment against the three definitions before looking at the order.

**Why the other options are wrong**

**B is incorrect.** Neither step rescales the overall dispersion of the alphas, and the capping step is trimming, not neutralisation.

**C is incorrect.** The first step removes exposures, not outliers; neither step is scaling.

**D is incorrect.** Right procedures, reversed order.

**LO:** IM-4 c: Describe neutralization and the different approaches used for refining alphas to be neutral; IM-4 b: Evaluate the motivation for and the methods used for refining alphas in the implementation process. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 and note 2

### Solution IM-4 c.6 Answer: C

New • built from GARP Ch.4, 4.3

**Step 1 – GARP's guidance**

"As a general principle, we should consider a priori how to neutralize our alphas. ... Do our alphas contain any information distinguishing one industry from another? If not, then industry-neutralize."

⇒ C applies the rule directly.

**Step 2 – Check the other proposals**

"The a priori approach works better than simply trying all possibilities and choosing the best performer" (against A); "it is possible to make the alphas both cash- and benchmark-neutral" (against B); "Neutralization ... isn't uniquely defined" (against D).

**Interpretation**

The principles are tested as often as the definitions: decide in advance from what the alphas contain, expect more than one way to reach neutrality, and combine neutralisations freely.

**Why the other options are wrong**

**A is incorrect.** Choosing by backtest is the "trying all possibilities" approach GARP rejects; it invites data mining.

**B is incorrect.** Cash and benchmark neutrality can be imposed together.

**D is incorrect.** Neutralisation is not uniquely defined: many portfolios can hedge out an active beta.

**LO:** IM-4 c: Describe neutralization and the different approaches used for refining alphas to be neutral. • **Reference:** Grinold and Kahn, Active Portfolio Management 2e, Ch.14 (GARP 2025 Ch.4), 4.3 Neutralization

IM-12 • Performing Due Diligence on Specific Managers and Funds • Vardeez: MODERATE

# IM-12 c

P3 CONCEPT

Identify themes and questions investors can consider when evaluating a hedge fund manager.

Workbook: A24 revision notes only • no workbook question assigned • Vardeez focus

**WHAT IS TESTED HERE**

**Why P3.** Vardeez flags this LO ("LO 94.c, Manager Evaluation") in a chapter rated Moderate. No workbook question is assigned; the A24 revision notes summarise the themes (Map #34). **Vardeez:** manager evaluation.

**Reported exam shapes:** none reported. The questions are built from Vardeez's focus, GARP 2025 Ch.10 (Mirabile) and Schweser Reading 94.

**What the questions look like:** all conceptual. A list of due diligence questions to sort into the four themes (strategy, ownership, track record, investment management); an "EXCEPT" item on desirable steps; or a short scenario in which an investor relies on the manager's own material and you pick the step that adds independent verification.

## 1. Due diligence in one picture (background for IM-12)

Due diligence is "the process of evaluation and analysis that an investor follows to get comfortable with a strategy, a manager, and a fund prior to making an investment". Investors should assess all three skill sets a modern hedge fund needs (investment, operational and business) before committing capital (Figure 1). Reputation and past performance were once the main criteria; today the investment process with its risk controls, and the operations with the business model, are "considered almost equally important". The work pairs a checklist, so that nothing obvious is missed, with open-ended questions that expose the firm's culture. Managers are rarely good at all three skills: founders who are CIO and CEO and also run risk and the business "are, quite frankly, considered dinosaurs today". Funds fail through bad decisions, fraud, leverage and tail events, withdrawals at the worst time, squeezes, weak supervision and the failure of others, such as a prime broker (IM-12 a, P4).

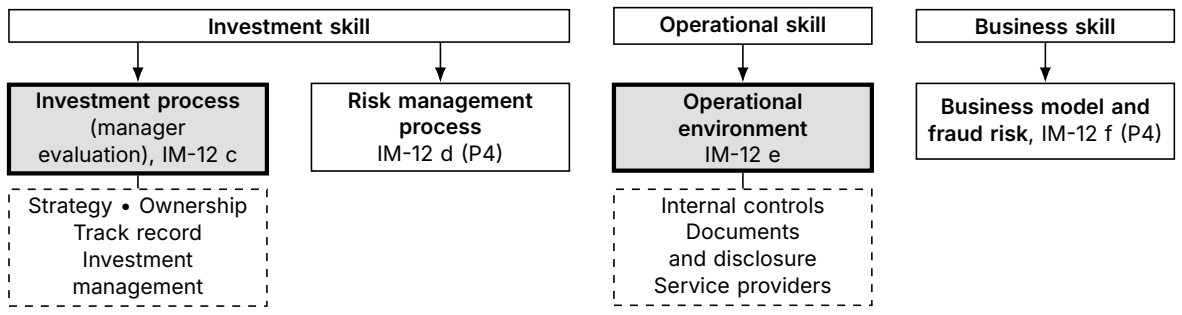



Figure 1. Mirabile's due diligence process: investment skill maps onto IM-12 c and d, operational skill onto IM-12 e, business skill onto IM-12 f. Heavy borders with grey fill mark the two Vardeez focus LOs built here; dashed boxes list their parts. The questionnaire (IM-12 g) gathers inputs for every box, and fraud risk can survive a complete process (IM-12 f).

## 2. The four themes of manager evaluation

**Strategy: "What is your strategy, and how does it work?"** GARP's fifteen opening questions cover the style and its classification; current themes and most concentrated positions; how the portfolio has evolved; gross and net exposure targets; turnover and days to liquidate; whether exit triggers are "hard coded into the risk management process or just guidelines"; stop losses; how models are built, tested and retired; short sales as hedges or alpha (and any squeeze, recall or buy-in); listed and OTC derivatives; the trading desk; capacity; private investments; return versus asset-growth orientation; and whether the fund has ever been soft or hard closed or returned capital.

**Ownership: "How is equity ownership allocated among the portfolio management, trading and research teams?"** Ownership helps align the team with investors and helps attract and retain staff (Schweser), but "one model is not necessarily better than the other" (GARP): understand the firm's philosophy and its effect on performance and retention.

**Track record: "Is the track record reliable?"** Is it comparable with similar strategies, audited or verified by a third party, long enough for statistical evaluation, affected by fund size, and produced by a team that is still in place? How did it hold up in market stress? The absolute level of past returns is the least informative item.

**Investment management: "Who are the principals, and are they trustworthy?"** Interview the actual decision makers, not just investor relations: "if you cannot get access to the actual decision makers when you are trying to invest, it is unlikely you will have access if something goes wrong!" *Reference checks* are a 360-degree review (former employers, colleagues, clients, investors). A new manager's references are verified, both those on the list and those found independently; with an established manager the focus is motivation, behaviour in stress, communication and other investors' experiences. *Background checks* are records: a background report (about $750), the Form ADV, SEC and court databases, personal credit and bankruptcy reports, verification of representations with auditors, administrators and prime brokers, and related-party activity.

> **Correction to the source notes.** Vardeez's strategy list ends with "has the fund ever been shut down". The reading asks whether the fund has been "soft or hard closed or returned capital to investors" (GARP 10.6): *closed* means closed to new money, a capacity decision, not wound up. Vardeez also lists "enquiries with past colleagues" as a background check; in the reading, colleagues are reference checks, while background checks verify the manager's representations with auditors and brokers (GARP adds administrators and prime brokers).

**Summary card: IM-12 c**

| Theme | What the questions reveal | Typical warning sign |
|---|---|---|
| **Strategy** | Whether the edge, the exposures and the loss controls are real | Triggers that are only guidelines; capacity ignored |
| **Ownership** | Alignment and retention of talent | None by itself: either model can work |
| **Track record** | Whether the record is real (verified), meaningful (comparable, long, size-adjusted) and repeatable (same team) | Unaudited record; the team that earned it has left |
| **Investment management** | Integrity of the principals and access to them | No access to decision makers; litigation, credit or related-party findings |

**Exam traps**

- **Theme swap.** Team continuity is a *track record* question; derivatives use, turnover and quantitative models are *strategy* questions.
- **Absolute returns.** A record is judged against peers, size and risk; the absolute level of past returns is the least relevant item.
- **Ownership as a pass/fail test.** Equity ownership helps alignment, but GARP says neither model is necessarily better.
- **Manager-supplied evidence.** A pitch book or answers from investor relations are not due diligence; independent references, background checks and third-party verification are.
- **References versus background checks.** References are what people say; background checks are records (Form ADV, courts, credit, bankruptcy).
- **The one-person firm.** A founder holding the CIO, CEO, risk and business roles alone is a weakness, not concentrated accountability. And **"closed"** means closed to new money, not shut down.

## Practice questions: IM-12 c

### Question IM-12 c.1

Bank Q512 • Mock 2 Q14 (rebuilt)

An analyst at a European fund of hedge funds organises a manager review under four themes: strategy, ownership, track record and investment management. Which of the following questions is LEAST likely to be asked when examining the manager's strategy?

**A.** Are most of the investment team members who generated the past results still employed by the firm?

**B.** Are derivatives used for hedging or for speculative purposes?

**C.** What is the fund's level of turnover and liquidity?

**D.** To what extent are quantitative models used in the investment process?

### Question IM-12 c.2

Bank Q513 • Schweser Module Quiz 94.1 Q4

Lisa Tahara, FRM, is considering an institutional investment in a hedge fund that has experienced volatile and generally positive returns in the past. Which of the following considerations about the fund's track record is least relevant for consideration in her investment decision?

**A.** Size of investment assets.

**B.** Absolute level of past returns.

**C.** Verification of returns by a third party.

**D.** Employment continuity of the investment team.

**Question IM-12 c.3** Bank Q510 • AnalystPrep Q.2597

During an on-site visit, the chief operating officer of a hedge fund tells a prospective investor that "a firm's ownership structure impacts performance". Which of the following is the most accurate statement?

**A.** A firm's ownership structure helps in aligning the interests of the fund managers with those of investors.

**B.** A firm's ownership does not affect its performance.

**C.** The presence of large personal investments on the part of fund managers increases the firm's returns.

**D.** The presence of significant personal investments on the part of fund managers decreases the firm's returns.

**Question IM-12 c.4** Bank Q511 • BT P2.T8.413.1 (rebuilt)

A consultant is drafting the manager evaluation section of a pension plan's hedge fund due diligence policy. In performing due diligence on a hedge fund manager, each of the following is a desirable step or characteristic EXCEPT:

**A.** Performing reference checks and background checks on the firm's principals.

**B.** Obtaining specific information on the firm's ownership structure.

**C.** Having access to the actual decision makers and risk takers of the fund.

**D.** The founder and CEO also serving as CIO with sole responsibility for investment, risk and business decisions, because this concentrates accountability.

**Question IM-12 c.5** Bank Q508 • AnalystPrep Q.2596 (rebuilt)

Gong Fang attended an investor meeting at which a large investment management firm launched a new fund and presented the past performance of its earlier funds. Fang's questions were answered by investor relations staff and the CEO, and he is about to invest. He has not contacted anyone outside the firm or verified any of the performance figures. Which advice is most consistent with a proper manager due diligence process?

**A.** Conduct independent reference checks and background checks on the manager and verify the presented track record with a third party before investing.

**B.** Invest, because attending the meeting and having his questions answered constitutes due diligence.

**C.** Invest, because a strong presentation of past performance is the most important part of due diligence.

**D.** Decline, because a firm's marketing presentation is never an acceptable source of information.

### Question IM-12 c.6

New • built from GARP Ch.10

A due diligence analyst at a Japanese pension fund is planning reference checks on two managers. Manager N launched her first fund eight months ago after ten years as a portfolio manager at a bank. Manager E has run the same flagship fund for fifteen years. Which plan is most consistent with the reading's guidance on evaluating a fund's principals?

**A.** For N, verify references from her former bank, both those on her list and those found independently; for E, focus on his motivation, his behaviour in stress periods, his communication with investors and other investors' experiences.

**B.** For N, rely on the reference list she supplies, since a new manager has no fund investors to contact; for E, re-verify every employer since the start of his career.

**C.** For both, skip reference checks and rely on background reports, since references come from the manager's own contacts and are biased.

**D.** For both, limit reference checks to whether the manager generated ideas or only implemented them, since that is the one question references can answer.

### Question IM-12 c.7

New • built from GARP Ch.10

A fund of hedge funds is completing its manager evaluation of a long/short equity manager. The analyst's notes record four findings. Which finding should most concern the analyst?

**A.** The firm does not share equity ownership with its portfolio managers and traders and pays them a share of fund profits instead.

**B.** Every meeting so far has been with investor relations staff, and requests to meet the portfolio managers and traders have been deferred until after the subscription.

**C.** Exit triggers for positions are hard coded into the risk management system rather than left to the portfolio manager's judgement.

**D.** The fund was soft closed last year to protect returns for existing investors and has since reopened with limited capacity.

## Solutions • IM-12 c

### Solution IM-12 c.1 — Answer: A

Bank Q512 • Mock 2 Q14

**Step 1 – Sort each question into its theme**

Strategy questions (GARP's "What is your strategy, and how does it work?"): style, turnover and liquidity, loss limits, quantitative models, short sales, derivatives, trading, capacity. Track record questions: peers, verification, length of history, size, and whether the team is still in place.
⇒ B, C and D are strategy questions; A is a track record question.

**Step 2 – Why team continuity belongs to the track record**

The Greenwich Roundtable list asks whether the team that produced the historical record is still in place. A record earned by people who have left says little about the returns the fund can earn next.

**Interpretation**

Sorting questions into themes is the most natural shape for this LO. Team continuity sounds like a "people" question, which tempts you towards investment management or strategy, but the reading files it under the track record.

**Why the other options are wrong**

**B is incorrect.** A strategy question: are derivatives used to hedge or to speculate?
**C is incorrect.** A strategy question: turnover and liquidity (days to liquidate) describe how the strategy is run.
**D is incorrect.** A strategy question: how quantitative the process is and how models are developed and tested.

**LO:** IM-12 c: Identify themes and questions investors can consider when evaluating a hedge fund manager. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.6; SchweserNotes 2026 Book 5, Reading 94, LO 94.c

### Solution IM-12 c.2 — Answer: B

Bank Q513 • Schweser Module Quiz 94.1 Q4

**Step 1 – List the track record questions**

Comparison with peers; audit or third-party verification; enough history for trend and attribution analysis, including downturns; returns relative to the size of the assets; continuity of the investment team.
⇒ A, C and D are on the list.

**Step 2 – Why the absolute level is least relevant**

The returns were volatile, and an absolute number carries no information about peers, risk or asset base. Past returns are also no assurance of future returns.
⇒ The relative level of returns matters more than the absolute level.

**Interpretation**

The track record is a reliability test, not a beauty contest: is the record real (verified), meaningful (comparable, long enough, size-adjusted) and repeatable (same team)?

**Why the other options are wrong**

**A is incorrect.** Returns relative to the size of the assets are on the list: a record earned on a small asset base may not survive growth.
**C is incorrect.** Third-party verification gives assurance that the returns were computed fairly

and accurately.

**D is incorrect.** Team continuity gives some assurance that the people who earned the record can repeat it.

**LO:** IM-12 c: Identify themes and questions investors can consider when evaluating a hedge fund manager. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.6 (Is the Track Record Reliable?); SchweserNotes 2026 Book 5, Reading 94, LO 94.c

### Solution IM-12 c.3 — Answer: A

Bank Q510 • AnalystPrep Q.2597

#### Step 1 – What ownership does

Ownership interests help align the interests of the investment team with those of investors, and help attract and retain quality staff (Schweser). GARP calls the investment team's participation in ownership "a critical area of differentiation among firms".

⇒ A states the reading's reason for asking about ownership.

#### Step 2 – What ownership does not do

Ownership does not mechanically raise or lower returns. GARP stresses that "one model is not necessarily better than the other": the investor should understand the firm's philosophy and its effect on performance, talent acquisition and retention.

#### Interpretation

Ownership is a question to ask, not a pass/fail test. Not sharing equity is not by itself a red flag (Question IM-12 c.7, option A).

#### Why the other options are wrong

**B is incorrect.** Denies any effect; the structure matters through alignment and retention.

**C is incorrect.** Absolute claim: alignment may help, but no ownership arrangement guarantees higher returns.

**D is incorrect.** No basis in the reading: personal stakes, if anything, strengthen alignment.

**LO:** IM-12 c: Identify themes and questions investors can consider when evaluating a hedge fund manager. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.6 (How Is Equity Ownership Allocated?); SchweserNotes 2026 Book 5, Reading 94, LO 94.c

### Solution IM-12 c.4 — Answer: D

Bank Q511 • BT P2.T8.413.1

#### Step 1 – Check A, B and C against the four themes

Reference and background checks are the investment management theme; ownership structure is its own theme; access to the people who take risk and make decisions is a stated requirement.

⇒ A, B and C are desirable.

#### Step 2 – Test D against the three skill sets

Mirabile: managers are rarely good at investment, operational and business skills at once, and founders who are CIO and CEO and also manage risk and the business model themselves "are, quite frankly, considered dinosaurs today". One person in every role means no independent check on risk and a large key person risk.

⇒ D is not desirable: it is the exception.

#### Interpretation

Investors want the organisation, not one founder, to supply all three skills at a high level, which usually means senior partners with specialised risk, operations or business skills.

**Why the other options are wrong**

**A is incorrect.** Desirable: references and background checks are the core of evaluating the principals.

**B is incorrect.** Desirable: ownership is one of the four themes.

**C is incorrect.** Desirable: investors should deal with the decision makers, not only investor relations staff.

**LO:** IM-12 c: Identify themes and questions investors can consider when evaluating a hedge fund manager. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), introduction and 10.6; SchweserNotes 2026 Book 5, Reading 94, LO 94.c

## Solution IM-12 c.5 — Answer: A

Bank Q508 • AnalystPrep Q.2596

### Step 1 – Identify where Fang's information came from

Everything he has seen (the presentation, the answers of investor relations and the CEO) came from the firm itself.

### Step 2 – Apply the manager evaluation themes

The reading calls for reference checks (former employers, colleagues, clients, investors), background checks (Form ADV, public databases, credit reports) and a track record that has been audited or verified by a third party.

⇒ Only A adds independent verification.

### Interpretation

The test in these scenarios is simple: does the step rely on the manager's own material, or does it check that material against an independent source?

### Why the other options are wrong

**B is incorrect.** Treats a firm-run meeting as due diligence; it verifies nothing.

**C is incorrect.** Past performance is a traditional input but "no longer sufficient on its own", and here it is unverified.

**D is incorrect.** Overreach: the presentation is a legitimate input; it must be checked, not discarded.

**LO:** IM-12 c: Identify themes and questions investors can consider when evaluating a hedge fund manager. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.6; SchweserNotes 2026 Book 5, Reading 94, LO 94.c

## Solution IM-12 c.6 — Answer: A

New • built from GARP Ch.10

### Step 1 – The new manager

GARP: new managers need their references from previous hedge funds or banks verified, "both those that are on the manager's reference list and those that the investor can obtain independently". Check whether she did what she claims: trigger puller or support role, generating or implementing ideas, alone or in a group.

### Step 2 – The established manager

Established managers do not need to be interrogated about distant employers. The dialogue moves to "motivation, behavior during crisis or stress periods, ability to communicate to investors, and experiences other investors have had".

⇒ A applies the right emphasis to each manager.

#### Interpretation

Reference checking is a 360-degree review: if enough people say the same thing, the investor takes comfort; real differences are vetted and then raised with the manager.

#### Why the other options are wrong

**B is incorrect.** Reverses both halves: the new manager's own list is not enough, and it is the established manager who needs less digging into the distant past.

**C is incorrect.** References and background checks are complements; GARP asks investors to allocate resources to both.

**D is incorrect.** Too narrow: the role question is one part of the review; motivation, behaviour in stress and investors' experiences matter too.

**LO:** IM-12 c: Identify themes and questions investors can consider when evaluating a hedge fund manager. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.6 (Who Are the Principals, and Are They Trustworthy?); SchweserNotes 2026 Book 5, Reading 94, LO 94.c

### Solution IM-12 c.7 Answer: B

New • built from GARP Ch.10

#### Step 1 – Test the three findings that look unusual

A: some firms do not share equity, and "one model is not necessarily better than the other". C: GARP asks whether exit triggers are hard coded or just guidelines; hard-coded triggers are the stronger control. D: GARP asks whether the fund has ever been soft or hard closed; closing to protect returns is a way of managing the conflict between returns and asset growth.

$\Rightarrow$ None of A, C or D is a concern by itself.

#### Step 2 – Test B

Investors must deal directly with the actual risk takers and decision makers, not just investor relations, sales or junior staff: "if you cannot get access to the actual decision makers when you are trying to invest, it is unlikely you will have access if something goes wrong!"

$\Rightarrow$ Being kept away from the decision makers until the money is in is the finding that should worry the analyst.

#### Interpretation

Restricted access is also one of the Madoff red flags (IM-14 b): investors were not allowed to visit BMIS or question the strategy.

#### Why the other options are wrong

**A is incorrect.** A legitimate compensation choice; the investor should understand it, not reject it.

**C is incorrect.** Reassuring, not alarming: hard-coded triggers limit discretion over losses.

**D is incorrect.** A soft close to protect existing investors is a sign of capacity discipline, not of trouble.

**LO:** IM-12 c: Identify themes and questions investors can consider when evaluating a hedge fund manager. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.6; SchweserNotes 2026 Book 5, Reading 94, LO 94.c

IM-12 • Performing Due Diligence on Specific Managers and Funds • Vardeez: MODERATE

# IM-12 e

P3 CONCEPT

Explain how due diligence can be performed on a hedge fund's operational environment.

Workbook: A24 A24.4 touches this LO (worked as Question IM-14 b.10) • Vardeez focus

**WHAT IS TESTED HERE**

**Why P3.** Vardeez flags this LO ("LO 94.e, Operational Due Diligence") in a chapter rated Moderate.
**Vardeez:** operational due diligence.

**Reported exam shapes:** none reported. Workbook A24.4 (worked as Question IM-14 b.10) tests the same lesson through the Madoff case: in-house custody and administration with a three-person auditor.

**What the questions look like:** conceptual. "Which statement about operational due diligence is correct" or "which is not true" (GARP PE2 Q1 is the model); a finding in the fund documents or financial statements to judge; or a choice of how to evaluate service providers. The bank holds two questions, so five are new, built from GARP Ch.10.

## 1. What operational due diligence is for

Recap from IM-12 c: the operational environment is the second of the three skill sets (Figure 1 there). Its purpose is to ensure that "no significant additional risk of loss is being created for investors related to the settlement of securities, process of corporate actions, misappropriation or theft by employees or agents, or any other breakdown in the manager's confirmation, verification, valuation, and reconciliation process". Schweser names three focus areas: **internal control assessment**, **documents and disclosure**, and **service provider evaluation**. "It is not enough to have a good plan. The people must be qualified, and the process must be followed every day."

## 2. Internal control assessment

- **People.** Does the CEO support a culture of control and compliance? Are operations, accounting, treasury, technology and compliance staff qualified and trained? Background checks reach the COO, CFO and accounting managers, not only the investment team.
- **Written procedures** show intent, not effectiveness: reading them gives "only a sense of whether the management of the hedge fund takes the whole control process seriously". A good sign is an outside audit firm's opinion that the controls are well designed and operating effectively.
- **Compliance**, in-house or outsourced: a code of ethics, a ban on related-party transactions, limits on employee trading.
- **Counterparties.** When Lehman, MF Global and Refco failed, funds with cash at the dealer recovered perhaps "30 or 40 percent" after a lengthy court battle. Spread exposure, move excess cash, monitor daily.
- **Governance** beyond the CEO, "with teeth": directors accountable for service provider choices, style drift or fraud.

## 3. Documents and disclosure, including the financial statements

- **Law firm.** Confirm it drafted the original documents and every update; look for changes made after the as-of date and raise them at once with the manager and the law firm; confirm the firm is still retained.
- **Consistency** (Figure 1). The offering memorandum (OM), which "ultimately governs an investor's rights and obligations", must agree with the other documents at the same point in time, especially on fees, liquidity, side pockets, gates, suspensions, subfunds or SPVs and offshore managers.
- **Conflicts, risks, powers.** Query vague conflict wording ("arm's-length basis" can hide a personal rebate from the prime broker). "Insufficient risk factors are more of a red flag than too many"; overly broad, irrelevant ones are a red flag too. Check the manager's powers and key man rights; indemnification "should not extend to the gross negligence, bad faith, fraud, or willful misconduct of the manager".
- **Financial statements.** The opinion must be *unqualified*. The statements must fit the strategy: an equity fund with low, falling leverage should not show high, rising interest expense; a macro fund usually earns interest income; a buy-and-hold fund shows unrealised, not large realised, gains. Read the footnotes, recalculate the fees ("never incentive fees earned in years when the fund lost money") and check the general partner's capital: withdrawals are "a big red flag".

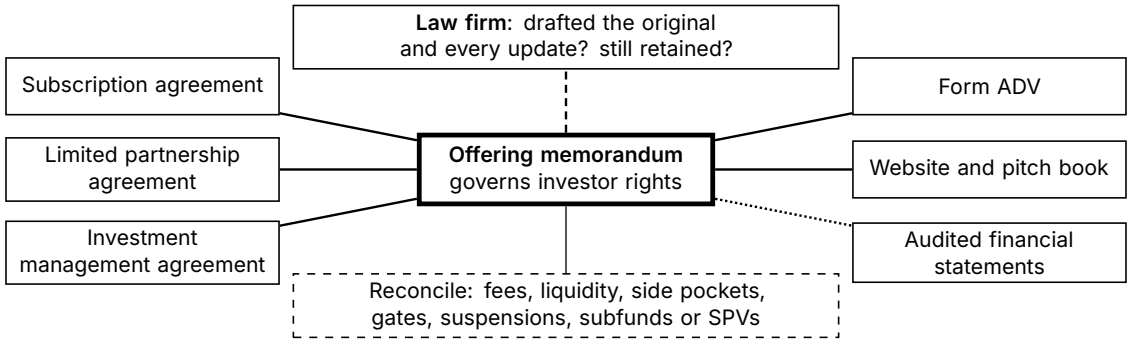



Figure 1. The documents cross-check. Solid lines: documents that must state the same terms as the OM at the same time. Dashed line: confirm authorship and retention with the law firm. Dotted line: recalculate the fees in the financial statements against the OM.

## 4. Service provider evaluation

Third parties execute trades, provide technology, value and verify positions and safeguard assets. The fund should make the key contacts available; the investor obtains each provider's **internal control letters** and **audited financial statements** (an independent check on the providers themselves) and **interviews** them about their role. Independent administration, audits and top-tier providers are "generally not negotiable" (GARP 10.4).

**Summary card: IM-12 e**

| Area | What the investor checks | Red flags |
|---|---|---|
| **Internal controls** | People and tone at the top; procedures plus an outside opinion; compliance; counterparties; governance | Policies with no evidence they work; all cash at one dealer |
| **Documents** | Law firm authorship and retention; consistency with the OM; conflicts; risk factors; powers; indemnity | Post-dated changes; vague conflicts; too few risk factors; indemnity for fraud |
| **Financial statements** | Unqualified opinion; fit with the strategy; footnotes; fees; GP capital | Incentive fee in a loss year; GP withdrawals |
| **Service providers** | Contacts, internal control letters, audited statements, interviews | Access refused; providers not independent |

**Exam traps**

- **Intent is not effectiveness.** Written procedures give little assurance; an outside opinion on design and operation gives more.
- **Compliance location.** In-house or outsourced are both fine; the investor tests the practices (GARP PE2 Q1).
- **Indemnification reversed.** Never for gross negligence, bad faith, fraud or willful misconduct.
- **Risk factor polarity.** Too few is a bigger red flag than too many; boilerplate is also a red flag.
- **Which document wins.** The OM governs; the website, pitch book or a verbal answer does not.
- **Sibling swap.** Operational due diligence is about loss from operations. The manager's edge is the investment review; fees against costs is business model risk; terms against strategy is a risk management criterion.

## Practice questions: IM-12 e

### Question IM-12 e.1

New • built from GARP Ch.10

The head of operational due diligence at a Canadian endowment explains her team's role to the investment committee. Which description of the primary purpose of operational due diligence on a hedge fund is most accurate?

**A.** To determine whether the manager has a repeatable edge that generates alpha relative to peers following the same strategy.

**B.** To ensure investors face no significant additional risk of loss from settlement, corporate actions, theft, or breakdowns in confirmation, verification, valuation and reconciliation.

**C.** To determine whether management fees cover operating expenses, so that the firm can survive a long period of poor performance.

**D.** To confirm that the fund's fees, lock-ups and redemption terms are in line with those of comparable funds.

### Question IM-12 e.2

Bank Q521 • GARP PE2 Q1

A group of newly hired investment analysts at a large wealth management firm is undergoing training on the company's investment practices. Part of the training focuses on hedge fund investments and addresses how the firm evaluates the risk management processes and procedures of a hedge fund being considered for investment. The analysts learn that the firm performs comprehensive due diligence on the hedge fund's investment environment, as well as on its operational environment and business practices. Which of the following is correct about a hedge fund investor's due diligence on the operational environment of a hedge fund?

**A.** An investor should assess the qualifications of the fund's investment team members but does not need to do the same for other fund personnel.

**B.** An investor should investigate the fund's compliance practices but does not need to ensure that the fund has an in-house compliance function.

**C.** An investor should review the fund documents but does not need to verify with the fund's law firm whether they are responsible for the content of these documents.

**D.** An investor should ask if the fund uses third-party service providers but does not need to inquire about the fund's relationships with these service providers.

### Question IM-12 e.3

Bank Q520 • BT P2.T8.414.1 (key corrected)

In regard to the hedge fund's operational environment, Mirabile advises each of the following as true for investors during due diligence, EXCEPT which is not true?

**A.** It is important to ensure that the fund offering memo, subscription agreement, limited partnership agreement, investment management agreement, Form ADV and website are all saying the same thing at the same point in time.

**B.** Insufficient risk factors are more of a red flag than too many. In addition, overly broad and irrelevant risk factors are also a red flag, as the law firm may be drafting the OM so broadly that it is looking primarily to protect itself, not the investor.

**C.** Investors should expect a hedge fund to make all of the key contacts at its service providers available so they can verify the scope of services, and can obtain internal control letters and audited financial statements from the service providers as an independent check on them.

**D.** The limited partnership agreement should include indemnification provisions such that the fund itself indemnifies the manager and all directors against gross negligence, bad faith, fraud, or willful misconduct of the manager.

**Question IM-12 e.4** New • built from GARP Ch.10

During a site visit, an operational due diligence analyst at a Swiss private bank reviews the control environment of a credit hedge fund. Which of the following conclusions is most appropriate?

**A.** The fund's detailed written procedures for trading and cash processing give strong assurance that its controls are operating effectively.

**B.** Background checks are needed only for the investment team, since operations and accounting staff are screened by the fund's own hiring process.

**C.** Holding all of the fund's cash and margin with a single large prime broker removes counterparty risk, since the broker is well capitalised.

**D.** An outside audit firm's opinion that the controls are well designed and operating effectively gives more assurance than a review of the written procedures.

**Question IM-12 e.5** New • built from GARP Ch.10

An analyst at a fund of hedge funds reviews a fund's legal documents before subscribing. She finds that:

- the offering memorandum (OM) offers quarterly liquidity with 90 days' notice, while the website and pitch book describe monthly liquidity;
- the limited partnership agreement is dated as of March 2025 but contains a gate provision that the manager says was added in October 2025;
- the law firm named in the OM confirms that it drafted the original documents.

Which action is most appropriate?

**A.** Rely on the website's monthly liquidity, since the website is the most recently updated statement of the fund's terms.

**B.** Treat the findings as red flags: reconcile the terms with the manager against the OM, and ask the law firm whether it drafted the later change and still represents the fund.

**C.** Accept the gate provision, since the law firm drafted the original documents and is therefore responsible for every later change.

**D.** Ignore the liquidity difference, since redemption terms are a business model issue rather than part of operational due diligence.

**Question IM-12 e.6** New • built from GARP Ch.10

An operational due diligence analyst reviews the latest audited financial statements of four hedge funds. Each received an unqualified audit opinion. Which finding is consistent with the fund's stated strategy and needs no follow-up with the manager?

**A.** A long/short equity fund that reports low and falling leverage shows interest expense that is high and has risen each year.

**B.** A buy-and-hold value fund reports large realised gains every year and only small unrealised gains.

**C.** A global macro fund reports interest income on the cash and short-term government bonds it holds.

**D.** A credit fund that lost money for the year paid its manager an incentive fee for that year.

**Question IM-12 e.7** New • built from GARP Ch.10

A pension consultant is evaluating the service providers of a multi-strategy hedge fund that uses an external administrator, two prime brokers and an independent valuation agent. Which approach is most consistent with the reading?

**A.** Rely on the offering memorandum's description of each provider's role, since contacting providers directly would breach the fund's confidentiality.

**B.** Obtain the providers' internal control letters and audited financial statements, and interview key contacts about their roles in execution, valuation and safeguarding assets.

**C.** Evaluate only the prime brokers, since an administrator and a valuation agent do not hold any of the fund's assets.

**D.** Accept the providers without review if they are large, well-known firms, since their size ensures their independence from the manager.

# Solutions • IM-12 e

## Solution IM-12 e.1

**Answer: B**

New • built from GARP Ch.10

### Step 1 – Recall the stated purpose

GARP: the primary purpose of operational due diligence is to ensure that no significant additional risk of loss is created for investors through settlement, corporate actions, misappropriation or theft, or a breakdown in confirmation, verification, valuation and reconciliation.
⇒ B is the purpose, nearly word for word.

### Step 2 – Place the other options in their own sections

A is the investment process (IM-12 c). C is business model risk (IM-12 f). D is "do the fund terms make sense for the strategy", a risk management criterion (IM-12 d).

### Interpretation

The four sections of due diligence answer four different questions. Operational due diligence asks whether the money can be lost through the plumbing, whatever the quality of the investment decisions.

### Why the other options are wrong

**A is incorrect.** Sibling swap: the manager's edge is the subject of the investment process review.
**C is incorrect.** Sibling swap: the gap between management fees and expenses is a business model risk test.
**D is incorrect.** Sibling swap: comparing terms with peers belongs to the risk management review of fund terms.

**LO:** IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.8; SchweserNotes 2026 Book 5, Reading 94, LO 94.e

## Solution IM-12 e.2

**Answer: B**

Bank Q521 • GARP PE2 Q1

### Step 1 – Test each "does not need to" clause

A: operations, accounting, treasury, technology and compliance staff must also be qualified, and background checks reach the COO, CFO and accounting managers. C: the investor must verify with the listed law firm that it was responsible for the original content and any updates. D: service providers are interviewed and their internal control letters and audited statements obtained.
⇒ A, C and D each drop a required step.

### Step 2 – Test B

GARP: "Most firms today have either their own in-house compliance function or an outsourced relationship with a compliance service provider." The investor verifies the practices (code of ethics, related-party prohibition, employee trading rules), not where the function sits.
⇒ B is correct.

### Interpretation

GARP's practice exam rewards precise scope: the qualifier "in-house" is what makes B true and the other three false.

### Why the other options are wrong

**A is incorrect.** Ignores the review of non-investment staff, including background checks on the COO and CFO.

**C is incorrect.** Skips the confirmation with the law firm of its role in drafting and updating the documents.
**D is incorrect.** Skips the evaluation of the relationships, the control letters and the interviews.

**LO:** IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.8 (Internal Control Assessment; Documents and Disclosures; Service Provider Evaluation); GARP Practice Exam 2, Q1

## Solution IM-12 e.3

**Answer: D**

Bank Q520 • BT P2.T8.414.1

### Step 1 – Check A, B and C against GARP 10.8

A is GARP's consistency requirement across the OM, subscription agreement, LPA, IMA, Form ADV and website. B is GARP's risk factor guidance word for word. C is GARP's service provider evaluation.
⇒ A, B and C are true.

### Step 2 – Check D

GARP: "Typical indemnifications should not extend to the gross negligence, bad faith, fraud, or willful misconduct of the manager." D says the fund should indemnify exactly those acts.
⇒ D is not true: it is the answer. (The bank marks this key as corrected; checked again here against GARP and confirmed.)

### Interpretation

Indemnifying the manager for ordinary errors is normal; indemnifying fraud or gross negligence would shift the cost of the manager's misconduct onto the investors.

### Why the other options are wrong

**A is incorrect.** True: the documents must agree at the same point in time; the OM governs.
**B is incorrect.** True: too few risk factors is the bigger red flag, and boilerplate ones are a red flag too.
**C is incorrect.** True: control letters and audited statements give an independent check on the providers themselves.

**LO:** IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.8 (Documents and Disclosures; Service Provider Evaluation); SchweserNotes 2026 Book 5, Reading 94, LO 94.e

## Solution IM-12 e.4

**Answer: D**

New • built from GARP Ch.10

### Step 1 – What written procedures can and cannot show

Reviewing procedures "only gives investors a sense of whether the management of the hedge fund takes the whole control process seriously". Schweser: they show "only the intention of having a strong control environment".
⇒ A overstates their value.

### Step 2 – What gives more assurance

GARP: some managers have an outside audit firm evaluate their procedures and controls periodically and give an opinion on whether they are well designed and operating effectively. Schweser: a proactive audit report is "usually a good sign".
⇒ D is correct.

### Step 3 – Dismiss B and C

Background checks extend to the COO, CFO, accounting managers and other back- and middle-office staff. Cash held with one dealer is exactly the exposure that cost funds money at Lehman, MF Global and Refco.

### Interpretation

The investor is looking for evidence that controls work every day, and that evidence is strongest when it comes from an independent party.

### Why the other options are wrong

**A is incorrect.** Intent is not effectiveness; documents alone give little assurance.

**B is incorrect.** Too narrow: the qualifications and backgrounds of operations and accounting staff are part of the internal control review.

**C is incorrect.** Polarity flip: concentration with one prime broker creates counterparty risk; recovery after a dealer failure may be only 30 or 40%.

**LO:** IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.8 (Internal Control Assessment); SchweserNotes 2026 Book 5, Reading 94, LO 94.e

## Solution IM-12 e.5

**Answer: B**

New • built from GARP Ch.10

### Step 1 – The liquidity mismatch

The OM, subscription agreement, LPA, IMA, Form ADV and website must say the same thing at the same time; documents often drift after launch, and the OM "ultimately governs an investor's rights and obligations". Liquidity terms are among the items GARP singles out.

⇒ The difference must be reconciled with the manager, using the OM as the reference.

### Step 2 – The post-dated gate

An investor "can actually check the document to see if any changes were made after the as-of date"; such changes "should be discussed with the manager immediately and with the law firm". Managers have been known to change documents without the law firm's knowledge, so the investor also confirms the firm is still retained.

⇒ B covers both findings.

### Interpretation

Document review is part of operational due diligence because inconsistent or quietly amended documents are how investors end up with terms (gates, side pockets, suspensions) they never agreed to.

### Why the other options are wrong

**A is incorrect.** The website does not govern investor rights; the OM does.

**C is incorrect.** Authorship of the original says nothing about a change made later, possibly without the law firm's knowledge.

**D is incorrect.** Liquidity terms are part of the document consistency check in operational due diligence.

**LO:** IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.8 (Documents and Disclosures); SchweserNotes 2026 Book 5, Reading 94, LO 94.e

### Solution IM-12 e.6 Answer: C

New • built from GARP Ch.10

#### Step 1 – Read the statements against the strategy

GARP: "Equity funds should look very different than global macro funds or fixed-income funds." A global macro fund "most likely has interest income earned in its income statement".
⇒ C is consistent with the strategy.

#### Step 2 – Test the other three

A: GARP's own example of an inconsistency: low, falling leverage cannot coexist with high, rising interest expense. B: a long-term buy-and-hold fund "should not be generating large amounts of realized gains"; it should show sizable unrealised gains or losses. D: "There should never be incentive fees earned in years when the fund lost money."
⇒ A, B and D each call for questions to the manager, or even the auditor.

#### Interpretation

An unqualified opinion is where the review starts, not where it ends: it says the statements are fairly presented, not that they make sense for the strategy the manager describes.

#### Why the other options are wrong

**A is incorrect.** Interest expense contradicts the leverage the manager represents; recalculate leverage from the balance sheet.
**B is incorrect.** Large realised gains do not fit a buy-and-hold strategy.
**D is incorrect.** An incentive fee in a loss year is a red flag; recalculate the fees against the OM.

**LO:** IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.8 (Documents and Disclosures: financial statements); SchweserNotes 2026 Book 5, Reading 94, LO 94.e

### Solution IM-12 e.7 Answer: B

New • built from GARP Ch.10

#### Step 1 – What the reading asks for

Investors should expect the fund to make all key contacts at its service providers available, obtain the providers' internal control letters and audited financial statements as an independent check on them, and interview them about their role in executing trades, providing technology, performing valuations or verification and safeguarding assets.
⇒ B is the full procedure.

#### Step 2 – Why every provider is in scope

Administrators and valuation agents produce the NAV and verify positions; a failure or a lack of independence there is how a reported value can be wrong even when the assets sit at a prime broker.

#### Interpretation

This is the check that would have exposed Madoff: BMIS kept execution, custody and administration in-house, so there was no independent provider to call (Workbook A24.4, in IM-14 b).

#### Why the other options are wrong

**A is incorrect.** Relies on the fund's own description; the fund is expected to make the contacts available.
**C is incorrect.** Too narrow: valuation and verification providers are part of the evaluation.
**D is incorrect.** Size is not evidence of independence or competence; the feeder funds' large auditors relied on BMIS's tiny auditor without checking its work.

**LO:** IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. • **Reference:** Mirabile, Hedge Fund Investing 2e, Ch.12 (GARP 2025 Ch.10), 10.8 (Service Provider Evaluation); SchweserNotes 2026 Book 5, Reading 94, LO 94.e

IM-13 • Private Capital Investing: Distress Symptoms and Remedies • Vardeez: UNTIERED

# IM-13 a

P3 CONCEPT

Explain different reasons why firms may experience financial distress and identify potential indicators of distress.

Workbook: no workbook section • A28 cross-reference note only • Vardeez focus

**WHAT IS TESTED HERE**

**Why P3.** Vardeez marks LO 95.a and LO 95.b as important ("both are Imp") in a reading it leaves untiered. There is no workbook section; the A28 revision notes carry a short cross-reference (Map #49).
**Vardeez pointer:** none beyond the flag.

**Content authority:** SchweserNotes 2026 Reading 95 (Ippolito, Ch.20), used in place of the missing GARP 2026 book. Vardeez's own notes for this reading do not follow the reading (see the correction box).

**Reported exam shapes:** none reported.

**What the questions look like:** conceptual. Classify an indicator as financial or nonfinancial; match a firm's story to its cause of distress; rank the leverage and business performance combinations by severity. The bank holds two questions, so four are new.

## 1. Spotting distress early: questions and indicators

Early detection is critical but hard: it is difficult to tell the *symptoms* of distress from the *underlying problem*, and a temporary, seasonal dip from a more permanent, cyclical one. The reading's seven accounting questions:

1. Are earnings generated from core operations?
2. Do accounting policies artificially increase profitability?
3. Any changes in reserve accounting policies?
4. Any restructuring or reorganisation adjustments?
5. Are creditors showing signs of concern?
6. Are bad debts being written off?
7. Does the tax charge look low relative to reported earnings?

The primary distinction among the symptoms of credit deterioration is between **financial** and **nonfinancial** indicators:

| Financial (in the numbers) | Nonfinancial (in behaviour and structure) |
|---|---|
| Covenant breaches | Unnecessarily complex corporate structure |
| Late submission of financial statements | Resignations of board members or senior management |
| Creative accounting: off-balance-sheet financing, contingent liabilities | Change in the fiscal year-end date |
| Earnings enhancements: sale-leasebacks, write-downs followed by gains on sale, accelerated revenue recognition, deferral of costs | Change in auditors; failure to meet commitments on time; material changes in risk tolerance |

## 2. Five causes of distress

Defaults arise from an unsuitable business model, too much leverage, an overly complex organisation, internal or external shocks, or a mix. The reading groups the causes in five families.

1. **Leverage and business performance** (Figure 1). There is always a trade-off between debt and operating performance. A solid business with too much debt suffers least; weak operations do the damage, and weak operations with too much debt are worst.
2. **Unexpected liabilities**, usually not debt. *Tort* liabilities (claims for harm to people or property, often class actions) are underestimated by management because they are low-probability events, awards often exceed expectations, insurance may or may not cover them, and markets "assume the worst" and discount the securities. *Contract* liabilities arise when contracts turn unfavourable, sometimes inherited in an acquisition; the terms are often unknown in advance, so they are hard to protect against.
3. **Performing well below expectations**: the most common cause. *Economic downturns* recur, and the key challenge is to tell whether the downturn itself or management's decisions and strategy drive the deterioration. *Uncompetitive products or services* lose demand as preferences shift.
4. **Crisis of confidence**, often fraud (Enron). Two areas of focus: the *reliability of historical information* (once fraud is found, all past information is suspect: assess the informational contagion and whether there is a practical cutoff point) and *liquidity and the likelihood of bankruptcy* (cash balances, available credit lines, risk of loan acceleration).
5. **Unrealistic business plans**: *leveraged buyouts* that assume cash from asset sales or operations will pay debt down; *roll-ups* that count on economies of scale that never arrive; *too many new entrants* drawn by high margins, so that supply outruns demand.

| | Average leverage | Excessive leverage |
|---|---|---|
| **Solid business model** | No distress from this source | Facilities fully drawn; tipped by a cyclical downturn. Cash flow stabilises about 10% below the point of default; revenue recovers within about 3 years. |
| **Weak operations** | Revenues and margins fall sharply; default with or without a downturn. Cash flow may fall more than 50% below the default point; revenue recovery 5 to 10 years. | **Worst case.** Facilities often fully drawn; default likely triggered by a cyclical event. Cash flow down 70% or more; extended recovery. |

Figure 1. Leverage against business performance (SchweserNotes 2026 Reading 95, LO 95.a). Darker grey means a deeper cash-flow fall and a longer recovery; the heavy border marks the worst case. The top-right cell is what Vardeez calls financial distress (a viable business with the wrong capital structure); the bottom row is closer to economic distress (the business itself is impaired).

**Correction to the source notes.** Vardeez's IM-13 notes come from "compiled notes", not from the reading, and do not reproduce its lists. Its indicators are model-based signals (Altman Z-score, Merton distance to default, KMV expected default frequency) and the claim that market indicators lead accounting ones; the reading instead gives seven accounting questions, a financial versus nonfinancial split, and five families of causes. The economic versus financial distress distinction is sound background and maps onto Figure 1, but exam answers should use the reading's lists. Bank items Q533 and Q534 and the A28 cross-reference note follow Schweser.

## Summary card: IM-13 a

| Cause | Mechanism | Examples in the reading |
|---|---|---|
| **Leverage and performance** | Debt load meets operating weakness; weak operations do the most damage | Three combinations of Figure 1 |
| **Unexpected liabilities** | Non-debt claims hit cash flow; hard to foresee or insure | Tort (class actions); contract (inherited in an acquisition) |
| **Below expectations (most common)** | The business earns less than planned | Economic downturn; uncompetitive products |
| **Crisis of confidence** | Trust in the numbers collapses, then liquidity | Fraud (Enron); loan acceleration |
| **Unrealistic business plans** | The plan's cash or savings never arrive | LBOs; roll-ups; too many new entrants |

## Exam traps

- **Financial or nonfinancial.** A change of auditor or of fiscal year-end is *nonfinancial*; a sale-leaseback (an earnings enhancement) is *financial*.
- **Severity order.** Excessive leverage with weak operations is worst (cash flow down 70% or more); a solid business with excessive leverage is mildest (about 10%, recovery within about 3 years).
- **Most common cause** is performing well below expectations, not fraud or leverage.
- **Tort polarity.** Management *underestimates* tort liabilities; markets assume the *worst*.
- **Crisis of confidence.** All past information becomes suspect, not only the fraud years, and liquidity is the immediate question.
- **Sibling swap.** Roll-ups, LBOs and crowded new industries are *unrealistic business plans*; obsolete products are *performing below expectations*.
- **Seasonal is not distress.** A dip in line with the usual seasonal pattern is not a warning signal by itself.

## Practice questions: IM-13 a

### Question IM-13 a.1

Bank Q533 • Schweser Module Quiz 95.1 Q1

A credit analyst at a private debt fund is screening a portfolio company for signs of credit deterioration. Which of the following is a clear financial indicator of credit deterioration?

**A.** Change in auditors.

**B.** Material changes in risk tolerance.

**C.** Use of a sale-leaseback arrangement.

**D.** Board or senior management resignations.

### Question IM-13 a.2

New • built from SchweserNotes 2026 Reading 95

A credit analyst at a regional bank reviews a borrower's latest annual report while preparing its annual credit review. Which observation is most consistent with an early warning signal of financial distress?

**A.** Revenue dipped in the first quarter, in line with the firm's usual seasonal pattern.

**B.** The tax charge appears low relative to the earnings the firm reports.

**C.** Almost all of the year's earnings were generated by the firm's core operations.

**D.** The firm kept the same auditor and the same fiscal year-end as in prior years.

### Question IM-13 a.3

New • built from SchweserNotes 2026 Reading 95

A distressed-debt analyst is preparing a training note on why firms default. Which statement about performing well below expectations as a cause of financial distress is correct?

**A.** It is the most common cause of distress, and in a recession the key challenge is to judge whether the downturn itself or management's decisions drive the deterioration.

**B.** It is a rare cause of distress, because recessions can be forecast and their effect on revenue hedged in advance.

**C.** It covers products that lose demand as preferences shift, which the reading classifies as a crisis of confidence.

**D.** In a recession, the deterioration should always be attributed to the business cycle rather than to management's strategy.

### Question IM-13 a.4

Bank Q534 • Schweser Module Quiz 95.1 Q2

A private credit portfolio manager is stress testing four borrowers. Which of the following scenarios is most likely to result in a rapid, deep cash flow decline and a prolonged period before revenue recovery?

**A.** Average leverage with weak business operations.

**B.** Minimal leverage and strong business performance.

**C.** Excessive leverage combined with a solid business model.

**D.** Excessive leverage combined with weak business operations.

**Question IM-13 a.5** New • built from SchweserNotes 2026 Reading 95

A distressed-debt analyst groups four troubled borrowers by the main cause of their distress. Which grouping is correct?

**A.** A roll-up of 40 regional dental practices whose expected economies of scale never materialised: performing well below expectations.

**B.** A chemicals maker facing a class action over groundwater contamination that its insurance may not cover: unexpected liabilities.

**C.** A retailer whose sales collapsed after its customers moved online: crisis of confidence.

**D.** A logistics firm whose former CFO is found to have inflated revenue for five years: unrealistic business plan.

**Question IM-13 a.6** New • built from SchweserNotes 2026 Reading 95

Shortly after a manufacturer discloses that its former CFO falsified three years of revenue, a credit analyst at an insurer that holds the firm's bonds is asked what to assess first. Which approach is most consistent with the reading?

**A.** Treat only the three restated years as unreliable, since information from outside the fraud period is unaffected.

**B.** Assess how far the fraud contaminates all previously reported information, and whether it could trigger a liquidity crisis through cash balances, credit lines and loan acceleration.

**C.** Focus on whether the firm's products remain competitive, since a crisis of confidence is mainly a problem of customer demand.

**D.** Wait for the next audited statements before acting, since market prices overreact to fraud news and then recover.

# Solutions • IM-13 a

## Solution IM-13 a.1 — Answer: C

Bank Q533 • Schweser Module Quiz 95.1 Q1

### Step 1 – Recall the two lists

Financial: covenant breaches, late financial statements, creative accounting, earnings enhancements (sale-leasebacks, write-downs followed by gains on sale, accelerated revenue recognition, cost deferral). Nonfinancial: complex structure, resignations, change of fiscal year-end or auditor, missed commitments, changes in risk tolerance.

### Step 2 – Classify each option

A sale-leaseback is an earnings enhancement, which sits in the financial list. A, B and D are all in the nonfinancial list.
$\Rightarrow$ C is the financial indicator.

### Interpretation

"Financial" here means visible in the accounts. A sale-leaseback books a gain or cash today against lease payments later, which flatters reported results.

### Why the other options are wrong

**A is incorrect.** A change in auditors is a nonfinancial (behavioural) indicator.
**B is incorrect.** A material change in risk tolerance is nonfinancial.
**D is incorrect.** Board or management resignations are a serious warning, but a nonfinancial one.

**LO:** IM-13 a: Explain different reasons why firms may experience financial distress and identify potential indicators of distress. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.a

## Solution IM-13 a.2 — Answer: B

New • built from SchweserNotes 2026 Reading 95

### Step 1 – Recall the early-warning questions

The reading's accounting questions include: are earnings generated from core operations, and does the tax charge appear low relative to reported earnings?
$\Rightarrow$ A tax charge that looks low against reported earnings suggests the earnings may be overstated: an early warning signal.

### Step 2 – Check the other observations

A dip that follows the usual seasonal pattern is the kind of temporary fluctuation the analyst must separate from a permanent deterioration. Earnings from core operations, and an unchanged auditor and fiscal year-end, are the healthy answers to the reading's questions.

### Interpretation

The tax line is hard to inflate along with profits: when reported earnings rise but the tax charge does not, ask what kind of earnings they are.

### Why the other options are wrong

**A is incorrect.** A seasonal dip is a temporary fluctuation, not by itself a distress signal.
**C is incorrect.** Earnings from core operations are the reassuring answer to the first question.
**D is incorrect.** Changes of auditor or fiscal year-end are warning signs; keeping them is not.

**LO:** IM-13 a: Explain different reasons why firms may experience financial distress and identify potential indicators of distress. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.a

### Solution IM-13 a.3 — Answer: A

New • built from SchweserNotes 2026 Reading 95

**Step 1 – Recall the reading's statement**

"Performing well below expectations. This is the most common cause of financial distress." Its two triggers are economic downturns and uncompetitive products or services.

**Step 2 – The diagnostic challenge**

For downturns, "the key challenge is determining whether the downturn itself is driving the deterioration, or whether management decisions and strategies are the underlying cause".
⇒ A states both points.

**Interpretation**

The diagnosis matters for the remedy: a cyclical problem may need time and liquidity; a management problem needs a new plan, and extra capital can simply give weak management more time to destroy value (IM-13 b).

**Why the other options are wrong**

**B is incorrect.** Polarity flip: it is the most common cause, and the business cycle "is unavoidable".
**C is incorrect.** Sibling swap: obsolete products are the second trigger of performing below expectations; a crisis of confidence is usually fraud.
**D is incorrect.** Absolute claim: the reading asks the analyst to separate the cycle from management's decisions, not to assume the answer.

**LO:** IM-13 a: Explain different reasons why firms may experience financial distress and identify potential indicators of distress. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.a

### Solution IM-13 a.4 — Answer: D

Bank Q534 • Schweser Module Quiz 95.1 Q2

**Step 1 – Recall the three outcomes**

Excessive leverage with a solid model: cash flow stabilises about 10% below the point of default; revenue recovers within about 3 years. Average leverage with weak operations: cash flow may fall more than 50%; recovery 5 to 10 years. Excessive leverage with weak operations: cash flow down 70% or more; an extended recovery.

**Step 2 – Pick the worst combination**

⇒ D: high leverage leaves no cushion while weak operations erode revenue and margins.

**Interpretation**

Operations drive the depth of the fall; leverage decides how little room there is to absorb it (Figure 1).

**Why the other options are wrong**

**A is incorrect.** Severe (more than 50%, 5 to 10 years) but less so than with excessive leverage added.
**B is incorrect.** A healthy firm, not a distress scenario.
**C is incorrect.** The mildest distress case: a solid business limits the fall to about 10%.

**LO:** IM-13 a: Explain different reasons why firms may experience financial distress and identify potential indicators of distress. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.a

### Solution IM-13 a.5 Answer: B

New • built from SchweserNotes 2026 Reading 95

**Step 1 – Recall the five families**

Leverage and business performance; unexpected liabilities (tort, contract); performing well below expectations (downturns, uncompetitive products); crisis of confidence (often fraud); unrealistic business plans (LBOs, roll-ups, too many entrants).

**Step 2 – Place each borrower**

Class action for environmental harm: a tort liability, under unexpected liabilities, often underestimated by management and possibly uninsured. Roll-up whose economies of scale never came: unrealistic business plan. Retailer losing customers to online rivals: uncompetitive products, under performing below expectations. CFO fraud: crisis of confidence.
$\Rightarrow$ Only B is grouped correctly.

**Interpretation**

Markets typically "assume the worst" about a tort claim and discount the firm's securities before the award is known.

**Why the other options are wrong**

**A is incorrect.** Roll-ups are one of the reading's three unrealistic business plans.
**C is incorrect.** Obsolete products are performing below expectations, not a crisis of confidence.
**D is incorrect.** Accounting fraud is the classic crisis of confidence (Enron), not an unrealistic plan.

**LO:** IM-13 a: Explain different reasons why firms may experience financial distress and identify potential indicators of distress. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.a

### Solution IM-13 a.6 Answer: B

New • built from SchweserNotes 2026 Reading 95

**Step 1 – Reliability of historical information**

"Once financial fraud is uncovered, all previously reported financial information becomes suspect." The analyst assesses the informational contagion and whether there is a practical cutoff point for reliable data.

**Step 2 – Liquidity and the likelihood of bankruptcy**

The analyst asks whether the fraud could trigger liquidity problems: cash balances, the availability of credit lines and the risk of loan acceleration.
$\Rightarrow$ B covers the two areas of focus the reading names.

**Interpretation**

A crisis of confidence turns a reporting problem into a funding problem: lenders who no longer trust the numbers can pull lines or accelerate loans.

**Why the other options are wrong**

**A is incorrect.** Too narrow: the contagion can reach information outside the restated years; that is why a cutoff point must be found.
**C is incorrect.** Sibling swap: competitiveness belongs to performing below expectations.
**D is incorrect.** Waiting ignores the immediate liquidity question, which is the more urgent of the two.

**LO:** IM-13 a: Explain different reasons why firms may experience financial distress and identify potential indicators of distress. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.a

IM-13 • Private Capital Investing: Distress Symptoms and Remedies • Vardeez: UNTIERED

# IM-13 b

P3 CONCEPT

Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method.

Workbook: no workbook section • A28 cross-reference note only • Vardeez focus

**WHAT IS TESTED HERE**

**Why P3.** Vardeez marks LO 95.a and LO 95.b as important in an untiered reading; no workbook section.
**Vardeez pointer:** none beyond the flag. Content authority: SchweserNotes 2026 Reading 95.

**Reported exam shapes:** none reported.

**What the questions look like:** conceptual. Match a firm's need to the right method (cash raising, debt reduction), name a method's implication or challenge, spot the practical limitation, and compare out-of-court with in-court restructuring. Bank Q539 is re-filed here from IM-13 c (P4, not built): choosing between the two restructuring routes is a remedy decision.

## 1. Three families of remedies, chosen by stage and cause

Solutions to distress fall into three families: **restructuring**, **raising additional capital** and **reducing leverage**. Which one fits depends on where the firm stands (recap of IM-13 a: the cause decides the cure):

- **Cause pathway** (Whitman and Diz, Schweser Figure 95.1): is the deterioration *temporary*, is this the firm's *first* episode of distress, or has a *previous turnaround already failed*? Each pathway calls for a different restructuring approach.
- **Debt trading level** (Moyer, Figure 95.2): the market price of the debt relative to its full repayment value reflects the market's expectation of how severe the distress is and which restructuring options are realistic. The further below full repayment value, the more severe the expected distress.

## 2. Raising capital: buys time, does not fix the problem

Extra cash can carry a firm through a bad period, but "it may also give ineffective management more time to further erode value".

- **Asset sales.** Price is critical; a sale above fair value helps. A good price is easier to get *before* a bankruptcy filing.
- **Secured financing** against unencumbered assets. Often limited in practice: by the time of distress the best assets are already pledged from earlier rescues.
- **Sale-leaseback.** Cash now while keeping use of the asset. The lease may be off-balance-sheet, making leverage look lower, so the analyst adjusts for it.
- **Equity sponsor.** A private owner with more than 50% is generally positive for distressed investors: equity gains only after debt is repaid, which aligns incentives. Harder to achieve with widely dispersed public ownership.

## 3. Reducing leverage

**More equity.** Lenders often require an equity injection in a restructuring; incumbent investors can provide it quickly, while new investors take longer because of their due diligence. **Less debt**:

- **Open market repurchases**: best for less than about 20% of the bonds; simple, fast, low cost, discreet; usually at a discount. If bank debt also trades at a discount, lenders often require cash to repay bank debt first.
- **Direct purchases** from individual bondholders, who often approach the firm hoping for more than the market price.
- **Cash tender offer**: to retire a large portion; open to all holders on strict terms (deadline, price, maximum amount).
- **Exchange offer**: for firms short of cash; new securities for the distressed debt. *Coercive*: non-participants are warned they may be disadvantaged. *Noncoercive*: participation is encouraged by the offer's attractiveness.

**Five practical limitations**: *liquidity* (no excess cash, no discounted buyback); *timing* (time to the next liquidity event; selling a unit takes time); *magnitude* (a short horizon and little liquidity make a small problem large); *complexity* (layered capital structures and restrictive bank covenants); *severability* (noncore units can be sold; a core-only firm lacks that option).

## 4. The last rungs: out of court or in court

When these remedies are not enough, the debt is restructured with or without the court (details in IM-13 c, P4). **Out of court (OCR)**: voluntary, faster and cheaper, but open to **holdouts**; coercive offers and the threat of a prepackaged bankruptcy are the answer. **In court (ICR), Chapter 11**: binds creditors class by class, but is slower, costlier and damaging to reputation.

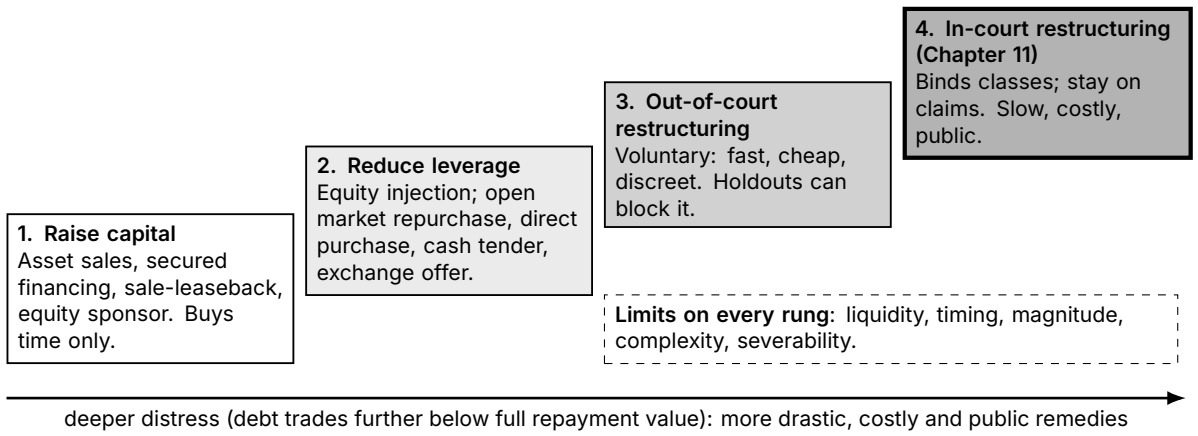



Figure 1. The remedy ladder. Each rung up is more drastic: fills darken from white to dark grey, and the heavy border marks the court process. The arrow below shows the direction of deeper distress. The ordering is a teaching device built from the reading's three families of remedies and its out-of-court versus in-court comparison, not a figure from the reading.

**Correction to the source notes.** Vardeez's remedy ladder (operational turnaround; out-of-court distressed exchange; formal insolvency; its stage 2 is missing) and its IM-13 c list (DIP priming liens, ipso facto clauses, the zone of insolvency, cross-border recognition) are not the reading's content. The reading organises remedies as restructuring, raising capital (four ways), reducing leverage (equity, or four debt-reduction methods) and five practical limitations; its three legal risks are the stay on claims, the

preference (twilight) period and upstream guarantees. Vardeez's points on rating agencies and the tax on forgiven debt lie outside the reading and are not needed for this LO.

## Summary card: IM-13 b

| Method | When it fits | Implication or challenge |
|---|---|---|
| **Asset sale** | Firm can spare the asset | Price is critical; better before a filing |
| **Secured financing** | Unencumbered assets exist | Best assets usually already pledged |
| **Sale-leaseback** | Needs cash, must keep using the asset | Off-balance-sheet lease flatters leverage |
| **Equity sponsor** | Private owner above 50% | Aligned with creditors; rare in dispersed public firms |
| **Equity injection** | Lenders require new equity | Incumbents fast; new investors slow |
| **Open market repurchase** | Under about 20% of bonds | Fast, cheap, discreet; bank debt may come first |
| **Cash tender offer** | Large portion of debt | Strict terms to all holders |
| **Exchange offer** | Little cash | Coercive or noncoercive |
| **Out of court** | Creditors can agree | Fast, cheap, private; holdouts |
| **In court (Chapter 11)** | Agreement impossible | Binds classes; slow, costly, public |

## Exam traps

- **Method swap.** Small buyback: open market. Large buyback on strict terms: cash tender offer. No cash: exchange offer. Holder calls the firm: direct purchase.
- **Keep the asset** with a sale-leaseback, not an asset sale; adjust leverage for an off-balance-sheet lease.
- **Timing polarity.** Asset sales fetch better prices *before* a bankruptcy filing.
- **Secured financing is often not available**: the best collateral is already pledged.
- **Capital is not a cure.** New money buys time and can let weak management destroy more value.
- **Complexity polarity.** Layered, not simple, capital structures make restructuring harder; non-core units help.
- **OCR versus ICR.** OCRs are faster and cheaper but face holdouts; ICRs are slower, costlier and public.

## Practice questions: IM-13 b

### Question IM-13 b.1

New • built from SchweserNotes 2026 Reading 95

A distressed-debt analyst notes that a retailer's senior bonds now trade at 45 per 100 of face value, down from 92 six months ago. Which interpretation of the debt trading level is most consistent with the reading?

**A.** The lower trading level signals that the market expects more severe distress, and it informs the analyst about which restructuring solutions are realistic.

**B.** The lower trading level signals that the market expects milder distress, because cheaper bonds make open market repurchases more attractive.

**C.** The trading level is informative only for bank loans, because bond prices mainly reflect changes in interest rates.

**D.** The trading level measures the bonds' coupon relative to market yields and says nothing about the restructuring options.

### Question IM-13 b.2

Bank Q535 • Schweser Module Quiz 95.2 Q1

The CFO of a distressed manufacturer needs cash urgently but must keep operating its main production plant. Which of the following methods most effectively provides cash to a distressed firm while allowing it to continue using a key operational asset?

**A.** Asset sale.

**B.** Equity sponsor.

**C.** Sale-leaseback.

**D.** Secured financing.

### Question IM-13 b.3

Bank Q536 • Schweser Module Quiz 95.2 Q2

A distressed firm's treasurer is considering how to reduce its bond debt. Which of the following scenarios is the most effective situation for using open market repurchases to reduce a company's debt?

**A.** The firm wants to offer alternative securities instead of cash.

**B.** The firm wants to repurchase only a small percentage of its outstanding bonds.

**C.** The firm needs to reduce a large proportion of its total debt under strict offer conditions.

**D.** Individual bondholders approach the firm directly, seeking smaller discounts than the market price.

**Question IM-13 b.4** New • built from SchweserNotes 2026 Reading 95

A distressed industrial firm with almost no cash wants to cut its bond debt. It offers bondholders new secured notes in exchange for their existing unsecured bonds, and warns that bondholders who do not take part may end up worse off than those who do. Which method is the firm using?

**A.** A noncoercive exchange offer.

**B.** A cash tender offer.

**C.** A coercive exchange offer.

**D.** A direct purchase from bondholders.

**Question IM-13 b.5** New • built from SchweserNotes 2026 Reading 95

A restructuring adviser briefs the board of a distressed company on the implications of different ways to raise capital and reduce leverage. Which of the following statements is correct?

**A.** A private equity sponsor holding more than 50% of the equity is generally positive for creditors, because the sponsor realises value only after the debt claims are satisfied.

**B.** Assets generally fetch better prices after a bankruptcy filing, so a distressed firm should delay asset sales until it has filed.

**C.** Secured financing is usually the easiest source of cash for a distressed firm, because its most valuable assets are rarely pledged.

**D.** An equity injection from new outside investors can usually be arranged faster than one from the firm's incumbent investors.

**Question IM-13 b.6** Bank Q537 • Schweser Module Quiz 95.2 Q3

A distressed firm's advisers have mapped out several restructuring strategies that work on paper. Which of the following statements best describes a key limitation that can prevent a distressed firm from executing a theoretical restructuring strategy?

**A.** A firm can always sell any business unit immediately to raise cash.

**B.** A firm with limited cash reserves may be unable to repurchase debt.

**C.** The presence of noncore business units always reduces restructuring risk.

**D.** Firms with simple capital structures typically face significant restructuring challenges.

**Question IM-13 b.7** Bank Q539 • Schweser Module Quiz 95.2 Q4 • re-filed from IM-13 c

A distressed firm's creditors are deciding whether to negotiate directly with the firm or let it file for court protection. Which of the following statements best captures a key difference between out-of-court restructurings (OCRs) and in-court restructurings (ICRs)?

**A.** OCRs are typically faster, while ICRs can be slower.

**B.** OCRs eliminate the risk of holdouts among creditors.

**C.** ICRs avoid reputational damage for the distressed firm.

**D.** ICRs are always preferred by creditors because they offer more flexibility.

# Solutions • IM-13 b

## Solution IM-13 b.1

**Answer: A**

New • built from SchweserNotes 2026 Reading 95

### Step 1 – What the trading level is

The debt trading level is the current market price of the debt relative to its full repayment value.

### Step 2 – What it tells the analyst

It "serves as an indicator of the market's expectations about distress severity and potential restructuring options". A fall from 92 to 45 means the market expects far less to be repaid.
⇒ A: deeper expected distress, and a guide to the realistic solutions.

### Interpretation

The price is the market's verdict on the stage of distress. Near full value, the problem looks like liquidity; far below it, the market is pricing in a restructuring of the claim itself.

### Why the other options are wrong

**B is incorrect.** Polarity flip: a lower price signals more severe expected distress. Cheaper bonds can make a discounted buyback attractive, but only for a firm with cash to spare.
**C is incorrect.** The reading applies the trading level to a firm's actively traded debt; a fall from 92 to 45 is a credit signal, not a rate move.
**D is incorrect.** The trading level is a price relative to repayment value, and the reading links it directly to the restructuring options.

**LO:** IM-13 b: Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20 (after Moyer, 2005); SchweserNotes 2026 Book 5, Reading 95, LO 95.b

## Solution IM-13 b.2

**Answer: C**

Bank Q535 • Schweser Module Quiz 95.2 Q1

### Step 1 – State the two requirements

Cash now, and continued use of the plant.

### Step 2 – Test each method

A sale-leaseback sells the asset to an investor and leases it straight back: cash, with the asset still in use. An outright sale gives up the asset. Secured financing needs unencumbered collateral, which distressed firms usually lack. An equity sponsor provides capital but is not tied to the asset and may be hard to find.
⇒ C meets both requirements.

### Interpretation

The implication to remember: the lease may be off-balance-sheet, making leverage look lower than it is, so an analyst adds it back.

### Why the other options are wrong

**A is incorrect.** Raises cash but the firm loses the use of the plant.
**B is incorrect.** Not tied to a specific asset, and often unavailable to a distressed public firm with dispersed owners.
**D is incorrect.** Usually infeasible: the most valuable assets were pledged in earlier rescues.

**LO:** IM-13 b: Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.b

### Solution IM-13 b.3 — Answer: B

Bank Q536 • Schweser Module Quiz 95.2 Q2

#### Step 1 – When open market repurchases work

They are "most effective when the firm intends to repurchase less than about 20% of the bonds outstanding", and they are simple, fast, low cost and discreet.
⇒ B fits.

#### Step 2 – Match the other scenarios to their methods

Alternative securities instead of cash: an exchange offer. A large reduction on strict terms: a cash tender offer. Holders approaching the firm: direct purchases.

#### Interpretation

Buying a large share in the open market would push the price up against the firm and could not stay discreet; that is when a tender offer takes over.

#### Why the other options are wrong

**A is incorrect.** Describes an exchange offer.
**C is incorrect.** Describes a cash tender offer.
**D is incorrect.** Describes direct purchases from bondholders.

**LO:** IM-13 b: Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.b

### Solution IM-13 b.4 — Answer: C

New • built from SchweserNotes 2026 Reading 95

#### Step 1 – Identify the family

The firm lacks cash and offers new securities for its distressed debt: an *exchange offer*, not a cash tender or a direct purchase.

#### Step 2 – Coercive or noncoercive

Exchange offers "may be coercive, meaning participants are warned that nonparticipants may be disadvantaged"; noncoercive offers encourage participation by highlighting the opportunity's attractiveness.
⇒ The warning makes it a coercive exchange offer.

#### Interpretation

Coercion is the firm's answer to the holdout problem: if staying out looks worse than taking part, fewer creditors hold out (IM-13 c).

#### Why the other options are wrong

**A is incorrect.** A noncoercive offer persuades with attractive terms, not with a warning.
**B is incorrect.** A tender offer pays cash; this firm has almost none.
**D is incorrect.** A direct purchase is a negotiated deal with individual holders, usually for cash.

**LO:** IM-13 b: Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.b

### Solution IM-13 b.5 Answer: A

New • built from SchweserNotes 2026 Reading 95

#### Step 1 – The equity sponsor

"A private owner with more than a 50% stake is generally positive for distressed investors. A majority equity holder realizes value only after debt claims are satisfied, which aligns incentives."
⇒ A is correct.

#### Step 2 – Test B, C and D

Asset sales: a favourable price is "generally easier to obtain before a bankruptcy filing". Secured financing: "often limited", since the most valuable assets were pledged in earlier restructurings. Equity: incumbent capital "can typically be arranged quickly", while new investors take longer because of due diligence.
⇒ B, C and D each reverse the reading.

#### Interpretation

Each method's implication follows from timing and incentives: sell early, expect the collateral to be gone, and look to existing owners first for fast equity.

#### Why the other options are wrong

**B is incorrect.** Polarity flip on timing: sell before filing.
**C is incorrect.** Polarity flip: the best assets are usually already pledged.
**D is incorrect.** Reversed: incumbents are faster; newcomers need time for due diligence.

**LO:** IM-13 b: Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.b

### Solution IM-13 b.6 Answer: B

Bank Q537 • Schweser Module Quiz 95.2 Q3

#### Step 1 – Recall the five limitations

Liquidity, timing, magnitude, complexity and severability.

#### Step 2 – Match B

Liquidity: "the easiest path to deleveraging, such as buying back debt at a discount, may not be feasible if the firm lacks sufficient excess cash".
⇒ B is a genuine limitation.

#### Interpretation

Limitations turn a theoretical menu into a practical one: the cheapest remedy, a discounted buy-back, needs exactly what a distressed firm lacks.

#### Why the other options are wrong

**A is incorrect.** Absolute and wrong: selling a unit takes time (the timing limitation).
**C is incorrect.** Absolute: noncore units add options (severability), but "always" overstates it, and it is not a limitation.
**D is incorrect.** Reversed: complex, layered structures (not simple ones) make restructuring harder.

**LO:** IM-13 b: Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.b

**Solution IM-13 b.7** **Answer: A**

Bank Q539 • Schweser Module Quiz 95.2 Q4

**Step 1 – Out of court**

Voluntary negotiation between the company and its creditors; "generally faster and less costly to execute", but exposed to holdouts.

**Step 2 – In court**

Court supervision, which "can be slower, more expensive, and potentially damaging to the firm's reputation".

$\Rightarrow$ A states the difference.

**Interpretation**

The choice is consent against cost: the cheap route can be blocked by holdouts, and the route that cannot be blocked is the expensive one. Prepackaged bankruptcies combine the two.

**Why the other options are wrong**

**B is incorrect.** Reversed: OCRs create the holdout risk, since participation is voluntary.

**C is incorrect.** Reversed: court proceedings are public and can damage reputation; OCRs are more discreet.

**D is incorrect.** Absolute claim: creditors do not always prefer court, which is more rigid, slower and costlier.

**LO:** IM-13 b: Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method. • **Reference:** Ippolito, Private Capital Investing (2020), Ch.20; SchweserNotes 2026 Book 5, Reading 95, LO 95.c (re-filed)

IM-15 • Market-Driven Scenarios: An Approach for Plausible Scenario Construction • Vardeez: VERY IMPORTANT

# IM-15 a

P3 CONCEPT

Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios.

Workbook: C9 Stress-Testing Practices and Scenario Design • Tier C • Map #38

**WHAT IS TESTED HERE**

**Why P3.** A Tier C workbook topic (C9, Map #38, 3 of 16 recall sheets) with no Vardeez pointer for this LO. Vardeez's focus in IM-15 is on the z-scores (IM-15 b) and on building a scenario (IM-15 c), both in P2.

**Reported exam shapes**

- **S2** (C9) Market-driven scenarios: plausible scenario construction, z-scores and covariance choice. Workbook C9.2 touches this LO; it is worked as Question IM-15 b.8 (Question IM-15 b.8).

**What the questions look like:** statement questions on why the approach exists (the pre-crisis models it replaces), what it is (a structured way of aggregating expert views into forward-looking scenarios, then disciplining them statistically), and how to evaluate it (subjectivity remains but becomes transparent; it is not a tail-loss measure). The mechanics (policy variables, implied shocks, Mahalanobis distance) are in IM-15 b, Section 1, and the five steps are in IM-15 c; this block recaps them only as far as the evaluation needs.

## 1. The problem MDS was built to solve

- **Before the Global Financial Crisis (GFC)**, stress testing was an *add-on* to statistical models such as VaR or ex ante tracking error, used to estimate the P&L impact of tail risks. The models and their "model-consistent" stress tests depended heavily on historical data.
- **In the GFC** the historical relationships between key market variables broke down entirely, and these models became unreliable.
- **Scenario analysis** emerged as the alternative because it can describe forward-looking events. Its weakness: no established framework for constructing scenarios, so they risk being ad hoc.

The MDS approach supplies that framework: a structured process designed to **reduce subjectivity** in scenario analysis.

## 2. The wisdom of crowds

James Surowiecki (*The Wisdom of Crowds*) argues that decisions improve when information is aggregated from many individuals, provided three criteria hold. MDS is designed to meet all three:

| Surowiecki's criterion | How MDS meets it |
|---|---|
| 1 Each individual has specialized knowledge not available to others | Input from a diverse group of investment professionals across asset classes, regions and areas of expertise |
| 2 Opinions are independent, not shaped by groupthink | Views are proposed separately and then made explicit, so they can be challenged |
| 3 An objective mechanism aggregates private views into a collective assessment | The risk management department runs a centralized process that aggregates the ideas, constrained by statistical tools |

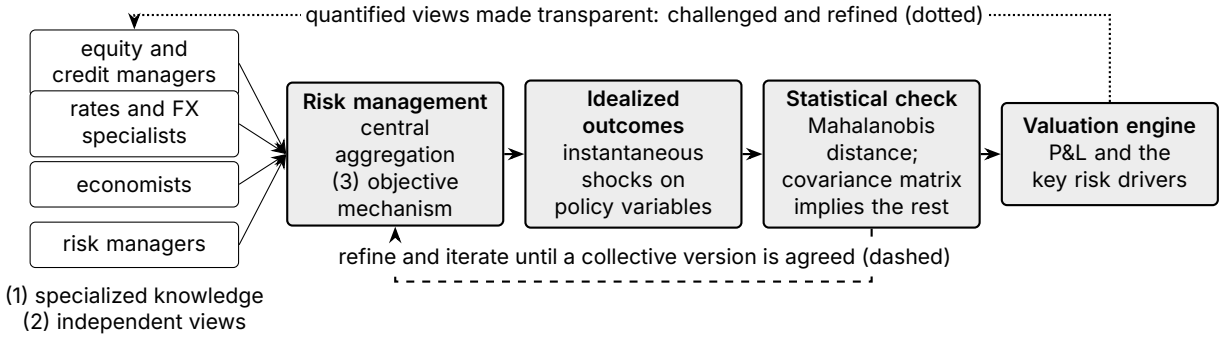



Figure 1. The MDS process as a wisdom-of-crowds mechanism. Creativity enters on the left (experts envision outcomes); statistics constrain it in the middle (plausibility check, covariance matrix); the P&L on the right is fed back to the experts. Solid arrows: the forward flow; dashed: the refinement loop; dotted: the challenge loop.

## 3. How the process runs (recap)

Risk managers, investors and economists periodically identify current issues that could affect markets (inflation, trade policy, monetary policy). For each issue they define a set of **idealized outcomes** covering a broad spectrum of plausible possibilities, and translate them into **instantaneous shocks on policy variables** (equity, bond or commodity indices). The **Mahalanobis distance** assesses the plausibility and magnitude of each shock vector. The shocks are passed through a **covariance matrix** that is itself reviewed for plausibility, the scenario is refined over several iterations until a collective version is agreed, and the final shocks go into a **valuation engine** that produces the P&L and shows the key risk drivers. The process needs creativity first, then statistical tools to constrain and refine the ideas. Details: the conditional framework and the statistics in IM-15 b; the five steps and the Brexit example in IM-15 c.

## 4. Evaluating the approach

**What it does well**

- **Forward-looking.** It does not depend on historical relationships holding, the failure that the GFC exposed.
- **Structured and systematic.** Diverse, independent views are aggregated through a consistent statistical framework into one coherent collective scenario.
- **Transparent.** Risk takers must quantify their subjective views. Once quantified, the views can be challenged and refined, and this iterative feedback is an essential part of effective portfolio risk management.
- **Disciplined by statistics.** The Mahalanobis distance and z-scores keep the shocks plausible; the covariance matrix keeps the other factors consistent with them.
- **A tool for thinking, not a loss measure.** Its goal is not to quantify tail losses or make managers target outcomes, but to integrate insights from different perspectives. Seeing returns under scenarios of varying plausibility makes managers reassess their strategic convictions.

**Where it is weak** (the challenges of IM-15 d, a P4 LO)

- **Subjectivity remains.** The approach rests on expert judgment; there is no universally accepted method and little research on best practice, so firms build their own frameworks, with a risk of bias and groupthink.

- **Translating events into shocks** is itself subjective and may rest on educated guesses rather than structural models.
- **Simplicity versus accuracy**, the **choice of covariance matrix**, and the **judgment** about what is "too extreme" all shape the result.

**Correction to the source notes.** Vardeez frames IM-15 a as a contest between historical replay, macro-narrative stress testing ("the regulatory norm") and MDS, and says MDS has "no macro-to-market mapping to defend". Neither point is the reading's argument. In the reading (per SchweserNotes), MDS answers the failure of pre-GFC stress tests that were add-ons to VaR and tracking-error models built on historical relationships, and its distinctive feature is the structured, wisdom-of-crowds aggregation of expert views. The macro-to-market translation does not disappear: converting economic events into policy-variable shocks is listed as one of the approach's challenges (IM-15 d). Vardeez's point that MDS states its shocks directly in market variables is fair, but answer evaluation questions from the reading's framing.

| | Pre-GFC stress testing | Market-driven scenarios |
|---|---|---|
| **Role** | Add-on to VaR or ex ante tracking error | A framework in its own right for scenario analysis |
| **Basis** | Historical data and relationships | Forward-looking expert views, then statistics |
| **Who supplies the view** | The model | A diverse group of professionals, aggregated centrally by risk management |
| **Role of statistics** | Generates the stress | Constrains and refines it (Mahalanobis distance, covariance matrix) |
| **Subjectivity** | Hidden in model choices | Explicit, quantified, open to challenge |
| **Failure mode** | Historical relationships break down (GFC) | Bias, groupthink, poor covariance choice |
| **Output** | Tail-loss estimate | P&L under scenarios of varying plausibility; the risk drivers |

**Exam traps**

- **"Removes subjectivity" (absolute claim).** MDS reduces and exposes subjectivity; it does not eliminate it.
- **"Relies on historical relationships" (wrong model).** That describes the pre-GFC approach MDS was built to replace. The covariance matrix is historical, but the scenarios are forward-looking.
- **"No covariance matrix needed" (polarity).** The policy shocks are passed through a covariance matrix to imply every other factor's shock.
- **Nonpolicy shocks set by the experts (sibling swap).** Experts shock only the policy variables; the rest are implied by regression.
- **Criterion mix-ups.** Diversity of experts gives specialized knowledge; the centralized process run by risk management is the objective aggregation mechanism; adopting a house view would breach independence.
- **A tail-loss or target tool (purpose error).** MDS integrates views and challenges convictions; it does not quantify tail losses or set loss targets.

## Practice questions: IM-15 a

### Question IM-15 a.1

Bank Q548 • Schweser Module Quiz 97.1 Q1

Which of the following statements best describes the key advantage of the market-driven scenario (MDS) approach compared with traditional stress testing methods used before the Global Financial Crisis (GFC) of 2007 to 2009? The MDS approach:

**A.** removes subjectivity entirely by relying on fully automated statistical models.

**B.** relies solely on historical relationships between market variables to estimate P&L impacts.

**C.** eliminates the need for covariance matrices by focusing only on individual asset-level shocks.

**D.** provides a structured framework for aggregating diverse expert opinions into forward-looking scenarios.

### Question IM-15 a.2

New • built from SchweserNotes 2026 Reading 97

The head of risk at a global asset manager tells the board that the firm's market-driven scenario (MDS) process is designed to meet James Surowiecki's three criteria for good collective decisions. Which of the following correctly matches a feature of the MDS process with the criterion it satisfies?

**A.** Requiring every expert to adopt the firm's house economic view before proposing shocks satisfies the criterion that opinions be independent.

**B.** A centralized process run by the risk management department to aggregate the ideas into a collective view satisfies the criterion of an objective aggregation mechanism.

**C.** Passing the policy shocks through a covariance matrix satisfies the criterion that each individual possess specialized knowledge not available to others.

**D.** Letting the experts settle the scenario among themselves, without the risk function, satisfies the criterion of an objective aggregation mechanism.

### Question IM-15 a.3

Bank Q550 • New, written for the bank (GARP style)

A risk analyst at a multi-asset fund applies the MDS approach as a conditional stress testing framework. The analyst specifies shocks to a small number of policy variables, such as an equity index and 10-year bond yields, and the portfolio also holds a book of equity options. Which of the following statements about the framework is correct?

**A.** Shocks to the nonpolicy factors are set independently by the experts, and the covariance matrix is used only to check them afterward.

**B.** The option positions can be handled with the same linear factor-exposure calculation as the cash positions because the policy shocks are small in number.

**C.** Shocks to the nonpolicy factors are implied from the policy variable shocks using multivariate regression, and the option positions are fully revalued using the full perturbation vector.

**D.** Keeping the number of policy variables small increases the arbitrariness of the perturbation vector because fewer factors are directly shocked.

**Question IM-15 a.4** Bank Q549 • New, written for the bank (GARP style)

A chief risk officer at an asset manager is presenting the firm's new market-driven scenario (MDS) process to the investment committee. A portfolio manager objects that because the scenarios rely on views from investors and economists, the output is just opinion and has no value. Which of the following responses is most consistent with the MDS approach?

**A.** The objection is valid, so MDS output should be used only after replacing expert views with a purely historical covariance model.

**B.** The process still contains subjectivity, but it forces risk takers to quantify their views, which makes them transparent and open to challenge and refinement.

**C.** The objection is invalid because aggregating views from many experts through a statistical framework removes subjectivity entirely.

**D.** The objection is valid for the policy variables but not for the nonpolicy factors, because only the nonpolicy factor shocks are determined subjectively.

**Question IM-15 a.5** Bank Q557 • Schweser Module Quiz 97.2 Q2

Which of the following statements best describes a key challenge in implementing an MDS approach?

**A.** Scenario development often depends on expert judgment.

**B.** Using only historical data is recommended when constructing scenarios.

**C.** Using a large number of policy variables guarantees more accurate and reliable scenario outputs.

**D.** Translating economic events into quantitative policy shocks relies solely on objective statistical models.

**Question IM-15 a.6** New • built from SchweserNotes 2026 Reading 97

At an investment committee meeting, the CIO of a balanced fund proposes replacing the fund's 99% VaR limit with the loss from the firm's Soft Brexit market-driven scenario, which he describes as "our new measure of tail loss". Which of the following responses is most consistent with the purpose of the MDS approach?

**A.** The proposal is appropriate, because the MDS approach is designed to quantify the tail loss of a portfolio at a stated confidence level.

**B.** The proposal is appropriate, because a scenario's P&L can be converted into an exact probability of loss without any further assumptions.

**C.** The proposal misreads the approach: MDS does not quantify tail losses or set targets; it integrates views in a structured way and prompts managers to reassess their convictions.

**D.** The proposal misreads the approach: MDS scenarios are built only from historical episodes, so they cannot inform any forward-looking decision.

# Solutions • IM-15 a

## Solution IM-15 a.1

**Answer: D**

Bank Q548 • Schweser Module Quiz 97.1 Q1

### Step 1 – What failed before the GFC

Stress tests were add-ons to VaR or tracking-error models and relied on historical relationships between market variables, which broke down during the crisis.

### Step 2 – What MDS adds

It gathers views from professionals across asset classes, regions and areas of expertise and aggregates them through a centralized risk management process, meeting Surowiecki's criteria. The policy shocks are then constrained with statistical tools (Mahalanobis distance, covariance matrix), refined over several iterations and run through a valuation engine.
$\Rightarrow$ A structured, forward-looking framework that aggregates diverse views: D.

### Interpretation

The advantage is the structure, not the absence of judgment: subjective views are channeled into a disciplined, transparent process.

### Why the other options are wrong

**A is incorrect.** Absolute claim: MDS reduces subjectivity and makes it transparent; it does not remove it, and it is not fully automated.
**B is incorrect.** Wrong model: reliance on historical relationships describes the pre-GFC approach MDS was built to overcome.
**C is incorrect.** Polarity flip: policy shocks are passed through a covariance matrix to derive every other factor's move, so covariances are central.

**LO:** IM-15 a: Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.a; Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.5

## Solution IM-15 a.2

**Answer: B**

New • built from SchweserNotes 2026 Reading 97

### Step 1 – The three criteria

(1) Each individual has specialized knowledge not available to others; (2) opinions are independent and not influenced by groupthink; (3) there is an objective mechanism for aggregating private views into a collective assessment.

### Step 2 – Match the MDS features

The diverse group of professionals supplies (1). The firm's risk management department "develops a centralized process to aggregate these ideas into a collective view": that is (3).
$\Rightarrow$ B.

### Interpretation

The mapping explains why the aggregation sits with risk management rather than with any one investment team: an aggregator that has no stake in a particular view is what makes the mechanism objective.

### Why the other options are wrong

**A is incorrect.** Polarity flip: imposing a house view is the groupthink that the independence criterion rules out.
**C is incorrect.** Sibling swap: the covariance matrix is a statistical tool that implies the nonpolicy

shocks; specialized knowledge comes from the diverse experts.
**D is incorrect.** Removing the central risk function removes the objective aggregator; a consensus reached among the experts themselves invites groupthink.

**LO:** IM-15 a: Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.a (Surowiecki, The Wisdom of Crowds)

### Solution IM-15 a.3 — Answer: C

Bank Q550 • New, written for the bank

#### Step 1 – Who sets which shocks

The experts specify shocks only for the $K$ policy variables. The remaining $M - K$ nonpolicy shocks are implied by multivariate regression on the policy shocks, which is equivalent to using the covariance matrix.

#### Step 2 – How the P&L is computed

Linear positions use exposures times shocks. Positions with embedded optionality have nonlinear payoffs and are fully revalued with the full perturbation vector.
$\Rightarrow$ C.

#### Interpretation

The conditional design keeps the expert input small and the rest consistent with history. The equity options are the reason the full perturbation vector is needed, not just the P&L shortcut.

#### Why the other options are wrong

**A is incorrect.** Reverses the mechanics: nonpolicy shocks are implied from the policy shocks, not set by the experts and checked later.
**B is incorrect.** The number of policy variables has nothing to do with linearity; options need full revaluation.
**D is incorrect.** Polarity flip: a small number of policy variables *minimizes* the arbitrariness of the perturbation vector.

**LO:** IM-15 a: Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.a and 97.b, Conditional Stress Testing Framework

### Solution IM-15 a.4 — Answer: B

Bank Q549 • New, written for the bank

#### Step 1 – Concede what is true

The approach does contain an element of subjectivity: the scenarios start from expert views.

#### Step 2 – State where the value lies

It requires risk takers to work through a structured process that quantifies their subjective views. Once quantified, the views are transparent and can be challenged and refined, and that iterative feedback is an essential component of effective portfolio risk management.
$\Rightarrow$ B.

#### Interpretation

The portfolio manager's objection assumes the alternative is objective. The pre-GFC alternative was a model built on historical relationships that failed when they broke down.

#### Why the other options are wrong

**A is incorrect.** Replacing views with a purely historical model returns to the pre-GFC weakness

that MDS addresses.
**C is incorrect.** Absolute claim: the reading says an element of subjectivity remains.
**D is incorrect.** Swaps the roles: policy shocks come from expert judgment; nonpolicy shocks are implied through the covariance matrix.

**LO:** IM-15 a: Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.a

### Solution IM-15 a.5 — Answer: A

Bank Q557 • Schweser Module Quiz 97.2 Q2

#### Step 1 – Where judgment enters

Choosing the event, translating it into policy-variable shocks and judging plausibility all need expert judgment. There is no universally accepted method for constructing scenarios, which can lead to inconsistencies, ad hoc decisions, bias and groupthink.

#### Step 2 – Test the options

Only A describes a genuine challenge; B, C and D contradict the reading's guidance.
⇒ A.

#### Interpretation

This is the evaluation side of IM-15 a: the same expert input that makes MDS forward-looking is its main weakness, which is why the structure, the transparency and the statistical checks matter.

#### Why the other options are wrong

**B is incorrect.** MDS exists because purely historical scenarios cannot capture events that have not happened; the covariance data are historical, the shocks are not.
**C is incorrect.** Absolute claim and polarity: more policy variables add redundancy and multicollinearity; the reading recommends a small set.
**D is incorrect.** Converting economic events into policy shocks is inherently subjective; it does not rely solely on objective models.

**LO:** IM-15 a: Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.d (filed there in the bank; used here for the evaluation)

### Solution IM-15 a.6 — Answer: C

New • built from SchweserNotes 2026 Reading 97

#### Step 1 – What the approach is for

"The goal of the MDS approach is not to quantify tail losses or force risk managers to target specific outcomes. Instead, it is a tool for integrating insights from different perspectives in a structured way."

#### Step 2 – What it gives instead

By quantifying returns under scenarios of varying plausibility, it encourages managers to reassess their strategic convictions and increases awareness of alternative outcomes.
⇒ The CIO's proposal misreads the approach: C.

#### Interpretation

A scenario loss is the outcome of one coherent story, with no confidence level attached. It complements a VaR limit; it does not replace it.

#### Why the other options are wrong

**A is incorrect.** Purpose error: quantifying tail losses at a confidence level is what the reading

says MDS is not for.

**B is incorrect.** Absolute claim: scenario z-scores give only relative probabilities; an absolute probability needs further assumptions (IM-15 b).

**D is incorrect.** Right verdict, wrong reason: MDS scenarios are forward-looking, not built only from historical episodes.

**LO:** IM-15 a: Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios. • **Reference:** SchweserNotes 2026 Book 5, Reading 97, LO 97.a and LO 97.c (closing paragraph of the Brexit example)

IM-17 • Illiquid Assets • Vardeez: VERY IMPORTANT

# IM-17 d

P3 CONCEPT

Explain the process of unsmoothing returns and the effects of unsmoothing.

Workbook: D9 Illiquid Assets • Tier D • Map #48

**WHAT IS TESTED HERE**

**Why P3.** Tier D workbook topic: section D9 (Map #48, an "illiquid asset question" on 2 of 16 recall sheets).
**Vardeez:** no pointer for this LO; the reading is rated VERY IMPORTANT, with LO 99.e and 99.f flagged Important.

**Reported exam shape**

- **D9·S1** Biases in reported illiquid asset returns: survivorship, selection, infrequent trading, unsmoothing. This block covers the unsmoothing half: what the filter does, what it changes and what it leaves alone. The biases themselves are drilled in IM-17 c (P1).

**What the questions look like:** statements about the filter. Does volatility (and correlation, and beta) go up or down? Does the mean move? What is the role of the autocorrelation, and what happens when it is zero? Which bias does unsmoothing fix, and which does it not? No calculation has been reported: the formula matters for its properties, not for number crunching.

## 1. Recap: why reported returns are smooth

IM-17 c, Section 5, set out the infrequent trading bias. Illiquid assets are valued by appraisal or at rare transactions. Appraisers anchor on the last valuation and on stale comparable sales, so only part of each true price change reaches the reported index in the quarter it happens; the rest leaks in over the following quarters. Reported returns are therefore **positively autocorrelated** (NCREIF: 0.78), the fingerprint of smoothing, and the volatility, correlations and betas computed from them are **too low**. Unsmoothing (or de-smoothing) undoes this. Ang calls it a *filter problem*: go from a series that is too smooth to the true, noisier one.

## 2. The process (Geltner, Ross and Zisler)

1. **Model the reported returns.** Treat them as an AR(1) process and estimate $\phi$, the first-order autocorrelation:
$$r_t^* = c + \phi\, r_{t-1}^* + \varepsilon_t, \qquad |\phi| < 1.$$
Equivalently, the reported return is a weighted average of the true return $r_t$ and last period's reported return:
$$r_t^* = (1-\phi)\, r_t + \phi\, r_{t-1}^*.$$

2. **Invert the smoothing.** Solve for the true return:
$$\boxed{r_t = \frac{1}{1-\phi}\, r_t^* - \frac{\phi}{1-\phi}\, r_{t-1}^* = \frac{r_t^* - \phi\, r_{t-1}^*}{1-\phi}}$$
Removing the carried-over part $\phi\, r_{t-1}^*$ and scaling up by $1/(1-\phi) > 1$ restores the full size of each shock.

3. **Check the output.** The recovered returns should be close to IID, with no autocorrelation left. If autocorrelation remains at some lag, the model is misspecified.

**A filter that works backwards.** An ordinary filter removes noise to reveal a signal. Unsmoothing *adds noise back* to recover the true, noisier returns. If true returns are IID with volatility $\sigma$, the smoothed series has variance $\sigma^2(1-\phi)/(1+\phi)$, so the true volatility is the larger one whenever $0 < \phi < 1$:

$$\frac{\sigma_{\text{true}}}{\sigma_{\text{reported}}} = \sqrt{\frac{1+\phi}{1-\phi}}: \qquad \phi = 0.50 \Rightarrow 1.73, \quad \phi = 0.75 \Rightarrow 2.65, \quad \phi = 0.78 \Rightarrow 2.84.$$

## 3. The effects of unsmoothing: Ang's three properties

1. **It affects risk estimates, not expected returns.** Smoothing spreads each shock over several periods but still counts all of it, so the first and last prices are unchanged and so is the mean. Exactly, summing the filter over $T$ periods gives

$$\sum_{t=1}^{T} r_t = \sum_{t=1}^{T} r_t^* + \frac{\phi}{1-\phi}\left(r_T^* - r_0^*\right),$$

   a single end term that is negligible over a long sample (Figure 1, lower panel). Individual period returns, by contrast, change a lot: the worst quarters become much deeper.

2. **It has no effect if the observed returns are uncorrelated.** With $\phi = 0$ the filter returns $r_t = r_t^*$: there is no smoothing to undo. Unsmoothing does not "always" raise volatility.

3. **It is an art.** Real estate is reappraised about once a year, which puts a fourth-order lag into quarterly returns, so a first-order filter is not enough. A good procedure fits an ARMA model and draws on economic knowledge of how the market is valued.

**What moves:** volatility, correlations and betas up; Sharpe ratio down (same mean, larger denominator); drawdowns deeper; the mean-variance allocation smaller.

## 4. The evidence

| NCREIF (US institutional real estate), quarterly | Reported | Unsmoothed |
|---|---|---|
| First-order autocorrelation | 0.78 | close to 0 (by design) |
| Volatility (stocks: about 7.5%) | 2.25% | 6.26% |
| Correlation with the S&P 500 | 9.2% | 15.8% |
| Worst quarter of the early 1990s downturn (December 1991) | −5.3% | −22.6% |
| December 2008 | −8.3% | −36.3% |

Art indexes tell the same story: volatility rises from 6.5% to 11.5% after unsmoothing. As a cross-check, a plain AR(1) filter with $\phi = 0.78$ would scale volatility by 2.84, to about 6.4%, close to the 6.26% reported; Ang's procedure is richer than a single lag, as property 3 requires.

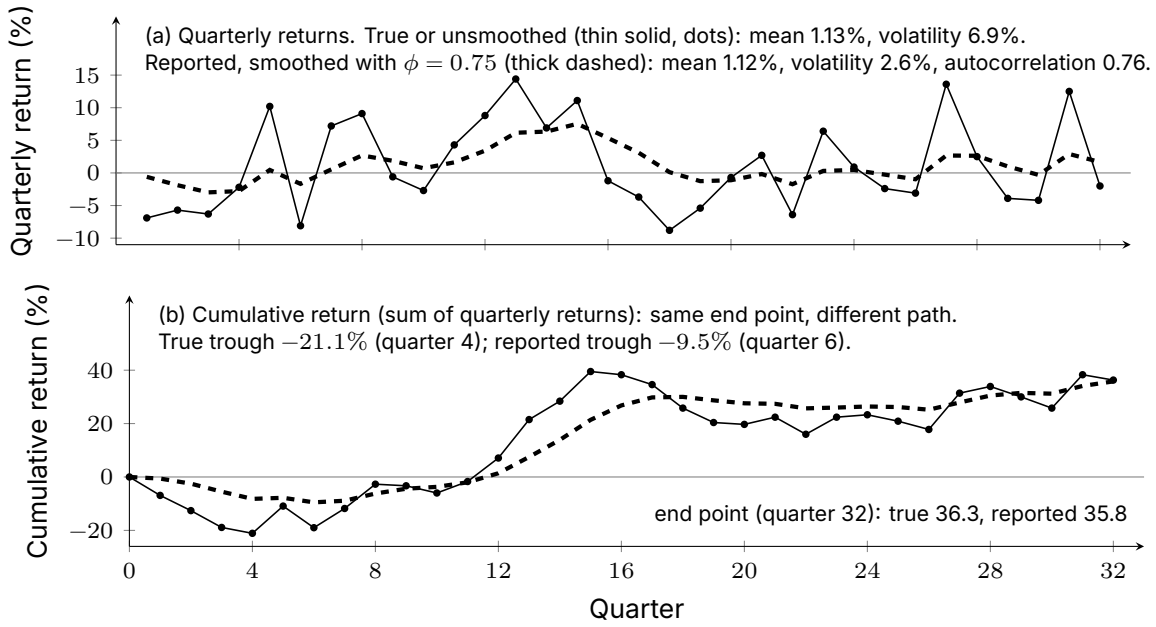



Figure 1. A simulated appraisal index: each reported return keeps 75% of last quarter's reported return and adds 25% of the true return. Unsmoothing the dashed series with $\phi = 0.75$ recovers the solid series exactly. (a) Volatility rises by a factor of 2.6 (theory 2.65). (b) The mean, and so the end point, is essentially unchanged; only the path, and every risk measure built on it, changes.

**Correction to the source notes.**

- SchweserNotes (LO 99.d and its key concept) says unsmoothing "affects risk and return estimates, and could have a dramatic effect on returns". It changes *individual period* returns (December 1991: $-5.3\%$ becomes $-22.6\%$), but not the *expected* return: summing the filter leaves the mean unchanged apart from one end term (Section 3). Ang's first property, as Vardeez and Workbook D9 report it, stands: "unsmoothing affects only risk estimates and not expected returns". Use that on the exam.
- Schweser reports "the standard deviation of the raw returns was 2.25% during the same quarter compared to 6.26% for unsmoothed returns". A standard deviation is measured over the whole sample, not within one quarter: 2.25% and 6.26% are the quarterly volatilities of the full NCREIF series.

## Formula and comparison card: IM-17 d

| | |
|---|---|
| Smoothing model | $r_t^* = (1-\phi)\, r_t + \phi\, r_{t-1}^*$, equivalently AR(1): $r_t^* = c + \phi\, r_{t-1}^* + \varepsilon_t$, $\lvert\phi\rvert < 1$ |
| Unsmoothing filter | $r_t = \left(r_t^* - \phi\, r_{t-1}^*\right)/(1-\phi)$ |
| Volatility (IID true returns) | $\sigma_{\text{true}}/\sigma_{\text{reported}} = \sqrt{(1+\phi)/(1-\phi)}$; no change when $\phi = 0$ |

| | Reported (smoothed) | Unsmoothed |
|---|---|---|
| **Autocorrelation** | Positive (NCREIF 0.78) | Close to zero |
| **Volatility** | Too low (2.25% a quarter) | Higher (6.26%) |
| **Correlation, beta with equities** | Too low (9.2%) | Higher (15.8%) |
| **Mean return** | Unbiased by smoothing | Essentially the same |
| **Worst quarter** | Shallow ($-8.3\%$) | Deep ($-36.3\%$) |
| **Sharpe ratio** | Too high | Lower |
| **Mean-variance allocation** | Too large | Smaller |

**Exam traps**

- **Direction (polarity flip).** Unsmoothing raises volatility, correlation and beta; it never lowers them.
- **"Unsmoothing changes the expected return."** False: risk only. A deeper worst quarter is not a lower mean.
- **"Unsmoothing always raises volatility."** False: with uncorrelated reported returns ($\phi = 0$) it does nothing.
- **Sign of the autocorrelation.** Smoothing creates *positive* autocorrelation; the fix uses it.
- **Adds noise, does not remove it.** An ordinary filter removes noise; unsmoothing adds it back.
- **Remedy mismatch (sibling swap).** Unsmoothing fixes infrequent trading only. It cannot recover funds that stopped reporting (survivorship) or values that were never recorded (selection, which needs Heckman models).
- **"A longer history fixes it."** Smoothing is a population problem, not a small-sample problem.
- **One lag is not always enough.** Annual reappraisals put a fourth-order lag into quarterly returns.

## Practice questions: IM-17 d

### Question IM-17 d.1

Bank Q607 • AnalystPrep Q.2435 (rebuilt)

A fund holds illiquid assets that trade infrequently, so its reported returns are smoothed. An analyst applies an unsmoothing (de-smoothing) procedure to the reported return series. Which of the following best describes the effect of unsmoothing?

**A.** It adds noise back to the reported returns, so estimated volatility rises and correlations with liquid asset classes typically rise.

**B.** It removes noise from the reported returns, so estimated volatility falls and the returns look more stable.

**C.** It leaves volatility unchanged but raises the mean return to offset survivorship bias.

**D.** It eliminates sample selection bias by including the returns of funds that stopped reporting.

### Question IM-17 d.2

Bank Q610 • GARP PE1 Q60 • Mock 3 Q21 (rebuilt, key corrected in bank)

An investor is performing due diligence on a hedge fund that invests in illiquid assets. The investor obtains monthly return data on the fund but is concerned about possible bias in the data, which may provide a misleading impression of the fund's risk profile. Which of the following is the correct assessment?

**A.** Volatility will be artificially high, giving the appearance of high total risk, which can be corrected by taking into account the resulting positive autocorrelation of returns.

**B.** Correlations with other asset classes will be artificially high, giving the appearance of high systematic risk, which can be corrected by unsmoothing the returns.

**C.** Volatility will be artificially low, giving the appearance of low total risk, which can be corrected by taking into account the resulting positive autocorrelation of returns.

**D.** Volatility will be artificially low, giving the appearance of low total risk, which can be corrected by taking into account the resulting negative autocorrelation of returns.

### Question IM-17 d.3

Bank Q609 • BT P2.T8.707.3 (rebuilt, key corrected in bank)

Reported returns of an illiquid asset are smoothed according to $r_t^* = (1-\phi)\, r_t + \phi\, r_{t-1}^*$, where $r_t$ is the true unobserved return, $r_t^*$ is the reported return and $0 < \phi < 1$. Unsmoothing recovers the true returns. Each of the following statements about unsmoothing is correct EXCEPT:

**A.** The unsmoothed returns have a higher variance than the reported returns.

**B.** If $\phi = 0$, unsmoothing leaves the reported returns unchanged.

**C.** The unsmoothed true returns should show approximately zero autocorrelation.

**D.** Unsmoothing lowers the estimated volatility of the asset and its correlation with liquid asset classes.

**Question IM-17 d.4** New • built from SchweserNotes 2026 Reading 99

The head of real estate at a Canadian pension plan unsmooths the quarterly returns of the plan's appraisal-based property index before the annual asset allocation review. The reported series has a first-order autocorrelation of 0.70, and the properties are reappraised once a year. Which of the following statements about the unsmoothed series is correct?

**A.** Its average quarterly return will be well below the reported average, because unsmoothing restores the losses that the appraisers failed to record.

**B.** Its average quarterly return will be close to the reported average, because smoothing spreads each shock over several quarters without removing it.

**C.** A first-order AR(1) filter is sufficient for this index, because the annual reappraisal cycle affects the level of the index but not the lag structure of its returns.

**D.** Its volatility will be lower than the reported volatility, because the filter removes the noise that appraisal errors add to the reported returns.

**Question IM-17 d.5** New • built from SchweserNotes 2026 Reading 99

A risk manager at a Dutch insurer presents the following statistics for a private real estate index to the investment committee. Over the same period the quarterly volatility of US equities was about 7.5%.

| | Reported | Unsmoothed |
|---|---|---|
| First-order autocorrelation | 0.78 | close to 0 |
| Quarterly volatility | 2.25% | 6.26% |
| Correlation with the S&P 500 | 9.2% | 15.8% |
| Worst quarter (December 2008) | −8.3% | −36.3% |

Which of the following conclusions should the risk manager draw?

**A.** Real estate's expected return was overstated by about 28 percentage points, so the index mean must be lowered before it enters the optimiser.

**B.** Real estate's Sharpe ratio is lower on unsmoothed data, and an optimiser fed the unsmoothed statistics will allocate less to real estate.

**C.** Because the unsmoothed correlation is still below 20%, smoothing biased the volatility estimate but not the correlation estimate.

**D.** Because reported volatility is less than a third of equity volatility, real estate is a low-risk diversifier whose allocation can be raised.

**Question IM-17 d.6** Workbook D9.1 • S1 • Map #48

A member of the investment committee of a Canadian university endowment is reviewing a consultant's report that relies on two data sets: an appraisal-based institutional real estate index whose quarterly returns show a first-order autocorrelation of 0.75, and a commercial database of venture capital returns to which reporting is voluntary. Which of the following statements is correct to include in the committee's response to the consultant?

**A.** Unsmoothing the real estate index with an AR(1) filter will raise its estimated volatility, correlation and beta with equities, but will leave its estimated expected return essentially unchanged.

**B.** Survivorship bias in the venture capital database makes reported returns understate true returns, because funds that drop out of the database tend to be those with above-average returns.

**C.** Selection bias in the venture capital database, which arises because values are recorded mainly when they are high, is corrected by applying the same Geltner-Ross-Zisler unsmoothing filter.

**D.** The downward bias in the real estate index's volatility is a small-sample problem that will disappear once the consultant uses a sufficiently long history of reported quarterly returns.

## Solutions • IM-17 d

### Solution IM-17 d.1 — Answer: A

Bank Q607 • AnalystPrep Q.2435

**Step 1 – What smoothing did**

Infrequent trading spread each shock over several periods: reported volatility, correlations and betas are too low, and the series is positively autocorrelated.

**Step 2 – What unsmoothing does**

It inverts the smoothing filter, stripping out the carried-over part of last period's return and scaling the rest up by $1/(1-\phi)$.

$\Rightarrow$ The recovered series is noisier: estimated volatility rises, and so do correlations with liquid assets (NCREIF: 9.2% to 15.8%).

**Interpretation**

Unsmoothing is an unusual filter: it adds noise instead of removing it. It leaves the mean return essentially where it was.

**Why the other options are wrong**

**B is incorrect.** Polarity flip: removing noise is what an ordinary filter does; unsmoothing does the opposite.

**C is incorrect.** Volatility does change, the mean does not, and unsmoothing has nothing to do with survivorship bias.

**D is incorrect.** Unsmoothing works on the returns that were reported. It cannot create observations for funds that stopped reporting (survivorship) or values that were never recorded (selection).

**LO:** IM-17 d: Explain the process of unsmoothing returns and the effects of unsmoothing. • **Reference:** Ang, Asset Management (2014), Ch.13, Unsmoothing; SchweserNotes 2026 Book 5, Reading 99, LO 99.d

### Solution IM-17 d.2 — Answer: C

Bank Q610 • GARP PE1 Q60 • Mock 3 Q21

**Step 1 – The bias in the data**

A fund holding illiquid assets reports returns based on stale or appraised prices, so its monthly returns are smoothed.

$\Rightarrow$ Volatility (and correlation and beta) is artificially low: the fund looks less risky than it is.

**Step 2 – Its fingerprint and the fix**

Smoothing makes each reported return depend on the previous one: positive autocorrelation. Unsmoothing estimates that autocorrelation ($\phi$) and uses it to recover the true returns, $r_t = (r_t^* - \phi\, r_{t-1}^*)/(1-\phi)$.

$\Rightarrow$ "Taking into account the positive autocorrelation" is unsmoothing, and it raises the volatility estimate.

**Interpretation**

In a pure AR(1) the correction scales volatility by $\sqrt{(1+\phi)/(1-\phi)}$: with $\phi = 0.5$, by 1.73. The bank had to correct this question's key once; C is the only option that pairs the right direction of bias with the right sign of autocorrelation.

**Why the other options are wrong**

**A is incorrect.** Polarity flip: smoothing makes volatility artificially low, not high.

**B is incorrect.** Polarity flip: smoothing makes correlations artificially low, not high.

**D is incorrect.** Right direction of bias, wrong sign: smoothing creates positive, not negative, autocorrelation.

**LO:** IM-17 d: Explain the process of unsmoothing returns and the effects of unsmoothing. • **Reference:** Ang, Asset Management (2014), Ch.13, Infrequent Sampling and Unsmoothing; SchweserNotes 2026 Book 5, Reading 99, LO 99.c and 99.d; GARP Practice Exam 1, Q60

### Solution IM-17 d.3 Answer: D

Bank Q609 • BT P2.T8.707.3

#### Step 1 – Invert the model

$$r_t = \frac{r_t^* - \phi\, r_{t-1}^*}{1-\phi}$$

#### Step 2 – Test A and B

With IID true returns, $\text{var}(r^*) = \sigma^2(1-\phi)/(1+\phi) < \sigma^2$ for $0 < \phi < 1$, so the unsmoothed series has the higher variance: A is correct. With $\phi = 0$, $r_t = r_t^*$: B is correct.

#### Step 3 – Test C

The model assumes the true returns are uncorrelated; removing the carried-over term $\phi\, r_{t-1}^*$ removes the autocorrelation. The unsmoothed series should look close to IID: C is correct.

#### Step 4 – Test D

Higher variance means a *higher* volatility estimate, and correlations with liquid assets rise too. $\Rightarrow$ D is the incorrect statement.

#### Interpretation

Statement C doubles as the diagnostic: if autocorrelation remains in the output (for example at lag 4 for annually reappraised property), the filter is misspecified.

#### Why the other options are wrong

**A is incorrect.** Correct statement: the true series is noisier than the reported one.
**B is incorrect.** Correct statement: with no smoothing there is nothing to undo.
**C is incorrect.** Correct statement: unsmoothing removes the carry-over that created the autocorrelation.

**LO:** IM-17 d: Explain the process of unsmoothing returns and the effects of unsmoothing. • **Reference:** Ang, Asset Management (2014), Ch.13, Unsmoothing (Geltner, Ross and Zisler filter); SchweserNotes 2026 Book 5, Reading 99, LO 99.d

### Solution IM-17 d.4 Answer: B

New • built from SchweserNotes 2026 Reading 99

#### Step 1 – What smoothing does to the level

Each true shock is spread over several quarters but still counted in full, so the index ends where the true index ends. Summing the filter gives $\sum r_t = \sum r_t^* + \frac{\phi}{1-\phi}(r_T^* - r_0^*)$, a single end term. $\Rightarrow$ The average return is essentially unchanged: Ang's first property.

#### Step 2 – What it does to dispersion

The filter adds noise back, so volatility rises: with $\phi = 0.70$ a pure AR(1) filter would scale it by $\sqrt{1.7/0.3} = 2.38$.

#### Step 3 – What the reappraisal cycle does

Annual reappraisals put a fourth-order lag into quarterly returns. A first-order filter misses it; Ang's third property is that unsmoothing is an art, best done with an ARMA model and knowl-

edge of how the market is valued.

### Interpretation

Unsmoothing changes the path, not the destination. Individual bad quarters become much worse, which is easy to mistake for a lower mean.

### Why the other options are wrong

**A is incorrect.** Confuses deeper individual quarters with a lower mean: unsmoothing affects risk estimates, not expected returns.

**C is incorrect.** Annual reappraisal creates a fourth-order lag in quarterly data; one lag is not enough.

**D is incorrect.** Polarity flip: unsmoothing adds noise back and raises volatility.

**LO:** IM-17 d: Explain the process of unsmoothing returns and the effects of unsmoothing. • **Reference:** Ang, Asset Management (2014), Ch.13, Unsmoothing (three properties), as reported in Vardeez IM-17 d and Workbook D9; SchweserNotes 2026 Book 5, Reading 99, LO 99.d

## Solution IM-17 d.5 Answer: B

New • built from SchweserNotes 2026 Reading 99

### Step 1 – Read the table

Unsmoothing multiplies quarterly volatility by $6.26/2.25 = 2.8$, to a level close to equities (7.5%), raises the correlation with the S&P 500 from 9.2% to 15.8%, and turns a $-8.3\%$ quarter into $-36.3\%$.

### Step 2 – Effect on the Sharpe ratio

The mean is unchanged by unsmoothing, so the excess return stays the same while volatility is 2.8 times larger.

$\Rightarrow$ The Sharpe ratio falls to about a third of its reported value.

### Step 3 – Effect on the allocation

Higher volatility and higher correlation with equities both reduce the optimal mean-variance weight.

$\Rightarrow$ The optimiser allocates less to real estate on unsmoothed data.

### Interpretation

Reported appraisal data make real estate look like a near-riskless diversifier. The committee should put unsmoothed statistics into the optimiser.

### Why the other options are wrong

**A is incorrect.** The $-36.3\%$ is one quarter's unsmoothed return, not a bias in the mean; the expected return is unchanged by unsmoothing. The 28 points is $36.3 - 8.3$, a gap in one quarter's return.

**C is incorrect.** The correlation rose from 9.2% to 15.8%: smoothing biases correlations downward too.

**D is incorrect.** The 2.25% is the smoothed figure. Unsmoothed, real estate volatility (6.26%) is close to that of equities.

**LO:** IM-17 d: Explain the process of unsmoothing returns and the effects of unsmoothing. • **Reference:** Ang, Asset Management (2014), Ch.13, Unsmoothing (NCREIF example); SchweserNotes 2026 Book 5, Reading 99, LO 99.c and 99.d

### Solution IM-17 d.6

**Answer: A**

Workbook D9.1 • S1

#### Step 1 – Read the autocorrelation

A first-order autocorrelation of 0.75 signals appraisal smoothing. Infrequent trading makes "estimates of risk, volatilities, correlations, and betas" too low; unsmoothing adds noise back to recover the true returns.

#### Step 2 – Apply Ang's first property

"Unsmoothing affects only risk estimates and not expected returns", because smoothing spreads shocks over several periods but still counts all of them, so the first and last prices are unchanged. $\Rightarrow$ A is correct. For NCREIF, volatility rose from 2.25% to 6.26% a quarter and the correlation with the S&P 500 from 9.2% to 15.8%.

#### Interpretation

The committee should ask for unsmoothed risk estimates before the real estate allocation enters a mean-variance optimiser; raw appraisal data make the asset look like a near-riskless diversifier. With $\phi = 0.75$ a pure AR(1) filter would scale the reported volatility by $\sqrt{1.75/0.25} = 2.65$.

#### Why the other options are wrong

**B is incorrect.** Polarity flip: survivorship bias "results from the tendency of poorly performing funds to stop reporting", so reported returns overstate true returns.

**C is incorrect.** Sibling swap: unsmoothing corrects infrequent trading. Selection bias is addressed with Heckman (1979) selection models, which let the threshold above which returns are observed vary over time.

**D is incorrect.** Scope error: Ang states that the infrequent-observation bias "is not a 'small sample' problem, which goes away when our sample becomes very large; it is a 'population' problem".

**LO:** IM-17 c: Assess the impact of biases on reported returns for illiquid assets. IM-17 d: Explain the process of unsmoothing returns and the effects of unsmoothing. • **Reference:** Ang, Asset Management, Ch.13 (LTR source text Ch.19), 19.4